



Digitized Automation for a Changing World

Delta SD3 Series Smart Screwdriving Controller (10.1-inch) User Manual

Preface

Thank you for purchasing this product. This manual provides information about the Delta SD3 series smart screwdriving system.

This manual includes:

- Installation and inspection of smart screwdriving controller and transducerized smart screwdriver
- Structure and wiring diagram of smart screwdriving controller
- Description of functions overview
- Description of parameters
- Description of sequence
- Description of sources
- Description of results
- Description of controller
- Description of tool
- Description of reports
- Troubleshooting
- Function codes of Modbus communication
- Function codes of TCP/IP communication
- Description of positioning arms

Product features:

- Multiple tightening strategies: you can customize different tightening strategies for different tightening scenarios to achieve quality requirements.
- New generation motors: the compact design meets the need for reducing the size and weight of the equipment structure.

How to use this manual:

Use this manual as a reference when installing, setting up, using, and maintaining the smart screwdriving system.

Delta technical services:

Consult your Delta equipment distributor or Delta Customer Service Center if you encounter any problems.

Safety precautions

Pay special attention to the following safety precautions at all times during installation, operation, maintenance, and inspection of the smart screwdriving system.

The symbols of “STOP”, “DANGER”, and “WARNING” indicate:



Absolutely prohibited activities. May cause serious damage or even malfunction of the product if the instructions are not followed.



Danger. May cause severe or fatal injuries to personnel if the instructions are not followed.



Warning. May cause moderate injury to personnel, or lead to severe damage or even malfunction of the product if the instructions are not followed.

Installation



- Do not expose the product to an environment containing water vapor, corrosive gas, inflammable gas, or other foreign matter, and do not use it in an environment where liquid leakage may occur. Otherwise, it may cause equipment breakdown, electric shock, or fire.
- The system must be properly connected to a grounded outlet.
- Before connecting or disconnecting the tool cable, make sure to turn off the controller power.

Operation



- During the operation, do not touch any rotating tool parts, or it may cause personal injury.
- Do not hit or drop the tool. Impact force may damage the tool and cause communication and mechanical failures (e.g., the screw bit cannot be installed or removed).



- Only personnel receiving professional training on industrial assembly can operate this product.
- This product is designed for intermittent use. The general duty cycle is to run for 1 second and then stop for 4 seconds. This product includes a protection feature to ensure that the allowable duty cycle (depending on the torque value and operation time) is not exceeded.

Maintenance and inspection



- Do not use cleaning fluids containing solvents to clean the tool parts or cables.
- Only authorized personnel can repair or restart the system.

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Product Overview

Before using the Delta smart screwdriving system, pay attention to the descriptions of the components of the system, nameplate, and model type. You can find suitable products in the table in Section 1.3.

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1.1 Components of the system

A complete Delta smart screwdriving system includes:

- (1) A smart screwdriving controller.
- (2) A transducerized smart screwdriver (two screwdrivers for dual-tool controllers).
- (3) A connection cable (two cables for low torque dual-tool controllers) for connecting the screwdriver to the controller.

Low torque models: with M16 8-pin circular connectors on both ends of the cable.

Medium torque models: with M24 8-pin circular connectors on both ends of the cable.

- (4) Power supply for the controller:

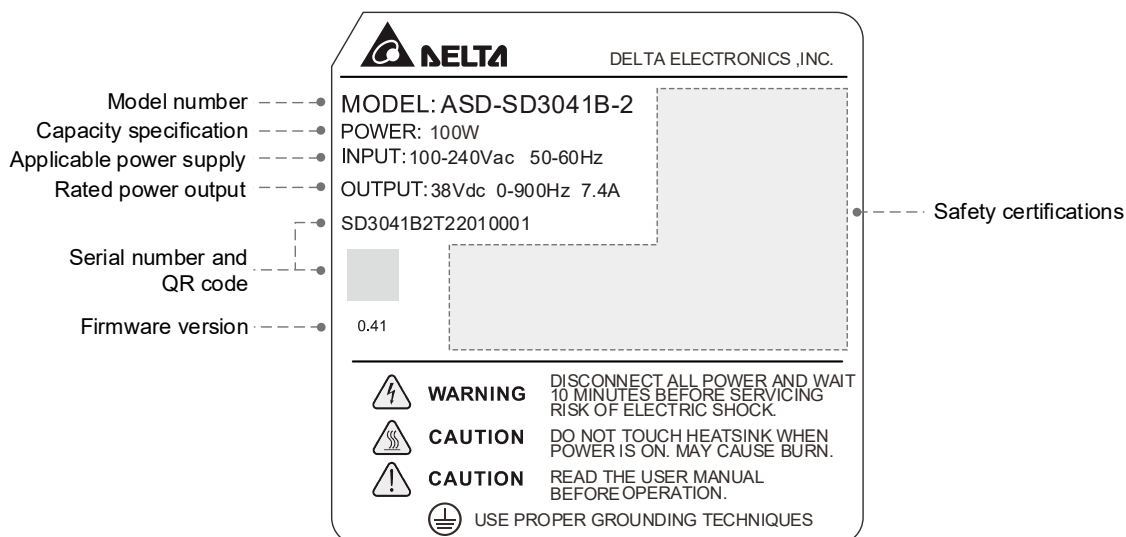
Model	Model number	Control circuit power input
Low torque	ASD-SD3021B-1	100 to 240 V _{AC}
	ASD-SD3041B-2	100 to 240 V _{AC}
Medium torque	ASD-SD3031B-1	100 to 240 V _{AC}

1.2 Model overview

1.2.1 Nameplate information

Delta smart screwdriving system – smart screwdriving controller

■ Nameplate information

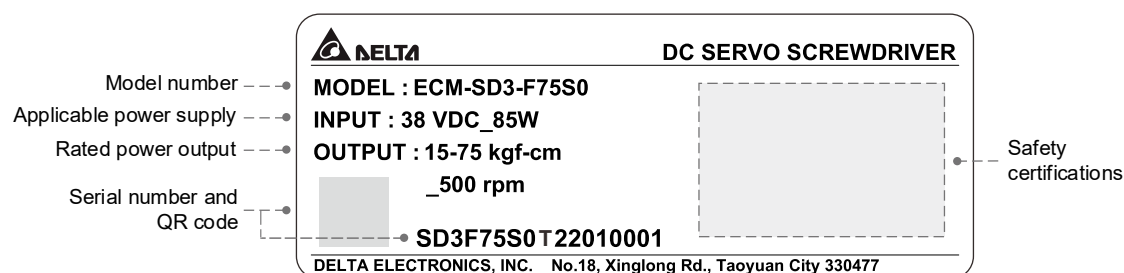


■ Serial number

SD3041B2	T	22	01	0001	
(1)	(2)	(3)	(4)	(5)	
					(1) Model number
					(2) Manufacturing plant (T: Taoyuan; W: Wujiang)
					(3) Year of production (22: year 2022)
					(4) Week of production (from 1 to 52)
					(5) Production sequence in a week (starting from 0001)

Delta smart screwdriving system – transducerized smart screwdriver

■ Nameplate information



■ Serial number

SD3F75S0	T	22	01	0001	
(1)	(2)	(3)	(4)	(5)	
					(1) Model number
					(2) Manufacturing plant (T: Taoyuan; W: Wujiang)
					(3) Year of production (22: year 2022)
					(4) Week of production (from 1 to 52)
					(5) Production sequence in a week (starting from 0001)

1.2.2 Model explanation

Delta smart screwdriving system – smart screwdriving controller

Low torque	Medium torque										
$\frac{\text{ASD}}{(1)} - \frac{\text{SD3}}{(2)} - \frac{02}{(3)} \frac{1\text{B}}{(4)} - \frac{1}{(5)}$	$\frac{\text{ASD}}{(1)} - \frac{\text{SD3}}{(2)} - \frac{03}{(3)} \frac{1\text{B}}{(4)} - \frac{1}{(5)}$										
(1) Product name ASD: AC Servo Drive	(1) Product name ASD: AC Servo Drive										
(2) Series SD3: Screw Driver 3 series	(2) Series SD3: Screw Driver 3 series										
(3) Rated power output <table><tr><th>Code</th><th>Specification</th></tr><tr><td>02</td><td>100 W</td></tr><tr><td>04</td><td>200 W</td></tr></table>	Code	Specification	02	100 W	04	200 W	(3) Rated power output <table><tr><th>Code</th><th>Specification</th></tr><tr><td>03</td><td>150 W</td></tr></table>	Code	Specification	03	150 W
Code	Specification										
02	100 W										
04	200 W										
Code	Specification										
03	150 W										
(4) Power supply 1B: 100 to 240 V _{AC} , 50 / 60 Hz, single-phase	(4) Power supply 1B: 100 to 240 V _{AC} , 50 / 60 Hz, single-phase										
(5) Supported number of tools 1: single-tool 2: dual-tool	(5) Supported number of tools 1: single-tool										

Delta smart screwdriving system – transducerized smart screwdriver

Low torque	Medium torque																		
ECM – SD3 – F 12 S 0 (1) (2) (3) (4) (5) (6)	ECM – SD3 – M 1C S 3 (1) (2) (3) (4) (5) (6)																		
(1) Product name ECM: electronically commutated motor	(1) Product name ECM: electronically commutated motor																		
(2) Series SD3: Screw Driver 3 series	(2) Series SD3: Screw Driver 3 series																		
(3) Shaft type F: straight R: angle	(3) Shaft type F: straight M: fixtured R: angle																		
(4) Torque output <table> <tr> <th>Code</th><th>Specification</th></tr> <tr> <td>12</td><td>2.4 to 12 kgf-cm</td></tr> <tr> <td>30</td><td>6 to 30 kgf-cm</td></tr> <tr> <td>50</td><td>10 to 50 kgf-cm</td></tr> <tr> <td>75</td><td>15 to 75 kgf-cm</td></tr> </table>	Code	Specification	12	2.4 to 12 kgf-cm	30	6 to 30 kgf-cm	50	10 to 50 kgf-cm	75	15 to 75 kgf-cm	(4) Torque output <table> <tr> <th>Code</th><th>Specification</th></tr> <tr> <td>1C</td><td>24 to 120 kgf-cm</td></tr> <tr> <td>1H</td><td>34 to 170 kgf-cm</td></tr> <tr> <td>2F</td><td>50 to 250 kgf-cm</td></tr> </table>	Code	Specification	1C	24 to 120 kgf-cm	1H	34 to 170 kgf-cm	2F	50 to 250 kgf-cm
Code	Specification																		
12	2.4 to 12 kgf-cm																		
30	6 to 30 kgf-cm																		
50	10 to 50 kgf-cm																		
75	15 to 75 kgf-cm																		
Code	Specification																		
1C	24 to 120 kgf-cm																		
1H	34 to 170 kgf-cm																		
2F	50 to 250 kgf-cm																		
(5) Special code S: general type D: for Asia M: for Mexico	(5) Special code S: standard type																		
(6) Bit specifications 0: 6.35 mm (1/4") hex bit 1: 5 mm hex bit	(6) Bit specifications 3: 9.525 mm (3/8") square bit																		

1.2.3 Product specifications

Smart screwdriving controller

Model		Low torque		Medium torque
		ASD-SD3021B-1	ASD-SD3041B-2	ASD-SD3031B-1
Display	Panel type	10.1" TFT LCD (IPS full viewing angle)		
	Touchscreen	4-wire resistive touchscreen Operation expectance > 1,000,000		
	Resolution	800 x 1280, 24-bit RGB		
	Backlight	LED backlight (when room temperature < 25°C (77°F), half-life > 50,000 hours)		
	Brightness	500 cd/m ² (Typ.)		
Serial communication port		1 set of RS-232 port, 2 sets of RS-485 ports		
Network interface		1 port, 10/100 Mbps auto-detection (with built-in isolated circuit)		
HDMI		1 port, HDMI 1.4a transmitter		
USB		1 port, USB 2.0		
Power supply		50 / 60 Hz, 100 to 240 V _{AC}		
Supported number of tools (screwdrivers)		1	2	1
DC power output		1 set of 24 V _{DC} ; max. output: 1 A per tool		
Digital input / output		8 DIs (max. 30 V _{DC}) and 8 DOs (max. output current 150 mA) per tool		
Output voltage		38 V _{DC}		
Continuous output current		3.7 Arms per tool		5.5 Arms per tool
Max. instantaneous output current		15 Arms per tool		16 Arms per tool
Weight		4.06 kg (8.95 lb)		
Ambient temperature		0°C to 40°C (32°F to 104°F)		
Storage temperature		0°C to 60°C (32°F to 140°F)		
Humidity		0% to 90% RH (non-condensing)		
IP rating		IP20		

Transducerized smart screwdriver

Model ECM-SD3-	Low torque (2.4 to 75 kgf-cm)				Medium torque (24 to 250 kgf-cm)		
	F12S0	F30S0	F50S0	F75S0	F1CS3	F1HS3	F2FS3
	F12S1	F30S1	F50S1	F75S1	M1CS3	M1HS3	M2FS3
	R12S0	R30S0	R50S0	R75S0	R1CS3	R1HS3	R2FS3
Torque output (kgf-cm)	2.4 to 12	6 to 30	10 to 50	15 to 75	24 to 120	34 to 170	50 to 250
Maximum speed (rpm)	2000	1100	700	500	1000	700	500
Torque precision	3%						
Weight	F: 560 g (1.23 lb) R: 830 g (1.83 lb)				F: 2000 g (4.41 lb) M: 2400 g (5.29 lb) R: 2330 g (5.14 lb)		
Shaft type	F: straight R: angle				F: straight M: fixtured R: angle		
Bit specifications	Model number that ends with 0: 6.35 mm (1/4") hex bit Model number that ends with 1: 5 mm hex bit				Model number that ends with 3: 9.525 mm (3/8") square bit		
Ambient temperature	0°C to 40°C (32°F to 104°F)						
Storage temperature	0°C to 60°C (32°F to 140°F)						
Humidity	0% to 90% RH (non-condensing)						
IP rating	IP40						

1

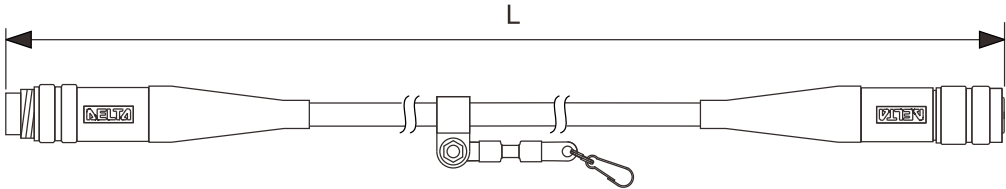
1.3 Controller and screwdriver combination table

Model	Smart screwdriving controller			Transducerized smart screwdriver					
	Model number	Supported number of tools	Power supply	Model number	Torque output (kgf-cm)	Max. speed (rpm)			
Low torque (2.4 to 75 kgf-cm)	ASD-SD3021B-1	1	100 to 240 V _{AC}	ECM-SD3-F12S0 ECM-SD3-F12S1 ECM-SD3-R12S0	2.4 to 12	2000			
				ECM-SD3-F30S0 ECM-SD3-F30S1 ECM-SD3-R30S0	6 to 30	1100			
				ECM-SD3-F50S0 ECM-SD3-F50S1 ECM-SD3-R50S0	10 to 50	700			
				ECM-SD3-F75S0 ECM-SD3-F75S1 ECM-SD3-R75S0	15 to 75	500			
	ASD-SD3041B-2	2		ECM-SD3-F12S0 ECM-SD3-F12S1 ECM-SD3-R12S0	2.4 to 12	2000			
				ECM-SD3-F30S0 ECM-SD3-F30S1 ECM-SD3-R30S0	6 to 30	1100			
				ECM-SD3-F50S0 ECM-SD3-F50S1 ECM-SD3-R50S0	10 to 50	700			
				ECM-SD3-F75S0 ECM-SD3-F75S1 ECM-SD3-R75S0	15 to 75	500			
				Medium torque (24 to 250 kgf-cm)	ASD-SD3031B-1	1	ECM-SD3-F1CS3 ECM-SD3-M1CS3 ECM-SD3-R1CS3	24 to 120	1000
							ECM-SD3-F1HS3 ECM-SD3-M1HS3 ECM-SD3-R1HS3	34 to 170	700
ECM-SD3-F2FS3 ECM-SD3-M2FS3 ECM-SD3-R2FS3	50 to 250	500							

Optional accessories

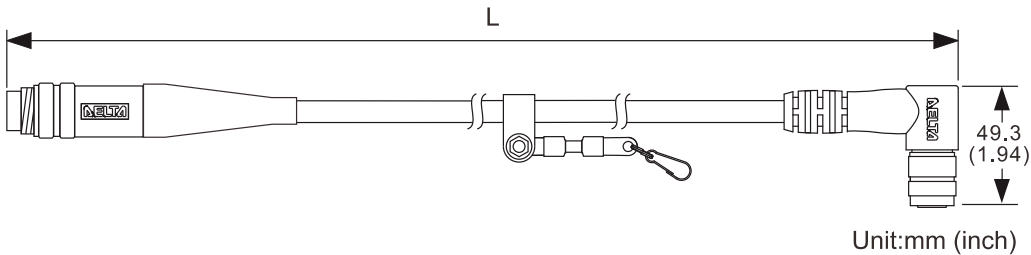
- **Connection cable between low torque screwdriver and controller** (with M16 8-pin circular connectors on both ends)

For screwdriver (straight):



Model number	Length (L)	
	m	feet
ACS3-CASDFH03	3	9.84
ACS3-CASDFH05	5	16.4
ACS3-CASDFH07	7	22.97
ACS3-CASDFH10	10	32.81

For screwdriver (angle):



Model number	Length (L)	
	m	feet
ACS3-CRSDFH03	3	9.84
ACS3-CRSDFH05	5	16.4
ACS3-CRSDFH07	7	22.97
ACS3-CRSDFH10	10	32.81

- **Connection cable between medium torque screwdriver and controller** (with M24 8-pin circular connectors on both ends)

For screwdriver (straight):

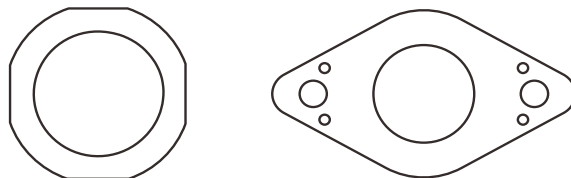


Model number	Length (L)	
	m	feet
ACS3-CASDPB03	3	9.84
ACS3-CASDPB05	5	16.4
ACS3-CASDPB07	7	22.97
ACS3-CASDPB10	10	32.81

- **Flange module** (including a flange and a fixing nut): only applicable to low torque screwdriver (straight)

For fixing the screwdriver to a screw tightening machine or a robotic arm.

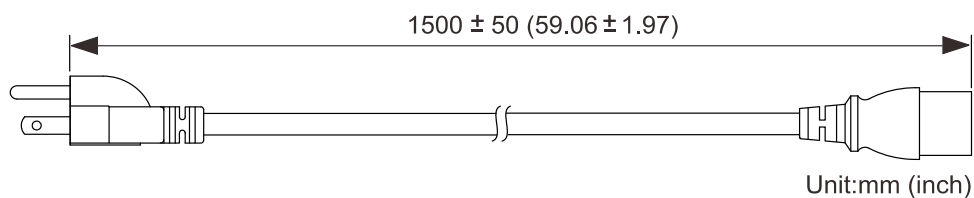
Delta model number: ECM-SD3-ACSSR



Note: refer to Sections 2.4 and 2.5 for the installation method and dimensions.

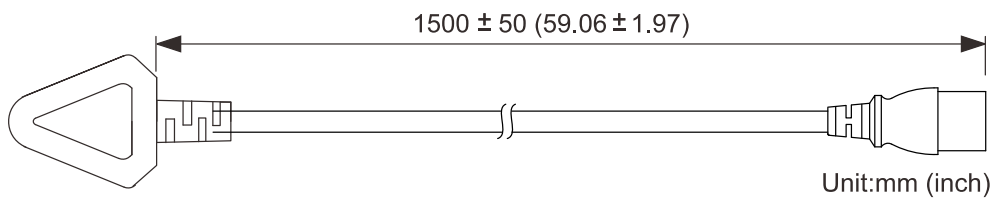
- **AC power cable** (125V / 10 A; 1.5 m; Type B plug, US standard)

Delta model number: ACS3-CAPW311A



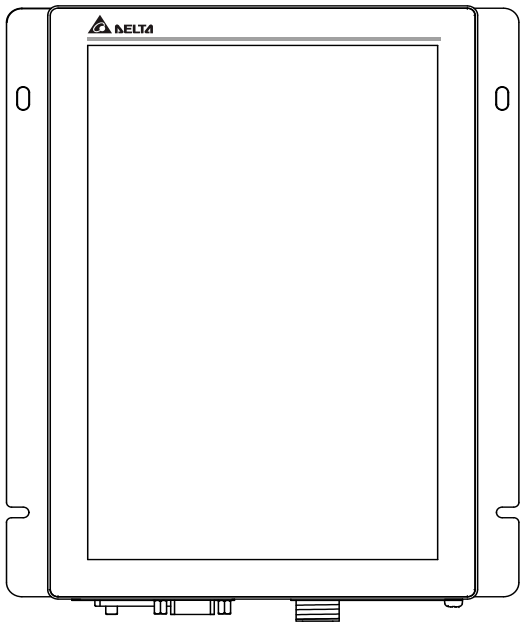
- **AC power cable** (250V / 10 A; 1.5 m; Type I plug, CN standard)

Delta model number: ACS3-CAPW311C

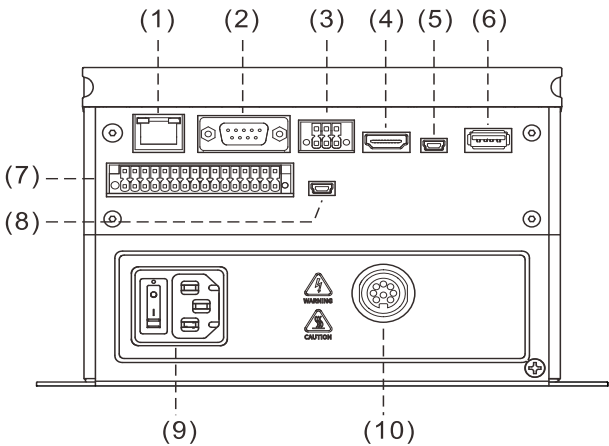


1.4 Description of the smart screwdriving controller interface

Low torque single-tool controller (ASD-SD3021B-1)



Front view

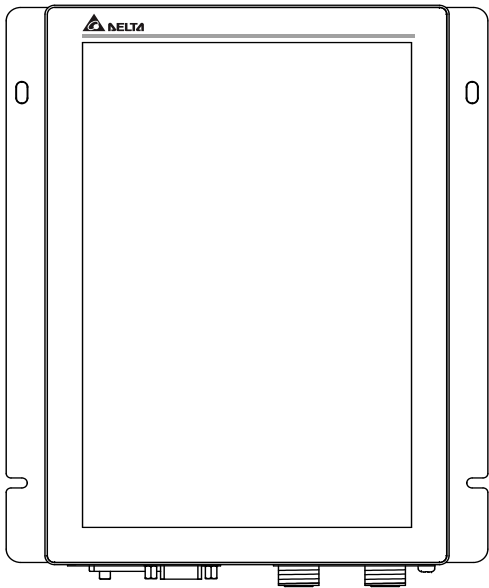


Bottom view

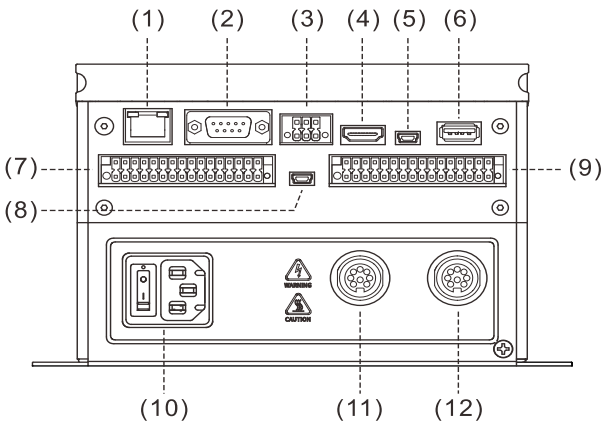
No.	Description	No.	Description
(1)	Standard RJ45 connector – connects to the host system (PC / PLC / HMI)	(6)	Host connector ■ Accesses the tightening data (parameters and results). ■ Connects to the Barcode Scanner.
(2)	Standard RS-232 connector	(7)	Digital I/O port
(3)	Two RS-485 connectors – connect to extra RS-485 devices	(8)	SLAVE-B connector – for the controller firmware update
(4)	HDMI output – synchronously projects the Result screen to an external monitor	(9)	AC connector and switch
(5)	SLAVE-A connector – for the HMI software and firmware update	(10)	Connector for Tool 1

Low torque dual-tool controller (ASD-SD3041B-2)

1



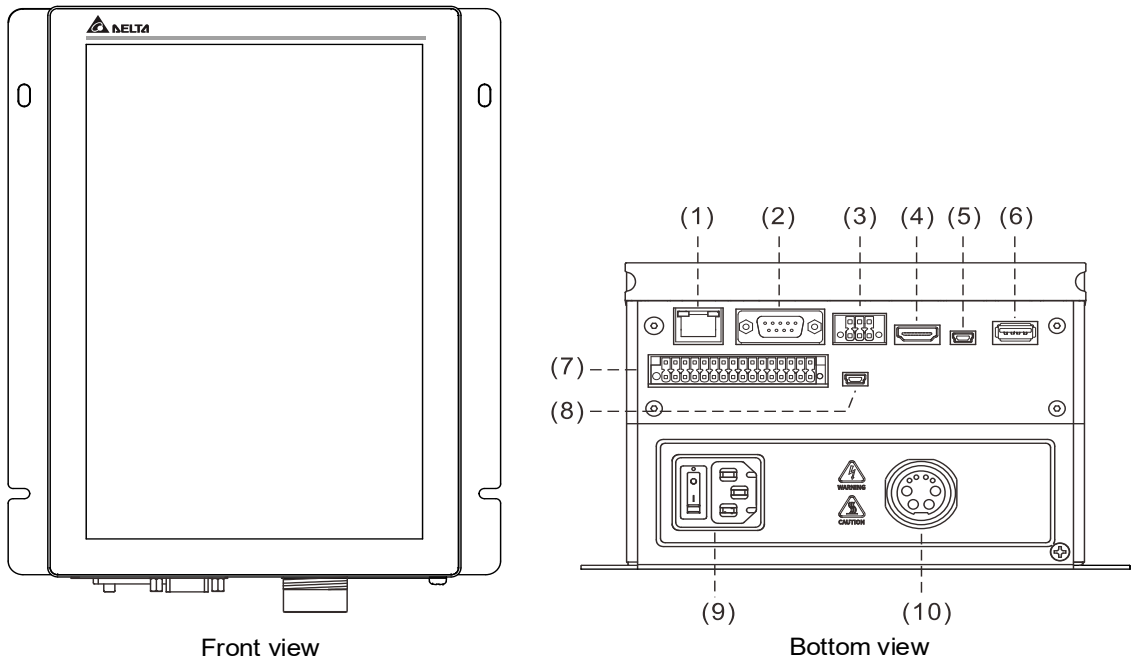
Front view



Bottom view

No.	Description	No.	Description
(1)	Standard RJ45 connector – connects to the host system (PC / PLC / HMI)	(7)	Digital I/O port 1
(2)	Standard RS-232 connector	(8)	SLAVE-B connector – for the controller firmware update
(3)	Two RS-485 connectors – connect to extra RS-485 devices	(9)	Digital I/O port 2
(4)	HDMI output – synchronously projects the Result screen to an external monitor	(10)	AC connector and switch
(5)	SLAVE-A connector – for the HMI software and firmware update	(11)	Connector for Tool 1
(6)	Host connector ■ Accesses the tightening data (parameters and results). ■ Connects to the Barcode Scanner.	(12)	Connector for Tool 2

Medium torque controller (ASD-SD3031B-1)



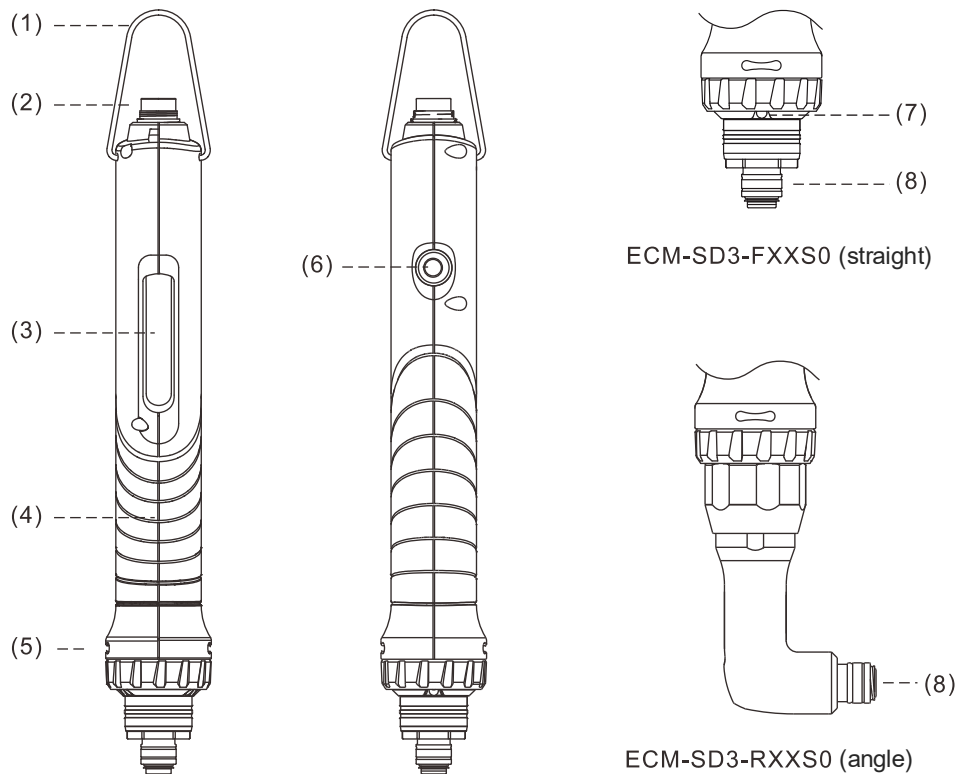
No.	Description	No.	Description
(1)	Standard RJ45 connector – connects to the host system (PC / PLC / HMI)	(6)	Host connector ■ Accesses the tightening data (parameters and results). ■ Connects to the Barcode Scanner.
(2)	Standard RS-232 connector	(7)	Digital I/O port
(3)	Two RS-485 connectors – connect to extra RS-485 devices	(8)	SLAVE-B connector – for the controller firmware update
(4)	HDMI output – synchronously projects the Result screen to an external monitor	(9)	AC connector and switch
(5)	SLAVE-A connector – for the HMI software and firmware update	(10)	Connector for Tool 1

Pay special attention to the following when wiring:

1. The controller must be properly grounded to prevent the large power supply noise from causing precision error of the screwdriver.
2. When the power is off, do not touch the main power cable since the large capacitor inside the controller still contains a dangerously large amount of electric charge.
3. Keep the connection cable between screwdriver and controller at least 30 cm (11.8 inches) away from the cables of other peripheral devices.
4. Do not use any external capacitor, otherwise it may damage the controller.

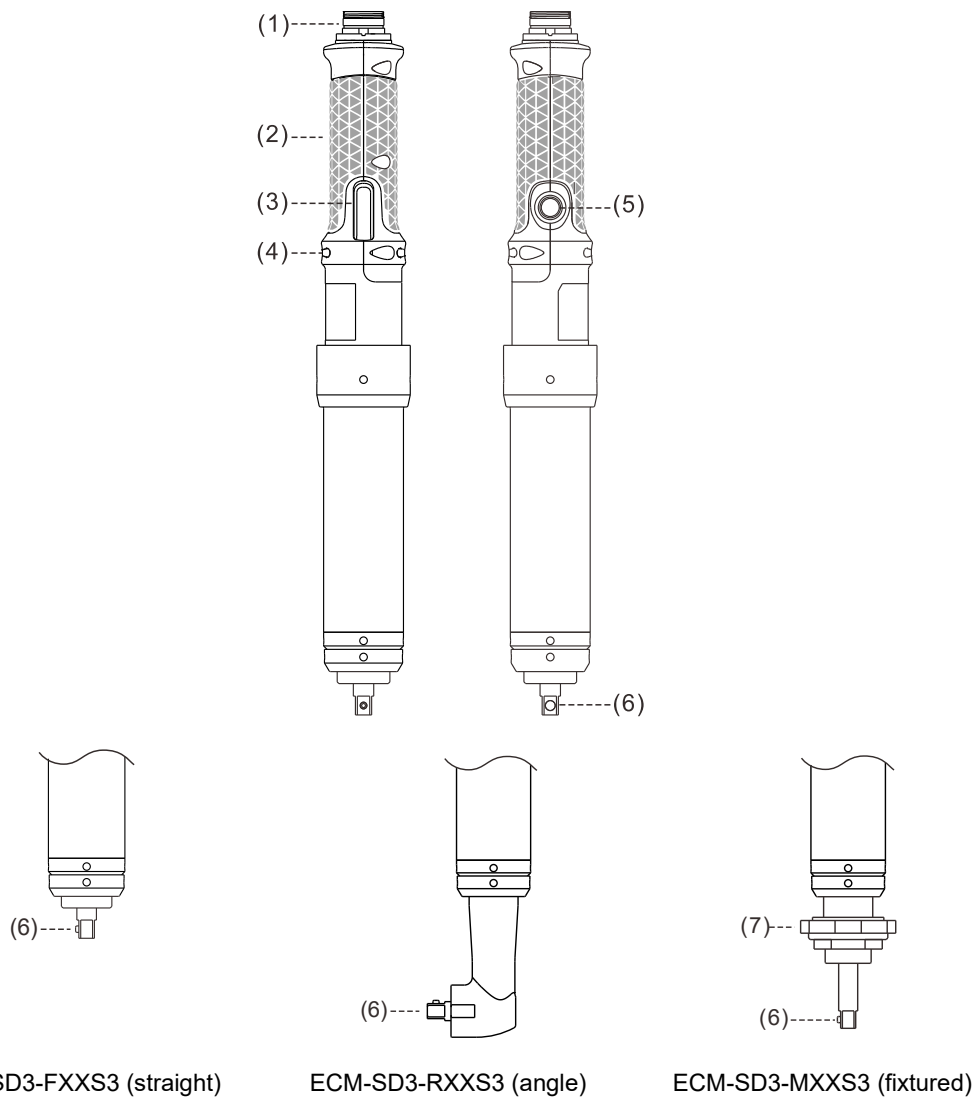
1.5 Description of the transducerized smart screwdriver interface

Low torque screwdriver:



No.	Item	Description
(1)	Hanging ring	Used for hanging the screwdriver up for easy access.
(2)	Connector	Connects the screwdriver and controller using the connection cable.
(3)	Lever start	Press this lever to start the screwdriver.
(4)	Non-slip handle	Prevents the screwdriver from slipping out during handheld use.
(5)	Status LED indicator	■ Displays the current status of the screwdriver. The displayed status can be adjusted through LED Light Settings. ■ Default status of light indicators: Yellow: Ready Red: Tightening NOK Green: Tightening OK
(6)	CW / CCW switch	Switches the screwdriver rotation direction between clockwise (CW) and counterclockwise (CCW) to tighten or loosen the screws.
(7)	Work light LED indicator	Only applicable to straight models. Displays solid white light when power is on. The brightness can be adjusted through Tool Settings.
(8)	Shaft of the screwdriver	Straight: used for installing a 6.35 mm (1/4") or 5 mm hex bit. Angle: used for installing a 6.35 mm (1/4") hex bit.

Medium torque screwdriver:



No.	Item	Description
(1)	Connector	Connects the screwdriver and controller using the connection cable.
(2)	Non-slip handle	Prevents the screwdriver from slipping out during handheld use.
(3)	Lever start	Press this lever to start the screwdriver.
(4)	Status LED indicator	■ Displays the current status of the screwdriver. The displayed status can be adjusted through LED Light Settings. ■ Default status of light indicators: Yellow: Ready Red: Tightening NOK Green: Tightening OK
(5)	CW / CCW switch	Switches the screwdriver rotation direction between clockwise (CW) and counterclockwise (CCW) to tighten or loosen the screws.
(6)	Shaft of the screwdriver	Used for installing a 9.525 mm (3/8") square bit.
(7)	Diamond flange	Only applicable to fixtured models. Used for fixing the screwdriver to a screw tightening machine or a robotic arm.

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1

Installation

2

Observe the ambient storage conditions, ambient installation conditions, and safety precautions in this chapter during installation.

2.1	Ambient storage conditions	2-2
2.2	Ambient installation conditions	2-2
2.3	Mounting direction and space of the controller	2-3
2.4	The installation and use of the screwdriver	2-4
2.4.1	Installing the bit	2-4
2.4.2	Installing the connection cable	2-5
2.4.3	Clamping position	2-6
2.5	Dimensions	2-9
2.6	Safety precautions	2-12
2.7	Maintenance	2-14

2

Precaution

The smart screwdriving controller must be properly grounded to prevent the large power supply noise from causing precision error to the transducerized smart screwdriver.

2.1 Ambient storage conditions

Before installation, this product must be kept in the shipping carton. In order to retain the warranty coverage and for future maintenance, follow these instructions for storage.

While the product is temporarily not in use:

- The storage temperature should be 0°C (32°F) to 60°C (140°F).
- The storage relative humidity should be 0% to 90% (non-condensing).
- Avoid storing the product in an environment containing corrosive gas.

2.2 Ambient installation conditions



Delta smart screwdriving system: the ambient temperature should be between 0°C (32°F) and 40°C (104°F). The environment should be free of water, water vapor, dust, oily dust, corrosive and inflammable gas or liquids, airborne dust or metal particles, electromagnetic interference, or devices that generate excessive heat. And the environment should be solid without vibration.

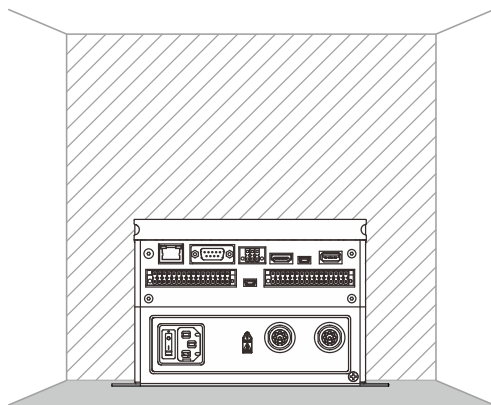
If the ambient temperature is over 30°C (86°F), place the product in a well-ventilated environment. During long-term operation, the ambient temperature should be kept at 25°C (77°F) or below to ensure product performance.

2.3 Mounting direction and space of the controller

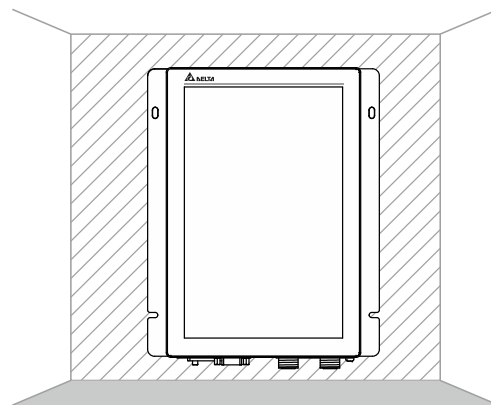
Precautions:

- Mount the controller in the correct direction according to the following illustrations, with the controller placed flat on a desk or mounted perpendicular to a wall. Incorrect mounting direction may cause malfunction.
- For better ventilation and cooling, allow sufficient clearance (at least 5 cm or 2 inches) between the controller and the adjacent objects or walls, otherwise the overheating may cause malfunction.

Correct

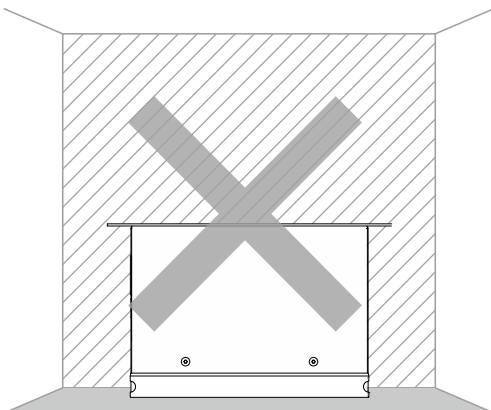


Placed flat on a desk -
with the panel display facing up

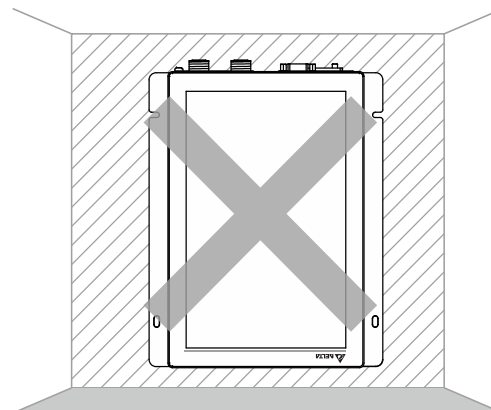


Mounted perpendicular to a wall -
with the connectors at the bottom
(install with M4 screws)

Incorrect



Placed flat on a desk -
with the panel display facing down



Mounted perpendicular to a wall -
with the connectors on top

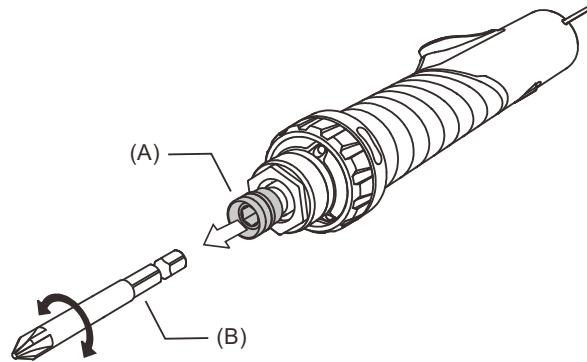
2

2.4 The installation and use of the screwdriver

2.4.1 Installing the bit

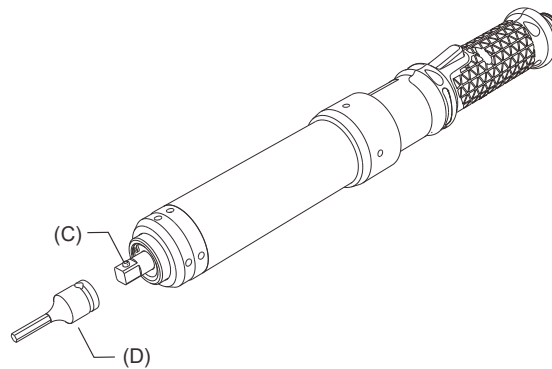
Low torque screwdriver (straight):

1. Pull out the collar (A).
2. Insert the bit (B). If unsuccessful, rotate the bit until it can be inserted.



Medium torque screwdriver (straight):

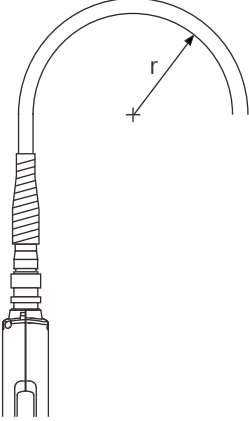
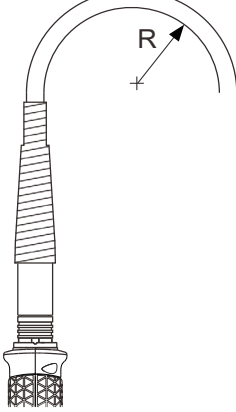
1. Push the shaft (C) inwards.
2. Insert the bit (D).



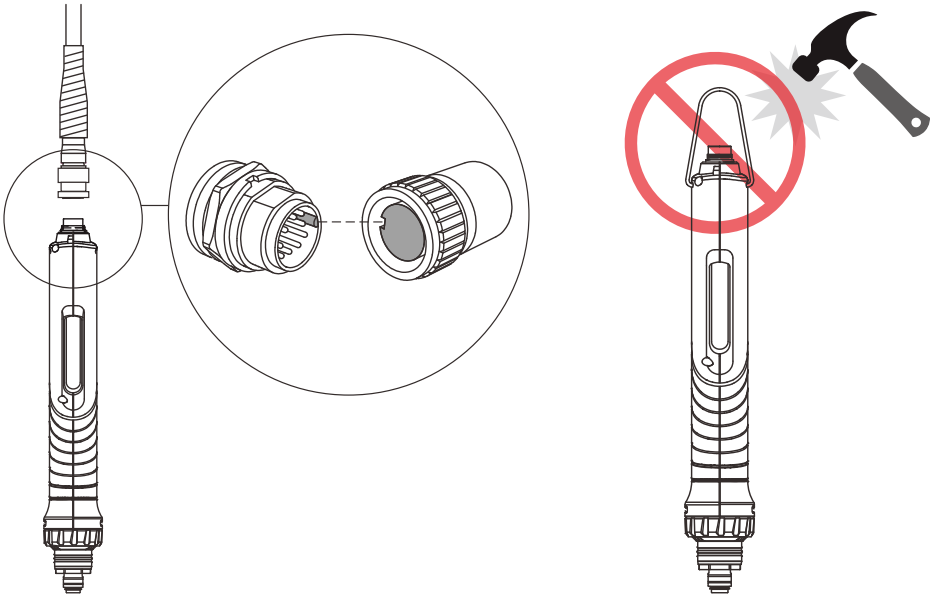
2.4.2 Installing the connection cable

The connection cable is used for connecting the screwdriver to the controller.

- Make sure to follow the minimum bend radius specifications.

Low torque screwdriver	Medium torque screwdriver
The minimum bend radius (r) is 60 mm (2.36").	The minimum bend radius (R) is 72 mm (2.83").
	

- When attaching the connection cable to the screwdriver, align the poka-yoke notch on the cable connector with the poka-yoke bump on the screwdriver connector (as shown on the left as follows), and then tighten the nut on the cable connector clockwise. While attaching, avoid impacting the pins of screwdriver connector (as shown on the right as follows), otherwise, it may cause malfunction of the screwdriver.



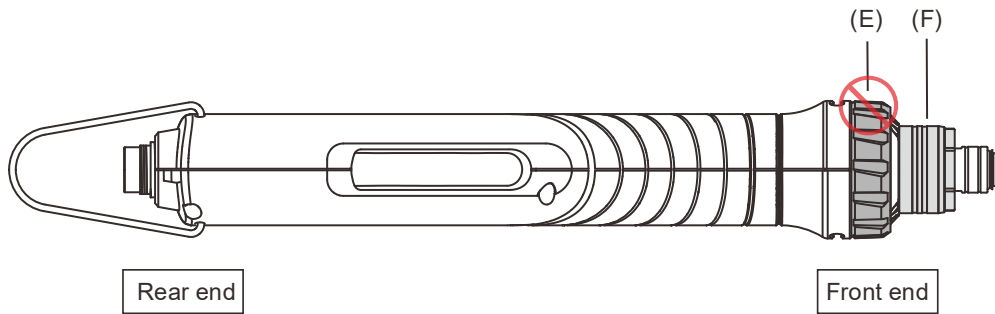
2.4.3 Clamping position

Low torque screwdriver (straight):

The screwdriver can be used as a stationary or handheld device.

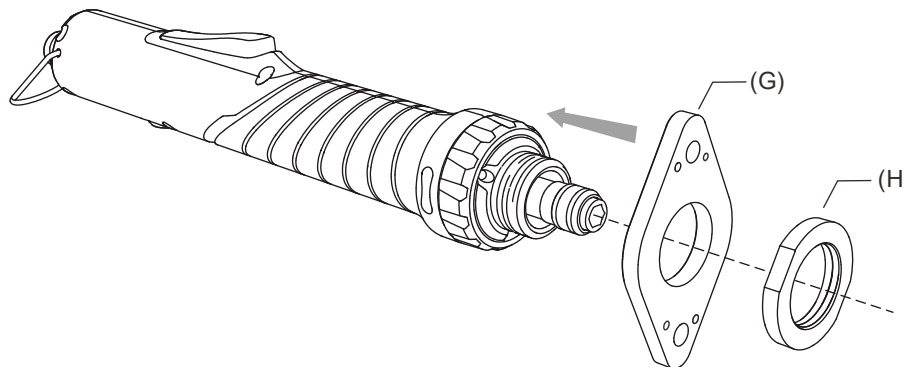
Installation precautions:

- The front end and rear end of the screwdriver are marked as follows.
- Do not remove the part (E) shown as follows.
- When installing accessories to the screwdriver, remove the protective cover from the part (F).

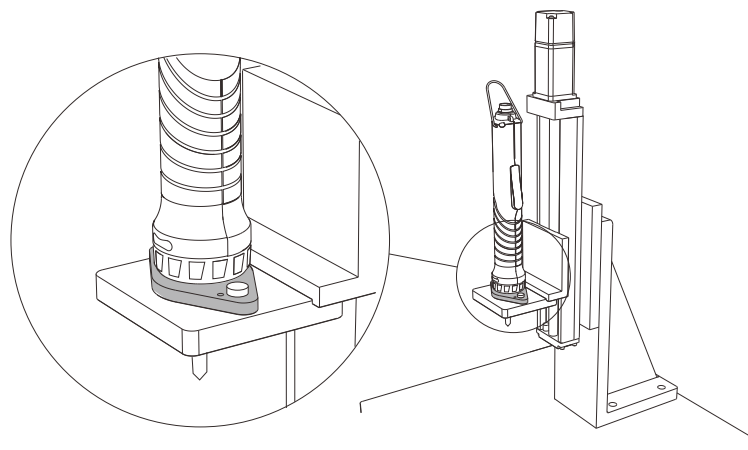


For use as a stationary device

Remove the protective cover from the part (F). Attach the flange (G) of the flange module to the part (F), and then tighten the fixing nut (H) with the torque of 55 to 60 kgf-cm.

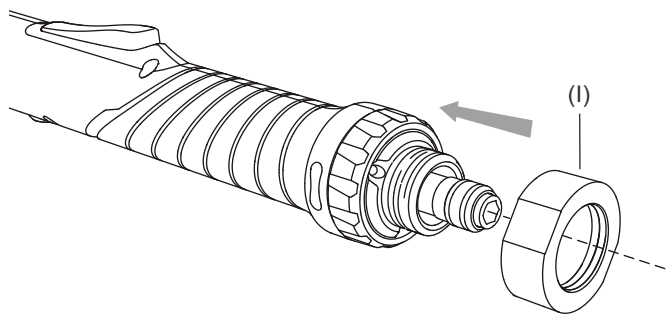


The screwdriver can be fixed to a screw tightening machine or a robotic arm.

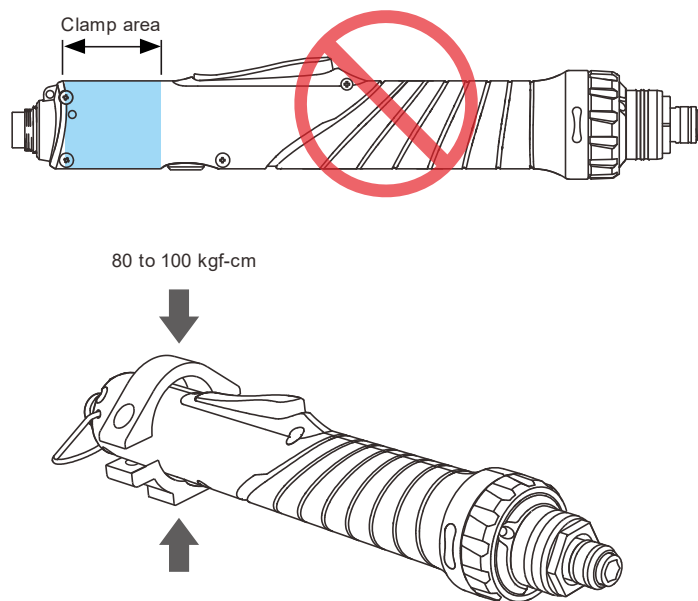


For use as a handheld device (with a torque arm)**■ Clamping the front end of the screwdriver**

Do not mount the screwdriver on the tool holder of the torque arm directly by the part (F). Refer to Section 2.5 for the front-end dimensions of the screwdriver and select an adapting nut (I) according to the tool holder dimensions of the torque arm. Next, remove the protective cover from the part (F) of the screwdriver, and then attach the adapting nut (I) to the part (F).

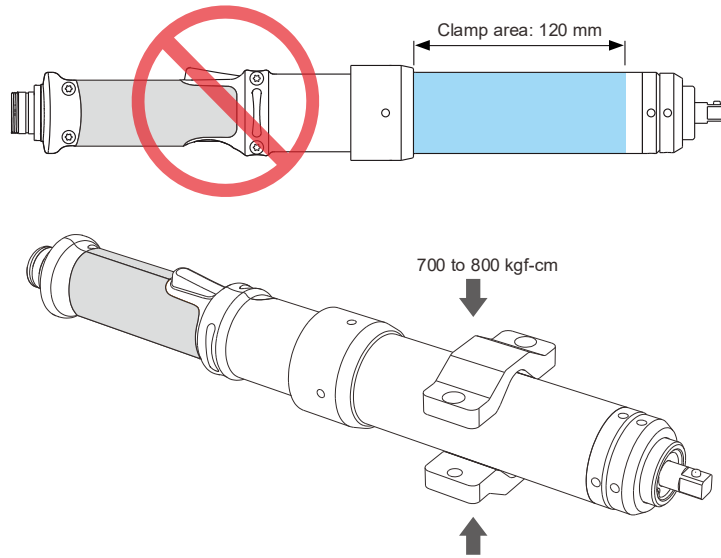
**■ Clamping the rear end of the screwdriver**

The suggested clamping torque is 80 to 100 kgf-cm. Avoid the part of the CW / CCW switch or the lever start of the screwdriver.



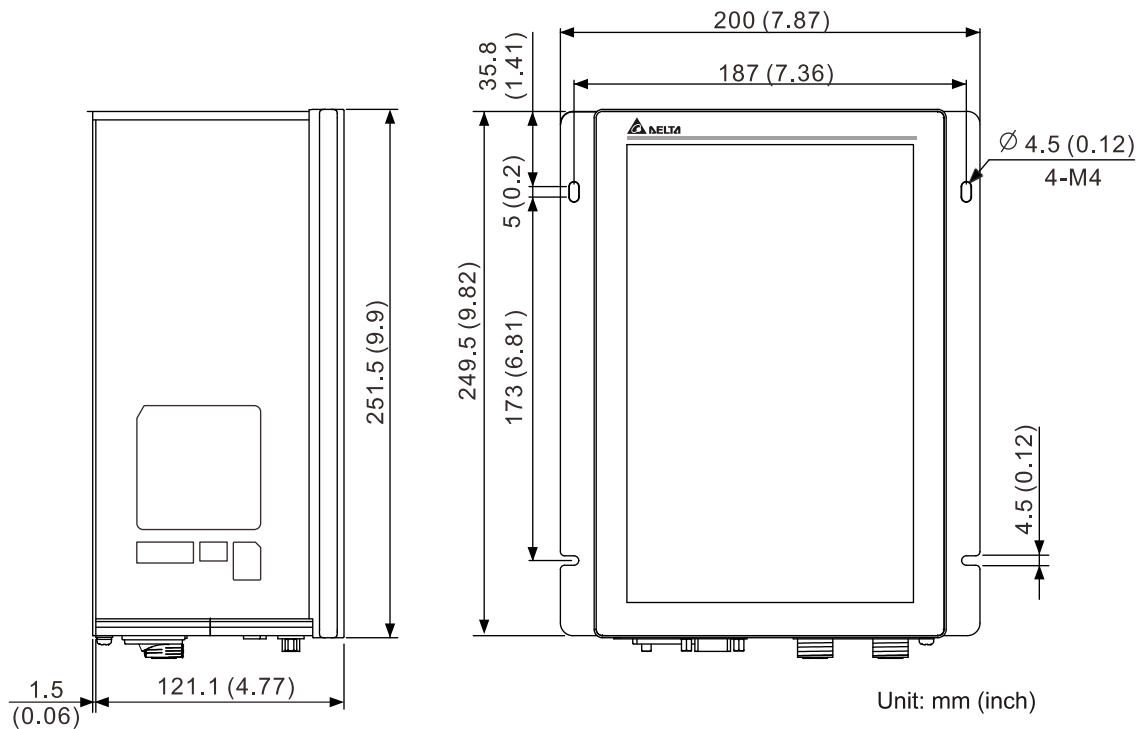
Medium torque screwdriver (straight):

The screwdriver can be clamped by the area marked as follows. The suggested clamping torque is 700 to 800 kgf-cm.



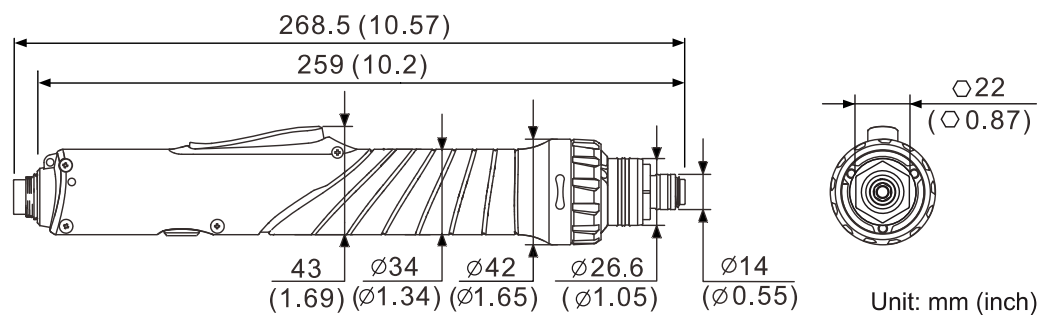
2.5 Dimensions

Smart screwdriving controller

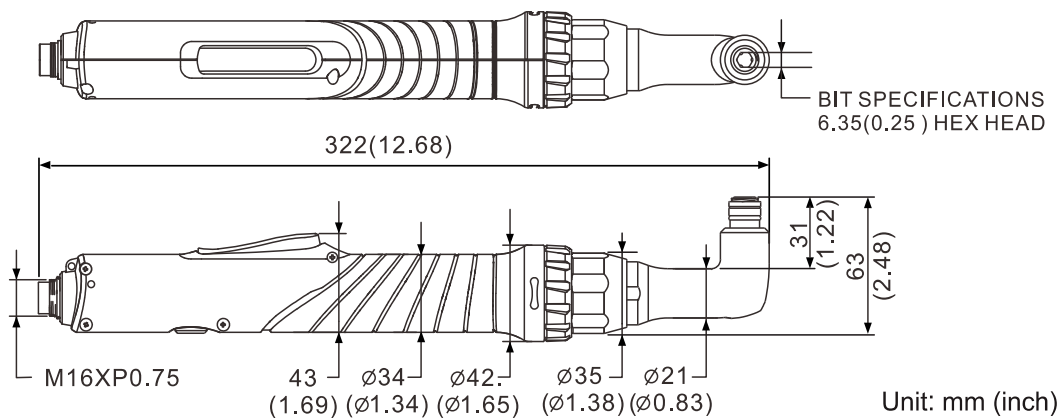


Transducerized smart screwdriver

Low torque (straight)

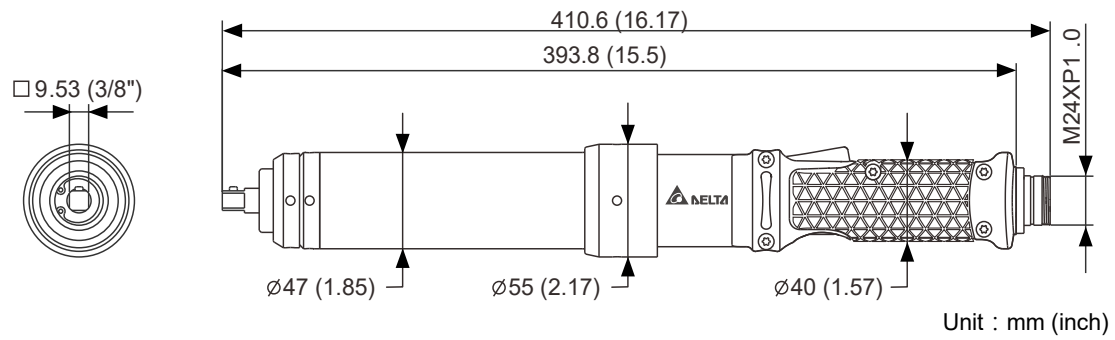


Low torque (angle)

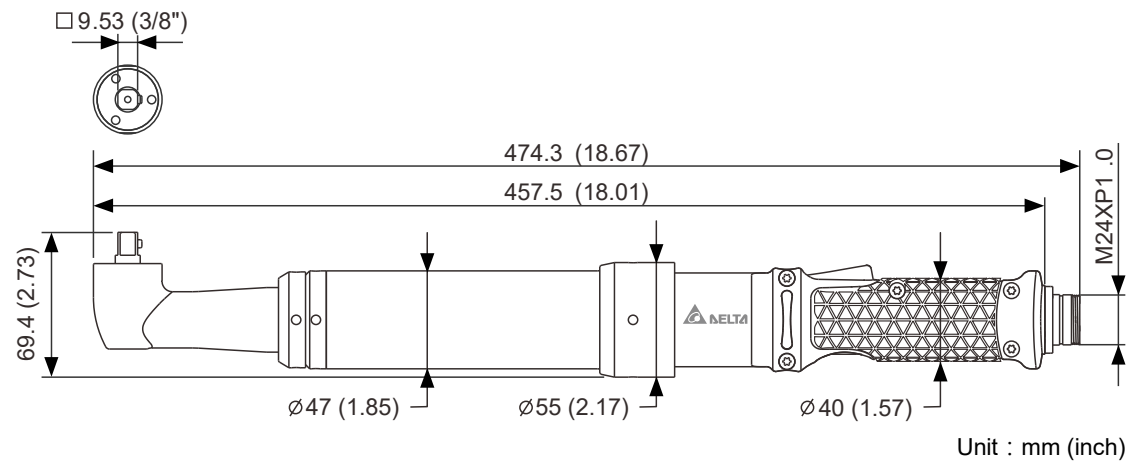


2

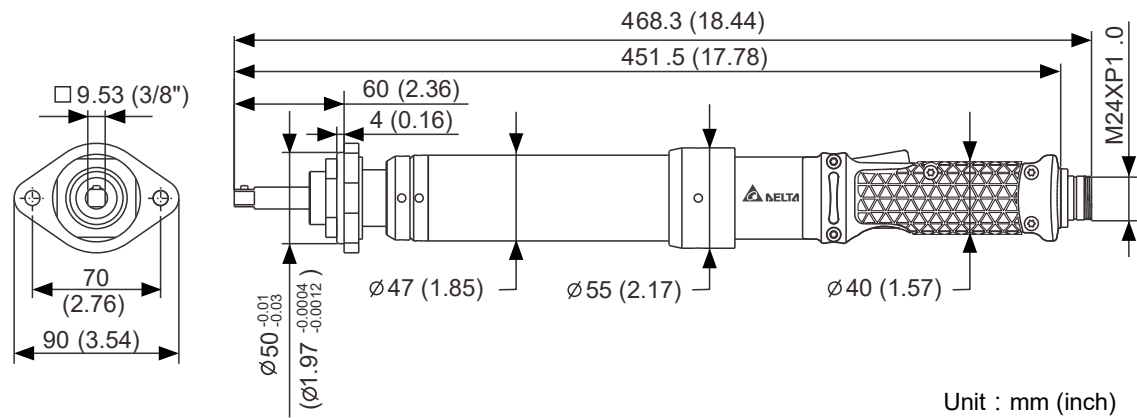
Medium torque (straight)



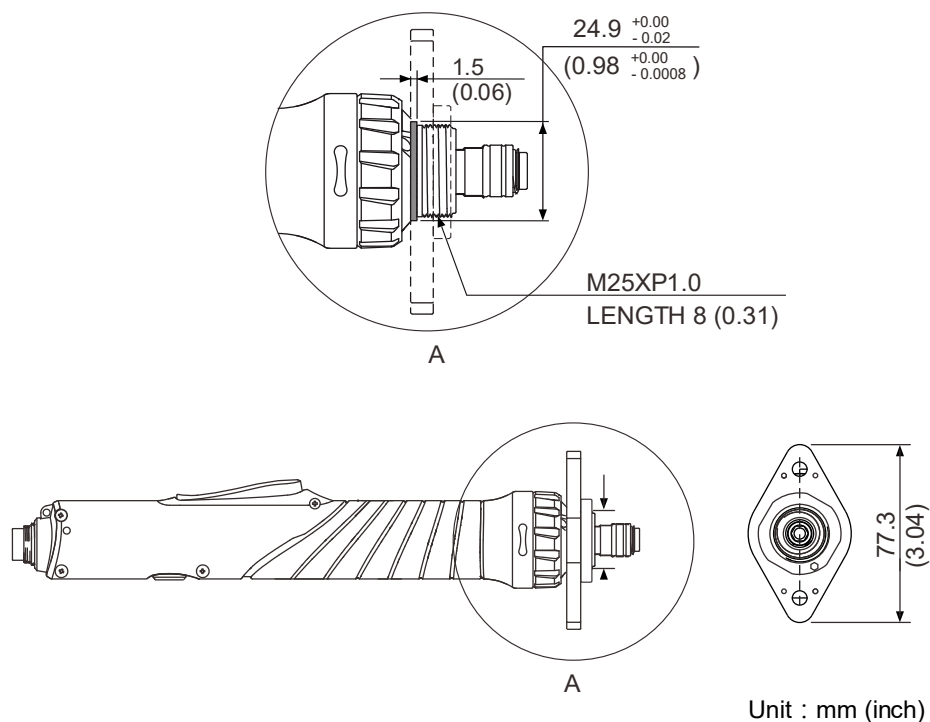
Medium torque (angle)



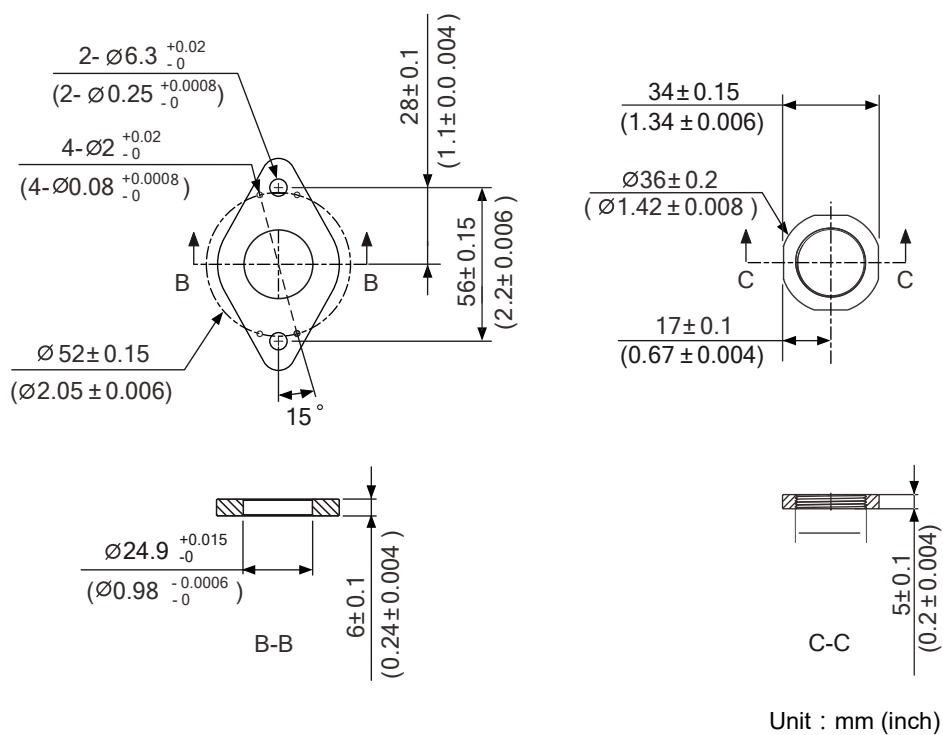
Medium torque (fixtured)



Transducerized smart screwdriver + flange module: only applicable to low torque screwdriver (straight)



Flange module (optional purchase; a flange and a fixing nut are included): only applicable to low torque screwdriver (straight)



2.6 Safety precautions

The Delta smart screwdriving system is designed for industrial applications. It is necessary that you fully understand the system specifications and the content of this user manual. For your safety and correct use, read the safety precautions for the system carefully before installing.

The safety precautions are as follows:

Handling, mounting, and storage

- When handling the transducerized smart screwdriver, hold the body of the screwdriver instead of holding the cable or the shaft only.
- Do not impact the shaft of the screwdriver. Impact force will damage the shaft and the motor that is attached at the rear end of the shaft.
- The shaft of the screwdriver is not waterproof or oil-proof. Do not use, install, or store the screwdriver in an environment with high humidity, moisture, oily liquids, or corrosive or inflammable gases.
- The material of the screwdriver shaft is not rustproof. Although rustproof oil has been applied to the shaft during the manufacturing process, you must check the shaft condition every three months and apply rustproof oil timely if storing the motor for more than six months.
- Ensure that the environmental conditions for storing the product conform to the specifications written in this manual.

Operation

- The operation of the transducerized smart screwdriver is controlled by the smart screwdriving controller. Do not directly connect a commercial power supply (100/220V, 50/60 Hz) to the circuit of the screwdriver, otherwise the screwdriver will not operate normally and may be permanently damaged.
- Follow the specifications when using the product. The ambient temperature must not exceed the specified range.
- The material of the screwdriver shaft is not rustproof. To ensure a longer service life, apply rustproof oil during operation.
- If any odor, noise, smoke, heat, or abnormal vibration occurs during operation, stop the screwdriver and turn off the power immediately.
- Before replacing the screwdriver or the connection cable, be sure to first turn off the power to the controller. Otherwise, it may damage the screwdriver or the connection cable.
- Keep the ambient environment clean and tidy to ensure smooth operation of the screwdriver and controller.
- The connection cable must be properly attached and fixed to avoid cable damage due to entangling or tripping hazard due to loose cables.
- Wear safety goggles when using electric assembly tools.

Others

- Use the transducerized smart screwdriver, smart screwdriving controller, and connection cable only for screwdriving operations. Do not use them for other purposes (such as hole drilling or wall renovation). Illegal or improper use may cause personal injury or damage the product and its parts; this will also void the warranty.
- Do not disassemble or attempt to repair the controller, the screwdriver or their parts without authorization, otherwise it may cause personal injury and will void the warranty.
- Use the screwdriver with a torque arm as much as possible to reduce the reaction force. Do not use the screwdriver as a handheld device when the required torque is over 30 kgf-cm.

2

2.7 Maintenance

The recommended maintenance cycle and maintenance measures for the smart screwdriving system under normal use are as follows:

Maintenance cycle	Maintenance measures
Every 250,000 counts of tightening + loosening	It is advisable to send the system back for precision calibration and maintenance.
One year of use or every 1,000,000 counts of tightening + loosening	Send the system back for precision calibration and maintenance.

Wiring

3

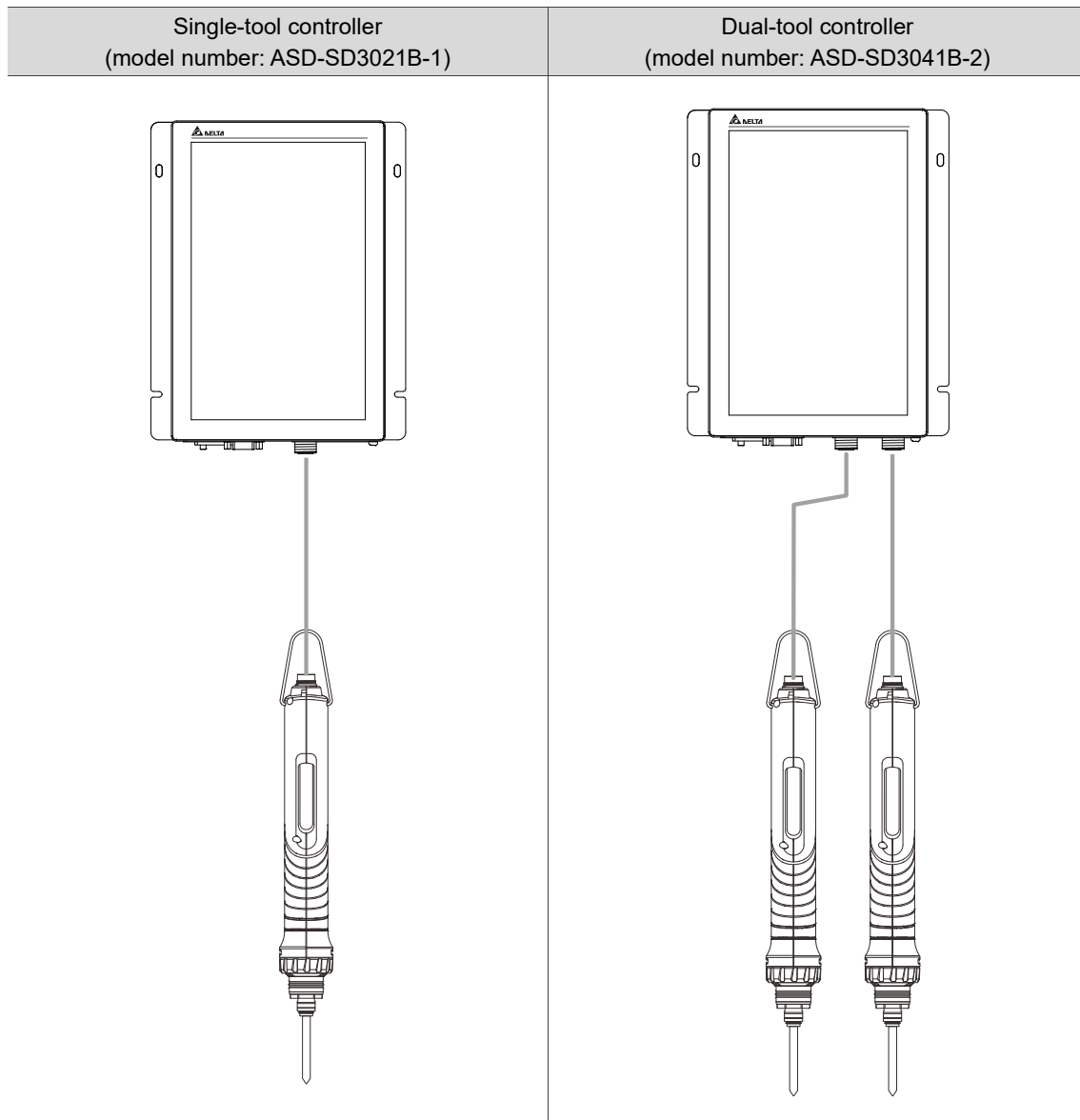
This chapter illustrates the power supply circuit, connectors, and wiring of the Delta smart screwdriving system.

3.1	System connection	3-2
3.1.1	Connection diagrams (low torque models)	3-2
3.1.2	Connection diagram (medium torque model)	3-3
3.2	Wiring for I/O connectors.....	3-4
3.2.1	Pin assignments for I/O connectors	3-4
3.2.2	Signal description for I/O connectors.....	3-6
3.2.3	Wiring diagrams	3-7
3.3	Wiring for RS-485 connector (for extension)	3-9

3.1 System connection

3.1.1 Connection diagrams (low torque models)

3

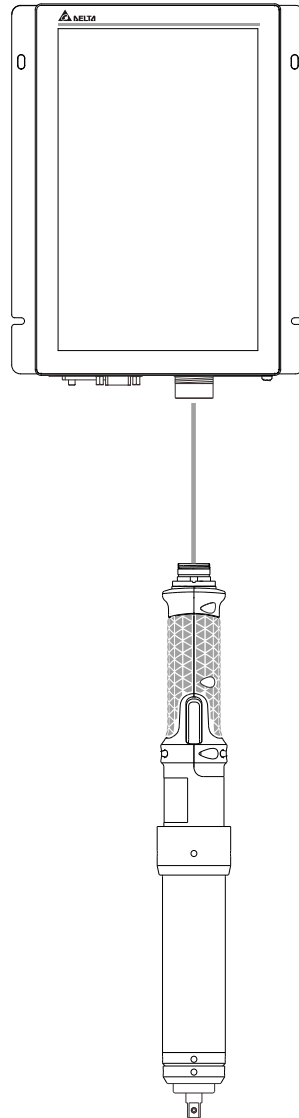


Installation precautions:

1. Make sure the power supply and connection cable(s) (with M16 8-pin connectors) are correct.
2. When an alarm or emergency stop occurs, turn off the switch next to the AC socket on the smart screwdriving controller to power off the controller and the transducerized smart screwdriver.

3.1.2 Connection diagram (medium torque model)

Single-tool controller
(model number: ASD-SD3031B-1)



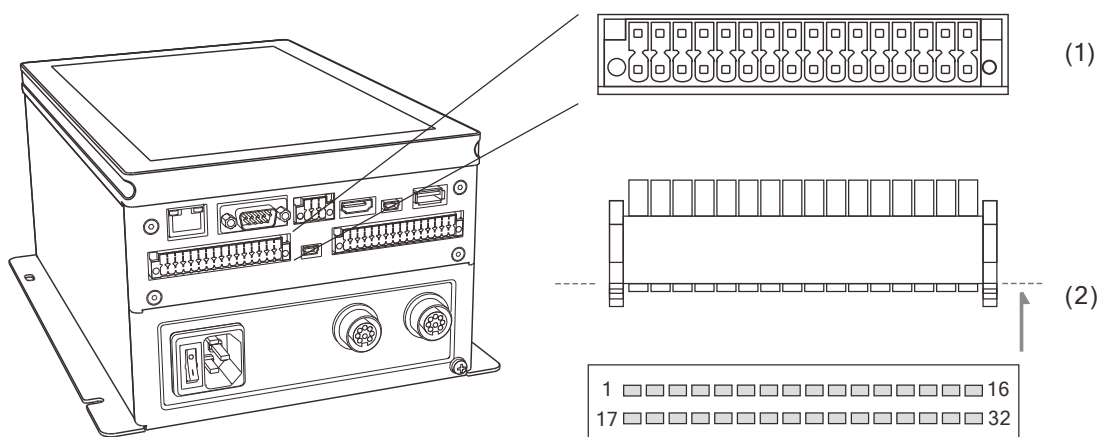
Installation precautions:

1. Make sure the power supply and connection cable (with M24 8-pin connectors) are correct.
2. When an alarm or emergency stop occurs, turn off the switch next to the AC socket on the smart screwdriving controller to power off the controller and the transducerized smart screwdriver.

3.2.1 Pin assignments for I/O connectors

3.2.1 Pin assignments for I/O connectors

Define 8 digital inputs (DI) and 8 digital outputs (DO) for the I/O connector to achieve higher flexibility in the communication between the host system and smart screwdriving controller. Refer to Section 3.2.3 for the wiring diagrams. The pin assignments for the digital I/O ports of the controllers are shown as follows: for the single-tool controller, refer to the diagram of DIGITAL I/O-1; for the dual-tool controller, refer to the diagram of DIGITAL I/O-2.



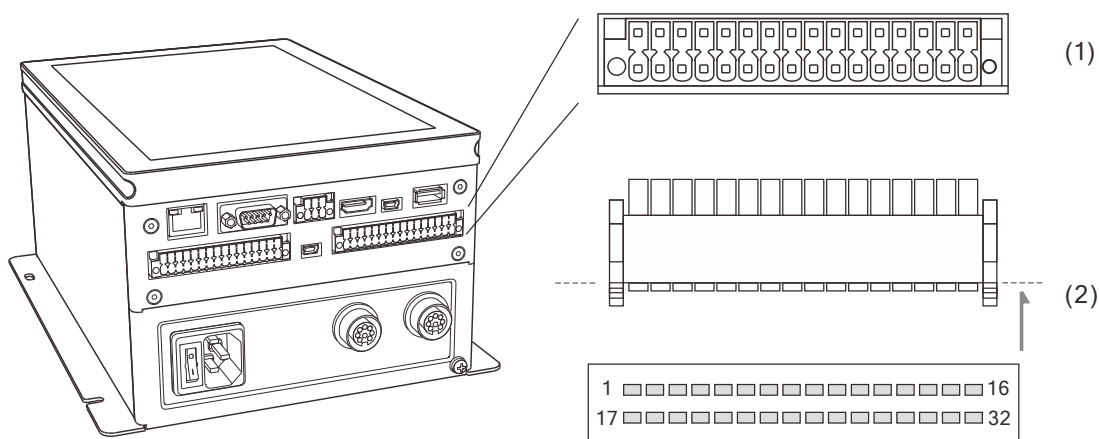
(1) DIGITAL I/O-1 connector (female); (2) DIGITAL I/O-1 connector (male)

Pin assignment:

Pin	Signal	Description	Pin	Signal	Description
1	DO_24V_A	Power output (+) (24V $\pm$ 10%)	17	NC*1	-
2	DO_A1+	Digital output (+)	18	DO_A8+	Digital output (+)
3	DO_A1-	Digital output (-)	19	DO_A8-	Digital output (-)
4	DO_A2+	Digital output (+)	20	DI_A1	Digital input
5	DO_A2-	Digital output (-)	21	DI_A2	Digital input
6	DO_A3+	Digital output (+)	22	DI_A3	Digital input
7	DO_A3-	Digital output (-)	23	DI_A4	Digital input
8	DO_A4+	Digital output (+)	24	DI_A5	Digital input
9	DO_A4-	Digital output (-)	25	DI_A6	Digital input
10	DO_A5+	Digital output (+)	26	COM_A+	Common reference pin for DI_A1 to DI_A6
11	DO_A5-	Digital output (-)	27	DI_A7+	Digital input (+)
12	DO_A6+	Digital output (+)	28	DI_A7-	Digital input (-)
13	DO_A6-	Digital output (-)	29	DI_A8+	Digital input (+)
14	DO_A7+	Digital output (+)	30	DI_A8-	Digital input (-)
15	DO_A7-	Digital output (-)	31	NC*1	-
16	NC*1	-	32	PGND	Power output (-) (ground for 24V $\pm$ 10%)

Note:

1. NC indicates the terminal is for internal use only. Do not connect to NC, or it may damage the controller.
2. You can wire the bit selector to Pin 20 - 25 and 27 - 30, and then refer to Sections 9.2 and 9.3 for the settings.



(1) DIGITAL I/O-2 connector (female); (2) DIGITAL I/O-2 connector (male)

Pin assignment:

Pin	Signal	Description	Pin	Signal	Description
1	DO_24V_B	Power output (+) (24V $\pm$ 10%)	17	NC*1	-
2	DO_B1+	Digital output (+)	18	DO_B8+	Digital output (+)
3	DO_B1-	Digital output (-)	19	DO_B8-	Digital output (-)
4	DO_B2+	Digital output (+)	20	DI_B1	Digital input
5	DO_B2-	Digital output (-)	21	DI_B2	Digital input
6	DO_B3+	Digital output (+)	22	DI_B3	Digital input
7	DO_B3-	Digital output (-)	23	DI_B4	Digital input
8	DO_B4+	Digital output (+)	24	DI_B5	Digital input
9	DO_B4-	Digital output (-)	25	DI_B6	Digital input
10	DO_B5+	Digital output (+)	26	COM_B+	Common reference pin for DI_B1 to DI_B6
11	DO_B5-	Digital output (-)	27	DI_B7+	Digital input (+)
12	DO_B6+	Digital output (+)	28	DI_B7-	Digital input (-)
13	DO_B6-	Digital output (-)	29	DI_B8+	Digital input (+)
14	DO_B7+	Digital output (+)	30	DI_B8-	Digital input (-)
15	DO_B7-	Digital output (-)	31	PGND	Power output (-) (ground for 24V $\pm$ 10%)
16	NC*1	-	32	PGND	

Note:

1. NC indicates the terminal is for internal use only. Do not connect to NC, or it may damage the controller.
2. You can wire the bit selector to Pin 20 - 25 and 27 - 30, and then refer to Sections 9.2 and 9.3 for the settings.

3.2.2 Signal description for I/O connectors

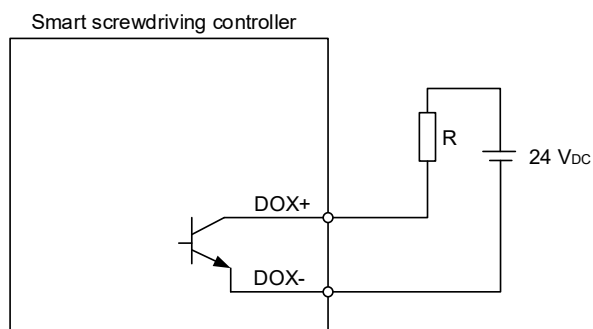
The signals listed in the previous section are further described in the following table.

Signal		Pin	Description	Wiring method (refer to Section 3.2.3)
DI	DI_A1 to A6	DIGITAL I/O-1 20 to 25	For voltage input; requires an external power supply (24V ± 10%). Supports both NPN and PNP wiring (must be used with COM_A+).	C3 / C4
	DI_B1 to B6	DIGITAL I/O-2 20 to 25	For voltage input; requires an external power supply (24V ± 10%). Supports both NPN and PNP wiring (must be used with COM_B+).	
	DI_A7+ DI_A7-	DIGITAL I/O-1 27, 28	DI_A7+ / A8+ / B7+ / B8+ are for voltage input; requires an external power supply (24V ± 10%).	C5 / C6
	DI_A8+ DI_A8-	DIGITAL I/O-1 29, 30	DI_A7- / A8- / B7- / B8- are reference pins for voltage input.	
	DI_B7+ DI_B7-	DIGITAL I/O-2 27, 28	NPN: connect DI_A7- / A8- / B7- / B8- to the positive (+) terminal of the external power supply (24V ± 10%).	
	DI_B8+ DI_B8-	DIGITAL I/O-2 29, 30	PNP: connect DI_A7- / A8- / B7- / B8- to the negative (-) terminal of the external power supply (24V ± 10%).	
DO	DO_A1+ to A8+ DO_A1- to A8-	DIGITAL I/O-1 2 to 15, 18, 19	Perform wiring based on the load type. Require an external power supply (24V ± 10%).	C1 / C2
	DO_B1+ to B8+ DO_B1- to B8-	DIGITAL I/O-2 2 to 15, 18, 19		
Power	COM_A+	DIGITAL I/O-1 26	The common reference pins for voltage input for DI_A1 to A6 / DI_B1 to B6. NPN: connect COM_A+ / B+ to the positive (+) terminal of the external power supply (24V ± 10%). PNP: connect COM_A+ / B+ to the negative (-) terminal of the external power supply (24V ± 10%).	-
	COM_B+	DIGITAL I/O-2 26		
	DO_24V_A	DIGITAL I/O-1 1	Power output (+) (24V ± 10%)	-
	DO_24V_B	DIGITAL I/O-2 1		
	PGND		DIGITAL I/O-1 32	Power output (-) (ground for 24V ± 10%)
DIGITAL I/O-2 31 or 32				
Other	NC	DIGITAL I/O-1 16, 17, 31	These terminals are for internal use only. Do not connect to NC, or it may damage the controller.	-
		DIGITAL I/O-2 16, 17		

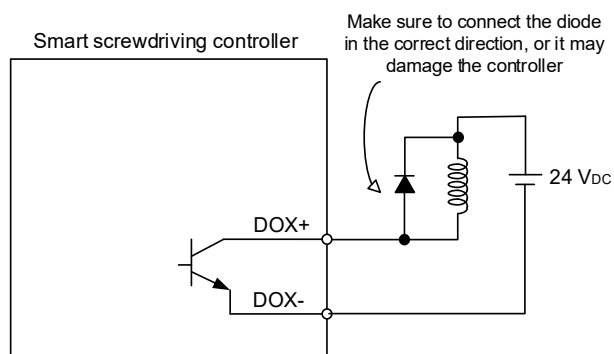
3.2.3 Wiring diagrams

DO wiring:

C1: external power supply, general load



C2: external power supply, inductive load



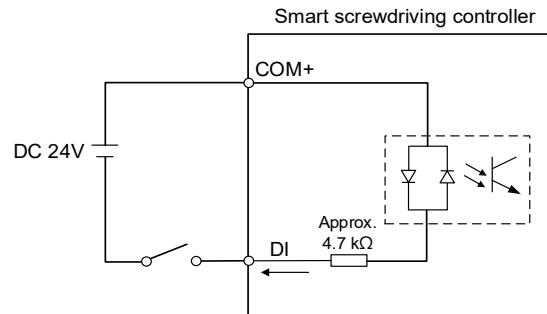
DI wiring: input signals by relay or open collector transistor.

Conditions of DI signal On / Off:

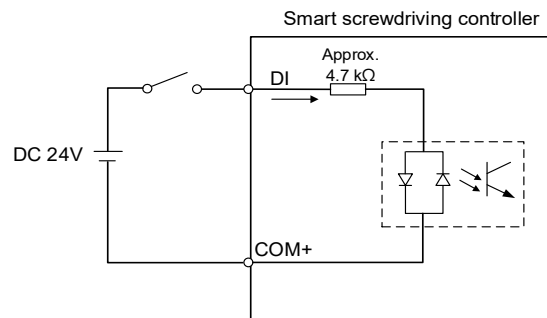
On: 15V - 24V; input current of 8 mA.

Off: 5V or lower; input current must not be higher than 0.5 mA.

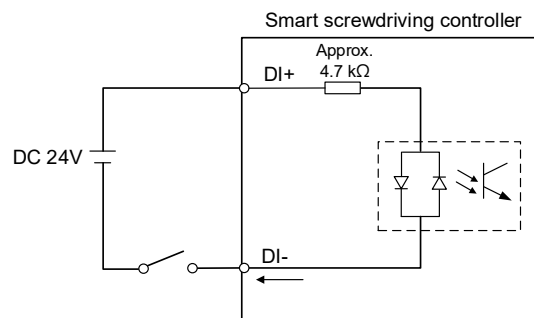
C3: NPN transistor, SINK mode (DI_A1 to A6, DI_B1 to B6)



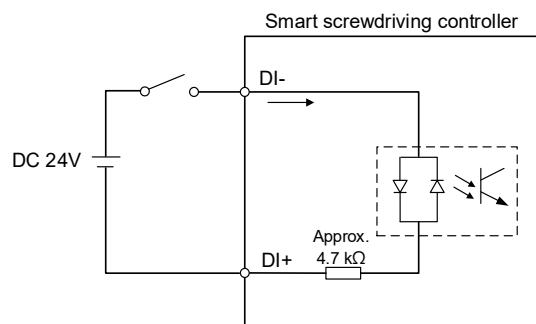
C4: PNP transistor, SOURCE mode (DI_A1 to A6, DI_B1 to B6)



C5: NPN transistor, SINK mode, not using COM+ for input (DI_A7, A8, B7, B8)



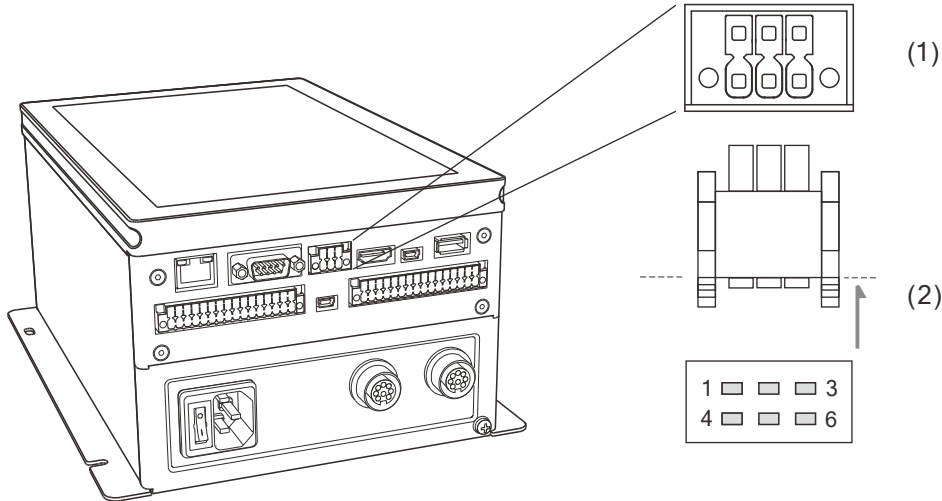
C6: PNP transistor, SOURCE mode, not using COM+ for input (DI_A7, A8, B7, B8)



3.3 Wiring for RS-485 connector (for extension)

Connect the equipment (such as a PLC) with an RS-485 communication interface to the smart screwdriving controller through the RS-485 connector.

There are 2 signal pairs for RS-485: D1- & D1+ and D2- & D2+.



Pin assignment:

Pin	Signal	Description	Pin	Signal	Description
1	D1-	First pair of differential signals Tx- / Rx-	4	D1+	First pair of differential signals Tx+ / Rx+
2	GND	Ground for +12V	5	NC*	-
3	D2-	Second pair of differential signals Tx- / Rx-	6	D2+	Second pair of differential signals Tx+ / Rx+

Note: NC indicates the terminal is for internal use only. Do not connect to NC, or it may damage the controller.

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3

Functions Overview

4

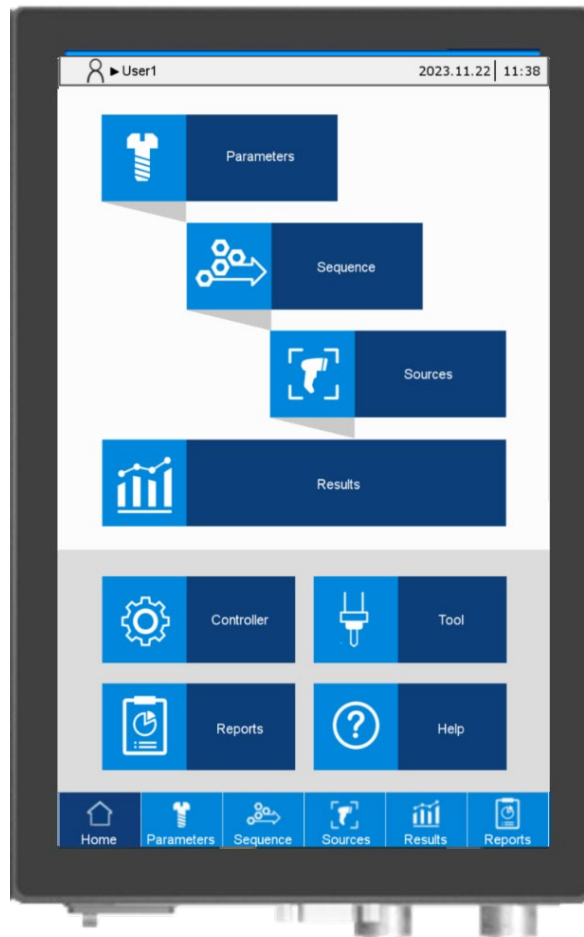
This chapter introduces the functions on the main screen of the smart screwdriving controller.



4.1 Functions on the main screen..... 4-2

4.1 Functions on the main screen

There are eight functions on the main screen of the smart screwdriving controller:
Parameters, Sequence, Sources, Results, Controller, Tool, Reports, and Help.



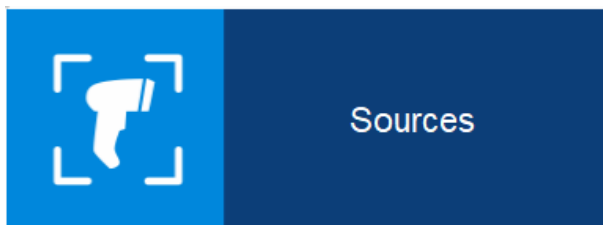
Each function is described as follows:



Parameters: configure the appropriate parameters and various screw tightening strategies for different tightening scenarios.



Sequence: create the screw tightening sequences for different tightening requirements.



Sources: specify the operating mode and switching method of the smart screwdriving system.



Results: obtain information of the current tightening operation and the operating curves.



Controller: access and configure the smart screwdriving controller information and settings, including the system settings, DI/DO, peripheral device, communication (Modbus), and service station.

4



Tool: access and configure the transducerized smart screwdriver information and settings, including the tool info, tool settings, LED light settings, and tool calibration.



Reports: access the reports of all operational statuses, including production reports, error reports, warning reports, and button reports.



Help: obtain the documents for operational support.

Parameters (Tightening parameters)

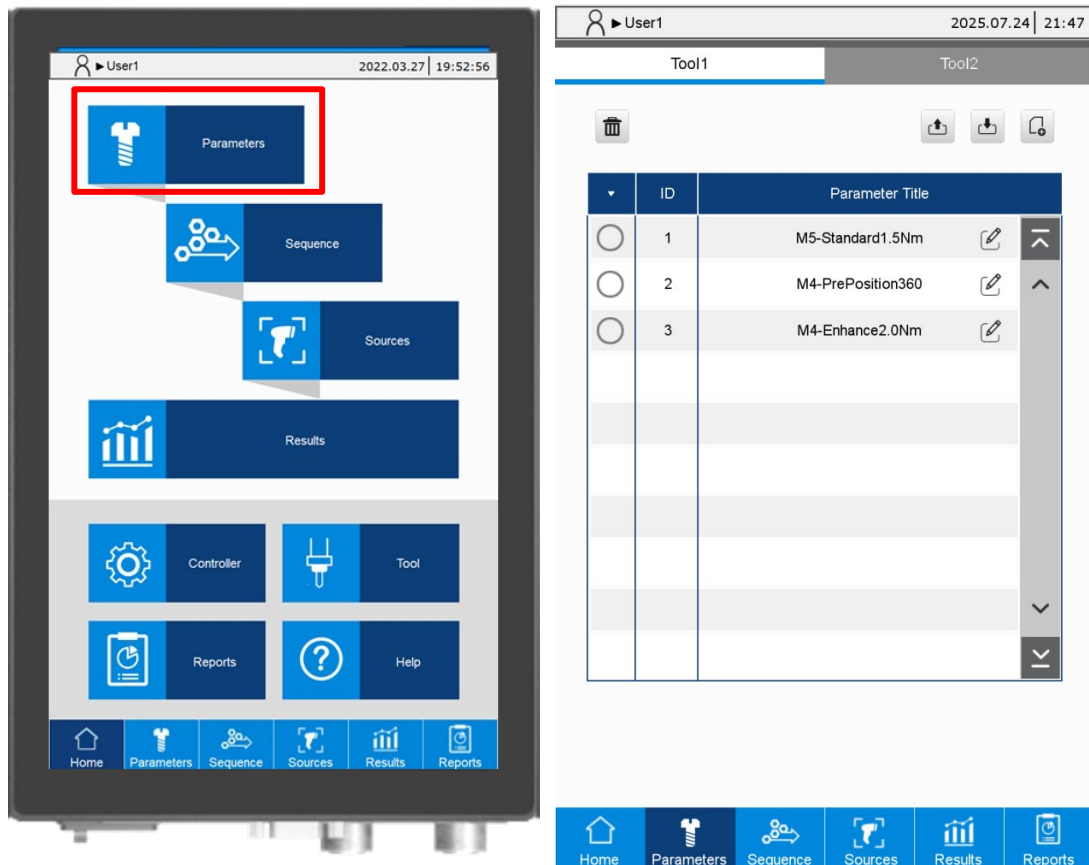
5

This chapter introduces the **Parameters** page of the smart screwdriving controller. You can configure the appropriate parameters for different tightening scenarios to achieve the quality requirements of products. You can also use the screw tightening strategies to customize various tightening processes.

5.1	Parameters page overview	5-2
5.2	Creating a tightening parameter	5-3
5.3	Tightening strategies and the Settings tabs	5-4
5.3.1	General Settings	5-5
5.3.2	Tightening Settings	5-7
5.3.2.1	Standard strategy	5-7
5.3.2.2	Enhanced strategy	5-14
5.3.2.3	Pre-position strategy	5-15
5.3.2.4	Self-defined strategy	5-17
5.3.2.4.1	Synchronization settings in tightening parameters	5-17
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5.3.3	Loosening settings	5-28
5.4	Saving a tightening parameter	5-29
5.5	Copying / pasting a tightening parameter	5-29
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5.1 Parameters page overview

Create different tightening parameters for transducerized smart screwdrivers of different specifications. Up to 500 sets of parameters can be created for a single tool. Each row in the following parameter list represents one set of tightening parameter.



5.2 Creating a tightening parameter

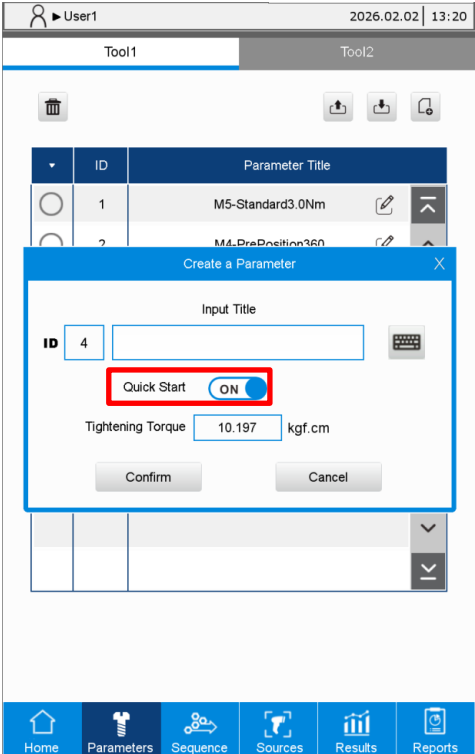
Steps to create a tightening parameter are as follows:

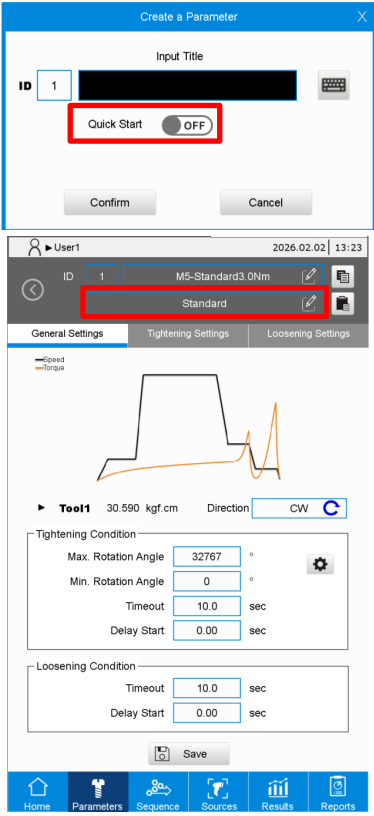
- 5
1. Click , and then the system searches for an available ID and sets it as the new parameter ID. If desiring to specify an ID, directly set it in the **ID** field of the Create a Parameter window.

2. Input the parameter title.

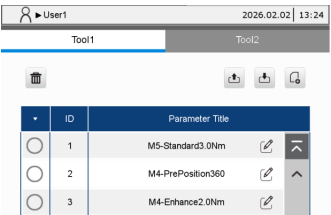
3. Choose whether to use the Quick Start function for the parameter setting.

Quick Start function On	Quick Start function Off
■ The Quick Start function is On by default. (a) Input the final Tightening Torque and click Confirm to create the parameter. Default values are applied to the remaining settings.	■ Switch off the Quick Start function and click Confirm to enter the parameter editing screen. (a) Set the tightening strategy as Standard, Enhanced, Pre-position, or Self-defined. (b) Set the contents of the General Settings, Tightening Settings, and Loosening Settings tabs in sequence. (c) Click Save to save the settings.





4. The parameter list displays the ID and Parameter Title of the parameters that are successfully created.



5.3 Tightening strategies and the Settings tabs

The system provides four tightening strategies for you to customize the tightening processes based on different needs.

1. Standard: there are four stages, including Start, Rundown, Pre-tightening, and Tightening. The settings in each stage can be adjusted individually.
2. Enhanced: further tightening with the Tightening stage from the Standard strategy.
3. Pre-position: drive in the screw to reach the specified angle, torque, or torque rate with the Start and Rundown stages from the Standard strategy.

Stage Strategy	Start	Rundown	Pre-tightening	Tightening
Standard	Start angle	Rundown angle Rundown torque Torque rate	Pre-tightening torque Torque rate	Target angle Target torque Clamp torque Clamp angle
Enhanced	-	-	-	Target angle Target torque
Pre-position	Start angle	Rundown angle Rundown torque Torque rate	-	-

4. Self-defined: arrange the tightening process as needed with a maximum of six stages. Use self-defined strategies for special scenarios.

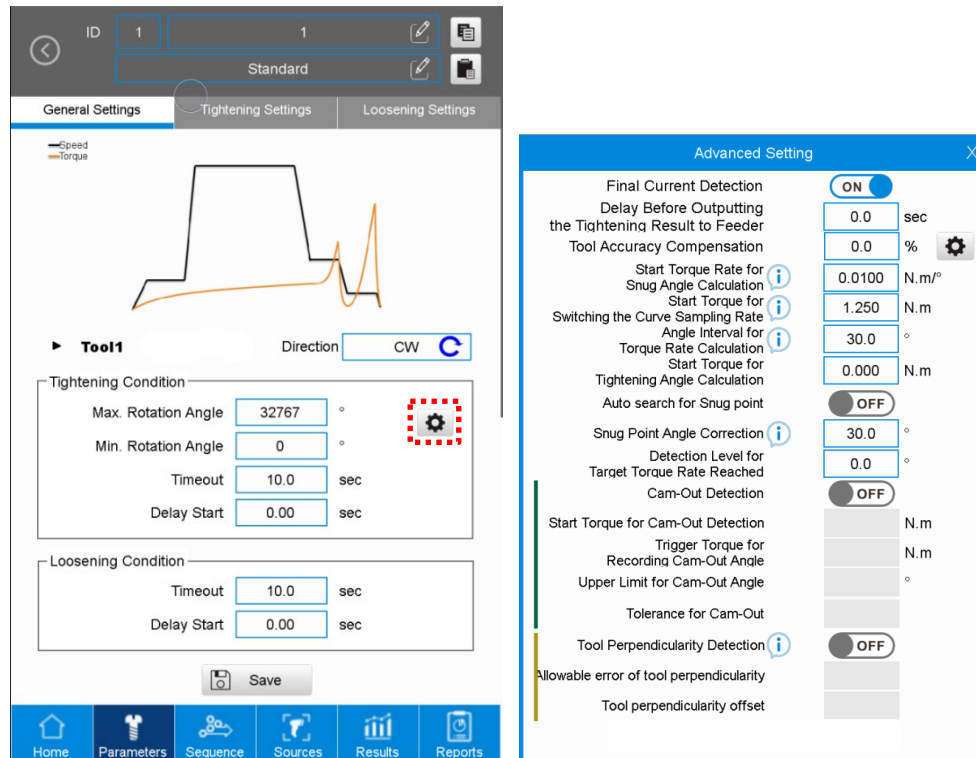
Stage Strategy	Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6
Self-defined	Angle Control Torque Control Torque Rate Control Clamp Torque Control Clamp Angle Control Yield point control					

Each strategy contains three setting tabs: **General Settings**, **Tightening Settings**, and **Loosening Settings**. The following illustrates the setting screens of the three tabs under different tightening strategies.

5.3.1 General Settings

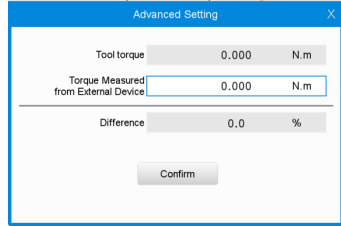
The screen of the **General Settings** tab is the same (except the curve graphs) for all tightening strategies. In this tab, you can set the general tightening and loosening conditions.

This section takes the General Settings screen of Standard strategy as an example.



Item		Function description	
Tool 1 / Tool 2		Tool ID for the current parameter.	
Direction		Specify the rotation direction of the tool.	
Tightening Condition	Max. Rotation Angle	Set the maximum and minimum rotation angle of the tightening (1 turn = 360°).	For checking whether the screw length is suitable.
	Min. Rotation Angle		For checking whether there is an under-driven screw.
	Timeout	Limit the total time of the tightening process.	
	Delay Start	Set the delay time before the tightening starts.	
Loosening Condition	Timeout	Limit the total time of the loosening process.	
	Delay Start	Set the delay time before the loosening starts.	
Advanced Setting	Final Current Detection	This function is On by default. It can determine if the screwdriver is degraded by using the controller current sensor and the tool torque sensor to measure the current values at the final tightening stage and then comparing them.	
	Delay Before Outputting the Tightening Result to Feeder	If you have set a DO function as "Tightening Result Sent to Feeder", after the tightening task is complete, the system does not send the DO signal to the feeder for feeding the next screw until this delay time has elapsed.	

5

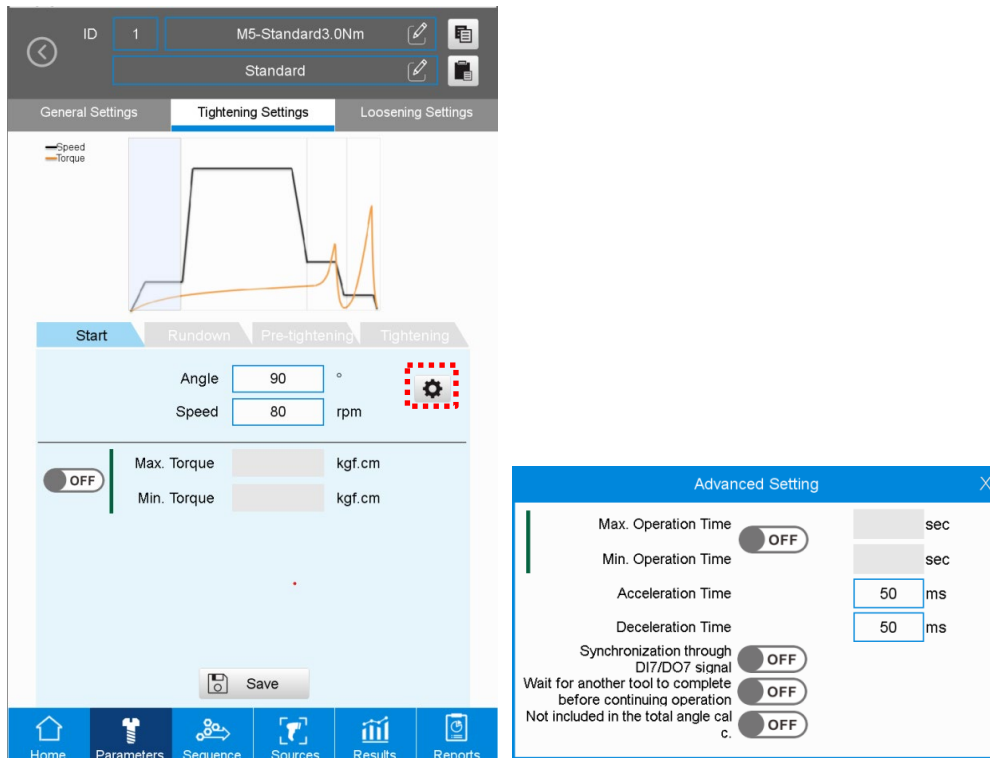
Item		Function description	
Advanced Setting	Tool Accuracy Compensation	Calibrate the tool torque with a third-party calibration device. This function is the same as the Tool Calibration in Section 10.4, except that it only calibrates the parameter being set at present.	
	Start Torque Rate for Snug Angle Calculation	When the set torque rate is reached, the system starts recording the rotation angle of the tool from this time point until the end of the final tightening.	
	Start Torque for Switching the Curve Sampling Rate	The curve sampling rate changes by referring to this torque value. When the tightening torque reaches this value, the sampling rate changes from 10 ms to 4 ms.	
	Angle Interval for Torque Rate Calculation	Torque rate is a curve described by the calculation of dividing the unit torque by the unit angle. Adjusting this interval can filter out the power surge during operation, but when the setting is too high, the curve is more likely to become distorted.	
	Start Torque for Tightening Angle Calculation	If the tool start torque > 0 and reaches this setting value, the rotation angle of the tool from this time point until the end of the final tightening is recorded as the tightening angle. If the tool start torque = 0, the rotation angle in the final stage is recorded as the tightening angle.	
	Auto search for Snug point	In Torque mode, you can set this function to On for calculating the snug point torque after the tightening process is complete.	
	Snug Point Angle Correction	Correct the angle by which the system determines the snug point is reached. For example, if the angle determined by the system is 100° and you set the Snug Point Angle Correction to 10°, then the angle is corrected to 90° (= 100° - 10°).	
	Detection Level for Target Torque Rate Reached	Setting this function can avoid misjudgment of reaching the target torque rate caused by bit camming-out and can effectively filter out the temporary surge in the torque rate during operation.	
	Cam-Out Detection	This function is Off by default. When this function is On, the following four functions can be set.	
	Start Torque for Cam-Out Detection	Set the torque at which the system starts to detect for cam-out. The default setting is 5% of the tool's maximum torque specification.	
	Trigger Torque for Recording Cam-Out Angle	When the tool torque is lower than this setting, the system starts recording the accumulated rotation angle. Once the tool torque is higher than this setting, the record is reset to 0.	
	Upper Limit for Cam-Out Angle	Set the upper limit for the rotation angle. Once the accumulated rotation angle exceeds this setting, it means the cam-out count increments by 1.	
	Tolerance for Cam-Out	Set the tolerance count for cam-out. Once the setting is exceeded, an alarm is triggered.	
	Tool Perpendicularity Detection	This function is Off by default. When this function is On, the following two functions can be set.	
	Allowable error of tool perpendicularity	Set the perpendicularity tolerance zone. If the perpendicularity error exceeds this setting, an alarm is triggered.	
	Tool perpendicularity offset	Calibrate the perpendicularity deviation at system start-up, and then the perpendicularity detection is started from this angle.	

5.3.2 Tightening Settings

5.3.2.1 Standard strategy

When you set the tightening strategy as Standard, the **Tightening Settings** tab contains four stages: Start, Rundown, Pre-tightening, and Tightening.

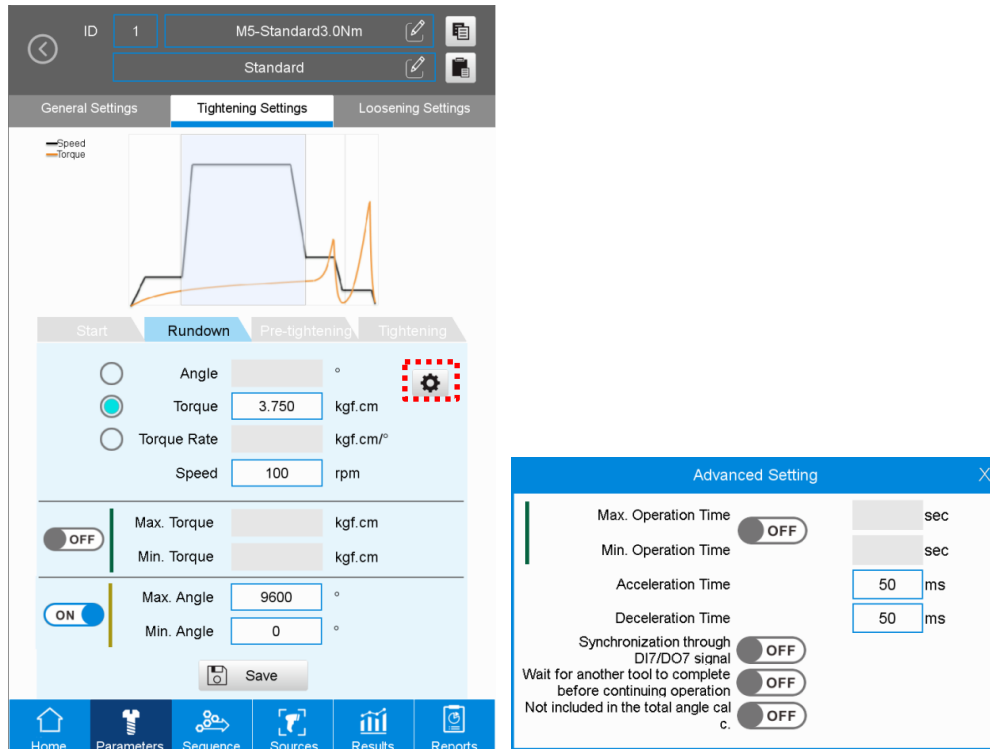
A. Start: drive in the screw at low speed to make sure it aligns with the screw hole.



Item		Function description	
Angle		After the set angle is reached, the system switches to the Rundown stage. The recommended angle is 90° to 360°. (0: skip the Start stage and switch to the Rundown stage).	
Speed		The speed should not be set too high; the recommended speed is 80 rpm.	
Max. Torque*		Set the upper and lower limits for the torque of this stage. You can turn this function on or off.	For checking whether the screw is already tightened.
Min. Torque*			For identifying whether the tool is successfully started.
Advanced Setting	Max. Operation Time	Set the upper and lower limits for the operation time of this stage. You can turn this function on or off.	
	Min. Operation Time		
	Acceleration Time	Adjust the acceleration time for this stage.	
	Deceleration Time	Adjust the deceleration time for this stage.	
	Synchronization through DI7/DO7 signal	For synchronization with tool(s) of the other controller. When this function is On, the functions of DI7 and DO7 signals are force switched to “tool synchronization” and waits for triggering. Meanwhile, the system sends out the DO7 signal to inform the other controller that it can continue operating.	
	Wait for another tool to complete before continuing operation	For tool synchronization operation by a dual-tool controller. When this function is On, the current tool will wait after finishing its operation at this stage until another tool has finished.	
	Not included in the total angle calc.	When this function is On, the rotation angle at this stage does not count in the total angle.	

Note: the torque values set for this stage cannot exceed the torque values set in the Tightening stage.

B. Rundown: drive in the screw at high speed.



Determine if the Rundown stage is complete by the set Angle, Torque, or Torque Rate.

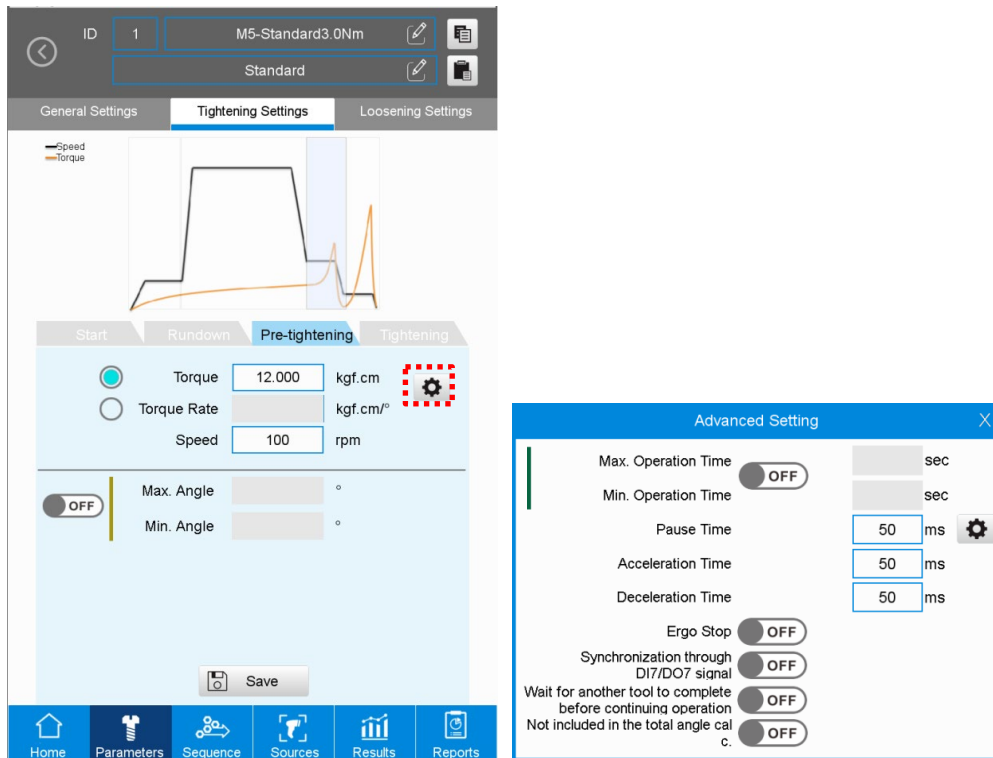
Item	Function description	
Angle	After the set angle is reached, the system switches to the Pre-tightening stage. For example, if the number of threads on the screw is 10, the rundown angle must be smaller than 3600°; the recommended angle is around 2880°, so some margin is kept for the Pre-tightening and Tightening stages.	
Torque*	After the set torque is reached, the system switches to the Pre-tightening stage. The recommended torque is 25% of the target torque.	
Torque Rate	After the set torque rate is reached, the system switches to the Pre-tightening stage. Torque rate is a curve described by the calculation of dividing the unit torque by the unit angle, often used to observe the drastic torque changes during the tightening process. You can refer to the torque rate curve to find when snug point is reached and then set the torque rate as the threshold.	
Speed	It is recommended that you set the speed lower than the maximum speed of the tool specification; the default value is 70% of the tool max. speed.	
Max. Torque*	Set the upper and lower limits of the torque for this stage. You can turn this function on or off.	When there are foreign objects or burrs in the screw holes, the friction increases and leads to excessive torque. Limit the maximum torque by setting the Max. Torque.
Min. Torque*		
Max. Angle	Set the upper and lower limits of the angle for this stage. You can turn this function on or off.	
Min. Angle		

Item		Function description
Advanced Setting	Max. Operation Time	Set the upper and lower limits of the operation time for this stage. You can turn this function on or off.
	Min. Operation Time	
	Acceleration Time	Adjust the acceleration time for this stage.
	Deceleration Time	Adjust the deceleration time for this stage.
	Synchronization through DI7/DO7 signal	For synchronization with tool(s) of the other controller. When this function is On, the functions of DI7 and DO7 signals are force switched to “tool synchronization” and waits for triggering. Meanwhile, the system sends out the DO7 signal to inform the other controller that it can continue operating.
	Wait for another tool to complete before continuing operation	For tool synchronization operation by a dual-tool controller. When this function is On, the current tool will wait after finishing its operation at this stage until another tool has finished.
	Not included in the total angle calc.	When this function is On, the rotation angle at this stage does not count in the total angle.

Note: the torque values set in this stage cannot exceed the torque values set in the Tightening stage, otherwise an alarm occurs when you save the settings.

- C. Pre-tightening: have the screw head sit flush with the object surface, and reach the specified torque requirement.

5



Determine if the Pre-tightening stage is complete by the set Torque or Torque Rate.

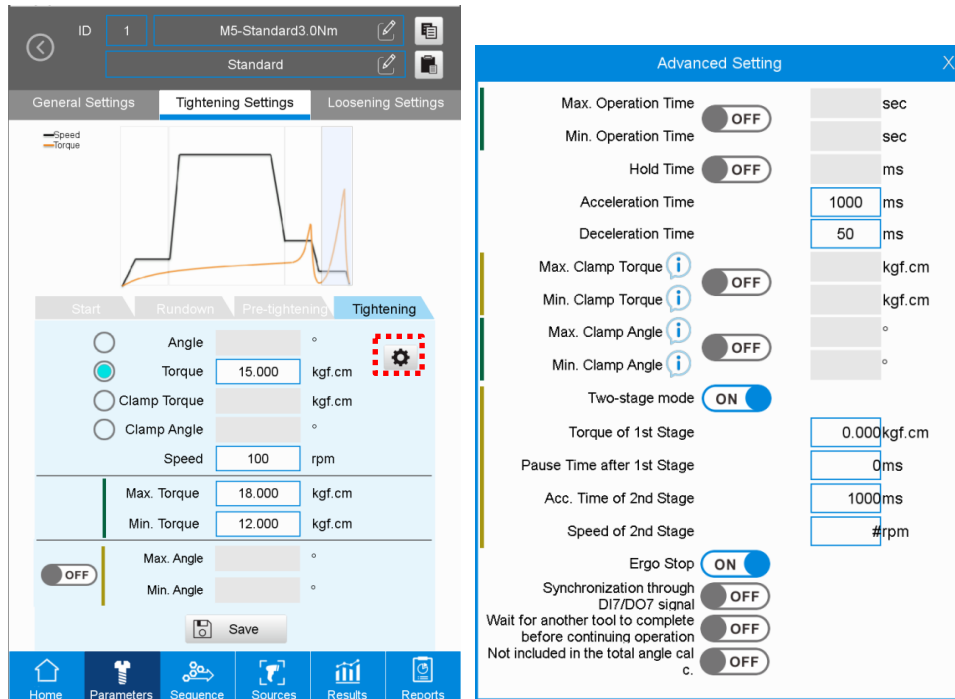
Item		Function description
Torque* ¹		After the set torque is reached, the system switches to the Tightening stage. The recommended torque is 80% of the target torque.
Torque Rate		After the set torque rate is reached, the system switches to the Tightening stage. Torque rate is a curve described by the calculation of dividing the unit torque by the unit angle, often used to observe the drastic torque changes during the tightening process. You can refer to the torque rate curve to find when snug point is reached and then set the torque rate as the threshold.
Speed* ²		In this stage, the screw head is tightened closer to the object surface, so the speed should not be too high; the recommended setting is < 200 rpm.
Max. Angle		Set the upper and lower limits of the angle for this stage. You can turn this function on or off.
Min. Angle		
Advanced Setting	Max. Operation Time	Set the upper and lower limits of the operation time for this stage. You can turn this function on or off.
	Min. Operation Time	
	Pause Time	After the set torque or torque rate is reached, the tool stops for a period of time to release stress. The recommended setting is 50 ms.
	Acceleration Time	Adjust the acceleration time for this stage.
	Deceleration Time	Adjust the deceleration time for this stage.
	Ergo Stop	Enabling this function reduces the discomfort caused by stress released during tool operation.

Item		Function description
Advanced Setting	Synchronization through DI7/DO7 signal	For synchronization with tool(s) of the other controller. When this function is On, the functions of DI7 and DO7 signals are force switched to "tool synchronization" and waits for triggering. Meanwhile, the system sends out the DO7 signal to inform the other controller that it can continue operating.
	Wait for another tool to complete before continuing operation	For tool synchronization operation by a dual-tool controller. When this function is On, the current tool will wait after finishing its operation at this stage until another tool has finished.
	Not included in the total angle calc.	When this function is On, the rotation angle at this stage does not count in the total angle.

Note:

1. The torque value set in this stage cannot exceed the torque value set in the Tightening stage, otherwise an alarm occurs when you save the settings.
2. The speed set for this stage cannot exceed the speed set in the Tightening stage.

D. Tightening: tighten the screw at a low speed to achieve the final tightening results.



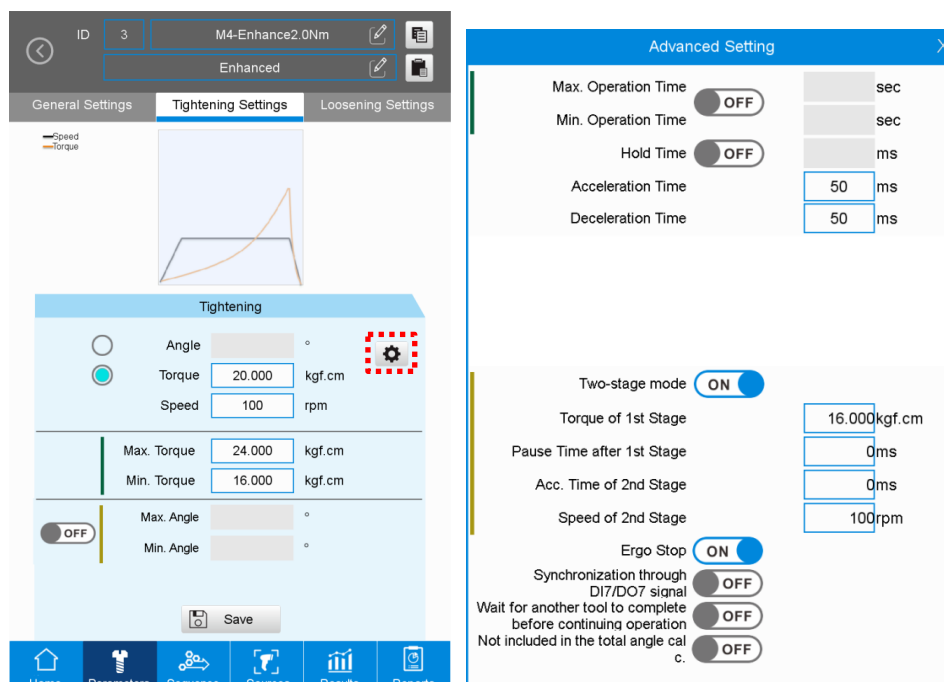
Determine if the Tightening stage is complete by the set Angle, Torque, Clamp Torque, or Clamp Angle.

Item	Function description	
Angle	After the set angle is reached, the Tightening operation ends.	
Torque	After the set torque is reached, the Tightening operation ends.	
Clamp Torque	Set a relative torque, and after the torque value is reached, the Tightening operation ends. - When the Rundown stage is in Torque Rate control: Relative torque = (Final torque) - (Torque when snug point is reached) - When the Rundown stage is not in Torque Rate control: Relative torque = (Final torque) - (Pre-tightening torque)	
Clamp Angle	Set a relative angle, and after the angle value is reached, the Tightening operation ends. - When the Rundown stage is in Torque Rate control: Relative angle = (Final tightening angle) - (Angle when snug point is reached) - When the Rundown stage is not in Torque Rate control: Relative angle = (Final tightening angle) - (Angle when Pre-tightening is complete)	
Speed	During the Tightening stage, do not set the speed too high so as to avoid object surface damage caused by excessive tightening speed. The default setting is 100 rpm.	
Max. Torque	Set the upper and lower limits of the torque for this stage.	For checking whether the tightening results meet the product specification.
Min. Torque		
Max. Angle	Set the upper and lower limits of the angle for this stage. You can turn this function on or off.	The Max. angle is for monitoring the final torque to determine whether the tightening results meet the product specification. The final tightening angle is different according to the hardness of the object material. The softer the material, the larger the angle.
Min. Angle		

Item		Function description	
Advanced Setting	Max. Operation Time	Set the upper and lower limits of the operation time for this stage. You can turn this function on or off.	The Min. Operation Time is for checking whether the actual operation time is too short and determining whether there is an under-driven screw.
	Min. Operation Time		
	Hold Time	Set the duration of keeping the Servo ON state after the tightening process is complete to enhance the tightening effect. The default value is 0 ms. You can turn this function on or off.	
	Acceleration Time	Adjust the acceleration time for this stage.	
	Deceleration Time	Adjust the deceleration time for this stage.	
	Max. Clamp Torque	Set the upper and lower limits of the torque for the duration from the time when snug point is reached to the Tightening stage. You can turn this function on or off.	
	Min. Clamp Torque		
	Max. Clamp Angle	Set the upper and lower limits of the angle for the duration from the time when snug point is reached to the Tightening stage. You can turn this function on or off.	
	Min. Clamp Angle		
	Two-stage mode	Further divide this stage into two stages. You can turn this function on or off.	
		Torque of 1st Stage	Torque of the 1 st stage. The default is 0.8 times of the target torque.
		Pause Time after 1st Stage	The pause before entering the 2 nd stage from the 1 st stage. The default is 0 ms.
		Acc. Time of 2nd Stage	The acceleration time of the 2 nd stage. The default is 1000 ms.
		Speed of 2nd Stage	The rotation speed of the 2 nd stage. The default is 100 rpm.
	Ergo Stop	Enabling this function reduces the discomfort caused by stress released during tool operation.	
	Synchronization through DI7/DO7 signal	For synchronization with tool(s) of the other controller. When this function is On, the functions of DI7 and DO7 signals are force switched to “tool synchronization” and waits for triggering. Meanwhile, the system sends out the DO7 signal to inform the other controller that it can continue operating.	
	Wait for another tool to complete before continuing operation	For tool synchronization operation by a dual-tool controller. When this function is On, the current tool will wait after finishing its operation at this stage until another tool has finished.	
	Not included in the total angle calc.	When this function is On, the rotation angle at this stage does not count in the total angle.	

5.3.2.2 Enhanced strategy

When you set the tightening strategy as Enhanced, the **Tightening Settings** tab only contains the Tightening stage.



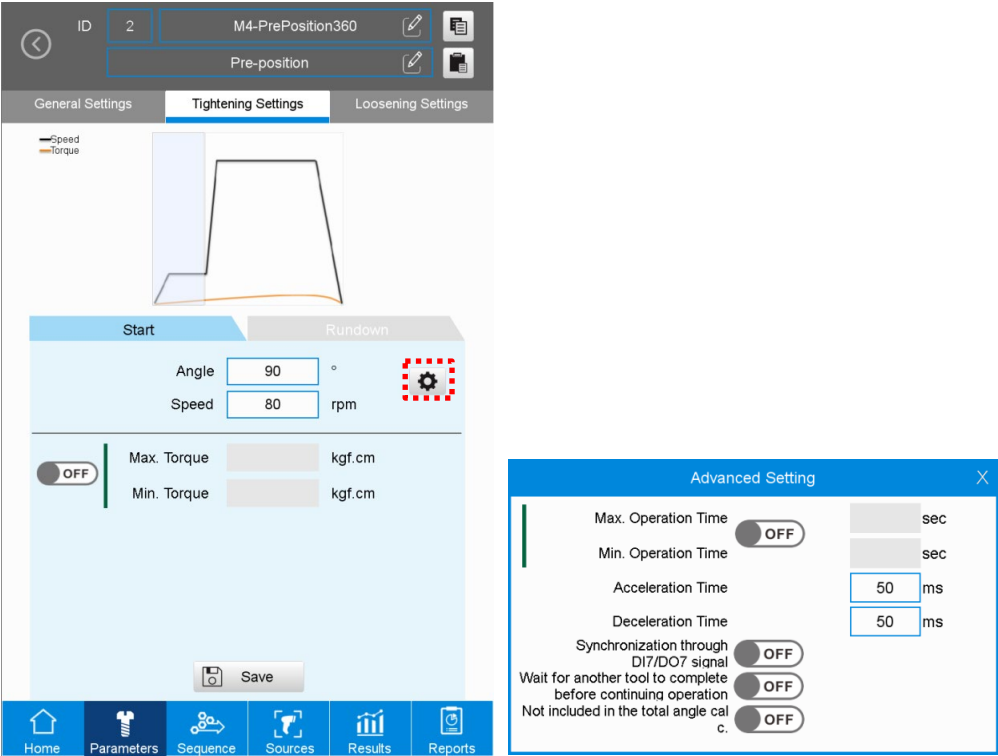
Determine whether the Tightening is complete by the set Angle or Torque.

Item		Function description		
Angle		After the set angle is reached, the Tightening operation ends.		
Torque		After the set torque is reached, the Tightening operation ends.		
Speed		This strategy is for further tightening the screw, so the speed should not be too high. The default speed is 100 rpm.		
Max. Torque		Refer to Section 5.3.2.1 for the Tightening stage under Standard strategy.		
Min. Torque				
Max. Angle				
Min. Angle				
Advanced Setting	Max. Operation Time	Refer to Section 5.3.2.1 for the Tightening stage under Standard strategy.		
	Min. Operation Time			
	Hold Time			
	Acceleration Time			
	Deceleration Time			
	Two-stage mode	Torque of 1st Stage	Refer to Section 5.3.2.1 for the Tightening stage under Standard strategy.	
		Pause Time after 1st Stage		
		Acc. Time of 2nd Stage		
		Speed of 2nd Stage		
	Ergo Stop	Refer to Section 5.3.2.1 for the Tightening stage under Standard strategy.		
Synchronization through DI7/DO7 signal				
Wait for another tool to complete before continuing operation				
Not included in the total angle calc.				

5.3.2.3 Pre-position strategy

When you set the tightening strategy as Pre-position, the **Tightening Settings** tab contains two stages: Start and Rundown.

A. Start



Item		Function description
Angle		After the set angle is reached, the system switches to the Rundown stage. The recommended setting is 90° to 360°. (0: skip the Start stage and switch to the Rundown stage)
Speed		The speed should not be set too high; the recommended setting is < 80 rpm.
Max. Torque		Refer to Section 5.3.2.1 for the Start stage under Standard strategy.
Min. Torque		
Advanced Setting	Max. Operation Time	
	Min. Operation Time	
	Acceleration Time	
	Deceleration Time	
	Synchronization through DI7/DO7 signal	
	Wait for another tool to complete before continuing operation	
Not included in the total angle calc.		

B. Rundown

5

ID2M4-PrePosition360Pre-position

General SettingsTightening SettingsLoosening Settings

SpeedTorque

StartRundown

Angle2880°

Torquekgf.cm

Torque Ratekgf.cm/°

Speed490rpm

Max. Torquekgf.cm

Min. Torquekgf.cm

OFF

Save

Advanced Setting

Max. Operation Timesec

Min. Operation Timesec

Acceleration Time50ms

Deceleration Time50ms

Synchronization through DI7/DO7 signal

Wait for another tool to complete before continuing operation

Not included in the total angle calc.

Determine whether the Rundown stage is complete by the set Angle, Torque, or Torque Rate.

Item		Function description	
Angle		After the set angle is reached, the Rundown operation ends.	
Torque		After the set torque is reached, the Rundown operation ends.	
Torque Rate		After the set torque rate is reached, the Rundown operation ends.	
Speed		It is recommended that you set the speed lower than the maximum speed of the tool specification. The recommended setting is 70% of the tool max. speed.	
Max. Torque		Refer to Section 5.3.2.1 for the Rundown stage under Standard strategy.	
Min. Torque			
Advanced Setting	Max. Operation Time		
	Min. Operation Time		
	Acceleration Time		
	Deceleration Time		
	Synchronization through DI7/DO7 signal		
	Wait for another tool to complete before continuing operation		
	Not included in the total angle calc.		

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5.3.2.4 Self-defined strategy

5.3.2.4.1 Synchronization settings in tightening parameters

Using the functions of “Synchronization through DI7/DO7 signal” and “Wait for another tool to complete before continuing operation” to perform synchronous tightening operations using two controllers. The descriptions are as follows.

1. Create the same tightening parameters: use the Self-defined strategy and set the following four stages; each stage is available for setting the synchronization.

Stage Strategy	Stage 1	Stage 2	Stage 3	Stage 4
Self-defined	Angle Control (synchronization setting is available)	Angle Control (synchronization setting is available)	Torque Rate Control (synchronization setting is available)	Torque Control (synchronization setting is available)

2. Set the operating mode of the system to “Single tool” in the **Sources** page, and then call the tightening parameter set in Step 1.
3. In the **DI/DO** tab of the **Controller** page, set the function of DO7 to “Tighten to waiting stage” and the function of DI7 to “Wait for external signal before continuing tighten”.

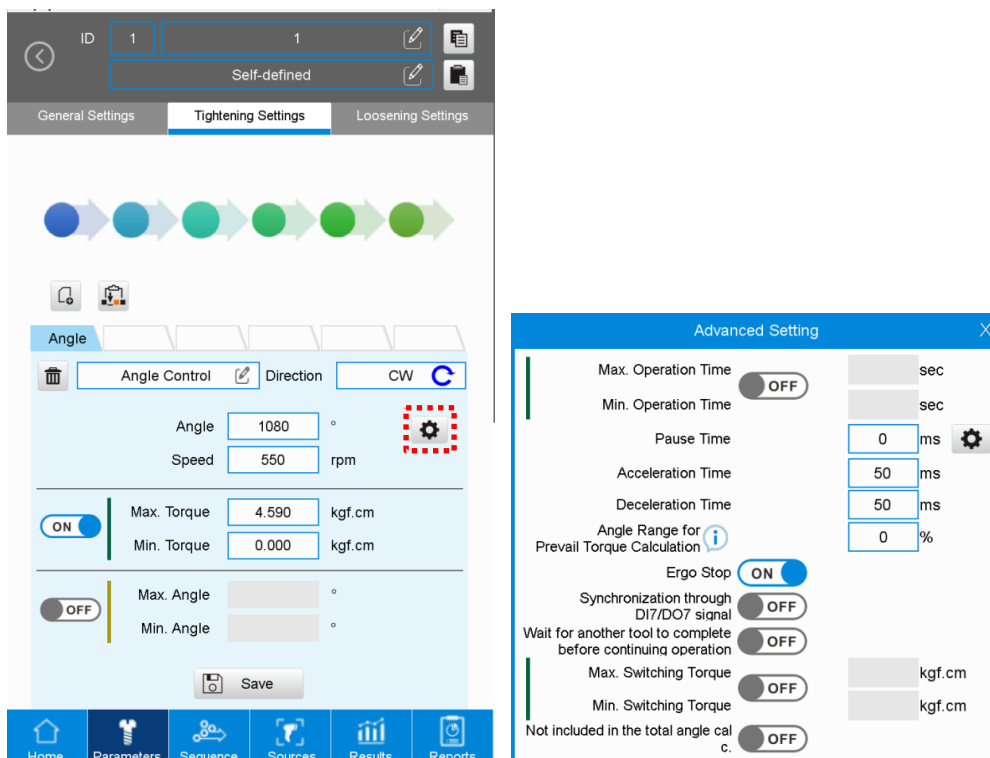
The following describes the synchronization settings for different application scenarios:

Scenario		Dual-tool synchronization (one dual-tool controller and two tools)		Dual-tool synchronization (two single-tool controllers, each with one tool)		Four-tool synchronization (two dual-tool controllers, each with two tools)	
Function		Wait for another tool to complete before continuing operation	Synchronization through DI7/DO7 signal	Wait for another tool to complete before continuing operation	Synchronization through DI7/DO7 signal	Wait for another tool to complete before continuing operation	Synchronization through DI7/DO7 signal
Controller A	Tool 1	On	Off	N/A	On; connect DO7 of Controller A to DI7 of Controller B	On	On; connect DO7 of Controller A to DI7 of Controller B
	Tool 2	On	Off	-	-	On	Off
Controller B	Tool 1	-	-	N/A	On; connect DO7 of Controller B to DI7 of Controller A	On	On; connect DO7 of Controller B to DI7 of Controller A
	Tool 2	-	-	-	-	On	Off

5.3.2.4.2 Control methods descriptions

When you set the tightening strategy as Self-defined, up to six stages can be set in the **Tightening Settings** tab. You can specify the tool rotation direction and control method for each stage. The control methods are detailed as follows.

A. Angle Control: when tightening, the tool stops after the specified angle is reached.



Item		Function description
Angle		After the set angle is reached, the operation for this stage ends.
Speed		Set the operation speed of this stage.
Max. Torque		Set the upper and lower limits for the torque of this stage. You can turn this function on or off.
Min. Torque		
Max. Angle		Set the upper and lower limits for the angle of this stage. You can turn this function on or off.
Min. Angle		
Advanced Setting	Max. Operation Time	Set the upper and lower limits for the operation time of this stage. You can turn this function on or off.
	Min. Operation Time	
	Pause Time	Set the tool pause time after the set angle is reached. After the operation for this stage ends, the system does not enter the next stage until the pause time has elapsed. (0: disable the function) ■ Release torque during pause: if this function is On during the pause time, the torque is released. Advanced Setting Release torque during pause OFF
	Acceleration Time	Adjust the acceleration time of this stage.
	Deceleration Time	Adjust the deceleration time of this stage.

Item		Function description
Advanced Setting	Angle Range for Prevail Torque Calculation	Set the angle range for which you want to calculate the average torque. For example, if the total rotation angle in this stage is 1000°, and the Angle Range for Prevail Torque Calculation is set as 70%, then the average torque of the last 700° in this stage is automatically calculated during the tightening process.
	Ergo Stop	Enabling this function reduces the discomfort caused by stress released during tool operation.
	Synchronization through DI7/DO7 signal	For synchronization with tool(s) of the other controller. When this function is On, the functions of DI7 and DO7 signals are force switched to “tool synchronization” and waits for triggering. Meanwhile, the system sends out the DO7 signal to inform the other controller that it can continue operating.
	Wait for another tool to complete before continuing operation	For tool synchronization operation by a dual-tool controller. When this function is On, the current tool will wait after finishing its operation at this stage until another tool has finished.
	Max. Switching Torque	Set the upper and lower limits for the ending torque of this stage. You can turn this function on or off.
	Min. Switching Torque	
	Not included in the total angle calc.	When this function is On, the rotation angle at this stage does not count in the total angle.

B. Torque Control: when tightening, the tool stops after the specified torque is reached.

The screenshot displays the SD3 Series Controller interface. On the left, the 'Torque Control' tab is active, showing a sequence of colored circles (blue, green, yellow, red) representing different stages. Below this, the 'Torque Control' section is set to 'CW' (Clockwise). The 'Torque' is set to 2.550 kgf.cm and 'Speed' is 550 rpm. There are also fields for 'Max. Torque' (4.590 kgf.cm) and 'Min. Torque' (0.000 kgf.cm), and 'Max. Angle' (9600 °) and 'Min. Angle' (0 °). A 'Save' button is at the bottom. On the right, the 'Advanced Setting' window is open, showing various parameters: 'Max. Operation Time' (OFF), 'Min. Operation Time' (OFF), 'Hold Time' (OFF), 'Pause Time' (0 ms), 'Acceleration Time' (50 ms), 'Deceleration Time' (50 ms), 'Link to Prevail Torque' (OFF), 'Max. Clamp Torque' (OFF), 'Min. Clamp Torque' (OFF), 'Max. Clamp Angle' (OFF), 'Min. Clamp Angle' (OFF), 'Two-stage mode' (ON), 'Torque of 1st Stage' (2.039 kgf.cm), 'Pause Time after 1st Stage' (0 ms), 'Acc. Time of 2nd Stage' (181 ms), 'Speed of 2nd Stage' (550 rpm), 'Ergo Stop' (ON), 'Synchronization through DI7/DO7 signal' (OFF), 'Wait for another tool to complete before continuing operation' (OFF), 'Max. Switching Torque' (OFF), 'Min. Switching Torque' (OFF), and 'Not included in the total angle calculation' (OFF).

Item		Function description	
Torque		After the set torque is reached, the operation for this stage ends.	
Speed		Set the operation speed of this stage.	
Max. Torque		Refer to Section 5.3.2.4.2 for the Angle Control under Self-defined strategy.	
Min. Torque			
Max. Angle			
Min. Angle			
Advanced Setting	Max. Operation Time		
	Min. Operation Time		
	Hold Time	Refer to Section 5.3.2.1 for the Tightening stage under Standard strategy.	
	Pause Time	After the set torque is reached, the tool stops for a period of time to release stress.	
	Acceleration Time	Refer to Section 5.3.2.4.2 for the Angle Control under Self-defined strategy.	
	Deceleration Time		
	Link to Prevail Torque	Link to the prevail torque saved in different tightening parameters to compensate the final torque in this stage. You can turn this function on or off.	
	Max. Clamp Torque	Refer to Section 5.3.2.1 for the Tightening stage under Standard strategy.	
	Min. Clamp Torque		
	Max. Clamp Angle		
	Min. Clamp Angle		
	Two-stage mode	Torque of 1st Stage	Refer to Section 5.3.2.1 for the Tightening stage under Standard strategy.
		Pause Time after 1st Stage	
Acc. Time of 2nd Stage			
Speed of 2nd Stage			

Item		Function description
Advanced Setting	Ergo Stop	Refer to Section 5.3.2.4.2 for the Angle Control under Self-defined strategy.
	Synchronization through DI7/DO7 signal	
	Wait for another tool to complete before continuing operation	
	Max. Switching Torque	
	Min. Switching Torque	
	Not included in the total angle calc.	

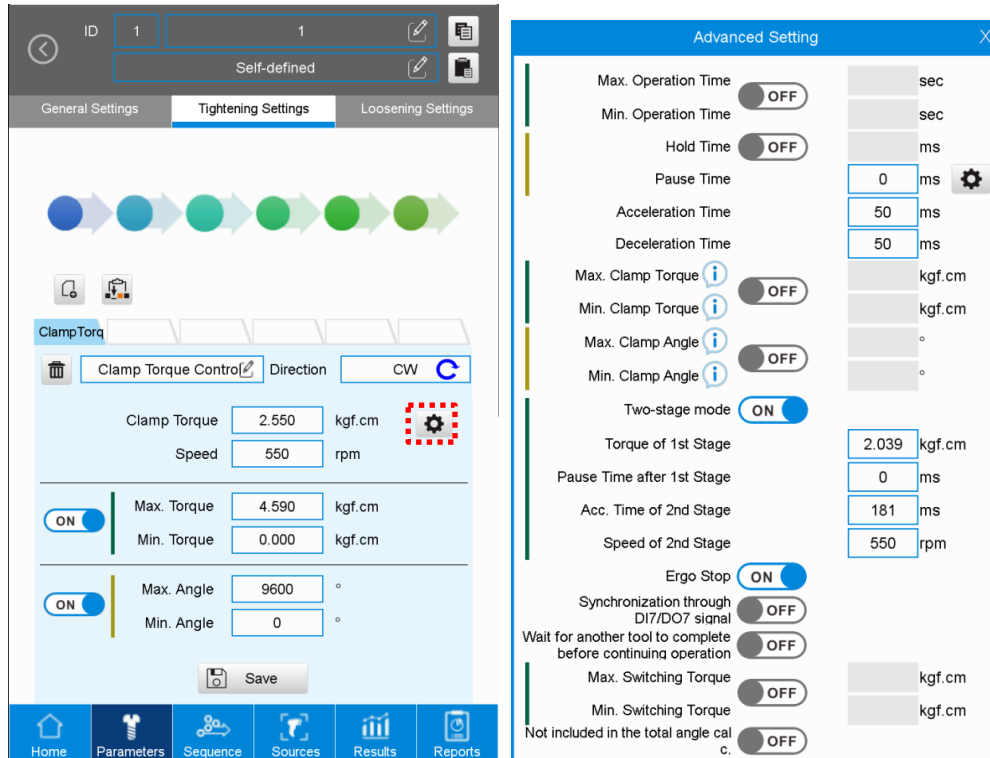
- C. Torque Rate Control: when tightening, the tool stops after the specified torque rate is reached.

5

The screenshot displays the 'Tightening Settings' window of the SD3 Series Controller. The 'Torque Rate Control' is selected, and the 'Direction' is set to 'CW'. The 'Torque Rate' is set to 0.1020 kgf.cm/r, and the 'Speed' is set to 550 rpm. The 'Max. Torque' is 4.590 kgf.cm, and the 'Min. Torque' is 0.000 kgf.cm. The 'Max. Angle' is 9600 degrees, and the 'Min. Angle' is 0 degrees. The 'Advanced Setting' window is also visible, showing various parameters like 'Max. Operation Time', 'Min. Operation Time', 'Pause Time', 'Acceleration Time', 'Deceleration Time', 'Ergo Stop', 'Synchronization through DI7/DO7 signal', 'Max. Switching Torque', 'Min. Switching Torque', and 'Not included in the total angle calc.'.

Item		Function description
Torque Rate		After the set torque rate is reached, the operation for this stage ends.
Speed		Set the operation speed of this stage.
Max. Torque		Refer to Section 5.3.2.4.2 for the Angle Control under Self-defined strategy.
Min. Torque		
Max. Angle		
Min. Angle		
Advanced Setting	Max. Operation Time	
	Min. Operation Time	
	Pause Time	After the set torque rate is reached, the tool stops for a period of time to release stress.
	Acceleration Time	Refer to Section 5.3.2.4.2 for the Angle Control under Self-defined strategy.
	Deceleration Time	
	Ergo Stop	
	Synchronization through DI7/DO7 signal	
	Wait for another tool to complete before continuing operation	
	Max. Switching Torque	
	Min. Switching Torque	
	Not included in the total angle calc.	

D. Clamp Torque Control: when tightening, the tool stops after the relative torque is reached.



Item		Function description
Clamp Torque		Set a relative torque, and after the torque value is reached, the operation for this stage ends. ■ When the previous stage is in Torque Rate Control: Relative torque = (Final torque) - (Torque when snug point is reached) ■ When the previous stage is not in Torque Rate Control: Relative torque = (Final torque) - (Torque of the previous stage) ■ If this is the first stage, set the Clamp Torque to the target torque, which must not exceed the Max. Torque of this stage.
Speed		Set the operation speed of this stage.
Max. Torque		Refer to Section 5.3.2.4.2 for the Angle Control under Self-defined strategy.
Min. Torque		
Max. Angle		
Min. Angle		
Advanced Setting	Max. Operation Time	Refer to Section 5.3.2.1 for the Tightening stage under Standard strategy.
	Min. Operation Time	
	Hold Time	Refer to Section 5.3.2.1 for the Tightening stage under Standard strategy.
	Pause Time	Refer to Section 5.3.2.4.2 for the Angle Control under Self-defined strategy.
	Acceleration Time	Refer to Section 5.3.2.1 for the Tightening stage under Standard strategy.
	Deceleration Time	
	Max. Clamp Torque	
	Min. Clamp Torque	
	Max. Clamp Angle	
	Min. Clamp Angle	

Item		Function description	
Advanced Setting	Two-stage mode	Torque of 1st Stage	Refer to Section 5.3.2.1 for the Tightening stage under Standard strategy.
		Pause Time after 1st Stage	
		Acc. Time of 2nd Stage	
		Speed of 2nd Stage	
	Ergo Stop	Refer to Section 5.3.2.4.2 for the Angle Control under Self-defined strategy.	
	Synchronization through DI7/DO7 signal		
	Wait for another tool to complete before continuing operation		
	Max. Switching Torque		
	Min. Switching Torque		
	Not included in the total angle calc.		

E. Clamp Angle Control: when tightening, the tool stops after the relative angle is reached.

1

1

Self-defined

General Settings

Tightening Settings

Loosening Settings

Clamp Angle

Clamp Angle Control

Direction

CW

Clamp Angle

1080

°

Speed

550

rpm

Max. Torque

4.590

kgf.cm

Min. Torque

0.000

kgf.cm

Max. Angle

Min. Angle

Save

Home

Parameters

Sequence

Sources

Results

Reports

Advanced Setting

Max. Operation Time

Min. Operation Time

Pause Time

Acceleration Time

Deceleration Time

Max. Clamp Torque

Min. Clamp Torque

Max. Clamp Angle

Min. Clamp Angle

Ergo Stop

Synchronization through DI7/DO7 signal

Wait for another tool to complete before continuing operation

Max. Switching Torque

Min. Switching Torque

Not included in the total angle cal c.

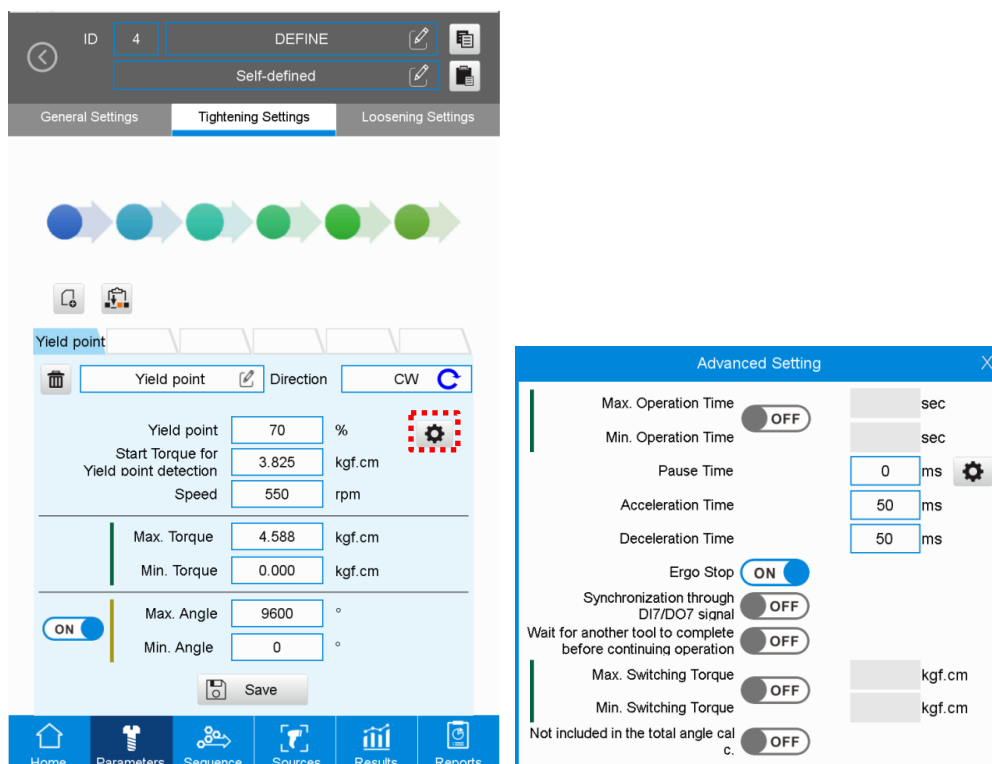
Item		Function description	
Clamp Angle		Set a relative angle, and after the angle value is reached, the operation for this stage ends. - When the previous stage is in Torque Rate Control: Relative angle = (Final tightening angle) - (Angle when snug point is reached) - When the previous stage is not in Torque Rate Control: Relative angle = (Final tightening angle) - (Angle of the previous stage) - If this is the first stage, set the Clamp Angle to the target angle, which must not exceed the Max. Angle of this stage.	
Speed		Set the operation speed of this stage.	
Max. Torque			
Min. Torque			
Max. Angle			
Min. Angle			
Advanced Setting	Max. Operation Time	Refer to Section 5.3.2.4.2 for the Angle Control under Self-defined strategy.	
	Min. Operation Time		
	Pause Time		
	Acceleration Time		
	Deceleration Time		
	Max. Clamp Torque	Refer to Section 5.3.2.1 for the Tightening stage under Standard strategy.	
	Min. Clamp Torque		
	Max. Clamp Angle		
	Min. Clamp Angle		

5

5-25

Item		Function description
Advanced Setting	Ergo Stop	Refer to Section 5.3.2.4.2 for the Angle Control under Self-defined strategy.
	Synchronization through DI7/DO7 signal	
	Wait for another tool to complete before continuing operation	
	Max. Switching Torque	
	Min. Switching Torque	
	Not included in the total angle calc.	

F. Yield point control: when tightening, the tool stops after reaching the specified yield point.

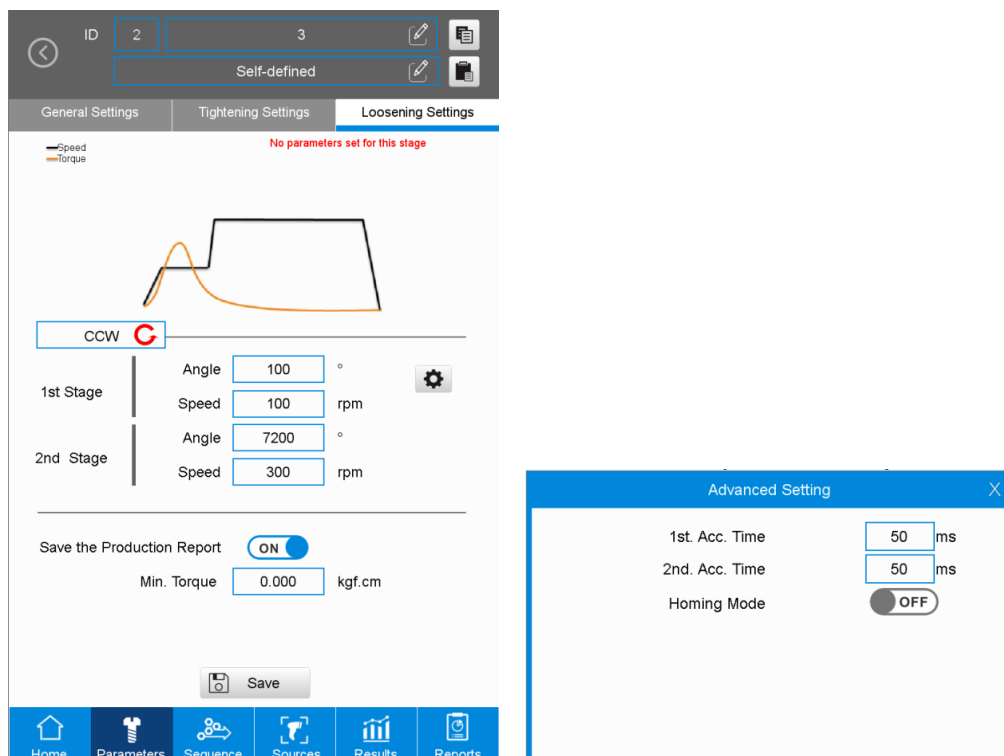


Item		Function description
Yield point		The tool stops the operation of this stage after reaching the maximum torque rate (%) set in this field.
Start Torque for Yield point detection		Set the start torque at which the yield point is being detected.
Speed		Set the operation speed of this stage.
Max. Torque		Refer to Section 5.3.2.4.2 for the Angle Control under Self-defined strategy.
Min. Torque		
Max. Angle		
Min. Angle		
Advanced Setting	Max. Operation Time	
	Min. Operation Time	
	Pause Time	
	Acceleration Time	
	Deceleration Time	
	Ergo Stop	
	Synchronization through DI7/DO7 signal	
	Wait for another tool to complete before continuing operation	
	Max. Switching Torque	
	Min. Switching Torque	
Not included in the total angle calc.		

5.3.3 Loosening settings

The screen of the **Loosening Settings** tab is nearly the same for all tightening strategies, except for the exclusive “CW/CCW” option in the Self-defined strategy and the curve graphs. In this tab, you can set the loosening parameters of two stages.

This section takes the Loosening Settings screen of Self-defined strategy as an example.



Item		Function description	
CW / CCW		You can select the rotation direction in the Loosening stage under Self-defined strategy.	
1st Stage	Angle	After the set angle is reached, the system switches to the second stage. The angle set for this stage must be sufficient for the screwdriver to remove the screw properly.	
	Speed	The torque in the 1 st stage is higher than that in the 2 nd Stage, so the speed should not be set too high. The default speed is 100 rpm.	
2nd Stage	Angle	Set a fixed reverse angle, and the loosening operation automatically stops when the set angle is reached. Note: the system regards that the loosening operation has ended when the set angle is reached or the servo screwdriver is withdrawn during the loosening process.	
	Speed	This can be set higher than the speed of the 1 st stage, but should not exceed the max. speed for the tool. The default value is 70% of the tool max. speed.	
Save the Production Report		Save the result of loosening operation, which can be checked in the Reports page.	
Min. Torque		Set the lower limit of the torque for this stage. You can turn this function on or off.	For identifying whether the screw is properly loosened. When the actual torque is lower than this torque setting, it triggers an alarm.
Advanced Setting	1st. Acc. time	Adjust the 1 st acceleration time.	
	2nd. Acc. time	Adjust the 2 nd acceleration time.	
	Homing Mode	When this function is On, the tool returns to the original angle at system start-up.	

5.4 Saving a tightening parameter

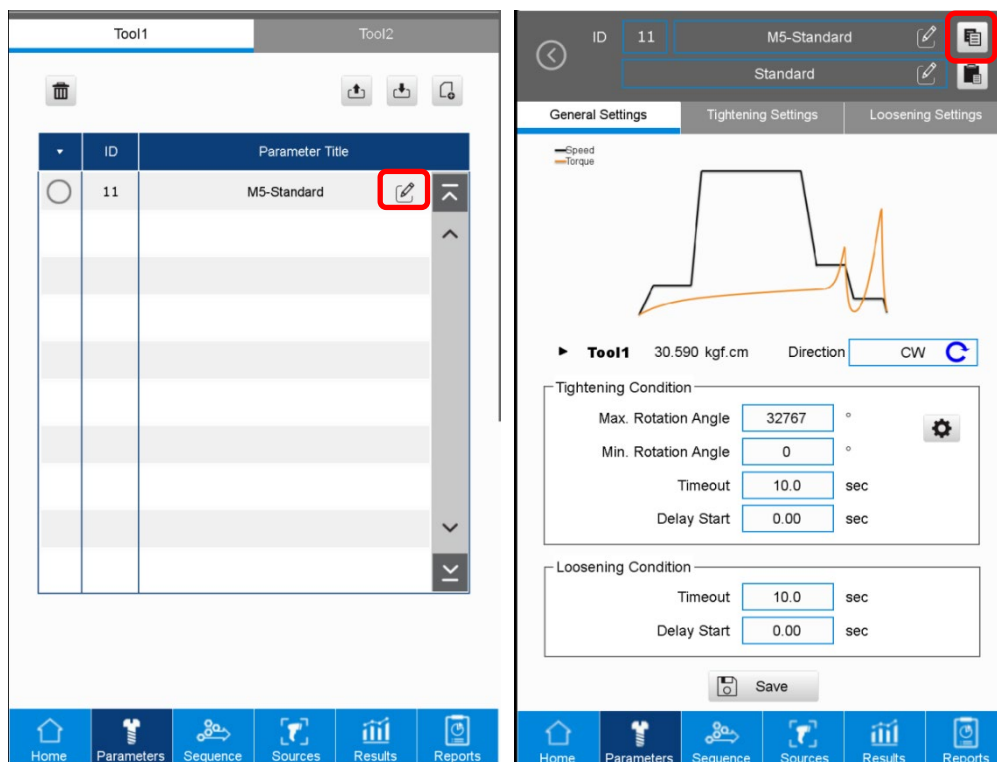
The **Save** button is available in the following situations:

1. After creating a tightening parameter and completing the settings.
2. After modifying an existing tightening parameter.
3. If you have edited a running tightening parameter and then you click **Save**, the **Results** page refreshes the tightening parameter contents.

5.5 Copying / pasting a tightening parameter

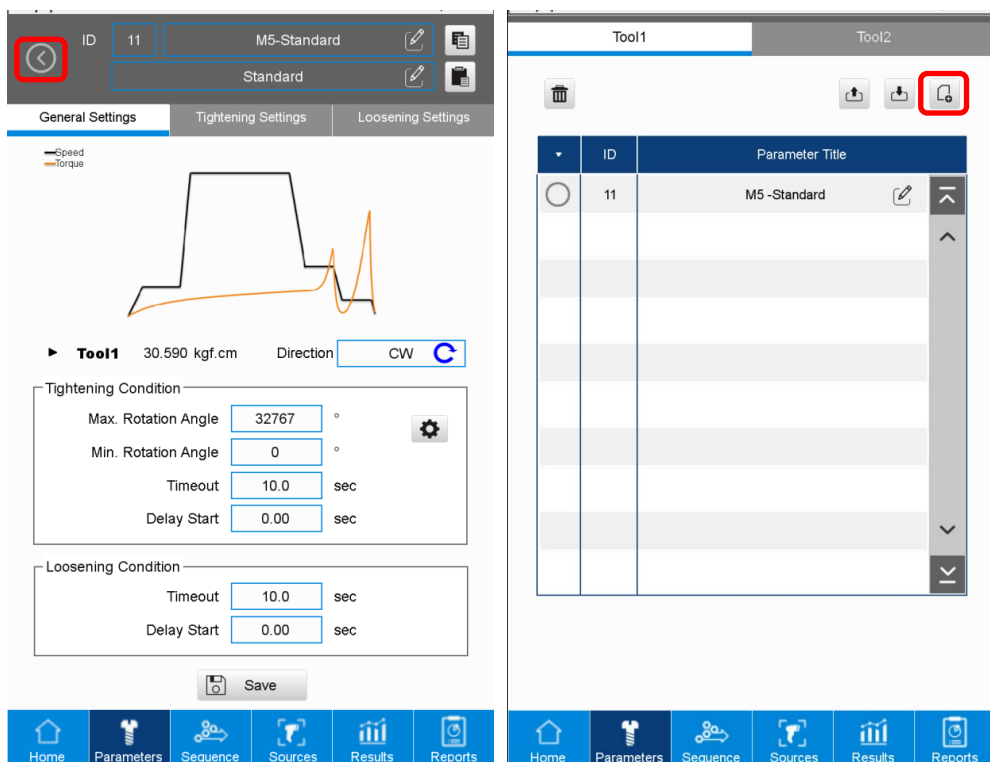
The system provides the Copy (📄) and Paste (📄) functions for editing the tightening parameters more efficiently. The following example illustrates the operation steps.

1. Click the  button to the right of the parameter title to enter the parameter editing screen, and then click  to copy the contents of the tightening parameter.

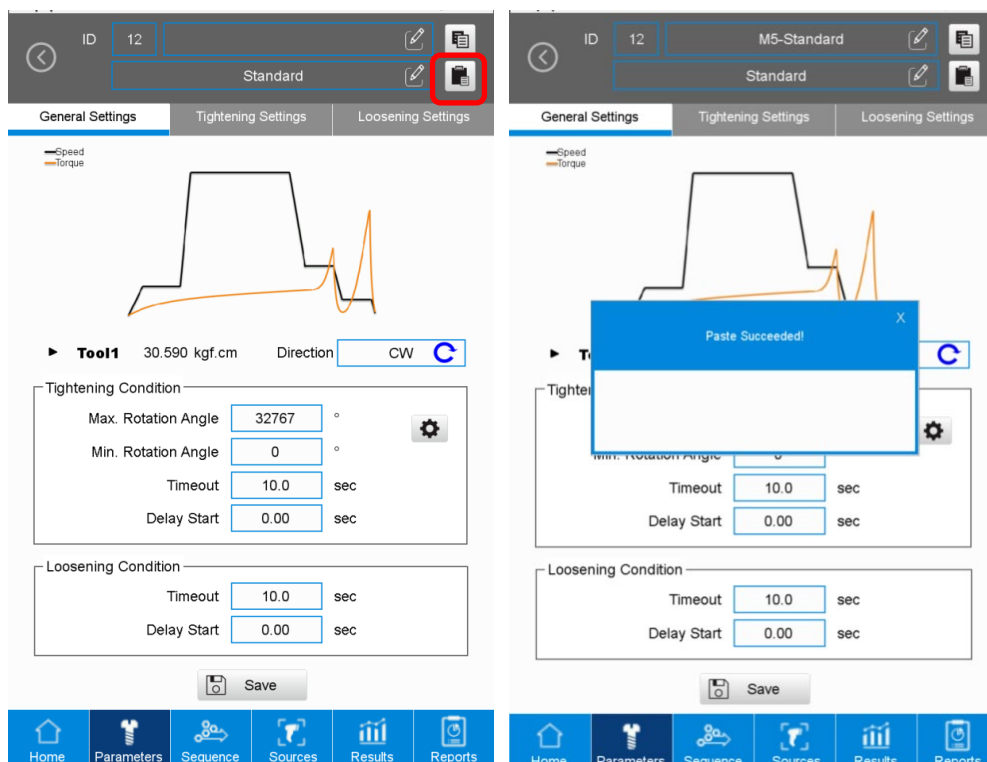


2. Click  to return to the parameter list, and then click  to create a new tightening parameter.

5

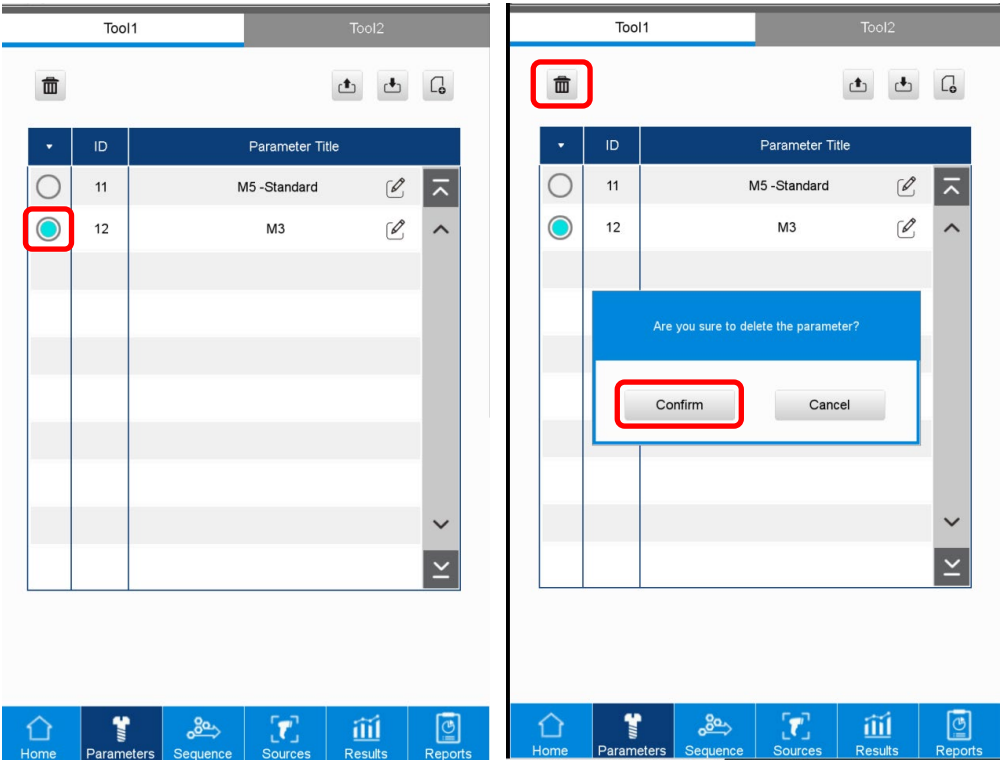


3. Enter the editing screen of the newly created parameter. Click  to paste the copied contents, and the screen displays a window showing pasting successful.



5.6 Deleting a tightening parameter

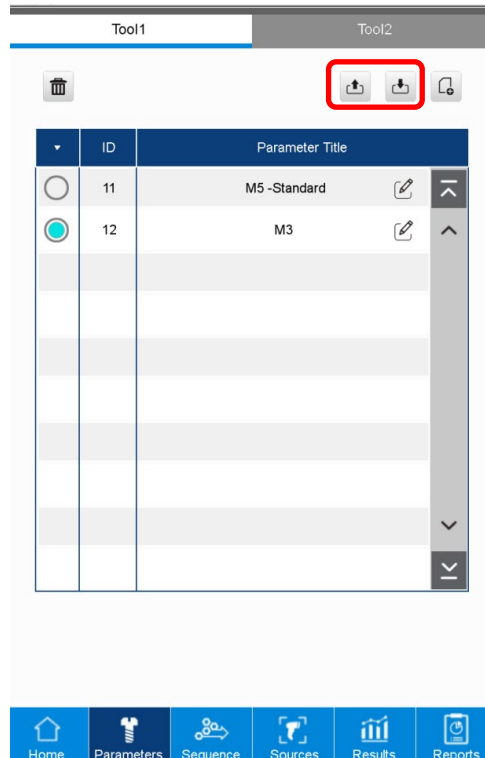
In the parameter list, click the radio button to the left of the parameter, click  , and then click **Confirm** in the confirmation window to delete the parameter.



5

5.7 Exporting / importing a tightening parameter

You can use a USB flash drive or the HMI Disk with the export () and import () functions for data backup and management.



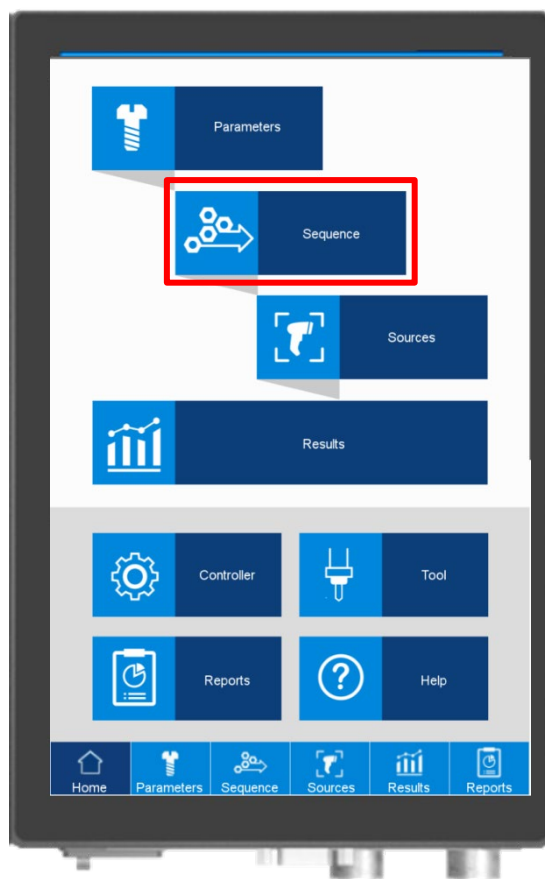
Sequence (Tightening sequence)

6

This chapter introduces the **Sequence** page of the smart screwdriving controller. You can create the screw tightening sequences for different tightening requirements, and export data to / import data from the USB flash drive or HMI Disk.

6.1	Sequence page overview	6-2
6.2	Creating a tightening sequence	6-4
6.3	Saving a tightening sequence	6-8
6.4	Copying / pasting a tightening sequence	6-9
6.5	Deleting a tightening sequence	6-11
6.6	Exporting / importing a tightening sequence	6-12

6.1 Sequence page overview



You can create and configure the tightening sequences in the **Sequence** page. Up to 500 sets of tightening sequences can be created; each set can store a maximum of 100 tightening parameters.

The **Sequence** page displays the ID, Mode, and Sequence Title fields, and the content of the Mode field indicates the combination of tool parameters in the tightening sequence.

- Tool 1: the tightening sequence simply includes the tightening parameters of Tool 1.
- Tool 2: the tightening sequence simply includes the tightening parameters of Tool 2.
- Mix: the tightening sequence includes the tightening parameters of both Tool 1 and Tool 2.

The example of screw tightening with a tightening sequence is as follows:

User12025.07.2508:09

	ID	Mode	Sequence Title
☐	1	Tool1	ControllerA
☐	2	Tool1	ControllerB
☐	3	Tool1	ControllerC
☐	4	Tool2	ControllerD
☐	5	Mix	ControllerE

HomeParametersSequenceSourcesResultsReports

User12025.07.2508:12

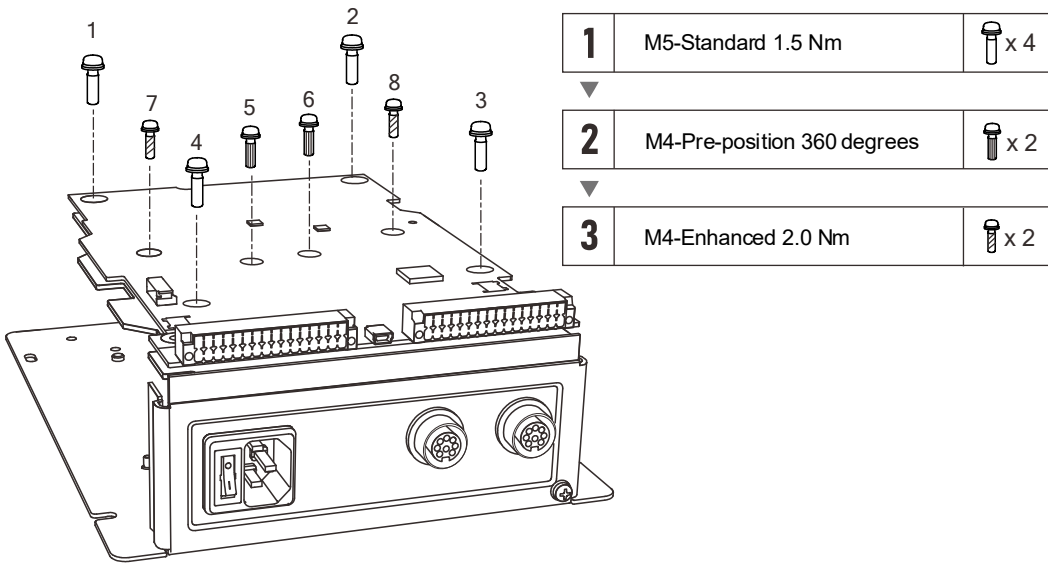
ID5ControllerE

GeneralNavigator

	ID	Tool	Parameter Title	Qty.	Bit ID
☐	1	Tool1	M5-Standard1.5Nm	4	1
☐	2	Tool2	M4-PrePosition360	2	2
☐	3	Tool1	M4-Enhance2.0Nm	2	2

Save

HomeParametersSequenceSourcesResultsReports

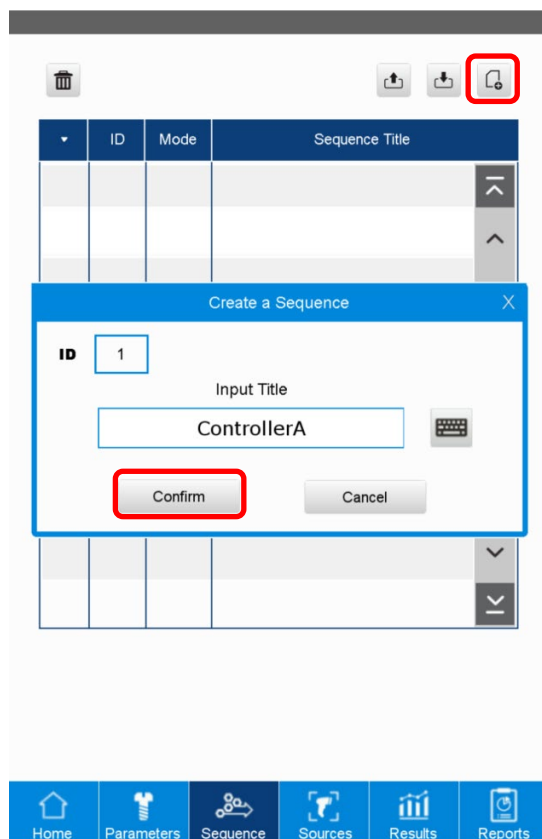


6.2 Creating a tightening sequence

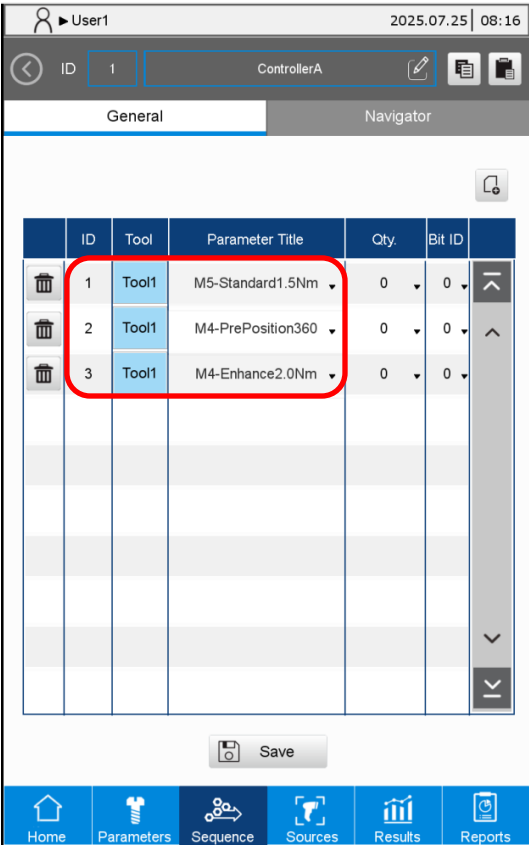
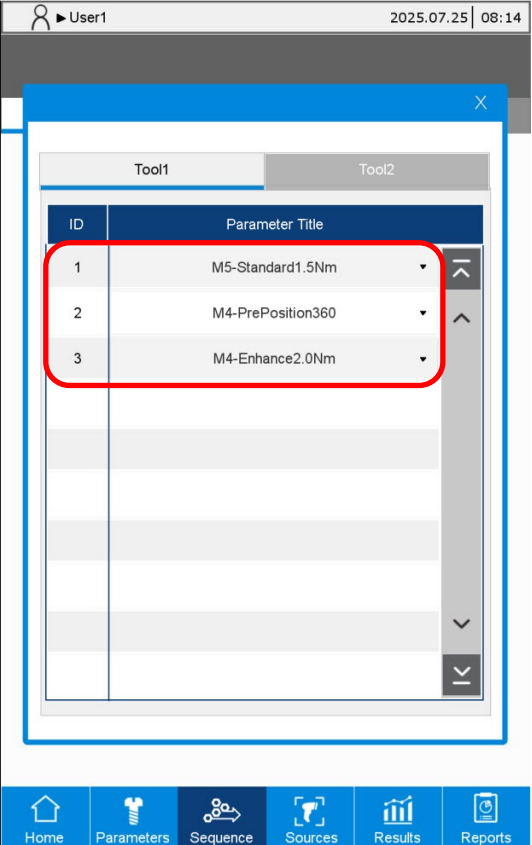
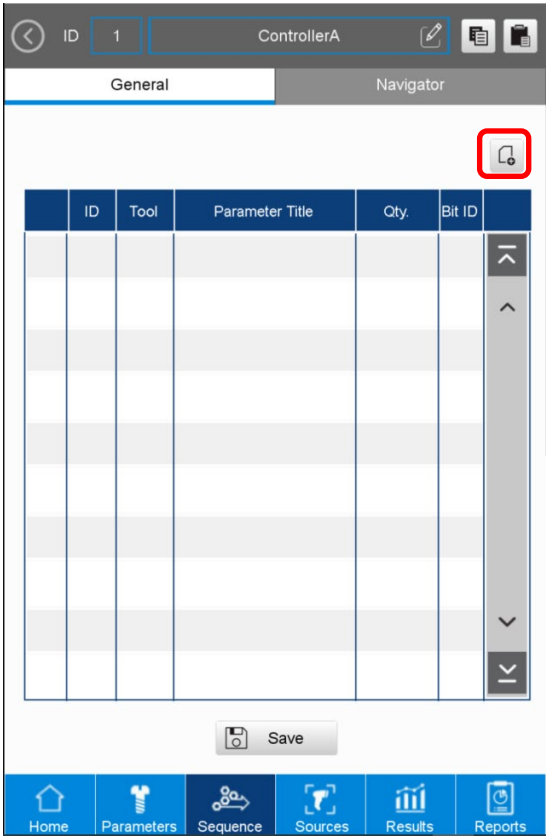
Each tightening sequence can configure a maximum of 100 sets of tightening parameters.

Steps for creating a tightening sequence are detailed as follows:

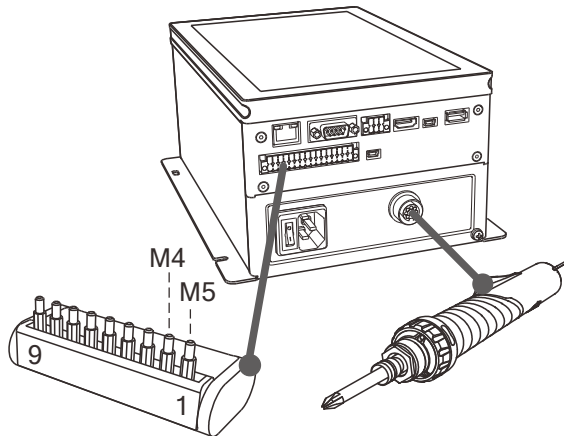
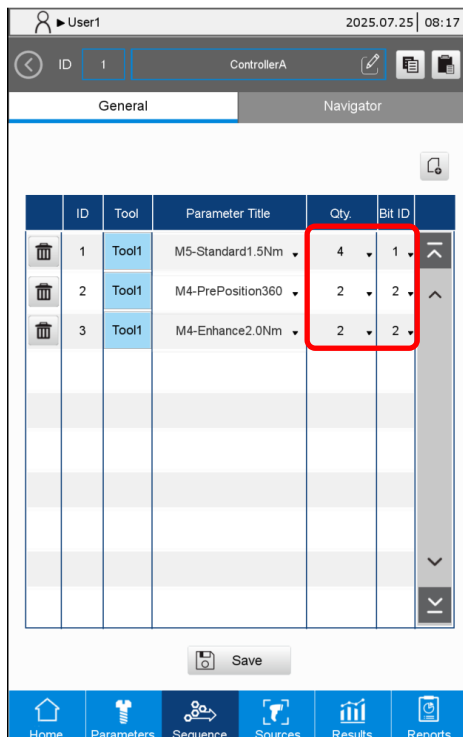
1. Click , input the title of the tightening sequence, and then click **Confirm**.



2. After entering the tightening sequence configuration screen, click  to show the list of all tightening parameters. Click the desired parameter to add it to the configuration screen.

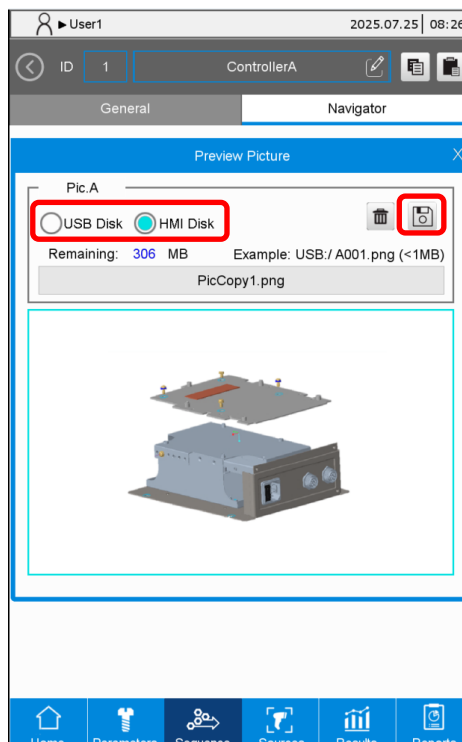
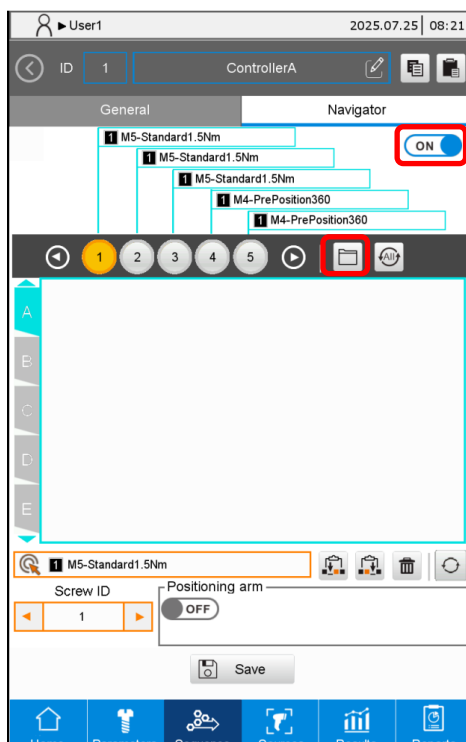


3. Set the fields of Qty. (the number of times to execute this parameter) and Bit ID (the corresponding bit ID number, 0: no corresponding bit) for each tightening parameter.

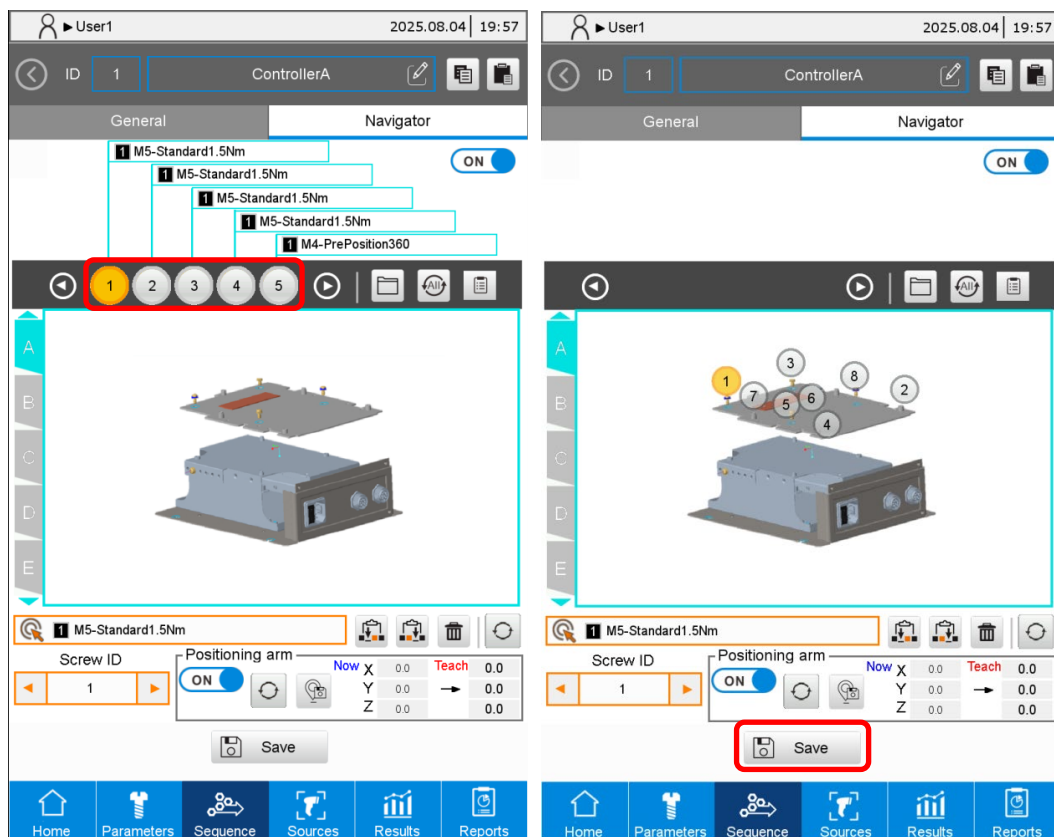


4. Go to the **Navigator** tab. This optional function supports PNG or JPG image files with the size no more than 1 MB. One tightening sequence can store up to 30 image files, and the maximum number of Screw ID is 100.

Click to open the Preview Picture window. Then, select an image from the USB flash drive or the HMI internal memory (need to load images to the HMI through FTP in advance) by clicking **USB Disk** or **HMI Disk**. And then, click to save the image. If the available space is smaller than the image file size, the image cannot be saved.



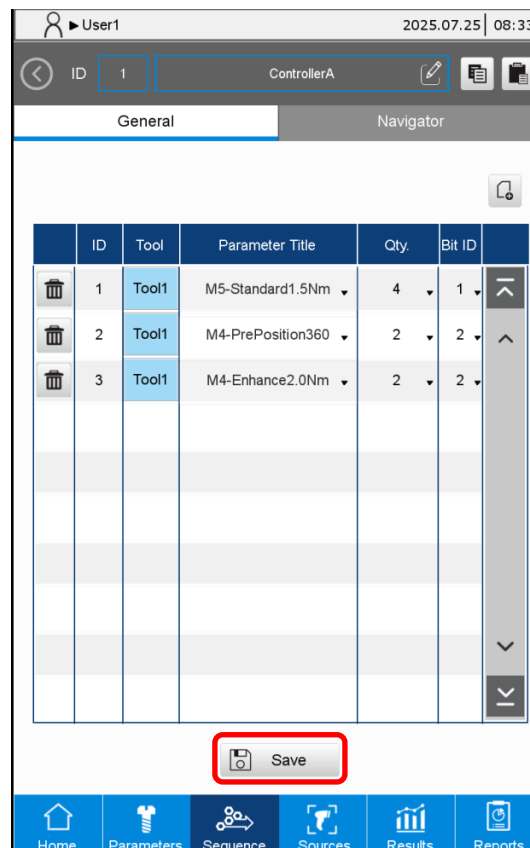
5. Directly enter a screw ID or click  or  to specify a screw, and then drag the screw ID to the desired positions on the image.
6. Switch on the **Positioning arm** function. Use this optional function together with the positioning arm setting under **Controller > Peripheral Device > RS485_1** to set the coordinate tolerance of the positioning arm for the current screw (You can also directly enter a screw ID or click  or  to specify a screw). For the current screw, clicking  saves the coordinates of positioning arm; clicking  clears the coordinates of positioning arm. Click  to view the XYZ coordinates of positioning arm for all screws. Refer to Appendix C for detailed description of positioning arm settings.
7. Click **Save** to save the settings.



6.3 Saving a tightening sequence

The **Save** button in the tightening sequence configuration screen is available in the following situations:

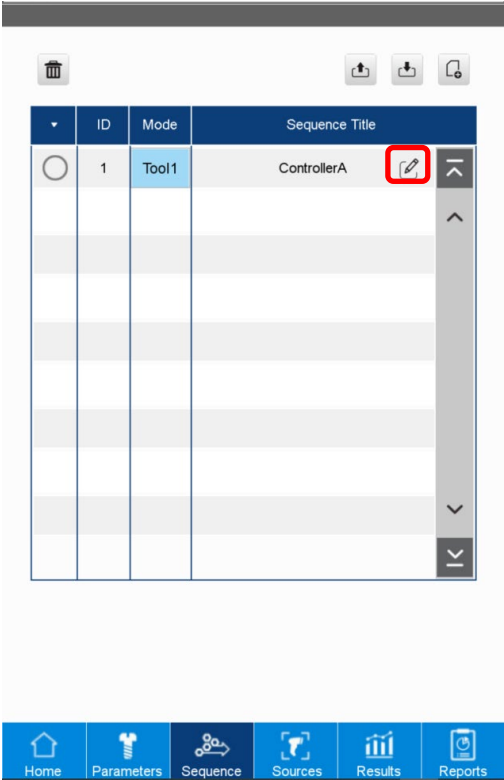
- After creating a tightening sequence and completing the settings.
- After modifying an existing tightening sequence.
- If you have edited a running tightening sequence and then you click **Save**, the **Results** page refreshes the tightening sequence contents and resets the total screw quantity, parameter quantity, and screw quantity of the parameter.



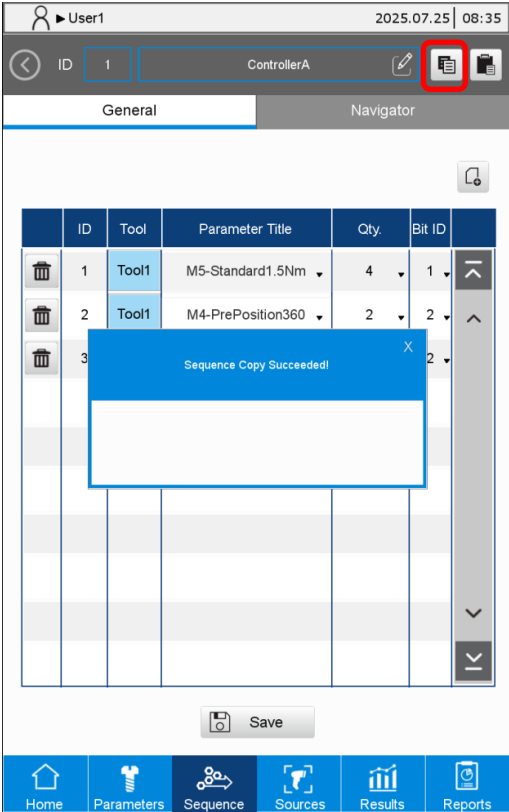
6.4 Copying / pasting a tightening sequence

The system provides the Copy (📄) and Paste (📄) functions for editing the tightening sequences more efficiently. The following illustrates the operation steps.

1. Click the  button to the right of the sequence title.

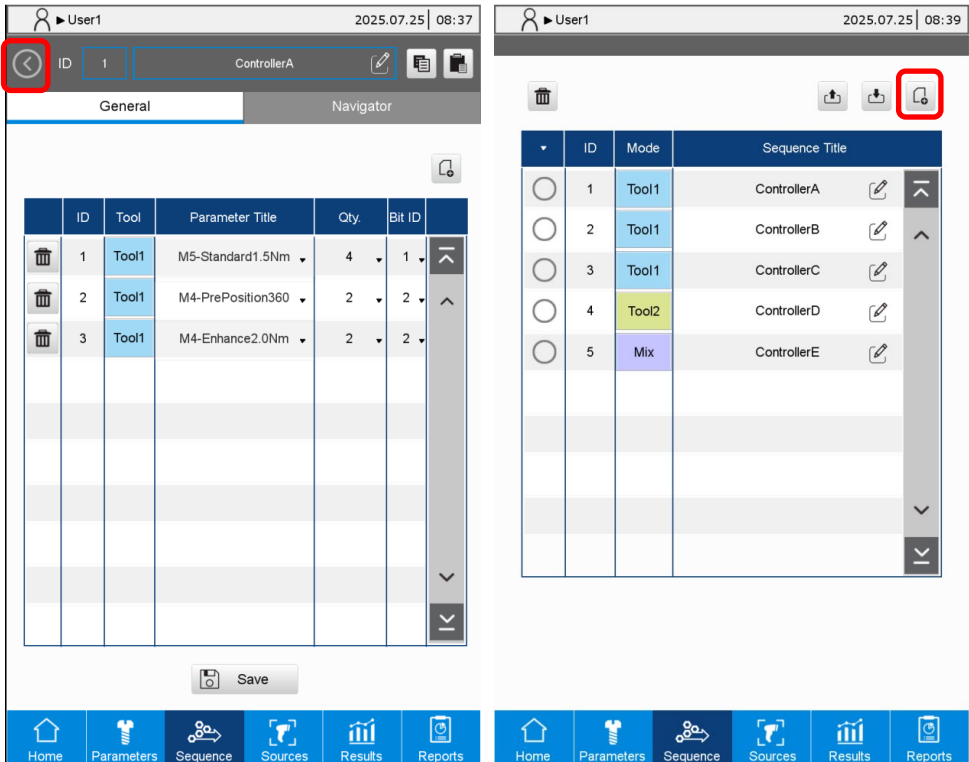


2. Click  to copy the contents of the selected tightening sequence.



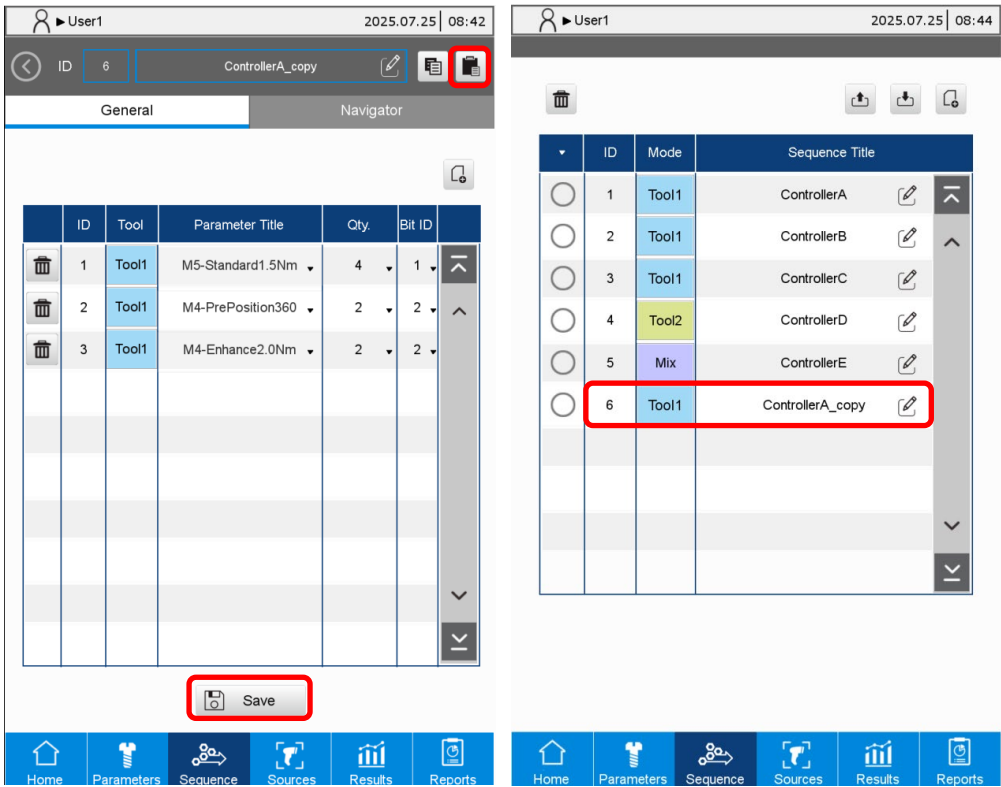
6

3. Click  to return to the list of tightening sequences, and then click  to create a new tightening sequence and input a sequence title.



The left screenshot shows the 'General' tab of a sequence configuration screen. The top bar displays 'User1', '2025.07.25', and '08:37'. Below the top bar, there is a navigation bar with 'General' and 'Navigator' tabs. The main area contains a table with columns: ID, Tool, Parameter Title, Qty, Bit ID, and a delete icon. The table has three rows of data. A red box highlights the back arrow icon in the top left corner. The right screenshot shows the 'Sequence' list screen. The top bar displays 'User1', '2025.07.25', and '08:39'. Below the top bar, there is a navigation bar with 'Sequence' and 'Navigator' tabs. The main area contains a table with columns: ID, Mode, and Sequence Title. The table has five rows of data. A red box highlights the 'New sequence' icon in the top right corner.

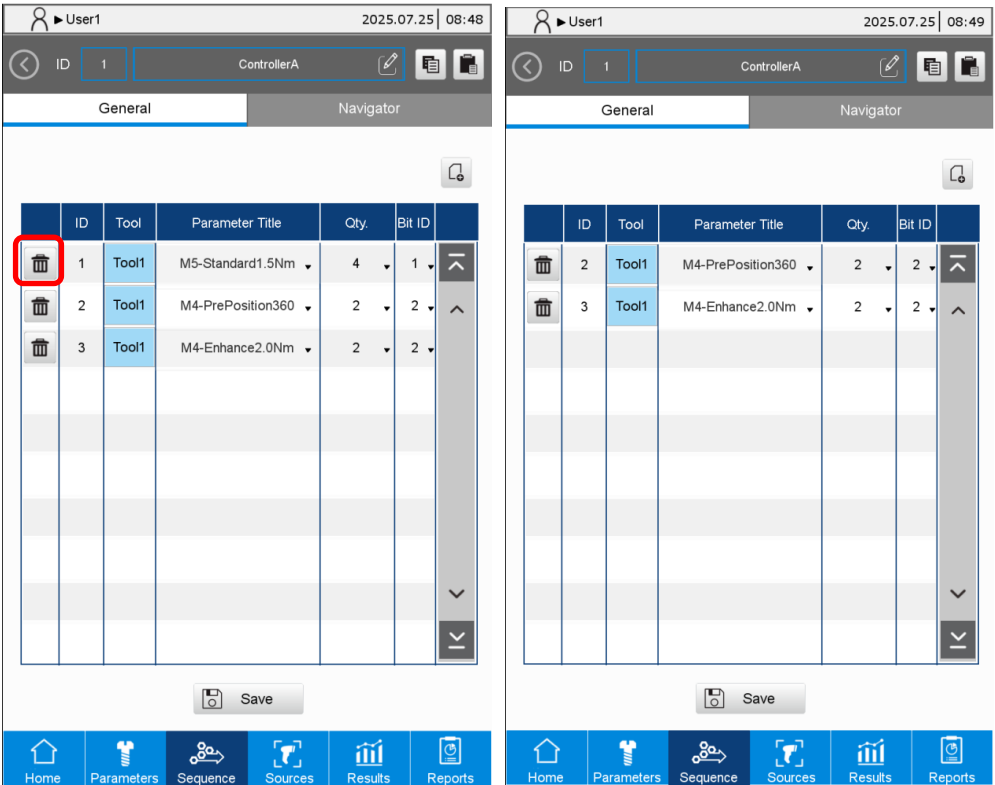
4. Click the  button to the right of the newly created tightening sequence. Then, click  to paste the copied contents in the configuration screen of the new sequence, and then click **Save**. Now, the copied sequence is shown in the list of tightening sequences.



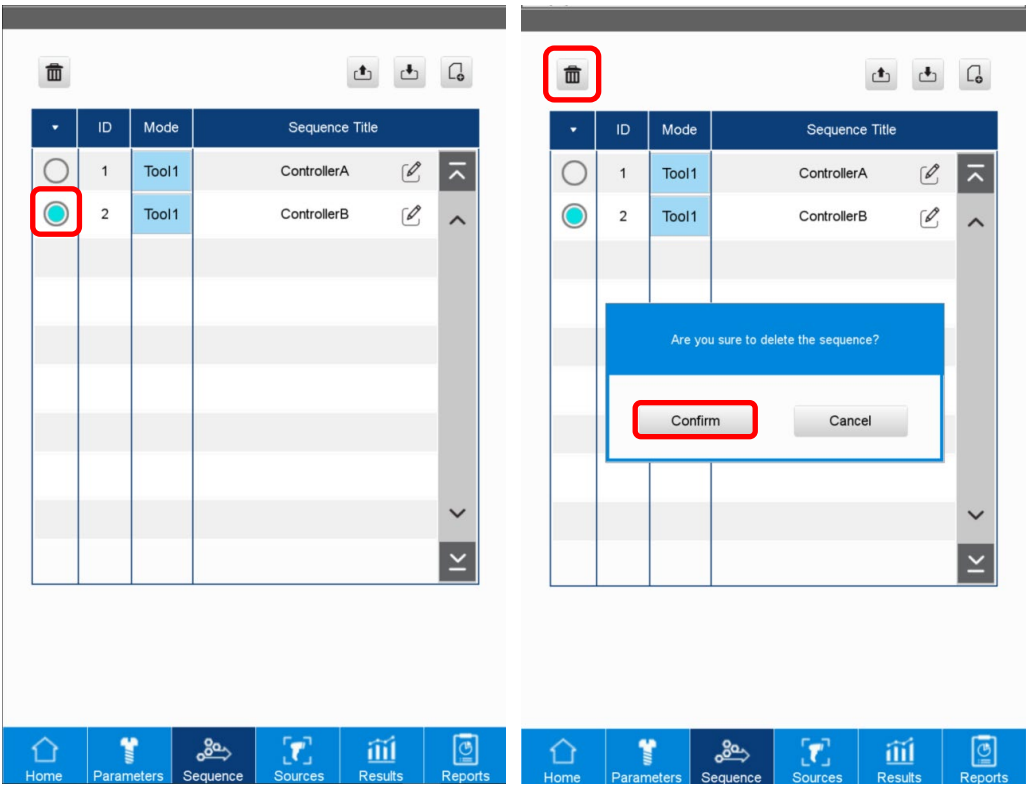
The left screenshot shows the 'General' tab of a sequence configuration screen. The top bar displays 'User1', '2025.07.25', and '08:42'. Below the top bar, there is a navigation bar with 'General' and 'Navigator' tabs. The main area contains a table with columns: ID, Tool, Parameter Title, Qty, Bit ID, and a delete icon. The table has three rows of data. A red box highlights the copy icon in the top right corner. The right screenshot shows the 'Sequence' list screen. The top bar displays 'User1', '2025.07.25', and '08:44'. Below the top bar, there is a navigation bar with 'Sequence' and 'Navigator' tabs. The main area contains a table with columns: ID, Mode, and Sequence Title. The table has six rows of data. A red box highlights the newly added sequence 'ControllerA_copy' in the bottom row.

6.5 Deleting a tightening sequence

- To delete a single parameter in the tightening sequence, click the  button to the left of the specific parameter.

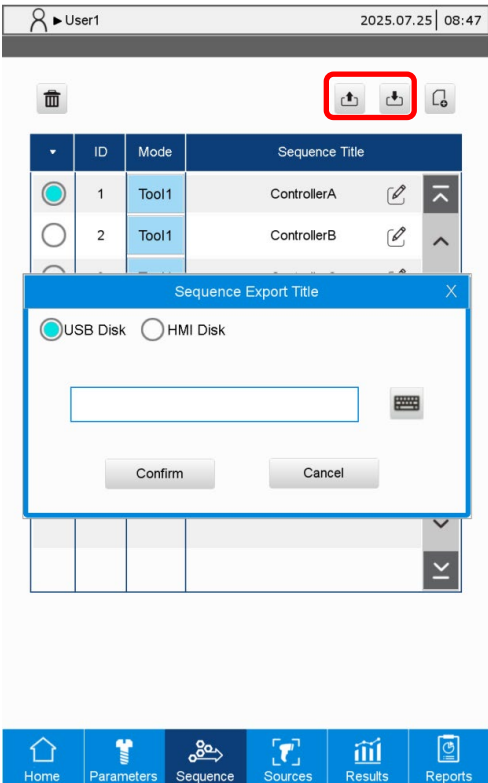


- To delete the entire set of tightening sequence, click the radio button on the left, click , and then click **Confirm** in the confirmation window.



6.6 Exporting / importing a tightening sequence

You can use the export () and import () functions with a USB flash drive or the HMI Disk to create and manage the tightening sequence file backups.



Sources

(Tightening source)

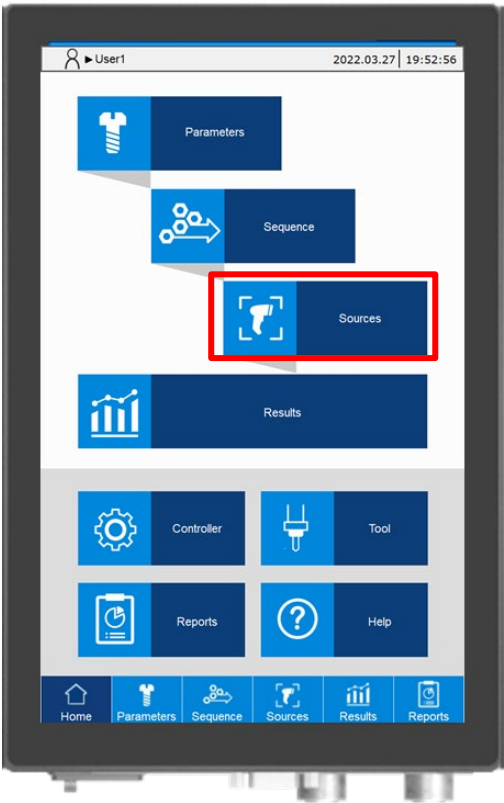
7

This chapter introduces the **Sources** page of the smart screwdriving controller. You can specify the operating mode of the system, and choose the switching method, which allows you to manually set or use the screw bit selector or the barcode scanner to call the tightening source (tightening parameters or tightening sequence) for execution.

7.1	Operating mode	7-2
7.2	Switching Method	7-4
7.2.1	Manual (Manual setting)	7-5
7.2.2	Screw Bit Selector	7-7
7.2.3	Barcode Scanner	7-8
7.3	Copying / pasting a tightening source	7-9
7.4	Deleting a tightening source	7-11
7.5	Exporting / importing a tightening source	7-12

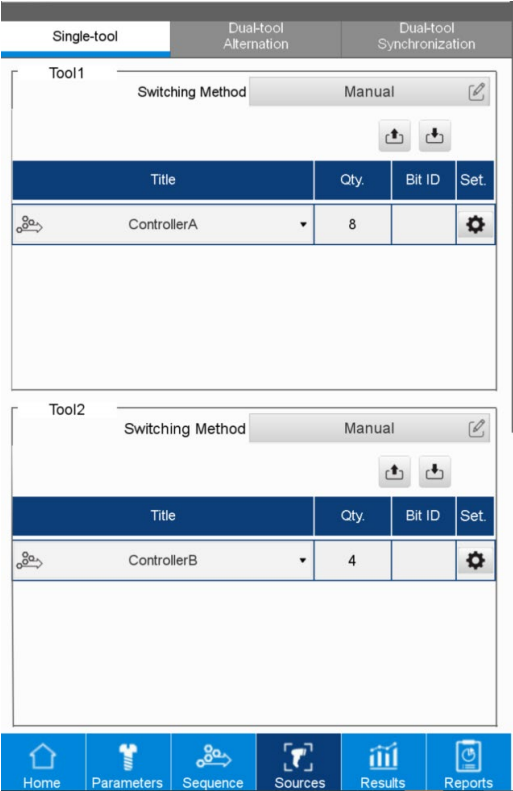
7.1 Operating mode

7



In the **Sources** page, set the operating mode of the system to one of the following:

- Single-tool: set the tightening source for Tool 1 and Tool 2 separately. The two tools operate independently.



- Dual-tool Alternation: select the tightening sequence that includes tightening parameters of both Tool 1 and Tool 2. The two tools operate alternately according to the configuration of the sequence.

Single-tool **Dual-tool Alternation** Dual-tool Synchronization

Alternation

Switching Method: Manual

Title	Qty.	Bit ID	Set
ControllerC	6		

Home Parameters Sequence Sources Results Reports

- Dual-tool Synchronization: select any of the tightening sources. Tool 1 and Tool 2 operate synchronously according to the configuration of the tightening source.

Single-tool Dual-tool Alternation **Dual-tool Synchronization**

Synchronization

Switching Method: Manual

Title	Qty.	Bit ID	Set
ControllerA	8		

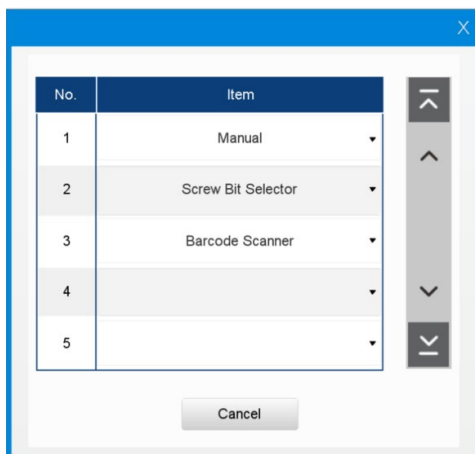
Home Parameters Sequence Sources Results Reports

7

7.2 Switching Method

In each operating mode, choose one of the following methods to switch the tightening source.

- Manual (Manual setting): manually specify the tightening source to be operated.
- Screw Bit Selector: pick up the screw bit to switch to the corresponding tightening source.
- Barcode Scanner: scan the barcode to switch to the corresponding tightening source.



7.2.1 Manual (Manual setting)

When you set the Switching Method to **Manual**, the operation interface is as follows.

Tool1

Switching MethodManual

Title	Qty.	Bit ID	Set.
M5-Standard1.5Nm	4	0	

Tool1

Switching MethodManual

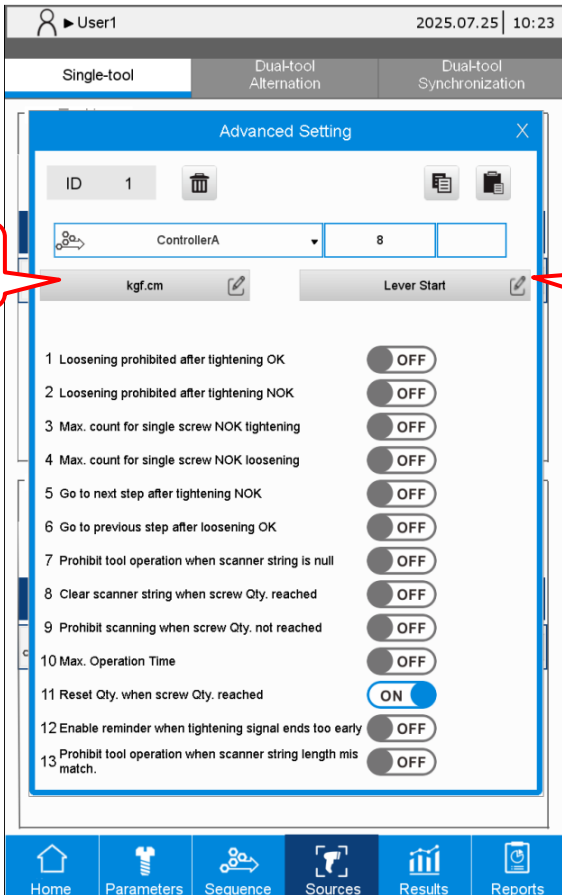
Title	Qty.	Bit ID	Set.
ControllerA	8		

Follow these steps:

- 1. Click the button to the right of the Title field; select one tightening source at a time.
- 2. Set the operating conditions of the tightening source. Refer to the following table.

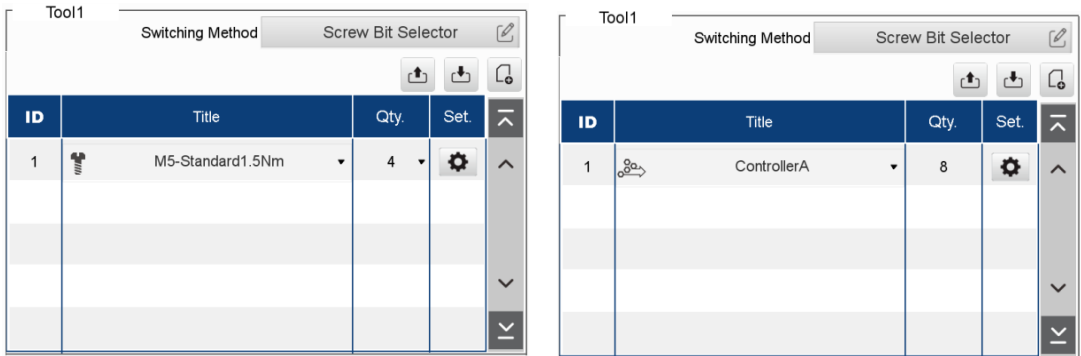
Item	Description	
Title	Select a tightening parameter. 	Select a tightening sequence.
	After a tightening source is set, the software automatically returns to the previous screen for you to set the quantity, bit ID, and advanced setting.	
Qty.	Set the number of times to execute the parameter.	Displays the total screw quantity required for the tightening sequence.
Bit ID	Set the corresponding bit ID number (0: no corresponding bit).	Displays the corresponding bit ID number set in the tightening sequence.

7

Item	Description																															
Set.	The advanced setting for the operating conditions. 																															
	<table border="1"> <thead> <tr> <th>Item</th><th>Description</th></tr> </thead> <tbody> <tr> <td>Torque unit</td><td>N.m, kgf.cm, lbf.ft, lbf.in, ozf.ft, ozf.in, cNm</td></tr> <tr> <td>Start condition</td><td>Push Start, DI, Lever Start, Push Start or Lever Start, Push Start and Lever Start, DI or Push Start, DI or Lever Start</td></tr> <tr> <td>Loosening prohibited after tightening OK</td><td>Protection condition switch.</td></tr> <tr> <td>Loosening prohibited after tightening NOK</td><td>Protection condition switch.</td></tr> <tr> <td>Max. count for single screw NOK tightening</td><td>Protection condition switch with a maximum count setting.</td></tr> <tr> <td>Max. count for single screw NOK loosening</td><td>Protection condition switch with a maximum count setting.</td></tr> <tr> <td>Go to next step after tightening NOK</td><td>Function switch.</td></tr> <tr> <td>Go to previous step after loosening OK</td><td>Function switch.</td></tr> <tr> <td>Prohibit tool operation when scanner string is null</td><td>Protection condition switch.</td></tr> <tr> <td>Clear scanner string when screw Qty. reached</td><td>Function switch.</td></tr> <tr> <td>Prohibit scanning when screw Qty. not reached</td><td>Function switch.</td></tr> <tr> <td>Max. Operation Time</td><td>Protection condition switch with a maximum setting of the total operation time (unit: second).</td></tr> <tr> <td>Reset Qty. when screw Qty. reached</td><td>Function switch.</td></tr> <tr> <td>Enable reminder when tightening signal ends too early</td><td>Function switch.</td></tr> <tr> <td>Prohibit tool operation when scanner string length mismatch</td><td>Protection condition switch with a maximum setting of the string length.</td></tr> </tbody> </table>	Item	Description	Torque unit	N.m, kgf.cm, lbf.ft, lbf.in, ozf.ft, ozf.in, cNm	Start condition	Push Start, DI, Lever Start, Push Start or Lever Start, Push Start and Lever Start, DI or Push Start, DI or Lever Start	Loosening prohibited after tightening OK	Protection condition switch.	Loosening prohibited after tightening NOK	Protection condition switch.	Max. count for single screw NOK tightening	Protection condition switch with a maximum count setting.	Max. count for single screw NOK loosening	Protection condition switch with a maximum count setting.	Go to next step after tightening NOK	Function switch.	Go to previous step after loosening OK	Function switch.	Prohibit tool operation when scanner string is null	Protection condition switch.	Clear scanner string when screw Qty. reached	Function switch.	Prohibit scanning when screw Qty. not reached	Function switch.	Max. Operation Time	Protection condition switch with a maximum setting of the total operation time (unit: second).	Reset Qty. when screw Qty. reached	Function switch.	Enable reminder when tightening signal ends too early	Function switch.	Prohibit tool operation when scanner string length mismatch
Item	Description																															
Torque unit	N.m, kgf.cm, lbf.ft, lbf.in, ozf.ft, ozf.in, cNm																															
Start condition	Push Start, DI, Lever Start, Push Start or Lever Start, Push Start and Lever Start, DI or Push Start, DI or Lever Start																															
Loosening prohibited after tightening OK	Protection condition switch.																															
Loosening prohibited after tightening NOK	Protection condition switch.																															
Max. count for single screw NOK tightening	Protection condition switch with a maximum count setting.																															
Max. count for single screw NOK loosening	Protection condition switch with a maximum count setting.																															
Go to next step after tightening NOK	Function switch.																															
Go to previous step after loosening OK	Function switch.																															
Prohibit tool operation when scanner string is null	Protection condition switch.																															
Clear scanner string when screw Qty. reached	Function switch.																															
Prohibit scanning when screw Qty. not reached	Function switch.																															
Max. Operation Time	Protection condition switch with a maximum setting of the total operation time (unit: second).																															
Reset Qty. when screw Qty. reached	Function switch.																															
Enable reminder when tightening signal ends too early	Function switch.																															
Prohibit tool operation when scanner string length mismatch	Protection condition switch with a maximum setting of the string length.																															

7.2.2 Screw Bit Selector

When you set the Switching Method to **Screw Bit Selector**, the operation interface is as follows.



Follow these steps:

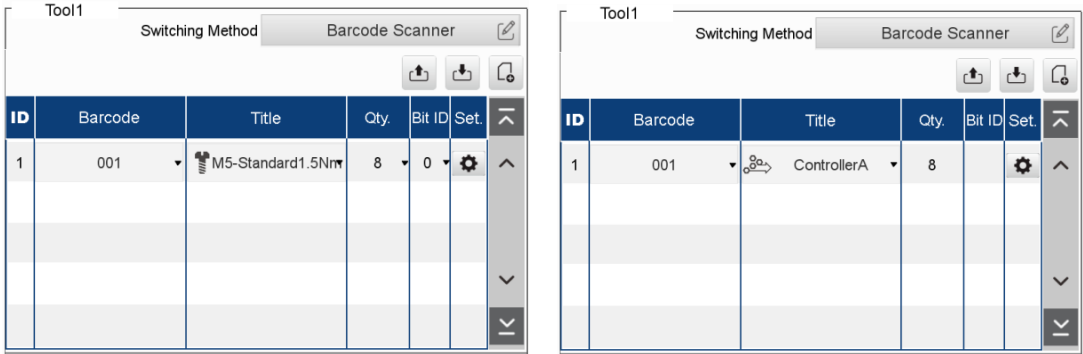
1. Click to specify a bit ID number.
2. Click the button to the right of the Title field to select a corresponding tightening source.
3. Set the operating conditions of the tightening source. Refer to the following table.
4. After the settings are complete, when you pick up the bit from the screw bit selector, the system automatically switches to the corresponding tightening source.

Note: the system switches the tightening source by referring to the Bit ID Output Table. Refer to Section 9.3 Peripheral Device.

Item	Description																			
ID	The bit ID number.																			
Title	Select a tightening parameter. <table><thead><tr><th>Parameter</th><th>Sequence</th></tr></thead><tbody><tr><td>1</td><td>M5-Standard1.5Nm</td></tr><tr><td>2</td><td>M4-PrePosition360</td></tr><tr><td>3</td><td>M4-Enhance2.0Nm</td></tr></tbody></table>	Parameter	Sequence	1	M5-Standard1.5Nm	2	M4-PrePosition360	3	M4-Enhance2.0Nm	Select a tightening sequence. <table><thead><tr><th>Parameter</th><th>Sequence</th></tr></thead><tbody><tr><td>1</td><td>ControllerA</td></tr><tr><td>2</td><td>ControllerB</td></tr><tr><td>3</td><td>ControllerC</td></tr><tr><td>6</td><td>ControllerA_copy</td></tr></tbody></table> Note: this window only shows the tightening sequences that include parameters set with the same bit ID.	Parameter	Sequence	1	ControllerA	2	ControllerB	3	ControllerC	6	ControllerA_copy
	Parameter	Sequence																		
1	M5-Standard1.5Nm																			
2	M4-PrePosition360																			
3	M4-Enhance2.0Nm																			
Parameter	Sequence																			
1	ControllerA																			
2	ControllerB																			
3	ControllerC																			
6	ControllerA_copy																			
	After a tightening source is set, the software automatically returns to the previous screen for you to set the quantity and advanced setting.																			
Qty.	Set the number of screws required for the operation.	Displays the total screw quantity required for the tightening sequence.																		
Set.	The advanced setting for the operating conditions. Refer to the description in Section 7.2.1.																			

7.2.3 Barcode Scanner

When you set the Switching Method to **Barcode Scanner**, the operation interface is as follows.



Follow these steps:

- 1. Click to add an ID, and then click the button to the right of the Title field to select a tightening source.
- 2. Set the operating conditions of the tightening source. Refer to the following table.
- 3. After completing the settings, use a scanner to scan the barcode. The system then compares the string and switches to the corresponding tightening source.

Item	Description	
ID	The sequential numbering of the tightening source in the list.	
Barcode	Set the barcode string corresponding to the tightening source.	
Title	Select a tightening parameter. 	Select a tightening sequence.
	After a tightening source is set, the software automatically returns to the previous screen for you to set the quantity, bit ID, and advanced setting.	
Qty.	Set the number of screws required for the operation.	Displays the total screw quantity required for the tightening sequence.
Bit ID	Set the corresponding bit ID number (0: no corresponding bit).	Displays the corresponding bit ID number set in the tightening sequence.
Set.	The advanced setting for the operating conditions. Refer to the description in Section 7.2.1.	

7.3 Copying / pasting a tightening source

The system provides the Copy (📄) and Paste (📄) functions for editing the tightening sources more efficiently. The following examples illustrate the operation steps.

1. Click  to create a new ID (ID 2).

Tool1

Switching MethodBarcode Scanner

ID	Barcode	Title	Qty.	Bit ID	Set.
1	001	ControllerA	8		
2		M5-Standard1.5Nm	0	0	

Tool1

Switching MethodBarcode Scanner

ID	Barcode	Title	Qty.	Bit ID	Set.
1	001	ControllerA	8		
2		M5-Standard1.5Nm	0	0	

2. Click the  of ID 1 to open the Advanced Setting window, and then click  to copy the Title, Qty., and Advanced Setting contents of ID 1.

User12025.07.2510:54

Single-toolDual-tool AlternationDual-tool Synchronization

Tool1

Switching MethodBarcode Scanner

ID	Barcode	Title	Qty.	Bit ID	Set.
1	001	ControllerA	8		
2		M5-Standard1.5Nm	0	0	

Tool2

Switching MethodManual

Title	Qty.	Bit ID	Set.
ControllerD	4		

HomeParametersSequenceSourcesResultsReports

User12025.07.2510:55

Single-toolDual-tool AlternationDual-tool Synchronization

Advanced Setting

ID 1

001

ControllerA

8

kgf.cm

Push Start or Lever Start

1 Loosening prohibited after tightening OK

ON

2 Loosening prohibited after tightening NOK

ON

3 Max. count for single screw NOK tightening

ON

999999

4 Max. count for single screw NOK loosening

ON

999999

5 Go to next step after tightening NOK

OFF

6 Go to previous step after loosening OK

OFF

7 Prohibit tool operation when scanner string is null

OFF

8 Clear scanner string when screw Qty. reached

OFF

9 Prohibit scanning when screw Qty. not reached

OFF

10 Max. Operation Time

ON

9999999 sec

11 Reset Qty. when screw Qty. reached

OFF

12 Enable reminder when tightening signal ends too early

OFF

13 Prohibit tool operation when scanner string length mismatch.

ON

1

HomeParametersSequenceSourcesResultsReports

3. Return to the Sources list, click the  of ID 2 to open the Advanced Setting window, and then click  to paste the copied contents.

7

User12025.07.2510:54

Single-toolDual-tool AlternationDual-tool Synchronization

Tool1

Switching MethodBarcode Scanner

ID	Barcode	Title	Qty.	Bit ID	Set.
1	001	ControllerA	8		
2		M5-Standard1.5Nm	0	0	

Tool2

Switching MethodManual

Title	Qty.	Bit ID	Set.
ControllerD	4		

HomeParametersSequenceSourcesResultsReports

User12025.07.2510:55

Single-toolDual-tool AlternationDual-tool Synchronization

Advanced Setting

ID2

M5-Standard1.5Nm00

kgf.cmPush Start or Lever Start

1 Loosening prohibited after tightening OKOFF

2 Loosening prohibited after tightening NOKOFF

3 Max. count for single screw NOK tighteningOFF

4 Max. count for single screw NOK looseningOFF

5 Go to next step after tightening NOKOFF

6 Go to previous step after loosening OKOFF

7 Prohibit tool operation when scanner string is nullON

8 Clear scanner string when screw Qty. reachedOFF

9 Prohibit scanning when screw Qty. not reachedON

10 Max. Operation TimeOFF

11 Reset Qty. when screw Qty. reachedOFF

12 Enable reminder when tightening signal ends too earlyOFF

13 Prohibit tool operation when scanner string length mis match.OFF

HomeParametersSequenceSourcesResultsReports

4. Return to the Sources list, where you can see the copy and paste is complete.

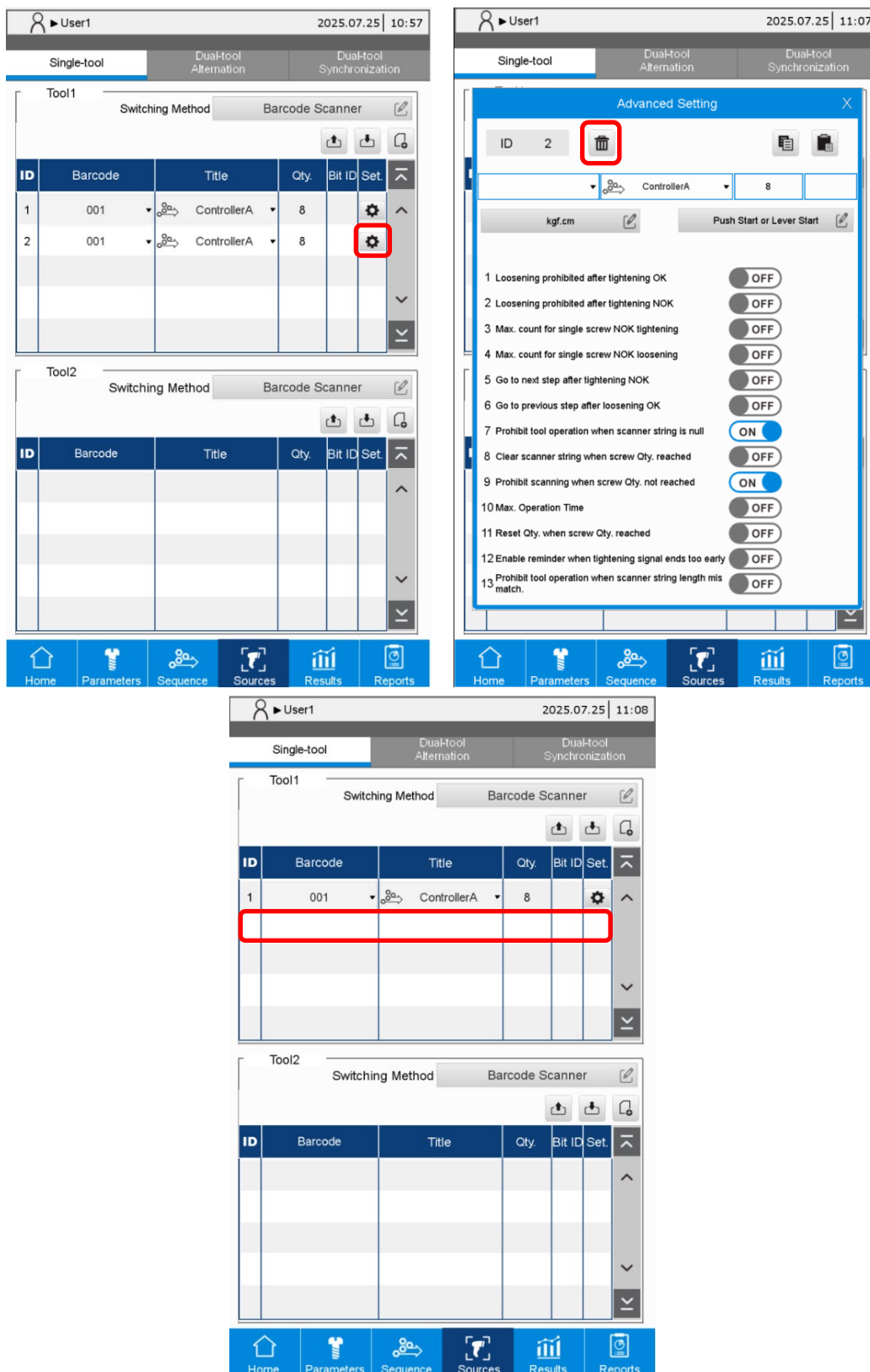
Tool1

Switching MethodBarcode Scanner

ID	Barcode	Title	Qty.	Bit ID	Set.
1	001	ControllerA	8		
2	001	ControllerA	8		

7.4 Deleting a tightening source

Click the  of a tightening source to open its Advanced Setting window, and then click  to delete the tightening source from the list.



The interface shows the 'Sources' tab with a list of tools. Tool 1 and Tool 2 are listed. Tool 2 is selected, and its 'Advanced Setting' window is open. In the 'Advanced Setting' window, the 'ID' field is set to 2, and the 'trash' icon is highlighted with a red box. The bottom screenshot shows the 'Sources' tab after deletion, with Tool 1 selected and its 'Advanced Setting' window open. The 'ID' field is set to 1, and the 'trash' icon is highlighted with a red box.

Tool 1 Data:

ID	Barcode	Title	Qty.	Bit ID	Set
1	001	ControllerA	8		

Tool 2 Data:

ID	Barcode	Title	Qty.	Bit ID	Set
2	001	ControllerA	8		

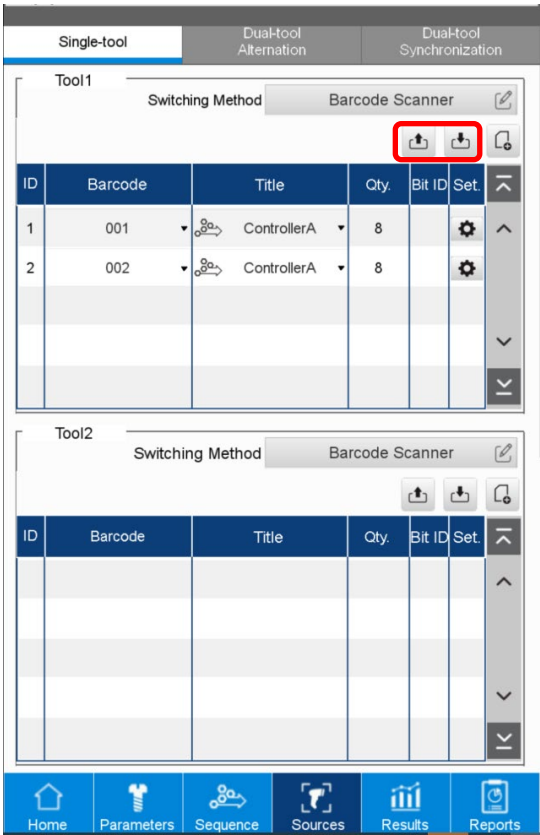
Advanced Setting (Tool 2):

Setting	Value
1 Loosening prohibited after tightening OK	OFF
2 Loosening prohibited after tightening NOK	OFF
3 Max. count for single screw NOK tightening	OFF
4 Max. count for single screw NOK loosening	OFF
5 Go to next step after tightening NOK	OFF
6 Go to previous step after loosening OK	OFF
7 Prohibit tool operation when scanner string is null	ON
8 Clear scanner string when screw Qty. reached	OFF
9 Prohibit scanning when screw Qty. not reached	ON
10 Max. Operation Time	OFF
11 Reset Qty. when screw Qty. reached	OFF
12 Enable reminder when tightening signal ends too early	OFF
13 Prohibit tool operation when scanner string length mis match.	OFF

7.5 Exporting / importing a tightening source

You can use the Export () and Import () functions with a USB flash drive or HMI Disk for the tightening source data backup and management.

7



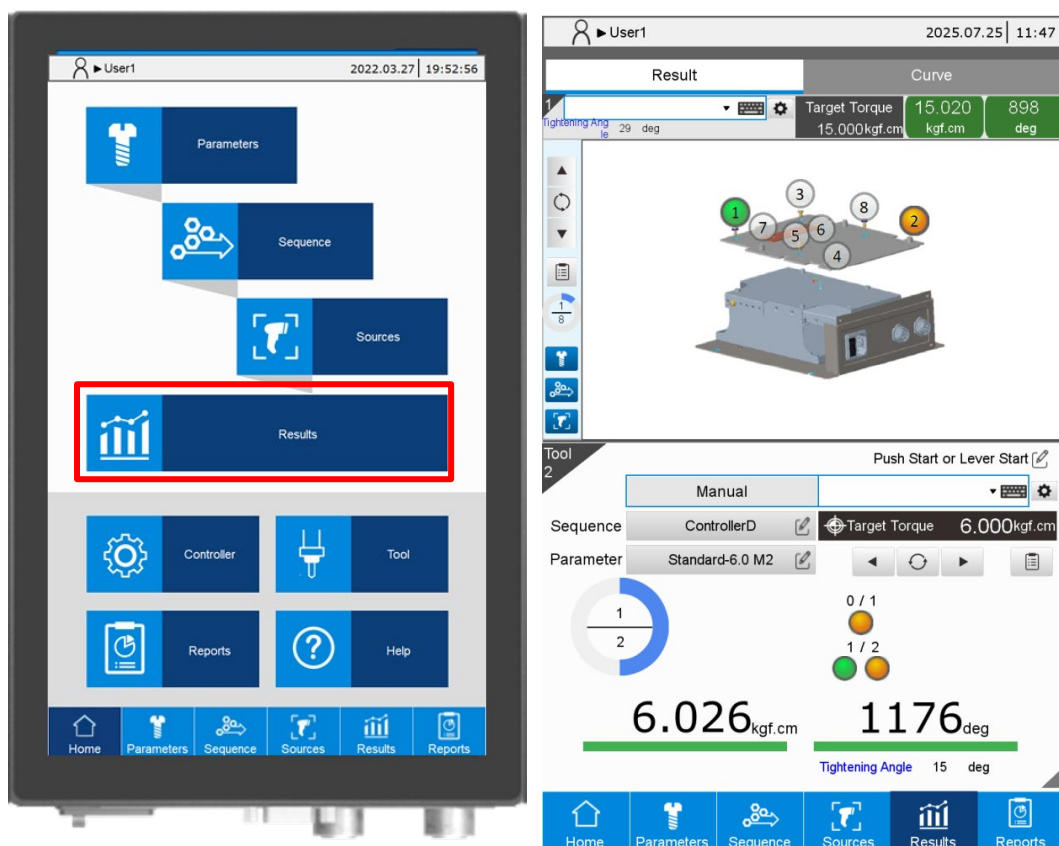
Results

8

This chapter introduces the **Results** page of the smart screwdriving controller. This function provides information of the current tightening operation and the operating curves.

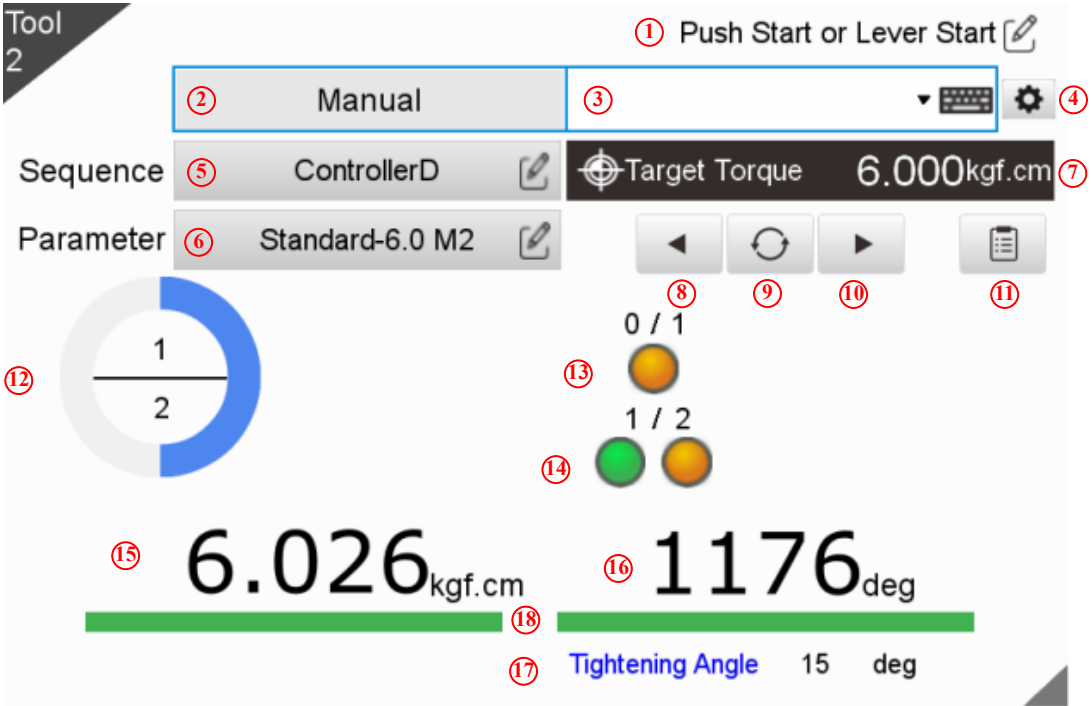
8.1	Operation results (Result).....	8-2
8.2	Operating curve (Curve).....	8-9
8.3	HDMI output.....	8-10
8.4	VNC remote connection	8-10

8.1 Operation results (Result)



The **Results** page provides information of the current tightening operation. After each tightening operation is completed, the screen shows information such as the screw tightening status, final torque, and total angle, so that you can easily check the operation status.

See the item descriptions of the **Result** tab in the following table.

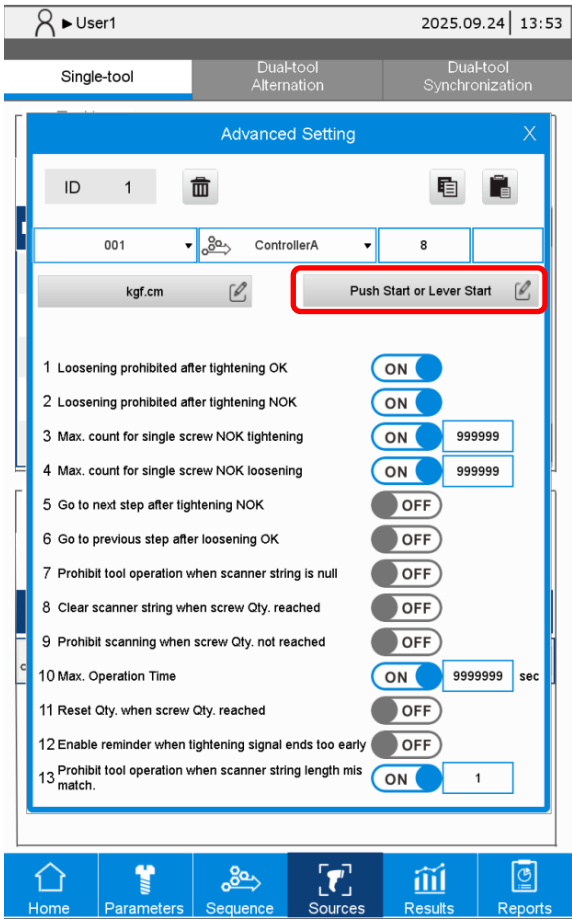


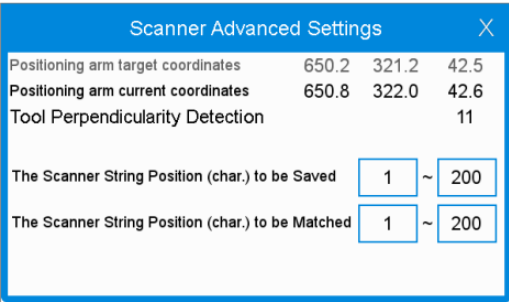
Navigator function Off



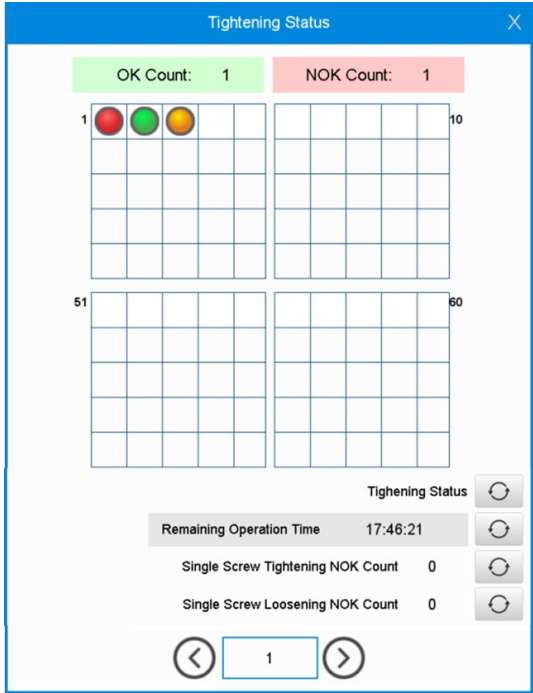
Navigator function On

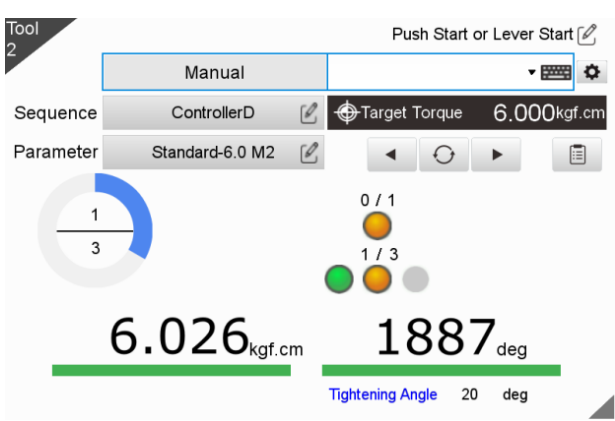
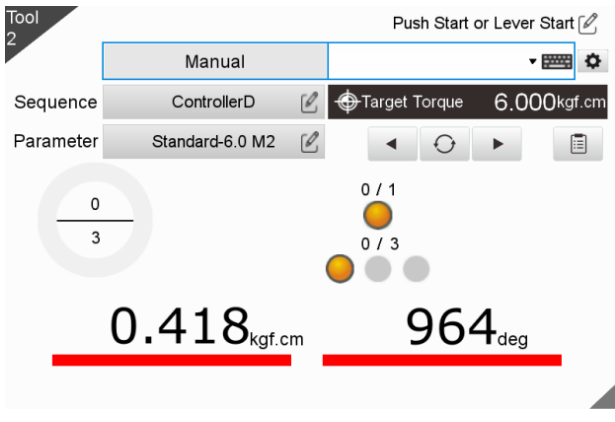
8

Item	Name	Description																																										
1	Tool start condition	Displays the start condition of the transducerized smart screwdriver. Click to open the Advanced Setting window of the tightening source (shown as follows), and then click the field marked with a red box to change the tool start condition.  The screenshot shows the 'Advanced Setting' window for a tool. At the top, it displays 'User1' and the date/time '2025.09.24 13:53'. Below this are three tabs: 'Single-tool', 'Dua-tool Alternation', and 'Dua-tool Synchronization'. The 'Single-tool' tab is selected. The window contains a list of 13 settings, each with a toggle switch and a numeric input field. The 'Push Start or Lever Start' option is highlighted with a red box. The settings are as follows: <table border="1"> <thead> <tr> <th>ID</th> <th>Setting</th> <th>Value</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>Loosening prohibited after tightening OK</td> <td>ON</td> </tr> <tr> <td>2</td> <td>Loosening prohibited after tightening NOK</td> <td>ON</td> </tr> <tr> <td>3</td> <td>Max. count for single screw NOK tightening</td> <td>999999</td> </tr> <tr> <td>4</td> <td>Max. count for single screw NOK loosening</td> <td>999999</td> </tr> <tr> <td>5</td> <td>Go to next step after tightening NOK</td> <td>OFF</td> </tr> <tr> <td>6</td> <td>Go to previous step after loosening OK</td> <td>OFF</td> </tr> <tr> <td>7</td> <td>Prohibit tool operation when scanner string is null</td> <td>OFF</td> </tr> <tr> <td>8</td> <td>Clear scanner string when screw Qty. reached</td> <td>OFF</td> </tr> <tr> <td>9</td> <td>Prohibit scanning when screw Qty. not reached</td> <td>OFF</td> </tr> <tr> <td>10</td> <td>Max. Operation Time</td> <td>9999999 sec</td> </tr> <tr> <td>11</td> <td>Reset Qty. when screw Qty. reached</td> <td>OFF</td> </tr> <tr> <td>12</td> <td>Enable reminder when tightening signal ends too early</td> <td>OFF</td> </tr> <tr> <td>13</td> <td>Prohibit tool operation when scanner string length mis match.</td> <td>ON</td> </tr> </tbody> </table>	ID	Setting	Value	1	Loosening prohibited after tightening OK	ON	2	Loosening prohibited after tightening NOK	ON	3	Max. count for single screw NOK tightening	999999	4	Max. count for single screw NOK loosening	999999	5	Go to next step after tightening NOK	OFF	6	Go to previous step after loosening OK	OFF	7	Prohibit tool operation when scanner string is null	OFF	8	Clear scanner string when screw Qty. reached	OFF	9	Prohibit scanning when screw Qty. not reached	OFF	10	Max. Operation Time	9999999 sec	11	Reset Qty. when screw Qty. reached	OFF	12	Enable reminder when tightening signal ends too early	OFF	13	Prohibit tool operation when scanner string length mis match.	ON
ID	Setting	Value																																										
1	Loosening prohibited after tightening OK	ON																																										
2	Loosening prohibited after tightening NOK	ON																																										
3	Max. count for single screw NOK tightening	999999																																										
4	Max. count for single screw NOK loosening	999999																																										
5	Go to next step after tightening NOK	OFF																																										
6	Go to previous step after loosening OK	OFF																																										
7	Prohibit tool operation when scanner string is null	OFF																																										
8	Clear scanner string when screw Qty. reached	OFF																																										
9	Prohibit scanning when screw Qty. not reached	OFF																																										
10	Max. Operation Time	9999999 sec																																										
11	Reset Qty. when screw Qty. reached	OFF																																										
12	Enable reminder when tightening signal ends too early	OFF																																										
13	Prohibit tool operation when scanner string length mis match.	ON																																										
2	Switching method	Displays the switching method of the tightening source in operation.																																										
3	Barcode string	Scan the barcode to show the string or manually input the string.																																										

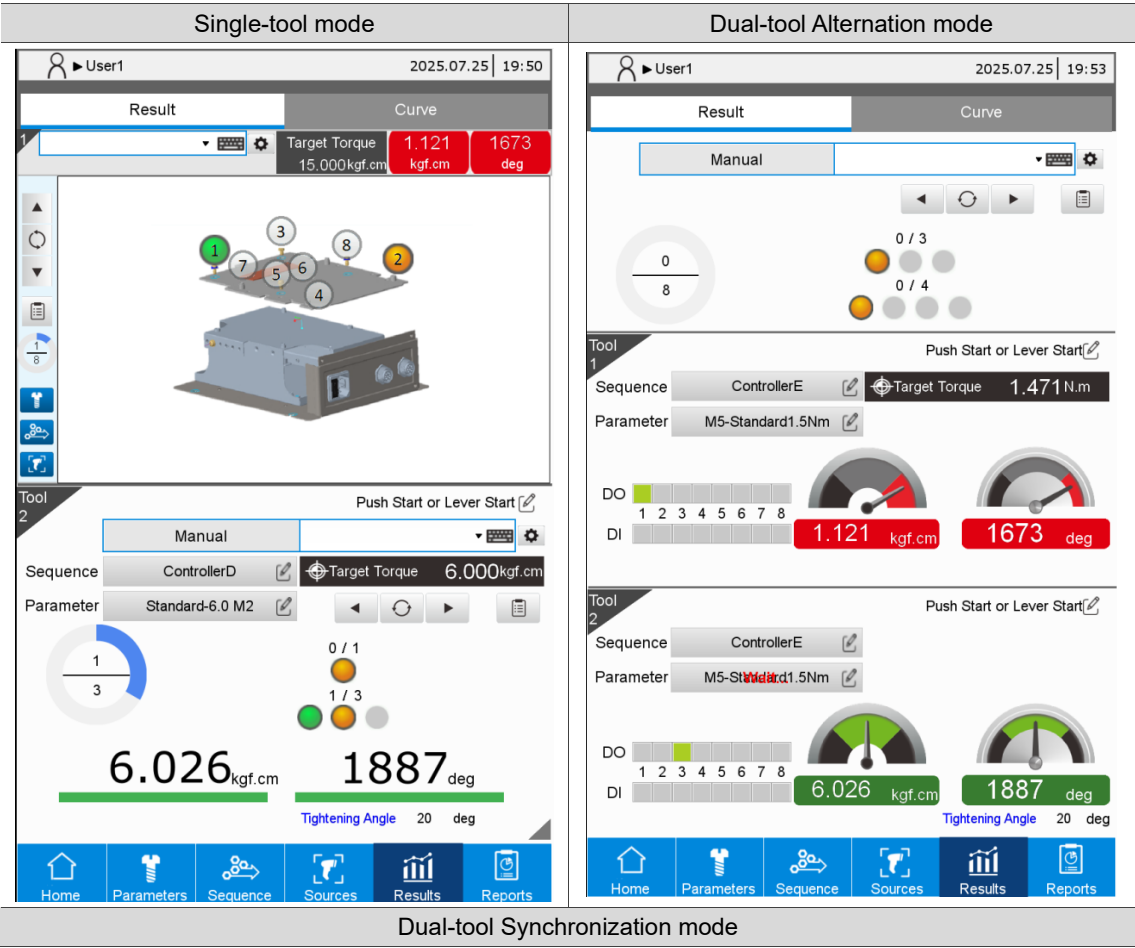
Item	Name	Description
4	Scanner Advanced Settings	 ■ Positioning arm target coordinates (XYZ axes): if the positioning arm function is enabled for the tightening sequence in execution, the target coordinates of the positioning arm are displayed. ■ Positioning arm current coordinates (XYZ axes): if the positioning arm function is enabled for the tightening sequence in execution, the current coordinates of the positioning arm are displayed. ■ Tool Perpendicularity Detection: if the detection is enabled for the tightening parameter in execution, the tool perpendicularity error (0° to 180°) is displayed. ■ The Scanner String Position (char.) to be Saved: save the characters of the specified positions in the scanner string and record the characters in the reports. ■ The Scanner String Position (char.) to be Matched: when the switching method is Barcode Scanner, the system matches the characters of the specified positions in the scanner string against the tightening source strings. If finding a match, the system calls the corresponding parameter or sequence. For example, when the scanner string is SD3F30S0T23010001, if you set The Scanner String Position (char.) to be Saved as 9 - 17 and The Scanner String Position (char.) to be Matched as 1 - 8, the system matches "SD3F30S0" against the tightening source strings and records "T23010001" in the reports.
5	Sequence	The tightening sequence in operation. Click the sequence title to directly link to the editing window of the sequence.
6	Parameter	The tightening parameter in operation. Click the parameter title to directly link to the editing window of the parameter.
7	Target torque or angle	Displays the final target torque or target angle of the operation.
8	Previous step	Force to execute the parameter of the previous screw.
9	Progress reset	Reset the total screw quantity, parameter quantity, and screw quantity of the parameter.
10	Next step	Force to execute the parameter of the next screw.

8

Item	Name	Description
11	Tightening Status	 The lights indicate the screw tightening status. ●: tightening OK. ●: tightening NOK. ●: ready for tightening or tightening in progress. The following items and their reset buttons (⌂) can only be displayed after you log in as Admin. (Reset the) Tightening Status: click ⌂ to clear all the completed screws (● and ●) Remaining Operation Time: the remaining time to execute tightening. Single Screw Tightening NOK Count: the cumulative tightening NOK counts of a single screw. Single Screw Loosening NOK Count: the cumulative loosening NOK counts of a single screw.
12	Total screw quantity	The quantity of completed screws / total quantity of screws (999999 represents an unlimited number), displayed with the progress ring.
13	Parameter quantity	The quantity of completed tightening parameters / total quantity of tightening parameters, displayed with the light indicators.
14	Screw quantity of the parameter	The completed screw quantity of the tightening parameter / total screw quantity of the tightening parameter, displayed with the light indicators.
15	Final torque	Displays the final torque of the tightening task.
16	Total angle	Displays the final total angle of the tightening task.
17	Tightening angle	Displays the tightening angle of the final stage.

Item	Name	Description
18	Color of light indicators and meter	Displays the screw tightening status: Green: tightening OK 
		Red: tightening NOK 

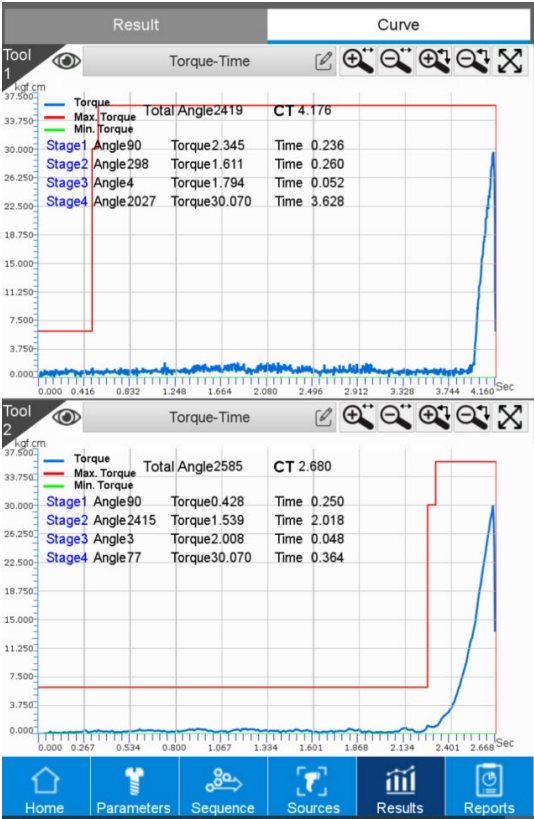
The **Result** tab displays changes depending on the operating mode set in the **Sources** page.



8.2 Operating curve (Curve)

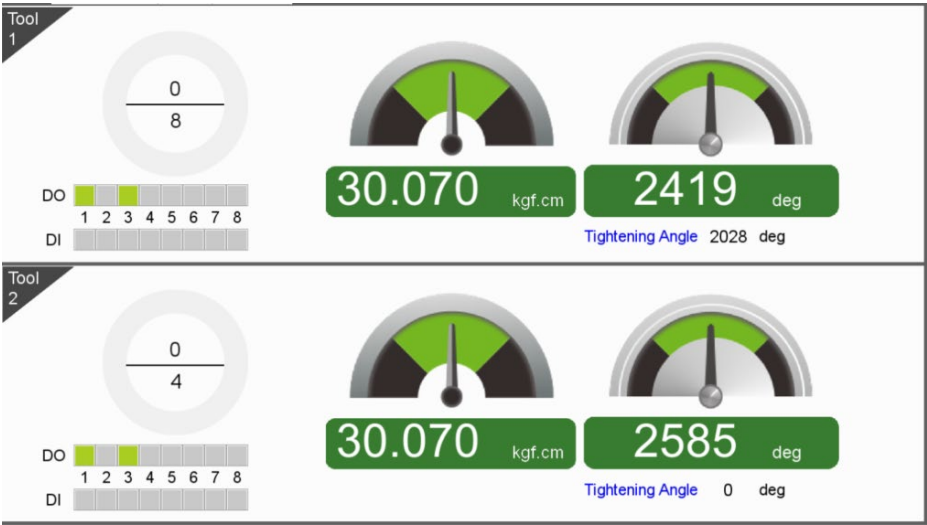
You can check the operating curve in the **Curve** tab of the **Results** page each time the tightening is complete. This function displays the tightening results intuitively and records the values (including angle, torque, and time) at each stage of the tightening process, facilitating the fine-tuning of key parameters.

8



8.3 HDMI output

Through the HDMI output port on the controller, you can synchronously project the simplified Result screen (shown as follows) to an external monitor.



8.4 VNC remote connection

You can establish a remote connection to the controller through VNC software to synchronously project the controller screen to the computer for operation. Refer to Section 9.1 for the details of Ethernet Settings.

Controller

9

This chapter introduces the smart screwdriving controller information and settings, including the system settings, DI/DO, peripheral device, communication (Modbus), and service station.

9.1	System Settings	9-2
9.2	DI/DO	9-16
9.3	Peripheral Device	9-16
9.4	Communication (Modbus)	9-20
9.5	Service Station	9-21

9.1 System Settings

9



User12025.07.2519:56

System Settings

DI/DO

Peripheral Device

Communication

Service Station

Language SettingEnglish

Default Angle UnitDegree

Default torque units of sourceskgf.cm

Default activation conditions of sourcesPush Start or Lever Start

Default Loosening 2nd Stage Speed300

Screen Settings

PermissionsLog In

Page Permissions

Ethernet Settings

Modbus RS485 Settings

System Menu

Factory Reset

Export / Import

Firmware Version1.01.00.0171.0339(125)

1

Home

Parameters

Sequence

Sources

Results

Reports

User12025.07.2519:57

System Settings

DI/DO

Peripheral Device

Communication

Service Station

Two-stage Mode under Self-defined Torque ControlEnable

Display Operation Warning WindowEnable

Tightening signal ends too early windowAbnormal

Display the Limits of All Stages for CurvesEnable

Sampling Rate for CurvesChanges from 10 ms to 4 ms when Start Torque is reached

The curve values are all positiveDisable

Torque curve coordinates displayed from zeroDisable

Torque rate curve replaced by speed curveDisable

Record curve cutoff pointAll Processes

Export Result File for Each ScrewBIN file(HMI Disk)

Send Result TCP for Each ScrewDisable

Check that tightening parameter do not match tool spec.Enable

Prohibit Tool Operation after each tighten or LoosenDisable

Prohibit tool operation after tightening fail, and restore operations by alarm clearDisable

DI response filter time20ms

2

Home

Parameters

Sequence

Sources

Results

Reports

User12025.07.2519:58

System Settings

DI/DO

Peripheral Device

Communication

Service Station

Controller Model Named 3041B2W24190001

Controller Temperature49°C

Always Monitor the Tool CurrentEnable

Compensation for Tool Temperature RiseEnable

Tool parameter range check during power-upEnable

Tool temperature errorEnable

3

Home

Parameters

Sequence

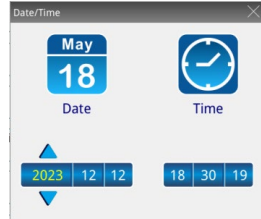
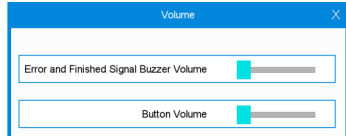
Sources

Results

Reports

The **System Settings** tab contains the following settings.

1. Language Setting: Traditional Chinese (繁體中文), English, Simplified Chinese (简体中文), Vietnamese (Tiếng Việt), Japanese (日本語), or Korean (한국어).
2. Default Angle Unit: Degree or Turn.
3. Default torque units of sources: N.m, kgf.cm, lbf.ft, lbf.in, ozf.ft, ozf.in, or cNm.
4. Default activation conditions of sources: Push Start, DI, Lever Start, Push Start or Lever Start, Push Start and Lever Start, DI or Push Start, DI or Lever Start.
5. Default Loosening 2nd Stage Speed: simultaneously set the 2nd stage default loosening speed (default: 300 rpm) for all parameters, but do not exceed the tool's maximum speed.
6. Screen Settings:

Item	Description										
Brightness	Adjust the screen brightness. 										
Date / Time	Set the date and time of the system. 										
Error Signal Buzzer	Set the buzzer sound patterns for the corresponding signals.										
Finished Signal Buzzer for A Single Screw	<table border="1"> <thead> <tr> <th>Diagram</th><th>Buzzer sound pattern</th></tr> </thead> <tbody> <tr> <td></td><td>Mute</td></tr> <tr> <td></td><td>One long beep</td></tr> <tr> <td></td><td>One short beep</td></tr> <tr> <td></td><td>Two short beeps</td></tr> </tbody> </table>	Diagram	Buzzer sound pattern		Mute		One long beep		One short beep		Two short beeps
Diagram	Buzzer sound pattern										
	Mute										
	One long beep										
	One short beep										
	Two short beeps										
Finished Signal Buzzer for Total Screws											
Buzzer Volume	Set the volume for Error Signal Buzzer and Finished Signal Buzzers, and the buttons. 										
Home Screen	Set the system home screen to the Home or Results page.										
Keyboard cursor blink in results page	Set the Barcode string field in the Result tab of Results page as a default selected field. The field blinks when this function is enabled.										
Product ID	ID of the product.										

9

7. Permissions

- Log In: log in to accounts of different permission levels or change the password.

Log In

Change Password

User

User2

Password

Scanner Login

Remember Password

Log In

LogOut

Log In

Change Password

User

User2

Current Password

Scanner Login

New Password

Change

User	Default password	Description
Admin	99	The highest permission level that can access and edit all pages of the controller. Log in as Admin to set the permissions of page access and edit for each account in the Permissions window.
User1	1	
User2	2	
User3	3	
User4	4	
User5	5	

■ Page Permissions

Logged in as Admin: you can modify the items that can be accessed and edited by other users.	Not logged in (same as User1 permissions): the controller main screen shows prohibition symbols on the items that cannot be accessed.
--	---

Permissions

User

User1User2User3User4User5Admin

Parameter:

☐

☒

☒

☒

☒

☒

Sequence:

☐

☒

☒

☒

☒

☒

Sources:

☐

☒

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☒

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☒

Controller:

☒

☒

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Tool:

☒

☒

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☒

☒

☒

Operate the screw progress:

☒

☒

☒

☒

☒

☒

Delete Production Report:

☒

☒

☒

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Delete Error/Warning Report:

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Sources Advanced:

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User12025.06.1721:35

Parameters

Sequence

Sources

Results

Controller

Tool

Reports

Help

Home

Parameters

Sequence

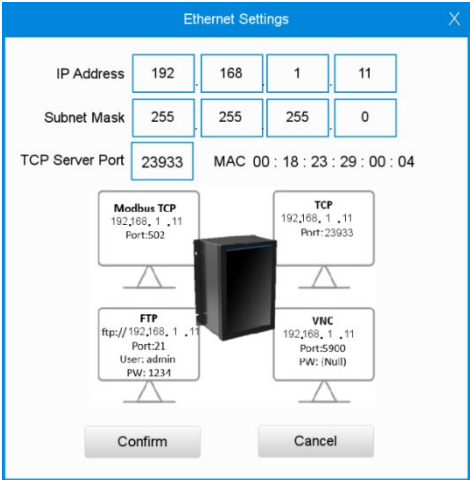
Sources

Results

Reports

When permissions are restricted, the **Parameters**, **Sequence**, **Sources**, and **Results** pages can be viewed but cannot be set; the **Controller** and **Tool** pages can neither be viewed nor set.

8. Ethernet Settings: set the network IP address (default: 192.168.1.11) and subnet mask (default: 255.255.255.0) of the smart screwdriving system. You can also set the TCP Server Port and check the MAC address of the controller.



If you want to operate the controller through the PC, use either of the following methods to connect to the controller.

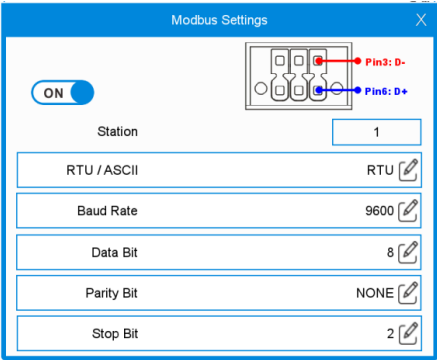
- VNC remote connection: install a VNC software, add a connecting device, enter the controller IP address and the port number (5900), and log in without a password. Then, the controller screen is synchronously projected to the PC.
- FTP connection: install FileZilla, enter the controller IP address in the Host field, “admin” in the Username field, “1234” in the Password field, “21” in the Port field, and then click **Quickconnect** to connect to the HMI internal memory for operation.

Note: the default port of each connection is listed as follows.

Connection	Default port
Modbus TCP	502
TCP	1001
VNC	5900
FTP	21

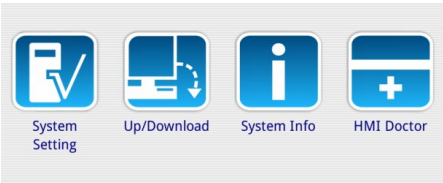
9

9. Modbus RS485 Settings: set the RS-485 communication parameters.



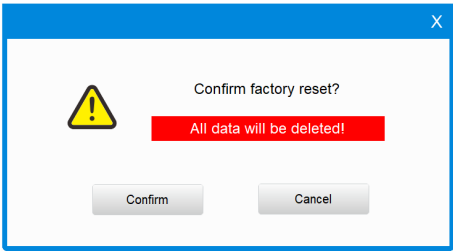
Item	Description
	ON: function On OFF: function Off
Station	1 to 255
RTU / ASCII	RTU or ASCII
Baud Rate	9600, 19200, or 38400
Data Bit	8 (8-bit) or 7 (7-bit)
Parity Bit	NONE, ODD, or EVEN
Stop Bit	2 (2-bit) or 1 (1-bit)

10. System Menu: switch to the BIOS system menu of the controller.



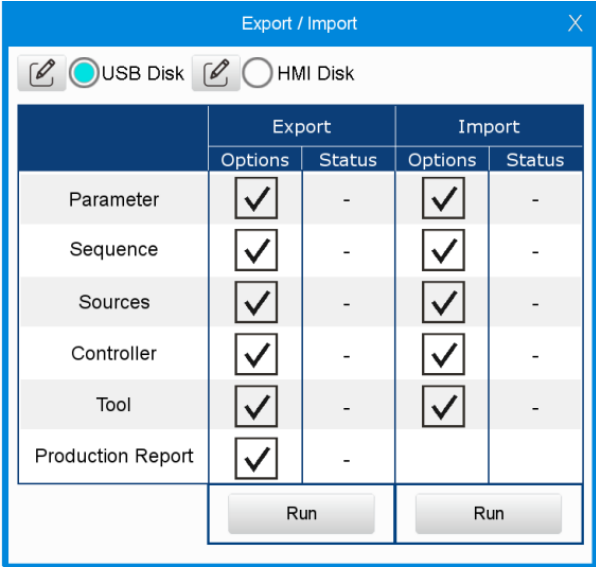
Note: operation by professional personnel is recommended.

11. Factory Reset: clear all system data and restore the Ethernet settings to the default.



Note: if you use the account of the highest permissions (Admin) for factory reset, the passwords for all accounts will be restored to the default.

12. Export / Import: export or import the selected system data (as CSV files).



Item	Export	Import
USB Disk	Export the selected system data to the USB flash drive.	To copy the data of controller A to controller B, first export the data of controller A to the USB flash drive, and then perform the Import function on controller B to update the data to controller B.
HMI Disk	Export the selected system data to the HMI internal memory, and then obtain the system data through FTP.	To copy the data of controller A to controller B, first export the data of controller A to the HMI internal memory, obtain the data through FTP, and then perform the Import function on controller B to update the data to controller B.

13. Firmware Version: displays the firmware version of the smart screwdriving system. The example and descriptions are as follows.

1.01.00.0171.0339(125)

(1)

(2)

(3)

(1) HMI software version

(2) Motor control firmware version

(3) HMI kernel version

14. Two-stage Mode under Self-defined Torque Control: this function is enabled by default when the final stage of self-defined strategy is set to torque control or clamp torque control.

15. Display Operation Warning Window

- Enable: the system displays a prompt window when a warning occurs.
- Disable: the system does not display a prompt window when a warning occurs.

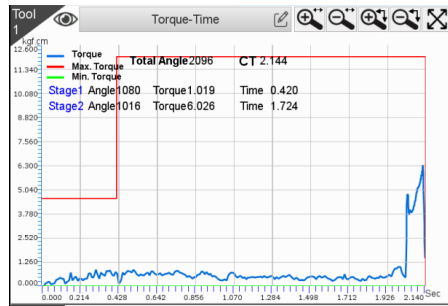
16. Tightening signal ends too early window

- Warning: when the tightening signal ends too early during operation, the system displays a warning window and records the event in the **Warning Report** tab.
- Abnormal: when the tightening signal ends too early during operation, the system displays an error window and records the event in the **Production Report** and **Error Report** tabs.

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17. Display the Limits of All Stages for Curves

- Enable: display the upper and lower limits of all stages.



- Disable: only display the upper and lower limits of the final stage.

18. Sampling Rate for Curves

Item	Sampling rate	Max. data points of the curve	Max. recording time of each tightening
Changes from 10 ms to 4 ms when Start Torque is reached	10 ms (when tightening torque < start torque)	2000	20 s (when the sampling rate is always 10 ms)
	4 ms (when tightening torque ≥ start torque)		8 s (when the sampling rate is always 10 ms)
2 ms; max. 4 s per Production Report Entry	2 ms	2000	4 s
2 ms; max. 8 s per Production Report Entry	2 ms	4000	8 s
1 ms; max. 4 s per Production Report Entry	1 ms	4000	4 s
1 ms; max. 8 s per Production Report Entry	1 ms	8000	8 s

19. The curve values are all positive

- Enable: the torque and tightening angle values displayed in the operating curves and saved data are all positive values.
- Disable: the torque and tightening angle values displayed in the operating curves and saved data can be positive or negative values.

20. Torque curve coordinates displayed from zero

- Disable: the tightening torque curve coordinates displayed in the operating torque curves and saved data start from the minimum value (0 or a negative value).
- Enable: the tightening torque curve coordinates displayed in the operating torque curves and saved data start from 0.

21. Torque rate curve replaced by speed curve

- Disable: save the torque rate curves.
- Enable: save the speed curves.

22. Record curve cutoff point

- All Processes: the curve records the full tightening process.
- Final Peak: the curve records the maximum torque peak at the end of the tightening process.

23. Export Result File for Each Screw: export the production report of each tightening result as a CSV or BIN file.

Item	Disable	CSV file (USB Disk)	BIN file (USB Disk) (the HMI kernel version must be 124 or later)	CSV file (HMI Disk)	BIN file (HMI Disk) (the HMI kernel version must be 124 or later)
File generation time after each screw tightening is finished	-	3 s	1.5 s	2.5 s	1 s
How to read the file	-	Insert a USB flash drive into the Host connector on the controller		Files are read through FTP in C#; for the C# example program, refer to Note 1 at the end of this section. File storage path: <code>ftp://Controller IP/ScrewInfo/</code>	
Filename	-	Production report ID (starts overwriting from the 1 st entry when 200,000 entries are reached)		CSV: last two digits of production report ID BIN: last two digits of production report ID - 1 For example, when the last two digits of production report ID is 01, the CSV filename is ID01.csv ; the BIN filename is ID00.bin . (starts overwriting from the 1 st entry when 100 entries are reached)	

Note:

1. SDSofT can decode the BIN files into the CSV files with header fields.
2. For the CSV file field descriptions, refer to Appendix A #750 and #751 Query type 10 & 11.
For the BIN file field descriptions, refer to Note 2 at the end of this section.

24. Send Result TCP for Each Screw: the host system must act as the TCP Client, while the controller act as the TCP Server. After the host system establishes a connection with the controller, it automatically sends the TCP packets of tightening result when each screw is finished (refer to Appendix B #60 for the TCP content).

No.	Item
1	Disable
2	Scale, Time, Torque
3	Scale, Angle, Torque
4	Scale, Angle, Torque Rate
5	Result, Scale, Time, Torque
6	Result, Scale, Angle, Torque
7	Result, Scale, Angle, Torque Rate
8 - 12	Reserved
13	Result, Scale (tightening or loosening)
14	Result, Scale (tightening)
15	Result, Scale, All Curve Data (tightening or loosening)
16	Result, Scale, All Curve Data (tightening)

Note: the HMI kernel version must be 118 or later.

25. Check that tightening parameter do not match tool spec.: for monitoring if the tool specification in operation matches that of the parameter setting.

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26. Prohibit Tool Operation after each tighten or Loosen: set whether to automatically prohibit tool operation after tightening or loosening OK; the restriction can only be removed after you execute function code #406 or #407.
27. Prohibit tool operation after tightening fail, and restore operations by alarm clear: the tool operation is automatically prohibited after tightening NOK; the restriction can only be removed after you click on the alarm clear window or execute function code #402.
28. DI response filter time: the DI will only be successfully triggered after the trigger duration has reached the set time (setting range: 0 - 1000 ms). This prevents false DI triggering caused by electrical noise.
29. Controller Model Name: displays the serial number of the controller.
30. Controller Temperature: displays the current temperature of the controller in units of °C.
31. (Admin login required) Always Monitor the Tool Current: always monitor the output current of the tool during the tightening process. When this function is disabled, the warnings WN5007, WN500B, WN500C, WN6007, WN600B, and WN600C are not displayed.
32. (Admin login required) Compensation for Tool Temperature Rise: calibrate the corresponding setting values according to the tool temperature. When this function is disabled, the alarms AL1894, AL1895, AL2894, and AL2895 are not displayed.
33. (Admin login required) Tool parameter range check during power-up: check the parameter range when the tool is powered-up to detect sensor component errors. When this function is disabled, the alarms AL1112, AL1113, AL1114, AL1115, AL2112, AL2113, AL2114, and AL2115 are not displayed.
34. (Admin login required) Tool temperature error: monitor if the tool temperature is in error. When this function is disabled, the alarms AL1902, AL1903, AL2902, and AL2903 are not displayed.

Note:

1. C# example program for reading through FTP:

```
public void UseFTPGetFile(int uType)
{
    // uType 1 = Get CSV File (HMI Disk);
    // uType 3 = Get Bin File (HMI Disk)
    int[] ReportIDArray = new int[2];
    ReportIDArray = ModB.ReadHoldingRegisters(0x68, 2); // Use Modbus TCP Get Report ID

    int ReportID = ReportIDArray[1] * 65536 + ReportIDArray[0];
    string url = "";
    if (uType == 1) // Save CSV
        url = "ftp://192.168.1.11/ScrewInfo/ReportCSV/ID" + (ReportID % 100).ToString() + ".csv";
    else // Save Bin
        url = "ftp://192.168.1.11/ScrewInfo/BIN/ID" + ((ReportID - 1) % 100).ToString() + ".bin";

    WebClient request = new WebClient();
    request.Credentials = new NetworkCredential("admin", "1234");

    try
    {
        if (!Directory.Exists(@".\ScrewInfo"))
        {
            Directory.CreateDirectory(@".\ScrewInfo");
        }

        byte[] RawBin = request.DownloadData(new Uri(url));
        if (uType == 1) // Save CSV
        {
            string csvContent = System.Text.Encoding.UTF8.GetString(RawBin);
            File.WriteAllText(@".\ScrewInfo\ID" + ReportID.ToString() + ".csv", csvContent);
        }
        else // Save Bin
        {
            using (BinaryWriter PicW = new BinaryWriter(File.Open(@".\ScrewInfo\ID" + ReportID.ToString() + ".bin", FileMode.Create)))
            {
                for (int i = 0; i < RawBin.Length; i++)
                {
                    PicW.Write(RawBin[i]);
                }
            }
        }
    }
    catch (WebException e){}
}
```

2. The BIN file field descriptions are as follows:

Byte	BIN file field	Description
1 - 200	Saved barcode string	ASCII code (100 words)
201 - 202	Year, month, day	The Unix timestamp that has elapsed since 1970-01-01 00:00:00 (UTC).
203 - 204	Hour, minute, second (L)	
205 - 206	Hour, minute, second (H)	
207 - 208	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
209 - 210	Screw ID (L)	—
211 - 212	Screw ID (H)	—
213 - 214	Sequence ID	—
215 - 216	Parameter ID	—
217 - 218	Target torque	Unit: 0.001 N.m
219 - 220	Target angle	Unit: degree

Byte	BIN file field	Description
221 - 222	Target torque rate	Unit: 0.0001 N.m/degree
223 - 224	Final torque	Unit: 0.001 N.m
225 - 226	Tightening angle	Unit: degree
227 - 228	Rotation angle	Unit: degree
229 - 230	Status	1: tightening OK; 2: tightening NOK; 3: loosening OK; 4: loosening NOK; 5: pass
231 - 232	Operation time	Unit: 0.001 second
233 - 234	AL / NG / WN code	Refer to Chapter 12.
235 - 236	Max. angle of final stage	Unit: degree
237 - 238	Min. angle of final stage	Unit: degree
239 - 240	Max. torque of final stage	Unit: 0.001 N.m
241 - 242	Min. torque of final stage	Unit: 0.001 N.m
243 - 244	Torque unit	0: N.m; 1: kgf.cm 2: lbf.ft; 3: lbf.in 4: ozf.ft; 5: ozf.in; 6: cN.m
245 - 246	Torque specification of the tool	Unit: 0.001 N.m
247 - 248	Max. torque of the tool	Unit: 0.001 N.m
249 - 250	Pre-tightening torque	Unit: 0.001 N.m
251 - 252	The set total operation time	Timeout setting for a single tightening parameter.
253 - 254	Max. total angle	–
255 - 256	Max. torque	Unit: 0.001 N.m
257 - 258	Min. torque	Unit: 0.001 N.m
259 - 260	Max. angle	Angle upper limit when the current stage finishes.
261 - 262	Min. angle	Angle lower limit when the current stage finishes.
263 - 264	Max. operation time	Time upper limit when the current stage finishes. Unit: 0.01 second
265 - 266	Min. operation time	Time lower limit when the current stage finishes. Unit: 0.01 second
267 - 268	Prevail torque	Unit: 0.001 N.m
269 - 270	Final + Prevail torque	Unit: 0.001 N.m
271 - 272	Actual current	Unit: 0.01 Amp
273 - 274	Clamp torque	Unit: 0.001 N.m
275 - 276	Max. clamp torque	Unit: 0.001 N.m
277 - 278	Min. clamp torque	Unit: 0.001 N.m
279 - 280	Clamp angle	Unit: degree
281 - 282	Max. clamp angle	Unit: degree
283 - 284	Min. clamp angle	Unit: degree
285 - 286	Min. total angle	Unit: degree
287 - 288	Permissions user	0: Unknown (only displayed in the HMI software versions before 165) 1: User1 2: User2 3: User3 4: User4 5: User5 6: Admin
289 - 290	Torque scale factor in the system record	For system use only.
291 - 292	Target yield point	Target yield point (unit: 0.01%)
293 - 300	Reserved	–
301 - 4300	Time scale data (1 to 2000)	Unit: ms
4301 - 8300	Angle scale data (1 to 2000)	Unit: degree

Byte	BIN file field	Description
8301 - 12300	Torque scale data (1 to 2000)	Unit: 0.001 N.m
12301 - 16300	Torque rate scale data (1 to 2000)	Unit: 0.0001 N.m/degree
16301 - 16302	Running angle of stage 1	Unit: degree
16303 - 16304	Running angle of stage 2	
16305 - 16306	Running angle of stage 3	
16307 - 16308	Running angle of stage 4	
16309 - 16310	Running angle of stage 5	
16311 - 16312	Running angle of stage 6	
16313 - 16314	Running angle of loosening stage 1	
16315 - 16316	Running angle of loosening stage 2	
16317 - 16318	Max. torque of stage 1	Unit: 0.001 N.m
16319 - 16320	Max. torque of stage 2	
16321 - 16322	Max. torque of stage 3	
16323 - 16324	Max. torque of stage 4	
16325 - 16326	Max. torque of stage 5	
16327 - 16328	Max. torque of stage 6	
16329 - 16330	Max. torque of loosening stage 1	
16331 - 16332	Max. torque of loosening stage 2	
16333 - 16334	Operation time of stage 1	Unit: 0.001 second
16335 - 16336	Operation time of stage 2	
16337 - 16338	Operation time of stage 3	
16339 - 16340	Operation time of stage 4	
16341 - 16342	Operation time of stage 5	
16343 - 16344	Operation time of stage 6	
16345 - 16346	Operation time of loosening stage 1	
16347 - 16348	Operation time of loosening stage 2	
16349 - 16350	Max. time on the scale	Unit: 0.001 second
16351 - 16352	Max. angle on the scale	Unit: degree
16353 - 16354	Max. torque on the scale	Unit: 0.001 N.m
16355 - 16356	Max. torque rate on the scale	Unit: 0.0001 N.m/degree
16357 - 16358	Total number of curve coordinates	–
16359 - 16360	Max. torque	Unit: 0.001 N.m
16361 - 16362	Min. torque	Unit: 0.001 N.m
16363 - 16364	Max. torque rate	Unit: 0.0001 N.m/degree
16365 - 16366	Max. angle	Unit: degree
16367 - 16368	Min. angle	Unit: degree
16369 - 16370	Version of the curve scale	–
16371 - 16372	Sampling rate for curves	0: changes from 10 ms to 4 ms when Start Torque is reached (max. 2000 data points per curve). Up to 200,000 entries can be saved.
16373 - 16374	Max. torque rate of curve	Unit: 0.0001 N.m
16375 - 16376	Min. time on the scale	Unit: ms
16377 - 16378	Min. angle on the scale	Unit: degree
16379 - 16380	Min. torque on the scale	Unit: 0.001 N.m
16381 - 16382	Min. torque rate on the scale	Unit: 0.0001 N.m

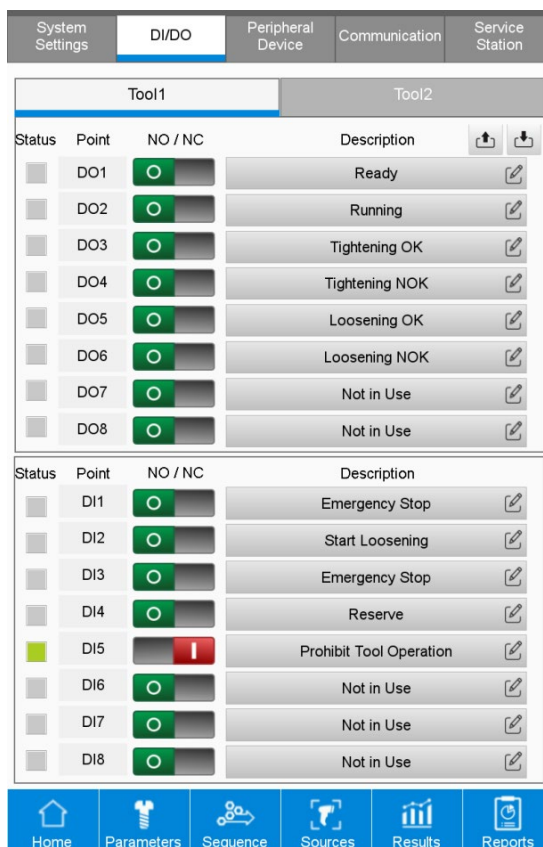
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Byte	BIN file field	Description
16383 - 16384	Switching torque of stage 1	Unit: 0.001 N.m
16385 - 16386	Switching torque of stage 2	
16387 - 16388	Switching torque of stage 3	
16389 - 16390	Switching torque of stage 4	
16391 - 16392	Switching torque of stage 5	
16393 - 16394	Switching torque of stage 6	
16395 - 16400	Reserved	–
16401 - 17500	Information of the current tightening parameter (refer to Appendix B #751 Query type 11 version number 1)	
17501 - 21500	Time scale data (2001 to 4000)	Unit: ms
21501 - 25500	Angle scale data (2001 to 4000)	Unit: degree
25501 - 29500	Torque scale data (2001 to 4000)	Unit: 0.001 N.m
29501 - 33500	Torque rate scale data (2001 to 4000)	Unit: 0.0001 N.m/degree
33501 - 37500	Time scale data (4001 to 6000)	Unit: ms
37501 - 41500	Angle scale data (4001 to 6000)	Unit: degree
41501 - 45500	Torque scale data (4001 to 6000)	Unit: 0.001 N.m
45501 - 49500	Torque rate scale data (4001 to 6000)	Unit: 0.0001 N.m/degree
49501 - 53500	Time scale data (6001 to 8000)	Unit: ms
53501 - 57500	Angle scale data (6001 to 8000)	Unit: degree
57501 - 61500	Torque scale data (6001 to 8000)	Unit: 0.001 N.m
61501 - 65500	Torque rate scale data (6001 to 8000)	Unit: 0.0001 N.m/degree
65501 - 65540	Parameter title	ASCII code (20 words)
65541 - 65580	Sequence title	ASCII code (20 words)
65581 - 65600	Model name	ASCII code (10 words)
65601 - 65620	Production number	ASCII code (10 words)
65621 - 65622	Tightening + loosening count (L)	Tool life (total)
65623 - 65624	Tightening + loosening count (H)	
65625 - 65626	Tightening + loosening count (L)	Tool life (maintenance)
65627 - 65628	Tightening + loosening count (H)	
65629 - 65630	Last maintenance date - year, month, day	The Unix timestamp that has elapsed since 1970-01-01 00:00:00 (UTC).
65631 - 65632	Last maintenance date - hour, minute, second (L)	
65633 - 65634	Last maintenance date - hour, minute, second (H)	
65635 - 65636	Max. speed	Unit: rpm
65637 - 65638	Target position of positioning arm X axis (L)	–
65639 - 65640	Target position of positioning arm X axis (H)	–
65641 - 65642	Target position of positioning arm Y axis (L)	–
65643 - 65644	Target position of positioning arm Y axis (H)	–
65645 - 65646	Target position of positioning arm Z axis (L)	–
65647 - 65648	Target position of positioning arm Z axis (H)	–
65649 - 65650	Current position of positioning arm X axis (L)	–
65651 - 65652	Current position of positioning arm X axis (H)	–
65653 - 65654	Current position of positioning arm Y axis (L)	–

Byte	BIN file field	Description
65655 - 65656	Current position of positioning arm Y axis (H)	—
65657 - 65658	Current position of positioning arm Z axis (L)	—
65659 - 65660	Current position of positioning arm Z axis (H)	—
65661 - 65700	Controller model name	ASCII code (10 words)
65701 - 65702	HMI main version	—
65703 - 65704	HMI subversion	—
65705 - 65706	Second development platform version	—
65707 - 65708	Kernel version	—
65709 - 65800	Reserved	—

9.2 DI/DO

You can customize the functions of DO1 to DO8 and DI1 to DI8 and set the DI/O as NO (normally open)  or NC (normally closed)  contacts.



Tool1			
Status	Point	NO / NC	Description
	DO1		Ready
	DO2		Running
	DO3		Tightening OK
	DO4		Tightening NOK
	DO5		Loosening OK
	DO6		Loosening NOK
	DO7		Not in Use
	DO8		Not in Use

Tool2			
Status	Point	NO / NC	Description
	DI1		Emergency Stop
	DI2		Start Loosening
	DI3		Emergency Stop
	DI4		Reserve
	DI5		Prohibit Tool Operation
	DI6		Not in Use
	DI7		Not in Use
	DI8		Not in Use

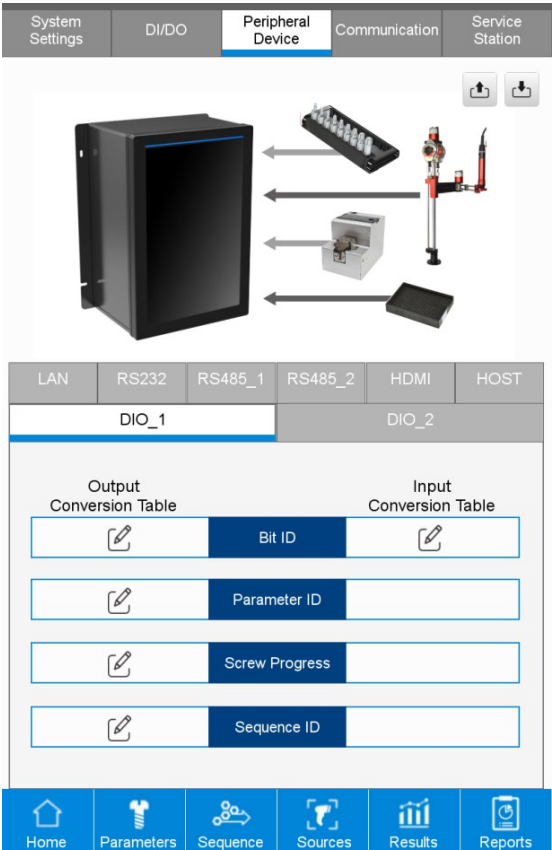
Function code	DO function	Function code	DI function
1	Not In Use	1	Not In Use
2	Ready	2	Start Tightening
3	Running	3	Start Loosening
4	Tightening OK and Report is being saved	4	Emergency Stop (Dual-tool)
5	Tightening NOK and Report is being saved	5	Clear Error and Clear Status (Dual-tool)
6	Loosening OK and Report is being saved	6	Reset Total Screw Quantity
7	Loosening NOK and Report is being saved	7	Next Step
8	Total Screw Quantity Reached	8	Previous Step
9	Device Alarms or Errors or Warning	9	Start Tightening (Pulse Signal)
10	Tighten to waiting stage (only available in DO7; you need to enable the related option in the Advanced Setting of Parameters page)	10	Start Loosening (Pulse Signal)
11	Modbus handshake end signal	11	Start Tightening (Dual-tool)
12	Device Alarms	12	Start Loosening (Dual-tool)
13	Reserved	13	Reset Operation Time
14	Tool reverse switch state	14	Reset Single Screw Tightening + Loosening NOK Count

Function code	DO function	Function code	DI function
15	Rotation direction	15	Prohibit Tool Operation
16	Tightening Result Sent to Feeder	16	Clear Communication Flag
17 - 24	Parameter ID Output 1 - 8	17	Wait for external signal before continuing tighten (only available in DI7; you need to enable the related option in the Advanced Setting of Parameters page)
25 - 32	Screw Progress Output 1 - 8	18	Emergency Stop
33 - 40	Sequence ID Output 1 - 8	19	Prohibit Tool Operation (Pulse Signal)
41 - 48	Bit ID Output Bit 1 - 8	20	Rotation direction
49 - 52	Running tightening Stage Output 1 - 4	21	Start Rotation
53	Tightening OK and Lever Signal End	22	Clear Error and Clear Status
54	Double-click the Lever button	23 - 40	Reserved
55	Tightening OK and the report is saved	41 - 48	Bit ID Input 1 - 8
56	Tightening NOK and the report is saved	-	-
57	Loosening OK and the report is saved	-	-
58	Loosening NOK and the report is saved	-	-
59	Busy	-	-
60 - 64	Reserved	-	-

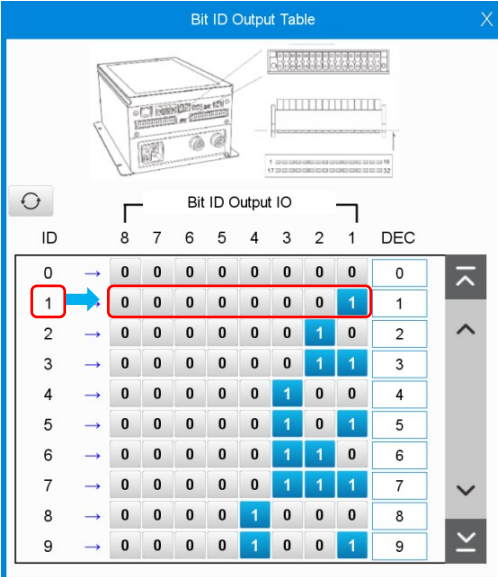
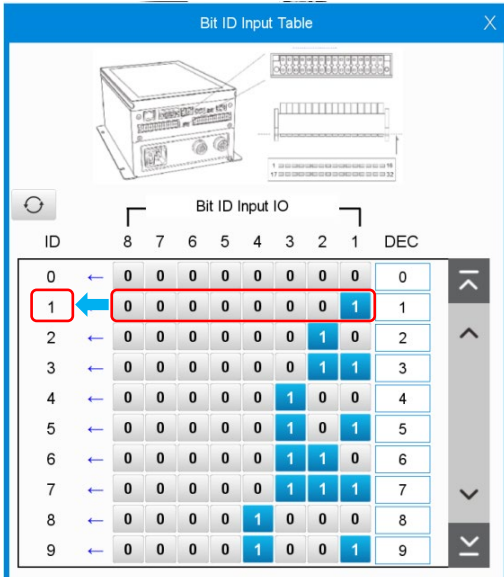
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9.3 Peripheral Device

When connecting external devices to the controller, use the conversion tables in the **Peripheral Device** tab to check or change the IO settings since the indexing method of DI/O varies from manufacturers. Make sure to first set the corresponding DO / DI functions (refer to Section 9.2) before using the output and input conversion tables.



1. Bit ID

Output Conversion Table	Input Conversion Table
Check the "Bit ID Output Bit 1 - 8" setting value of each Bit ID, and then output the values with the DO signals.	After the DI signals are input, check the "Bit ID Input Bit 1 - 8" setting value to find out the corresponding Bit ID.
Example: According to the Bit ID Output Table, the "Bit ID Output Bit 1 - 8" setting value of Bit ID 1 is 00000001.	Example: According to the Bit ID Input Table, the "Bit ID Input Bit 1 - 8" setting value is 00000001, so you can find that the corresponding Bit ID is 1.
	

2. Parameter ID

Output Conversion Table: check the "Parameter ID Output Bit 1 - 8" setting value of each tightening parameter ID, and then output the values with the DO signals.

3. Screw Progress

Output Conversion Table: check the "Screw Progress Output Bit 1 - 8" setting value of each screw, and then output the values with the DO signals.

4. Sequence ID

Output Conversion Table: check the "Sequence ID Output Bit 1 - 8" setting value of each tightening sequence ID, and then output the values with the DO signals.

9.4 Communication (Modbus)

The **Communication** tab provides the handshake data of Modbus TCP and Modbus RTU communication for you to debug during communication test. Refer to Appendix A for an overview of the Modbus communication interface and the function code descriptions.

There are eight subtabs which you can switch by clicking ◀ or ▶ on the screen.

- The **Status** subtab displays the real-time operational status of the system.
- The other seven subtabs display handshake data between the host system and the screwdriving system main functions of **Parameters**, **Sequence**, **Sources**, **Results**, **Controller**, **Tool**, and **Reports** respectively.

User1		2025.07.25 20:09	
System Settings	DI/DO	Peripheral Device	Communication
Status			
Real-time Status		Common Status	
Tightening ID set for cur. switch method 00	1 32 1	Tool 1ALN/WH Code 64	0
PAR/SEQ set for cur. switch method 01	1 33 1	Tool 2ALN/WH Code 65	0
SEQ ID of current switching method 02	1 34 1	DO status (BH1-BTool 1; BH5-16Tool 2) 66	1025
PAR ID of current switching method 03	1 35 1	DI status (BH1-BTool 1; BH5-16Tool 2) 67	0
Current target torque 04	1471	DI control(BH1-BTool 1; BH5-16Tool 2) 68	0
Current target angle 05	0 37 0	Current No. of error report entries 69	35
Current parameter progress 06	0 38 0	Current No. of warning report entries 68	23
Screw progress of current parameter (L) 07	0 39 0	Current No. of production report entries(L) 68	322
Screw progress of current parameter (H) 08	0 3A 0	Current No. of production report entries(H) 6C	0
All screw progress of current sequence (L) 09	0 3B 0	Current No. of button report entries(L) 6B	160
All screw progress of current sequence (H) 0A	0 3C 0	Current No. of button report entries(H) 6E	0
Tightening OK count (L) 0B	0 3D 0	Production report entry creation finished 6F	1
Tightening OK count (H) 0C	0 3E 0	Common flag cleared 70	0
Single screw tightening NOK count (L) 0D	0 3F 0	Common flag 71	0
Single screw tightening NOK count (H) 0E	0 40 0	All production report entries cleared 72	1
Loosening OK count (L) 0F	0 41 0	Product report is overwritten start from ID 173	1
Loosening OK count (H) 10	0 42 0	Error report is overwritten start from ID 174	1
Single screw loosening NOK count (L) 11	0 43 0	Warning report is overwritten start from ID 175	1
Single screw loosening NOK count (H) 12	0 44 0	Button report is overwritten start from ID 176	1
Final stage max. torque 13	1765	Keypad 77	197
Final stage min. torque 14	1176	Single-tool/Alternative/Synchronization 78	2
Current torque unit 15	0 47 0	Tool 1 1 Switching Method 79	0
Executing tightening / Executing loosening 16	0 48 0	Tool 2 1 Switching Method 7A	0
Current BH ID 17	1 49 1	Tool 1 curve refresh times 7B	33
Communication Firmware Version 18	4A 1	Tool 2 curve refresh times 7C	19
Flag cleared 19	0 4B 0	The running tool number in dual-tool 7D	0
Total screw Qty. of current sequence (L) 1A	8 4C 8	HMI screen refresh flag 7E	9500
Total screw Qty. of current sequence (H) 1B	0 4D 0	COM4 communication error flag 7F	0
Parameter Qty. of current sequence 1C	3 4E 3	COM4 communication error counter 80	0
Total screw Qty. of current parameter (L) 1D	4 4F 4	HMI init status(98/Finish) 81	99
Total screw Qty. of current parameter (H) 1E	0 50 0	User ID 82	1
All screws of current parameter finished 1F	0 51 1	HMI Main Ver. 83	171
Current parameter finished 20	0 52 1	HMI Sub Ver. 84	3
Current screw finished 21	1 53 1	HMI BIOS Ver. 85	293
Setting parameters (Waiting...) 22	0 54 0	2ndPlant Ver. 86	331
Final + prevail torque 23	1121	FW Ver. 87	339
Total angle 24	1673	Tool1 MCU Ver. 88	34
Tightening angle 25	67 20	Tool2 MCU Ver. 89	29
Loosening result (1 OK; 2 NOK; 3 Pass) 26	68 1	Pop-up Windows ID 8A	0
Curve creation finished 28	1 5A 1	8B	0
Tightening operation restricted status 29	0 5B 0	8C	0
Parameter settings OK / NOK 2A	0 5C 0	8D	0
Final torque 2B	1121	8E	0
Prevail torque 2C	0 5E 0	8F	0
Final current 2D	465	90	0
Cause to restrict tightening operation 2E	0 60 128	91	0
Cause to restrict loosening operation 2F	0 61 0	92	0
Remaining operation time (L) 30	0 62 0	93	0
Remaining operation time (H) 31	0 63 0	94	0
		95	0

User1		2025.07.25 20:10	
System Settings	DI/DO	Peripheral Device	Communication
Parameters			
#100 - #150 Parameter		D3-E5 Parameter title	
Code C8	0	CMD Resp. CF	0
Ver C9	0	1 OK 2 NOK DO	0
Tool CA	0	Error Code D1	0
ID CB	0		
CC	0		
CD	0		
CMD Req CE	0		
Min. tightening angle E6	0		
Persist time E7	0		
Prevail time to be inhold E8	0		
Max. tightening time E9	0		
Max. loosening time EA	0		
Max. tightening angle EB	0		
Max. loosening angle EC	0		
Delay before tightening starts ED	0		
Delay before loosening starts EE	0		
Start time for Curve Sampling Rate EF	0		
Start time for sang angle calc. F0	0		
Sang Point Angle Correction F1	0		
Final Current Detection Switch F2	0		
Delay Before Outputting to Fender F3	0		
Tool Accuracy Compensation F4	0		
Trigger Torque for Recording Cam-Out Angle F5	0		
Upper Limit for Cam-Out Angle F6	0		
Tolerance for Cam-Out F7	0		
Control mode FA	0 12C 0 19E 0 190 0 1C2 0 1F4 0		
Tightening direction FB	0 12D 0 19F 0 191 0 1C3 0 1F5 0		
Retraction speed FC	0 12E 0 160 0 192 0 1C4 0 1F6 0		
Target torque FD	0 12F 0 161 0 193 0 1C5 0 1F7 0		
Target angle FE	0 130 0 162 0 194 0 1C6 0 1F8 0		
Target torque rate FF	0 131 0 163 0 195 0 1C7 0 1F9 0		
Angle interval for torque rate calc. 100	0 132 0 164 0 196 0 1C8 0 1FA 0		
Acceleration time 101	0 133 0 165 0 197 0 1C9 0 1FB 0		
Max. angle 102	0 134 0 166 0 198 0 1CA 0 1FC 0		
Min. angle 103	0 135 0 167 0 199 0 1CB 0 1FD 0		
Max. torque 104	0 136 0 168 0 19A 0 1CC 0 1FE 0		
Min. torque 105	0 137 0 169 0 19B 0 1CD 0 1FF 0		
Min. operation time 106	0 138 0 16A 0 19C 0 1CE 0 200 0		
Min. operation time 107	0 139 0 16B 0 19D 0 1CF 0 201 0		
Prevail torque On / Off 108	0 13A 0 16C 0 19E 0 1D0 0 202 0		
Angle range for prevail torque calc. 109	0 13B 0 16D 0 19F 0 1D1 0 203 0		
Pause time 10A	0 13C 0 16E 0 1A0 0 1D2 0 204 0		
Max. Clamp Torque 10B	0 13D 0 16F 0 1A1 0 1D3 0 205 0		
Min. Clamp Torque 10C	0 13E 0 170 0 1A2 0 1D4 0 206 0		
Max. Clamp Angle 10D	0 13F 0 171 0 1A3 0 1D5 0 207 0		
Min. Clamp Angle 10E	0 140 0 172 0 1A4 0 1D6 0 208 0		
1st Target Torque 10F	0 141 0 173 0 1A5 0 1D7 0 209 0		
1st Pause Time 110	0 142 0 174 0 1A6 0 1D8 0 210 0		
Final Acc. Time 111	0 143 0 175 0 1A7 0 1D9 0 211 0		
Final Speed 112	0 144 0 176 0 1A8 0 1DA 0 212 0		
Dec. Time 113	0 145 0 177 0 1A9 0 1DB 0 213 0		
Advanced Set. 1 114	0 146 0 178 0 1AA 0 1DC 0 214 0		
Advanced Set. H 115	0 147 0 179 0 1AB 0 1DD 0 215 0		
Max switching time 116	0 148 0 17A 0 1AC 0 1DE 0 216 0		
Min switching time 117	0 149 0 17B 0 1AD 0 1DF 0 217 0		
Target Yield Point 118	0 14A 0 17C 0 1AE 0 1E0 0 218 0		
Start torque of yield detection 119	0 14B 0 17D 0 1AF 0 1E1 0 219 0		

9.5 Service Station

The **Service Station** tab displays the operating parameters of Tool 1 and Tool 2, which provides a clear view of the operational status.

User12025.07.2520:10

System SettingsDI/DOPeripheral DeviceCommunicationService Station

Tool1Tool2

General

Stage 1-6

Stage 7-8

1	5	21	21	17	0	0	89	64
0	0	0	0	4	0	0	0	0
0	50	250	800	1471	0	0	1000	1000
0	80	395	159	100	0	0	100	200
0	90	1080	0	10	0	0	100	7200
0	0	0	50	0	0	0	0	0
100	50	50	50	1000	0	0	50	50
7200	50	50	50	50	0	0	50	50
0	0	0	0	0	0	0	0	0
32768	0	0	0	0	0	0	0	0
0	0	100	100	0	0	0	0	0
12	0	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
300	0	0	0	0	0	0	0	0
300	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
5000	0	0	0	0	0	0	0	0
64	0	0	0	0	0	0	0	0
300	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0
32767	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	1800	9600	9600	360	0	0	9999	9999
0	0	0	0	0	0	0	0	0
0	250	450	1000	1050	0	0	0	0
0	0	0	0	0	0	0	0	0
32768	250	450	1000	1745	0	0	5999	5999
20294	0	0	0	1176	0	0	0	0
100	250	450	1000	1050	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
5174	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	100	1000	500	200	0	0	200	200
0	0	0	0	0	0	0	0	0
0	360	360	360	360	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
300	0	0	0	0	0	0	0	0
100	0	0	0	0	0	0	0	0
0	0	4	0	0	0	0	1	0
0	1	1	0	1	0	0	0	0
100	0	0	0	0	0	0	0	0
300	10	20	30	40	0	0	90	91

1

User12025.07.2520:11

System SettingsDI/DOPeripheral DeviceCommunicationService Station

Tool1Tool2

Max. count for single screw NOK tightening

The scanner string is null. Tightening prohibited.

Total Screw Quantity Reached

Exceeded max. operation time

Incorrect Parameter Sequence

Tightening prohibited by remote control or DI

Parameter Range Error

Parameters do not match tool

Prohibit Tightening by SDSoft

The controller cannot operate this tool model

Loosening prohibited after tightening OK

Loosening prohibited after tightening NOK

Exceeded the max. count for NOK loosening

Incorrect Parameter Sequence

Loosening prohibited by remote control or DI

Parameter Range Error

Prohibit Tightening by SDSoft

The controller cannot operate this tool model

S3

Tool MCU status monitoring

OFF

DD

HH

MM

0

0

0

Disable Check

Motion Firmware

HMI

FW

AL

0

0

0

HMI

Show

2

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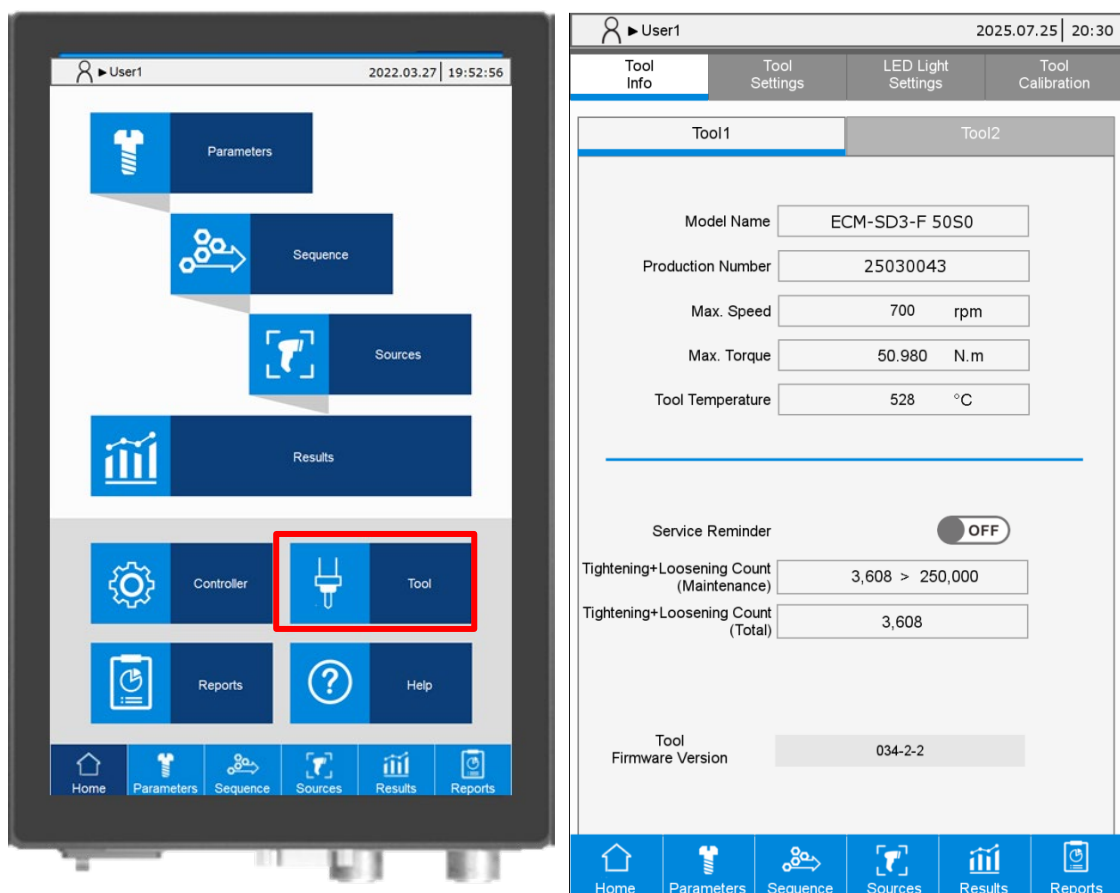
9

This chapter introduces the **Tool** page of the smart screwdriving controller, which allows you to access and configure the transducerized smart screwdriver information and settings, including the Tool Info, Tool Settings, LED Light Settings, and Tool Calibration.

10.1	Tool Info	10-2
10.2	Tool Settings	10-3
10.3	LED Light Settings	10-4
10.4	Tool Calibration	10-5

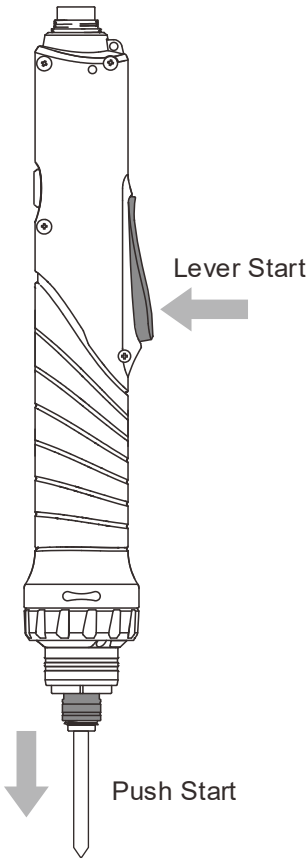
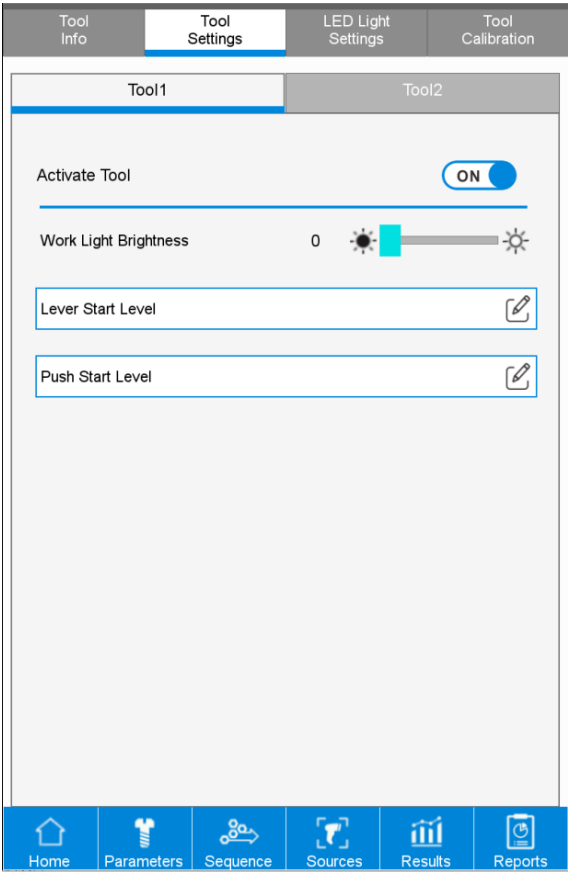
10.1 Tool Info

10

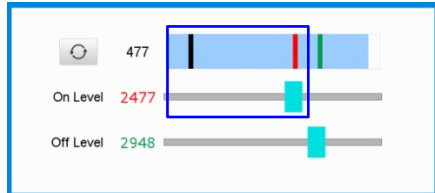
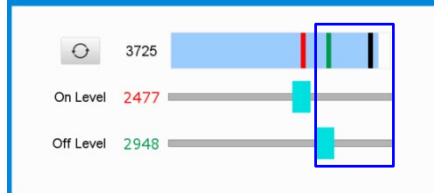
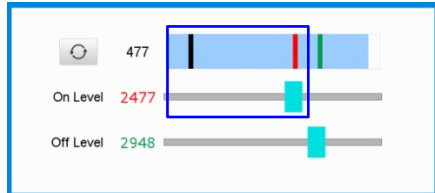
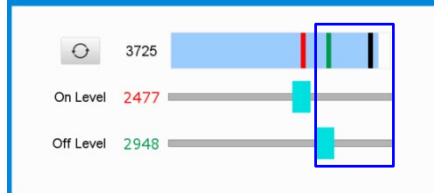
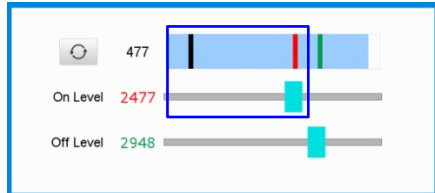
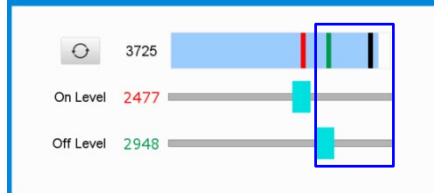


Item	Description
Model Name	Displays the information of the smart screwdriver in use.
Production Number	
Max. Speed	
Max. Torque	
Tool Temperature	Displays the current temperature (°C) of the smart screwdriver.
Service Reminder	If the function is switched to On, when the total tightening count + total loosening count exceeds the suggested count for maintenance (250,000 counts), the system displays the alarm WN5002 or WN6002 (Send the tool back for service). (If logged in as Admin (highest permissions), you can change the suggested count for maintenance.)
Tightening + Loosening Count (Maintenance)	Displays the total tightening count + total loosening count of the smart screwdriver. The count will be reset after you send the screwdriver back to the factory for maintenance.
Tightening + Loosening Count (Total)	Displays the total tightening count + total loosening count of the smart screwdriver after shipment from the factory.
Tool Firmware Version	Displays the firmware version number of the smart screwdriver.

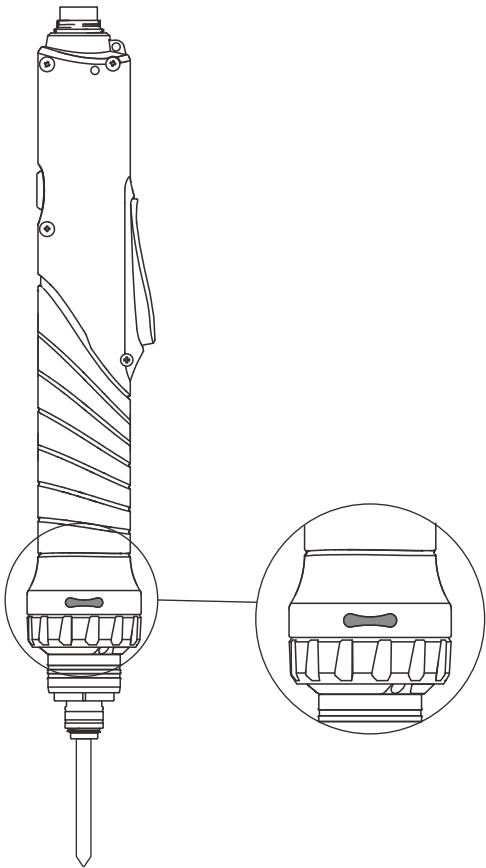
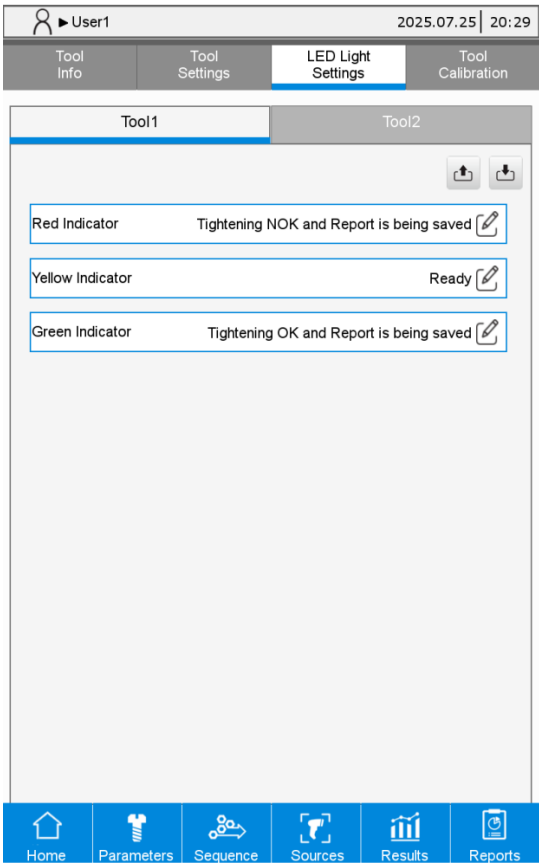
10.2 Tool Settings



10

Item	Description													
Activate Tool	Activate the smart screwdriver to have it ready for the tightening operation.													
Work Light Brightness	Adjust the brightness for the work light LED indicator of the smart screwdriver (this function only supports low-torque models). 0: Off 100: maximum brightness													
Lever Start Level	Adjust the On Level and Off Level for the lever start and push start signals of the smart screwdriver. Click  to reset the On Level and Off Level.													
Push Start Level (only supports low torque and specific medium torque models)	<table><tr><th colspan="2">On Level</th><th colspan="2">Off Level</th></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td colspan="2">Press the lever button or push the screwdriver all the way down, and the signal level (black) moves leftward. Then, set the On Level (red) higher than the signal level.</td><td colspan="2">Leave the lever button unpressed or the screwdriver unpushed, and the signal level (black) is at the rightmost end. Then, set the Off Level (green) lower than the signal level.</td></tr></table>	On Level		Off Level						Press the lever button or push the screwdriver all the way down, and the signal level (black) moves leftward. Then, set the On Level (red) higher than the signal level.		Leave the lever button unpressed or the screwdriver unpushed, and the signal level (black) is at the rightmost end. Then, set the Off Level (green) lower than the signal level.		
On Level		Off Level												
														
Press the lever button or push the screwdriver all the way down, and the signal level (black) moves leftward. Then, set the On Level (red) higher than the signal level.		Leave the lever button unpressed or the screwdriver unpushed, and the signal level (black) is at the rightmost end. Then, set the Off Level (green) lower than the signal level.												

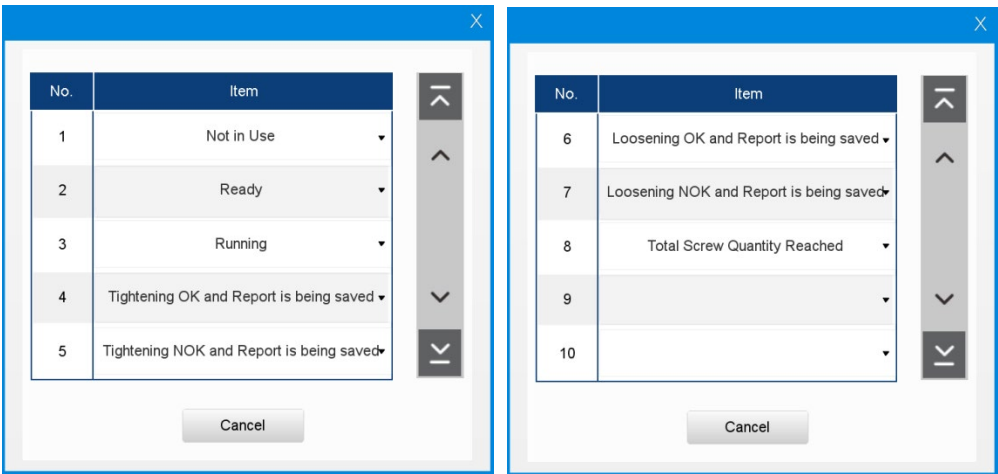
10.3 LED Light Settings



In the **LED Light Settings** tab, you can set the displayed status of each indicator light for the status LED indicator of the smart screwdriver.

If the red, yellow, and green indicator lights are all set to indicate the same status, since the three indicator lights cannot be On simultaneously, the LED indicator will be On in the sequence of red → yellow → green.

The indicator lights can be set to the following statuses:



10.4 Tool Calibration

Use the **Tool Calibration** tab along with a third-party calibration device for the torque calibration and precision adjustment of the smart screwdriver.

Select a tightening parameter with the tightening strategy set as Enhanced. Operate the screwdriver with this parameter along with the third-party calibration device. After completion, the Tool torque field displays the final torque of the tightening task. Then, manually input the torque value measured by the calibration device into the Torque Measured from External Device field. Click **Save**, and the system automatically calculates the calibrated tool torque value. Finally, click **Confirm** to complete the calibration procedure.

User1 2025.08.08 11:04

Tool Info Tool Settings LED Light Settings **Tool Calibration**

Tool1

Tool torque 0.000 N.m

Torque Measured from External Device 0.000 N.m

Difference 0.00 %

Factory Reset **Save**



Date	Tool torque	Torque Measured from External Device	Difference

 1 

Home Parameters Sequence Sources Results Reports

Save Parameter

 Apply the calibrated parameter?

Before: 0.020 After: 0.021

Gain: 3412 Gain: 3582

Confirm Cancel

10

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10

Reports

11

This chapter introduces the **Reports** page of the smart screwdriving controller, including the Production Report, Error Report, Warning Report, and Button Report.



11.1	Production Report	11-2
11.2	Error Report	11-4
11.3	Warning Report	11-5
11.4	Button Report	11-6

11.1 Production Report

The system records and displays the information of each tightening task in the **Production Report** tab. Up to 200,000 entries can be saved; once the limit is reached, the system will proceed to overwrite the records from the first entry. In addition, you can export the entries as CSV files* with the  (Export all operating results) and  (Export the operating result and curve of a single screw) buttons.

Note: for the CSV file field descriptions, refer to Appendix A #750 and #751 Query type 10 & 11.

User12025.07.2421:21

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User12025.07.2421:22

Production Report

Error Report

Warning Report

Button Report





	Date / Time	Tool	Final Angle	Tightening Angle	Final Torque	Status
☐	2025/07/19 12:26:37	Tool1	-1582 degrees	0 degrees	0.519 kgf.cm	▲ OK
☐	2025/07/19 12:26:40	Tool1	1641 degrees	20 degrees	10.013 kgf.cm	▼ OK
☐	2025/07/19 12:26:41	Tool1	-1452 degrees	0 degrees	0.503 kgf.cm	▲ OK
☐	2025/07/19 12:26:43	Tool1	1523 degrees	22 degrees	10.156 kgf.cm	▼ OK
☐	2025/07/19 12:26:45	Tool1	-1384 degrees	0 degrees	0.508 kgf.cm	▲ OK
☐	2025/07/19 12:26:47	Tool1	1451 degrees	19 degrees	10.013 kgf.cm	▼ OK
☐	2025/07/19 15:54:17	Tool1	0 degrees	0 degrees	0.000 kgf.cm	▼ NG

<

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>

Home

Parameters

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Sources

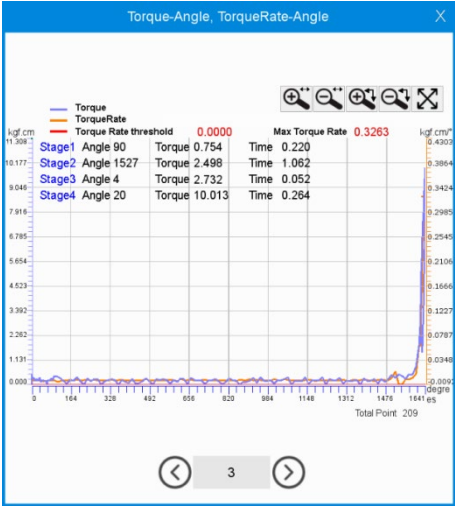
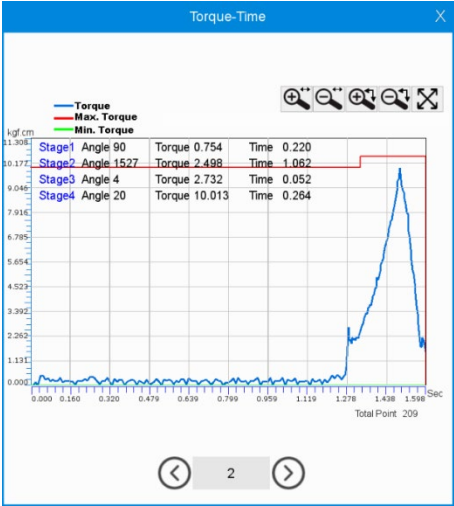
Results

Reports

To view the operating result and operating curve of a single entry, use the  button to search for a specific entry or directly click any of the entries. The example display content is shown as follows.

11

Report Information			
Production Report ID	282		
Date / Time	2025/07/19 12:26:40		
Tool	Tool1		
Status	OK		
Saved Scanner String			
Screw ID	2	Final Angle	1641 degrees
Sequence ID	0	Tightening Angle	20 degrees
Parameter ID	8	Final+Prevail Torque	10.013 kgf.cm
Operation time	1.598	Final Torque	10.013 kgf.cm
Final Current	2.87	Prevail Torque	0.000 kgf.cm
Description			



11.2 Error Report

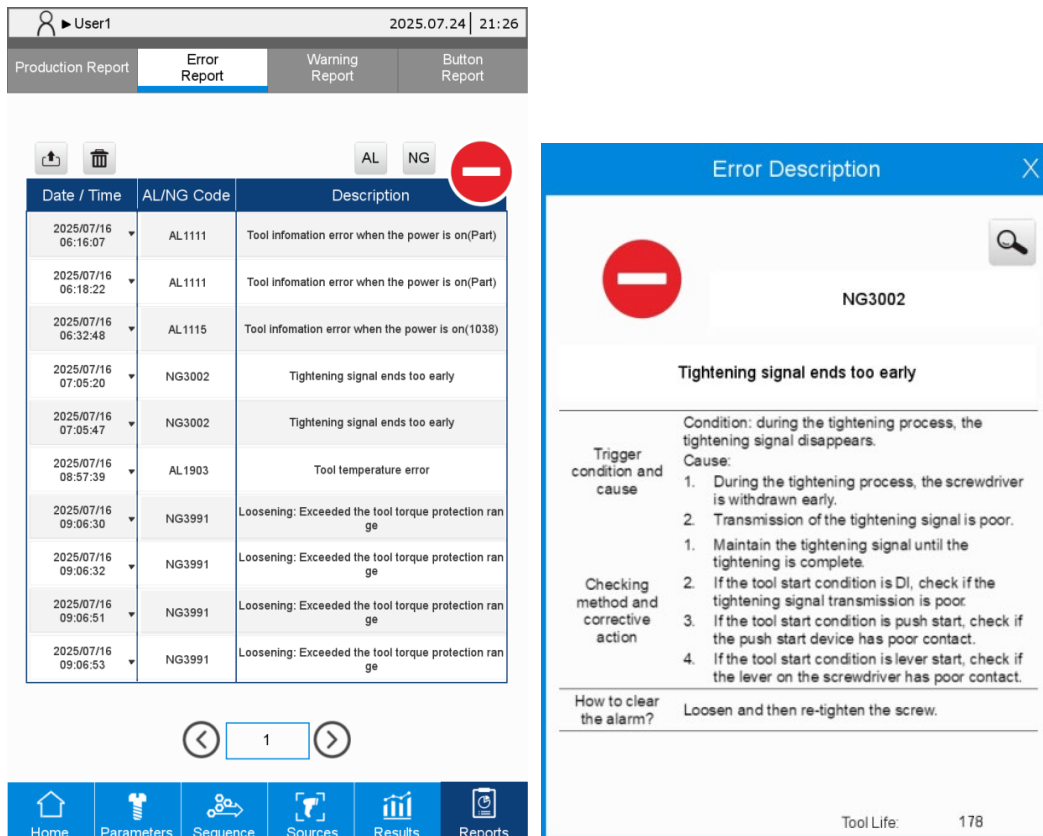
The **Error Report** tab displays the entries of device alarm (AL) and tightening error (NG) in the system, with which you can analyze the cause of the frequently occurring errors.

The **AL** button: only displays the device alarm entries (AL).

The **NG** button: only displays the tightening error entries (NG).

In addition, you can export all entries with the  button.

Click each entry to view the detailed description and the tool life count when the error occurs.



The screenshot displays the 'Error Report' tab in the SD3 Series Controller interface. The top bar shows 'User1' and the date '2025.07.24' at '21:26'. Below the top bar are four tabs: 'Production Report', 'Error Report' (selected), 'Warning Report', and 'Button Report'. The 'Error Report' tab contains a table with columns 'Date / Time', 'AL/NG Code', and 'Description'. The table lists several errors, including 'Tool information error when the power is on(Part)' (AL 1111, 1115) and 'Tightening signal ends too early' (NG3002). A red circular button with a white minus sign is visible next to the table. Below the table is a pagination bar showing '1' of 1 entries. At the bottom of the interface are six icons: Home, Parameters, Sequence, Sources, Results, and Reports. To the right of the table is a detailed view of the selected error, NG3002, titled 'Error Description'. This view includes a red circular button with a white minus sign, the error code 'NG3002', and the description 'Tightening signal ends too early'. It also provides a 'Trigger condition and cause', 'Checking method and corrective action', and 'How to clear the alarm?'. The 'Tool Life' is shown as 178.

Date / Time	AL/NG Code	Description
2025/07/16 06:16:07	AL 1111	Tool information error when the power is on(Part)
2025/07/16 06:18:22	AL 1111	Tool information error when the power is on(Part)
2025/07/16 06:32:48	AL 1115	Tool information error when the power is on(1038)
2025/07/16 07:05:20	NG3002	Tightening signal ends too early
2025/07/16 07:05:47	NG3002	Tightening signal ends too early
2025/07/16 08:57:39	AL 1903	Tool temperature error
2025/07/16 09:06:30	NG3991	Loosening: Exceeded the tool torque protection range
2025/07/16 09:06:32	NG3991	Loosening: Exceeded the tool torque protection range
2025/07/16 09:06:51	NG3991	Loosening: Exceeded the tool torque protection range
2025/07/16 09:06:53	NG3991	Loosening: Exceeded the tool torque protection range

Error Description

NG3002

Tightening signal ends too early

Trigger condition and cause

Condition: during the tightening process, the tightening signal disappears.

Cause:

1. During the tightening process, the screwdriver is withdrawn early.
2. Transmission of the tightening signal is poor.

Checking method and corrective action

1. Maintain the tightening signal until the tightening is complete.
2. If the tool start condition is DI, check if the tightening signal transmission is poor.
3. If the tool start condition is push start, check if the push start device has poor contact.
4. If the tool start condition is lever start, check if the lever on the screwdriver has poor contact.

How to clear the alarm?

Loosen and then re-tighten the screw.

Tool Life: 178

11.3 Warning Report

The **Warning Report** tab displays the entries of operation warning (WN) in the system, with which you can check the warnings caused by misoperations. Click any of the entries to view the detailed description.

11

User12025.07.2421:29

Production ReportError ReportWarning ReportButton Report

Date / Time	Code	Description
2025/07/16 06:20:55	WN5057	Tightening parameters exceeded the tool spec.
2025/07/16 06:20:59	WN5057	Tightening parameters exceeded the tool spec.
2025/07/16 06:21:22	WN5057	Tightening parameters exceeded the tool spec.
2025/07/16 07:03:49	WN5058	Tightening parameters do not match the tool,Please rebuild tightening parameter
2025/07/16 07:04:03	WN5058	Tightening parameters do not match the tool,Please rebuild tightening parameter
2025/07/16 07:12:35	WN5057	Tightening parameters exceeded the tool spec.
2025/07/16 07:13:49	WN5086	Loosening parameters exceed the tool spec.
2025/07/16 07:14:27	WN5057	Tightening parameters exceeded the tool spec.
2025/07/16 07:14:28	WN5086	Loosening parameters exceed the tool spec.
2025/07/16 08:50:53	WN5058	Tightening parameters do not match the tool,Please rebuild tightening parameter

<1>

HomeParametersSequenceSourcesResultsReports

Warning Report

WN5058

Tightening parameters do not match the tool,Please rebuild tightening parameter

Trigger condition and cause	The operating parameter does not match the tool, so tightening is prohibited.
Checking method and corrective action	Modify the content of the parameter.
How to clear the alarm?	Modify the parameter and then re-tighten the screw.

11.4 Button Report

The **Button Report** tab displays the main button operations in the system.

11

Production Report	Error Report	Warning Report	Button Report	
Date / Time	Button ID	User	Before	After
2023/11/24 10:51:24	4000	User1	0	1
2023/12/06 13:24:38	4000	User1	0	1
2023/12/06 13:31:00	4000	User1	0	1
2023/12/06 13:34:54	4000	User1	0	1
2023/12/06 13:35:49	4000	User1	0	1
2023/12/06 13:37:07	4002	User1	0	1
2023/12/06 13:44:50	4000	User1	0	1

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Reports

The corresponding operations of each Button ID are as follows:

Button ID		Corresponding operation
Tool 1	Tool 2	
04000	14000	Reset the operation progress
04001	14001	Force execute the previous step
04002	14002	Force execute the next step
04003	14003	Reset the operation time
04004	14004	Clear the single screw tightening NOK count
04005	14005	Clear the single screw loosening NOK count
04006	14006	Clear the single screw tightening status
07000		Clear the production report entries
00010		Controller power-on timestamp (automatically recorded by the system; the User field displays "SystemRecord")

Troubleshooting

12

This chapter provides alarm descriptions and the corrective actions you can use for troubleshooting.

12.1	Alarm list	12-2
12.1.1	Device alarms	12-2
12.1.2	Tightening errors	12-4
12.1.3	Operation warnings	12-8
12.2	Causes and corrective actions	12-9
12.2.1	Description of device alarms	12-9
12.2.2	Description of tightening errors	12-17
12.2.3	Description of operation warnings	12-49

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The alarms include device alarms, tightening errors, and operation warnings. The definition of each is as follows (the nnn in the alarm display indicates numbers).

Alarm types	Alarm display		Description
	Tool 1	Tool 2	
Device alarms	AL1nnn	AL2nnn	The alarm messages for hardware signal and encoder signal.
Tightening errors	NG3nnn	NG4nnn	The errors generated during tightening.
Operation warnings	WN5nnn	WN6nnn	The warning messages caused by misoperation.

12.1 Alarm list

12.1.1 Device alarms

Alarm display	Alarm name
AL1001 & AL2001	Overcurrent
AL1004 & AL2004	Tool combination error
AL1006 & AL2006	Overload
AL1007 & AL2007	Excessive control error of Speed command
AL1011 & AL2011	Tool communication error
AL1013 & AL2013	Emergency stop
AL1016 & AL2016	MOSFET overheating
AL1036 & AL2036	No response from tool encoder
AL1037 & AL2037	Tool encoder communication error
AL1038 & AL2038	Tool encoder alarm
AL1039 & AL2039	Tool encoder warning
AL1070 & AL2070	Tool control board did not complete the read / write procedure
AL1083 & AL2083	Controller outputs excessive current
AL1087 & AL2087 AL1887 & AL2887	Tool torque exceeded the sensor spec.
AL1088 & AL2088 AL1888 & AL2888	Tool torque sensor error
AL1090 & AL2090 AL1890 & AL2890	No response from tool torque sensor
AL1091 & AL2091 AL1891 & AL2891	Tool torque sensor communication error
AL1092 & AL2092 AL1892 & AL2892	Tool torque sensor alarm
AL1093 & AL2093 AL1893 & AL2893	Tool torque sensor initialization failed
AL1094 & AL2094 AL1894 & AL2894	Tool temperature error when the power is on
AL1095 & AL2095 AL1895 & AL2895	Tool temperature error
AL1099 & AL2099	EEPROM not reset after firmware update
AL1111 & AL2111	Tool data error when the power is on (partial)
AL1112 & AL2112	Tool data error when the power is on (1030)
AL1113 & AL2113	Tool data error when the power is on (1078)
AL1114 & AL2114	Tool data error when the power is on (1060)

Alarm display	Alarm name
AL1115 & AL2115	Tool data error when the power is on (1038)
AL1896 & AL2896	Cam-out detection error
AL1897 & AL2897	Cam-out or tool perpendicularity detection sensor is in error
AL1900 & AL2900	Tool parameter write-in failed
AL1901 & AL2901	Key parameter reading error
AL1902 & AL2902	Tool temperature is too high
AL1903 & AL2903	Tool temperature is abnormal
AL1C31 & AL2C31	Tool cable error (UVW)

Note: if the triggered alarm is not on the list, contact your local distributor or technical staff.

12.1.2 Tightening errors

Alarm display	Alarm name
NG3000 & NG4000	Unknown parameter
NG3002 & NG4002	Tightening signal ends too early
NG3003 & NG4003	Bit camming-out
NG3011 & NG4011	Tightening: lower than the min. rotation angle
NG3012 & NG4012	Tightening: exceeded the max. rotation angle
NG3013 & NG4013	Tightening: timeout
NG3014 & NG4014	Tightening: exceeded tool max. current
NG3015 & NG4015	Tightening: error occurs during parameter setting
NG3032 & NG4032	Loosening: exceeded the max. rotation angle
NG3033 & NG4033	Loosening: timeout
NG3034 & NG4034	Loosening: exceeded tool max. current
NG3035 & NG4035	Loosening: error occurs during parameter setting
NG3110 & NG4110	Start stage: operation error
NG3111 & NG4111	Start stage: exceeded the tool torque protection range Start stage: exceeded the tightening torque protection range
NG3112 & NG4112	Start stage: exceeded the max. operation time
NG3113 & NG4113	Start stage: lower than the min. operation time
NG3114 & NG4114	Start stage: exceeded the switching torque
NG3115 & NG4115	Start stage: lower than the switching torque
NG3120 & NG4120	Start stage: exceeded tool max. current
NG3121 & NG4121	Start stage: lower than tool min. current
NG3122 & NG4122	Start stage: exceeded the max. angle
NG3123 & NG4123	Start stage: lower than the min. angle
NG3124 & NG4124	Start stage: exceeded the max. torque
NG3125 & NG4125	Start stage: lower than the min. torque
NG3210 & NG4210	Rundown stage: operation error
NG3211 & NG4211	Rundown stage: exceeded the tool torque protection range Rundown stage: exceeded the tightening torque protection range
NG3212 & NG4212	Rundown stage: exceeded the max. operation time
NG3213 & NG4213	Rundown stage: lower than the min. operation time
NG3214 & NG4214	Rundown stage: exceeded the switching torque
NG3215 & NG4215	Rundown stage: lower than the switching torque
NG3220 & NG4220	Rundown stage: exceeded tool max. current
NG3221 & NG4221	Rundown stage: lower than tool min. current
NG3222 & NG4222	Rundown stage: exceeded the max. angle
NG3223 & NG4223	Rundown stage: lower than the min. angle
NG3224 & NG4224	Rundown stage: exceeded the max. torque
NG3225 & NG4225	Rundown stage: lower than the min. torque
NG3310 & NG4310	Pre-tightening stage: operation error
NG3311 & NG4311	Pre-tightening stage: exceeded the tool torque protection range Pre-tightening stage: exceeded the tightening torque protection range
NG3312 & NG4312	Pre-tightening stage: exceeded the max. operation time
NG3313 & NG4313	Pre-tightening stage: lower than the min. operation time
NG3314 & NG4314	Pre-tightening stage: exceeded the switching torque
NG3315 & NG4315	Pre-tightening stage: lower than the switching torque
NG3320 & NG4320	Pre-tightening stage: exceeded tool max. current

Alarm display	Alarm name
NG3321 & NG4321	Pre-tightening stage: lower than tool min. current
NG3322 & NG4322	Pre-tightening stage: exceeded the max. angle
NG3323 & NG4323	Pre-tightening stage: lower than the min. angle
NG3324 & NG4324	Pre-tightening stage: exceeded the max. torque
NG3325 & NG4325	Pre-tightening stage: lower than the min. torque
NG3410 & NG4410	Tightening stage: operation error
NG3411 & NG4411	Tightening stage: exceeded the tool torque protection range Tightening stage: exceeded the tightening torque protection range
NG3412 & NG4412	Tightening stage: exceeded the max. operation time
NG3413 & NG4413	Tightening stage: lower than the min. operation time
NG3414 & NG4414	Tightening stage: exceeded the switching torque
NG3415 & NG4415	Tightening stage: lower than the switching torque
NG3420 & NG4420	Tightening stage: exceeded tool max. current
NG3421 & NG4421	Tightening stage: lower than tool min. current
NG3422 & NG4422	Tightening stage: exceeded the max. angle
NG3423 & NG4423	Tightening stage: lower than the min. angle
NG3424 & NG4424	Tightening stage: exceeded the max. torque
NG3425 & NG4425	Tightening stage: lower than the min. torque
NG3426 & NG4426	Tightening stage: exceeded the max. clamp torque
NG3427 & NG4427	Tightening stage: lower than the min. clamp torque
NG3428 & NG4428	Tightening stage: exceeded the max. clamp angle
NG3429 & NG4429	Tightening stage: lower than the min. clamp angle
NG3510 & NG4510	Rundown stage (torque rate): operation error
NG3511 & NG4511	Rundown stage (torque rate): exceeded the tightening torque protection range
NG3512 & NG4512	Rundown stage (torque rate): exceeded the max. operation time
NG3513 & NG4513	Rundown stage (torque rate): lower than the min. operation time
NG3514 & NG4514	Rundown stage (torque rate): exceeded the switching torque
NG3515 & NG4515	Rundown stage (torque rate): lower than the switching torque
NG3520 & NG4520	Rundown stage (torque rate): exceeded tool max. current
NG3521 & NG4521	Rundown stage (torque rate): lower than tool min. current
NG3522 & NG4522	Rundown stage (torque rate): exceeded the max. angle
NG3523 & NG4523	Rundown stage (torque rate): lower than the min. angle
NG3524 & NG4524	Rundown stage (torque rate): exceeded the max. torque
NG3525 & NG4525	Rundown stage (torque rate): lower than the min. torque
NG3881 & NG4881 & NG3981 & NG4981	Loosening stage: lower than the set torque
NG3891 & NG4891 & NG3991 & NG4991	Loosening stage: exceeded the tool torque protection range
NG3A10 & NG4A10	Stage 1: operation error
NG3A11 & NG4A11	Stage 1: exceeded the tightening torque protection range
NG3A12 & NG4A12	Stage 1: exceeded the max. operation time
NG3A13 & NG4A13	Stage 1: lower than the min. operation time
NG3A14 & NG4A14	Stage 1: exceeded the switching torque
NG3A15 & NG4A15	Stage 1: lower than the switching torque
NG3A20 & NG4A20	Stage 1: exceeded tool max. current
NG3A21 & NG4A21	Stage 1: lower than tool min. current
NG3A22 & NG4A22	Stage 1: exceeded the max. angle
NG3A23 & NG4A23	Stage 1: lower than the min. angle

Alarm display	Alarm name
NG3A24 & NG4A24	Stage 1: exceeded the max. torque
NG3A25 & NG4A25	Stage 1: lower than the min. torque
NG3A26 & NG4A26	Stage 1: exceeded the max. clamp torque
NG3A27 & NG4A27	Stage 1: lower than the min. clamp torque
NG3A28 & NG4A28	Stage 1: exceeded the max. clamp angle
NG3A29 & NG4A29	Stage 1: lower than the min. clamp angle
NG3B10 & NG4B10	Stage 2: operation error
NG3B11 & NG4B11	Stage 2: exceeded the tightening torque protection range
NG3B12 & NG4B12	Stage 2: exceeded the max. operation time
NG3B13 & NG4B13	Stage 2: lower than the min. operation time
NG3B14 & NG4B14	Stage 2: exceeded the switching torque
NG3B15 & NG4B15	Stage 2: lower than the switching torque
NG3B20 & NG4B20	Stage 2: exceeded tool max. current
NG3B21 & NG4B21	Stage 2: lower than tool min. current
NG3B22 & NG4B22	Stage 2: exceeded the max. angle
NG3B23 & NG4B23	Stage 2: lower than the min. angle
NG3B24 & NG4B24	Stage 2: exceeded the max. torque
NG3B25 & NG4B25	Stage 2: lower than the min. torque
NG3B26 & NG4B26	Stage 2: exceeded the max. clamp torque
NG3B27 & NG4B27	Stage 2: lower than the min. clamp torque
NG3B28 & NG4B28	Stage 2: exceeded the max. clamp angle
NG3B29 & NG4B29	Stage 2: lower than the min. clamp angle
NG3C10 & NG4C10	Stage 3: operation error
NG3C11 & NG4C11	Stage 3: exceeded the tightening torque protection range
NG3C12 & NG4C12	Stage 3: exceeded the max. operation time
NG3C13 & NG4C13	Stage 3: lower than the min. operation time
NG3C14 & NG4C14	Stage 3: exceeded the switching torque
NG3C15 & NG4C15	Stage 3: lower than the switching torque
NG3C20 & NG4C20	Stage 3: exceeded tool max. current
NG3C21 & NG4C21	Stage 3: lower than tool min. current
NG3C22 & NG4C22	Stage 3: exceeded the max. angle
NG3C23 & NG4C23	Stage 3: lower than the min. angle
NG3C24 & NG4C24	Stage 3: exceeded the max. torque
NG3C25 & NG4C25	Stage 3: lower than the min. torque
NG3C26 & NG4C26	Stage 3: exceeded the max. clamp torque
NG3C27 & NG4C27	Stage 3: lower than the min. clamp torque
NG3C28 & NG4C28	Stage 3: exceeded the max. clamp angle
NG3C29 & NG4C29	Stage 3: lower than the min. clamp angle
NG3D10 & NG4D10	Stage 4: operation error
NG3D11 & NG4D11	Stage 4: exceeded the tightening torque protection range
NG3D12 & NG4D12	Stage 4: exceeded the max. operation time
NG3D13 & NG4D13	Stage 4: lower than the min. operation time
NG3D14 & NG4D14	Stage 4: exceeded the switching torque
NG3D15 & NG4D15	Stage 4: lower than the switching torque
NG3D20 & NG4D20	Stage 4: exceeded tool max. current
NG3D21 & NG4D21	Stage 4: lower than tool min. current
NG3D22 & NG4D22	Stage 4: exceeded the max. angle

Alarm display	Alarm name
NG3D23 & NG4D23	Stage 4: lower than the min. angle
NG3D24 & NG4D24	Stage 4: exceeded the max. torque
NG3D25 & NG4D25	Stage 4: lower than the min. torque
NG3D26 & NG4D26	Stage 4: exceeded the max. clamp torque
NG3D27 & NG4D27	Stage 4: lower than the min. clamp torque
NG3D28 & NG4D28	Stage 4: exceeded the max. clamp angle
NG3D29 & NG4D29	Stage 4: lower than the min. clamp angle
NG3E10 & NG4E10	Stage 5: operation error
NG3E11 & NG4E11	Stage 5: exceeded the tightening torque protection range
NG3E12 & NG4E12	Stage 5: exceeded the max. operation time
NG3E13 & NG4E13	Stage 5: lower than the min. operation time
NG3E14 & NG4E14	Stage 5: exceeded the switching torque
NG3E15 & NG4E15	Stage 5: lower than the switching torque
NG3E20 & NG4E20	Stage 5: exceeded tool max. current
NG3E21 & NG4E21	Stage 5: lower than tool min. current
NG3E22 & NG4E22	Stage 5: exceeded the max. angle
NG3E23 & NG4E23	Stage 5: lower than the min. angle
NG3E24 & NG4E24	Stage 5: exceeded the max. torque
NG3E25 & NG4E25	Stage 5: lower than the min. torque
NG3E26 & NG4E26	Stage 5: exceeded the max. clamp torque
NG3E27 & NG4E27	Stage 5: lower than the min. clamp torque
NG3E28 & NG4E28	Stage 5: exceeded the max. clamp angle
NG3E29 & NG4E29	Stage 5: lower than the min. clamp angle
NG3F10 & NG4F10	Stage 6: operation error
NG3F11 & NG4F11	Stage 6: exceeded the tightening torque protection range
NG3F12 & NG4F12	Stage 6: exceeded the max. operation time
NG3F13 & NG4F13	Stage 6: lower than the min. operation time
NG3F14 & NG4F14	Stage 6: exceeded the switching torque
NG3F15 & NG4F15	Stage 6: lower than the switching torque
NG3F20 & NG4F20	Stage 6: exceeded tool max. current
NG3F21 & NG4F21	Stage 6: lower than tool min. current
NG3F22 & NG4F22	Stage 6: exceeded the max. angle
NG3F23 & NG4F23	Stage 6: lower than the min. angle
NG3F24 & NG4F24	Stage 6: exceeded the max. torque
NG3F25 & NG4F25	Stage 6: lower than the min. torque
NG3F26 & NG4F26	Stage 6: exceeded the max. clamp torque
NG3F27 & NG4F27	Stage 6: lower than the min. clamp torque
NG3F28 & NG4F28	Stage 6: exceeded the max. clamp angle
NG3F29 & NG4F29	Stage 6: lower than the min. clamp angle

Note: if the triggered alarm is not on the list, contact your local distributor or technical staff.

12.1.3 Operation warnings

Alarm display	Alarm name
WN5001 & WN6001	Quantity not reached. String scanning prohibited.
WN5002 & WN6002	Send the tool back for service
WN5003 & WN6003	Parameters not set
WN5004 & WN6004	Unknown tool model and specification. Update the firmware.
WN5005 & WN6005	Cannot perform tightening and loosening at the same time
WN5006 & WN6006	Parameter setting prohibited when tool is in operation
WN5007 & WN6007	Tool final current is lower than the range. Check the tightening parameter.
WN5008 & WN6008	Positioning arm is not configured
WN5009 & WN6009	Tightening signal ends too early
WN500A & WN600A	Disconnection of positioning arm communication
WN500B & WN600B	Tool final current is lower than the range. Check the tightening parameter.
WN500C & WN600C	Tool final current exceeds the range. Check the tool.
WN500D & WN600D	Tool perpendicularity detection error
WN500E & WN600E	Disable the tool operation restriction in the DI/DO tab
WN500F & WN600F	Tool temperature is abnormal
WN5051 & WN6051	Exceeded the max. count for NOK tightening
WN5052 & WN6052	The scanner string is null or the string length is incorrect. Tightening prohibited.
WN5053 & WN6053	Screw quantity reached. Tightening prohibited.
WN5054 & WN6054	Exceeded max. operation time
WN5055 & WN6055	Incorrect parameter setting. Tightening prohibited.
WN5056 & WN6056	Tightening prohibited by remote communication or DI
WN5057 & WN6057	Tightening: parameter exceeded the tool specification. Check the torque or speed setting.
WN5058 & WN6058	Tightening: parameter does not match the tool. Recreate a tightening parameter.
WN5059 & WN6059	Tightening operation prohibited by SDSofT
WN505A & WN605A	The controller cannot operate this tool model (customized)
WN5081 & WN6081	Loosening prohibited after tightening OK
WN5082 & WN6082	Loosening prohibited after tightening NOK
WN5083 & WN6083	Exceeded the max. count for NOK loosening
WN5084 & WN6084	Incorrect parameter setting. Loosening prohibited.
WN5085 & WN6085	Loosening prohibited by remote communication or DI
WN5086 & WN6086	Loosening: parameter exceeded the tool specification. Check the torque or speed setting.
WN5087 & WN6087	Tightening operation prohibited by SDSofT
WN5088 & WN6088	The controller cannot operate this tool model (customized)

Note: if the triggered alarm is not on the list, contact your local distributor or technical staff.

12.2 Causes and corrective actions

12.2.1 Description of device alarms

AL1001 & AL2001 Overcurrent	
Trigger condition and cause	Condition: main circuit current is greater than 1.5 times the maximum instantaneous current of the controller. Cause: 1. The controller output is short-circuited. 2. The screwdriver wiring is in error.
Checking method and corrective action	1. Check the connection between the screwdriver and the controller and make sure the wire is not short-circuited. Do not expose the metal conductor. 2. If the alarm occurs once the controller is powered on or the screwdriver starts operating, send your controller back to the distributor or contact Delta.
How to clear the alarm?	Cycle power on the controller.

AL1004 & AL2004 Tool combination error	
Trigger condition and cause	Condition: an incorrect screwdriver is used with the controller. Cause: 1. Combination error of the controller and screwdriver. The screwdriver specification cannot be identified. 2. The encoder of the screwdriver is loose. 3. The encoder of the screwdriver is damaged. 4. The firmware version is outdated so the screwdriver specification cannot be identified.
Checking method and corrective action	1. Perform a cross-check by using other working screwdrivers to see if the specification of the current screwdriver is incorrect. If so, replace the current screwdriver with a correct one. 2. Re-connect the controller and the screwdriver, make sure the connectors are firmly connected, and then cycle power on the controller. 3. Check for and update the controller firmware. If the issue persists, send your screwdriver back to the distributor or contact Delta.
How to clear the alarm?	1. Cycle power on the controller. 2. Update the controller firmware.

AL1006 & AL2006 Overload	
Trigger condition and cause	Condition: screwdriver and controller are overloaded. Cause: 1. The controller is overloaded and is in a continuous overload condition. 2. Improper settings of the controller system parameters.
Checking method and corrective action	1. Use a screwdriver with higher capacity, or increase the interval between each operation. 2. Check if the parameters are set improperly. A. The speed set for the tightening stage is too low. B. Acceleration / deceleration setting (constant) is too high. 3. Check if the cable between the controller and the screwdriver is properly connected. Perform a cross-check by using other working cables of the same specifications. 4. Check if the shaft of the screwdriver is stuck. If the issue persists, send your controller back to the distributor or contact Delta.
How to clear the alarm?	Alarm reset.

AL1007 & AL2007 Excessive control error of Speed command

Trigger condition and cause	Condition: the difference between speed command and speed feedback exceeded the allowable range. Cause: 1. Drastic change in the input speed command. 2. Improper setting of the speed error warning condition.
Checking method and corrective action	1. Check if the acceleration / deceleration setting (constant) has caused drastic change in the command. 2. Perform a cross-check by using other working screwdrivers of the same specifications. If the current screwdriver is in error, send your screwdriver back to the distributor or contact Delta.
How to clear the alarm?	Alarm reset.

AL1011 & AL2011 Tool communication error

Trigger condition and cause	Condition: the encoder signal is in error. Cause: 1. The encoder wiring is incorrect. 2. The connectors are loose. 3. The encoder wiring is poor. 4. Encoder communication is cut off due to interference. 5. The encoder or the MCU board is damaged.
Checking method and corrective action	1. Check if the wiring follows the instructions in the user manual. If not, connect the wire correctly. 2. Check if the cable between the controller and screwdriver is firmly connected. If not, correctly connect the cable again. 3. Check the cable between the controller and screwdriver to see if there is any poor wiring or damaged cable. If so, replace the connector and the cable. 4. Perform a cross-check by using other working controllers or screwdrivers of the same specifications. If the issue persists, send your screwdriver back to the distributor or contact Delta.
How to clear the alarm?	Cycle power on the controller.

AL1013 & AL2013 Emergency stop

Trigger condition and cause	The emergency stop (DI) is triggered.
Checking method and corrective action	Make sure the emergency stop (DI) is off.
How to clear the alarm?	Set the emergency stop (DI) to off and then perform alarm reset.

AL1016 & AL2016 MOSFET overheating

Trigger condition and cause	Condition: MOSFET error. Cause: 1. The controller is overloaded and is in a continuous overload condition. 2. The controller output is short-circuited.
Checking method and corrective action	1. Check if the average load of the screwdriver is too high. 2. Check if the ambient temperature is too high. 3. Check if the connection cable of the screwdriver is damaged.
How to clear the alarm?	Cycle power on the controller.

AL1036 & AL2036 No response from tool encoder

Trigger condition and cause	Condition: encoder internal error or the digital torque sensor is in error. Cause: 1. The encoder sends out an alarm signal, but the controller reads back the encoder alarm status without showing an error. 2. The CRC error has occurred to the encoder 10 consecutive times.
Checking method and corrective action	1. Check if the screwdriver and the controller are properly grounded. 2. Check if there is any foreign object inside the screwdriver. If so, send your screwdriver back to the distributor or contact Delta. 3. Check if the cable between the controller and screwdriver is firmly connected. If not, correctly connect the cable again. If the issue persists, send your screwdriver back to the distributor or contact Delta.
How to clear the alarm?	Alarm reset or cycle power on the controller.

AL1037 & AL2037 Tool encoder communication error

Trigger condition and cause	The CRC error has occurred to the encoder 2 cumulative times.
Checking method and corrective action	Check if the screwdriver and the controller are properly grounded.
How to clear the alarm?	Cycle power on the controller.

AL1038 & AL2038 Tool encoder alarm

Trigger condition and cause	The encoder returns an alarm.
Checking method and corrective action	Send your screwdriver back to the distributor or contact Delta.
How to clear the alarm?	Cycle power on the controller.

AL1039 & AL2039 Tool encoder warning

Trigger condition and cause	The encoder returns a warning.
Checking method and corrective action	Send your screwdriver back to the distributor or contact Delta.
How to clear the alarm?	Cycle power on the controller.

AL1070 & AL2070 Tool control board did not complete the read / write procedure

Trigger condition and cause	When the barcode string is written or other related actions are performed, the related commands are incomplete.
Checking method and corrective action	Check if the wiring is correct or if the connector is loose, and then connect the wire correctly.
How to clear the alarm?	Cycle power on the controller.

AL1083 & AL2083 Controller outputs excessive current

Trigger condition and cause	When the controller output current exceeds the threshold defined by the firmware, this alarm is triggered to protect the MOSFET from overheating or burning due to the high current.
Checking method and corrective action	1. Check the connection cable of the screwdriver for exposed metal wires or torn wires, which can cause a short circuit. In such a case, replace the cable and prevent the metal conductor from being exposed to troubleshoot the short circuit. 2. Check if the controller output wiring is correct.
How to clear the alarm?	Cycle power on the controller.

AL1087 & AL2087 & AL1887 & AL2887 Tool torque exceeded the sensor spec.

Trigger condition and cause	Condition: the controller does not detect a screwdriver connection, or the torque sensor is in error. Cause: 1. No screwdriver is connected to the controller. 2. The connection cable of the screwdriver is loose on both ends. 3. The connection cable of the screwdriver is not working. 4. The screwdriver is not working. 5. The actual torque exceeded the range of the torque sensor.
Checking method and corrective action	1. Check if the screwdriver is connected to the controller. 2. Reinstall the connection cable of the screwdriver. 3. Perform a cross-check by using other working cables of the same specifications to check for a faulty cable. If so, replace the cable. 4. If the alarm occurs once the controller is powered on, perform a cross-check by using other working screwdrivers of the same specifications to check for a faulty screwdriver. If so, replace the screwdriver. If the issue persists, send your screwdriver back to the distributor or contact Delta.
How to clear the alarm?	Cycle power on the controller.

AL1088 & AL2088 & AL1888 & AL2888 Tool torque sensor error

Trigger condition and cause	Condition: the torque sensor is in error. Cause: 1. Use the controller current sensor and the tool torque sensor to measure the current values and compare them to check if the tool torque sensor is in error. 2. There are records of AL1011 & AL2011 alarms (Tool communication error) in the Error Report tab of the Reports page, which means the both ends of the screwdriver connection cable are loose.
Checking method and corrective action	1. The torque sensor is degraded; send your screwdriver back to the distributor or contact Delta. 2. Perform a cross-check by using other working cables. If the current cable is in error, replace the cable.
How to clear the alarm?	Cycle power on the controller.

AL1090 & AL2090 & AL1890 & AL2890 No response from tool torque sensor

Trigger condition and cause	The CRC error has occurred to the torque sensor 10 consecutive times.
Checking method and corrective action	Check if the cable between the controller and screwdriver is firmly connected. If not, correctly connect the cable again.
How to clear the alarm?	Cycle power on the controller.

AL1091 & AL2091 & AL1891 & AL2891 Tool torque sensor communication error

Trigger condition and cause	The CRC error has occurred to the torque sensor 2 cumulative times.
Checking method and corrective action	Check if the screwdriver and the controller are properly grounded.
How to clear the alarm?	Cycle power on the controller.

AL1092 & AL2092 & AL1892 & AL2892 Tool torque sensor alarm

Trigger condition and cause	The torque sensor returns an alarm.
Checking method and corrective action	Send your screwdriver back to the distributor or contact Delta.
How to clear the alarm?	Cycle power on the controller.

AL1093 & AL2093 & AL1893 & AL2893 Tool torque sensor initialization failed

Trigger condition and cause	The internal parameter of the torque sensor is in error when the controller is powered on.
Checking method and corrective action	Send your screwdriver back to the distributor or contact Delta.
How to clear the alarm?	Cycle power on the controller.

AL1094 & AL2094 & AL1894 & AL2894 Tool temperature error when the power is on

Trigger condition and cause	When the controller is powered on, the read initial temperature of the torque sensor EEPROM exceeded the range of 5 to 45°C (41 to 113°F).
Checking method and corrective action	Send your screwdriver back to the distributor or contact Delta.
How to clear the alarm?	Cycle power on the controller.

AL1095 & AL2095 & AL1895 & AL2895 Tool temperature error

Trigger condition and cause	The read internal temperature of the torque sensor exceeded the range of -20 to +80°C (-4 to +176°F).
Checking method and corrective action	Send your screwdriver back to the distributor or contact Delta.
How to clear the alarm?	Cycle power on the controller.

AL1099 & AL2099 EEPROM not reset after firmware update

Trigger condition and cause	After the firmware is updated, EEPROM is not reset yet.
Checking method and corrective action	Make sure the firmware update is complete, then remove the SLAVE-B USB cable, and then cycle power on the controller.
How to clear the alarm?	Cycle power on the controller.

AL1111 & AL2111 Tool data error when the power is on (partial)

Trigger condition and cause	Condition: when the controller is powered on, encoder signal error is detected (same as AL1011 & AL2011). Cause: 1. The encoder wiring is incorrect. 2. The connectors are loose. 3. The encoder wiring is poor. 4. Encoder communication is cut off due to interference. 5. The encoder or the MCU board is damaged.
Checking method and corrective action	1. Check if the wiring follows the instructions in the user manual. If not, connect the wire correctly. 2. Check if the cable between the controller and screwdriver is firmly connected. If not, correctly connect the cable again. 3. Check the cable between the controller and screwdriver to see if there is any poor wiring or damaged cable. If so, replace the connector and the cable. 4. Perform a cross-check by using other working controllers or screwdrivers of the same specifications. If the issue persists, send your screwdriver back to the distributor or contact Delta.
How to clear the alarm?	Cycle power on the controller.

AL1112 & AL2112 Tool data error when the power is on (1030)

Trigger condition and cause	When the controller is powered on, the torque sensor precision exceeds the specification.
Checking method and corrective action	1. Check for and update the controller firmware. 2. Perform a cross-check by using other working screwdrivers of the same specifications. If the current screwdriver is in error, send it back to the distributor or contact Delta.
How to clear the alarm?	Cycle power on the controller.

AL1113 & AL2113 Tool data error when the power is on (1078)

Trigger condition and cause	When the controller is powered on, the calibrated value of the torque sensor deviates from the default by more than $\pm 20\%$.
Checking method and corrective action	1. Check if the cable between the controller and screwdriver is firmly connected. If not, correctly connect the cable again. 2. Check for and update the controller firmware. 3. Perform a cross-check by using other working screwdrivers of the same specifications. If the current screwdriver is in error, send it back to the distributor or contact Delta.
How to clear the alarm?	Cycle power on the controller.

AL1114 & AL2114 Tool data error when the power is on (1060)

Trigger condition and cause	When the controller is powered on, the motor current exceeds the specification.
Checking method and corrective action	1. Check for and update the controller firmware. 2. Perform a cross-check by using other working screwdrivers of the same specifications. If the current screwdriver is in error, send it back to the distributor or contact Delta.
How to clear the alarm?	Cycle power on the controller.

AL1115 & AL2115 Tool data error when the power is on (1038)

Trigger condition and cause	When the controller is powered on, the torque sensor zero offset is excessive. The alarm is triggered to avoid the occurrence of AL1887 & AL2887.
Checking method and corrective action	1. Check for and update the controller firmware. 2. Perform a cross-check by using other working screwdrivers of the same specifications. If the current screwdriver is in error, send it back to the distributor or contact Delta.
How to clear the alarm?	Cycle power on the controller.

AL1896 & AL2896 Cam-out detection error

Trigger condition and cause	Condition: tool rotation exceeds the threshold. Cause: enabled the "Cam-Out Detection" function in the Advanced Setting of the Parameters page to avoid bit camming-out.
Checking method and corrective action	1. Check if the tool is fixed to the platform. 2. Check if cam-out occurs when the tool is used as a handheld device. 3. Evaluate if the "Cam-Out Detection" function is appropriate for the operating condition. If not, disable the function.
How to clear the alarm?	Firmly secure the tool to the platform and restart the tightening operation.

AL1897 & AL2897 Cam-out or tool perpendicularity detection sensor is in error

Trigger condition and cause	The cam-out or tool perpendicularity detection sensor is in error.
Checking method and corrective action	Perform a cross-check by using other working screwdrivers of the same specifications. If the current screwdriver is in error, send it back to the distributor or contact Delta.
How to clear the alarm?	1. Cycle power on the controller. 2. Disable the "Cam-Out Detection" or "Tool Perpendicularity Detection" function.

AL1900 & AL2900 Tool parameter write-in failed

Trigger condition and cause	Check if the tool life is correctly written.
Checking method and corrective action	Send your screwdriver back to the distributor or contact Delta.
How to clear the alarm?	Cycle power on the controller.

AL1901 & AL2901 Key parameter reading error

Trigger condition and cause	When the system is powered on, the read parameter settings in the working area and the buffer area do not match.
Checking method and corrective action	Send your screwdriver back to the distributor or contact Delta.
How to clear the alarm?	Cycle power on the controller.

AL1902 & AL2902 Tool temperature is too high

Trigger condition and cause	Condition: the motor temperature exceeds 120°C (248°F). Cause: 1. Excessive duty cycle or insufficient dwell time. 2. The temperature sensor circuit is in error.
Checking method and corrective action	1. Check if the tightening parameters are improperly set. 2. Send your screwdriver back to the distributor or contact Delta.
How to clear the alarm?	Cycle power on the controller.

AL1903 & AL2903 Tool temperature is abnormal

Trigger condition and cause	Condition: the motor temperature exceeds 150°C (302°F) or is below -20°C (-4°F). Cause: the temperature sensor circuit is in error.
Checking method and corrective action	Send your screwdriver back to the distributor or contact Delta.
How to clear the alarm?	Cycle power on the controller.

AL1C31 & AL2C31 Tool cable error (UVW)

Trigger condition and cause	Disconnection of the tool power cable (UVW) and ground wire (GND).
Checking method and corrective action	Check if the tool cable is disconnected. If so, replace the cable.
How to clear the alarm?	Cycle power on the controller.

12.2.2 Description of tightening errors

NG3000 & NG4000 Unknown parameter	
Trigger condition and cause	Condition: when the tightening process starts, the parameter configuration is incorrect. Cause: the controller cannot identify the parameter.
Checking method and corrective action	1. Make sure the parameter is not being configured when the screwdriver is started. 2. Check if the controller is configuring any parameter. 3. Make sure the parameter is already configured before you perform tightening or loosening.
How to clear the alarm?	Modify the parameter configuration and perform tightening or loosening again.

NG3002 & NG4002 Tightening signal ends too early	
Trigger condition and cause	Condition: during the tightening process, the tightening signal disappears. Cause: 1. During the tightening process, the screwdriver is withdrawn early. 2. Transmission of the tightening signal is poor. 3. The tightening signal is interfered by noise.
Checking method and corrective action	1. Maintain the tightening signal until the tightening is complete. 2. If the tool start condition is DI, check if the tightening signal transmission is poor. 3. If the tool start condition is push start, check if there is poor contact in the push start device. 4. If the tool start condition is lever start, check if there is poor contact in the lever on the screwdriver. 5. Set the DI response filter time in the System Settings tab of the Controller page.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3003 & NG4003 Bit camming-out	
Trigger condition and cause	Condition: bit cam-out occurs during the tightening process. Cause: 1. The tightening fails because of bit camming-out during the tightening process. 2. The cam-out related parameters are not properly set.
Checking method and corrective action	1. Correct the target condition in the Tightening Settings tab. 2. Correct the settings related to bit camming-out.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3011 & NG4011 Tightening: lower than the min. rotation angle	
Trigger condition and cause	Condition: during the tightening process, the rotation angle is smaller than the minimum tightening angle setting. Cause: 1. The tightening fails because the set tightening condition is not met during the tightening process. 2. The minimum tightening angle is not properly set.
Checking method and corrective action	1. Correct the tightening condition. 2. Correct the setting of the minimum tightening angle.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3012 & NG4012 Tightening: exceeded the max. rotation angle

Trigger condition and cause	Condition: during the tightening process, the rotation angle exceeded the maximum tightening angle setting. Cause: 1. The tightening fails because the set tightening condition is not met during the tightening process. 2. The maximum tightening angle is not properly set.
Checking method and corrective action	1. Correct the tightening condition. 2. Correct the setting of the maximum tightening angle.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3013 & NG4013 Tightening: timeout

Trigger condition and cause	Condition: the total operation time is exceeded before the tightening is complete. Cause: 1. The tightening fails because the set tightening condition is not met during the tightening process. 2. The total operation time is not properly set.
Checking method and corrective action	1. Correct the tightening condition. 2. Correct the timeout setting for tightening.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3014 & NG4014 Tightening: exceeded tool max. current

Trigger condition and cause	During the tightening process, the tool current exceeded its maximum current specification.
Checking method and corrective action	1. Check if the tool current has exceeded the maximum current during the tightening process. 2. Check if the controller current feedback is abnormal.
How to clear the alarm?	Cycle power on the controller.

NG3015 & NG4015 Tightening: error occurs during parameter setting

Trigger condition and cause	When the tightening process starts, an error occurs in the controller.
Checking method and corrective action	Call the parameter of tightening source again.
How to clear the alarm?	Cycle power on the controller.

NG3032 & NG4032 Loosening: exceeded the max. rotation angle

Trigger condition and cause	Condition: during the loosening process, the rotation angle exceeded the maximum loosening angle setting. Cause: the maximum loosening angle is not properly set.
Checking method and corrective action	Correct the setting of the maximum loosening angle.
How to clear the alarm?	Re-loosen the screw.

NG3033 & NG4033 Loosening: timeout

Trigger condition and cause	Condition: the total operation time is exceeded before the loosening is complete. Cause: the total operation time is not properly set.
Checking method and corrective action	Correct the timeout setting for loosening.
How to clear the alarm?	Re-loosen the screw.

NG3034 & NG4034 Loosening: exceeded tool max. current

Trigger condition and cause	During the loosening process, the tool current exceeded its maximum current specification.
Checking method and corrective action	1. Check if the tool current has exceeded the maximum current during the loosening process. 2. Check if the controller current feedback is abnormal.
How to clear the alarm?	Cycle power on the controller.

NG3035 & NG4035 Loosening: error occurs during parameter setting

Trigger condition and cause	During the process of setting the loosening parameter, an error occurs in the controller.
Checking method and corrective action	Call the loosening parameter again.
How to clear the alarm?	Cycle power on the controller.

NG3110 & NG4110 Start stage: operation error

Trigger condition and cause	Condition: the controller returns an error during the tightening process. Cause: 1. The torque of the start stage exceeded the tool maximum torque. 2. During the tightening process, the tool is in the Servo Off state.
Checking method and corrective action	Check if the torque of the start stage is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

**NG3111 & NG4111 Start stage: exceeded the tool torque protection range
Start stage: exceeded the tightening torque protection range**

Trigger condition and cause	Condition: the torque of the start stage exceeded the protection range. Cause: 1. Exceeded the tightening torque protection range (standard strategy). 2. Exceeded the tool maximum torque (pre-position strategy).
Checking method and corrective action	1. Check if the tool has reached a foreign object before this stage is complete. 2. Check if the speed for the start stage is too high. 3. Check if the setting of the start torque and start angle is appropriate. ■ The torque setting in the start stage cannot be greater than that in the tightening stage (standard strategy). ■ The torque setting in the start stage cannot be greater than the tool maximum torque (pre-position strategy).
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3112 & NG4112 Start stage: exceeded the max. operation time

Trigger condition and cause	Exceeded the maximum operation time of the start stage.
Checking method and corrective action	Check if the setting for the maximum operation time of the start stage is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3113 & NG4113 Start stage: lower than the min. operation time

Trigger condition and cause	Shorter than the minimum operation time of the start stage.
Checking method and corrective action	Check if the setting for the minimum operation time of the start stage is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3114 & NG4114 Start stage: exceeded the switching torque

Trigger condition and cause	The ending torque of this stage exceeds the set maximum switching torque.
Checking method and corrective action	Check if the setting for the maximum switching torque is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3115 & NG4115 Start stage: lower than the switching torque

Trigger condition and cause	The ending torque of this stage is lower than the set minimum switching torque.
Checking method and corrective action	Check if the setting for the minimum switching torque is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3120 & NG4120 Start stage: exceeded tool max. current

Trigger condition and cause	After the final tightening torque value is converted into an electric current value, it exceeds the tool maximum current.
Checking method and corrective action	1. Use the controller current sensor and the tool torque sensor to measure the current values and then compare them to check if the screwdriver is degraded. 2. Check if the torque sensor is damaged. 3. Check if the speed of the start stage is too high. 4. Check if the torque of the start stage is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3121 & NG4121 Start stage: lower than tool min. current

Trigger condition and cause	After the final tightening torque value is converted into an electric current value, it is lower than the tool minimum current.
Checking method and corrective action	1. Use the controller current sensor and the tool torque sensor to measure the current values and then compare them to check if the screwdriver is degraded. 2. Check if the torque sensor is damaged. 3. Check if the speed of the start stage is too high. 4. Check if the torque of the start stage is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3122 & NG4122 Start stage: exceeded the max. angle

Trigger condition and cause	The angle of the start stage exceeded the Max. Rotation Angle set in the General Settings tab.
Checking method and corrective action	Check if the Max. Rotation Angle set in the General Settings tab is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3123 & NG4123 Start stage: lower than the min. angle

Trigger condition and cause	The angle of the start stage is lower than the Min. Rotation Angle set in the General Settings tab.
Checking method and corrective action	Check if the Min. Rotation Angle set in the General Settings tab is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3124 & NG4124 Start stage: exceeded the max. torque

Trigger condition and cause	Exceeded the maximum torque of the start stage.
Checking method and corrective action	Check if the setting for the maximum torque of the start stage is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3125 & NG4125 Start stage: lower than the min. torque

Trigger condition and cause	Lower than the minimum torque of the start stage.
Checking method and corrective action	Check if the setting for the minimum torque of the start stage is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3210 & NG4210 Rundown stage: operation error

Trigger condition and cause	Condition: the controller returns an error during the tightening process. Cause: 1. The torque of the rundown stage exceeded the tool maximum torque. 2. During the tightening process, the tool is in the Servo Off state.
Checking method and corrective action	Check if the torque of the rundown stage is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3211 & NG4211 Rundown stage: exceeded the tool torque protection range	
Rundown stage: exceeded the tightening torque protection range	
Trigger condition and cause	Condition: the torque of the rundown stage exceeded the protection range. Cause: 1. Exceeded the tightening torque protection range (standard strategy). 2. Exceeded the tool maximum torque (pre-position strategy).
Checking method and corrective action	1. Check if the tool has reached a foreign object before this stage is complete. 2. Check if the speed of the rundown stage is too high. 3. Check if the setting of the angle, torque, or torque rate for the rundown stage is appropriate. 4. Check if the speed of the previous stage is too high. 5. Check if the angle of the previous stage is set too large. ■ The torque setting in the rundown stage cannot be greater than that in the tightening stage (standard strategy). ■ The torque setting in the rundown stage cannot be greater than the tool maximum torque (pre-position strategy).
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3212 & NG4212 Rundown stage: exceeded the max. operation time	
Trigger condition and cause	Exceeded the maximum operation time of the rundown stage.
Checking method and corrective action	Check if the setting for the maximum operation time of the rundown stage is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3213 & NG4213 Rundown stage: lower than the min. operation time	
Trigger condition and cause	Shorter than the minimum operation time of the rundown stage.
Checking method and corrective action	Check if the setting for the minimum operation time of the rundown stage is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3214 & NG4214 Rundown stage: exceeded the switching torque	
Trigger condition and cause	The ending torque of this stage exceeded the set maximum switching torque.
Checking method and corrective action	Check if the setting for the maximum switching torque is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3215 & NG4215 Rundown stage: lower than the switching torque	
Trigger condition and cause	The ending torque of this stage is lower than the set minimum switching torque.
Checking method and corrective action	Check if the setting for the minimum switching torque is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3220 & NG4220 Rundown stage: exceeded tool max. current

Trigger condition and cause	After the final tightening torque value is converted into an electric current value, it exceeds the tool maximum current.
Checking method and corrective action	1. Use the controller current sensor and the tool torque sensor to measure the current values and then compare them to check if the screwdriver is degraded. 2. Check if the torque sensor is damaged. 3. Check if the speed of the rundown stage is too high. 4. Check if the torque of the rundown stage is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3221 & NG4221 Rundown stage: lower than tool min. current

Trigger condition and cause	After the final tightening torque value is converted into an electric current value, it is lower than the tool minimum current.
Checking method and corrective action	1. Use the controller current sensor and the tool torque sensor to measure the current values and then compare them to check if the screwdriver is degraded. 2. Check if the torque sensor is damaged. 3. Check if the speed of the rundown stage is too high. 4. Check if the torque of the rundown stage is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3222 & NG4222 Rundown stage: exceeded the max. angle

Trigger condition and cause	Exceeded the maximum angle of the rundown stage.
Checking method and corrective action	Check if the setting for the maximum angle of the rundown stage is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3223 & NG4223 Rundown stage: lower than the min. angle

Trigger condition and cause	Lower than the minimum angle of the rundown stage.
Checking method and corrective action	Check if the setting for the minimum angle of the rundown stage is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3224 & NG4224 Rundown stage: exceeded the max. torque

Trigger condition and cause	Exceeded the maximum torque of the rundown stage.
Checking method and corrective action	1. Check if the setting for the maximum torque of the rundown stage is appropriate. 2. Check if the speed of the previous stage is too high. 3. Check if the angle of the previous stage is set too large.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3225 & NG4225 Rundown stage: lower than the min. torque

Trigger condition and cause	Lower than the minimum torque of the rundown stage.
Checking method and corrective action	Check if the setting for the minimum torque of the rundown stage is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3310 & NG4310 Pre-tightening stage: operation error

Trigger condition and cause	Condition: the controller returns an error during the tightening process. Cause: 1. The torque of the pre-tightening stage exceeded the tool maximum torque. 2. During the tightening process, the tool is in the Servo Off state.
Checking method and corrective action	Check if the torque of the pre-tightening stage is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

**NG3311 & NG4311 Pre-tightening stage: exceeded the tool torque protection range
Pre-tightening stage: exceeded the tightening torque protection range**

Trigger condition and cause	Condition: the torque of the pre-tightening stage exceeded the protection range. Cause: exceeded the tightening torque protection range (standard strategy).
Checking method and corrective action	1. Check if the tool has reached a foreign object before this stage is complete. 2. Check if the speed of the pre-tightening stage is too high. 3. Check if the setting for the torque or torque rate of the pre-tightening stage is appropriate. 4. Check if the speed of the previous stage is too high. 5. Check if the torque, angle, or torque rate of the previous stage is set too large. ■ The torque setting in the pre-tightening stage cannot be greater than that in the tightening stage.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3312 & NG4312 Pre-tightening stage: exceeded the max. operation time

Trigger condition and cause	Exceeded the maximum operation time of the pre-tightening stage.
Checking method and corrective action	Check if the setting for the maximum operation time of the pre-tightening stage is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3313 & NG4313 Pre-tightening stage: lower than the min. operation time

Trigger condition and cause	Shorter than the minimum operation time of the pre-tightening stage.
Checking method and corrective action	Check if the setting for the minimum operation time of the pre-tightening stage is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3314 & NG4314 Pre-tightening stage: exceeded the switching torque

Trigger condition and cause	The ending torque of this stage exceeded the set maximum switching torque.
Checking method and corrective action	Check if the setting for the maximum switching torque is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3315 & NG4315 Pre-tightening stage: lower than the switching torque

Trigger condition and cause	The ending torque of this stage is lower than the set minimum switching torque.
Checking method and corrective action	Check if the setting for the minimum switching torque is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3320 & NG4320 Pre-tightening stage: exceeded tool max. current

Trigger condition and cause	After the final tightening torque value is converted into an electric current value, it exceeds the tool maximum current.
Checking method and corrective action	1. Use the controller current sensor and the tool torque sensor to measure the current values and then compare them to check if the screwdriver is degraded. 2. Check if the torque sensor is damaged. 3. Check if the speed of the pre-tightening stage is too high. 4. Check if the torque of the pre-tightening stage is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3321 & NG4321 Pre-tightening stage: lower than tool min. current

Trigger condition and cause	After the final tightening torque value is converted into an electric current value, it is lower than the tool minimum current.
Checking method and corrective action	1. Use the controller current sensor and the tool torque sensor to measure the current values and then compare them to check if the screwdriver is degraded. 2. Check if the torque sensor is damaged. 3. Check if the speed of the pre-tightening stage is too high. 4. Check if the torque of the pre-tightening stage is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3322 & NG4322 Pre-tightening stage: exceeded the max. angle

Trigger condition and cause	Exceeded the maximum angle of the pre-tightening stage.
Checking method and corrective action	Check if the setting for the maximum angle of the pre-tightening stage is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3323 & NG4323 Pre-tightening stage: lower than the min. angle

Trigger condition and cause	Lower than the minimum angle of the pre-tightening stage.
Checking method and corrective action	Check if the setting for the minimum angle of the pre-tightening stage is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3324 & NG4324 Pre-tightening stage: exceeded the max. torque

Trigger condition and cause	The torque of the pre-tightening stage exceeded the tool maximum torque.
Checking method and corrective action	1. Check if the setting for the torque of the pre-tightening stage is appropriate. 2. Check if the speed of the previous stage is too high. 3. Check if the angle, torque, or torque rate of the previous stage is set too large.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3325 & NG4325 Pre-tightening stage: lower than the min. torque

Trigger condition and cause	The torque of the pre-tightening stage is lower than the tool minimum torque.
Checking method and corrective action	Check if the setting for the torque of the pre-tightening stage is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3410 & NG4410 Tightening stage: operation error

Trigger condition and cause	Condition: the controller returns an error during the tightening process. Cause: 1. The torque of the tightening stage exceeded the tool maximum torque. 2. During the tightening process, the tool is in the Servo Off state.
Checking method and corrective action	Check if the torque of the tightening stage is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

**NG3411 & NG4411 Tightening stage: exceeded the tool torque protection range
Tightening stage: exceeded the tightening torque protection range**

Trigger condition and cause	Condition: the torque of the tightening stage exceeded the protection range. Cause: 1. Exceeded the tightening torque protection range (standard strategy). 2. Exceeded the tool maximum torque (enhanced strategy).
Checking method and corrective action	1. Check if the setting of the tightening torque protection range is appropriate. 2. Check if the speed of the tightening stage is too high. 3. Check if the setting for the angle, torque, clamp torque, or clamp angle of the tightening stage is too large. 4. Check if the speed of the previous stage is too high (standard strategy). 5. Check if the torque or torque rate of the previous stage is set too large (standard strategy). ■ The torque setting in the tightening stage cannot be greater than the tool maximum torque.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3412 & NG4412 Tightening stage: exceeded the max. operation time

Trigger condition and cause	Exceeded the maximum operation time of the tightening stage.
Checking method and corrective action	Check if the setting for the maximum operation time of the tightening stage is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3413 & NG4413 Tightening stage: lower than the min. operation time

Trigger condition and cause	Shorter than the minimum operation time of the tightening stage.
Checking method and corrective action	Check if the setting for the minimum operation time of the tightening stage is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3414 & NG4414 Tightening stage: exceeded the switching torque

Trigger condition and cause	The ending torque of this stage exceeded the set maximum switching torque.
Checking method and corrective action	Check if the setting for the maximum switching torque is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3415 & NG4415 Tightening stage: lower than the switching torque

Trigger condition and cause	The ending torque of this stage is lower than the set minimum switching torque.
Checking method and corrective action	Check if the setting for the minimum switching torque is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3420 & NG4420 Tightening stage: exceeded tool max. current

Trigger condition and cause	After the final tightening torque value is converted into an electric current value, it exceeds the tool maximum current.
Checking method and corrective action	1. Use the controller current sensor and the tool torque sensor to measure the current values and then compare them to check if the screwdriver is degraded. 2. Check if the torque sensor is damaged. 3. Check if the speed of the tightening stage is too high. 4. Check if the torque of the tightening stage is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3421 & NG4421 Tightening stage: lower than tool min. current

Trigger condition and cause	After the final tightening torque value is converted into an electric current value, it is lower than the tool minimum current.
Checking method and corrective action	1. Use the controller current sensor and the tool torque sensor to measure the current values and then compare them to check if the screwdriver is degraded. 2. Check if the torque sensor is damaged. 3. Check if the speed of the tightening stage is too high. 4. Check if the torque of the tightening stage is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3422 & NG4422 Tightening stage: exceeded the max. angle

Trigger condition and cause	Exceeded the maximum angle of the tightening stage.
Checking method and corrective action	Check if the setting for the maximum angle of the tightening stage is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3423 & NG4423 Tightening stage: lower than the min. angle

Trigger condition and cause	Lower than the minimum angle of the tightening stage.
Checking method and corrective action	Check if the setting for the minimum angle of the tightening stage is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3424 & NG4424 Tightening stage: exceeded the max. torque

Trigger condition and cause	Exceeded the maximum torque of the tightening stage.
Checking method and corrective action	1. Check if the setting for the maximum torque of the tightening stage is appropriate. 2. Check if the speed of the previous stage is too high (standard strategy). 3. Check if the torque or torque rate of the previous stage is set too large (standard strategy).
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3425 & NG4425 Tightening stage: lower than the min. torque

Trigger condition and cause	Lower than the minimum torque of the tightening stage.
Checking method and corrective action	Check if the setting for the minimum torque of the tightening stage is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3426 & NG4426 Tightening stage: exceeded the max. clamp torque

Trigger condition and cause	Exceeded the maximum clamp torque of the tightening stage (standard strategy).
Checking method and corrective action	Check if the setting for the maximum clamp torque of the tightening stage is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3427 & NG4427 Tightening stage: lower than the min. clamp torque

Trigger condition and cause	Lower than the minimum clamp torque of the tightening stage (standard strategy).
Checking method and corrective action	Check if the setting for the minimum clamp torque of the tightening stage is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3428 & NG4428 Tightening stage: exceeded the max. clamp angle

Trigger condition and cause	Exceeded the maximum clamp angle of the tightening stage (standard strategy).
Checking method and corrective action	Check if the setting for the maximum clamp angle of the tightening stage is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3429 & NG4429 Tightening stage: lower than the min. clamp angle	
Trigger condition and cause	Lower than the minimum clamp angle of the tightening stage (standard strategy).
Checking method and corrective action	Check if the setting for the minimum clamp angle of the tightening stage is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.
NG3510 & NG4510 Rundown stage (torque rate): operation error	
Trigger condition and cause	Condition: the controller returns an error during the tightening process. Cause: 1. The torque of the rundown stage exceeded the tool maximum torque. 2. During the tightening process, the tool is in the Servo Off state.
Checking method and corrective action	Check if the torque of the rundown stage is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.
NG3511 & NG4511 Rundown stage (torque rate): exceeded the tightening torque protection range	
Trigger condition and cause	Condition: the torque of the rundown stage exceeded the protection range. Cause: 1. Exceeded the tightening torque protection range (standard strategy). 2. Exceeded the tool maximum torque (pre-position strategy).
Checking method and corrective action	1. Check if the tool has reached a foreign object before this stage is complete. 2. Check if the speed of the rundown stage is too high. 3. Check if the settings of the torque rate and the angle interval for torque rate calculation are appropriate. 4. Check if the speed of the previous stage is too high. 5. Check if the angle of the previous stage is set too large. ■ The torque setting in the rundown stage cannot be greater than that in the tightening stage (standard strategy). ■ The torque setting in the rundown stage cannot be greater than the tool maximum torque (pre-position strategy).
How to clear the alarm?	Loosen and then re-tighten the screw.
NG3512 & NG4512 Rundown stage (torque rate): exceeded the max. operation time	
Trigger condition and cause	Exceeded the maximum operation time of the rundown stage.
Checking method and corrective action	Check if the setting for the maximum operation time of the rundown stage is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.
NG3513 & NG4513 Rundown stage (torque rate): lower than the min. operation time	
Trigger condition and cause	Shorter than the minimum operation time of the rundown stage.
Checking method and corrective action	Check if the setting for the minimum operation time of the rundown stage is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3514 & NG4514 Rundown stage (torque rate): exceeded the switching torque

Trigger condition and cause	The ending torque of this stage exceeded the set maximum switching torque.
Checking method and corrective action	Check if the setting for the maximum switching torque is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3515 & NG4515 Rundown stage (torque rate): lower than the switching torque

Trigger condition and cause	The ending torque of this stage is lower than the set minimum switching torque.
Checking method and corrective action	Check if the setting for the minimum switching torque is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3520 & NG4520 Rundown stage (torque rate): exceeded tool max. current

Trigger condition and cause	After the final tightening torque value is converted into an electric current value, it exceeds the tool maximum current.
Checking method and corrective action	1. Use the controller current sensor and the tool torque sensor to measure the current values and then compare them to check if the screwdriver is degraded. 2. Check if the torque sensor is damaged. 3. Check if the speed of the rundown stage is too high. 4. Check if the torque of the rundown stage is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3521 & NG4521 Rundown stage (torque rate): lower than tool min. current

Trigger condition and cause	After the final tightening torque value is converted into an electric current value, it is lower than the tool minimum current.
Checking method and corrective action	1. Use the controller current sensor and the tool torque sensor to measure the current values and then compare them to check if the screwdriver is degraded. 2. Check if the torque sensor is damaged. 3. Check if the speed of the rundown stage is too high. 4. Check if the torque of the rundown stage is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3522 & NG4522 Rundown stage (torque rate): exceeded the max. angle

Trigger condition and cause	Exceeded the maximum angle of the rundown stage.
Checking method and corrective action	Check if the setting for the maximum angle of the rundown stage is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3523 & NG4523 Rundown stage (torque rate): lower than the min. angle

Trigger condition and cause	Lower than the minimum angle of the rundown stage.
Checking method and corrective action	Check if the setting for the minimum angle of the rundown stage is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3524 & NG4524 Rundown stage (torque rate): exceeded the max. torque

Trigger condition and cause	Exceeded the maximum torque of the rundown stage.
Checking method and corrective action	1. Check if the setting for the maximum torque of the rundown stage is appropriate. 2. Check if the speed of the previous stage is too high. 3. Check if the angle of the previous stage is set too large.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3525 & NG4525 Rundown stage (torque rate): lower than the min. torque

Trigger condition and cause	Lower than the minimum torque of the rundown stage.
Checking method and corrective action	Check if the setting for the minimum torque of the rundown stage is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3881 & NG4881 & NG3981 & NG4981 Loosening stage: lower than the set torque

Trigger condition and cause	The torque during loosening operation is lower than the minimum torque of the loosening stage.
Checking method and corrective action	Check if the setting for the minimum torque of the loosening stage is appropriate.
How to clear the alarm?	Loosen the screw again.

NG3891 & NG4891 & NG3991 & NG4991 Loosening stage: exceeded the tool torque protection range

Trigger condition and cause	The torque of the loosening stage exceeded the tool maximum torque.
Checking method and corrective action	The torque setting in the loosening stage cannot be greater than the tool maximum torque.
How to clear the alarm?	Loosen the screw again.

NG3A10 & NG4A10 Stage 1: operation error

Trigger condition and cause	Condition: the controller returns an error during the tightening process. Cause: 1. The torque of this stage exceeded the tool maximum torque. 2. During the tightening process, the tool is in the Servo Off state.
Checking method and corrective action	Check if the torque of this stage is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3A11 & NG4A11 Stage 1: exceeded the tightening torque protection range	
Trigger condition and cause	The torque of Stage 1 exceeded the protection range.
Checking method and corrective action	1. Check if the tool has reached a foreign object before this stage is complete. 2. Check if the speed of Stage 1 is too high. 3. Check if the setting of the torque rate or the angle interval for torque rate calculation is appropriate. ■ The torque setting in Stage 1 cannot be greater than the tool maximum torque.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3A12 & NG4A12 Stage 1: exceeded the max. operation time	
Trigger condition and cause	Exceeded the maximum operation time of Stage 1.
Checking method and corrective action	Check if the setting for the maximum operation time of Stage 1 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3A13 & NG4A13 Stage 1: lower than the min. operation time	
Trigger condition and cause	Shorter than the minimum operation time of Stage 1.
Checking method and corrective action	Check if the setting for the minimum operation time of Stage 1 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3A14 & NG4A14 Stage 1: exceeded the switching torque	
Trigger condition and cause	The ending torque of this stage exceeded the set maximum switching torque.
Checking method and corrective action	Check if the setting for the maximum switching torque is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3A15 & NG4A15 Stage 1: lower than the switching torque	
Trigger condition and cause	The ending torque of this stage is lower than the set minimum switching torque.
Checking method and corrective action	Check if the setting for the minimum switching torque is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3A20 & NG4A20 Stage 1: exceeded tool max. current	
Trigger condition and cause	After the final tightening torque value is converted into an electric current value, it exceeds the tool maximum current.
Checking method and corrective action	1. Use the controller current sensor and the tool torque sensor to measure the current values and then compare them to check if the screwdriver is degraded. 2. Check if the torque sensor is damaged. 3. Check if the speed of Stage 1 is too high. 4. Check if the torque of Stage 1 is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3A21 & NG4A21 Stage 1: lower than tool min. current

Trigger condition and cause	After the final tightening torque value is converted into an electric current value, it is lower than the tool minimum current.
Checking method and corrective action	1. Use the controller current sensor and the tool torque sensor to measure the current values and then compare them to check if the screwdriver is degraded. 2. Check if the torque sensor is damaged. 3. Check if the speed of Stage 1 is too high. 4. Check if the torque of Stage 1 is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3A22 & NG4A22 Stage 1: exceeded the max. angle

Trigger condition and cause	Exceeded the maximum angle of Stage 1.
Checking method and corrective action	Check if the setting for the maximum angle of Stage 1 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3A23 & NG4A23 Stage 1: lower than the min. angle

Trigger condition and cause	Lower than the minimum angle of Stage 1.
Checking method and corrective action	Check if the setting for the minimum angle of Stage 1 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3A24 & NG4A24 Stage 1: exceeded the max. torque

Trigger condition and cause	Exceeded the maximum torque of Stage 1.
Checking method and corrective action	Check if the setting for the maximum torque of Stage 1 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3A25 & NG4A25 Stage 1: lower than the min. torque

Trigger condition and cause	Lower than the minimum torque of Stage 1.
Checking method and corrective action	Check if the setting for the minimum torque of Stage 1 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3A26 & NG4A26 Stage 1: exceeded the max. clamp torque

Trigger condition and cause	Exceeded the maximum clamp torque of Stage 1.
Checking method and corrective action	Check if the setting for the maximum clamp torque of Stage 1 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3A27 & NG4A27 Stage 1: lower than the min. clamp torque

Trigger condition and cause	Lower than the minimum clamp torque of Stage 1.
Checking method and corrective action	Check if the setting for the minimum clamp torque of Stage 1 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3A28 & NG4A28 Stage 1: exceeded the max. clamp angle

Trigger condition and cause	Exceeded the maximum clamp angle of Stage 1.
Checking method and corrective action	Check if the setting for the maximum clamp angle of Stage 1 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3A29 & NG4A29 Stage 1: lower than the min. clamp angle

Trigger condition and cause	Lower than the minimum clamp angle of Stage 1.
Checking method and corrective action	Check if the setting for the minimum clamp angle of Stage 1 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3B10 & NG4B10 Stage 2: operation error

Trigger condition and cause	Condition: the controller returns an error during the tightening process. Cause: 1. The torque of this stage exceeded the tool maximum torque. 2. During the tightening process, the tool is in the Servo Off state.
Checking method and corrective action	Check if the torque of this stage is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3B11 & NG4B11 Stage 2: exceeded the tightening torque protection range

Trigger condition and cause	The torque of Stage 2 exceeded the protection range.
Checking method and corrective action	1. Check if the tool has reached a foreign object before this stage is complete. 2. Check if the speed of Stage 2 is too high. 3. Check if the setting of the torque rate or the angle interval for torque rate calculation is appropriate. 4. Check if the speed of the previous stage is too high. 5. Check if the torque, angle, or torque rate of the previous stage is set too large. ■ The torque setting in Stage 2 cannot be greater than the tool maximum torque.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3B12 & NG4B12 Stage 2: exceeded the max. operation time

Trigger condition and cause	Exceeded the maximum operation time of Stage 2.
Checking method and corrective action	Check if the setting for the maximum operation time of Stage 2 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3B13 & NG4B13 Stage 2: lower than the min. operation time

Trigger condition and cause	Shorter than the minimum operation time of Stage 2.
Checking method and corrective action	Check if the setting for the minimum operation time of Stage 2 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3B14 & NG4B14 Stage 2: exceeded the switching torque

Trigger condition and cause	The ending torque of this stage exceeded the set maximum switching torque.
Checking method and corrective action	Check if the setting for the maximum switching torque is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3B15 & NG4B15 Stage 2: lower than the switching torque

Trigger condition and cause	The ending torque of this stage is lower than the set minimum switching torque.
Checking method and corrective action	Check if the setting for the minimum switching torque is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3B20 & NG4B20 Stage 2: exceeded tool max. current

Trigger condition and cause	After the final tightening torque value is converted into an electric current value, it exceeds the tool maximum current.
Checking method and corrective action	1. Use the controller current sensor and the tool torque sensor to measure the current values and then compare them to check if the screwdriver is degraded. 2. Check if the torque sensor is damaged. 3. Check if the speed of Stage 2 is too high. 4. Check if the torque of Stage 2 is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3B21 & NG4B21 Stage 2: lower than tool min. current

Trigger condition and cause	After the final tightening torque value is converted into an electric current value, it is lower than the tool minimum current.
Checking method and corrective action	1. Use the controller current sensor and the tool torque sensor to measure the current values and then compare them to check if the screwdriver is degraded. 2. Check if the torque sensor is damaged. 3. Check if the speed of Stage 2 is too high. 4. Check if the torque of Stage 2 is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3B22 & NG4B22 Stage 2: exceeded the max. angle

Trigger condition and cause	Exceeded the maximum angle of Stage 2.
Checking method and corrective action	Check if the setting for the maximum angle of Stage 2 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3B23 & NG4B23 Stage 2: lower than the min. angle

Trigger condition and cause	Lower than the minimum angle of Stage 2.
Checking method and corrective action	Check if the setting for the minimum angle of Stage 2 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3B24 & NG4B24 Stage 2: exceeded the max. torque

Trigger condition and cause	Exceeded the maximum torque of Stage 2.
Checking method and corrective action	1. Check if the setting for the maximum torque of Stage 2 is appropriate. 2. Check if the speed of the previous stage is too high. 3. Check if the torque, angle, or torque rate of the previous stage is set too large.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3B25 & NG4B25 Stage 2: lower than the min. torque

Trigger condition and cause	Lower than the minimum torque of Stage 2.
Checking method and corrective action	Check if the setting for the minimum torque of Stage 2 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3B26 & NG4B26 Stage 2: exceeded the max. clamp torque

Trigger condition and cause	Exceeded the maximum clamp torque of Stage 2.
Checking method and corrective action	Check if the setting for the maximum clamp torque of Stage 2 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3B27 & NG4B27 Stage 2: lower than the min. clamp torque

Trigger condition and cause	Lower than the minimum clamp torque of Stage 2.
Checking method and corrective action	Check if the setting for the minimum clamp torque of Stage 2 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3B28 & NG4B28 Stage 2: exceeded the max. clamp angle

Trigger condition and cause	Exceeded the maximum clamp angle of Stage 2.
Checking method and corrective action	Check if the setting for the maximum clamp angle of Stage 2 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3B29 & NG4B29 Stage 2: lower than the min. clamp angle

Trigger condition and cause	Lower than the minimum clamp angle of Stage 2.
Checking method and corrective action	Check if the setting for the minimum clamp angle of Stage 2 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3C10 & NG4C10 Stage 3: operation error

Trigger condition and cause	Condition: the controller returns an error during the tightening process. Cause: 1. The torque of this stage exceeded the tool maximum torque. 2. During the tightening process, the tool is in the Servo Off state.
Checking method and corrective action	Check if the torque of this stage is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3C11 & NG4C11 Stage 3: exceeded the tightening torque protection range

Trigger condition and cause	The torque of Stage 3 exceeded the protection range.
Checking method and corrective action	1. Check if the tool has reached a foreign object before this stage is complete. 2. Check if the speed of Stage 3 is too high. 3. Check if the setting of the torque rate or the angle interval for torque rate calculation is appropriate. 4. Check if the speed of the previous stage is too high. 5. Check if the torque, angle, or torque rate of the previous stage is set too large. ■ The torque setting in Stage 3 cannot be greater than the tool maximum torque.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3C12 & NG4C12 Stage 3: exceeded the max. operation time

Trigger condition and cause	Exceeded the maximum operation time of Stage 3.
Checking method and corrective action	Check if the setting for the maximum operation time of Stage 3 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3C13 & NG4C13 Stage 3: lower than the min. operation time

Trigger condition and cause	Shorter than the minimum operation time of Stage 3.
Checking method and corrective action	Check if the setting for the minimum operation time of Stage 3 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3C14 & NG4C14 Stage 3: exceeded the switching torque

Trigger condition and cause	The ending torque of this stage exceeded the set maximum switching torque.
Checking method and corrective action	Check if the setting for the maximum switching torque is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3C15 & NG4C15 Stage 3: lower than the switching torque

Trigger condition and cause	The ending torque of this stage is lower than the set minimum switching torque.
Checking method and corrective action	Check if the setting for the minimum switching torque is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3C20 & NG4C20 Stage 3: exceeded tool max. current

Trigger condition and cause	After the final tightening torque value is converted into an electric current value, it exceeds the tool maximum current.
Checking method and corrective action	1. Use the controller current sensor and the tool torque sensor to measure the current values and then compare them to check if the screwdriver is degraded. 2. Check if the torque sensor is damaged. 3. Check if the speed of Stage 3 is too high. 4. Check if the torque of Stage 3 is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3C21 & NG4C21 Stage 3: lower than tool min. current

Trigger condition and cause	After the final tightening torque value is converted into an electric current value, it is lower than the tool minimum current.
Checking method and corrective action	1. Use the controller current sensor and the tool torque sensor to measure the current values and then compare them to check if the screwdriver is degraded. 2. Check if the torque sensor is damaged. 3. Check if the speed of Stage 3 is too high. 4. Check if the torque of Stage 3 is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3C22 & NG4C22 Stage 3: exceeded the max. angle

Trigger condition and cause	Exceeded the maximum angle of Stage 3.
Checking method and corrective action	Check if the setting for the maximum angle of Stage 3 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3C23 & NG4C23 Stage 3: lower than the min. angle

Trigger condition and cause	Lower than the minimum angle of Stage 3.
Checking method and corrective action	Check if the setting for the minimum angle of Stage 3 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3C24 & NG4C24 Stage 3: exceeded the max. torque

Trigger condition and cause	Exceeded the maximum torque of Stage 3.
Checking method and corrective action	1. Check if the setting for the maximum torque of Stage 3 is appropriate. 2. Check if the speed of the previous stage is too high. 3. Check if the torque, angle, or torque rate of the previous stage is set too large.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3C25 & NG4C25 Stage 3: lower than the min. torque

Trigger condition and cause	Lower than the minimum torque of Stage 3.
Checking method and corrective action	Check if the setting for the minimum torque of Stage 3 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3C26 & NG4C26 Stage 3: exceeded the max. clamp torque

Trigger condition and cause	Exceeded the maximum clamp torque of Stage 3.
Checking method and corrective action	Check if the setting for the maximum clamp torque of Stage 3 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3C27 & NG4C27 Stage 3: lower than the min. clamp torque

Trigger condition and cause	Lower than the minimum clamp torque of Stage 3.
Checking method and corrective action	Check if the setting for the minimum clamp torque of Stage 3 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3C28 & NG4C28 Stage 3: exceeded the max. clamp angle

Trigger condition and cause	Exceeded the maximum clamp angle of Stage 3.
Checking method and corrective action	Check if the setting for the maximum clamp angle of Stage 3 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3C29 & NG4C29 Stage 3: lower than the min. clamp angle

Trigger condition and cause	Lower than the minimum clamp angle of Stage 3.
Checking method and corrective action	Check if the setting for the minimum clamp angle of Stage 3 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3D10 & NG4D10 Stage 4: operation error

Trigger condition and cause	Condition: the controller returns an error during the tightening process. Cause: 1. The torque of this stage exceeded the tool maximum torque. 2. During the tightening process, the tool is in the Servo Off state.
Checking method and corrective action	Check if the torque of this stage is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3D11 & NG4D11 Stage 4: exceeded the tightening torque protection range

Trigger condition and cause	The torque of Stage 4 exceeded the protection range.
Checking method and corrective action	1. Check if the tool has reached a foreign object before this stage is complete. 2. Check if the speed of Stage 4 is too high. 3. Check if the setting of the torque rate or the angle interval for torque rate calculation is appropriate. 4. Check if the speed of the previous stage is too high. 5. Check if the torque, angle, or torque rate of the previous stage is set too large. ■ The torque setting in Stage 4 cannot be greater than the tool maximum torque.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3D12 & NG4D12 Stage 4: exceeded the max. operation time

Trigger condition and cause	Exceeded the maximum operation time of Stage 4.
Checking method and corrective action	Check if the setting for the maximum operation time of Stage 4 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3D13 & NG4D13 Stage 4: lower than the min. operation time

Trigger condition and cause	Shorter than the minimum operation time of Stage 4.
Checking method and corrective action	Check if the setting for the minimum operation time of Stage 4 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3D14 & NG4D14 Stage 4: exceeded the switching torque

Trigger condition and cause	The ending torque of this stage exceeded the set maximum switching torque.
Checking method and corrective action	Check if the setting for the maximum switching torque is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3D15 & NG4D15 Stage 4: lower than the switching torque

Trigger condition and cause	The ending torque of this stage is lower than the set minimum switching torque.
Checking method and corrective action	Check if the setting for the minimum switching torque is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3D20 & NG4D20 Stage 4: exceeded tool max. current

Trigger condition and cause	After the final tightening torque value is converted into an electric current value, it exceeds the tool maximum current.
Checking method and corrective action	1. Use the controller current sensor and the tool torque sensor to measure the current values and then compare them to check if the screwdriver is degraded. 2. Check if the torque sensor is damaged. 3. Check if the speed of Stage 4 is too high. 4. Check if the torque of Stage 4 is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3D21 & NG4D21 Stage 4: lower than tool min. current

Trigger condition and cause	After the final tightening torque value is converted into an electric current value, it is lower than the tool minimum current.
Checking method and corrective action	1. Use the controller current sensor and the tool torque sensor to measure the current values and then compare them to check if the screwdriver is degraded. 2. Check if the torque sensor is damaged. 3. Check if the speed of Stage 4 is too high. 4. Check if the torque of Stage 4 is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3D22 & NG4D22 Stage 4: exceeded the max. angle

Trigger condition and cause	Exceeded the maximum angle of Stage 4.
Checking method and corrective action	Check if the setting for the maximum angle of Stage 4 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3D23 & NG4D23 Stage 4: lower than the min. angle

Trigger condition and cause	Lower than the minimum angle of Stage 4.
Checking method and corrective action	Check if the setting for the minimum angle of Stage 4 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3D24 & NG4D24 Stage 4: exceeded the max. torque

Trigger condition and cause	Exceeded the maximum torque of Stage 4.
Checking method and corrective action	1. Check if the setting for the maximum torque of Stage 4 is appropriate. 2. Check if the speed of the previous stage is too high. 3. Check if the torque, angle, or torque rate of the previous stage is set too large.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3D25 & NG4D25 Stage 4: lower than the min. torque

Trigger condition and cause	Lower than the minimum torque of Stage 4.
Checking method and corrective action	Check if the setting for the minimum torque of Stage 4 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3D26 & NG4D26 Stage 4: exceeded the max. clamp torque

Trigger condition and cause	Exceeded the maximum clamp torque of Stage 4.
Checking method and corrective action	Check if the setting for the maximum clamp torque of Stage 4 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3D27 & NG4D27 Stage 4: lower than the min. clamp torque

Trigger condition and cause	Lower than the minimum clamp torque of Stage 4.
Checking method and corrective action	Check if the setting for the minimum clamp torque of Stage 4 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3D28 & NG4D28 Stage 4: exceeded the max. clamp angle

Trigger condition and cause	Exceeded the maximum clamp angle of Stage 4.
Checking method and corrective action	Check if the setting for the maximum clamp angle of Stage 4 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3D29 & NG4D29 Stage 4: lower than the min. clamp angle

Trigger condition and cause	Lower than the minimum clamp angle of Stage 4.
Checking method and corrective action	Check if the setting for the minimum clamp angle of Stage 4 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3E10 & NG4E10 Stage 5: operation error

Trigger condition and cause	Condition: the controller returns an error during the tightening process. Cause: 1. The torque of this stage exceeded the tool maximum torque. 2. During the tightening process, the tool is in the Servo Off state.
Checking method and corrective action	Check if the torque of this stage is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3E11 & NG4E11 Stage 5: exceeded the tightening torque protection range

Trigger condition and cause	The torque of Stage 5 exceeded the protection range.
Checking method and corrective action	1. Check if the tool has reached a foreign object before this stage is complete. 2. Check if the speed of Stage 5 is too high. 3. Check if the setting of the torque rate or the angle interval for torque rate calculation is appropriate. 4. Check if the speed of the previous stage is too high. 5. Check if the torque, angle, or torque rate of the previous stage is set too large. ■ The torque setting in Stage 5 cannot be greater than the tool maximum torque.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3E12 & NG4E12 Stage 5: exceeded the max. operation time

Trigger condition and cause	Exceeded the maximum operation time of Stage 5.
Checking method and corrective action	Check if the setting for the maximum operation time of Stage 5 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3E13 & NG4E13 Stage 5: lower than the min. operation time

Trigger condition and cause	Shorter than the minimum operation time of Stage 5.
Checking method and corrective action	Check if the setting for the minimum operation time of Stage 5 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3E14 & NG4E14 Stage 5: exceeded the switching torque

Trigger condition and cause	The ending torque of this stage exceeded the set maximum switching torque.
Checking method and corrective action	Check if the setting for the maximum switching torque is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3E15 & NG4E15 Stage 5: lower than the switching torque

Trigger condition and cause	The ending torque of this stage is lower than the set minimum switching torque.
Checking method and corrective action	Check if the setting for the minimum switching torque is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3E20 & NG4E20 Stage 5: exceeded tool max. current

Trigger condition and cause	After the final tightening torque value is converted into an electric current value, it exceeds the tool maximum current.
Checking method and corrective action	1. Use the controller current sensor and the tool torque sensor to measure the current values and then compare them to check if the screwdriver is degraded. 2. Check if the torque sensor is damaged. 3. Check if the speed of Stage 5 is too high. 4. Check if the torque of Stage 5 is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3E21 & NG4E21 Stage 5: lower than tool min. current

Trigger condition and cause	After the final tightening torque value is converted into an electric current value, it is lower than the tool minimum current.
Checking method and corrective action	1. Use the controller current sensor and the tool torque sensor to measure the current values and then compare them to check if the screwdriver is degraded. 2. Check if the torque sensor is damaged. 3. Check if the speed of Stage 5 is too high. 4. Check if the torque of Stage 5 is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3E22 & NG4E22 Stage 5: exceeded the max. angle

Trigger condition and cause	Exceeded the maximum angle of Stage 5.
Checking method and corrective action	Check if the setting for the maximum angle of Stage 5 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3E23 & NG4E23 Stage 5: lower than the min. angle

Trigger condition and cause	Lower than the minimum angle of Stage 5.
Checking method and corrective action	Check if the setting for the minimum angle of Stage 5 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3E24 & NG4E24 Stage 5: exceeded the max. torque

Trigger condition and cause	Exceeded the maximum torque of Stage 5.
Checking method and corrective action	1. Check if the setting for the maximum torque of Stage 5 is appropriate. 2. Check if the speed of the previous stage is too high. 3. Check if the torque, angle, or torque rate of the previous stage is set too large.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3E25 & NG4E25 Stage 5: lower than the min. torque

Trigger condition and cause	Lower than the minimum torque of Stage 5.
Checking method and corrective action	Check if the setting for the minimum torque of Stage 5 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3E26 & NG4E26 Stage 5: exceeded the max. clamp torque

Trigger condition and cause	Exceeded the maximum clamp torque of Stage 5.
Checking method and corrective action	Check if the setting for the maximum clamp torque of Stage 5 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3E27 & NG4E27 Stage 5: lower than the min. clamp torque

Trigger condition and cause	Lower than the minimum clamp torque of Stage 5.
Checking method and corrective action	Check if the setting for the minimum clamp torque of Stage 5 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3E28 & NG4E28 Stage 5: exceeded the max. clamp angle

Trigger condition and cause	Exceeded the maximum clamp angle of Stage 5.
Checking method and corrective action	Check if the setting for the maximum clamp angle of Stage 5 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3E29 & NG4E29 Stage 5: lower than the min. clamp angle

Trigger condition and cause	Lower than the minimum clamp angle of Stage 5.
Checking method and corrective action	Check if the setting for the minimum clamp angle of Stage 5 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3F10 & NG4F10 Stage 6: operation error

Trigger condition and cause	Condition: the controller returns an error during the tightening process. Cause: 1. The torque of this stage exceeded the tool maximum torque. 2. During the tightening process, the tool is in the Servo Off state.
Checking method and corrective action	Check if the torque of this stage is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3F11 & NG4F11 Stage 6: exceeded the tightening torque protection range

Trigger condition and cause	The torque of Stage 6 exceeded the protection range.
Checking method and corrective action	1. Check if the tool has reached a foreign object before this stage is complete. 2. Check if the speed of Stage 6 is too high. 3. Check if the setting of the torque rate or the angle interval for torque rate calculation is appropriate. 4. Check if the speed of the previous stage is too high. 5. Check if the torque, angle, or torque rate of the previous stage is set too large. ■ The torque setting in Stage 6 cannot be greater than the tool maximum torque
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3F12 & NG4F12 Stage 6: exceeded the max. operation time

Trigger condition and cause	Exceeded the maximum operation time of Stage 6.
Checking method and corrective action	Check if the setting for the maximum operation time of Stage 6 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3F13 & NG4F13 Stage 6: lower than the min. operation time

Trigger condition and cause	Shorter than the minimum operation time of Stage 6.
Checking method and corrective action	Check if the setting for the minimum operation time of Stage 6 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3F14 & NG4F14 Stage 6: exceeded the switching torque

Trigger condition and cause	The ending torque of this stage exceeded the set maximum switching torque.
Checking method and corrective action	Check if the setting for the maximum switching torque is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3F15 & NG4F15 Stage 6: lower than the switching torque	
Trigger condition and cause	The ending torque of this stage is lower than the set minimum switching torque.
Checking method and corrective action	Check if the setting for the minimum switching torque is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.
NG3F20 & NG4F20 Stage 6: exceeded tool max. current	
Trigger condition and cause	After the final tightening torque value is converted into an electric current value, it exceeds the tool maximum current.
Checking method and corrective action	1. Use the controller current sensor and the tool torque sensor to measure the current values and then compare them to check if the screwdriver is degraded. 2. Check if the torque sensor is damaged. 3. Check if the speed of Stage 6 is too high. 4. Check if the torque of Stage 6 is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.
NG3F21 & NG4F21 Stage 6: lower than tool min. current	
Trigger condition and cause	After the final tightening torque value is converted into an electric current value, it is lower than the tool minimum current.
Checking method and corrective action	1. Use the controller current sensor and the tool torque sensor to measure the current values and then compare them to check if the screwdriver is degraded. 2. Check if the torque sensor is damaged. 3. Check if the speed of Stage 6 is too high. 4. Check if the torque of Stage 6 is within the tool specification.
How to clear the alarm?	Loosen and then re-tighten the screw.
NG3F22 & NG4F22 Stage 6: exceeded the max. angle	
Trigger condition and cause	Exceeded the maximum angle of Stage 6.
Checking method and corrective action	Check if the setting for the maximum angle of Stage 6 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.
NG3F23 & NG4F23 Stage 6: lower than the min. angle	
Trigger condition and cause	Lower than the minimum angle of Stage 6.
Checking method and corrective action	Check if the setting for the minimum angle of Stage 6 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3F24 & NG4F24 Stage 6: exceeded the max. torque

Trigger condition and cause	Exceeded the maximum torque of Stage 6.
Checking method and corrective action	1. Check if the setting for the maximum torque of Stage 6 is appropriate. 2. Check if the speed of the previous stage is too high. 3. Check if the torque, angle, or torque rate of the previous stage is set too large.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3F25 & NG4F25 Stage 6: lower than the min. torque

Trigger condition and cause	Lower than the minimum torque of Stage 6.
Checking method and corrective action	Check if the setting for the minimum torque of Stage 6 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3F26 & NG4F26 Stage 6: exceeded the max. clamp torque

Trigger condition and cause	Exceeded the maximum clamp angle of Stage 6.
Checking method and corrective action	Check if the setting for the maximum clamp angle of Stage 6 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3F27 & NG4F27 Stage 6: lower than the min. clamp torque

Trigger condition and cause	Lower than the minimum clamp torque of Stage 6.
Checking method and corrective action	Check if the setting for the minimum clamp torque of Stage 6 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3F28 & NG4F28 Stage 6: exceeded the max. clamp angle

Trigger condition and cause	Exceeded the maximum clamp angle of Stage 6.
Checking method and corrective action	Check if the setting for the maximum clamp angle of Stage 6 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

NG3F29 & NG4F29 Stage 6: lower than the min. clamp angle

Trigger condition and cause	Lower than the minimum clamp angle of Stage 6.
Checking method and corrective action	Check if the setting for the minimum clamp angle of Stage 6 is appropriate.
How to clear the alarm?	Loosen and then re-tighten the screw.

12.2.3 Description of operation warnings

WN5001 & WN6001 Quantity not reached. String scanning prohibited.	
Trigger condition and cause	String setting is prohibited until the operation progress is complete.
Checking method and corrective action	1. Input the string after the settings for total screw quantity, parameter quantity, and screw quantity of the parameter are reached. 2. Input the string after progress reset.
How to clear the alarm?	After the total screw quantity is reached, set the string again.

WN5002 & WN6002 Send the tool back for service	
Trigger condition and cause	Condition: the total tightening count and loosening count exceeded the suggested count for maintenance. Cause: the tool service reminder is triggered.
Checking method and corrective action	Send your screwdriver back to Delta for maintenance.
How to clear the alarm?	Send your screwdriver back to Delta for maintenance.

WN5003 & WN6003 Parameters not set	
Trigger condition and cause	Condition: the parameter configuration is incomplete. Cause: 1. The tightening source is not configured. 2. The parameter configuration is not complete when the screwdriver starts to operate.
Checking method and corrective action	1. Check if the parameters of the tightening source are configured. 2. Make sure the Ready signal (DI) is on before you start tightening.
How to clear the alarm?	Restart the tightening operation.

WN5004 & WN6004 Unknown tool model and specification. Update the firmware.	
Trigger condition and cause	The controller cannot identify the specifications and model number of the screwdriver.
Checking method and corrective action	1. Check if the controller and the screwdriver are connected correctly. 2. Check if the controller supports the specifications of the connected screwdriver. 3. Check for and update the controller firmware.
How to clear the alarm?	1. Send your screwdriver back to Delta. 2. Update the controller firmware.

WN5005 & WN6005 Cannot perform tightening and loosening at the same time	
Trigger condition and cause	The controller receives both the Start Tightening signal and the Start Loosening signal at the same time.
Checking method and corrective action	1. Check if there is any incorrect operation of the tool. 2. Check if the timing for sending the Start Tightening signal is in conflict with that for sending the Start Loosening signal.
How to clear the alarm?	Check the performed operation and then restart the screwdriver.

WN5006 & WN6006 Parameter setting prohibited when tool is in operation

Trigger condition and cause	The tightening or loosening process starts before the parameter configuration is complete.
Checking method and corrective action	1. Make sure the parameter configuration is complete before you start tightening or loosening. 2. Avoid configuring any parameter during operation.
How to clear the alarm?	Reconfigure the parameter, and then restart the tightening or loosening.

WN5007 & WN6007 Tool final current is lower than the range. Check the tightening parameter.

Trigger condition and cause	After the final tightening torque value is converted into an electric current value, it is lower than the tool maximum current.
Checking method and corrective action	1. Use the controller current sensor and the tool torque sensor to measure the current values and then compare them to check if the screwdriver is degraded. 2. Check if the torque sensor is damaged. 3. Check if the tightening speed is too high.
How to clear the alarm?	1. Loosen and then re-tighten the screw. 2. Disable the "Final Current Detection" function under the Advanced Setting in the General Settings tab of the parameter editing page.

WN5008 & WN6008 Positioning arm is not configured

Trigger condition and cause	The positioning arm function in the Sequence page is enabled, but the positioning arm settings are not defined.
Checking method and corrective action	Connect the positioning arm to the controller and then complete the positioning arm settings in Controller > Peripheral Device > RS485_1 .
How to clear the alarm?	Check the performed operation and then restart the screwdriver.

WN5009 & WN6009 Tightening signal ends too early

Trigger condition and cause	Condition: during the tightening process, the tightening signal disappears. Cause: 1. During the tightening process, the screwdriver is withdrawn early. 2. Transmission of the tightening signal is poor. 3. The tightening signal is interfered by noise.
Checking method and corrective action	1. Maintain the tightening signal until the tightening is complete. 2. If the tool start condition is DI, check the tightening signal transmission is poor. 3. If the tool start condition is push start, check if there is poor contact in the push start device. 4. If the tool start condition is lever start, check if there is poor contact in the lever on the screwdriver. 5. Set the DI response filter time in the System Settings tab of the Controller page to avoid noise interference.
How to clear the alarm?	Loosen and then re-tighten the screw.

WN500A & WN600A Disconnection of positioning arm communication

Trigger condition and cause	Condition: the positioning arm coordinate values are not read. Cause: the positioning arm communication is disconnected.
Checking method and corrective action	1. Check if the positioning arm communication cable is loose or broken. 2. Check if the positioning arm communication is interfered by noise. 3. Make sure the +5V power input to the positioning arm is stable. 4. Check for and update the controller firmware.
How to clear the alarm?	Restart the tightening operation.

WN500B & WN600B Tool final current is lower than the range. Check the tightening parameter.

Trigger condition and cause	After the final tightening torque value is converted into an electric current value, it is lower than the tool maximum current.
Checking method and corrective action	Check if the tightening speed is too high.
How to clear the alarm?	1. Loosen and then re-tighten the screw. 2. Disable the "Final Current Detection" function under the Advanced Setting in the General Settings tab of the parameter editing page.

WN500C & WN600C Tool final current exceeds the range. Check the tool.

Trigger condition and cause	After the final tightening torque value is converted into an electric current value, it exceeds the tool maximum current.
Checking method and corrective action	1. Use the controller current sensor and the tool torque sensor to measure the current values and then compare them to check if the screwdriver is degraded. 2. Check if the torque sensor is damaged.
How to clear the alarm?	Send your screwdriver back to Delta for maintenance.

WN500D & WN600D Tool perpendicularity detection error

Trigger condition and cause	The tool perpendicularity error during operation exceeds the allowable range.
Checking method and corrective action	1. Check if the tool operation angle is correct. 2. Check if the tool perpendicularity detection is applicable to the operating condition.
How to clear the alarm?	1. Adjust the tool operation angle and then restart the tightening operation. 2. Disable the "Tool Perpendicularity Detection" function under the Advanced Setting in the General Settings tab of the parameter editing page.

WN500E & WN600E Disable the tool operation restriction in the DI/DO tab

Trigger condition and cause	The DI function in the DI/DO tab is set to "Prohibit Tool Operation", but no positioning arm is connected to the controller.
Checking method and corrective action	1. Use the Modbus communication function codes #406 and #407 to remove the restriction. 2. Write to the Modbus communication addresses 0x1F71 and 0x2039 to remove the restriction. 3. Disable the "Prohibit tool operation after each tighten or loosen" function in the System Setting tab of the Controller page. 4. Disable the "Prohibit tool operation after tightening fail, and restore operations by alarm clear" function in the System Setting tab of the Controller page. 5. Check if the positioning arm is enabled within the set position range.
How to clear the alarm?	Modify the setting and re-tighten the screw.

WN500F & WN600F Tool temperature is abnormal

Trigger condition and cause	Condition: the motor temperature exceeds the set threshold. Cause: 1. The temperature warning threshold is exceeded. 2. The temperature sensor circuit is in error.
Checking method and corrective action	1. Check if the temperature warning threshold is set too low. 2. Cycle power on the controller.
How to clear the alarm?	Modify the setting and re-tightening the screw.

WN5051 & WN6051 Exceeded the max. count for NOK tightening

Trigger condition and cause	Exceeded the allowable NOK count for tightening.
Checking method and corrective action	1. Identify the main cause of NOK tightening. If the failure is random, modify the "Max. count for NOK tightening" in the Advanced Setting of the Sources page. 2. Reset the screw progress.
How to clear the alarm?	Modify the setting and then re-tighten the screw.

WN5052 & WN6052 The scanner string is null or the string length is incorrect. Tightening prohibited.

Trigger condition and cause	The barcode string is null or the string length is incorrect, so tightening is prohibited.
Checking method and corrective action	1. Input the barcode string. 2. Disable the "Prohibit tool operation when the scanner string is null" function in the Advanced Setting of the Sources page.
How to clear the alarm?	Modify the setting and then re-tighten the screw.

WN5053 & WN6053 Screw quantity reached. Tightening prohibited.

Trigger condition and cause	The screw quantity to be tightened is reached, so tightening is prohibited.
Checking method and corrective action	1. Call a new tightening parameter or tightening sequence. 2. Reset the screw progress.
How to clear the alarm?	Modify the setting and then re-tighten the screw.

WN5054 & WN6054 Exceeded max. operation time

Trigger condition and cause	The tightening process exceeded the maximum operation time.
Checking method and corrective action	1. Modify the time set for "Max. Operation Time" in the Advanced Setting of the Sources page. 2. Reset the screw progress.
How to clear the alarm?	Modify the setting and then re-tighten the screw.

WN5055 & WN6055 Incorrect parameter setting. Tightening prohibited.

Trigger condition and cause	Condition: the parameter setting is incorrect, so tightening is prohibited. Cause: 1. The tightening source is not configured. 2. The tightening parameter or tightening sequence is deleted.
Checking method and corrective action	Go to the Sources page and modify the tightening parameter or tightening sequence.
How to clear the alarm?	Select another tightening parameter or modify the tightening sequence, and then re-tighten the screw.

WN5056 & WN6056 Tightening prohibited by remote communication or DI

Trigger condition and cause	1. Tightening operation is prohibited by remote communication. 2. Tightening operation is prohibited by DI.
Checking method and corrective action	1. Use the Modbus communication function code #406 to remove the restriction. 2. Check if the DI function is set to "Prohibit Tool Operation" and the contact is normally closed (NC).
How to clear the alarm?	Modify the restriction setting through communication or DI, and then re-tighten the screw.

WN5057 & WN6057 Tightening: parameter exceeded the tool specification. Check the torque or speed setting.

Trigger condition and cause	Condition: the parameter setting exceeded the tool specification, so tightening is prohibited. Cause: 1. The parameter setting exceeded the maximum specification of the tool. 2. The parameter setting configured through communication is not appropriate. 3. The parameter setting imported from the USB flash drive or the HMI disk is not appropriate.
Checking method and corrective action	Modify the content of the parameter.
How to clear the alarm?	Modify the parameter and then re-tighten the screw.

WN5058 & WN6058 Tightening: parameter does not match the tool. Recreate a tightening parameter.

Trigger condition and cause	The operating parameter does not match the tool, so tightening is prohibited.
Checking method and corrective action	1. Recreate a tightening parameter. 2. Use other screwdrivers of the same specification.
How to clear the alarm?	Create a new tightening parameter and then re-tighten the screw.

WN5059 & WN6059 Tightening operation prohibited by SDSofT

Trigger condition and cause	The SDSofT software has not completed reading the controller tightening data.
Checking method and corrective action	1. The software is disconnected. Restart the software. 2. Disable the "Export result file for each screw" function in the System Setting tab of the Controller page.
How to clear the alarm?	Create a new tightening parameter and then re-tighten the screw.

WN505A & WN605A The controller cannot operate this tool model (customized)

Trigger condition and cause	The controller cannot perform tightening operation with this tool model.
Checking method and corrective action	1. The parameters imported from the USB flash drive or the HMI disk cannot be used on this tool. 2. Use a compatible tool model for the controller.
How to clear the alarm?	Check the performed operations and then restart the screwdriver.

WN5081 & WN6081 Loosening prohibited after tightening OK

Trigger condition and cause	When the tightening result is OK, the loosening is prohibited.
Checking method and corrective action	Modify the Advanced Setting in the Sources page.
How to clear the alarm?	Modify the setting and then re-loosen the screw.

WN5082 & WN6082 Loosening prohibited after tightening NOK

Trigger condition and cause	When the tightening result is NOK, the loosening is prohibited.
Checking method and corrective action	Modify the Advanced Setting in the Sources page.
How to clear the alarm?	Modify the setting and then re-loosen the screw.

WN5083 & WN6083 Exceeded the max. count for NOK loosening

Trigger condition and cause	Exceeded the allowable NOK count for loosening.
Checking method and corrective action	1. Modify the Advanced Setting in the Sources page. 2. Reset the screw progress.
How to clear the alarm?	Modify the setting and then re-loosen the screw.

WN5084 & WN6084 Incorrect parameter setting. Loosening prohibited.

Trigger condition and cause	Condition: the parameter setting is incorrect, so loosening is prohibited. Cause: 1. The tightening source is not configured. 2. The tightening parameter or tightening sequence is deleted.
Checking method and corrective action	Modify the settings in the Sources page.
How to clear the alarm?	Modify the settings and then re-loosen the screw.

WN5085 & WN6085 Loosening prohibited by remote communication or DI

Trigger condition and cause	1. Loosening operation is prohibited by remote communication. 2. Loosening operation is prohibited by DI.
Checking method and corrective action	1. Use the Modbus communication function code #407 to remove the restriction. 2. Check if the DI function is set to "Prohibit Tool Operation" and the contact is normally closed (NC).
How to clear the alarm?	Modify the restriction setting through communication or DI, and then re-loosen the screw.

WN5086 & WN6086 Loosening: parameter exceeded the tool specification. Check the torque or speed setting.

Trigger condition and cause	Condition: the parameter setting exceeded the tool specification, so loosening is prohibited. Cause: 1. The parameter setting exceeded the maximum specification of the tool. 2. The parameter setting configured through communication is not appropriate. 3. The parameter setting imported from the USB flash drive or the HMI disk is not appropriate.
Checking method and corrective action	Modify the content of the parameter.
How to clear the alarm?	Modify the parameter and then re-loosen the screw.

WN5087 & WN6087 Tightening operation prohibited by SDSof

Trigger condition and cause	The SDSof software has not completed reading the controller tightening data.
Checking method and corrective action	1. The software is disconnected. Restart the software. 2. Disable the "Export result file for each screw" function in the System Setting tab of the Controller page.
How to clear the alarm?	Modify the parameter and then re-tighten the screw.

WN5088 & WN6088 The controller cannot operate this tool model (customized)

Trigger condition and cause	The controller cannot perform loosening operation with this tool model.
Checking method and corrective action	1. The parameters imported from the USB flash drive or the HMI disk cannot be used on this tool. 2. Use a compatible tool model for the controller.
How to clear the alarm?	Check the performed operations and then restart the screwdriver.

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12

Modbus Communication - Function Codes



This chapter introduces the handshake method and information of the Modbus TCP and Modbus RTU communication interfaces provided by the smart screwdriving controller. You can operate the controller remotely through these interfaces.

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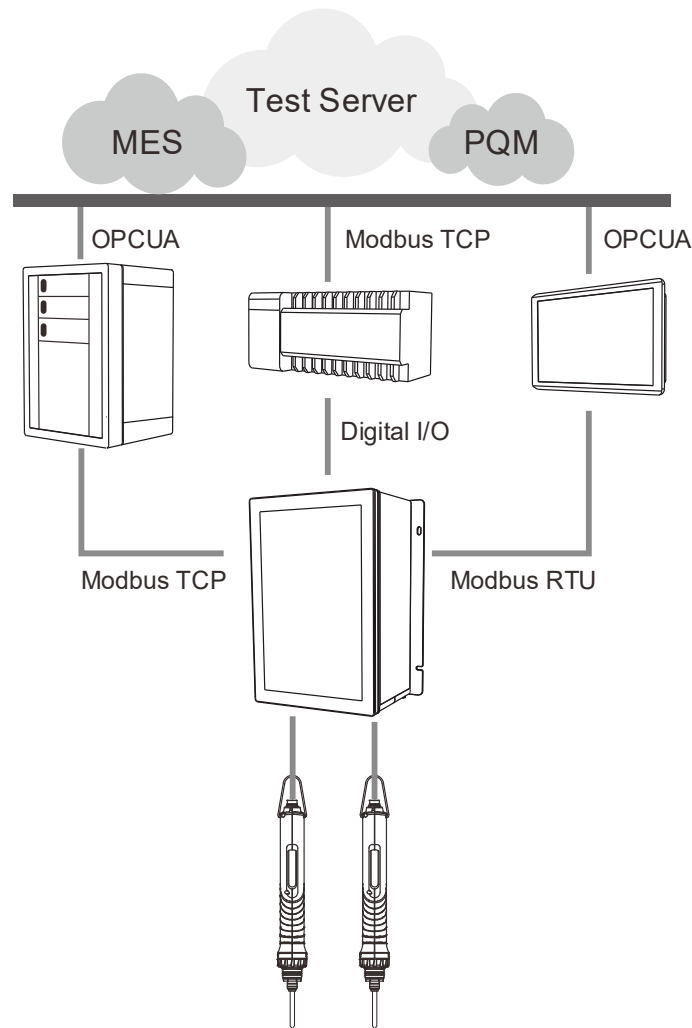
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A.1 System structure

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The Delta smart screwdriving system provides multiple communication protocols: Modbus TCP (Ethernet), Modbus RTU (RS-485) and TCP/IP. It allows peripheral IIoT equipment to obtain information from the smart screwdriving controller, and then collect data to construct host computer systems, such as MES and PQM. The connection between the IIoT equipment and the smart screwdriving system is not limited by the hardware platform. Therefore, any equipment that supports the protocol (such as PC, PLC, or HMI) is able to control the smart screwdriving system and obtain tightening data.

The Modbus handshake information in this chapter consists of:

1. Operational status of the smart screwdriving system.
2. Handshake data of the smart screwdriving system.

Modbus TCP Slave

The smart screwdriving system supports Modbus TCP communications with the default IP: 192.168.1.11; default port: 502. After successfully connected, the system starts detecting the keepalive time. You need to maintain periodic data handshaking within 30 seconds; if there is no data handshaking within 30 seconds, the system will be automatically disconnected.

Modbus RTU Slave

The smart screwdriving system supports Modbus RTU (Remote Terminal Unit) communication. The default station number is 1, transmission speed is 9600, data bit is 8-bit, parity bit is NONE, and stop bit is 2-bit.

The supported functions are: 03H (read multiple word data), 06H (write single word data), and 10H (write multiple word data).

Start	Minimum silent interval: 10 ms
Slave address	Communication address: 1 byte
Function	Function code: 1 byte
Data (n-1)	Data content: n word = 2n bytes n ≤ 10 (the maximum words of a single read/write is 10)
.....	
Data (0)	
CRC	Error check: 1 byte
End	Minimum silent interval: 10 ms

Note: in the RTU mode, there must be a minimum silent interval of 10 ms before and after transmission.

1. Function code 03H (read multiple word data)

The following example illustrates how the master issues a read command to slave no. 1.

The slave reads 3 consecutive word data starting from address 0x00CF. The slave then returns the content of 0x0096 read from address 0x00CF, 0x0001 read from address 0x00D0, and 0x0000 read from address 0x00D1. The maximum words of a single read is 10.

Command message (Master):

Slave address	01H
Function	03H
Starting data address	00H (high)
	CFH (low)
Number of data (in words)	00H
	03H
CRC Low	35H (low)
CRC High	F4H (high)

Response message (Slave):

Slave address	01H
Function	03H
Number of data (in bytes)	06H
Content of starting data address 00CFH	00H (high)
	96H (low)
Content of second data address 00D0H	00H (high)
	01H (low)
Content of third data address 00D1H	00H (high)
	00H (low)
CRC Low	38H (low)
CRC High	A8H (high)

2. Function code 06H (write single word data)

The following example illustrates how the master issues a write command to slave no. 1.

The slave writes the data 0x0064 to address 0x00C8, and then responds to the master once the writing is complete.

Command message (Master):

Slave address	01H
Function	06H
Starting data address	00H (high)
	C8H (low)
Data content	00H (high)
	64H (low)
CRC Low	09H (low)
CRC High	DFH (high)

Response message (Slave):

Slave address	01H
Function	06H
Starting data address	02H (high)
	00H (low)
Data content	00H (high)
	64H (low)
CRC Low	09H (low)
CRC High	DFH (high)

3. Function code 10H (write multiple word data)

The following example illustrates how the master issues a write command to slave no. 1.

The slave writes 7 word data (0x0096, 0x0000, 0x0000, 0x0001, 0x0000, 0x0000, and 0x0001) respectively to 7 addresses starting from 0x00C8. That is, the slave writes 0x0096 to address 0x00C8, 0x0000 to address 0x00C9, 0x0000 to address 0x00CA, 0x0001 to address 0x00CB, 0x0000 to address 0x00CC, 0x0000 to address 0x00CD, and 0x0001 to address 0x00CE.

The maximum words of a single write is 10. The slave responds to the master once the writing is complete.

Command message (Master):

Slave address	01H
Function	10H
Starting data address	00H (high)
	C8H (low)
Number of data (in words)	00H (high)
	07H (low)
Number of data (in bytes)	0EH
First data content	00H (high)
	96H (low)
Second data content	00H (high)
	00H (low)
Third data content	00H (high)
	00H (low)
Fourth data content	00H (high)
	01H (low)

Response message (Slave):

Slave address	01H
Function	10H
Starting data address	00H (high)
	C8H (low)
Number of data (in words)	00H (high)
	07H (low)
CRC Low	00H (low)
CRC High	35H (high)

Fifth data content	00H (high)
	00H (low)
Sixth data content	00H (high)
	00H (low)
Seventh data content	00H (high)
	01H (low)
CRC Low	A2H (low)
CRC High	A6H (high)

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4. CRC error check (RTU mode)

The following steps illustrate how to calculate the CRC value.

1. Load a 16-bit register with the content of FFFFH, and take it as the CRC register.
2. Perform Exclusive OR operation on the first byte of the command message with the low byte of the 16-bit CRC register, and then return the result to the CRC register.
3. Check the least significant bit (LSB) of the CRC register. If the LSB is 0, shift the CRC register value one bit to the right; if the LSB is 1, after shifting the CRC register value one bit to the right, perform Exclusive OR operation on the shifted value with A001H. Perform this step 8 times.
4. Repeat steps 2 to 3 until all bytes have been fully processed. The final content of the CRC register is the CRC value.

After the CRC value is calculated, first input the low byte of the CRC value, then the high byte into the command message.

For example, if the result of CRC calculation is 0xDF09, input 0x09 first and then 0xDF, as shown in the following table.

Slave address	01H
Function	06H
Starting data address	00H (high)
	C8H (low)
Number of data (in words)	00H (high)
	64H (low)
CRC Low	09H (low)
CRC High	DFH (high)

Example of CRC generating program

The following example uses the C language to generate the CRC value. This function requires two parameters:

```

unsigned char* data;
unsigned char length;
//This function returns the CRC value in unsigned integer.
unsigned int crc_chk(unsigned char* data, unsigned char length) {
    int j;
    unsigned int reg_crc=0xFFFF;

    while( length-- ) {
        reg_crc^= *data++;
        for (j=0; j<8; j++ ) {
            if( reg_crc & 0x01 ) { /*LSB(bit 0 ) = 1 */
                reg_crc = (reg_crc >> 1)^0xA001;
            } else {
                reg_crc = (reg_crc>>1);
            }
        }
    }
    return reg_crc;
}

```

Transmission process error report

The following is an example of the response message from the slave when an error occurs.

Slave address	01H
Error code	83H
Exception code	04H
CRC Low	40H
CRC High	F3H

Descriptions of exception codes:

Exception code (1 byte)	Description
01	Station number does not exist.
02	Unable to recognize the function code: not 03, 06, or 10.
03	Value error: the set value exceeded the maximum or minimum range.
04	Exceeded the maximum words: exceeded the maximum words (10) for a single access.
05	Number error: accessed number is set to 0.

Precautions: use either Modbus TCP Slave or Modbus RTU Slave; both of which share the same Modbus handshake addresses.

A.2 Operational status of the smart screwdriving system

Obtain real-time information of the smart screwdriving system from the operational status area.

The data refreshes every 0.3 seconds. Refer to the following Modbus address table.

Modbus (Hex)		Function code #21 (Dec)		Tool 1 status / Tool 2 status	R/W
Tool 1	Tool 2	Tool 1	Tool 2		
0	32	1	51	Tightening ID set for cur. switch method	R
1	33	2	52	PAR/SEQ set for cur. switch method	R
2	34	3	53	SEQ ID of current switching method	R
3	35	4	54	PAR ID of current switching method	R
4	36	5	55	Current target torque / Current start torque for detecting the yield point	R
5	37	6	56	Current target angle	R
6	38	7	57	Current parameter progress	R
7	39	8	58	Screw progress of current parameter (L)	R
8	3A	9	59	Screw progress of current parameter (H)	R
9	3B	10	60	Screw progress of current sequence (L)	R
A	3C	11	61	Screw progress of current sequence (H)	R
B	3D	12	62	Tightening OK count (L)	R
C	3E	13	63	Tightening OK count (H)	R
D	3F	14	64	Single screw tightening NOK count (L)	R
E	40	15	65	Single screw tightening NOK count (H)	R
F	41	16	66	Loosening OK count (L)	R
10	42	17	67	Loosening OK count (H)	R
11	43	18	68	Single screw loosening NOK count (L)	R
12	44	19	69	Single screw loosening NOK count (H)	R
13	45	20	70	Final stage max. torque	R
14	46	21	71	Final stage min. torque	R
15	47	22	72	Current torque unit ^{*1}	R
16	48	23	73	Tightening / Loosening in operation ^{*2}	R
17	49	24	74	Bit ID of current switching method	R
18	4A	25	75	Version of operational status ^{*3}	R
19	4B	26	76	Clear the flag ^{*4}	W
1A	4C	27	77	Total screw Qty. of current sequence (L)	R
1B	4D	28	78	Total screw Qty. of current sequence (H)	R
1C	4E	29	79	Parameter Qty. of current sequence	R
1D	4F	30	80	Screw Qty. of current parameter (L)	R
1E	50	31	81	Screw Qty. of current parameter (H)	R
1F	51	32	82	All screws of current parameter finished	R/W
20	52	33	83	Current parameter finished	R/W
21	53	34	84	Current screw finished	R/W
22	54	35	85	Setting parameters (Waiting...)	R
23	55	36	86	Final + Prevail torque	R
24	56	37	87	Total angle	R
25	57	38	88	Tightening angle	R
26	58	39	89	Tightening result (1: OK; 2: NOK; 5: Pass)	R/W
27	59	40	90	Loosening result (1: OK; 2: NOK)	R/W
28	5A	41	91	Curve creation finished	R/W

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Modbus (Hex)		Function code #21 (Dec)		Tool 1 status / Tool 2 status	R/W
Tool 1	Tool 2	Tool 1	Tool 2		
29	5B	42	92	Restrict tightening status	R
2A	5C	43	93	Parameter settings OK / NOK	R
2B	5D	44	94	Final torque	R
2C	5E	45	95	Prevail torque	R
2D	5F	46	96	Final current	R
2E	60	47	97	Cause to restrict tightening operation ^{*5}	R
2F	61	48	98	Cause to restrict loosening operation ^{*6}	R
30	62	49	99	Remaining operation time (L)	R
31	63	50	100	Remaining operation time (H)	R
1F40	2008	201	401	Current target torque (L) / Current start torque for detecting the yield point (L)	R
1F41	2009	202	402	Current target torque (H) / Current start torque for detecting the yield point (H)	R
1F42	200A	203	403	Final stage max. torque (L)	R
1F43	200B	204	404	Final stage max. torque (H)	R
1F44	200C	205	405	Final stage min. torque (L)	R
1F45	200D	206	406	Final stage min. torque (H)	R
1F46	200E	207	407	Final + Prevail torque (L)	R
1F47	200F	208	408	Final + Prevail torque (H)	R
1F48	2010	209	409	Final torque (L)	R
1F49	2011	210	410	Final torque (H)	R
1F4A	2012	211	411	Prevail torque (L)	R
1F4B	2013	212	412	Prevail torque (H)	R
1F4C	2014	213	413	Tightening OK count (L)	R
1F4D	2015	214	414	Tightening OK count (H)	R
1F4E	2016	215	415	Tightening NOK count (L)	R
1F4F	2017	216	416	Tightening NOK count (H)	R
1F50	2018	217	417	Tightening OK + NOK count (L)	R
1F51	2019	218	418	Tightening OK + NOK count (H)	R
1F52	201A	219	419	Ready	R
1F53	201B	220	420	Max. angle of final stage	R
1F54	201C	221	421	Min. angle of final stage	R
1F55	201D	222	422	Max. total angle	R
1F56	201E	223	423	Min. total angle	R
1F57	201F	224	424	Year	R
1F58	2020	225	425	Month	R
1F59	2021	226	426	Day	R
1F5A	2022	227	427	Hour	R
1F5B	2023	228	428	Minute	R
1F5C	2024	229	429	Second	R
1F5D	2025	230	430	Tightening result is 1: OK or 2: NOK; to be cleared in next operation	R
1F5E	2026	231	431	Loosening result is 1: OK or 2: NOK; to be cleared in next operation	R
1F5F	2027	232	432	Tightening result is 5: Pass; to be cleared in next operation	R
1F60	2028	233	433	Current position of positioning arm X axis (L)	R
1F61	2029	234	434	Current position of positioning arm X axis (H)	R

Modbus (Hex)		Function code #21 (Dec)		Tool 1 status / Tool 2 status	R/W
Tool 1	Tool 2	Tool 1	Tool 2		
1F62	202A	235	435	Current position of positioning arm Y axis (L)	R
1F63	202B	236	436	Current position of positioning arm Y axis (H)	R
1F64	202C	237	437	Current position of positioning arm Z axis (L)	R
1F65	202D	238	438	Current position of positioning arm Z axis (H)	R
1F66	202E	239	439	Target position of positioning arm X axis (L)	R
1F67	202F	240	440	Target position of positioning arm X axis (H)	R
1F68	2030	241	441	Target position of positioning arm Y axis (L)	R
1F69	2031	242	442	Target position of positioning arm Y axis (H)	R
1F6A	2032	243	443	Target position of positioning arm Z axis (L)	R
1F6B	2033	244	444	Target position of positioning arm Z axis (H)	R
1F6C	2034	245	445	Clamp torque (L)	R
1F6D	2035	246	446	Clamp torque (H)	R
1F6E	2036	247	447	Clamp angle	R
1F6F	2037	248	448	Snug torque (L)	R
1F70	2038	249	449	Snug torque (H)	R
1F71	2039	250	450	Tool operation prohibited ^{*7}	W
1F72	203A	251	451	Angle of Stage 1	R
1F73	203B	252	452	Angle of Stage 2	R
1F74	203C	253	453	Angle of Stage 3	R
1F75	203D	254	454	Angle of Stage 4	R
1F76	203E	255	455	Angle of Stage 5	R
1F77	203F	256	456	Angle of Stage 6	R
1F78	2040	257	457	Angle of first loosening stage	R
1F79	2041	258	458	Angle of second loosening stage	R
1F7A	2042	259	459	Torque of Stage 1 (L)	R
1F7B	2043	260	460	Torque of Stage 1 (H)	R
1F7C	2044	261	461	Torque of Stage 2 (L)	R
1F7D	2045	262	462	Torque of Stage 2 (H)	R
1F7E	2046	263	463	Torque of Stage 3 (L)	R
1F7F	2047	264	464	Torque of Stage 3 (H)	R
1F80	2048	265	465	Torque of Stage 4 (L)	R
1F81	2049	266	466	Torque of Stage 4 (H)	R
1F82	204A	267	467	Torque of Stage 5 (L)	R
1F83	204B	268	468	Torque of Stage 5 (H)	R
1F84	204C	269	469	Torque of Stage 6 (L)	R
1F85	204D	270	470	Torque of Stage 6 (H)	R
1F86	204E	271	471	Torque of first loosening stage (L)	R
1F87	204F	272	472	Torque of first loosening stage (H)	R
1F88	2050	273	473	Torque of second loosening stage (L)	R
1F89	2051	274	474	Torque of second loosening stage (H)	R
1F8A	2052	275	475	Time of Stage 1	R
1F8B	2053	276	476	Time of Stage 2	R
1F8C	2054	277	477	Time of Stage 3	R
1F8D	2055	278	478	Time of Stage 4	R
1F8E	2056	279	479	Time of Stage 5	R
1F8F	2057	280	480	Time of Stage 6	R

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Modbus (Hex)		Function code #21 (Dec)		Tool 1 status / Tool 2 status	R/W
Tool 1	Tool 2	Tool 1	Tool 2		
1F90	2058	281	481	Time of first loosening stage	R
1F91	2059	282	482	Time of second loosening stage	R
1F92	205A	283	483	Speed of Stage 1	R
1F93	205B	284	484	Speed of Stage 2	R
1F94	205C	285	485	Speed of Stage 3	R
1F95	205D	286	486	Speed of Stage 4	R
1F96	205E	287	487	Speed of Stage 5	R
1F97	205F	288	488	Speed of Stage 6	R
1F98	2060	289	489	Speed of first loosening stage	R
1F99	2061	290	490	Speed of second loosening stage	R
1F9A	2062	291	491	Total operation time	R
1F9B	2063	292	492	Switching parameters* ⁸	W
1F9C to 1FA2	2064 to 206B	293 to 299	493 to 499	Reserved	-
1FA3	206C	300	500	Current target yield point	R
1FA4 to 2007	206D to 20D0	301 to 400	501 to 600	Barcode string (ASCII code; 100 words)	R

Modbus (Hex)	Function code #21 (Dec)	Tool 1 / Tool 2 common status	R/W
64	101	Tool 1 AL / NG / WN code* ⁹	R
65	102	Tool 2 AL / NG / WN code* ⁹	R
66	103	DO status (Bits 1 - 8: Tool 1; Bits 9 - 16: Tool 2)	R
67	104	DI status (Bits 1 - 8: Tool 1; Bits 9 - 16: Tool 2)	R
68	105	DI control (Bits 1 - 8: Tool 1; Bits 9 - 16: Tool 2)	W
69	106	Current No. of error report entries	R
6A	107	Current No. of warning report entries	R
6B	108	Current No. of production report entries (L)	R
6C	109	Current No. of production report entries (H)	R
6D	110	Current No. of button report entries (L)	R
6E	111	Current No. of button report entries (H)	R
6F	112	Production report entry creation finished	R/W
70	113	Clear the common flag* ¹⁰	W
71	114	Reserved	-
72	115	All production report entries cleared	R/W
73	116	Start overwriting from production report ID 1	R/W
74	117	Start overwriting from error report ID 1	R/W
75	118	Start overwriting from warning report ID 1	R/W
76	119	Start overwriting from button report ID 1	R/W
77	120	Keepalive	R
78	121	Single-tool / Dual-tool alternation / Dual-tool synchronization	R
79	122	Switching method of Tool 1	R
7A	123	Switching method of Tool 2	R
7B	124	Operating curve count for Tool 1	R
7C	125	Operating curve count for Tool 2	R
7D	126	Tool No. under dual-tool alternation	R
7E	127	Notice of HMI screen refresh	R

Modbus (Hex)	Function code #21 (Dec)	Tool 1 / Tool 2 common status	R/W
7F	128	Flag of underlying communication error	R
80	129	Count of underlying communication errors	R
81	130	HMI initialization status ^{*11}	R
82	131	Current user ID	R
83	132	HMI main version	R
84	133	HMI sub-version	R
85	134	Second development platform version	R
86	135	Underlying version	R
87	136	MCU version of Tool 1	R
88	137	MCU version of Tool 2	R
89	138	Pop-up screen No.	R
8A to 95	139 to 150	Reserved	-

Note:

1. 0: N.m; 1: kgf.cm; 2: lbf.ft; 3: lbf.in; 4: ozf.ft; 5: ozf.in; 6: cNm
2. Bit 0: tightening in progress; Bit 1: loosening in progress
3. 0: does not support operational statuses 0x1FA4 to 0x2007 and 0x2008 to 0x20CF
1: supports operational statuses 0x1FA4 to 0x2007 and 0x2008 to 0x20CF
4. Bit 0: clear the "All screws of current parameter finished" flag (0x1F, 0x51)
Bit 1: clear the "Current parameter finished" flag (0x20, 0x52)
Bit 2: clear the "Current screw finished" flag (0x21, 0x53)
Bit 3: clear the "Curve creation finished" flag (0x28, 0x5A)
Bit 4: clear the "Tightening result" flag (0x26, 0x58) and "Loosening result" flag (0x27, 0x59)
5. Bit 0: max. count for single screw NOK tightening
Bit 1: prohibit tool operation when scanner string is null
Bit 2: prohibit tool operation when screw Qty. reached
Bit 3: max. operation time
Bit 4: incorrect parameter sequence
Bit 5: tightening operation restricted
Bit 6: parameter check protection
Bit 7: parameters do not match the tool
Bit 8: tightening operation is prohibited by SDsoft
Bit 9: the controller cannot operate this tool model
6. Bit 0: loosening prohibited after tightening OK
Bit 1: loosening prohibited after tightening NOK
Bit 2: max. count for single screw NOK loosening
Bit 3: incorrect parameter sequence
Bit 4: loosening operation restricted
Bit 5: parameter range error
Bit 6: tightening operation is prohibited by SDsoft
Bit 7: the controller cannot operate this tool model
7. 0: remove restriction for tightening; remove restriction for loosening
1001: restrict tightening operation; remove restriction for loosening
1002: remove restriction for tightening; restrict loosening operation
1003: restrict tightening operation; restrict loosening operation
8. When the system is in Single-tool operating mode and the switching method is Manual, you can set 1001 - 1500 to switch to the corresponding tightening parameters ID1 to ID500.
9. 0x1001 - 0x1999: Tool 1 device alarm (AL); 0x2000 - 0x2999: Tool 2 device alarm (AL)
0x3001 - 0x3999: Tool 1 tightening error (NG); 0x4000 - 0x4999: Tool 2 tightening error (NG)
0x5001 - 0x5999: Tool 1 operation warning (WN); 0x6000 - 0x6999: Tool 2 operation warning (WN)
10. Bit 0: clear the "Production report entry creation finished" flag (0x6F)
Bit 1: clear the "All production report entries cleared" flag (0x72)
Bit 2: clear the "Start overwriting from production report ID 1" flag (0x73)
Bit 3: clear the "Start overwriting from error report ID 1" flag (0x74)
Bit 4: clear the "Start overwriting from warning report ID 1" flag (0x75)
Bit 5: clear the "Start overwriting from button report ID 1" flag (0x76)
11. 0: underlying status is not yet initialized
10: underlying status initialization is complete; waiting for the HMI initialization
99: initialization of underlying status and HMI is complete

A.3 Handshake data of the smart screwdriving system

You can operate all functions of the smart screwdriving controller with the function code table through handshaking. All function codes are listed as follows:

Parameters			
Function code	Function name	Function code	Function name
#100	Write the parameter	#150	Read the parameter
#101	Write to set the tightening parameter through quick setting	#160	Read the created sets of tightening parameters
#110	Write to delete the parameter	-	-

Sequence			
Function code	Function name	Function code	Function name
#200	Write the sequence	#250	Read the sequence
#201	Write the navigator coordinates	#251	Read the navigator coordinates
#202	Write the navigator image codes	#252	Read the navigator image codes
#203	Write the coordinates of positioning arm	#253	Read the coordinates of positioning arm
#210	Write to delete the sequence	#260	Read the created sets of tightening sequences

Sources			
Function code	Function name	Function code	Function name
#300	Write the operating mode and switching method of source	#350	Read the operating mode and switching method of source
#301	Write the contents of a single source	#351	Read the contents of a single source
#302	Write to switch parameter under manual setting	-	-
#303	Write to switch sequence under manual setting	-	-
#310	Write to delete the source	-	-

Results			
Function code	Function name	Function code	Function name
#400	Write the switching method of source	#450	Read the switching method of source
#401	Write the barcode string	#451	Read the barcode string
#402	Write to clear all errors	#452	Read the scanner advanced settings
#403	Write to reset the operation progress	#453	Read the tightening status indicator
#404	Write to force execute the previous step	-	-
#405	Write to force execute the next step	-	-
#406	Write to restrict tightening operation	-	-
#407	Write to restrict loosening operation	-	-
#408	Write the scanner advanced settings	-	-
#409	Write to clear the single screw tightening NOK count	-	-

Results			
Function code	Function name	Function code	Function name
#410	Write to clear the single screw loosening NOK count	-	-
#411	Write to reset the operation time	-	-
#412	Write to reset the operational status	-	-

Controller			
Function code	Function name	Function code	Function name
#500	Write the request for permissions login	#550	Read the Ethernet settings
#501	Write the request for password change	#551	Read the page permissions
#502	Write the request for permissions logout	#552	Read the firmware version
#503	Write the page permissions	#553	Read the DI/DO functions
#504	Write the Ethernet settings	#554	Read the DI/DO conversion table
#505	Write the request for factory reset	#555	Read the default torque unit
#506	Write the buzzer sound pattern	#556	Read the default tool start condition
#507	Write the DI/DO functions	#557	Read NO / NC contact setting for a single DI/DO
#508	Write the DI/DO conversion table	#558	Read the default two-stage mode under self-defined torque control
#509	Write the default torque unit	#559	Read the display of upper limit for each stage of curves
#510	Write the default tool start condition	#560	Read the display of operation warning window
#511	Write NO / NC contact setting for a single DI/DO	#561	Read the file format of the exported result for each screw
#512	Write to export all system data	#562	Read the sampling rate for curves
#513	Write to import all system data	#563	Read the function of always monitoring the current
#514	Write the default two-stage mode under self-defined torque control	#564	Read the compensation for tool temperature rise
#515	Write the display of upper limit for each stage of curves	#565	Read the peripheral device settings
#516	Write the display of operation warning window	#566	Read the TCP data content of the result for each screw
#517	Write the file format of the exported result for each screw	#567	Read the monitor function when parameters do not match the tool
#518	Write the sampling rate for curves	#568	Read the default angle unit
#519	Write the function of always monitoring the current	#569	Read the volume of buzzer and operation screen buttons
#520	Write the compensation for tool temperature rise	#570	Read the display orientation of external screen
#521	Write the peripheral device settings	#571	Read the home screen
#522	Write the TCP data content of the result for each screw	#572	Read the RS-485 settings
#523	Write the monitor function when parameters do not match the tool	#573	Read the function of converting curve data to positive values

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Controller			
Function code	Function name	Function code	Function name
#524	Write the default angle unit	#574	Read the second stage default loosening speed
#525	Write the volume of buzzer and operation screen buttons	#575	Read the keyboard cursor blink in Results page
#526	Write the display orientation of external screen	#576	Read whether the torque rate curve is replaced by the speed curve
#527	Write the home screen	#577	Read whether tool operation is prohibited after each tightening or loosening
#528	Write the RS-485 settings	#578	Read the DI response filter time
#529	Write the function of converting curve data to positive values	#579	Read the MAC information
#530	Write the default loosening second stage speed	#580	Read the model number
#531	Write the keyboard cursor blink in Results page	#581	Read whether the torque curve coordinates are reset to zero
#532	Write to replace torque rate curve with speed curve	#582	Read whether parameter range check at tool power-on is enabled
#533	Write to prohibit tool operation after each tightening or loosening	#583	Read the DO timer
#534	Write the DI response filter time	#584	Read the popup error message type for tightening signal that ends too early
#535	Write the model number	#585	Read the buzzer sound pattern
#536	Write to reset the torque curve coordinates to zero	#586	Read the curve cutoff point recording mode
#537	Write to check parameter ranges when tool is powered on	#587	Read the notification setting for abnormal tool temperature
#538	Write the DO timer	#588	Read whether tool operation restoration after NOK tightening is limited to alarm reset
#539	Write the popup error message type for tightening signal that ends too early	-	-
#540	Write the curve cutoff point recording mode	-	-
#541	Write to enable the notification for abnormal tool temperature	-	-
#542	Write to limit tool operation restoration after NOK tightening to alarm reset	-	-

Tool			
Function code	Function name	Function code	Function name
#600	Write to activate the tool	#650	Read the tool information
#601	Write to enable service reminder	#651	Read the lever start level
#602	Write the lever start level	#652	Read the push start level
#603	Write the push start level	#653	Read the work light brightness
#604	Write the work light brightness	#655	Read the LED light settings
#606	Write the LED light settings	#656	Read the tool calibration setting
#607	Write to calibrate the tool	#657	Read the tool firmware version
#608	Write the service reminder threshold	#658	Read the status of service reminder
-	-	#659	Read the status of tool activation
-	-	#660	Read the service reminder threshold
-	-	#661	Read the tool calibration reports

Reports			
Function code	Function name	Function code	Function name
#700	Clear the production report entries	#750	Find and read the production report entries
#701	Clear the error and warning report entries	#751	Find and read curves
#702	Clear the production report files	#752	Find and read the error report entries
-	-	#753	Find and read the warning report entries
-	-	#754	Find and read the button report entries
-	-	#760	Read the sorted production reports

Others			
Function code	Function name	Function code	Function name
#21	Write to sort operational status into sequential data sets	#1449	Write to specify the screw progress

A

A.3.1 Parameters operations

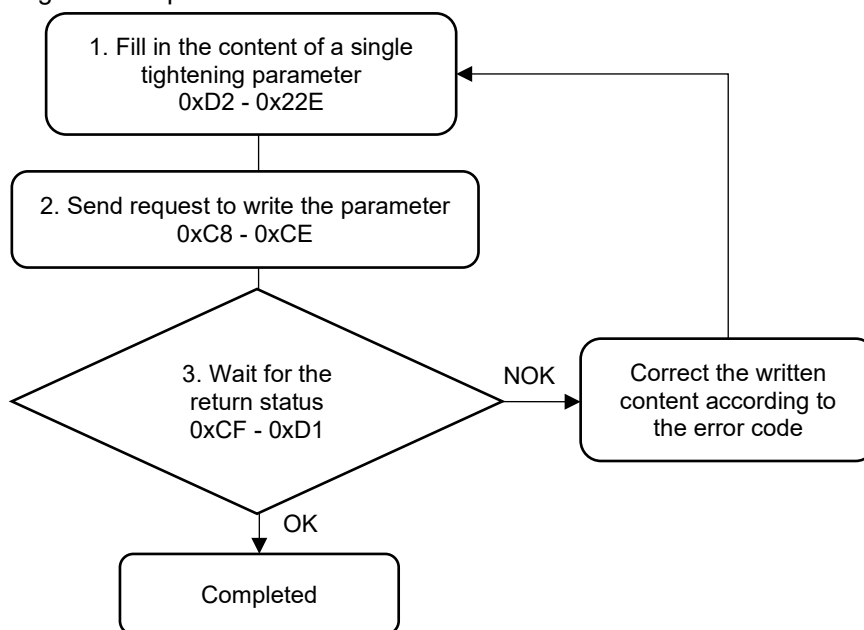
#100 Write the parameter

Content description:

Use function code #100 to set the tightening parameter.

- A. Add a tightening parameter.
- B. Modify an existing tightening parameter. First, read the parameter with function code #150. Then, modify and write the parameter with function code #100.

Handshake signal description:



1. Fill in the tightening parameter content to 0xD2 - 0x22E.

■ Version number: 0

Modbus (Hex)	Function	Description
D2 to E5	Parameter title	ASCII code (20 words)
E6	Min. tightening angle	Unit: degree (< 32767 degrees)
E7	Hold time switch of the final stage	Whether to stay in the Servo On state during the final stage (used with the Pause time setting)
E8	The prevail torque to be linked	The parameter ID to be linked
E9	Max. tightening time	Unit: 0.1 second (< 3276.7 seconds)
EA	Max. loosening time	Unit: 0.1 second (< 3276.7 seconds)
EB	Max. tightening angle	Unit: degree (< 32767 degrees)
EC	Reserved	—
ED	Delay before tightening starts	Unit: 0.01 second (< 327.67 seconds)
EE	Delay before loosening starts	Unit: 0.01 second (< 327.67 seconds)

Modbus (Hex)	Function	Description
EF	Start torque for switching the curve sampling rate / Start torque for cam-out detection	Unit: 0.001 N.m
F0	Start torque rate for snug angle calculation	Unit: 0.0001 N.m/degree
F1	Snug point angle correction	Unit: 0.1 degree
F2	Final current detection	0: Off; 1: On
F3	Delay before outputting the tightening result to feeder	Unit: 0.1 second
F4	Tool precision compensation	-10.0% to +10.0% Unit: 0.1%
F5	Angle delay for torque rate reached	Unit: 0.1 degree
F6	Trigger torque for recording cam-out angle	Unit: 0.001 N.m
F7	Upper limit for cam-out angle	Unit: degree
F8	Tolerance for cam-out	0 - 50
F9	Reserved	—

Self-defined strategy						Function	Description
Stage 1 (Hex)	Stage 2 (Hex)	Stage 3 (Hex)	Stage 4 (Hex)	Stage 5 (Hex)	Stage 6 (Hex)		
FA	12C	15E	190	1C2	1F4	Control method	0: angle; 1: torque; 2: torque rate; 3: clamp torque; 4: clamp angle; 5: yield point
FB	12D	15F	191	1C3	1F5	Tightening direction	0: CW; 1: CCW
FC	12E	160	192	1C4	1F6	Rotation speed	Unit: rpm
FD	12F	161	193	1C5	1F7	Target torque	Unit: 0.001 N.m
FE	130	162	194	1C6	1F8	Target angle	Unit: degree
FF	131	163	195	1C7	1F9	Target torque rate	Unit: 0.0001 N.m/degree
100	132	164	196	1C8	1FA	Angle interval for torque rate calculation	Unit: 0.1 degree
101	133	165	197	1C9	1FB	Acceleration time	Unit: ms
102	134	166	198	1CA	1FC	Max. angle	Unit: degree
103	135	167	199	1CB	1FD	Min. angle	Unit: degree
104	136	168	19A	1CC	1FE	Max. torque	Unit: 0.001 N.m
105	137	169	19B	1CD	1FF	Min. torque	Unit: 0.001 N.m
106	138	16A	19C	1CE	200	Max. operation time	Unit: 0.01 second
107	139	16B	19D	1CF	201	Min. operation time	Unit: 0.01 second
108	13A	16C	19E	1D0	202	Prevail torque On / Off	0: Off; 1: On
109	13B	16D	19F	1D1	203	Angle range for prevail torque calculation	0 to 100%
10A	13C	16E	1A0	1D2	204	Pause time	Unit: ms
10B	13D	16F	1A1	1D3	205	Max. clamp torque	Unit: 0.001 N.m
10C	13E	170	1A2	1D4	206	Min. clamp torque	Unit: 0.001 N.m
10D	13F	171	1A3	1D5	207	Max. clamp angle	Unit: degree
10E	140	172	1A4	1D6	208	Min. clamp angle	Unit: degree
10F	141	173	1A5	1D7	209	Torque of 1st stage	Unit: 0.001 N.m

Self-defined strategy						Function	Description
Stage 1 (Hex)	Stage 2 (Hex)	Stage 3 (Hex)	Stage 4 (Hex)	Stage 5 (Hex)	Stage 6 (Hex)		
110	142	174	1A6	1D8	20A	Pause time after 1st stage	Unit: ms
111	143	175	1A7	1D9	20B	Acc. time of 2nd stage	Unit: ms
112	144	176	1A8	1DA	20C	Speed of final stage	Unit: rpm
113	145	177	1A9	1DB	20D	Deceleration time	Unit: ms
114	146	178	1AA	1DC	20E	Advanced settings (L)	Bit 0: synchronization through DI7/DO7 signal Bit 1: wait for another tool to complete before continuing operation Bit 2: ergo stop (0: Off; 1: On) Bit 3: release torque during pause (0: Off; 1: On) Bit 4: not included in the total angle calculation (0: Off; 1: On)
115	147	179	1AB	1DD	20F	Advanced settings (H)	—
116	148	17A	1AC	1DE	210	Max. switching torque	Unit: 0.001 N.m
117	149	17B	1AD	1DF	211	Min. switching torque	Unit: 0.001 N.m
118	14A	17C	1AE	1E0	212	Target yield point	0 - 100%
119	14B	17D	1AF	1E1	213	Start torque for yield point detection	Unit: 0.001 N.m
11A to 12B	14C to 15D	17E to 18F	1B0 to 1C1	1E2 to 1F3	214 to 225	Reserved	—
226						1st stage loosening angle	Unit: degree
227						1st stage loosening speed	Unit: rpm
228						2nd stage loosening angle	Unit: degree
229						2nd stage loosening speed	Unit: rpm
22A						Loosening direction	0: CW; 1: CCW
22B						Detect loosening torque	Unit: 0.001 N.m
22C						Production report saving	0: Off; 1: On
22D						1st stage loosening acceleration time	Unit: ms
22E						2nd stage loosening acceleration time	Unit: ms
22F						Homing mode	0: Off; 1: On
230 to 243						Reserved	—
244						Bias instability of gyroscope	0 - 360 (0: Off)
245						Gyroscope error	—
246						Reserved	—
247						Start torque for tightening angle calculation	Unit: 0.001 N.m

■ Version number: 1

Modbus (Hex)	Function	Description
D2 to E5	Parameter title	ASCII code (20 words)
E6	Min. tightening angle	Unit: degree (< 32767 degrees)
E7	Hold time switch of the final stage	Whether to stay in the Servo On state during the final stage (used with the Pause time setting)
E8	The prevail torque to be linked	The parameter ID to be linked
E9	Max. tightening time	Unit: 0.1 second (< 3276.7 seconds)
EA	Max. loosening time	Unit: 0.1 second (< 3276.7 seconds)
EB	Max. tightening angle	Unit: degree (< 32767 degrees)
EC	Reserved	–
ED	Delay before tightening starts	Unit: 0.01 second (< 327.67 seconds)
EE	Delay before loosening starts	Unit: 0.01 second (< 327.67 seconds)
EF	Torque unit	0: N.m; 1: kgf.cm; 2: lb.ft; 3: lb.in; 4: ozf.ft; 5: ozf.in; 6: cN.m
F0	Angle interval for torque rate calculation	Unit: 0.1 degree
F1	Snug point angle correction	Unit: 0.1 degree
F2	Final current detection	0: Off; 1: On
F3	Delay before outputting the tightening result to feeder	Unit: 0.1 second
F4	Tool accuracy compensation	-100 to +100 Unit: 0.1%
F5	Detection level for target torque rate reached	Unit: 0.1 degree
F6	Start torque for switching the curve sampling rate (L) / Start torque for cam-out detection (L)	Unit: 0.001 torque unit
F7	Start torque for switching the curve sampling rate (H) / Start torque for cam-out detection (H)	Unit: 0.001 torque unit
F8	Start torque rate for snug angle calculation (L)	Unit: 0.0001 torque unit/degree
F9	Start torque rate for snug angle calculation (H)	Unit: 0.0001 torque unit/degree
FA	Trigger torque for recording cam-out angle (L)	Unit: 0.001 torque unit
FB	Trigger torque for recording cam-out angle (H)	Unit: 0.001 torque unit
FC	Upper limit for cam-out angle	Unit: degree
FD	Tolerance for cam-out	0 - 50
FE	Bias instability of gyroscope	0 - 360 (0: Off)
FF	Gyroscope error	–
100	Gyroscope advanced settings	–
101	Start torque for tightening angle calculation (L)	Unit: 0.001 torque unit

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Modbus (Hex)	Function	Description
102	Start torque for tightening angle calculation (H)	Unit: 0.001 torque unit
103 to 10D	Reserved	—
10E	1st stage loosening angle	Unit: degree
10F	1st stage loosening speed	Unit: rpm
110	2nd stage loosening angle	Unit: degree
111	2nd stage loosening speed	Unit: rpm
112	Loosening direction	0: CW; 1: CCW
113	Detect loosening torque (L)	Unit: 0.001 torque unit
114	Detect loosening torque (H)	Unit: 0.001 torque unit
115	Production report saving	0: Off; 1: On
116	1st stage loosening acceleration time	Unit: ms
117	2nd stage loosening acceleration time	Unit: ms
118	Homing mode	0: Off; 1: On
119 to 121	Reserved	—

Self-defined strategy						Function	Description
Stage 1 (Hex)	Stage 2 (Hex)	Stage 3 (Hex)	Stage 4 (Hex)	Stage 5 (Hex)	Stage 6 (Hex)		
122	154	186	1B8	1EA	21C	Control method	0: angle; 1: torque; 2: torque rate; 3: clamp torque; 4: clamp angle; 5: yield point
123	155	187	1B9	1EB	21D	Tightening direction	0: CW; 1: CCW
124	156	188	1BA	1EC	21E	Rotation speed	Unit: rpm
125	157	189	1BB	1ED	21F	Target torque (L)	Unit: 0.001 torque unit
126	158	18A	1BC	1EE	220	Target torque (H)	Unit: 0.001 torque unit
127	159	18B	1BD	1EF	221	Target angle	Unit: degree
128	15A	18C	1BE	1F0	222	Target torque rate (L)	Unit: 0.0001 torque unit/degree
129	15B	18D	1BF	1F1	223	Target torque rate (H)	Unit: 0.0001 torque unit/degree
12A	15C	18E	1C0	1F2	224	Acceleration time	Unit: ms
12B	15D	18F	1C1	1F3	225	Max. angle	Unit: degree
12C	15E	190	1C2	1F4	226	Min. angle	Unit: degree
12D	15F	191	1C3	1F5	227	Max. torque (L)	Unit: 0.001 torque unit
12E	160	192	1C4	1F6	228	Max. torque (H)	Unit: 0.001 torque unit
12F	161	193	1C5	1F7	229	Min. torque (L)	Unit: 0.001 torque unit
130	162	194	1C6	1F8	22A	Min. torque (H)	Unit: 0.001 torque unit
131	163	195	1C7	1F9	22B	Max. operation time	Unit: 0.01 second
132	164	196	1C8	1FA	22C	Min. operation time	Unit: 0.01 second
133	165	197	1C9	1FB	22D	Prevail torque On / Off	0: Off; 1: On
134	166	198	1CA	1FC	22E	Angle range for prevail torque calculation	0 - 100%
135	167	199	1CB	1FD	22F	Pause time	Unit: ms
136	168	19A	1CC	1FE	230	Max. clamp torque (L)	Unit: 0.001 torque unit
137	169	19B	1CD	1FF	231	Max. clamp torque (H)	Unit: 0.001 torque unit

Self-defined strategy						Function	Description
Stage 1 (Hex)	Stage 2 (Hex)	Stage 3 (Hex)	Stage 4 (Hex)	Stage 5 (Hex)	Stage 6 (Hex)		
138	16A	19C	1CE	200	232	Min. clamp torque (L)	Unit: 0.001 torque unit
139	16B	19D	1CF	201	233	Min. clamp torque (H)	Unit: 0.001 torque unit
13A	16C	19E	1D0	202	234	Max. clamp angle	Unit: degree
13B	16D	19F	1D1	203	235	Min. clamp angle	Unit: degree
13C	16E	1A0	1D2	204	236	Torque of 1st stage (L)	Unit: 0.001 torque unit
13D	16F	1A1	1D3	205	237	Torque of 1st stage (H)	Unit: 0.001 torque unit
13E	170	1A2	1D4	206	238	Pause time after 1st stage	Unit: ms
13F	171	1A3	1D5	207	239	Acc. time of 2nd stage	Unit: ms
140	172	1A4	1D6	208	23A	Speed of final stage	Unit: rpm
141	173	1A5	1D7	209	23B	Deceleration time	Unit: ms
142	174	1A6	1D8	20A	23C	Advanced settings (L)	Bit 0: synchronization through DI7/DO7 signal Bit 1: wait for another tool to complete before continuing operation Bit 2: ergo stop (0: Off; 1: On) Bit 3: release torque during pause (0: Off; 1: On) Bit 4: not included in the total angle calculation (0: Off; 1: On)
143	175	1A7	1D9	20B	23D	Advanced settings (H)	–
144	176	1A8	1DA	20C	23E	Max. switching torque (L)	Unit: 0.001 torque unit
145	177	1A9	1DB	20D	23F	Max. switching torque (H)	Unit: 0.001 torque unit
146	178	1AA	1DC	20E	240	Min. switching torque (L)	Unit: 0.001 torque unit
147	179	1AB	1DD	20F	241	Min. switching torque (H)	Unit: 0.001 torque unit
148	17A	1AC	1DE	210	242	Target yield point	0 - 100%
149	17B	1AD	1DF	211	243	Start torque for yield point detection (L)	Unit: 0.001 torque unit
14A	17C	1AE	1E0	212	244	Start torque for yield point detection (H)	Unit: 0.001 torque unit
14B to 153	17D to 185	1AF to 1B7	1E1 to 1E9	213 to 21B	245 to 24D	Reserved	–

2. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	100
C9	Version number	0 or 1
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB	Parameter ID	1 to 500
CC	Auto identifying the strategy	0: automatically identify the strategy 1: fixed in Self-defined strategy
CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

3. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	100
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

Code	Error description
1	Start: Max. torque < Min. torque
2	Start: Max. angle < Min. angle
3	Rundown: Max. torque < Min. torque
4	Rundown: Max. angle < Min. angle
5	Pre-tightening: Max. torque < Min. torque
6	Pre-tightening: Max. angle < Min. angle
7	Tightening: Max. torque < Min. torque
8	Tightening: Max. angle < Min. angle
9	Check if the stage sequence of the tightening settings is correct
10	No parameters set for this stage
11	Rundown: The set torque < Min. torque
12	Rundown: The set torque > Max. torque
13	Pre-tightening: The set torque < Min. torque
14	Pre-tightening: The set torque > Max. torque
15	Tightening: The set torque < Min. torque
16	Tightening: The set torque > Max. torque
17	The set angle > 32767
18	The set torque > Tool spec. torque The set min. torque > Tool spec. torque
19	The set torque > Tool max. torque
20	The set speed > Tool max. speed
21	Rundown: Torque cannot be 0
22	Pre-tightening: Torque cannot be 0
23	Tightening: Torque cannot be 0
24	The set rundown torque > The set pre-tightening torque
25	The set rundown torque > The set tightening torque
26	The set pre-tightening torque > The set tightening torque
27	The set clamp torque > Max. clamp torque
28	Max. clamp torque < Min. clamp torque
29	Max. clamp torque > Tool max. torque
30	Tightening: The set speed < Min. speed
32	General: The set timeout > 32767
33	General: The set max. total angle > 32767
34	Start: Min. operation time > Max. operation time
35	Rundown: Min. operation time > Max. operation time
36	Pre-tightening: Min. operation time > Max. operation time
37	Tightening: Min. operation time > Max. operation time

Code	Error description
38	General: The set tightening timeout > Max. operation time of the stage
39	Linked prevail torque cannot be 0
40	Cannot set multiple torque rates
41	Max. clamp angle < Min. clamp angle
42	Cannot set multiple clamp torques or angles
43	Max. switching torque < Min. switching torque
44	Max. switching torque > Tool max. torque
47	Tightening speed > Pre-tightening speed
48	Start speed > Rundown speed
49	Pre-tightening speed > Rundown speed
50	Tightening speed > Rundown speed
51	Start: Max. torque > Tightening torque
52	Start: Min. torque > Tightening torque
53	Rundown: Max. torque > Tightening torque
54	Rundown: Min. torque > Tightening torque
55	Rundown: Max. torque > Tightening torque
56	Rundown: Min. torque > Tightening torque
57	Pre-tightening: Max. torque > Tightening torque
58	Pre-tightening: Min. torque > Tightening torque
59	Parameter not applicable to this model
60	Setting multiple yield points is prohibited
61	The set torque < Min. torque
62	The set torque > Max. torque
63	Max. torque < Min. torque
64	Max. angle < Min. angle
65	Min. operation time > Max. operation time
100	Parameter ID exceeds the range of 1 to 500
101	Tool 1 / Tool 2 must be 0 or 1
102	Title string is null
103	Title already exists

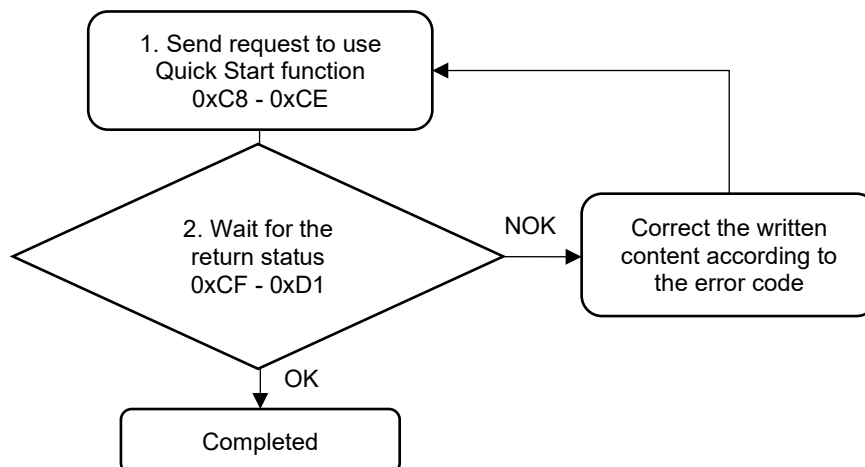
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#101 Write to set the tightening parameter through quick setting

Content description:

Use function code #101 to set the tightening parameter with the Quick Start function.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	101
C9	Version number	0
CA	Specifying tool and parameter	0: Tool 1; 1: Tool 2 1001 - 1500: Tool 1 parameter IDs 2001 - 2500: Tool 2 parameter IDs
CB	Target torque unit	0: N.m; 1: kgf.cm; 2: lbf.ft; 3: lbf.in; 4: ozf.ft; 5: ozf.in; 6: cNm
CC	Target torque (L)	Target torque x 1000
CD	Target torque (H)	
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	101
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

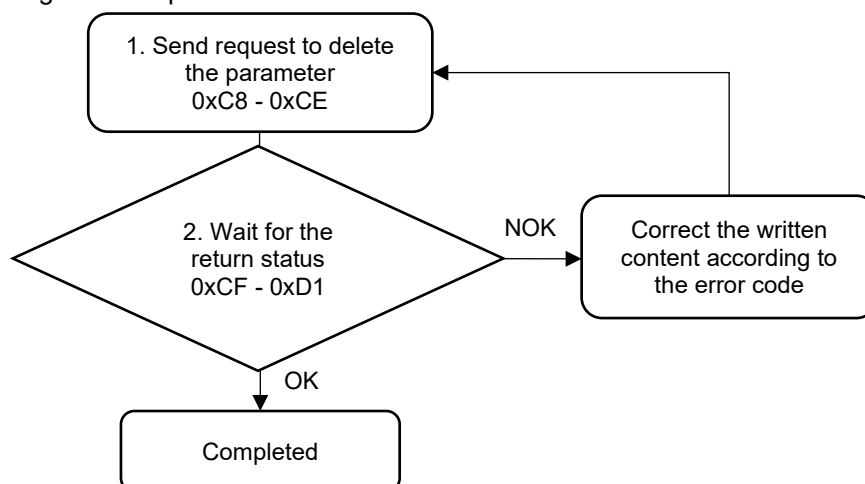
Code	Error description	Description
1	Tool 1 / Tool 2	Exceeds the range of 0 - 1, 1001 - 1500, or 2001 - 2500
2	Target torque unit	Exceeds the range of 0 - 6
3	Target torque	Exceeds the tool max. torque or lower than the tool min. torque

#110 Write to delete the parameter

Content description:

Use function code #110 to delete the tightening parameter.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	110
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB	The ID to be deleted	0: all the parameters 1 to 500: a single parameter
CC, CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	110
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	The ID to be deleted	Exceeds the range of 0 to 500

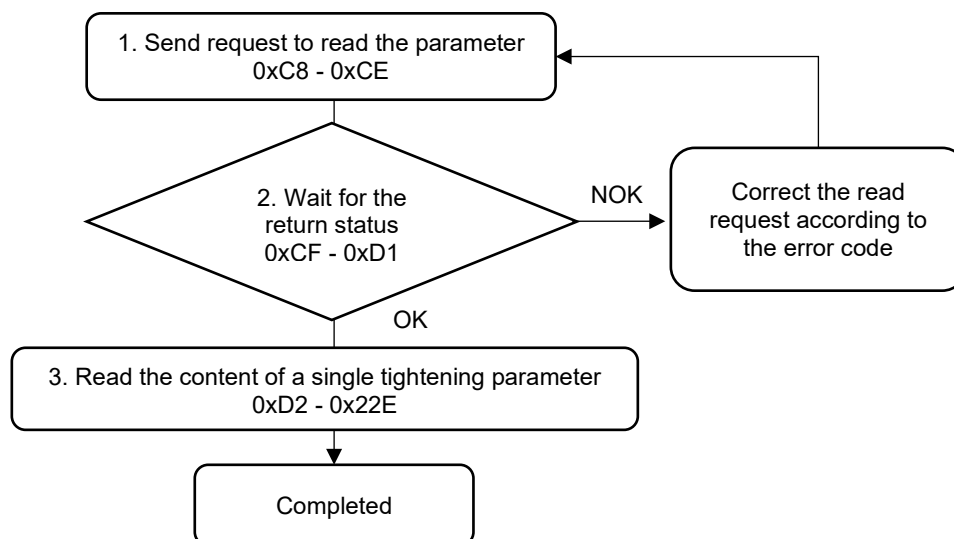
#150 Read the parameter

Content description:

Use function code #150 to read the tightening parameter.

- A. Obtain the content of an existing tightening parameter.
B. Check whether the tightening parameter has been set.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	150
C9	Version number	0 or 1
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB	Parameter ID	1 to 500
CC, CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	150
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

Code	Error description	Description
1	Parameter ID	Exceeds the range of 1 to 500
2	Tool 1 / Tool 2	Must be 0 or 1
3	Parameter content	Not set

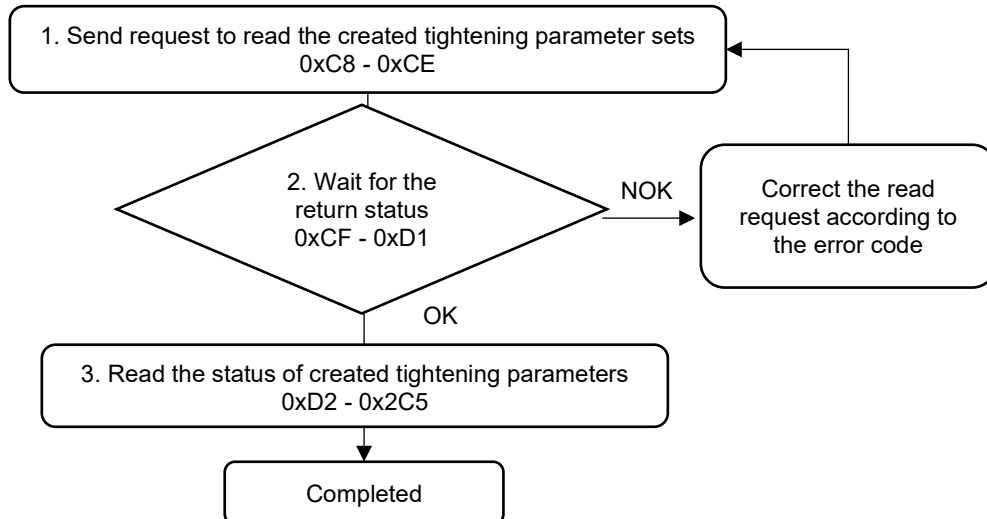
3. Read the parameter content in 0xD2 - 0x22E. Refer to function code #100 for the function list.

#160 Read the created sets of tightening parameters

Content description:

Use function code #160 to read the successfully created tightening parameters from the 1st to 500th sets.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	160
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	160
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

3. Read the tightening parameter creation status in 0xD2 - 0x2C5.

Code	Write / Read request	Read
D2	1 st set of tightening parameter	0: not yet created; > 0: created
D3	2 nd set of tightening parameter	0: not yet created; > 0: created
...	...	...
2C4	499 th set of tightening parameter	0: not yet created; > 0: created
2C5	500 th set of tightening parameter	0: not yet created; > 0: created

A

A.3.2 Sequence operations

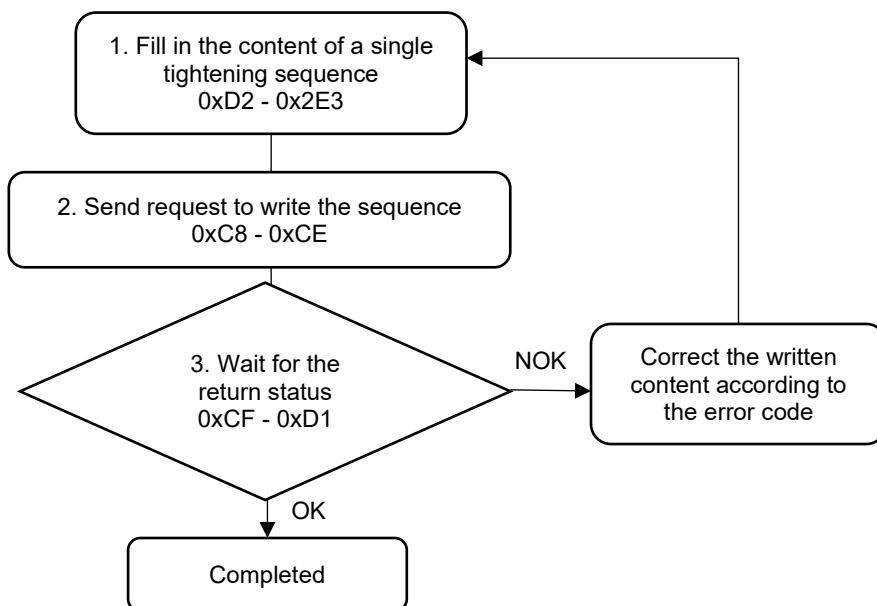
#200 Write the sequence

Content description:

Use function code #200 to set the tightening sequence.

- A. Add a tightening sequence.
- B. Modify an existing tightening sequence. First, read the sequence with function code #250. Then, modify and write the sequence with function code #200.

Handshake signal description:



1. Fill in the tightening sequence content to 0xD2 - 0x2E3.

In each sequence, you can configure up to 100 sets of parameters.

Modbus (Hex)	Function	Description
D2 to E5	Sequence title	ASCII code (20 words)
E6	General / Navigator mode	0: General mode 1: Navigator mode (must be used with #201 and #202)
E7	Positioning arm function	0: Off 1: On (must be used with #203)
E8 to EF	Reserved	—
F0	Tool ID for Set 1	0: Tool 1 1: Tool 2
F1	Tool ID for Set 2	
F2	Tool ID for Set 3	
...	Tool ID for Set N	
151	Tool ID for Set 98	
152	Tool ID for Set 99	
153	Tool ID for Set 100	
154	Parameter ID for Set 1	1 to 500
155	Parameter ID for Set 2	1 to 500
156	Parameter ID for Set 3	1 to 500
...	Parameter ID for Set N	1 to 500

Modbus (Hex)	Function	Description
1B5	Parameter ID for Set 98	1 to 500
1B6	Parameter ID for Set 99	1 to 500
1B7	Parameter ID for Set 100	1 to 500
1B8	Screw quantity for Set 1 (L)	1 to 999999
1B9	Screw quantity for Set 1 (H)	
1BA	Screw quantity for Set 2 (L)	1 to 999999
1BB	Screw quantity for Set 2 (H)	
1BC	Screw quantity for Set 3 (L)	1 to 999999
1BD	Screw quantity for Set 3 (H)	
...	Screw quantity for Set N (L)	1 to 999999
...	Screw quantity for Set N (H)	
27A	Screw quantity for Set 98 (L)	1 to 999999
27B	Screw quantity for Set 98 (H)	
27C	Screw quantity for Set 99 (L)	1 to 999999
27D	Screw quantity for Set 99 (H)	
27E	Screw quantity for Set 100 (L)	1 to 999999
27F	Screw quantity for Set 100 (H)	
280	Bit ID for Set 1	0 to 255
281	Bit ID for Set 2	0 to 255
282	Bit ID for Set 3	0 to 255
...	Bit ID for Set N	0 to 255
2E1	Bit ID for Set 98	0 to 255
2E2	Bit ID for Set 99	0 to 255
2E3	Bit ID for Set 100	0 to 255

2. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	200
C9	Version number	0
CA	Sequence ID	1 to 500
CB to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

3. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	200
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

Code	Error description	Description
1	Sequence ID	Exceeds the range of 1 to 500
2	The quantity is 0	Cannot be set as 0
3	Total screw quantity > 999999	Must not exceed 999999
100	Title string is null	–
101	Title already exists	–

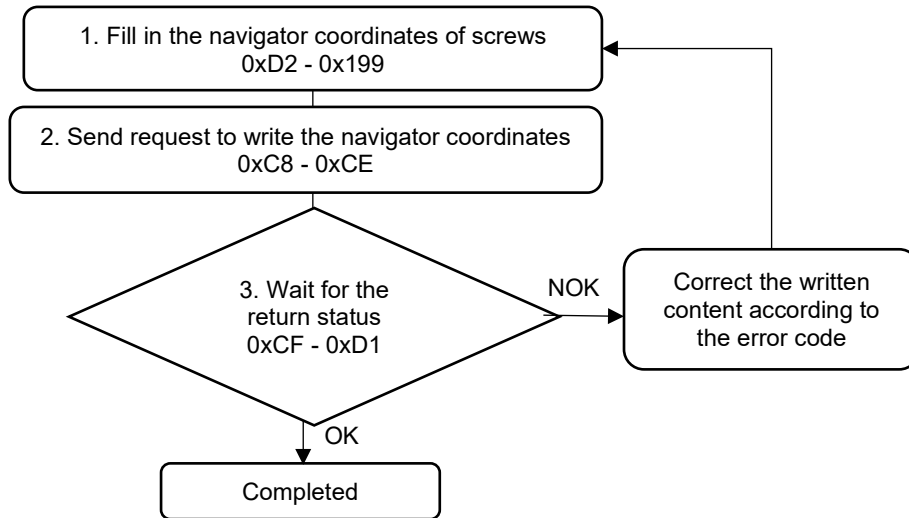
#201 Write the navigator coordinates

Content description:

Use function code #201 to set the navigator coordinates of screws.

Note: first, set the coordinates of screws with function code #201. Then, set the navigator image codes corresponding to screws with function code #202.

Handshake signal description:



1. Fill in the navigator coordinates of up to 100 screws to 0xD2 - 0x199.

Modbus (Hex)	Function	Description
D2	Screw ID 1 - X axis position	0 - 740
D3	Screw ID 1 - Y axis position	0 - 460
D4	Screw ID 2 - X axis position	0 - 740
D5	Screw ID 2 - Y axis position	0 - 460
...	Screw ID N - X axis position	0 - 740
...	Screw ID N - Y axis position	0 - 460
196	Screw ID 99 - X axis position	0 - 740
197	Screw ID 99 - Y axis position	0 - 460
198	Screw ID 100 - X axis position	0 - 740
199	Screw ID 100 - Y axis position	0 - 460

2. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	201
C9	Version number	0
CA	Sequence ID	1 to 500
CB to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

3. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	201
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

Code	Error description	Description
1	Sequence ID	Exceeds the range of 1 to 500
2	Axis position exceeds the range	X axis: exceeds the range of 0 to 740 Y axis: exceeds the range of 0 to 460

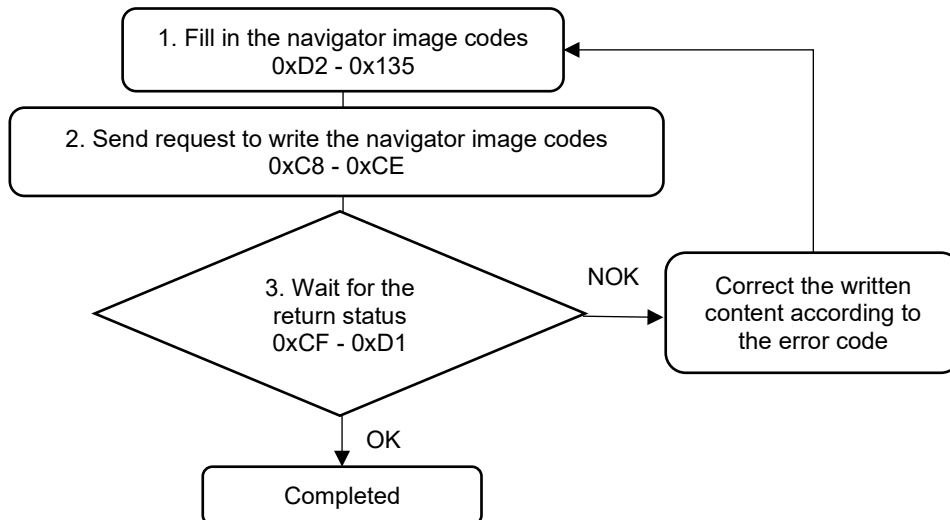
#202 Write the navigator image codes

Content description:

Use function code #202 to set the navigator image codes corresponding to the screws.

Note: first, set the coordinates of screws with function code #201. Then, set the navigator image codes corresponding to the screws with function code #202.

Handshake signal description:



1. Fill in the navigator image codes of up to 100 screws to 0xD2 - 0x135.

Modbus (Hex)	Function	Description
D2	Screw ID 1 - displayed image	0: no display 1: image A
D3	Screw ID 2 - displayed image	2: image B 3: image C 4: image D
D4	Screw ID 3 - displayed image	5: image E 6: image F 7: image G 8: image H
D5	Screw ID 4 - displayed image	9: image I 10: image J 11: image K
D6	Screw ID 5 - displayed image	12: image L 13: image M 14: image N
...	Screw ID N - displayed image	15: image O 16: image P 17: image Q
131	Screw ID 96 - displayed image	18: image R 19: image S 20: image T
132	Screw ID 97 - displayed image	21: image U

Modbus (Hex)	Function	Description
133	Screw ID 98 - displayed image	22: image V 23: image W 24: image X
134	Screw ID 99 - displayed image	25: image Y 26: image Z 27: image AA
135	Screw ID 100 - displayed image	28: image AB 29: image AC 30: image AD

2. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	202
C9	Version number	0
CA	Sequence ID	1 to 500
CB to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

3. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	202
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

Code	Error description	Description
1	Sequence ID	Exceeds the range of 1 to 500
2	Image code exceeds the range	Exceeds the range of 0 to 30

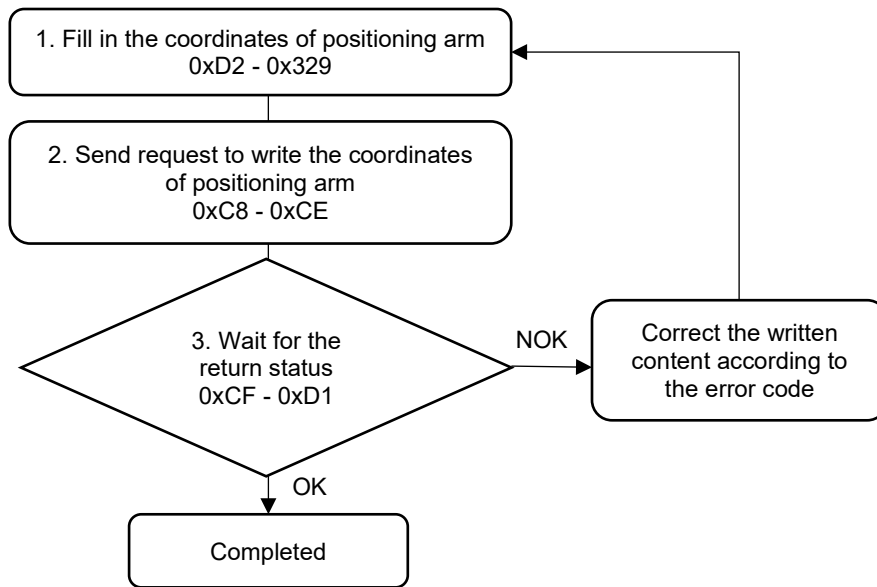
#203 Write the coordinates of positioning arm

Content description:

Use function code #203 to set the coordinates of positioning arm.

Note: first, set the tightening sequence with function code #200. Then, set the coordinates of positioning arm with function code #203.

Handshake signal description:



1. Fill in the coordinates of positioning arm for up to 100 screws to 0xD2 - 0x329.

Modbus (Hex)	Function	Description
D2	Screw ID 1 - positioning arm X axis position (L)	Unit: mm
D3	Screw ID 1 - positioning arm X axis position (H)	
D4	Screw ID 1 - positioning arm Y axis position (L)	Unit: mm
D5	Screw ID 1 - positioning arm Y axis position (H)	
D6	Screw ID 1 - positioning arm Z axis position (L)	Unit: mm
D7	Screw ID 1 - positioning arm Z axis position (H)	
D8	Screw ID 2 - positioning arm X axis position (L)	Unit: mm
D9	Screw ID 2 - positioning arm X axis position (H)	
DA	Screw ID 2 - positioning arm Y axis position (L)	Unit: mm
DB	Screw ID 2 - positioning arm Y axis position (H)	
DC	Screw ID 2 - positioning arm Z axis position (L)	Unit: mm
DD	Screw ID 2 - positioning arm Z axis position (H)	
DE	Screw ID 3 - positioning arm X axis position (L)	Unit: mm
DF	Screw ID 3 - positioning arm X axis position (H)	
E0	Screw ID 3 - positioning arm Y axis position (L)	Unit: mm
E1	Screw ID 3 - positioning arm Y axis position (H)	
E2	Screw ID 3 - positioning arm Z axis position (L)	Unit: mm
E3	Screw ID 3 - positioning arm Z axis position (H)	

Modbus (Hex)	Function	Description
...	Screw ID N - positioning arm X axis position (L)	Unit: mm
...	Screw ID N - positioning arm X axis position (H)	
...	Screw ID N - positioning arm Y axis position (L)	Unit: mm
...	Screw ID N - positioning arm Y axis position (H)	
...	Screw ID N - positioning arm Z axis position (L)	Unit: mm
...	Screw ID N - positioning arm Z axis position (H)	
318	Screw ID 98 - positioning arm X axis position (L)	Unit: mm
319	Screw ID 98 - positioning arm X axis position (H)	
31A	Screw ID 98 - positioning arm Y axis position (L)	Unit: mm
31B	Screw ID 98 - positioning arm Y axis position (H)	
31C	Screw ID 98 - positioning arm Z axis position (L)	Unit: mm
31D	Screw ID 98 - positioning arm Z axis position (H)	
31E	Screw ID 99 - positioning arm X axis position (L)	Unit: mm
31F	Screw ID 99 - positioning arm X axis position (H)	
320	Screw ID 99 - positioning arm Y axis position (L)	Unit: mm
321	Screw ID 99 - positioning arm Y axis position (H)	
322	Screw ID 99 - positioning arm Z axis position (L)	Unit: mm
323	Screw ID 99 - positioning arm Z axis position (H)	
324	Screw ID 100 - positioning arm X axis position (L)	Unit: mm
325	Screw ID 100 - positioning arm X axis position (H)	
326	Screw ID 100 - positioning arm Y axis position (L)	Unit: mm
327	Screw ID 100 - positioning arm Y axis position (H)	
328	Screw ID 100 - positioning arm Z axis position (L)	Unit: mm
329	Screw ID 100 - positioning arm Z axis position (H)	

2. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	203
C9	Version number	0
CA	Sequence ID	1 to 500
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

3. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	203
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

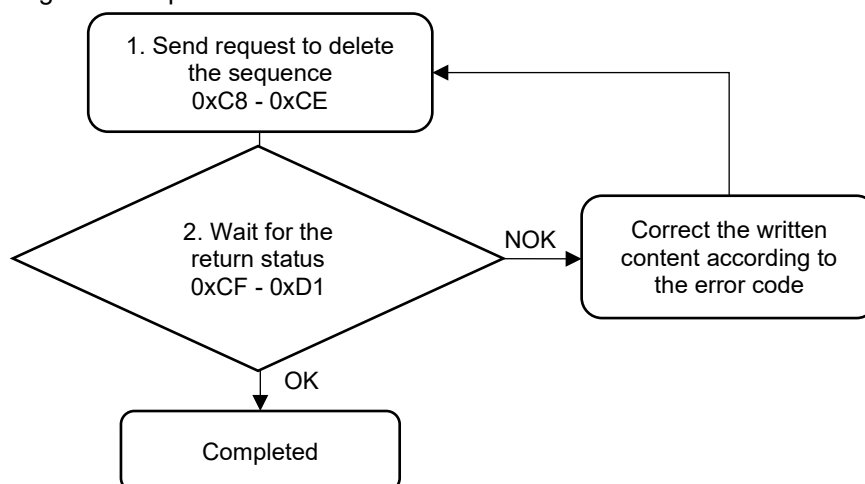
Code	Error description	Description
1	Sequence ID	Exceeds the range of 1 to 500

#210 Write to delete the sequence

Content description:

Use function code #210 to delete the tightening sequence.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	210
C9	Version number	0
CA	The ID to be deleted	0: all the sequences 1 to 500: a single sequence
CB to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	210
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	The ID to be deleted	Exceeds the range of 0 to 500

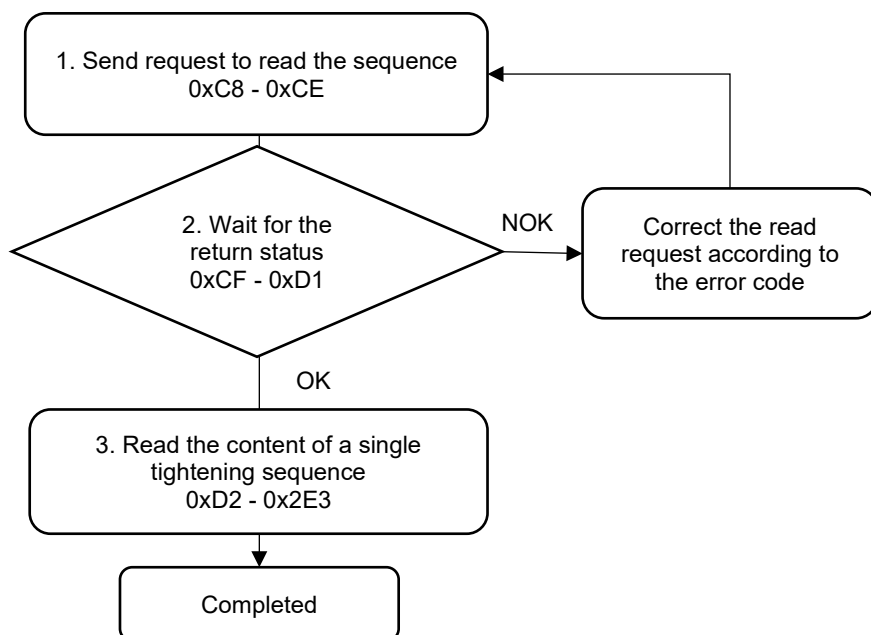
#250 Read the sequence

Content description:

Use function code #250 to read the tightening sequence.

- A. Obtain the content of an existing tightening sequence.
- B. Check whether the tightening sequence has been set.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	250
C9	Version number	0
CA	Sequence ID	1 to 500
CB to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	250
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Sequence ID	Exceeds the range of 1 to 500

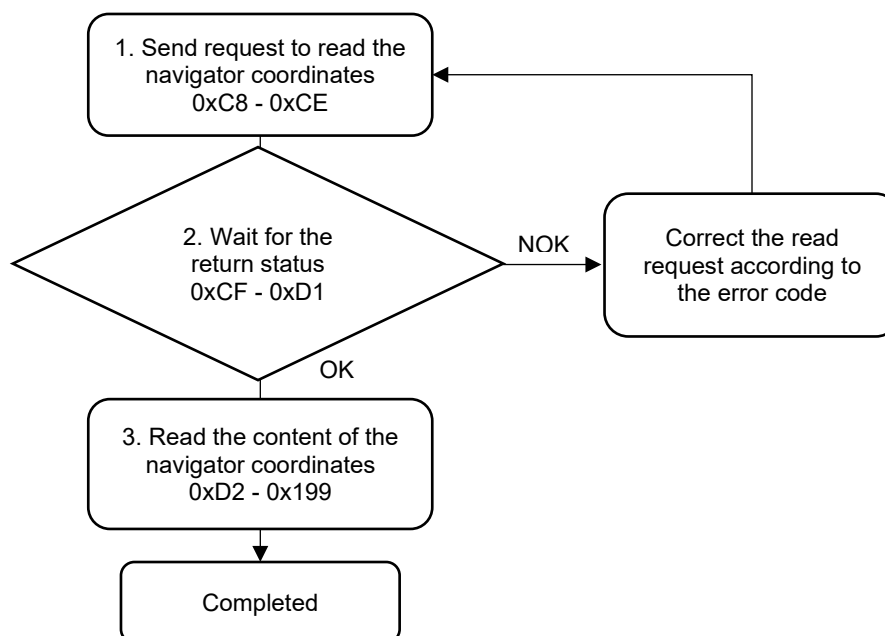
3. Read the content of the tightening sequence in 0xD2 - 0x2E3. Refer to function code #200 for the function list.

#251 Read the navigator coordinates

Content description:

Use function code #251 to read the navigator coordinates of screws.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	251
C9	Version number	0
CA	Sequence ID	1 to 500
CB to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	251
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Sequence ID	Exceeds the range of 1 to 500

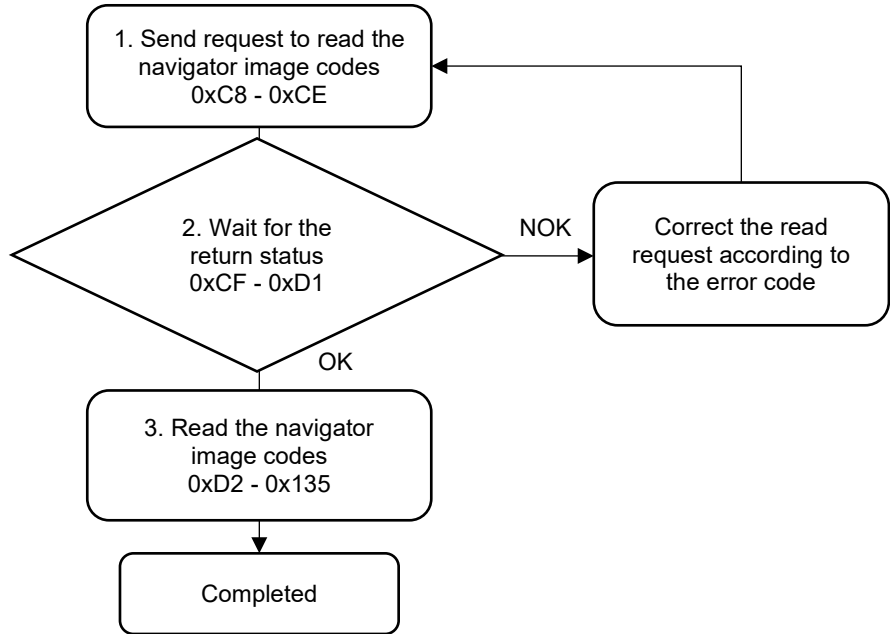
3. Read the content of the navigator coordinates in 0xD2 - 0x199. Refer to function code #201 for the function list.

#252 Read the navigator image codes

Content description:

Use function code #252 to read the navigator image codes corresponding to the screws.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	252
C9	Version number	0
CA	Sequence ID	1 to 500
CB to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	252
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Sequence ID	Exceeds the range of 1 to 500

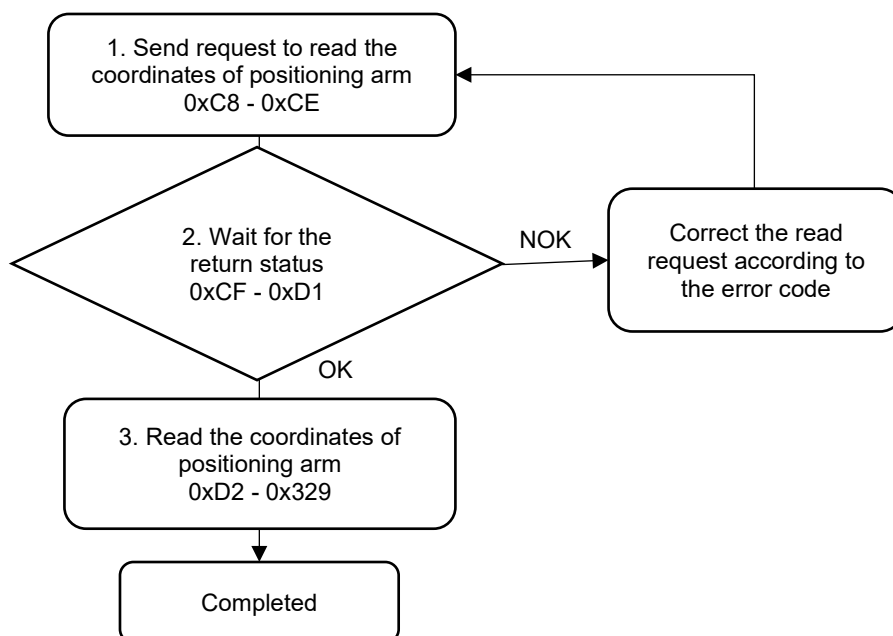
3. Read the content of the navigator image codes corresponding to the screws in 0xD2 - 0x135. Refer to function code #202 for the function list.

#253 Read the coordinates of positioning arm

Content description:

Use function code #253 to read the coordinates of positioning arm.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	253
C9	Version number	0
CA	Sequence ID	1 to 500
CB to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	253
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Sequence ID	Exceeds the range of 1 to 500

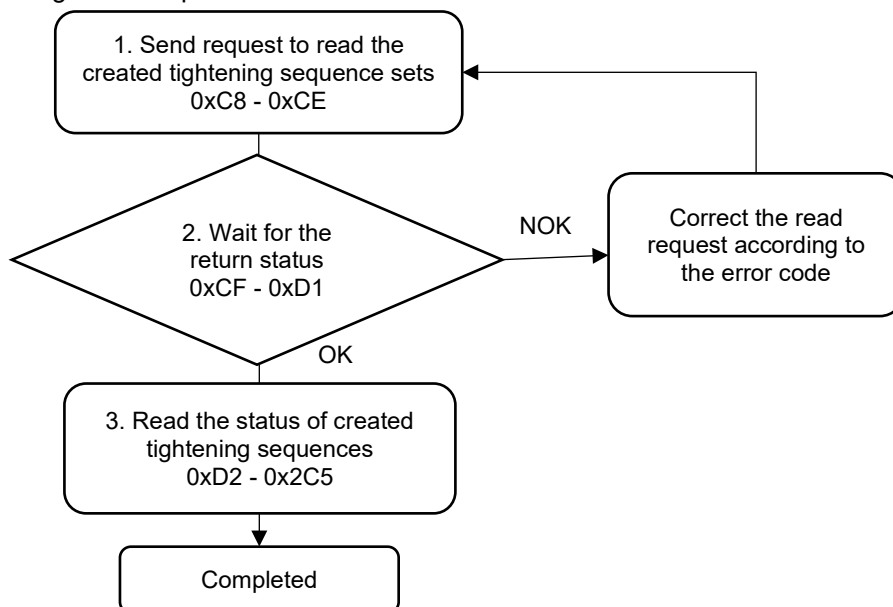
3. Read the content of the coordinates of positioning arm in 0xD2 - 0x329. Refer to function code #203 for the function list.

#260 Read the created sets of tightening sequences

Content description:

Use function code #260 to read the successfully created tightening sequences from the 1st to 500th sets.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	260
C9	Version number	0
CA	Sequence ID	1 to 500
CB to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	260
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Sequence ID	Exceeds the range of 1 to 500

3. Read the tightening sequence creation status in 0xD2 - 0x2C5.

Modbus (Hex)	Return status	Read
D2	1 st set of tightening sequence	0: not yet created; > 0: created
D3	2 nd set of tightening sequence	0: not yet created; > 0: created
D4	3 rd set of tightening sequence	0: not yet created; > 0: created
...	...	...
2C4	499 th set of tightening sequence	0: not yet created; > 0: created
2C5	500 th set of tightening sequence	0: not yet created; > 0: created

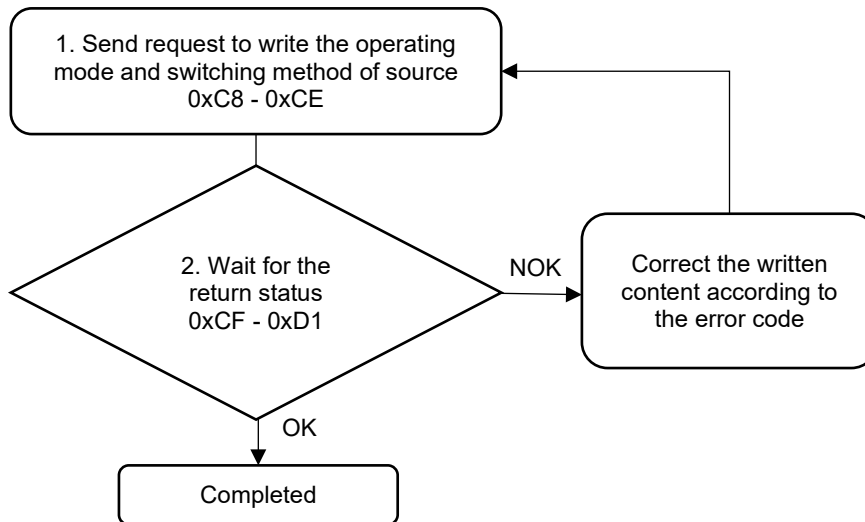
A.3.3 Sources operations

#300 Write the operating mode and switching method of source

Content description:

Use function code #300 to switch the operating mode and switching method of the controller.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	300
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB	Operating mode	0: single-tool 1: dual-tool alternation 2: dual-tool synchronization
CC	Switching method	0: manual 1: screw bit selector 2: barcode scanner
CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	300
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

Code	Error description	Description
1	Operating mode	Exceeds the range of 0 to 2
2	Switching method	Exceeds the range of 0 to 2
3	Tool 1 / Tool 2	Must be 0 or 1
4	Cannot set to dual-tool alternation or dual-tool synchronization when only one tool is activated	-
5	Cannot use the screw bit selector mode under dual-tool alternation	-

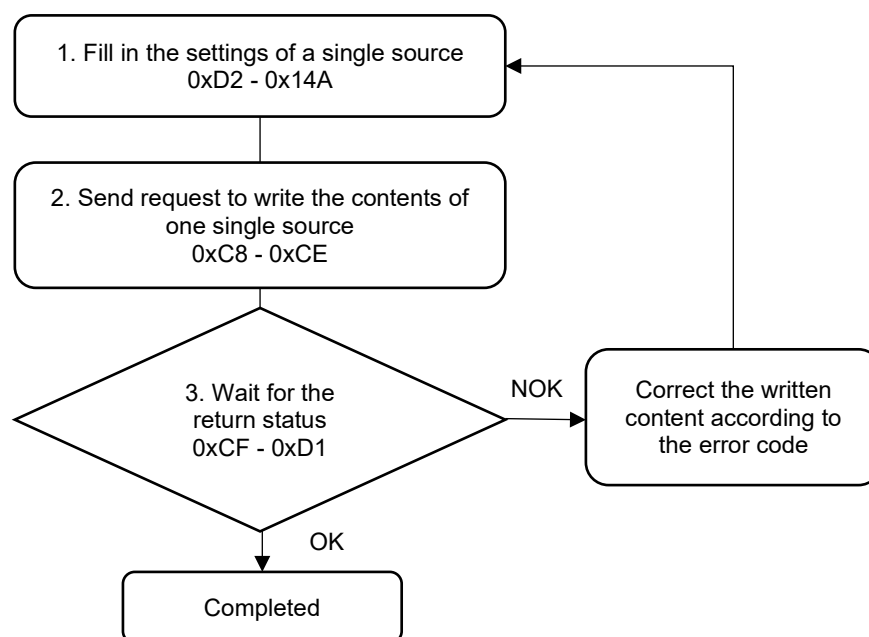
#301 Write the contents of a single source

Content description:

Use function code #301 to set the tightening source.

- A. Add a single tightening source for each tool.
- B. Modify the settings for an existing tightening source. First, read the settings of a single source with function code #351. Then, modify and write the settings with function code #301.

Handshake signal description:



1. Fill in the settings of a single source to 0xD2 - 0x14A.

Modbus (Hex)	Function	Description
D2 to 135	Corresponding barcode string	ASCII code (100 words); only applicable when the switching method is Barcode Scanner
136	Source for the switching method	0: parameter 1: sequence
137	Parameter / Sequence ID	1 to 500
138	Total screw quantity (L)	1 to 999999
139	Total screw quantity (H)	
13A	Bit ID	0 to 255

Modbus (Hex)	Function	Description
13B	Advanced settings (L)	Bit 0: loosening prohibited after tightening OK Bit 1: loosening prohibited after tightening NOK Bit 2: max. count for single screw NOK tightening Bit 3: max. count for single screw NOK loosening Bit 4: go to next step after tightening NOK Bit 5: go to previous step after loosening OK Bit 6: prohibit tool operation when scanner string is null
13C	Advanced settings (H)	Bit 7: clear scanner string when screw Qty. reached Bit 8: prohibit scanning when screw Qty. not reached Bit 9: max. operation time Bit 10: reset Qty. when screw Qty. reached Bit 11: enable reminder when tightening signal ends too early Bit 12: prohibit tool operation when scanner string length mismatch
13D	Single screw tightening NOK count (L)	1 to 999999
13E	Single screw tightening NOK count (H)	
13F	Single screw loosening NOK count (L)	1 to 999999
140	Single screw loosening NOK count (H)	
141 to 144	Reserved	-
145	Max. operation time (L)	1 to 9999999
146	Max. operation time (H)	
147	The parameters to be used under dual-tool alternation mode	0: use parameters of Tool 1 1: use parameters of Tool 2
148	Torque unit	0: N.m; 1: kgf.cm; 2: lbf.ft; 3: lbf.in; 4: ozf.ft; 5: ozf.in; 6:cNm
149	Start condition for Tool 1	0: push start 1: DI 2: lever start
14A	Start condition for Tool 2	3: push start or lever start 4: push start and lever start 5: DI or push start 6: DI or lever start

A

2. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	301
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB	Switching method ID	Manual: 1 Screw bit selector: 1 to 255 Barcode scanner: 1 to 500
CC, CD	Reserved	—
CE	Request to send command	1 (to be filled in last)

3. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	301
D0	Status	1: OK; 2: NOK
D1	Error code	—

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

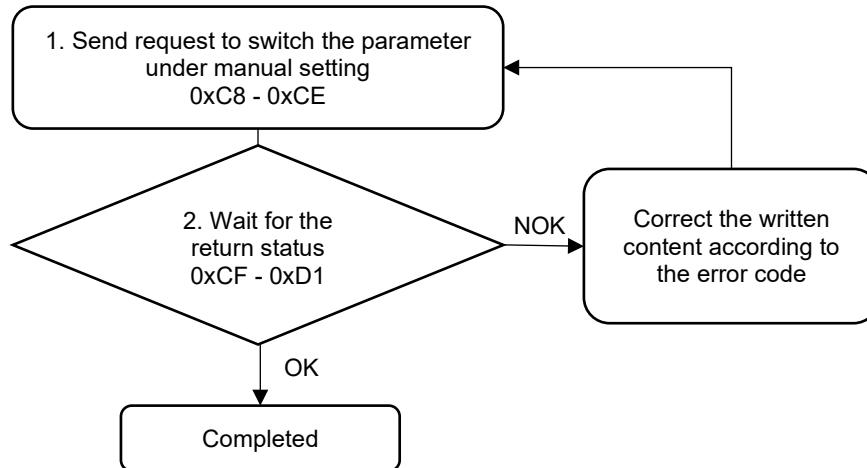
Code	Error description	Description
1	Switching method ID	Exceeds the range of 1 to 500
2	Tool 1 / Tool 2	Must be 0 or 1
10	Source for the switching method	Must be 0 or 1
11	Sequence / Parameter ID	Exceeds the range of 1 to 500
12	Total screw quantity	Exceeds the range of 1 to 999999
13	Bit ID	Exceeds the range of 0 to 255
20	Single screw tightening NOK count	Exceeds the range of 1 to 999999
21	Single screw loosening NOK count	Exceeds the range of 1 to 999999
40	Max. operation time	Exceeds the range of 1 to 9999999
50	Cannot specify the tightening parameter	Cannot select the tightening parameter under dual-tool alternation
51	Sequence error	Must select the sequence which includes tightening parameters of both Tool 1 and Tool 2 under dual-tool alternation
100	Title string is null	—
101	Title already exists	—

#302 Write to switch parameter under manual setting

Content description:

Use function code #302 to switch the parameter ID and screw quantity when the switching method is set to manual setting.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	302
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB	Parameter ID	1 to 500
CC	Total screw quantity (L)	0 to 999999
CD	Total screw quantity (H)	
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	302
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

Code	Error description	Description
1	Incorrect switching method	Must be manual setting. Using function code #302 is prohibited when the switching method is set to screw bit selector or barcode scanner
2	Tool 1 / Tool 2	Must be 0 or 1
11	Parameter ID	Exceeds the range of 1 to 500
12	Total screw quantity	Exceeds the range of 1 to 999999
50	Unsupported operating mode	Using function code #302 is prohibited under dual-tool alternation mode or dual-tool synchronization mode

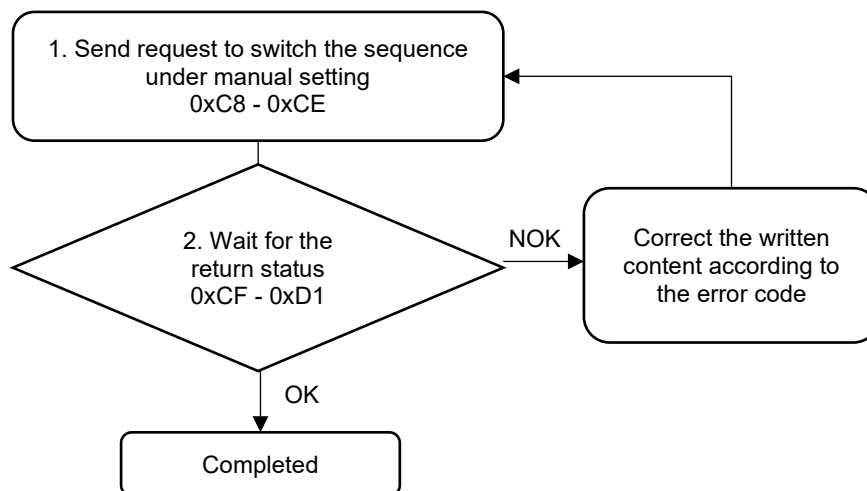
A

#303 Write to switch sequence under manual setting

Content description:

Use function code #303 to switch the sequence ID when the switching method is set to manual setting.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	303
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB	Sequence ID	1 to 500
CC, CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	303
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

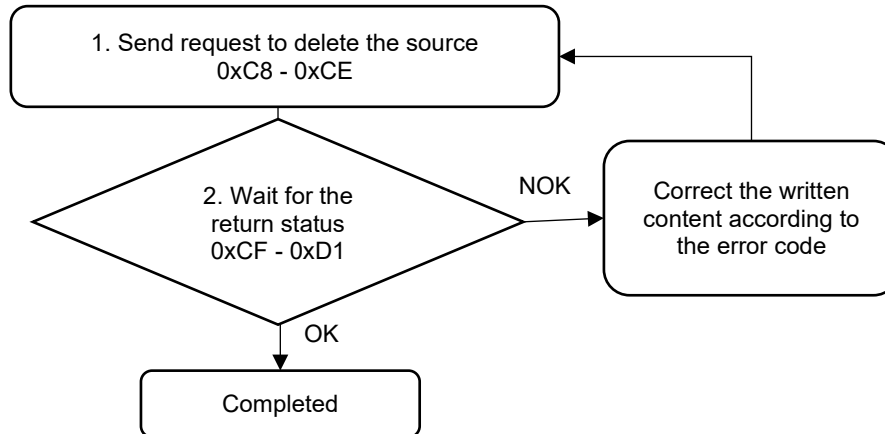
Code	Error description	Description
1	Incorrect switching method	Must be manual setting. Using function code #303 is prohibited when the switching method is set to screw bit selector or barcode scanner.
2	Tool 1 / Tool 2	Must be 0 or 1
11	Sequence ID	Exceeds the range of 1 to 500
50	Unsupported operating mode	Using function code #303 is prohibited under dual-tool alternation mode or dual-tool synchronization mode

#310 Write to delete the source

Content description:

Use function code #310 to delete the tightening source.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	310
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB	Operating mode	0: single-tool 1: dual-tool alternation 2: dual-tool synchronization
CC	Switching method	0: manual 1: screw bit selector 2: barcode scanner
CD	Switching method ID to be deleted	Manual: 1 Screw bit selector: 1 to 255 Barcode scanner: 1 to 500 Delete all: 0
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	310
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

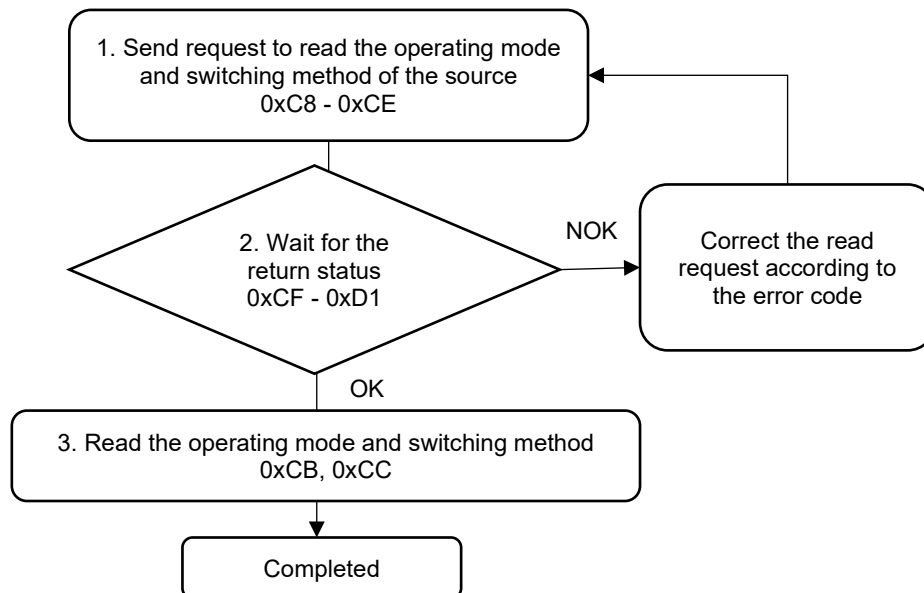
Code	Error description	Description
1	Operating mode	Exceeds the range of 0 to 2
2	Switching method	Exceeds the range of 0 to 2
3	Tool 1 / Tool 2	Must be 0 or 1
4	Cannot set to dual-tool alternation or dual-tool synchronization when only one tool is activated	-
5	Cannot use the screw bit selector mode under dual-tool alternation	-
6	Switching method ID to be deleted	Exceeds the range of 0 to 500

#350 Read the operating mode and switching method of source

Content description:

Use function code #350 to read the current operating mode and switching method of the smart screwdriving controller.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	350
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	350
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

3. Read the operating mode and switching method in 0xCB - 0xCC.

Modbus (Hex)	Write / Read request	Read
CB	Operating mode	0: single-tool 1: dual-tool alternation 2: dual-tool synchronization
CC	Switching method	0: manual 1: screw bit selector 2: barcode scanner

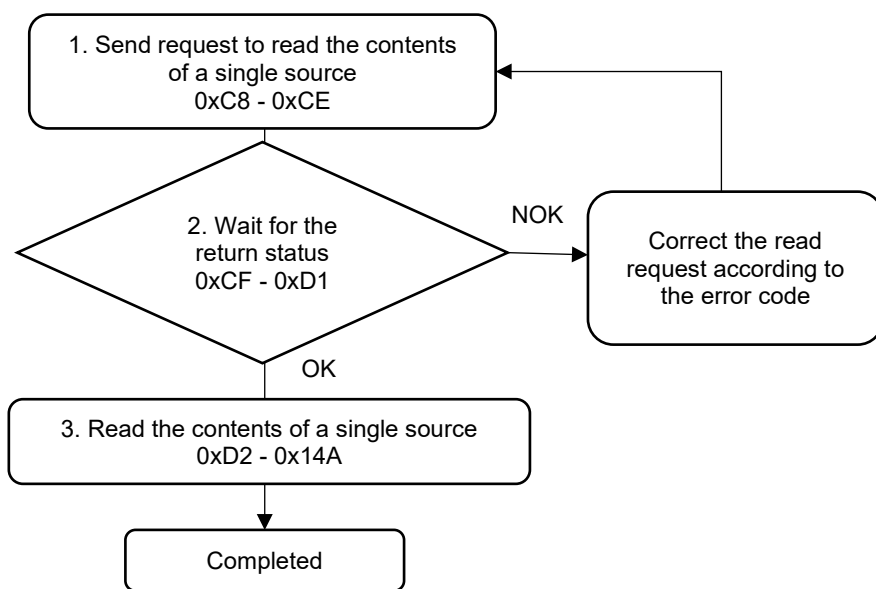
#351 Read the contents of a single source

Content description:

Use function code #351 to read the settings of a single tightening source.

- A. Obtain the contents of an existing tightening source.
- B. Check whether the tightening source has been set.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	351
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB	Switching method ID	Manual: 1 Screw bit selector: 1 to 255 Barcode scanner: 1 to 500
CC, CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	351
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

Code	Error description	Description
1	Switching method ID	Exceeds the range of 1 to 500
2	Tool 1 / Tool 2	Must be 0 or 1

3. Read the contents of a single source in 0xD2 - 0x14A. Refer to function code #301 for the function list.

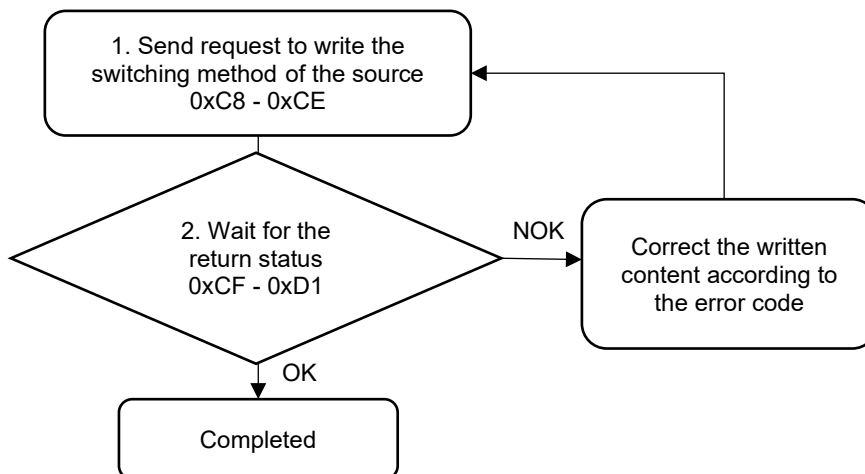
A.3.4 Results operations

#400 Write the switching method of source

Content description:

Use function code #400 to change the switching method of the tightening source in operation.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	400
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB	Switching method ID	Manual: 1 Screw bit selector: 1 to 255 Barcode scanner: 1 to 500
CC, CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	400
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	Switching method ID	Manual: exceeds 1 Screw bit selector: exceeds the range of 1 to 255 Barcode scanner: exceeds the range of 1 to 500

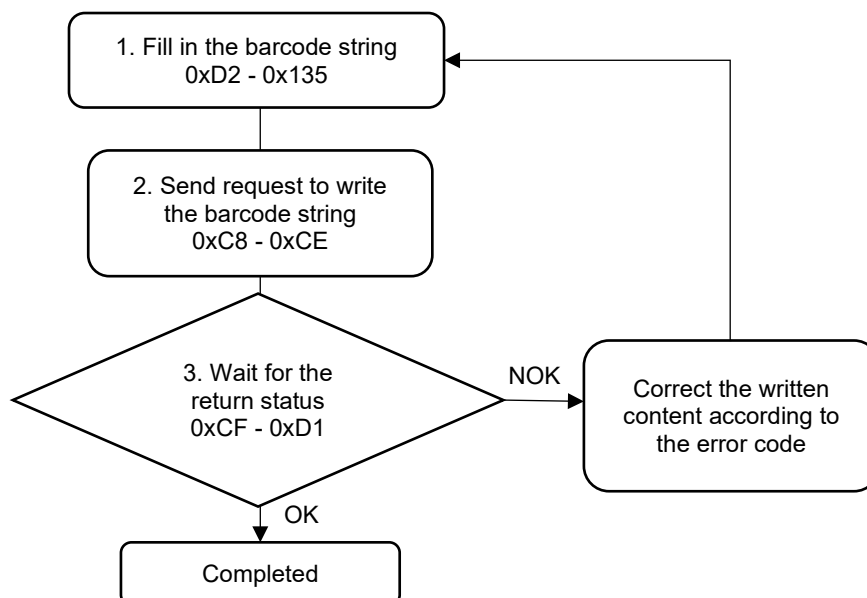
#401 Write the barcode string

Content description:

Use function code #401 to input the barcode string.

- A. Input the barcode string to the controller.
- B. If the switching method is Barcode Scanner, when you scan the barcode, the content of the string is immediately matched and switched to the corresponding parameter or sequence.

Handshake signal description:



1. Fill in the barcode string to 0xD2 - 0x135.

Modbus (Hex)	Function	Description
D2 to 135	Barcode string	ASCII code (100 words)

2. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	401
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

3. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	401
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

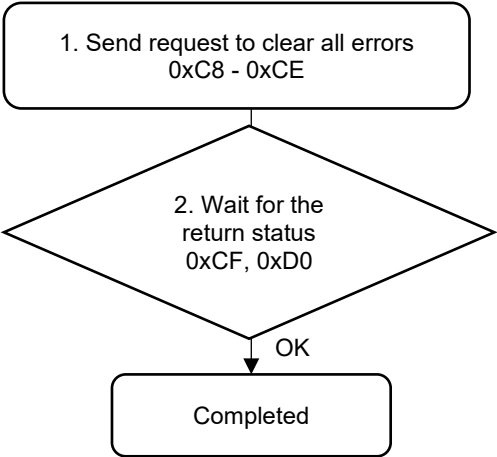
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	Matching failed	Failed to match the strings in Barcode Scanner mode

#402 Write to clear all errors

Content description:

Use function code #402 to clear all the AL, NG, and WN alarms.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	402
C9	Version number	0
CA to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF and 0xD0.

Modbus (Hex)	Return status	Description
CF	Function code	402
D0	Status	1: OK

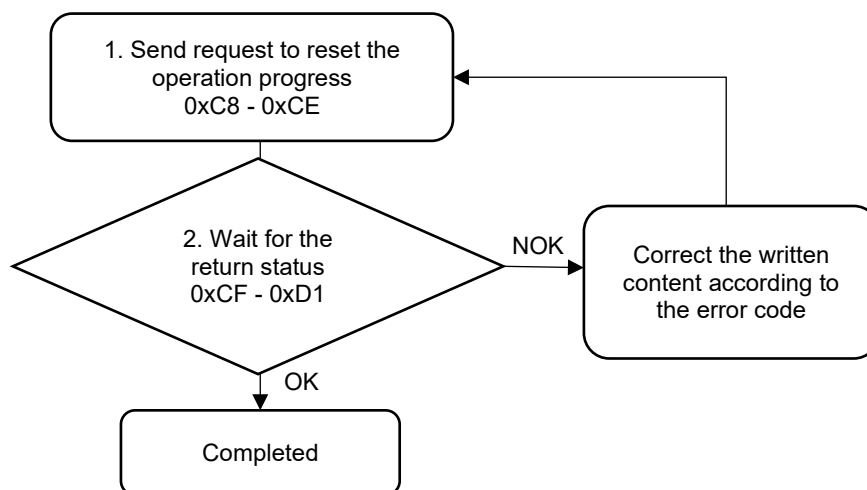
If successful, the read status (0xD0) is 1.

#403 Write to reset the operation progress

Content description:

Use function code #403 to reset the operation progress (total screw quantity, parameter quantity, and screw quantity of the parameter).

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	403
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	403
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

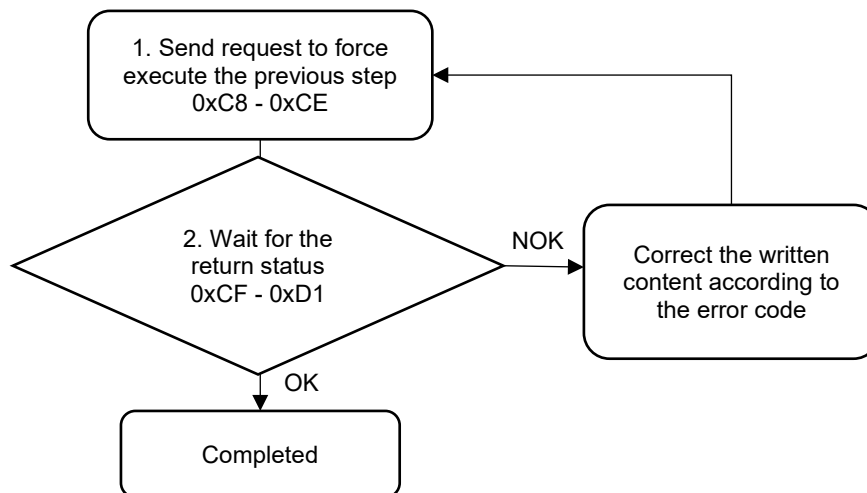
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

#404 Write to force execute the previous step

Content description:

Use function code #404 to execute the parameter of the previous screw.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	404
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	404
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

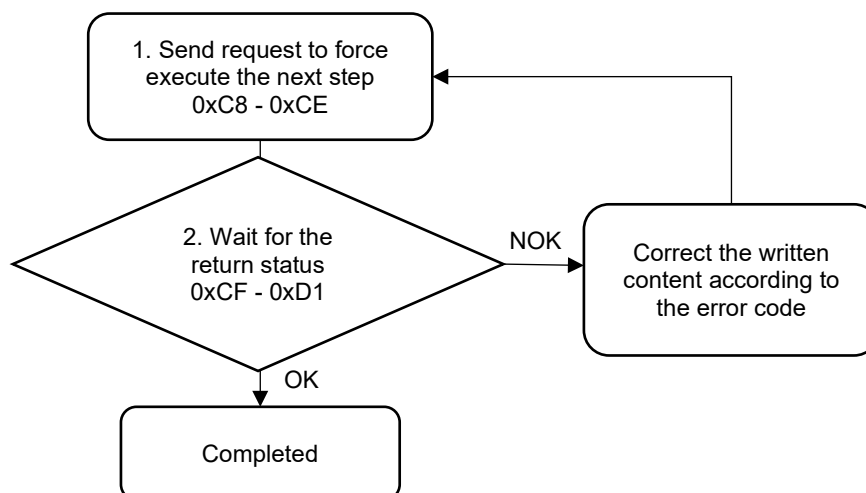
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

#405 Write to force execute the next step

Content description:

Use function code #405 to execute the parameter of the next screw.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	405
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	405
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

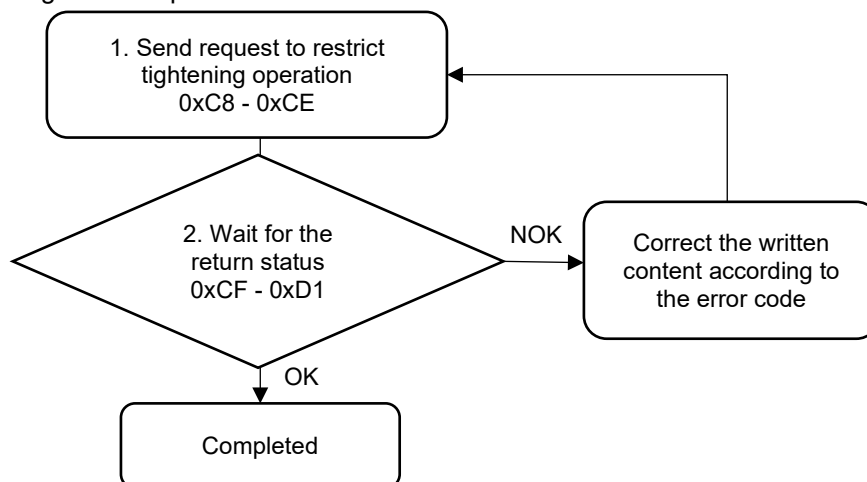
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

#406 Write to restrict tightening operation

Content description:

Use function code #406 to restrict the transducerized smart screwdriver from performing tightening operations until you use function code #406 again to remove the restriction.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	406
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB	Restrict operation / Remove restriction	0: remove restriction 1: restrict operation
CC, CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	406
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

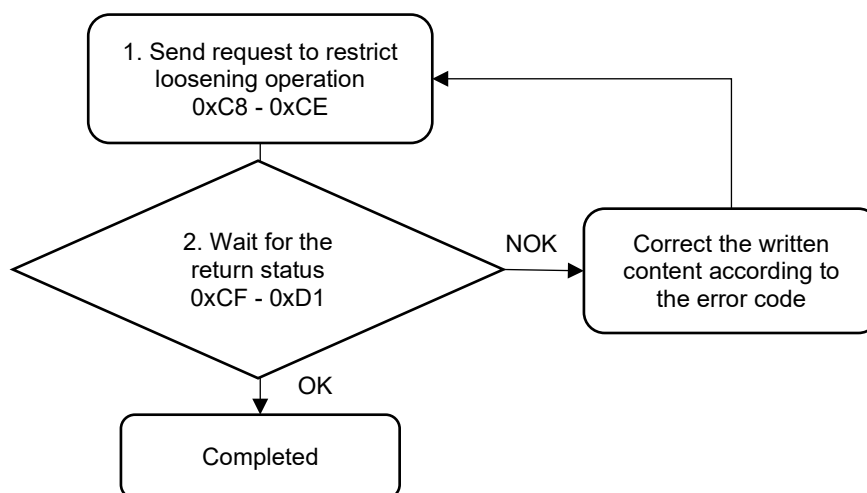
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

#407 Write to restrict loosening operation

Content description:

Use function code #407 to restrict the transducerized smart screwdriver from performing loosening operations until you use function code #407 again to remove the restriction.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	407
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB	Restrict operation / Remove restriction	0: remove restriction 1: restrict operation
CC, CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	407
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

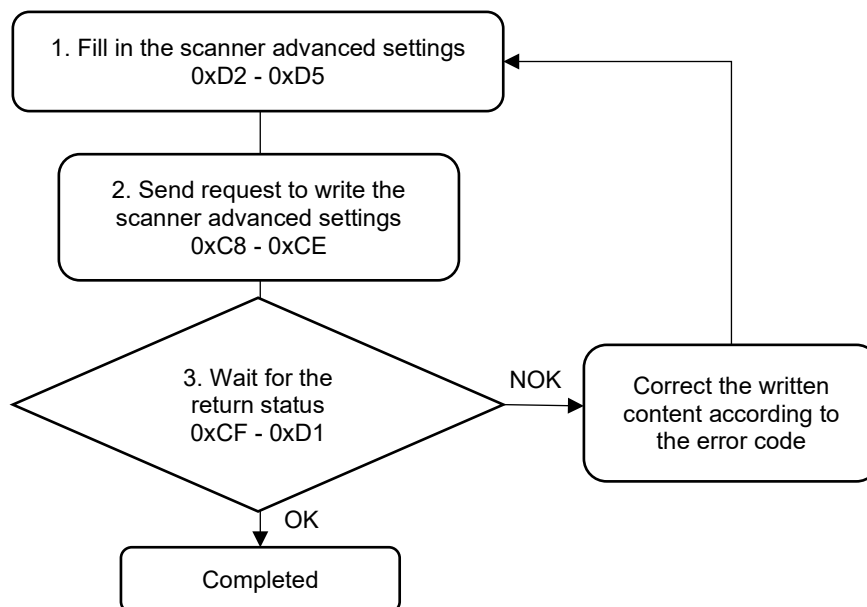
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

#408 Write the scanner advanced settings

Content description:

Use function code #408 to set the scanner advanced settings.

Handshake signal description:



1. Fill in the scanner advanced settings to 0xD2 - 0xD5.

Modbus (Hex)	Function	Description
D2	Start position (char.) for saving scanner string	1 to 200
D3	End position (char.) for saving scanner string	1 to 200
D4	Start position (char.) for matching scanner string	1 to 200
D5	End position (char.) for matching scanner string	1 to 200

Example 1: when the barcode string is "ABCDE", if you set 2 to 0xD2 and 4 to 0xD3, the system then saves the 3-character string "BCD" to the Results screen.

Example 2: when the barcode string is "ABCDE" and the switching method is Barcode Scanner, if you set 2 to 0xD4 and 4 to 0xD5, the system then matches the 3-character string "BCD" against the corresponding strings of the 500 sets of parameters or sequences.

2. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	408
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

3. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	408
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	Scanner advanced setting value out of range	Exceeds the range of 1 to 200

A

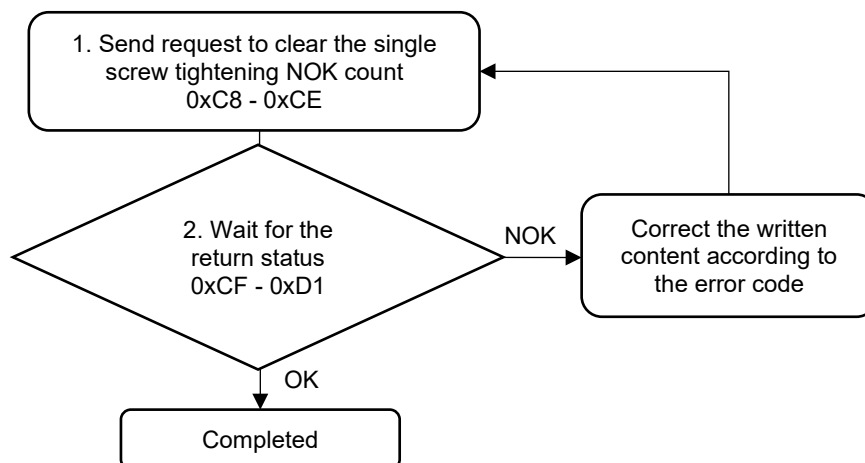
#409 Write to clear the single screw tightening NOK count

Content description:

Use function code #409 to clear the single screw tightening NOK count.

Modbus (Hex)		Function code #21 (Dec)		Tool 1 status / Tool 2 status	R/W
Tool 1	Tool 2	Tool 1	Tool 2		
D	3F	14	64	Single screw tightening NOK count (L)	R
E	40	15	65	Single screw tightening NOK count (H)	R

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	409
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	409
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

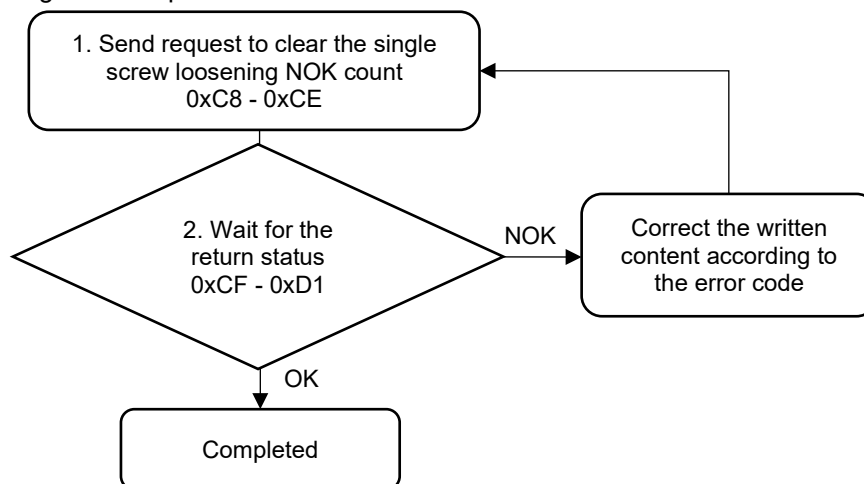
#410 Write to clear the single screw loosening NOK count

Content description:

Use function code #410 to clear the single screw loosening NOK count.

Modbus (Hex)		Function code #21 (Dec)		Tool 1 status / Tool 2 status	R/W
Tool 1	Tool 2	Tool 1	Tool 2		
11	43	18	68	Single screw loosening NOK count (L)	R
12	44	19	69	Single screw loosening NOK count (H)	R

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	410
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	410
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

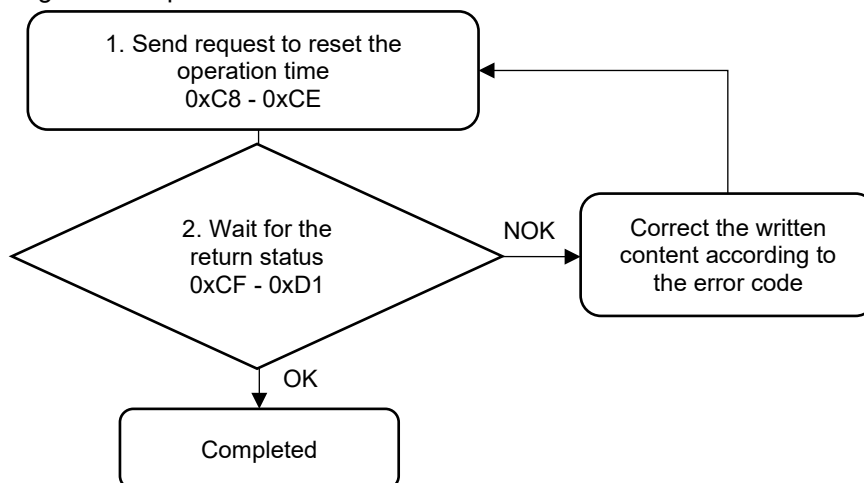
#411 Write to reset the operation time

Content description:

Use function code #411 to restart the timer for counting the remaining operation time.

Modbus (Hex)		Function code #21 (Dec)		Tool 1 status / Tool 2 status	R/W
Tool 1	Tool 2	Tool 1	Tool 2		
30	62	49	99	Remaining operation time (L)	R
31	63	50	100	Remaining operation time (H)	R

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	411
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	411
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

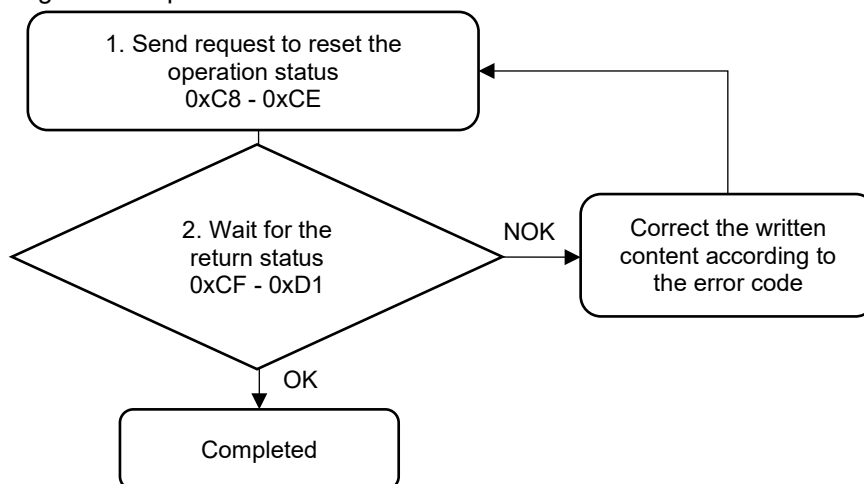
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

#412 Write to reset the operation status

Content description:

Use function code #412 to reset the operation status, including the operation progress and other tightening information.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	412
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	412
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

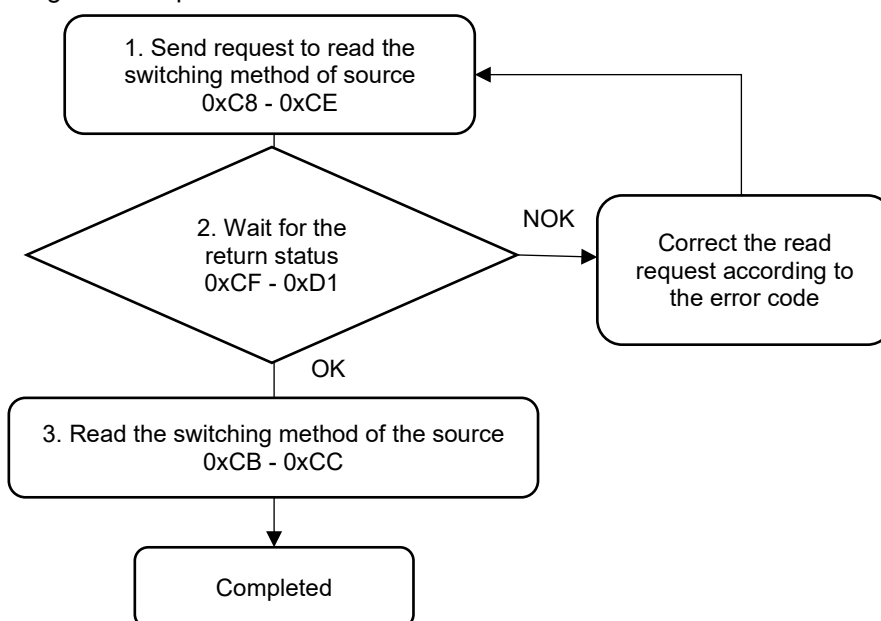
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

#450 Read the switching method of source

Content description:

Use function code #450 to read the switching method of the tightening source in operation.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	450
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	450
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

3. Read the switching method of the source in 0xCB.

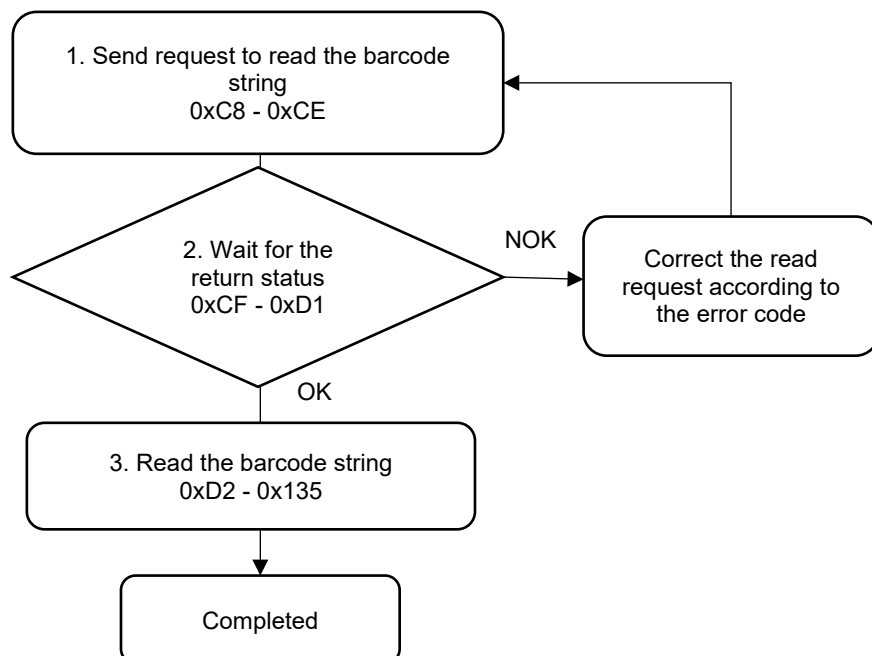
Modbus (Hex)	Write / Read request	Read
CB	Switching method ID	Manual: 1 Screw bit selector: 1 to 255 Barcode scanner: 1 to 500 (0: the controller is not running)

#451 Read the barcode string

Content description:

Use function code #451 to read the content of the barcode string.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	451
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	451
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

3. Read the barcode string in 0xD2 - 0x135.

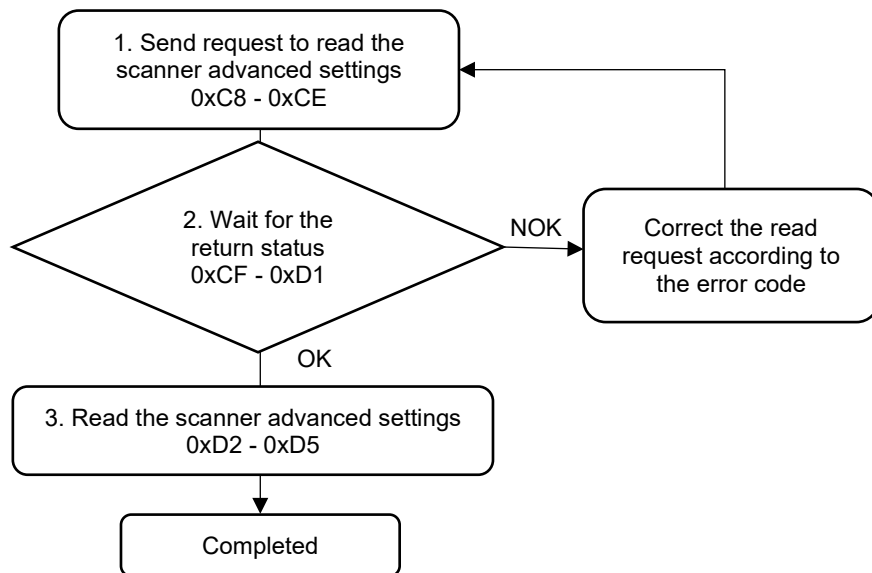
Modbus (Hex)	Function	Description
D2 to 135	Barcode string	ASCII code (100 words)

#452 Read the scanner advanced settings

Content description:

Use function code #452 to read the Scanner Advanced Settings.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	452
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	452
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

3. Read the scanner advanced settings in 0xD2 - 0xD5.

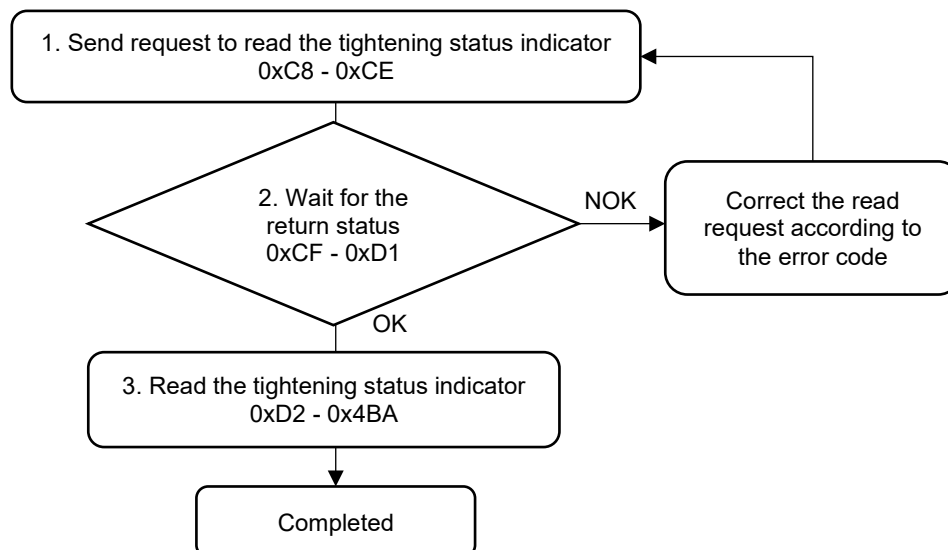
Modbus (Hex)	Function	Description
D2	Start position (char.) for saving scanner string	1 to 200
D3	End position (char.) for saving scanner string	1 to 200
D4	Start position (char.) for matching scanner string	1 to 200
D5	End position (char.) for matching scanner string	1 to 200

#453 Read the tightening status indicator

Content description:

Use function code #453 to read the tightening status indicator for up to 1000 screws.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	453
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	453
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

3. Read the tightening status indicator in 0xD2 - 0x4BA.

Modbus (Hex)	Function	Description
D2	Tightening status indicator for screw ID 1	2: green 8: red
D3	Tightening status indicator for screw ID 2	
D4	Tightening status indicator for screw ID 3	
...	Tightening status indicator for screw ID N	
4BA	Tightening status indicator for screw ID 1000	

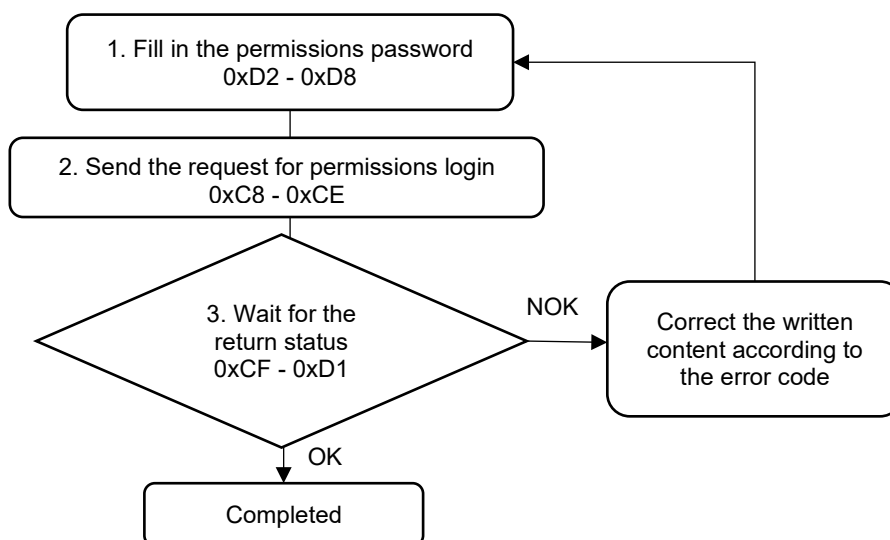
A.3.5 Controller operations

#500 Write the request for permissions login

Content description:

Use function code #500 to switch the controller permissions account through the host controller.

Handshake signal description:



1. Fill in the permissions password to 0xD2 - 0xD8.

Modbus (Hex)	Function	Description
D2 to D8	Permissions password	Numbers converted to ASCII code (10 words)

2. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	500
C9	Version number	0
CA	Permissions account	1: User1 2: User2 3: User3 4: User4 5: User5 6: Admin (highest level)
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

3. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	500
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

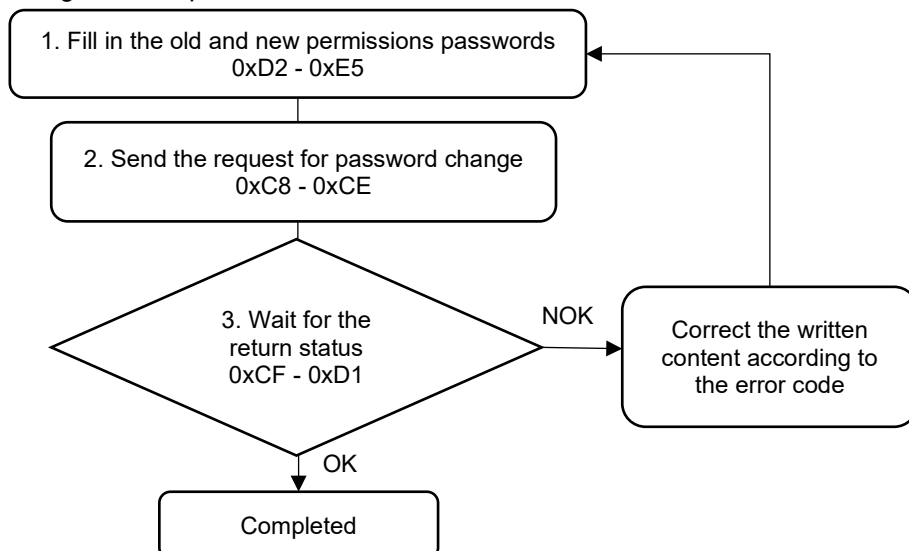
Code	Error description	Description
1	Permissions account	Exceeds the range of 1 to 6
2	Login failed	-

#501 Write the request for password change

Content description:

Use function code #501 to change the permissions password of controller through the host controller.

Handshake signal description:



1. Fill in the old permissions password to 0xD2 - 0xDB, and the new permissions password to 0xDC - 0xE5.

Modbus (Hex)	Function	Description
D2 to DB	Old password	Numbers converted to ASCII code (10 words)
DC to E5	New password	Numbers converted to ASCII code (10 words)

2. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	501
C9	Version number	0
CA	Permissions account	1: User1 2: User2 3: User3 4: User4 5: User5 6: Admin (highest level)
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

3. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	501
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

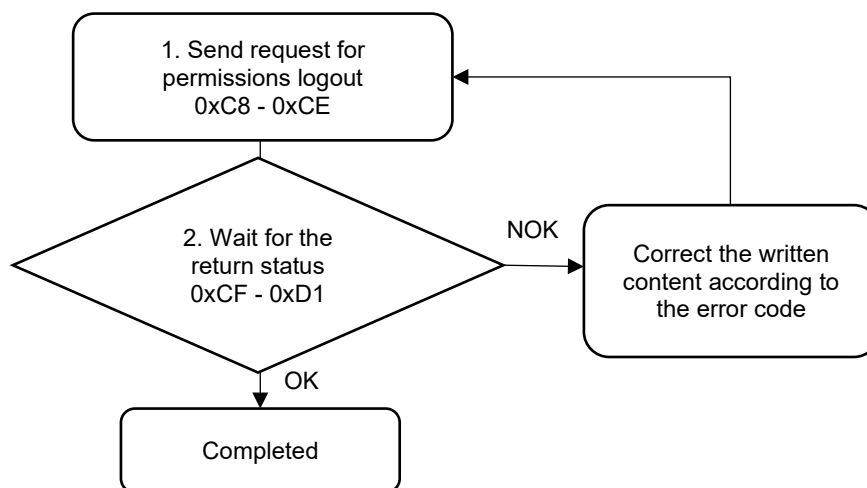
Code	Error description	Description
1	Permissions account	Exceeds the range of 1 to 6
2	Password change failed	-

#502 Write the request for permissions logout

Content description:

Use function code #502 to log out of the controller permissions account through the host controller. The default permissions account after logout is User1.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	502
C9	Version number	0
CA to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	502
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

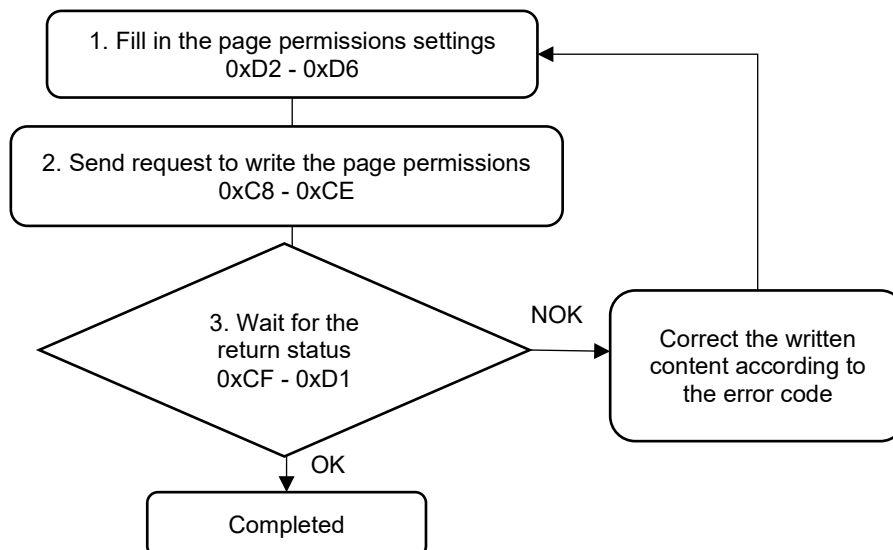
Code	Error description	Description
1	Logout failed	-

#503 Write the page permissions

Content description:

Use function code #503 to set the permissions of page access and edit for different accounts.

Handshake signal description:



1. Fill in the page permissions settings to 0xD2 - 0xD6.

Modbus (Hex)	Function	Description
D2	User1 page permissions	Bit 0: Parameters
D3	User2 page permissions	Bit 1: Sequence
D4	User3 page permissions	Bit 2: Sources
D5	User4 page permissions	Bit 3: Controller
D6	User5 page permissions	Bit 4: Tool
		Bit 5: operation for screw progress
		Bit 6: clearing of the production report entries
		Bit 7: clearing of the error, warning, and button report entries

2. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	503
C9	Version number	0
CA	Confirm command	99
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

3. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	503
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

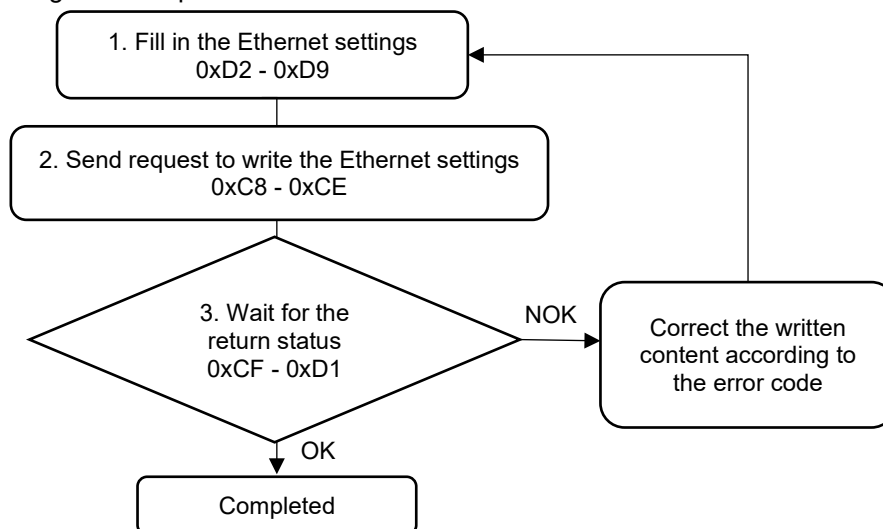
Code	Error description	Description
1	Confirm command	Must be 99

#504 Write the Ethernet settings

Content description:

Use function code #504 to write the Ethernet settings.

Handshake signal description:



1. Fill in the content of Ethernet settings to 0xD2 - 0xD9.

Modbus (Hex)	Function	Description
D2	IP1 address	Default: 192
D3	IP2 address	Default: 168
D4	IP3 address	Default: 1
D5	IP4 address	Default: 11
D6	Subnet mask 1	Default: 255
D7	Subnet mask 2	Default: 255
D8	Subnet mask 3	Default: 255
D9	Subnet mask 4	Default: 0

2. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	504
C9	Version number	0
CA	Confirm command	99
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

3. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	504
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

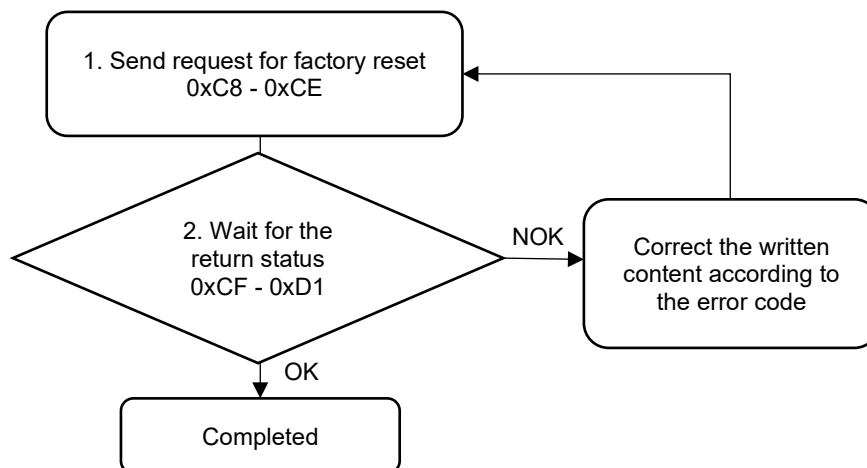
Code	Error description	Description
1	Confirm command	Must be 99

#505 Write the request for factory reset

Content description:

Use function code #505 to write the request for factory reset.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	505
C9	Version number	0
CA	Confirm command	99
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

Note:

- After factory reset, all data will be deleted and all settings will be restored to default.
- If you use the account of the highest permissions (Admin) for factory reset, the Ethernet settings will be restored to default, and the error reports and warning reports will be cleared.

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	505
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

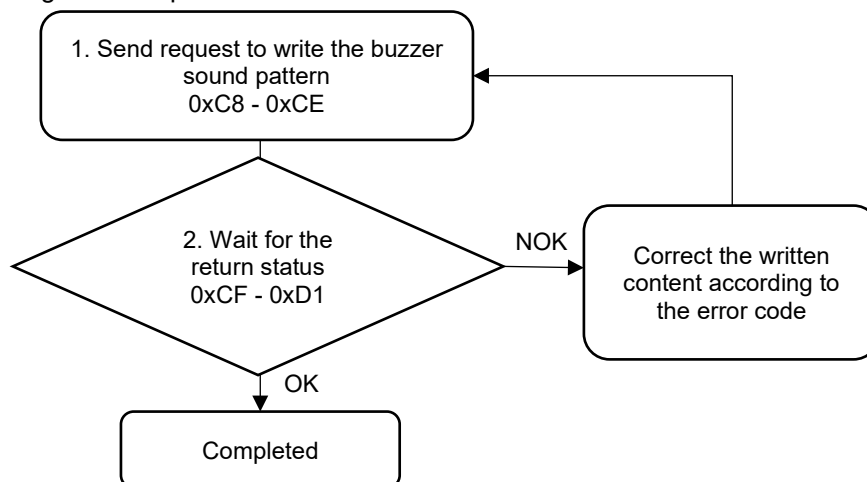
Code	Error description	Description
1	Confirm command	Must be 99

#506 Write the buzzer sound pattern

Content description:

Use function code #506 to set the buzzer sound patterns.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	506
C9	Version number	0
CA	Sound pattern of error signal buzzer	0: mute 1: one long beep 2: one short beep 3: two short beeps
CB	Sound pattern of finished signal buzzer for a single screw	
CC	Sound pattern of finished signal buzzer for total screws	
CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	506
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

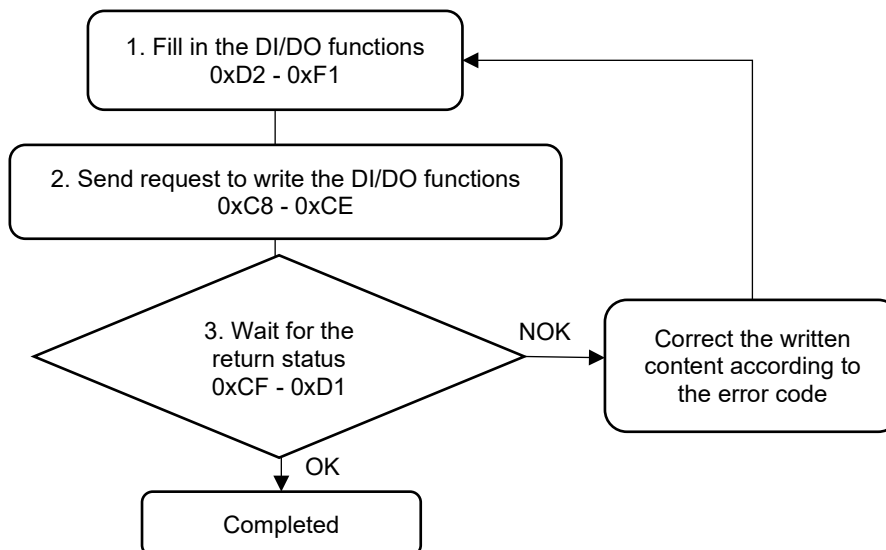
Code	Error description	Description
1	Buzzer sound pattern	Exceeds the range of 0 to 3

#507 Write the DI/DO functions

Content description:

Use function code #507 to set the DI/DO functions.

Handshake signal description:



1. Fill in the DI/DO functions for a single tool to 0xD2 - 0xF1.

Modbus (Hex)	Function	Description
D2	DO1 NO / NC	0: NO (normally open) 1: NC (normally closed)
D3	DO2 NO / NC	
D4	DO3 NO / NC	
D5	DO4 NO / NC	
D6	DO5 NO / NC	
D7	DO6 NO / NC	
D8	DO7 NO / NC	
D9	DO8 NO / NC	

Modbus (Hex)	Function	Description
DA	DO1 function	0: not in use 1: ready 2: running 3: tightening OK and report is being saved
DB	DO2 function	4: tightening NOK and report is being saved 5: loosening OK and report is being saved 6: loosening NOK and report is being saved 7: total screw quantity reached
DC	DO3 function	8: device alarms or errors or warning 9: tighten to waiting stage (only available for DO7; must enable the related option in the Advanced Setting window)
DD	DO4 function	10: Modbus handshake end signal 11: device alarms 12: reserved 13: tool reverse switch state
DE	DO5 function	14: rotation direction 15: tightening result sent to feeder 16 to 23: parameter ID output (Bits 1 - 8) 24 to 31: screw progress output (Bits 1 - 8)
DF	DO6 function	32 to 39: sequence ID output (Bits 1 - 8) 40 to 47: bit ID output (Bits 1 - 8) 48 to 51: running tightening stage output (Bits 1 - 4) 52: tightening OK and lever signal end
E0	DO7 function	53: double-click the lever button 54: tightening OK and the report is saved 55: tightening NOK and the report is saved
E1	DO8 function	56: loosening OK and the report is saved 57: loosening NOK and the report is saved 58: busy 59 - 63: reserved
E2	DI1 NO / NC	0: NO (normally open) 1: NC (normally closed)
E3	DI2 NO / NC	
E4	DI3 NO / NC	
E5	DI4 NO / NC	
E6	DI5 NO / NC	
E7	DI6 NO / NC	
E8	DI7 NO / NC	
E9	DI8 NO / NC	

Modbus (Hex)	Function	Description
EA	DI1 function	0: not in use 1: start tightening 2: start loosening
EB	DI2 function	3: emergency stop (dual-tool) 4: clear error and clear status (dual-tool) 5: reset total screw quantity
EC	DI3 function	6: next step 7: previous step 8: start tightening (pulse signal) 9: start loosening (pulse signal)
ED	DI4 function	10: start tightening (dual-tool) 11: start loosening (dual-tool) 12: reset operation time
EE	DI5 function	13: reset single screw tightening + loosening NOK counts 14: prohibit tool operation 15: clear communication flag
EF	DI6 function	16: wait for external signal before continuing tightening (only available for DI7; must enable the related option in the Advanced Setting window) 17: emergency stop
F0	DI7 function	18: prohibit tool operation (pulse signal) 19: rotation direction 20: start rotation
F1	DI8 function	21: clear error and clear status 22 to 39: reserved 40 to 47: bit ID input (Bits 1 - 8)

2. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	507
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

3. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	507
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

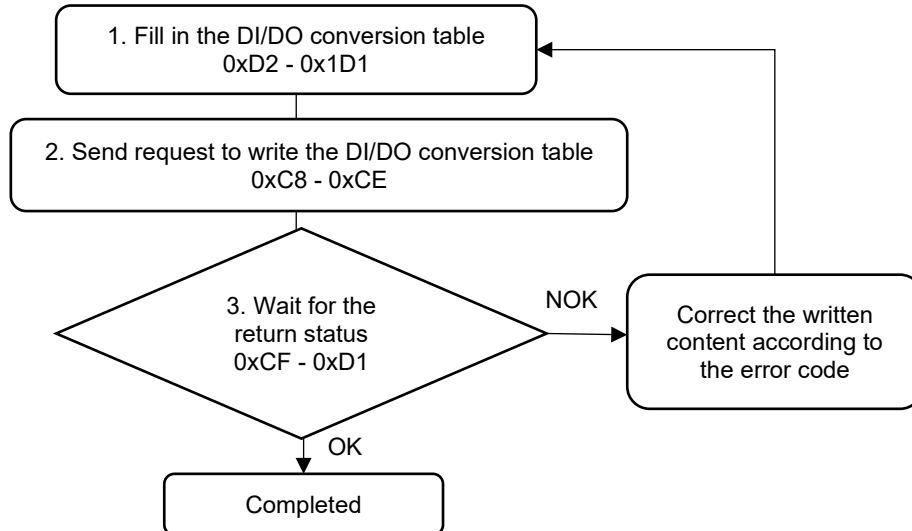
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

#508 Write the DI/DO conversion table

Content description:

Use function code #508 to set the DI/DO conversion table.

Handshake signal description:



1. Fill in the DI/DO conversion table to 0xD2 - 0x1D1.

Modbus (Hex)	Function	Description
D2	DEC value corresponding to ID 0	0 to 255
D3	DEC value corresponding to ID 1	0 to 255
D4	DEC value corresponding to ID 2	0 to 255
...	DEC value corresponding to ID N	0 to 255
1CF	DEC value corresponding to ID 253	0 to 255
1D0	DEC value corresponding to ID 254	0 to 255
1D1	DEC value corresponding to ID 255	0 to 255

2. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	508
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB	Conversion table	0: bit ID output table 1: bit ID input table 2: parameter ID output table 4: screw progress output table 6: sequence ID output table
CC, CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

3. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	508
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

A

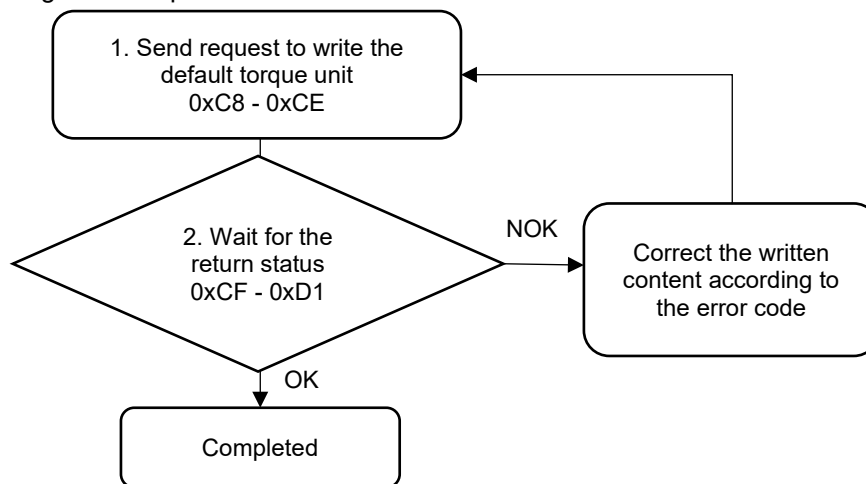
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	Conversion table	Must be 0, 1, 2, 4, or 6

#509 Write the default torque unit

Content description:

Use function code #509 to set the torque unit of the controller.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	509
C9	Version number	0
CA	Default torque unit	0: N.m 1: kgf.cm 2: lbf.ft 3: lbf.in 4: ozf.ft 5: ozf.in 6: cNm
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	509
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Default torque unit	Exceeds the range of 0 to 6

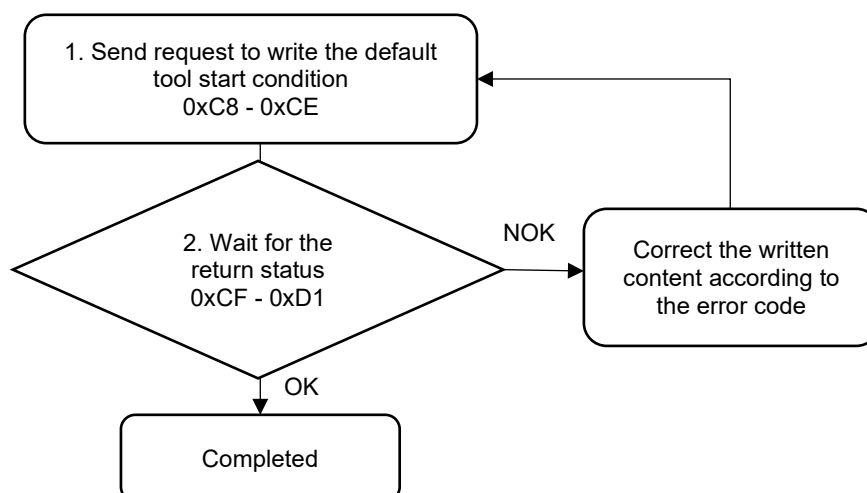
#510 Write the default tool start condition

Content description:

Use function code #510 to set the tool start condition.

Note: for example, first set the default tool start condition to “DI” with function code #510. Then, set the DI function to “start tightening” or “start loosening” with function code #507 or #511 to operate the tool through DI.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	510
C9	Version number	0
CA	Default tool start condition	0: push start 1: DI 2: lever start 3: push start or lever start 4: push start and lever start 5: DI or push start 6: DI or lever start
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	510
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

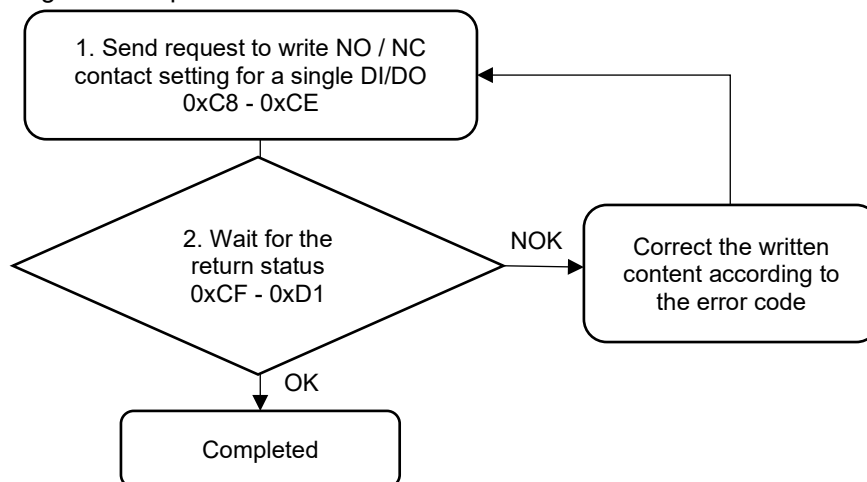
Code	Error description	Description
1	Tool start condition	Exceeds the range of 0 to 6

#511 Write NO / NC contact setting for a single DI/DO

Content description:

Use function code #511 to set a single DI/DO as an NO / NC contact.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	511
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB	DO / DI	0: DO; 1: DI
CC	Corresponding bit of DO / DI	0 to 7
CD	NO / NC	0: NO (normally open) 1: NC (normally closed)
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	511
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

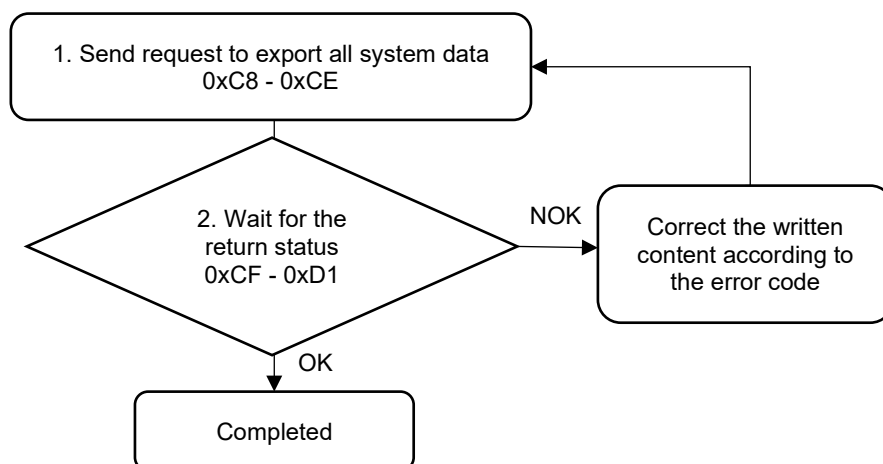
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	DO / DI	Must be 0 or 1
3	Corresponding bit of DO / DI	Exceeds the range of 0 to 7
4	NO / NC	Must be 0 or 1

#512 Write to export all system data

Content description:

Use function code #512 to export all system data, including contents of parameters, sequences, sources, controller, tool, and reports.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	512
C9	Version number	0
CA	Data storage location	0: USB Disk 1: HMI Disk
CB	Options	Bit 0: parameters Bit 1: sequences Bit 2: sources Bit 3: controller Bit 4: tool Bit 5: reports
CC, CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	512
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

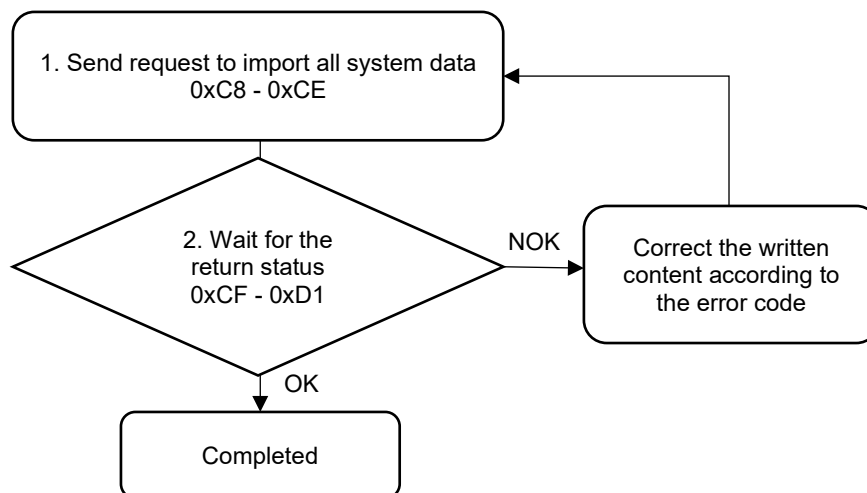
Code	Error description	Description
1	Data storage location	Must be 0 or 1

#513 Write to import all system data

Content description:

Use function code #513 to import all system data, including contents of parameters, sequences, sources, controller, and tool.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	513
C9	Version number	0
CA	Read data from	0: USB Disk 1: HMI Disk
CB	Options	Bit 0: parameters Bit 1: sequences Bit 2: sources Bit 3: controller Bit 4: tool
CC, CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	513
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

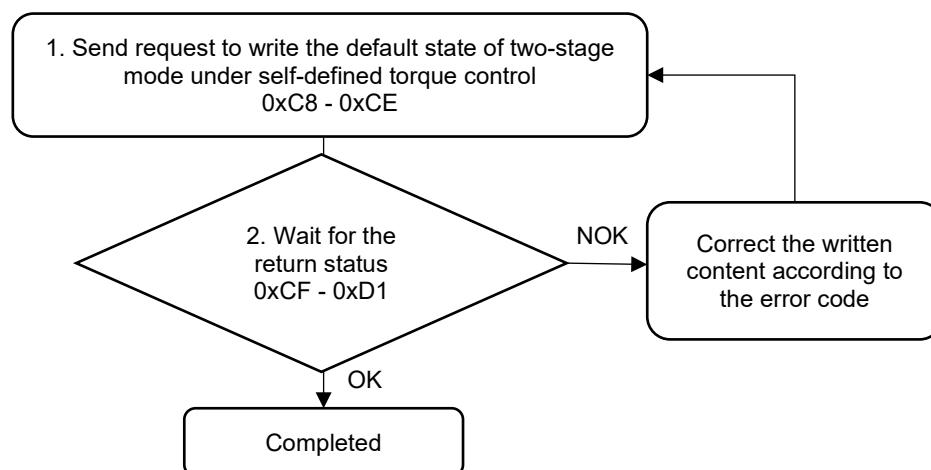
Code	Error description	Description
1	Read data from	Must be 0 or 1

#514 Write the default two-stage mode under self-defined torque control

Content description:

Use function code #514 to set the default state of two-stage mode under self-defined torque control.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	514
C9	Version number	0
CA	Two-stage mode	0: Off 1: On
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	514
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

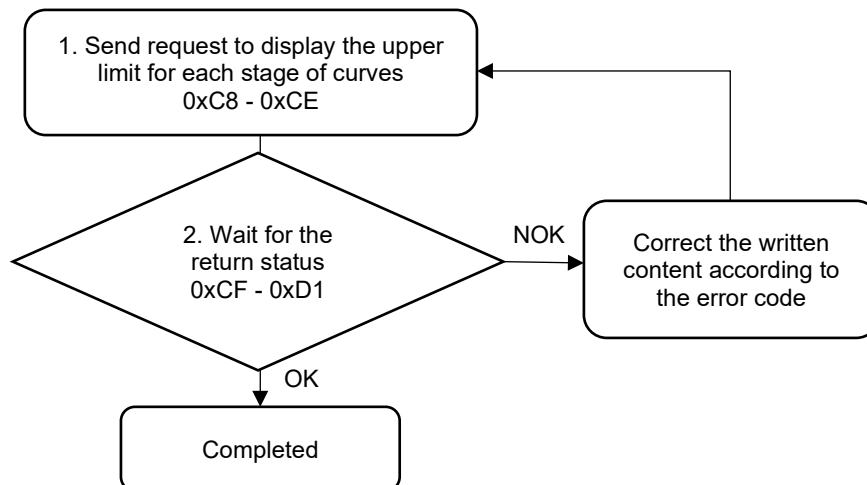
Code	Error description	Description
1	Two-stage mode	Must be 0 or 1

#515 Write the display of upper limit for each stage of curves

Content description:

Use function code #515 to set whether to display the upper limit for each stage of the curves.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	515
C9	Version number	0
CA	Display the upper limit for each stage of curves	0: On 1: Off
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	515
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

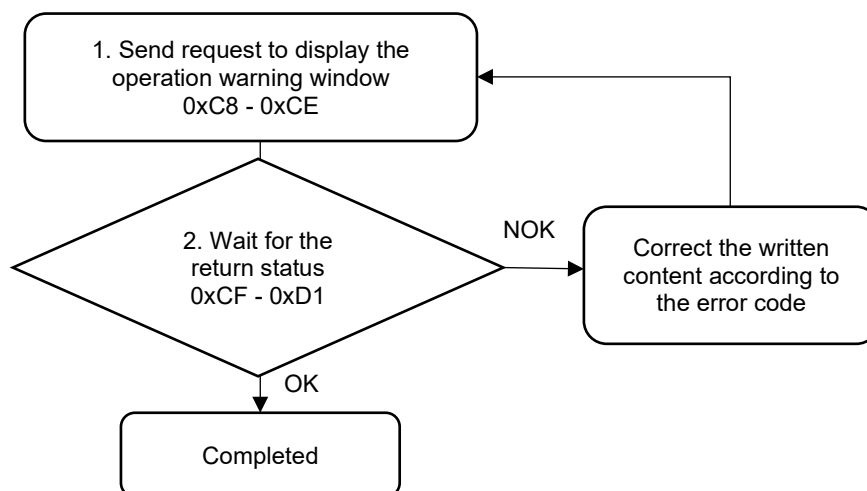
Code	Error description	Description
1	Display the upper limit for each stage of curves	Must be 0 or 1

#516 Write the display of operation warning window

Content description:

Use function code #516 to set whether to display the operation warning window.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	516
C9	Version number	0
CA	Display the operation warning window	0: On 1: Off
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	516
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

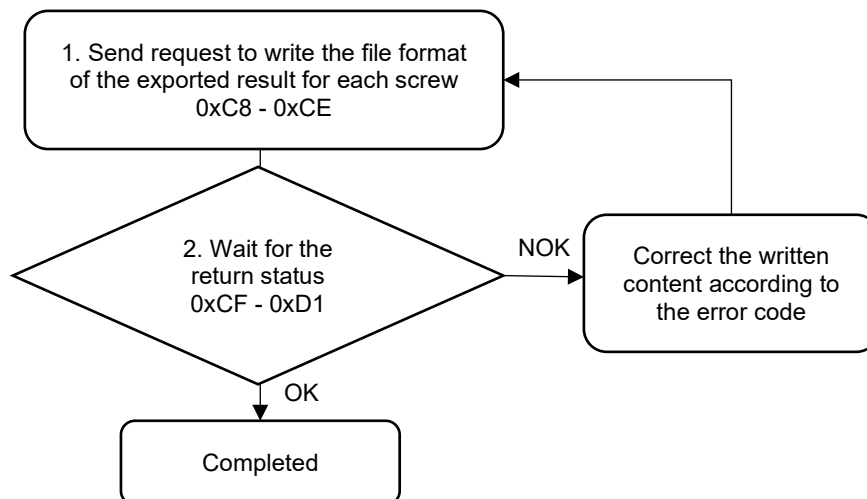
Code	Error description	Description
1	Display the operation warning window	Must be 0 or 1

#517 Write file format of the exported result for each screw

Content description:

Use function code #517 to set the file format of the exported result for each screw.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	517
C9	Version number	0
CA	File format of the exported result for each screw	0: disable 1: CSV file (HMI Disk) 2: CSV file (USB Disk) 3: BIN file (HMI Disk) 4: BIN file (USB Disk)
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	517
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

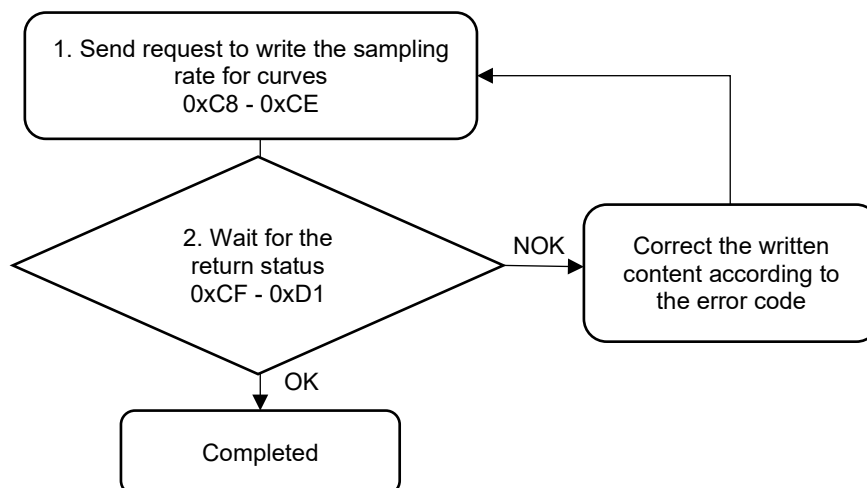
Code	Error description	Description
1	File format of the exported result for each screw	HMI kernel version 124 or later: exceeds the range of 0 to 4 HMI kernel version earlier than 124: exceeds the range of 0 to 2

#518 Write the sampling rate for curves

Content description:

Use function code #518 to set the sampling rate for curves.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	518
C9	Version number	0
CA	Sampling rate for curves	0: 10 ms→4 ms when start torque is reached 1: 2 ms; max. 4 seconds per production report entry 2: 2 ms; max. 8 seconds per production report entry 3: 1 ms; max. 4 seconds per production report entry 4: 1 ms; max. 8 seconds per production report entry
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	518
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

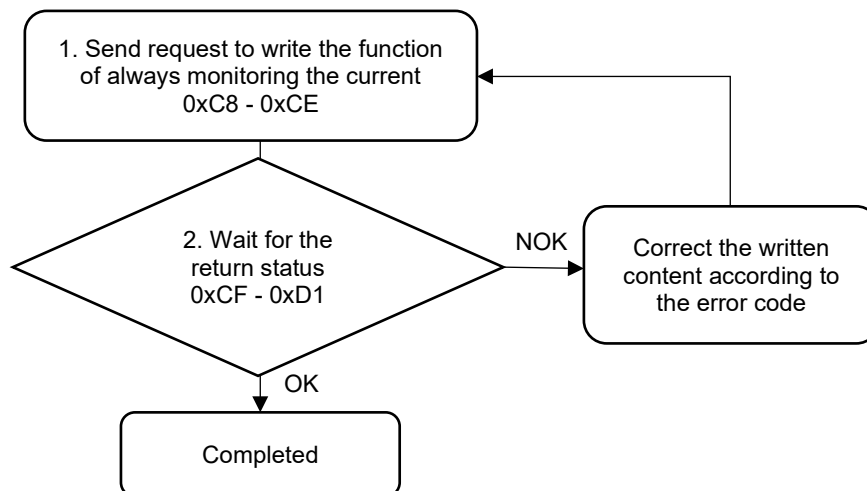
Code	Error description	Description
1	Sampling rate for curves	Exceeds the range of 0 to 4

#519 Write the function of always monitoring the current

Content description:

Use function code #519 to set whether to enable the function of always monitoring the current.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	519
C9	Version number	0
CA	Always monitor the current	0: On 1: Off
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	519
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

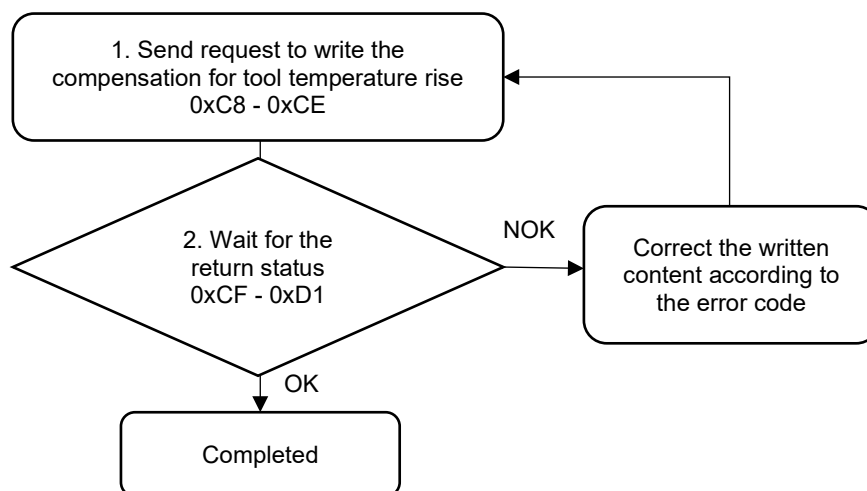
Code	Error description	Description
1	Always monitor the current	Must be 0 or 1

#520 Write the compensation for tool temperature rise

Content description:

Use function code #520 to set the compensation for tool temperature rise.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	520
C9	Version number	0
CA	Compensation for tool temperature rise	0: On 1: Off
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	520
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

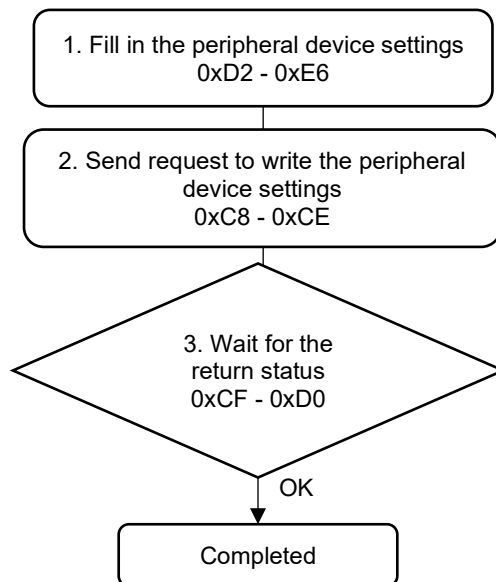
Code	Error description	Description
1	Compensation for tool temperature rise	Must be 0 or 1

#521 Write the peripheral device settings

Content description:

Use function code #521 to write the peripheral device settings.

Handshake signal description:



1. Fill in the settings of peripheral device to 0xD2 - 0xE6.

Modbus (Hex)	Function	Description
D2	Reserved	–
D3	RS-485 connector function	0: disabled 1: Tool 1 positioning arm (P65F0) 2: Tool 2 positioning arm (P65F0) 3: Tool 1 positioning arm (AM-TMTS-TA series) 4: Tool 2 positioning arm (AM-TMTS-TA series) 5: Tool 1 positioning arm (others) 6: Tool 2 positioning arm (others) 7: Tool 1 positioning arm (Modbus TCP slave) 8: Tool 2 positioning arm (Modbus TCP slave) 9: Tool 1 positioning arm (Modbus TCP master) 10: Tool 2 positioning arm (Modbus TCP master)
D4	Reserved	–
D5	Tool 1 positioning arm - target position tolerance (L)	Unit: mm
D6	Tool 1 positioning arm - target position tolerance (H)	
D7	Tool 1 - X axis position offset (L)	Unit: mm
D8	Tool 1 - X axis position offset (H)	
D9	Tool 1 - Y axis position offset (L)	Unit: mm
DA	Tool 1 - Y axis position offset (H)	
DB	Tool 1 - Z axis position offset (L)	Unit: mm
DC	Tool 1 - Z axis position offset (H)	
DD	Tool 2 positioning arm - target position tolerance (L)	Unit: mm

Modbus (Hex)	Function	Description
DE	Tool 2 positioning arm - target position tolerance (H)	
DF	Tool 2 - X axis position offset (L)	Unit: mm
E0	Tool 2 - X axis position offset (H)	
E1	Tool 2 - Y axis position offset (L)	Unit: mm
E2	Tool 2 - Y axis position offset (H)	
E3	Tool 2 - Z axis position offset (L)	Unit: mm
E4	Tool 2 - Z axis position offset (H)	
E5	L1 length of AM-TMTS-TA series	Unit: mm (default: 475 mm)
E6	L2 length of AM-TMTS-TA series	Unit: mm (default: 475 mm)

2. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	521
C9	Version number	0
CA to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

3. Check whether the reading is successful from the return status in 0xCF - 0xD0.

Modbus (Hex)	Return status	Description
CF	Function code	521
D0	Status	1: OK

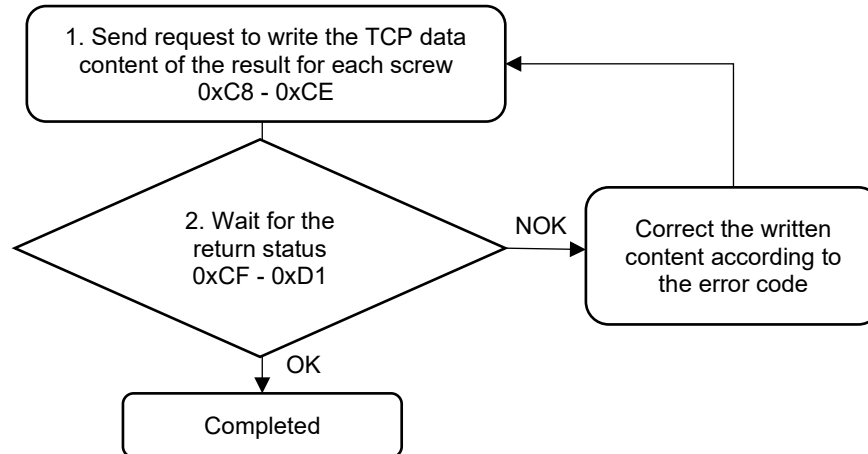
If successful, the read status (0xD0) is 1.

#522 Write the TCP data content of the result for each screw

Content description:

Use function code #522 to set the TCP data content of the result for each screw.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	522
C9	Version number	0
CA	TCP data content of the result for each screw	0: disable 1: scale, time, torque 2: scale, angle, torque 3: scale, angle, torque rate 4: result, scale, time, torque 5: result, scale, angle, torque 6: result, scale, angle, torque rate 7 - 11: reserved 12: result, scale (tightening or loosening) 13: result, scale (tightening) 14: result, scale, all curve data (tightening or loosening) 15: result, scale, all curve data (tightening)
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

Note:

1. When the data content includes result and scale, only packet 1 (Query type 10 in #750 and #751) is sent.

2. When the data content includes result, scale, and all curve data, the following packets are sent:

Packet 1: query type 10 in #750 and #751

Packet 2: time point data

Packet 3: angle point data

Packet 4: torque point data (double word)

Packet 5: torque rate point data (double word)

3. Tightening: one packet is sent only after a tightening operation is complete.

Loosening: one packet is sent only after a loosening operation is complete.

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	522
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

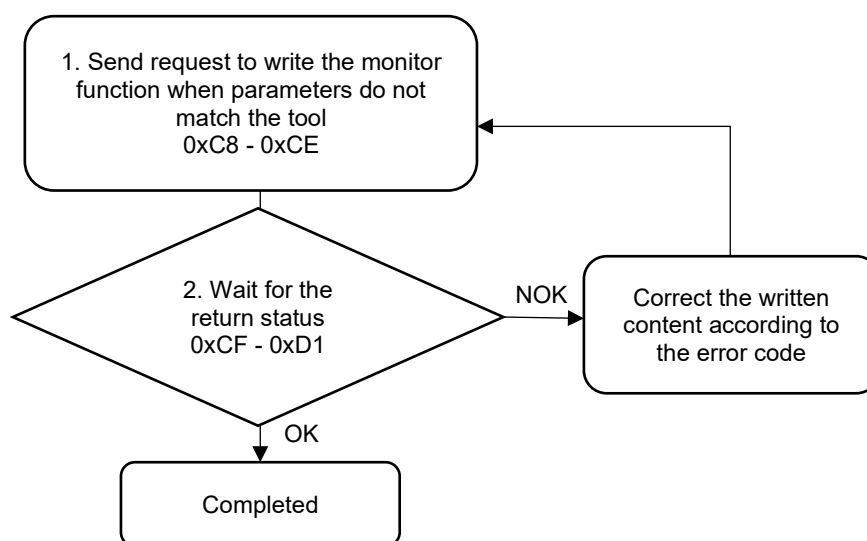
Code	Error description	Description
1	TCP data content of the result for each screw	Exceeds the range of 0 to 15

#523 Write the monitor function when parameters do not match the tool

Content description:

Use function code #523 to set the function for monitoring if the tool torque of the operating parameter matches that of the connected tool.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	523
C9	Version number	0
CA	Monitor function when parameters do not match the tool	0: On 1: Off
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	523
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

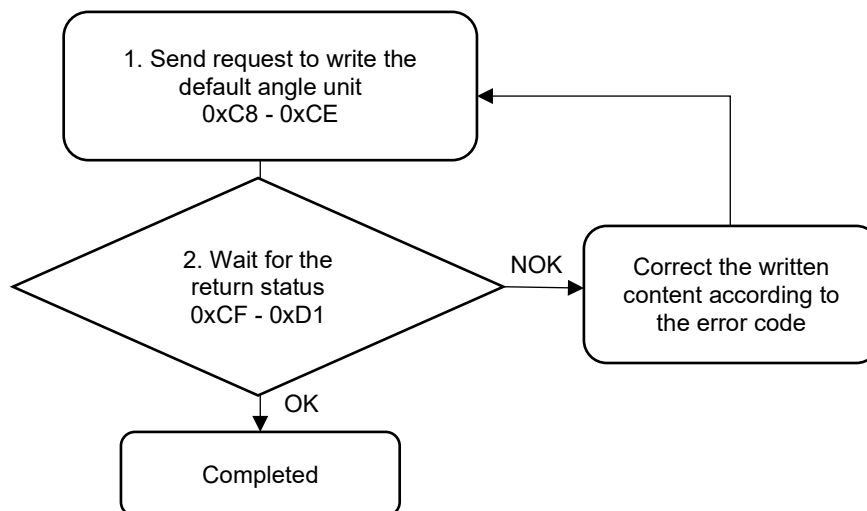
Code	Error description	Description
1	Monitor function when parameters do not match the tool	Must be 0 or 1

#524 Write the default angle unit

Content description:

Use function code #524 to set the default angle unit of the controller.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	524
C9	Version number	0
CA	Default angle unit	0: degree 1: turn
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	524
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

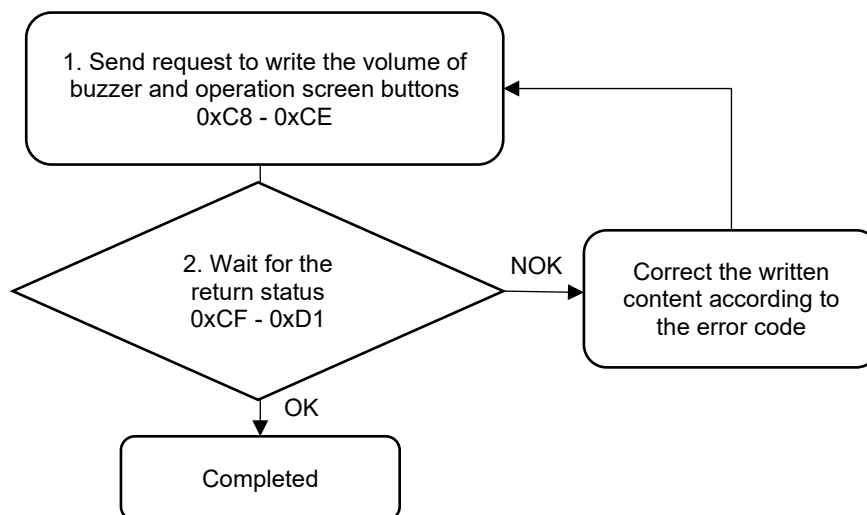
Code	Error description	Description
1	Default angle unit	Must be 0 or 1

#525 Write the volume of buzzer and operation screen buttons

Content description:

Use function code #525 to set the volume of buzzer and operation screen buttons.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	525
C9	Version number	0
CA	Volume of buzzer	0 to 100
CB	Volume of operation screen buttons	0 to 100
CC, CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	525
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

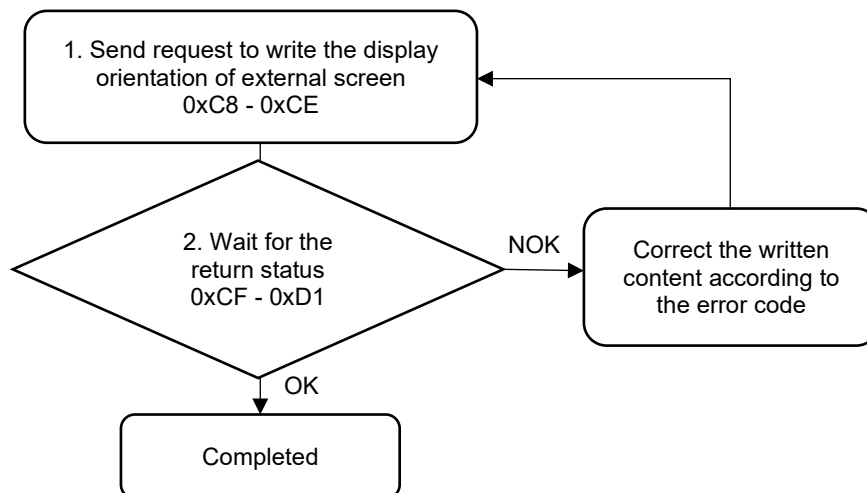
Code	Error description	Description
1	Volume of buzzer	Exceeds the range of 0 to 100
2	Volume of operation screen buttons	Exceeds the range of 0 to 100

#526 Write the display orientation of external screen

Content description:

Use function code #526 to set the display orientation of the external screen (HDMI).

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	526
C9	Version number	0
CA	Display orientation	0: portrait 1: landscape
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	526
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

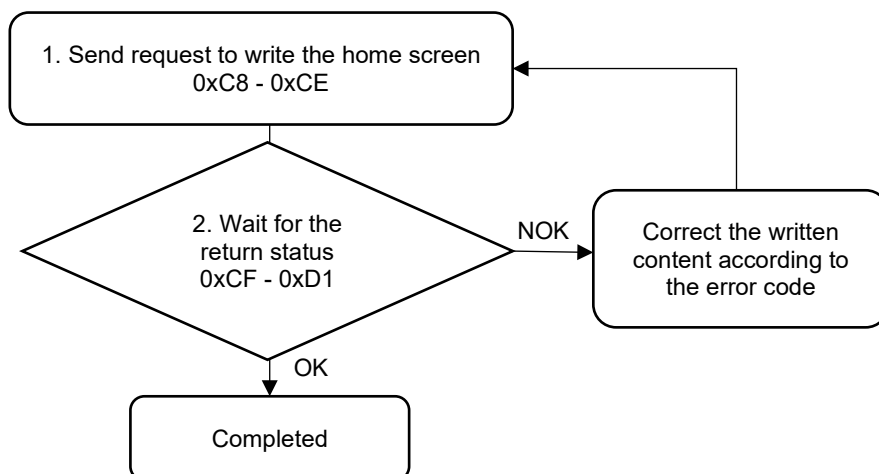
Code	Error description	Description
1	Display orientation	Must be 0 or 1

#527 Write the home screen

Content description:

Use function code #527 to set the home screen of the controller.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	527
C9	Version number	0
CA	Home screen	0: Home 1: Results
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	527
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

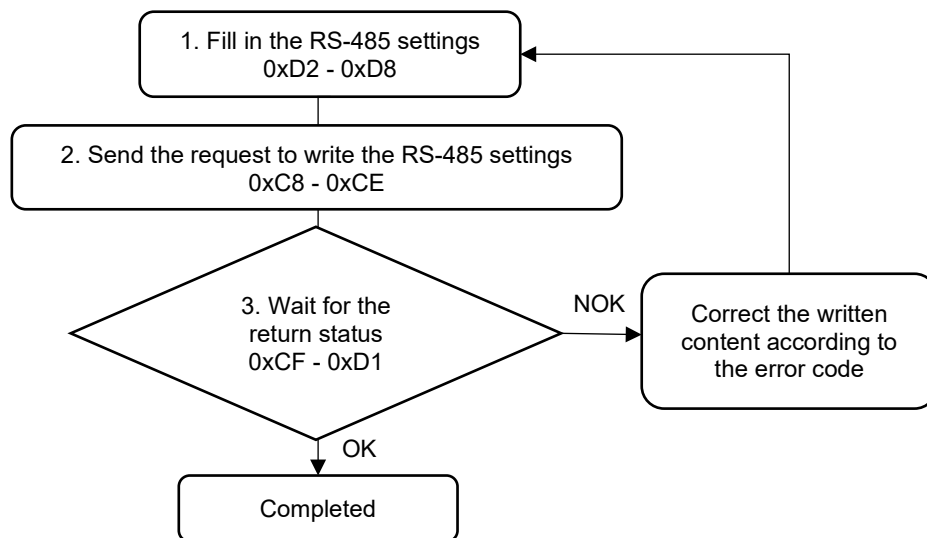
Code	Error description	Description
1	Home screen	Must be 0 or 1

#528 Write the RS-485 settings

Content description:

Use function code #528 to write the RS-485 settings.

Handshake signal description:



1. Fill in the RS-485 settings to 0xD2 - 0xD8.

Modbus (Hex)	Function	Description
D2	Function On / Off	0: Off; 1: On
D3	Station	0 to 255
D4	RTU / ASCII	0: RTU; 1: ASCII
D5	Baud rate	0: 9600; 1: 19200; 2: 38400
D6	Data bit	0: 8-bit; 1: 7-bit
D7	Parity bit	0: None; 1: Odd; 2: Even
D8	Stop bit	0: 2-bit; 1: 1-bit

2. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	528
C9	Version number	0
CA to CD	Reserved	-
CE	Request to send command	(to be filled in last)

3. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	528
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

Code	Error description	Description
1	Function On / Off	Must be 0 or 1
2	Station	Exceeds the range of 0 to 255
3	RTU / ASCII	Must be 0 or 1

Code	Error description	Description
4	Baud rate	Exceeds the range of 0 to 2
5	Data bit	Must be 0 or 1
6	Parity bit	Exceeds the range of 0 to 2
7	Stop bit	Must be 0 or 1

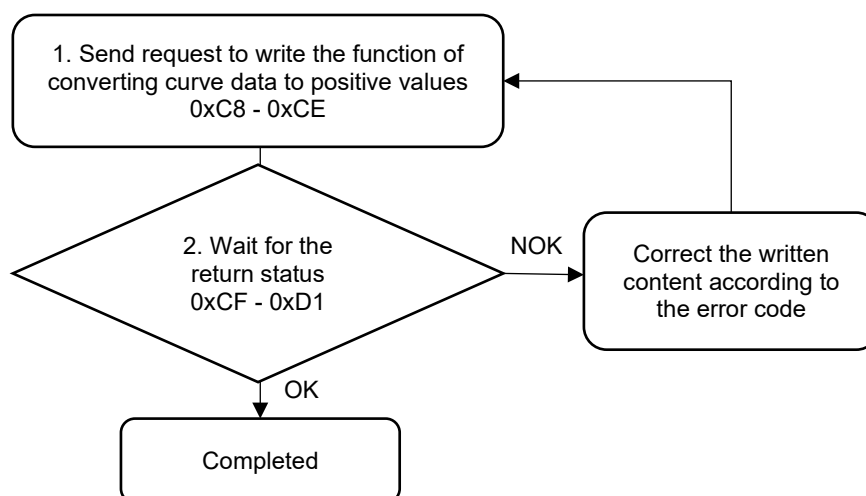
A

#529 Write the function of converting curve data to positive values

Content description:

Use function code #529 to set the function of converting curve data to positive values.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	529
C9	Version number	0
CA	Function of converting curve data to positive values	0: On 1: Off
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	529
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

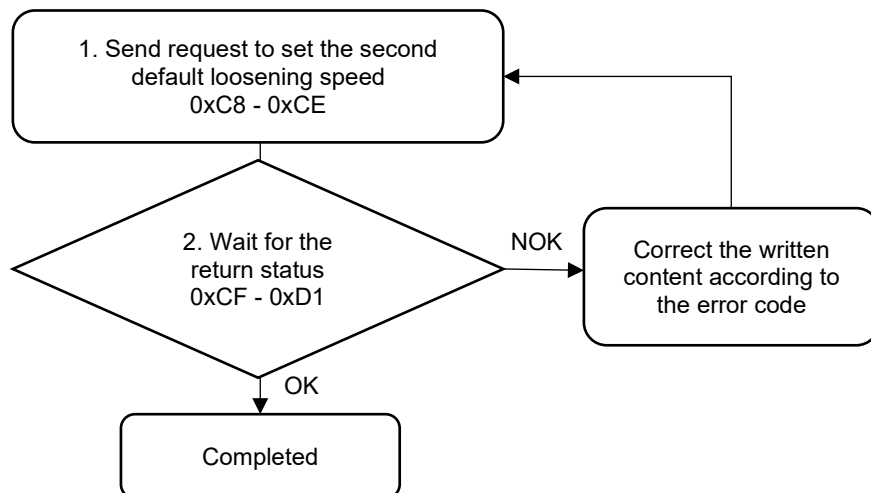
Code	Error description	Description
1	Function of converting curve data to positive values	Must be 0 or 1

#530 Write the default loosening second stage speed

Content description:

Use function code #530 to set the 2nd default loosening speed.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	530
C9	Version number	0
CA	2 nd default loosening speed	Unit: rpm
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	530
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

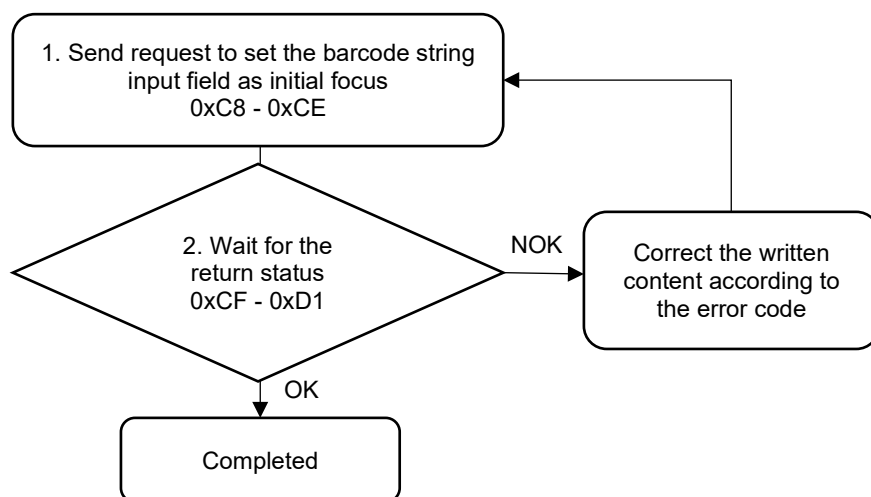
Code	Error description	Description
1	2 nd loosening speed	Exceeds the range of 10 - 2000

#531 Write the keyboard cursor blink in Results page

Content description:

Use function code #531 to set the barcode string input field as the initial focus in the **Results** tab of the **Results** page.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	531
C9	Version number	0
CA	Set the barcode string input field as the initial focus	0: Off 1: On
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	531
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

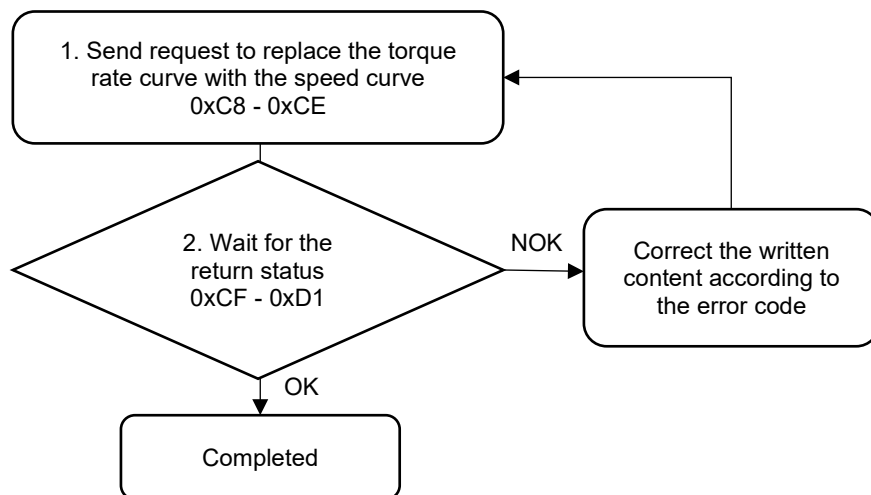
Code	Error description	Description
1	Set the barcode string input field as the initial focus	Exceeds the range of 0 - 1

#532 Write to replace torque rate curve with speed curve

Content description:

Use function code #532 to show speed curve and torque curve in the **Curve** tab of the **Results** page.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	532
C9	Version number	0
CA	Replace torque rate curve with speed curve	0: show torque rate curve and torque curve (default) 1: show speed curve and torque curve
CB to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	532
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

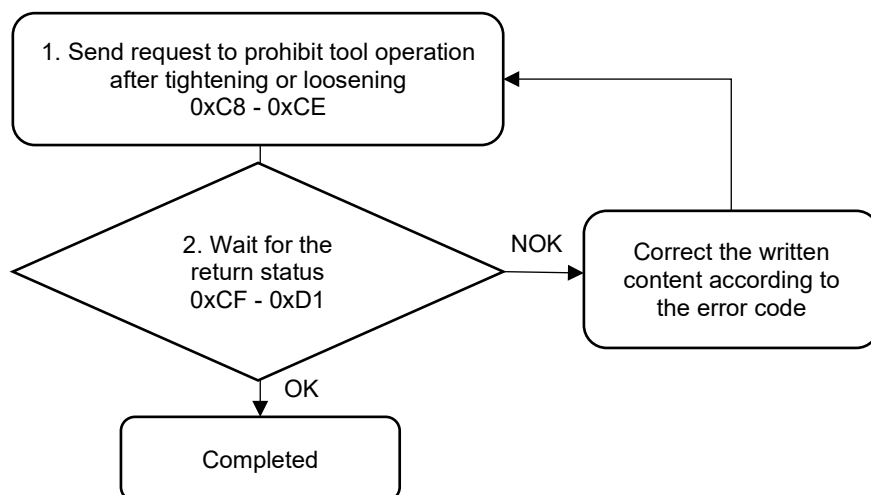
Code	Error description	Description
1	Replace torque rate curve with speed curve	Exceeds the range of 0 - 1

#533 Write to prohibit tool operation after each tightening or loosening

Content description:

Use function code #533 to prohibit tool operation after each tightening or loosening.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	533
C9	Version number	0
CA	Prohibit tool operation after tightening or loosening	0: Off 1: prohibit operation after tightening 2: prohibit operation after loosening 3: prohibit operation after tightening or loosening
CB to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	533
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

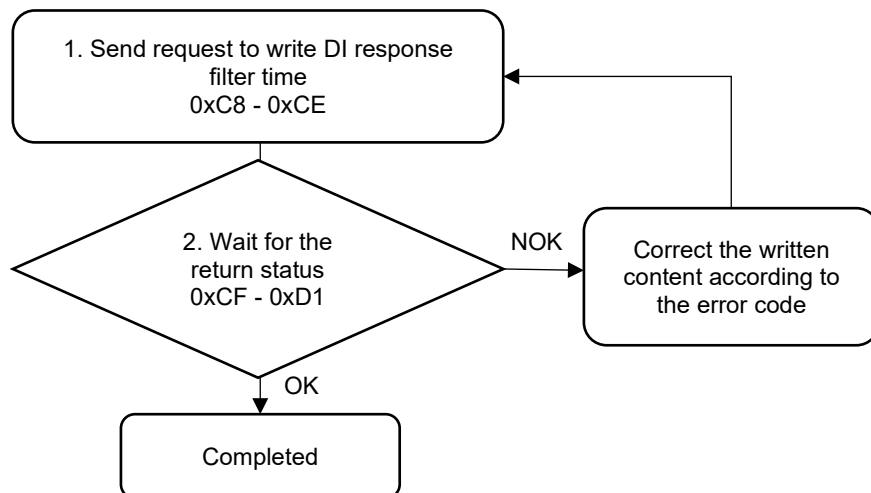
Code	Error description	Description
1	Prohibit tool operation after tightening or loosening	Exceeds the range of 0 - 3

#534 Write the DI response filter time

Content description:

Use function code #534 to set the DI response filter time.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	534
C9	Version number	0
CA	DI response filter time	Unit: ms
CB to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	534
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

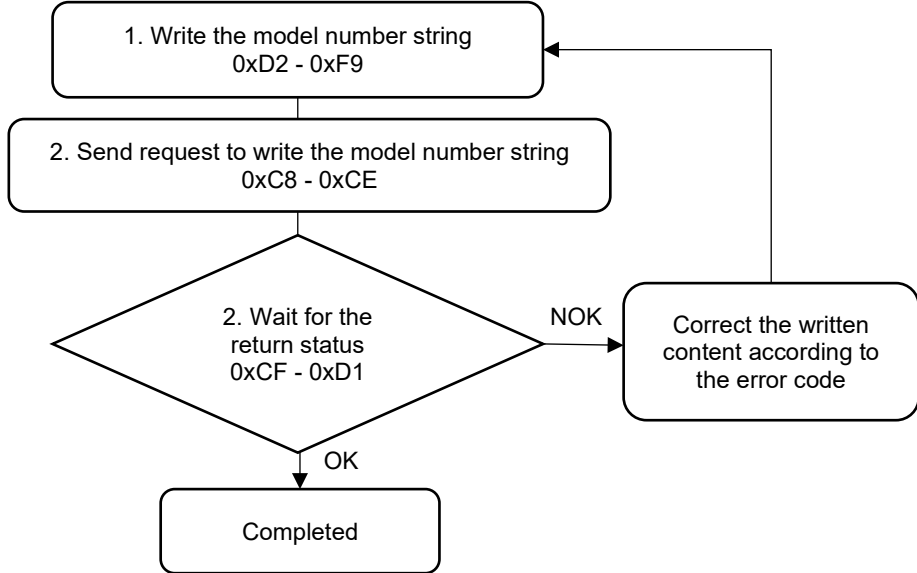
Code	Error description	Description
1	DI response filter time	Exceeds the range of 0 - 1000

#535 Write the model number

Content description:

Use function code #535 to input the model number.

Handshake signal description:



1. Write the model number string to 0xD2 - 0xF9.

Modbus (Hex)	Function	Description
D2 to F9	Model number string	ASCII code (40 words)

2. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	535
C9	Version number	0
CA to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

3. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	535
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

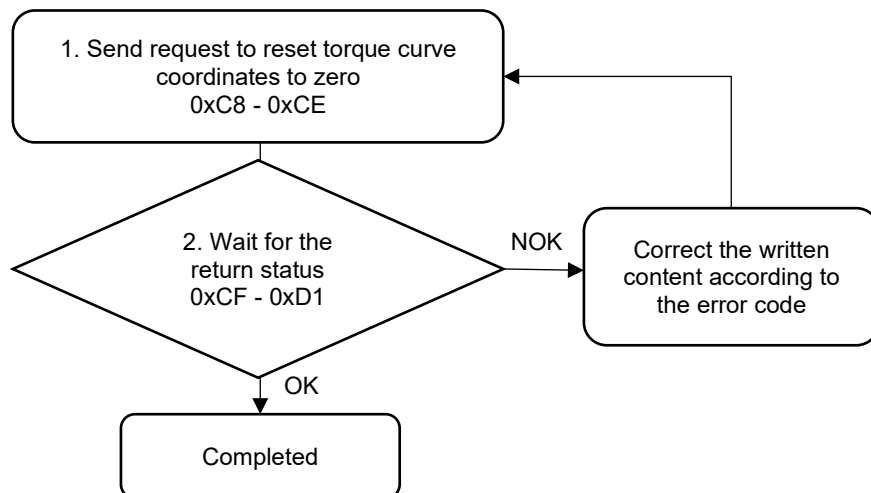
A

#536 Write to reset the torque curve coordinates to zero

Content description:

Use function code #536 to reset the torque curve coordinates to zero.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	536
C9	Version number	0
CA	Reset torque cruve coordinates to zero	0: Off (default) 1: On
CB to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	536
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

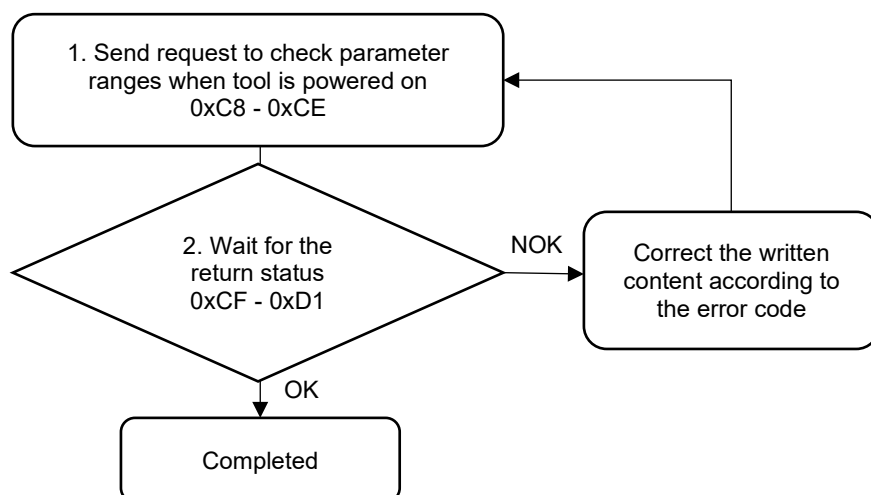
Code	Error description	Description
1	Reset torque curve coordinates to zero	Exceeds the range of 0 - 1

#537 Write to check parameter ranges when tool is powered on

Content description:

Use function code #537 to enable or disable the function of checking tool parameter ranges when the tool is powered on.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	537
C9	Version number	0
CA	Check parameter ranges when tool is powered on	0: On (default) 1: Off
CB to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	537
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

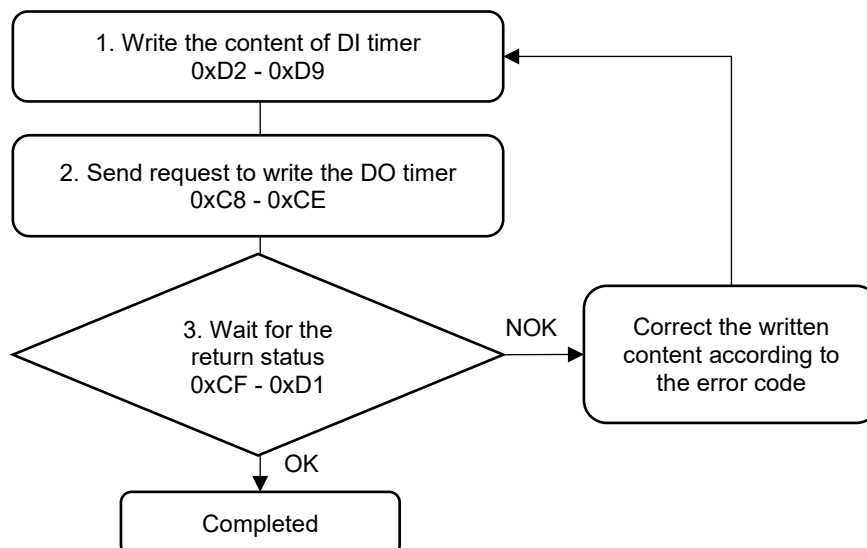
Code	Error description	Description
1	Check parameter ranges when tool is powered on	Exceeds the range of 0 - 1

#538 Write the DO timer

Content description:

Use function code #538 to enable or disable the DO timer. When the DO is On, the timer starts to count; when the set duration is reached, the DO automatically switches from On to Off.

Handshake signal description:



1. Write the contents of DO timer to 0xD2 - 0xD9.

Modbus (Hex)	Function	Description
D2	DO1 timer	0: Off 1 - 3000: timer duration in units of millisecond
D3	DO2 timer	
D4	DO3 timer	
D5	DO4 timer	
D6	DO5 timer	
D7	DO6 timer	
D8	DO7 timer	
D9	DO8 timer	

2. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	538
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

3. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	538
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

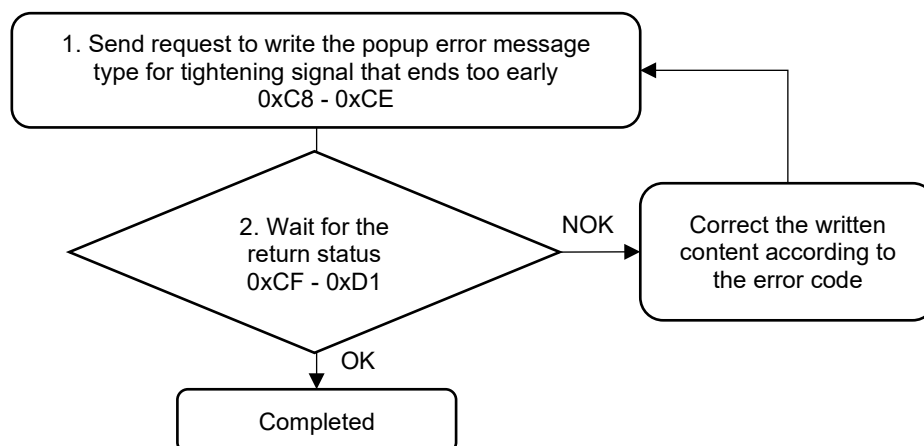
Code	Error description	Description
1	Tool 1 / Tool 2	Exceeds the range of 0 - 1

#539 Write the popup error message type for tightening signal that ends too early

Content description:

Use function code #539 to set the type of the popup error message for the tightening signal that ends too early.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	539
C9	Version number	0
CA	Popup error message type	0: error 1: warning
CB to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	539
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

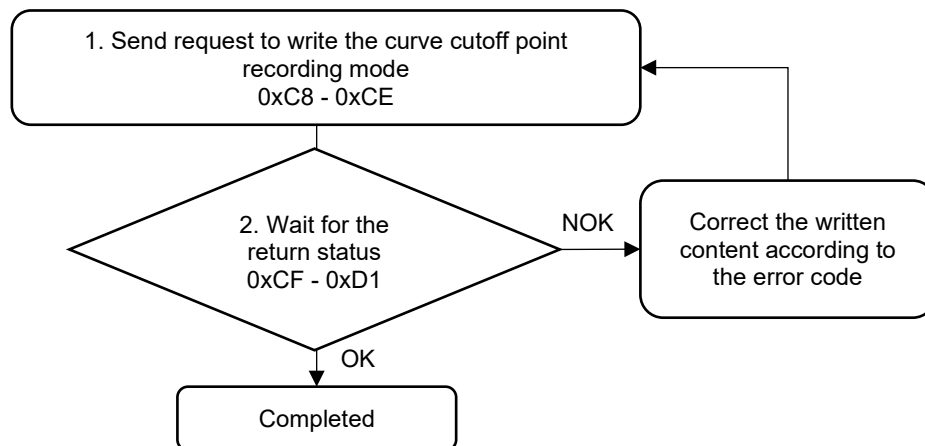
Code	Error description	Description
1	Popup error message type	Exceeds the range of 0 - 1

#540 Write the curve cutoff point recording mode

Content description:

Use function code #540 to set the curve cutoff point recording mode.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	540
C9	Version number	0
CA	Curve cutoff point recording mode	0: the entire process 1: peak value
CB to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	540
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

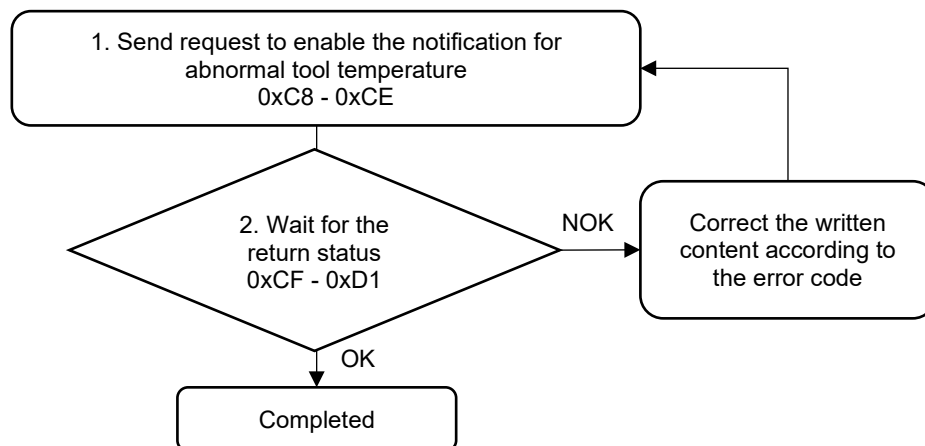
Code	Error description	Description
1	Curve cutoff point recording mode	Exceeds the range of 0 - 1

#541 Write to enable the notification for abnormal tool temperature

Content description:

Use function code #541 to enable or disable the notification for abnormal tool temperature.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	541
C9	Version number	0
CA	Notification for abnormal tool temperature	0: On (default) 1: Off
CB to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	541
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

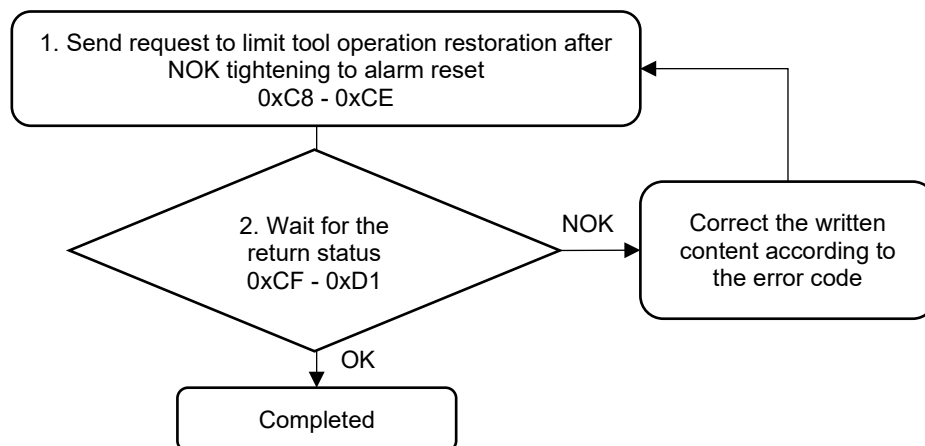
Code	Error description	Description
1	Notification for abnormal tool temperature	Exceeds the range of 0 - 1

#542 Write to limit tool operation restoration after NOK tightening to alarm reset

Content description:

Use function code #542 to set whether the tool operation after NOK tightening can be restored by alarm reset.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	542
C9	Version number	0
CA	Limit the tool operation restoration after NOK tightening to alarm reset	0: Off 1: On
CB to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	542
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

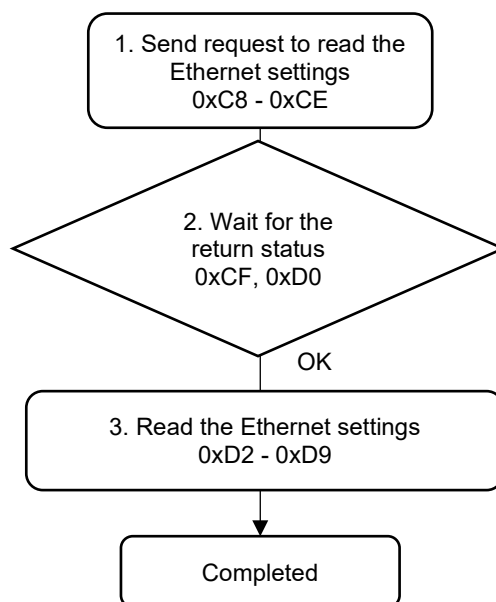
Code	Error description	Description
1	Limit the tool operation restoration after NOK tightening to alarm reset	Exceeds the range of 0 - 1

#550 Read the Ethernet settings

Content description:

Use function code #550 to read the Ethernet settings.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	550
C9	Version number	0
CA to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF and 0xD0.

Modbus (Hex)	Return status	Description
CF	Function code	550
D0	Status	1: OK

If successful, the read status (0xD0) is 1.

3. Read the Ethernet settings in 0xD2 - 0xD9.

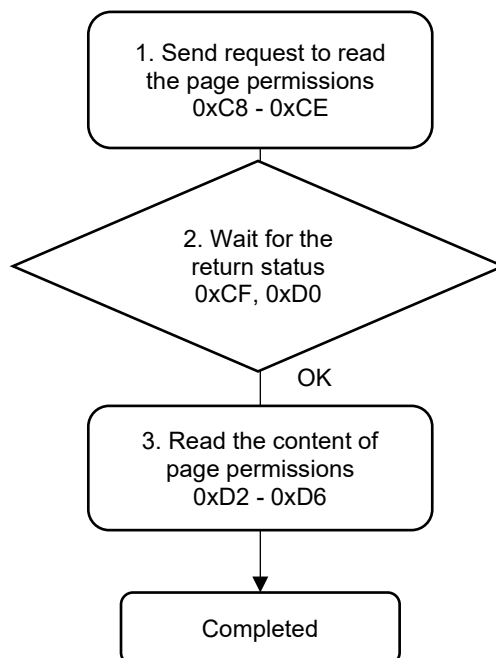
Modbus (Hex)	Function	Description
D2	IP1 address	Default: 192
D3	IP2 address	Default: 168
D4	IP3 address	Default: 1
D5	IP4 address	Default: 11
D6	Subnet mask 1	Default: 255
D7	Subnet mask 2	Default: 255
D8	Subnet mask 3	Default: 255
D9	Subnet mask 4	Default: 0

#551 Read the page permissions

Content description:

Use function code #551 to read the permissions of page access and edit for different accounts.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	551
C9	Version number	0
CA to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF and 0xD0.

Modbus (Hex)	Return status	Description
CF	Function code	551
D0	Status	1: OK

If successful, the read status (0xD0) is 1.

3. Read the page permissions in 0xD2 - 0xD6.

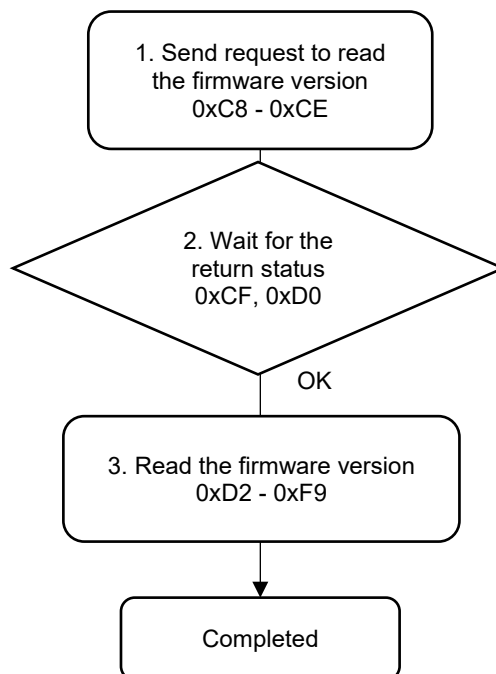
Modbus (Hex)	Function	Description
D2	User1 page permissions	Bit 0: Parameters Bit 1: Sequence
D3	User2 page permissions	Bit 2: Sources Bit 3: Controller
D4	User3 page permissions	Bit 4: Tool Bit 5: operation for screw progress
D5	User4 page permissions	Bit 6: clearing of the production report entries
D6	User5 page permissions	Bit 7: clearing of the error, warning, and button report entries

#552 Read the firmware version

Content description:

Use function code #552 to read the firmware version of the controller.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	552
C9	Controller firmware version	0
CA to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF and 0xD0.

Modbus (Hex)	Return status	Description
CF	Function code	552
D0	Status	1: OK

If successful, the read status (0xD0) is 1.

3. Read the controller firmware version in 0xD2 - 0xF9.

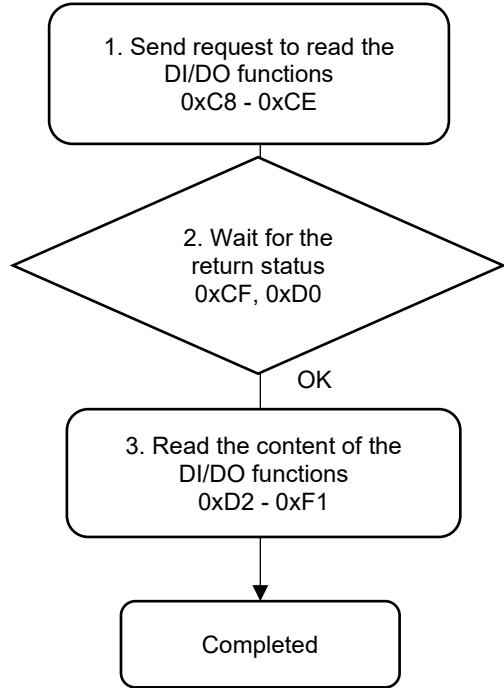
Modbus (Hex)	Function	Description
D2 to F9	Controller firmware version	ASCII code (40 words)

#553 Read the DI/DO functions

Content description:

Use function code #553 to read the DI/DO functions.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	553
C9	Version number	0
CA to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF and 0xD0.

Modbus (Hex)	Return status	Description
CF	Function code	553
D0	Status	1: OK

If successful, the read status (0xD0) is 1.

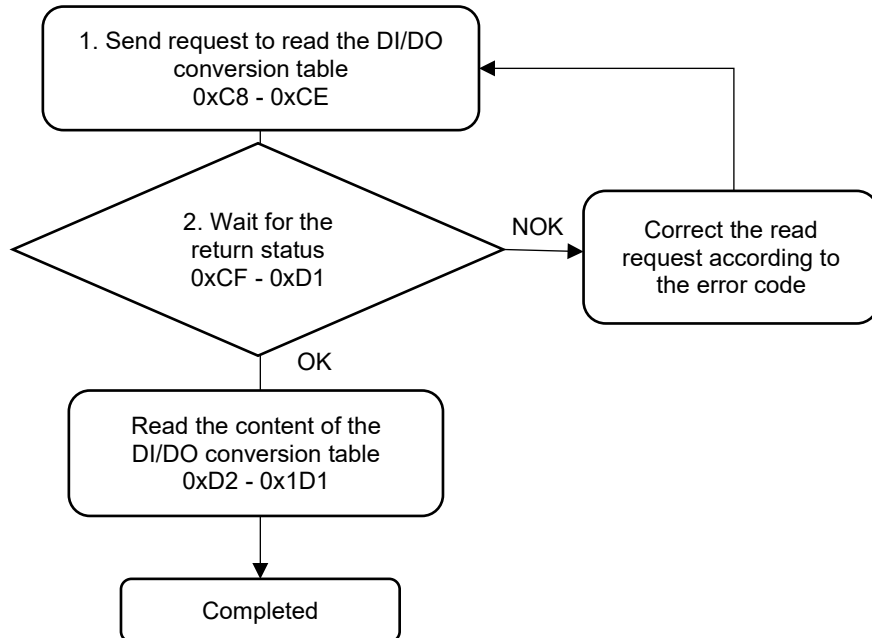
3. Read the content of the DI/DO functions in 0xD2 - 0xF1. Refer to function code #507 for the function list.

#554 Read the DI/DO conversion table

Content description:

Use function code #554 to read information about the DI/DO conversion table.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	554
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB	Conversion table	0: bit ID output table 1: bit ID input table 2: parameter ID output table 4: screw progress output table 6: sequence ID output table
CC, CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	554
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	Conversion table	Must be 0, 1, 2, 4, or 6

3. Read the content of the DI/DO conversion table in 0xD2 - 0xD1. Refer to function code #508 for the function list.

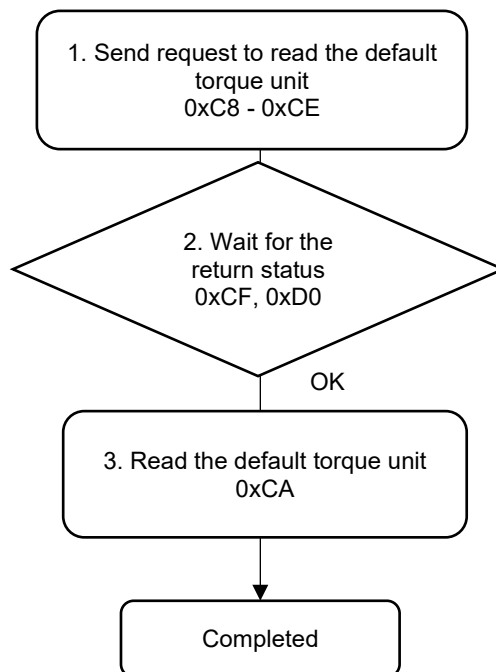
A

#555 Read the default torque unit

Content description:

Use function code #555 to read the default torque unit of the controller.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	555
C9	Version number	0
CA to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF and 0xD0.

Modbus (Hex)	Return status	Description
CF	Function code	555
D0	Status	1: OK

If successful, the read status (0xD0) is 1.

3. Read the default torque unit in 0xCA.

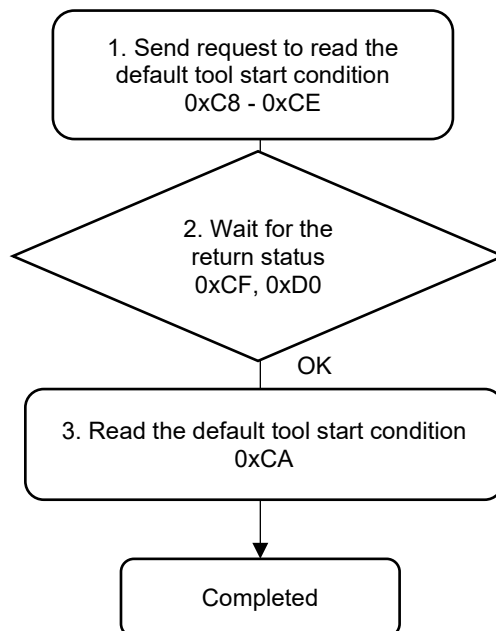
Modbus (Hex)	Write / Read request	Read
CA	Default torque unit	0: N.m 1: kgf.cm 2: lbf.ft 3: lbf.in 4: ozf.ft 5: ozf.in 6: cNm

#556 Read the default tool start condition

Content description:

Use function code #556 to read the default tool start condition.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	556
C9	Version number	0
CA to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF -and 0xD0.

Modbus (Hex)	Return status	Description
CF	Function code	556
D0	Status	1: OK

If successful, the read status (0xD0) is 1.

3. Read the default tool start condition in 0xCA.

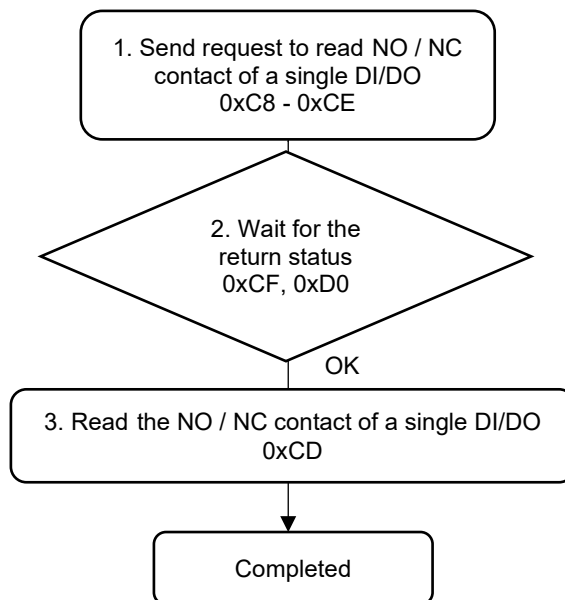
Modbus (Hex)	Write / Read request	Read
CA	Default tool start condition	0: push start 1: DI 2: lever start 3: push start or lever start 4: push start and lever start 5: DI or push start 6: DI or lever start

#557 Read NO / NC contact setting for a single DI/DO

Content description:

Use function code #557 to read the NO / NC contact of a single DI/DO.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	557
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB	DO / DI	0: DO; 1: DI
CC	Corresponding bit of DO / DI	0 to 7
CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF and 0xD0.

Modbus (Hex)	Return status	Description
CF	Function code	557
D0	Status	1: OK

If successful, the read status (0xD0) is 1.

3. Read the NO / NC contact of a single DI/DO in 0xCD.

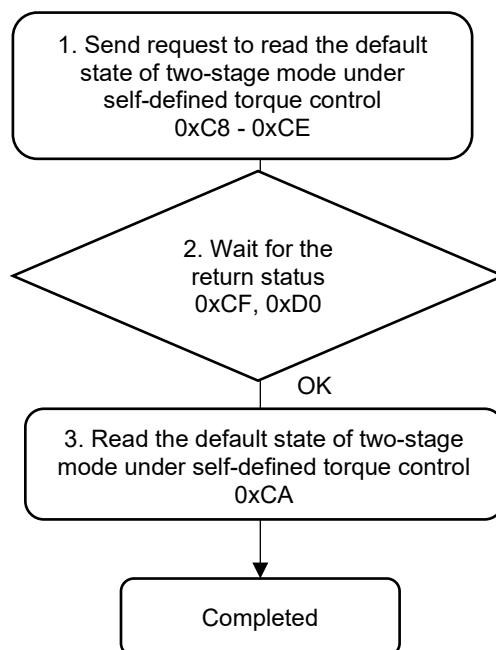
Modbus (Hex)	Write / Read request	Read
CD	NO / NC	0: NO (normally open) 1: NC (normally closed)

#558 Read the default two-stage mode under self-defined torque control

Content description:

Use function code #558 to read the default state of two-stage mode under self-defined torque control.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	558
C9	Version number	0
CA to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF and 0xD0.

Modbus (Hex)	Return status	Description
CF	Function code	558
D0	Status	1: OK

If successful, the read status (0xD0) is 1.

3. Read the default state of two-stage mode under self-defined torque control in 0xCA.

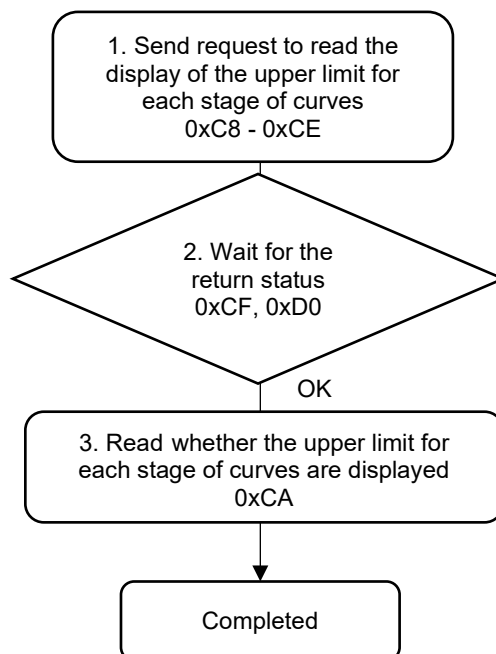
Modbus (Hex)	Write / Read request	Read
CA	Two-stage mode	0: Off 1: On

#559 Read the display of upper limit for each stage of curves

Content description:

Use function code #559 to read whether the upper limit for each stage of curves is displayed.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	559
C9	Version number	0
CA to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF and 0xD0.

Modbus (Hex)	Return status	Description
CF	Function code	559
D0	Status	1: OK

If successful, the read status (0xD0) is 1.

3. Read whether the upper limit for each stage of curves is displayed in 0xCA.

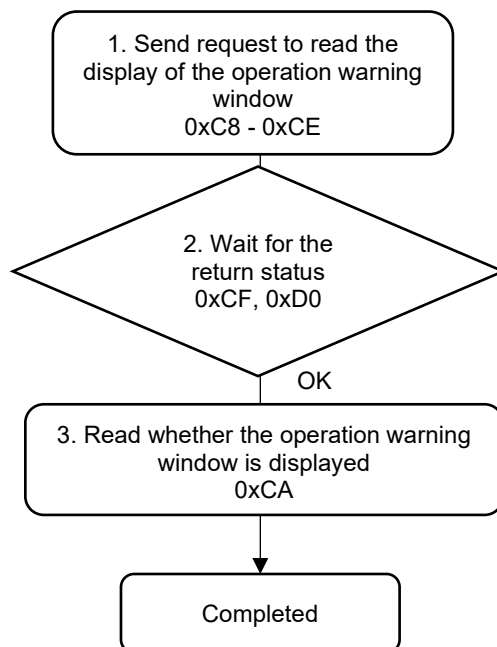
Modbus (Hex)	Write / Read request	Read
CA	Display the upper limit for each stage of curves	0: On 1: Off

#560 Read the display of operation warning window

Content description:

Use function code #560 to read whether the operation warning window is displayed.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	560
C9	Version number	0
CA to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF and 0xD0.

Modbus (Hex)	Return status	Description
CF	Function code	560
D0	Status	1: OK

If successful, the read status (0xD0) is 1.

3. Read whether the operation warning window is displayed in 0xCA.

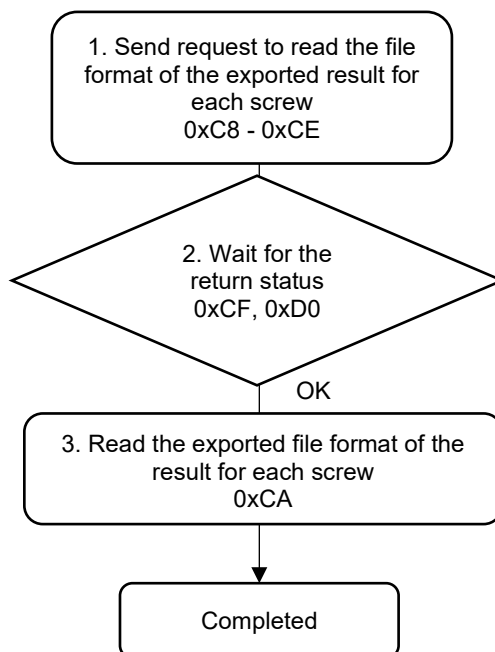
Modbus (Hex)	Write / Read request	Read
CA	Display the operation warning window	0: On 1: Off

#561 Read file format of the exported result for each screw

Content description:

Use function code #561 to read the file format of the exported result for each screw.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	561
C9	Version number	0
CA to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF and 0xD0.

Modbus (Hex)	Return status	Description
CF	Function code	561
D0	Status	1: OK

If successful, the read status (0xD0) is 1.

3. Read the file format of the exported result for each screw in 0xCA.

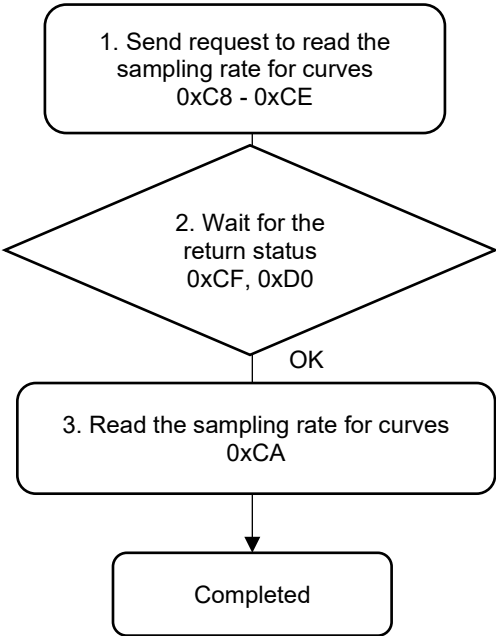
Modbus (Hex)	Write / Read request	Read
CA	File format of the exported result for each screw	0: disable 1: CSV file (HMI Disk) 2: CSV file (USB Disk) 3: BIN file (HMI Disk) 4: BIN file (USB Disk)

#562 Read the sampling rate for curves

Content description:

Use function code #562 to read the sampling rate for curves.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	562
C9	Version number	0
CA to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF and 0xD0.

Modbus (Hex)	Return status	Description
CF	Function code	562
D0	Status	1: OK

If successful, the read status (0xD0) is 1.

3. Read the sampling rate for curves in 0xCA.

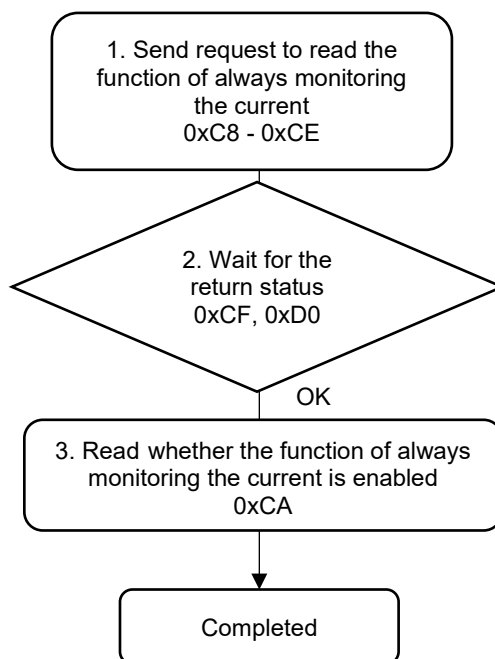
Modbus (Hex)	Write / Read request	Read
CA	Sampling rate for curves	0: 10 ms→4 ms when start torque is reached 1: 2 ms; max. 4 seconds per production report entry 2: 2 ms; max. 8 seconds per production report entry 3: 1 ms; max. 4 seconds per production report entry 4: 1 ms; max. 8 seconds per production report entry

#563 Read the function of always monitoring the current

Content description:

Use function code #563 to read whether the function of always monitoring the current is enabled.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	563
C9	Version number	0
CA to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF and 0xD0.

Modbus (Hex)	Return status	Description
CF	Function code	563
D0	Status	1: OK

If successful, the read status (0xD0) is 1.

3. Read whether the function of always monitoring the current is enabled in 0xCA.

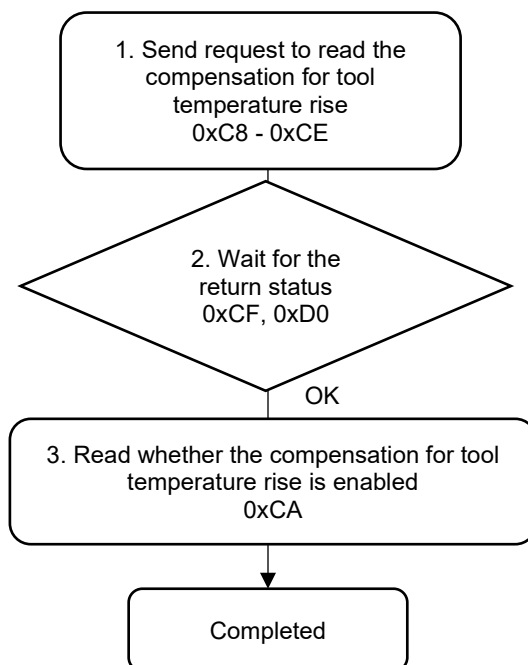
Modbus (Hex)	Write / Read request	Read
CA	Always monitor the current	0: On 1: Off

#564 Read the compensation for tool temperature rise

Content description:

Use function code #564 to read whether the compensation for tool temperature rise is enabled.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	564
C9	Version number	0
CA to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF and 0xD0.

Modbus (Hex)	Return status	Description
CF	Function code	564
D0	Status	1: OK

If successful, the read status (0xD0) is 1.

3. Read whether the compensation for tool temperature rise is enabled in 0xCA.

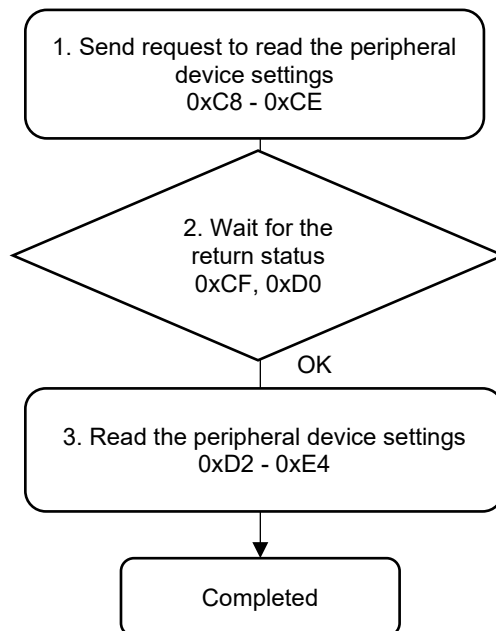
Modbus (Hex)	Write / Read request	Read
CA	Compensation for tool temperature rise	0: On 1: Off

#565 Read the peripheral device settings

Content description:

Use function code #565 to read the peripheral device settings.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	565
C9	Version number	0
CA to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF and 0xD0.

Modbus (Hex)	Return status	Description
CF	Function code	565
D0	Status	1: OK

If successful, the read status (0xD0) is 1.

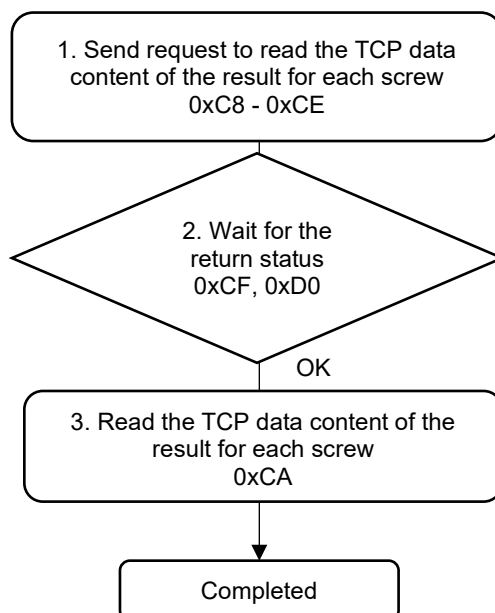
3. Read the peripheral device settings in 0xD2 - 0xE4. Refer to function code #521 for the function list.

#566 Read the TCP data content of the result for each screw

Content description:

Use function code #566 to read the TCP data content of the result for each screw.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	566
C9	Version number	0
CA to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF and 0xD0.

Modbus (Hex)	Return status	Description
CF	Function code	566
D0	Status	1: OK

If successful, the read status (0xD0) is 1.

3. Read the TCP data content of the result for each screw in 0xCA.

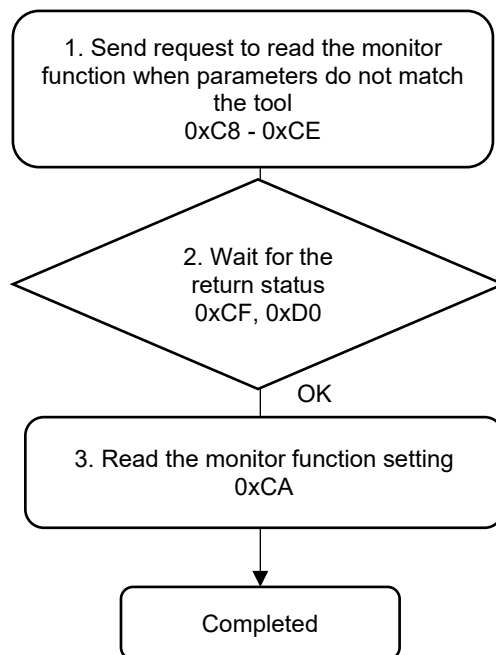
Modbus (Hex)	Write / Read request	Read
CA	TCP data content of the result for each screw	0: disable 1: scale, time, torque 2: scale, angle, torque 3: scale, angle, torque rate 4: result, scale, time, torque 5: result, scale, angle, torque 6: result, scale, angle, torque rate 7 - 11: reserved 12: result, scale (tightening or loosening) 13: result, scale (tightening) 14: result, scale, all curve data (tightening or loosening) 15: result, scale, all curve data (tightening)

#567 Read the monitor function when parameters do not match the tool

Content description:

Use function code #567 to read the function setting for monitoring if the tool torque of the operating parameter matches that of the connected tool.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	567
C9	Version number	0
CA to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF and 0xD0.

Modbus (Hex)	Return status	Description
CF	Function code	567
D0	Status	1: OK

If successful, the read status (0xD0) is 1.

3. Read the monitor function setting in 0xCA.

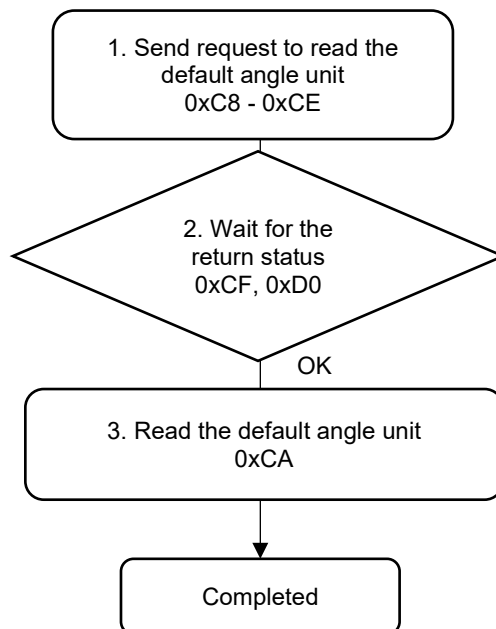
Modbus (Hex)	Write / Read request	Read
CA	Monitor function when parameters do not match the tool	0: On 1: Off

#568 Read the default angle unit

Content description:

Use function code #568 to read the default angle unit of the controller.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	568
C9	Version number	0
CA to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF and 0xD0.

Modbus (Hex)	Return status	Description
CF	Function code	568
D0	Status	1: OK

If successful, the read status (0xD0) is 1.

3. Read the default angle unit in 0xCA.

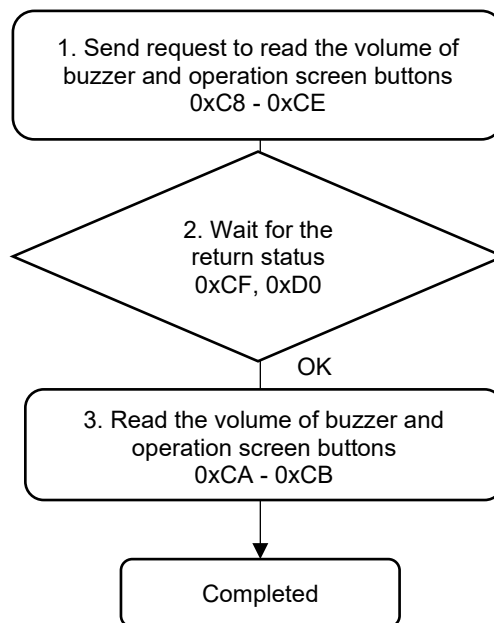
Modbus (Hex)	Write / Read request	Read
CA	Default angle unit	0: degree 1: turn

#569 Read the volume of buzzer and operation screen buttons

Content description:

Use function code #569 to read the volume of buzzer and operation screen buttons.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	569
C9	Version number	0
CA to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF and 0xD0.

Modbus (Hex)	Return status	Description
CF	Function code	569
D0	Status	1: OK

If successful, the read status (0xD0) is 1.

3. Read the volume of buzzer and operation screen buttons in 0xCA - 0xCB.

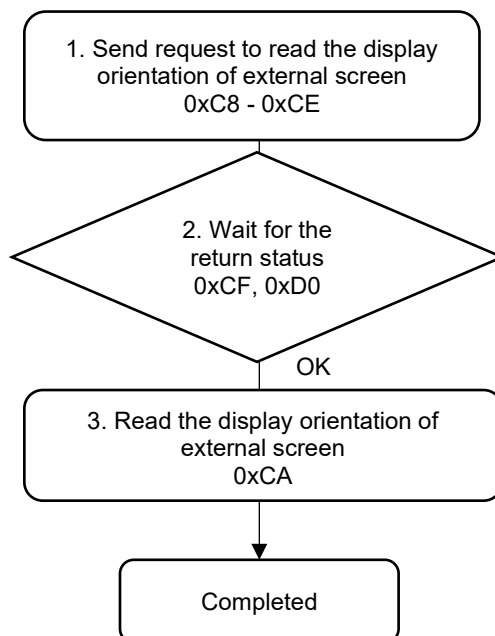
Modbus (Hex)	Write / Read request	Read
CA	Volume of buzzer	0 to 100
CB	Volume of operation screen buttons	0 to 100

#570 Read the display orientation of external screen

Content description:

Use function code #570 to read the display orientation of external screen (HDMI).

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	570
C9	Version number	0
CA to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF and 0xD0.

Modbus (Hex)	Return status	Description
CF	Function code	570
D0	Status	1: OK

If successful, the read status (0xD0) is 1.

3. Read the display orientation of external screen in 0xCA.

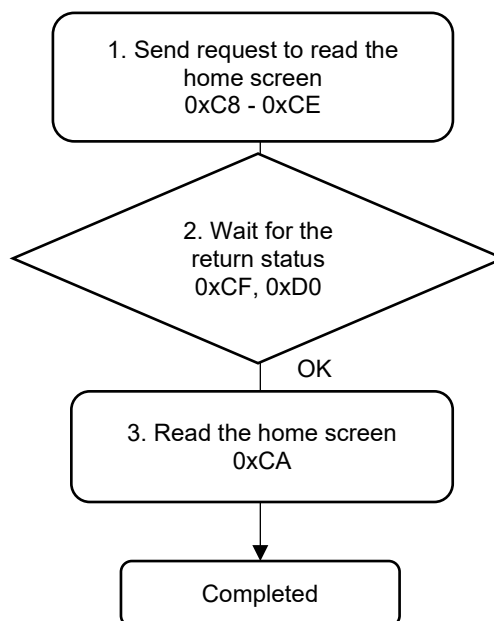
Modbus (Hex)	Write / Read request	Read
CA	Display orientation	0: portrait 1: landscape

#571 Read the home screen

Content description:

Use function code #571 to read the home screen of the controller.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	571
C9	Version number	0
CA to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF and 0xD0.

Modbus (Hex)	Return status	Description
CF	Function code	571
D0	Status	1: OK

If successful, the read status (0xD0) is 1.

3. Read the home screen in 0xCA.

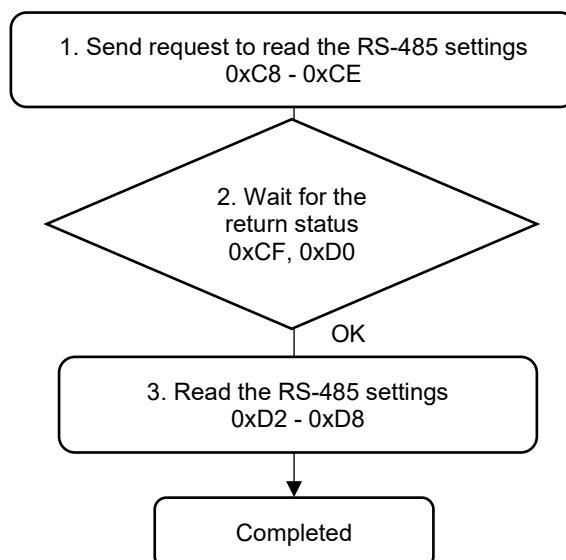
Modbus (Hex)	Write / Read request	Read
CA	Home screen	0: Home 1: Results

#572 Read the RS-485 settings

Content description:

Use function code #572 to read the RS-485 settings.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	572
C9	Version number	0
CA to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF and 0xD0.

Modbus (Hex)	Return status	Description
CF	Function code	572
D0	Status	1: OK

If successful, the read status (0xD0) is 1.

3. Read the RS-485 settings in 0xD2 - 0xD8.

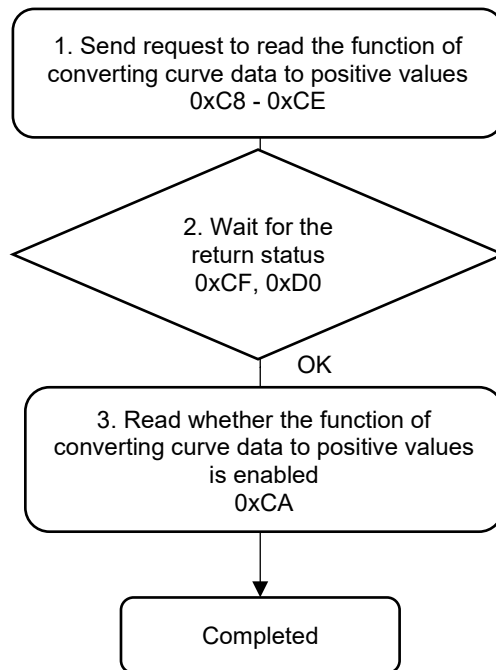
Modbus (Hex)	Write / Read request	Read
D2	Function On / Off	0: Off; 1: On
D3	Station	0 to 255
D4	RTU / ASCII	0: RTU; 1: ASCII
D5	Baud rate	0: 9600; 1: 19200; 2: 38400
D6	Data bit	0: 8-bit; 1: 7-bit
D7	Parity bit	0: None; 1: Odd; 2: Even
D8	Stop bit	0: 2-bit; 1: 1-bit

#573 Read the function of converting curve data to positive values

Content description:

Use function code #573 to read whether the function of converting curve data to positive values is enabled.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	573
C9	Version number	0
CA to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF and 0xD0.

Modbus (Hex)	Return status	Description
CF	Function code	573
D0	Status	1: OK

If successful, the read status (0xD0) is 1.

3. Read whether the function of converting curve data to positive values is enabled in 0xCA.

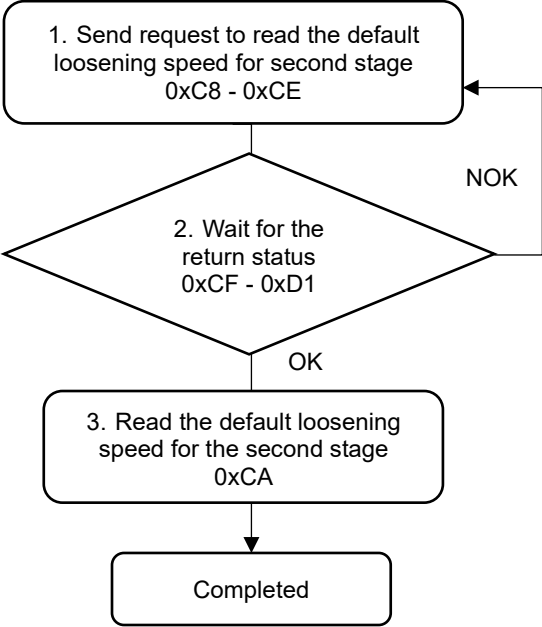
Modbus (Hex)	Write / Read request	Read
CA	Function of converting curve data to positive values	0: On 1: Off

#574 Read the second stage default loosening speed

Content description:

Use function code #574 to read the default loosening speed for the second stage.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	574
C9	Version number	0
CA to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	574
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2.

3. Read the default loosening speed for the second stage in 0xCA.

Modbus (Hex)	Write / Read request	Read
CA	Default loosening speed for the second stage	Unit: rpm

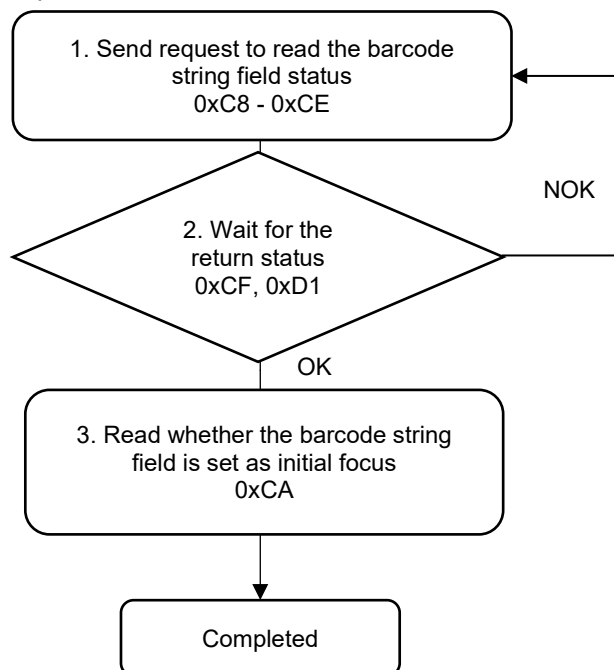
A

#575 Read the keyboard cursor blink in Results page

Content description:

Use function code #575 to read whether the barcode string field is set as the initial focus in the **Results** page.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	575
C9	Version number	0
CA to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	575
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2.

3. Read whether the barcode string field is set as initial focus in **Results** page in 0xCA.

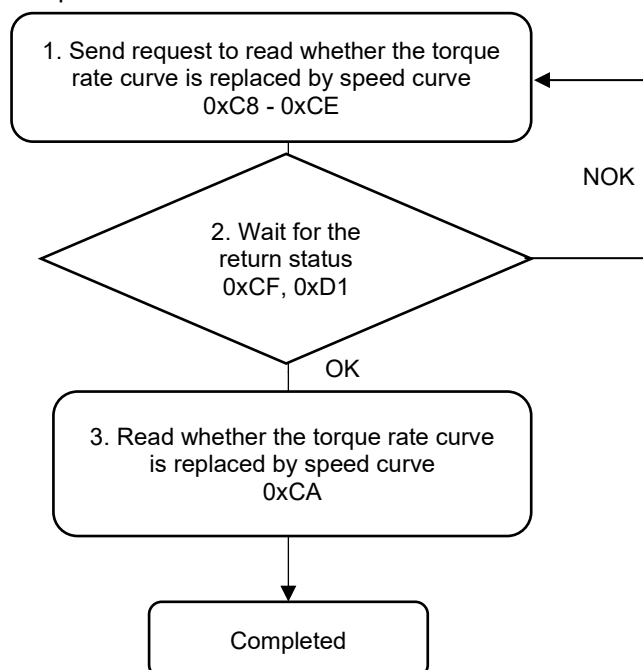
Modbus (Hex)	Write / Read request	Read
CA	Set the barcode string field as initial focus in Results page	0: Off 1: On

#576 Read whether the torque rate curve is replaced by the speed curve

Content description:

Use function code #576 to read whether the torque rate curve is replaced by the speed curve.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	576
C9	Version number	0
CA to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	576
D0	Status	1: OK; 2: NG
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2.

3. Read whether the torque rate curve is replaced by the speed curve in 0xCA.

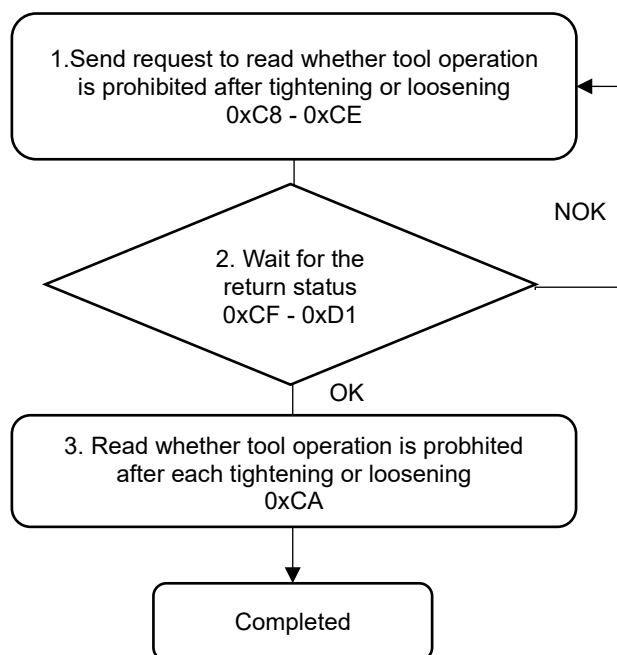
Modbus (Hex)	Write / Read request	Read
CA	Whether the torque rate curve is replaced by speed curve	0: torque rate curve and torque curve are shown (default) 1: speed curve and torque curve are shown

#577 Read whether tool operation is prohibited after each tightening or loosening

Content description:

Use function code #577 to read whether the tool operation is prohibited after each tightening or loosening.

Handshake signal description:



- Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	577
C9	Version number	0
CA to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

- Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	577
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2.

- Read whether tool operation is prohibited after each tightening or loosening in 0xCA.

Modbus (Hex)	Write / Read request	Read
CA	Whether tool operation is prohibited after each tightening or loosening	0: Off 1: tool operation prohibited after tightening 2: tool operation prohibited after loosening 3: tool operation prohibited after tightening or loosening

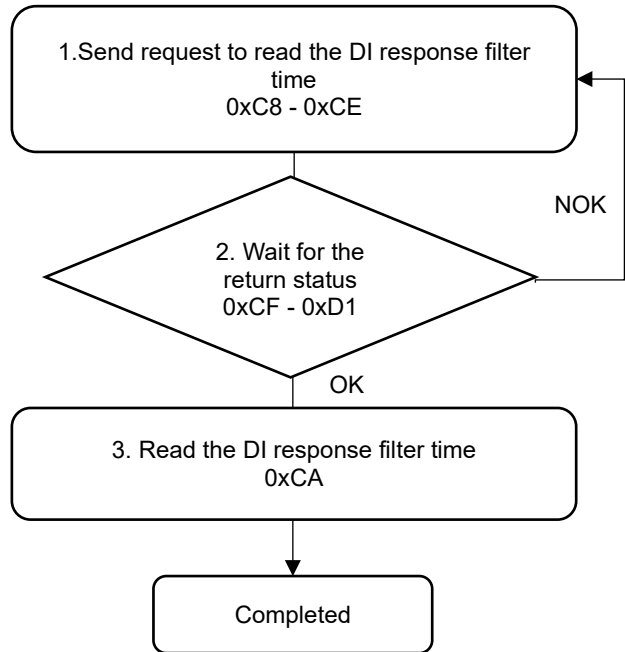
#578 Read the DI response filter time

Content description:

Use function code #578 to read the DI response filter time. Using this function helps mitigate the impact of noise on program execution.

Handshake signal description:

A



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	578
C9	Version number	0
CA to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	578
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2.

3. Read the DI response filter time in 0xCA.

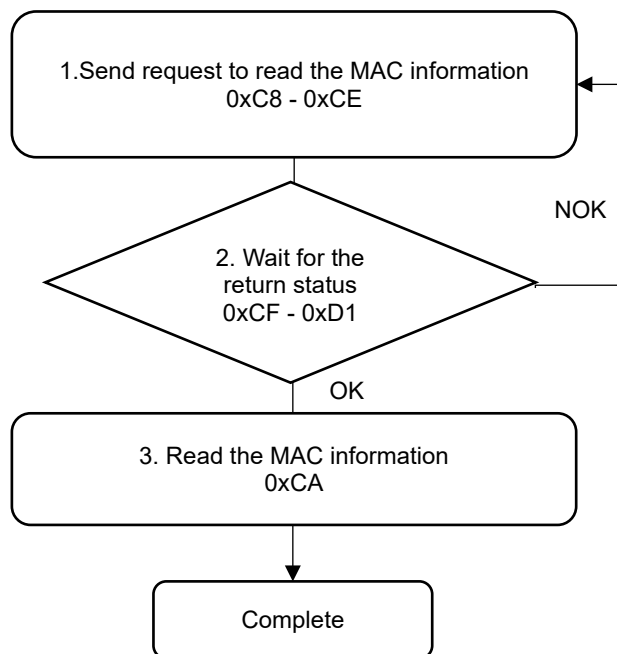
Modbus (Hex)	Write / Read request	Read
CA	DI response filter time	0 - 1000

#579 Read the MAC information

Content description:

Use function code #579 to read the MAC information.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	579
C9	Version number	0
CA to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	579
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2.

3. Read the MAC information in 0xD2 - 0xD7.

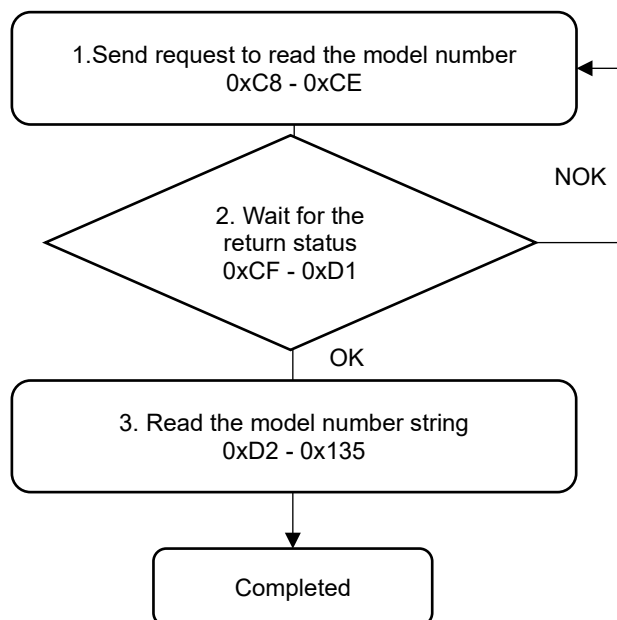
Modbus (Hex)	Write / Read request	Read
D2	MAC1	0 - 255
D3	MAC2	0 - 255
D4	MAC3	0 - 255
D5	MAC4	0 - 255
D6	MAC5	0 - 255
D7	MAC6	0 - 255

#580 Read the model number

Content description:

Use function code #580 to read the model number.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	580
C9	Version number	0
CA to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	580
D0	Status	1: OK; 2: NG
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2.

3. Read the model number string in 0xD2 - 0x135.

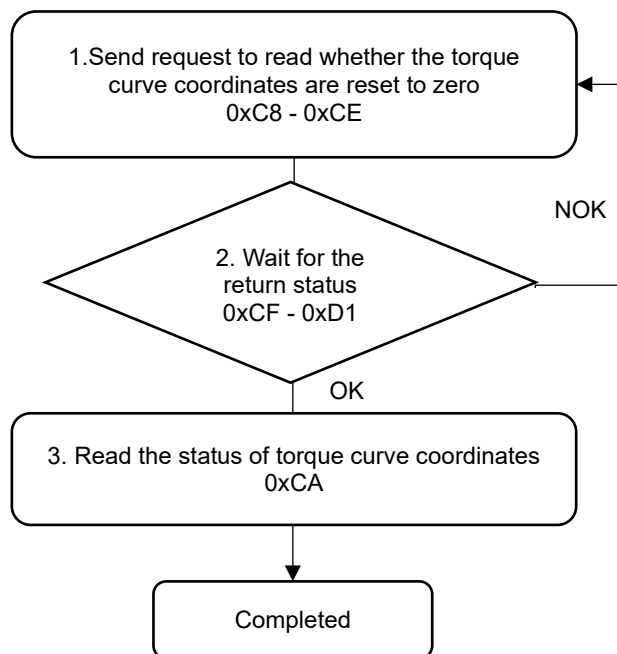
Modbus (Hex)	Write / Read request	Read
D2 to 135	Model number string	ASCII code (40 words)

#581 Read whether the torque curve coordinates are reset to zero

Content description:

Use function code #581 to read whether the torque curve coordinates are reset to zero.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	581
C9	Version number	0
CA to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	581
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2.

3. Read the status of torque curve coordinates in 0xCA.

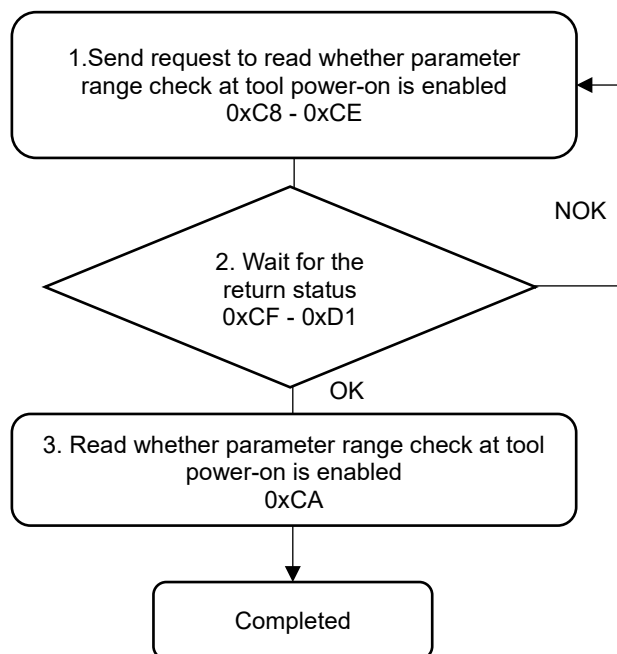
Modbus (Hex)	Write / Read request	Read
CA	Reset torque curve coordinates to zero	0: Off (default) 1: On

#582 Read whether parameter range check at tool power-on is enabled

Content description:

Use function code #582 to read whether parameter range check at tool power-on is enabled.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	582
C9	Version number	0
CA to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	582
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2.

3. Read whether parameter change check at tool power-on is enabled in 0xCA.

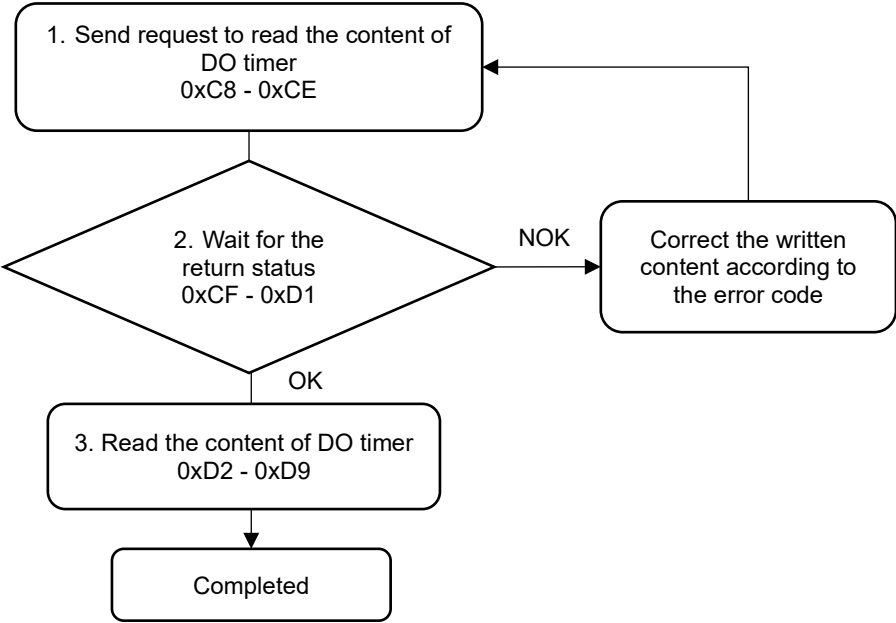
Modbus (Hex)	Write / Read request	Read
CA	Parameter range check at tool power-on	0: On (default) 1: Off

#583 Read the DO timer

Content description:

Use function code #583 to read the content of DO timer.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	583
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	583
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2.

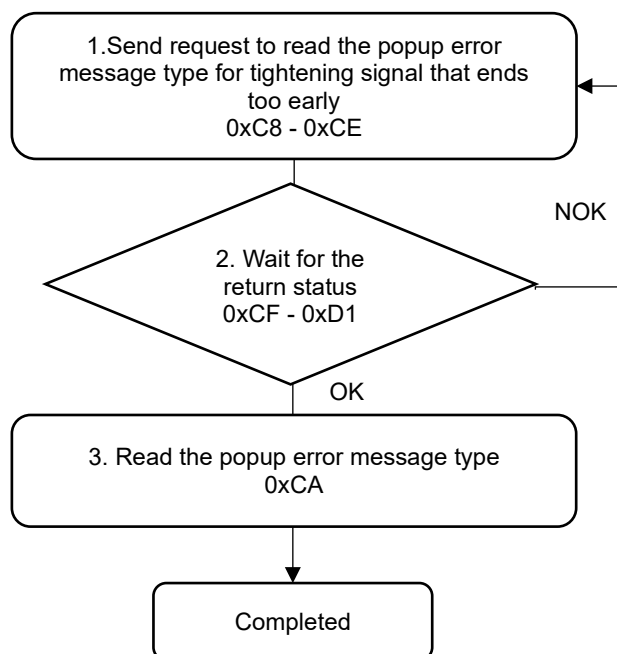
3. Read the content of DO timer in 0xD2 - 0xD9. Refer to function code #538 for the function list.

#584 Read the popup error message type for tightening signal that ends too early

Content description:

Use function code #584 to read the type of the popup error message for the tightening signal that ends too early.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	584
C9	Version number	0
CA to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	584
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2.

3. Read the popup error message type in 0xCA.

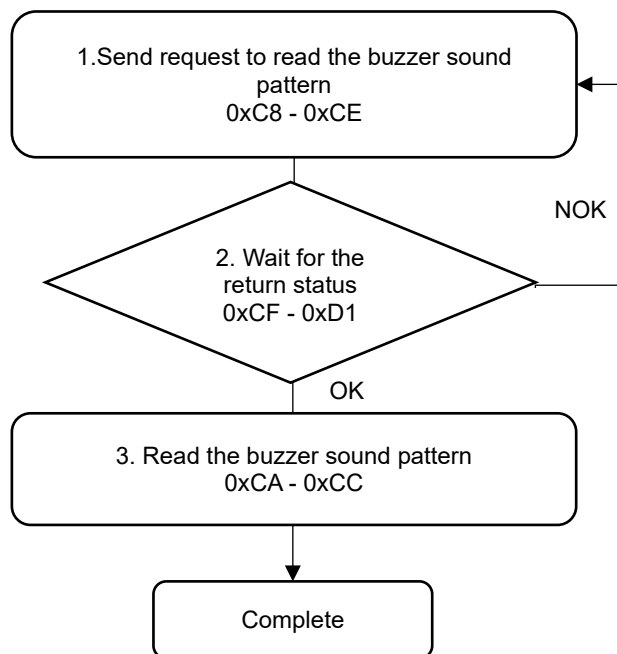
Modbus (Hex)	Write / Read request	Read
CA	Popup error message type	0: error 1: warning

#585 Read the buzzer sound pattern

Content description:

Use function code #585 to read the buzzer sound pattern.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	585
C9	Version number	0
CA to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	585
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2.

3. Read the buzzer sound pattern in 0xCA.

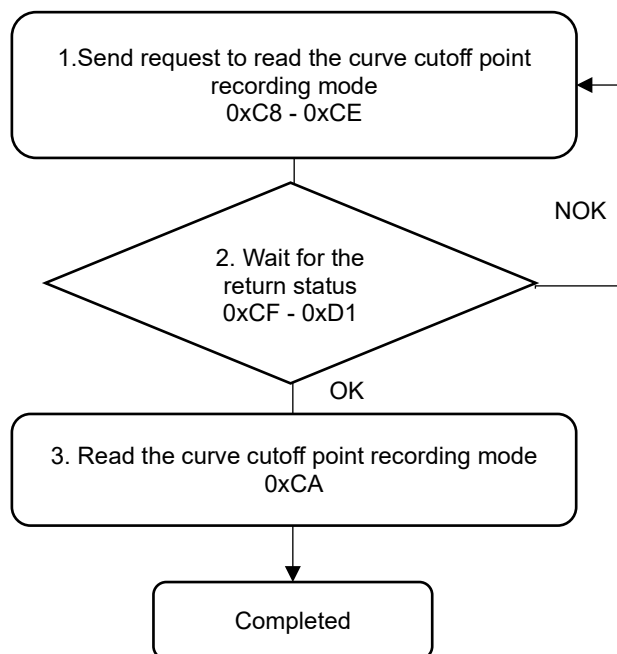
Modbus (Hex)	Write / Read request	Read
CA	Sound pattern of error signal buzzer	0: mute 1: one long beep 2: one short beep 3: two short beeps
CB	Sound pattern of finished signal buzzer for a single screw	
CC	Sound pattern of finished signal buzzer for total screws	

#586 Read the curve cutoff point recording mode

Content description:

Use function code #585 to read the curve cutoff point recording mode.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	586
C9	Version number	0
CA to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	586
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2.

3. Read the curve cutoff point recording mode in 0xCA.

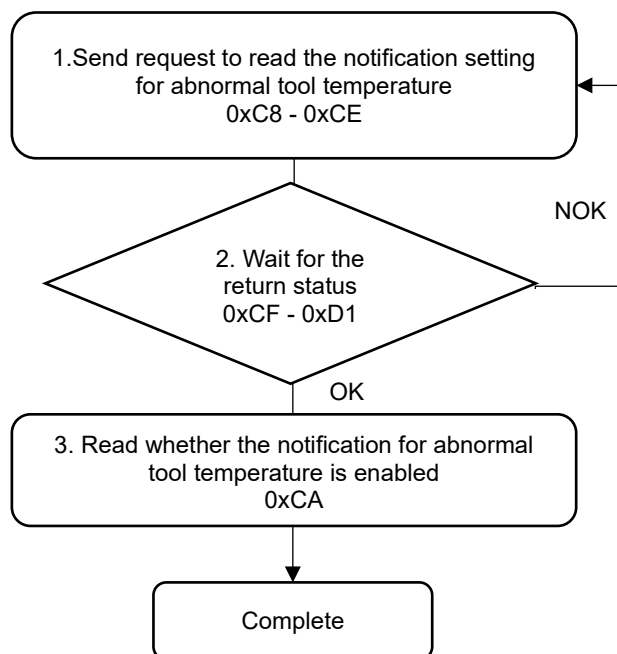
Modbus (Hex)	Write / Read request	Read
CA	Curve cutoff point recording mode	0: entire process 1: peak value

#587 Read the notification setting for abnormal tool temperature

Content description:

Use function code #587 to read whether the notification for abnormal tool temperature is enabled.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	587
C9	Version number	0
CA to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	587
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2.

3. Read whether the notification for abnormal tool temperature is enabled in 0xCA.

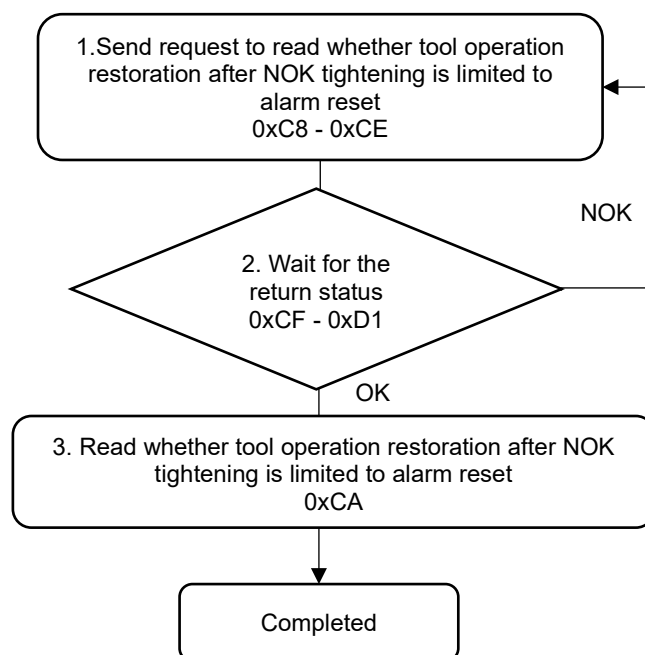
Modbus (Hex)	Write / Read request	Read
CA	Notification for abnormal tool temperature	0: On (default) 1: Off

#588 Read whether tool operation restoration after NOK tightening is limited to alarm reset

Content description:

Use function code #588 to read if tool operation restoration after NOK tightening is limited to alarm reset.

Handshake signal description:



- Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	588
C9	Version number	0
CA to CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

- Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	588
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2.

- Read whether tool operation restoration after NOK tightening is limited to alarm reset in 0xCA.

Modbus (Hex)	Write / Read request	Read
CA	Limit tool operation restoration after NOK tightening to alarm reset	0: Off 1: On

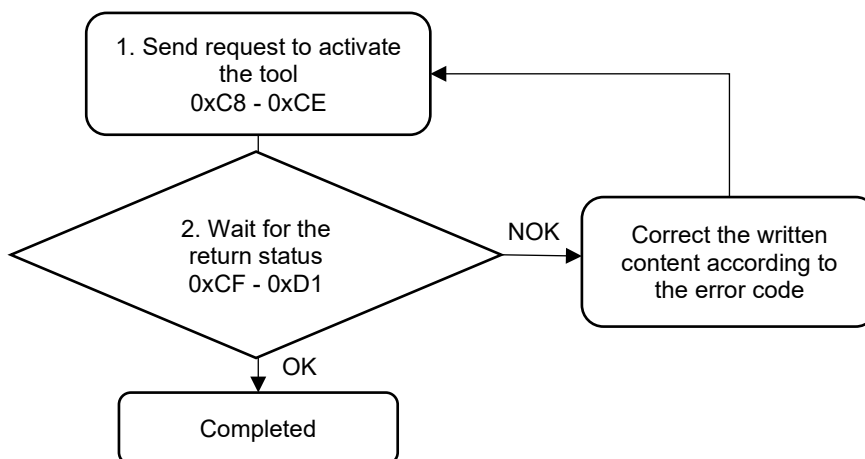
A.3.6 Tool operations

#600 Write to activate the tool

Content description:

Use function code #600 to activate Tool 1 or Tool 2.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	600
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB	Tool activation	0: Off; 1: On
CC, CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	600
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

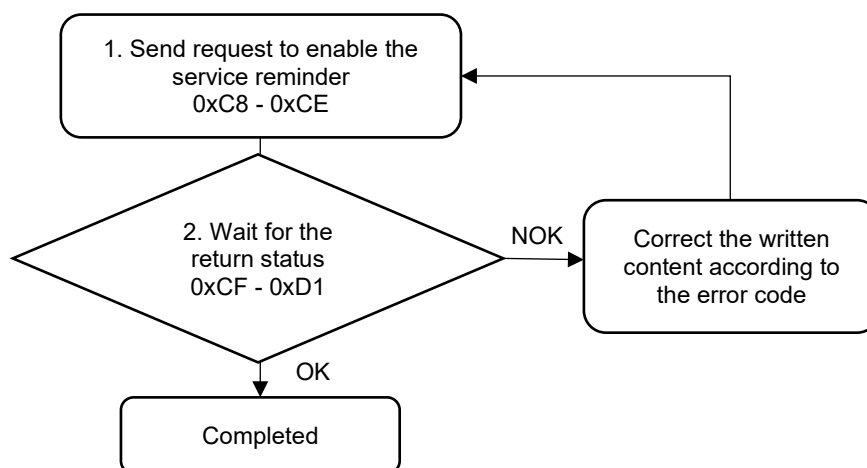
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	Tool activation	Must be 0 or 1

#601 Write to enable service reminder

Content description:

Use function code #601 to turn on or off the tool service reminder.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	601
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB	Service reminder	0: Off; 1: On
CC, CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	601
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

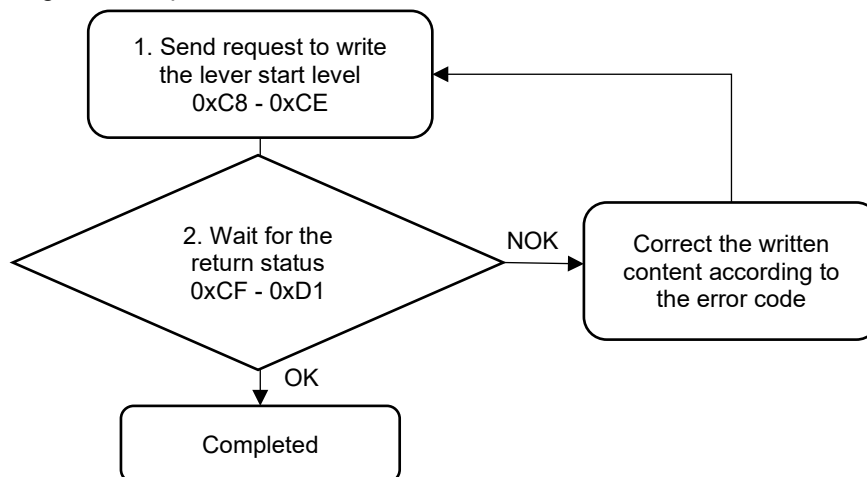
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	Service reminder	Must be 0 or 1

#602 Write the lever start level

Content description:

Use function code #602 to set the lever start level of the tool.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	602
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB	Threshold for On level	0 to 4095
CC	Threshold for Off level	0 to 4095
CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	602
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

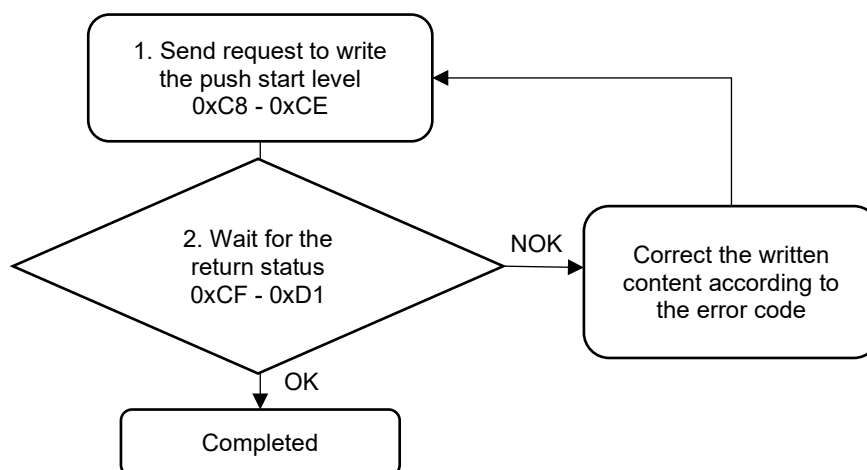
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	Threshold value	Exceeds the range of 0 to 4095

#603 Write the push start level

Content description:

Use function code #603 to set the push start level of the tool.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	603
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB	Threshold for On level	0 to 4095
CC	Threshold for Off level	0 to 4095
CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	603
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

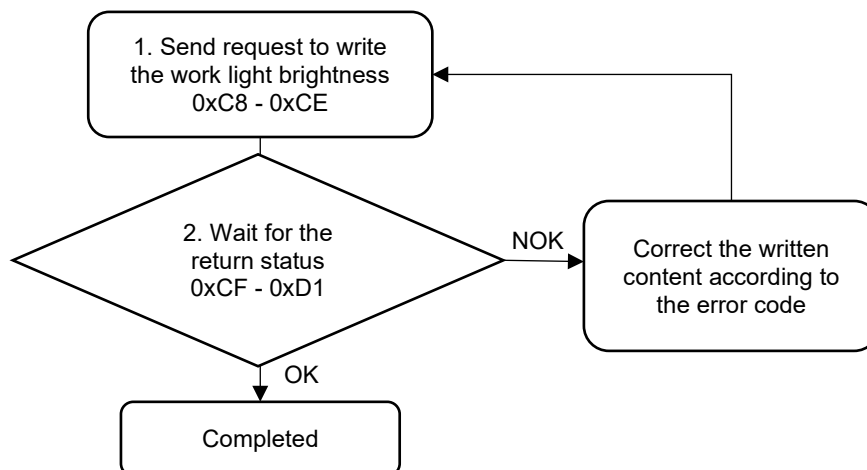
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	Threshold value	Exceeds the range of 0 to 4095

#604 Write the work light brightness

Content description:

Use function code #604 to set the work light brightness of the tool.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	604
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB	Brightness	0 to 50 (0: Off; 50: max. brightness)
CC, CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	604
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

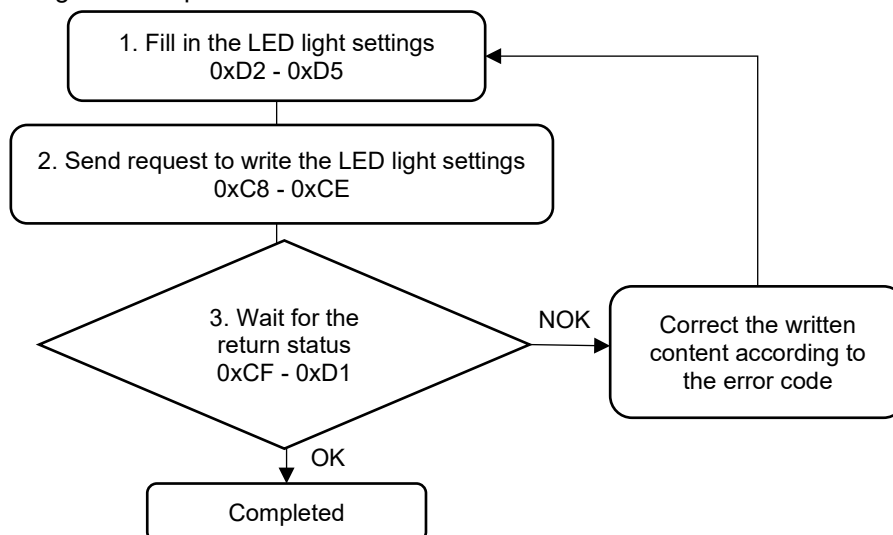
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	Brightness	Exceeds the range of 0 to 50

#606 Write the LED light settings

Content description:

Use function code #606 to set the LED light settings of the tool.

Handshake signal description:



1. Fill in the LED light settings to 0xD2 - 0xD5.

Modbus (Hex)	Function	Description
D2	NO / NC	0: NO (normally open) 1: NC (normally closed) (Bit 1 - 8: for Tool 1; Bit 9 - 16: for Tool 2)
D3	Status of red indicator	0: not in use
D4	Status of yellow indicator	1: ready 2: running
D5	Status of green indicator	3: tightening OK and report is being saved 4: tightening NOK and report is being saved 5: loosening OK and report is being saved 6: loosening NOK and report is being saved 7: total screw quantity reached

2. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	606
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

3. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	606
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

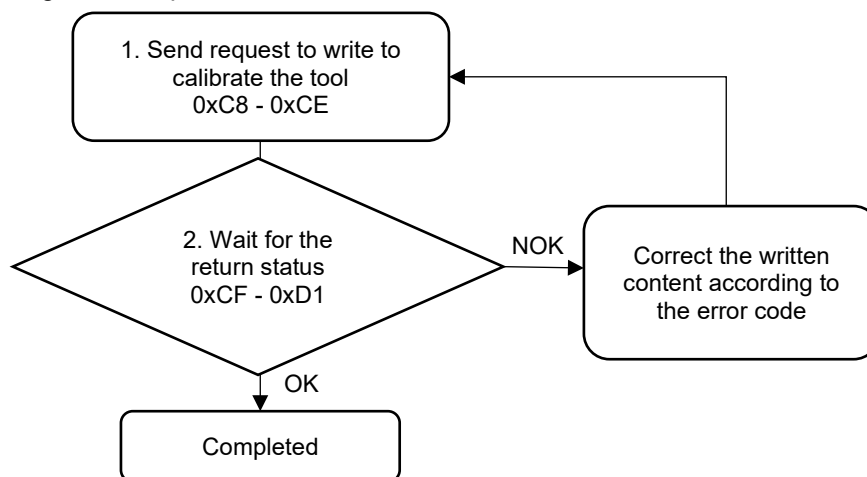
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

#607 Write to calibrate the tool

Content description:

Use function code #607 to set the precision of the tool.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

■ Mode 0

Modbus (Hex)	Write / Read request	Write
C8	Function code	607
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB	Precision	0 to 32767 (Default: 1400 for 12 kgf-cm models 3500 for 30 kgf-cm models 6000 for 50 kgf-cm models 5705 for 75 kgf-cm models)
CC	Reserved	-
CD	Mode	0
CE	Request to send command	1 (to be filled in last)

■ Mode 11

Modbus (Hex)	Write / Read request	Write
C8	Function code	607
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB	Final torque in Results page	Unit: 0.001 x N.m
CC	Torque value calibrated by third-party device	Unit: 0.001 x N.m
CD	Mode	11
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	607

Modbus (Hex)	Return status	Description
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

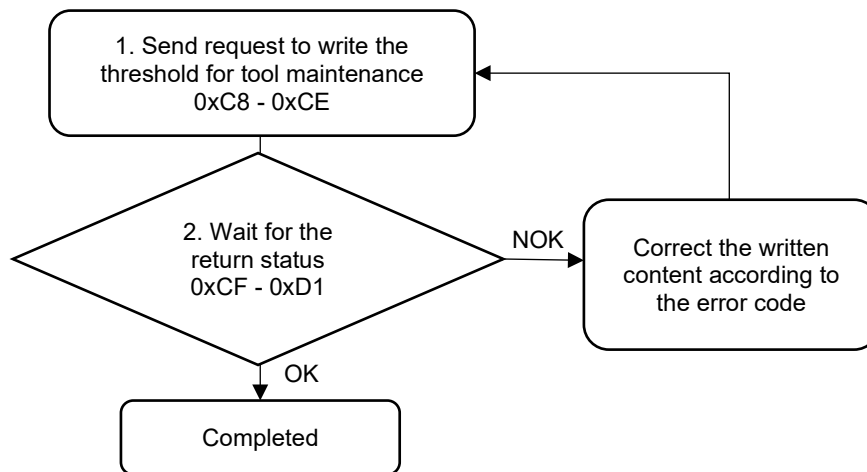
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	Precision	Exceeds the range of 0 to 32767

#608 Write the service reminder threshold

Content description:

Use function code #608 to set the threshold for tool maintenance.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	608
C9	Version number	0
CA	Threshold for tool service reminder (L)	0 - 4294967295 (default: 250000)
CB	Threshold for tool service reminder (H)	
CC, CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	608
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2.

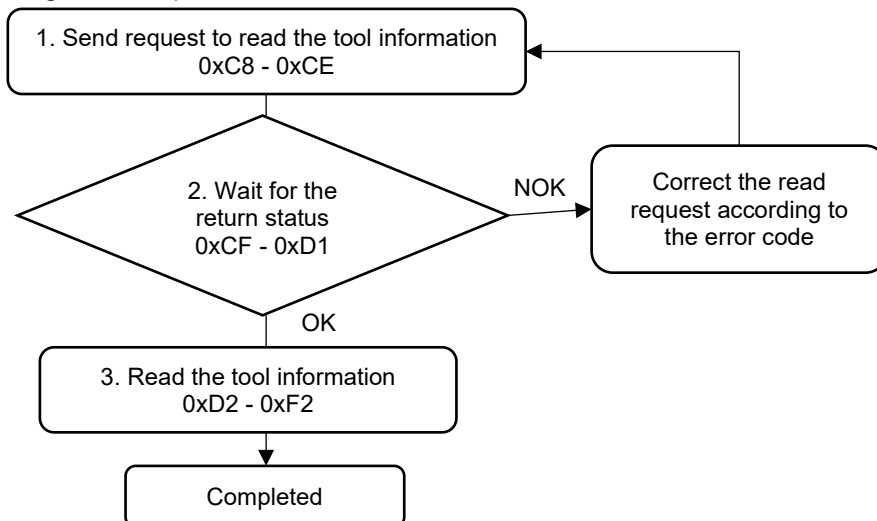
A

#650 Read the tool information

Content description:

Use function code #650 to read the tool information.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	650
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	650
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	Data is not ready	Reread the tool information.

3. Read the tool information in 0xD2 - 0xF2.

Modbus (Hex)	Function	Description
D2 to E5	Model name	ASCII code (20 words)
E6	Max. speed	Unit: rpm
E7	Max. torque	Unit: 0.001 N.m
E8	Tool temperature	Unit: °C
E9	Tightening + loosening count (L)	Tool life (maintenance)
EA	Tightening + loosening count (H)	
EB	Tightening + loosening count (L)	Tool life (total)
EC	Tightening + loosening count (H)	
ED	Recent service date - year	-

Modbus (Hex)	Function	Description
EE	Recent service date - month	–
EF	Recent service date - day	–
F0	Recent service date - hour	–
F1	Recent service date - minute	–
F2	Recent service date - second	–

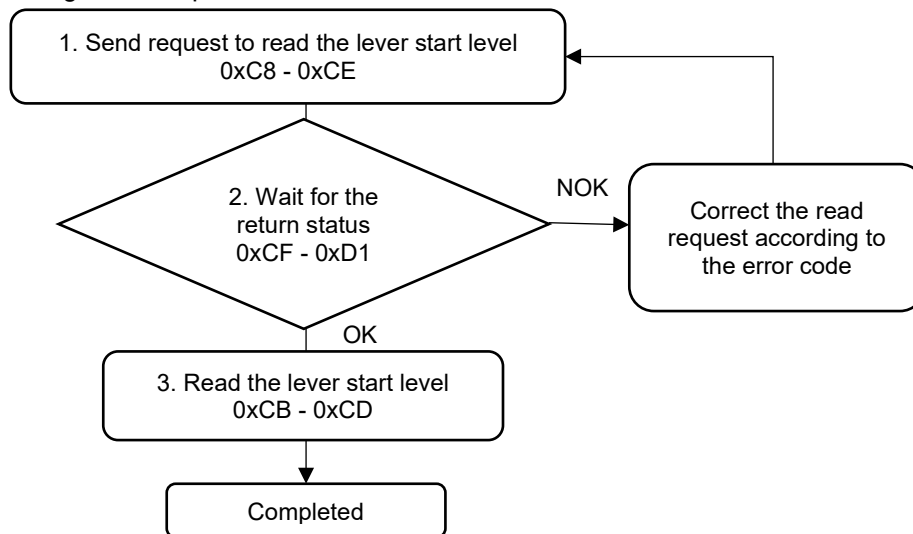
A

#651 Read the lever start level

Content description:

Use function code #651 to read the lever start level of the tool.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	651
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	651
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

3. Read the lever start level in 0xCB - 0xCD.

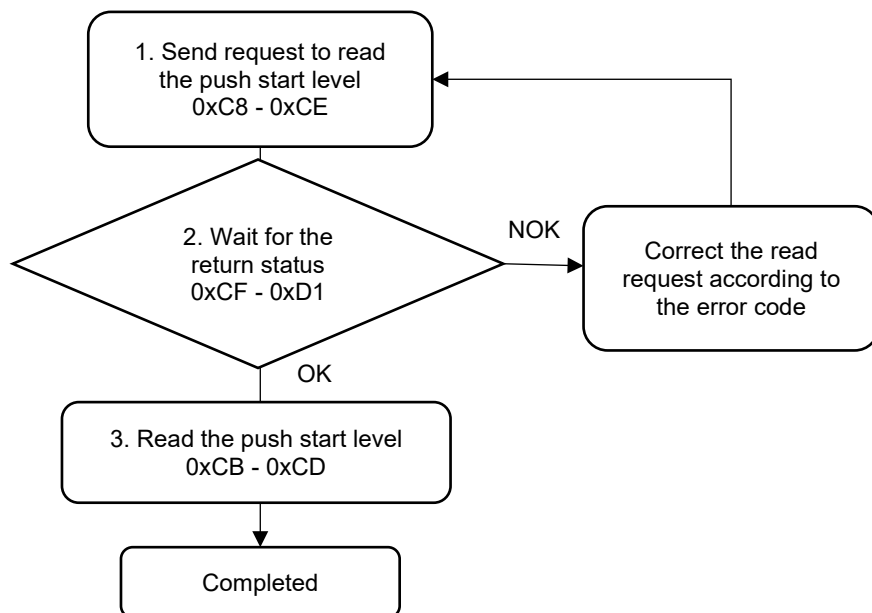
Modbus (Hex)	Write / Read request	Read
CB	Threshold of the current pressing	0 to 4095
CC	Threshold for On level	0 to 4095
CD	Threshold for Off level	0 to 4095

#652 Read the push start level

Content description:

Use function code #652 to read the push start level of the tool.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	652
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	652
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

3. Read the push start level in 0xCB - 0xCD.

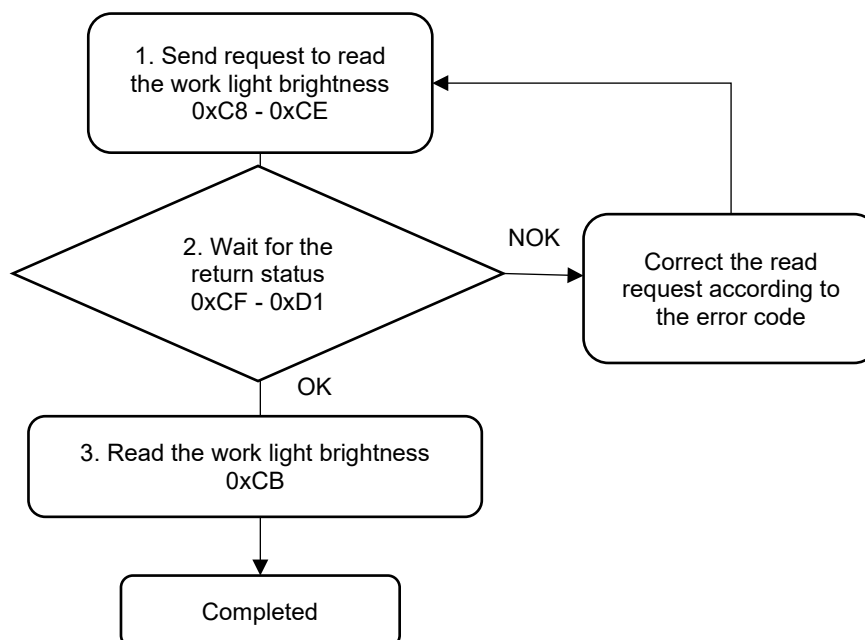
Modbus (Hex)	Write / Read request	Read
CB	Threshold of the current pressing	0 to 4095
CC	Threshold for On level	0 to 4095
CD	Threshold for Off level	0 to 4095

#653 Read the work light brightness

Content description:

Use function code #653 to read the work light brightness of the tool.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	653
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	653
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

3. Read the work light brightness in 0xCB.

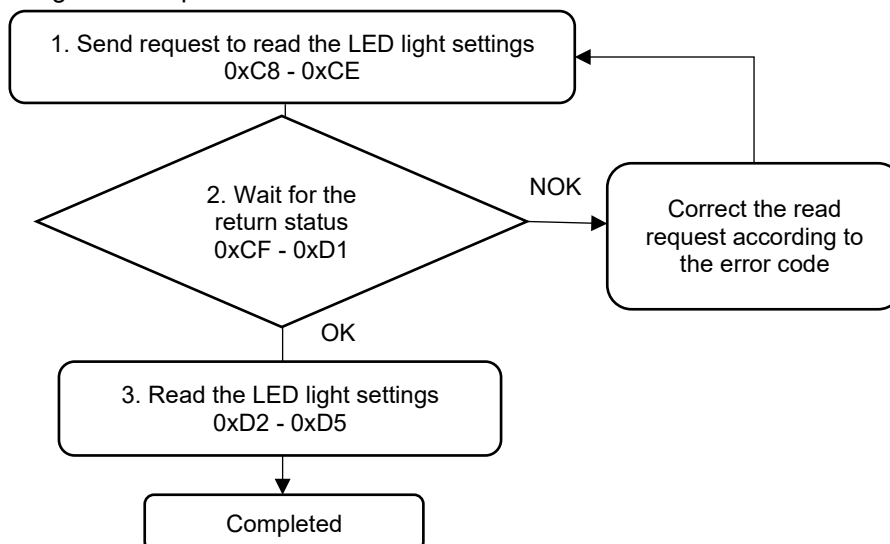
Modbus (Hex)	Write / Read request	Read
CB	Brightness	0 to 50 (0: Off; 50: max. brightness)

#655 Read the LED light settings

Content description:

Use function code #655 to read the LED light settings of the tool.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	655
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	655
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

3. Read the LED light settings in 0xD2 - 0xD5.

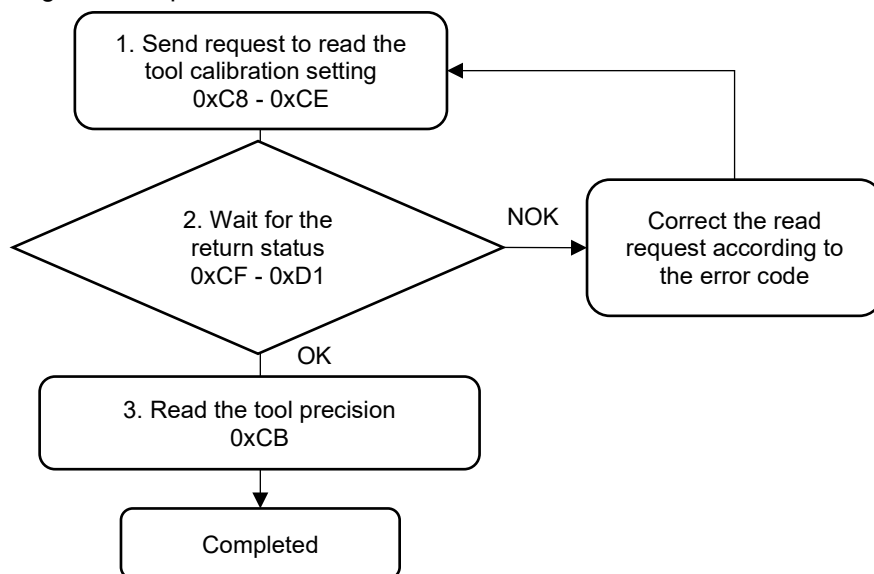
Modbus (Hex)	Write / Read request	Read
D2	Reserved	-
D3	Status of red indicator	0: not in use
D4	Status of yellow indicator	1: ready 2: running 3: tightening OK and report is being saved 4: tightening NOK and report is being saved 5: loosening OK and report is being saved 6: loosening NOK and report is being saved 7: total screw quantity reached
D5	Status of green indicator	

#656 Read the tool calibration setting

Content description:

Use function code #656 to read the precision of the tool.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	656
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	656
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

3. Read the precision of the tool in 0xCB.

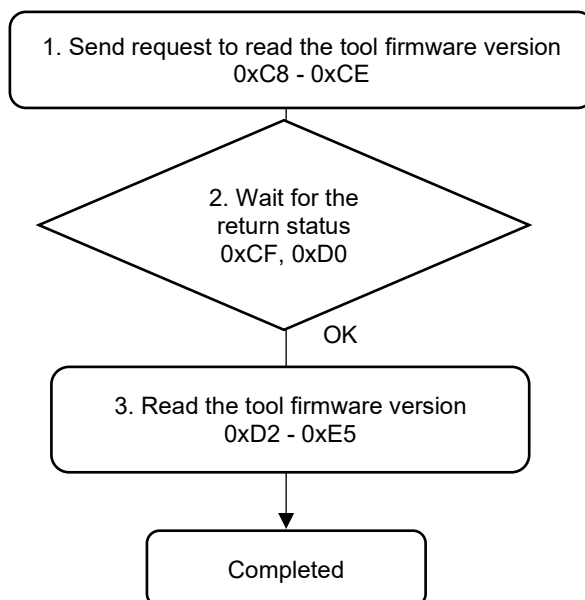
Modbus (Hex)	Write / Read request	Read
CB	Precision	0 to 32767 (Default: 1400 for 12 kgf-cm models 3500 for 30 kgf-cm models 6000 for 50 kgf-cm models 5705 for 75 kgf-cm models)

#657 Read the tool firmware version

Content description:

Use function code #657 to read the firmware version number of the tool.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	657
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD0.

Modbus (Hex)	Return status	Description
CF	Function code	657
D0	Status	1: OK

If successful, the read status (0xD0) is 1.

3. Read the firmware version number of the tool in 0xD2 to 0xE5.

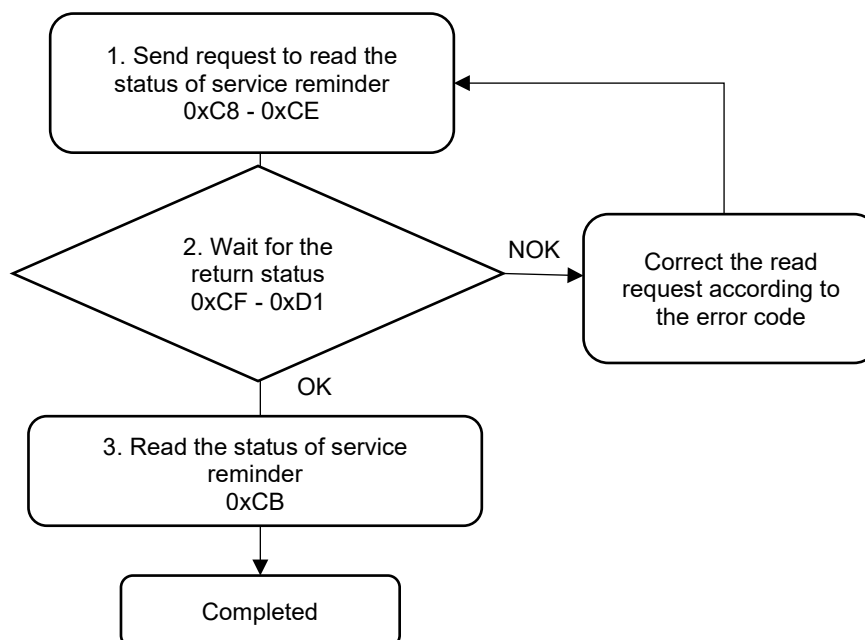
Modbus (Hex)	Write / Read request	Read
D2 to E5	Tool firmware version	ASCII code (20 words)

#658 Read the status of service reminder

Content description:

Use function code #658 to read the status of service reminder.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	658
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	658
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

3. Read the status of tool service reminder in 0xCB.

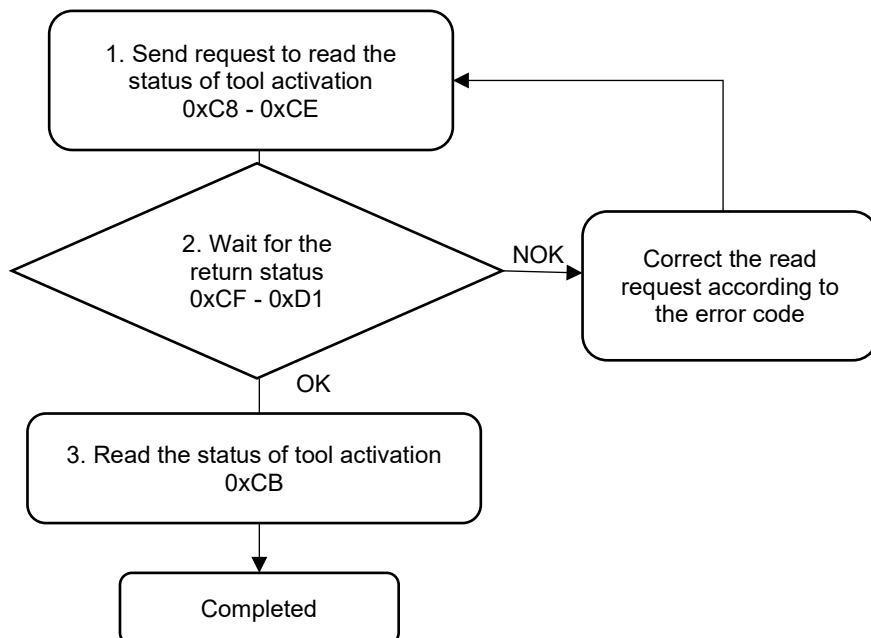
Modbus (Hex)	Write / Read request	Read
CB	Service reminder	0: Off; 1: On

#659 Read the status of tool activation

Content description:

Use function code #659 to read the status of the tool activation.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	659
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	659
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

3. Read the status of tool activation in 0xCB.

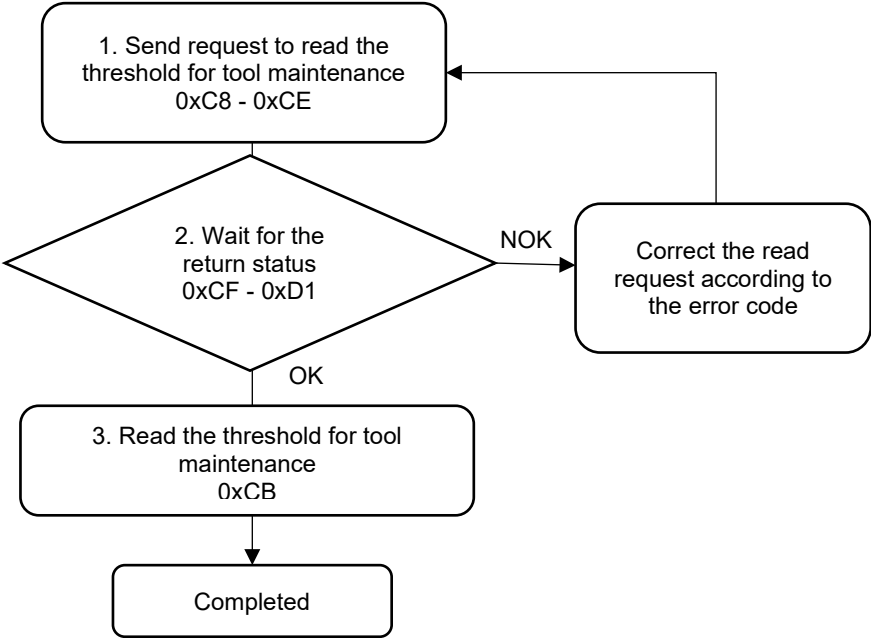
Modbus (Hex)	Write / Read request	Read
CB	Tool activation	0: Off; 1: On

#660 Read the service reminder threshold

Content description:

Use function code #660 to read the threshold for tool maintenance.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	660
C9	Version number	0
CA to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	660
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2.

3. Read the threshold for tool service reminder in 0xCA and 0xCB.

Modbus (Hex)	Write / Read request	Read
CA	Threshold for tool service reminder (L)	Default: 250000
CB	Threshold for tool service reminder (H)	

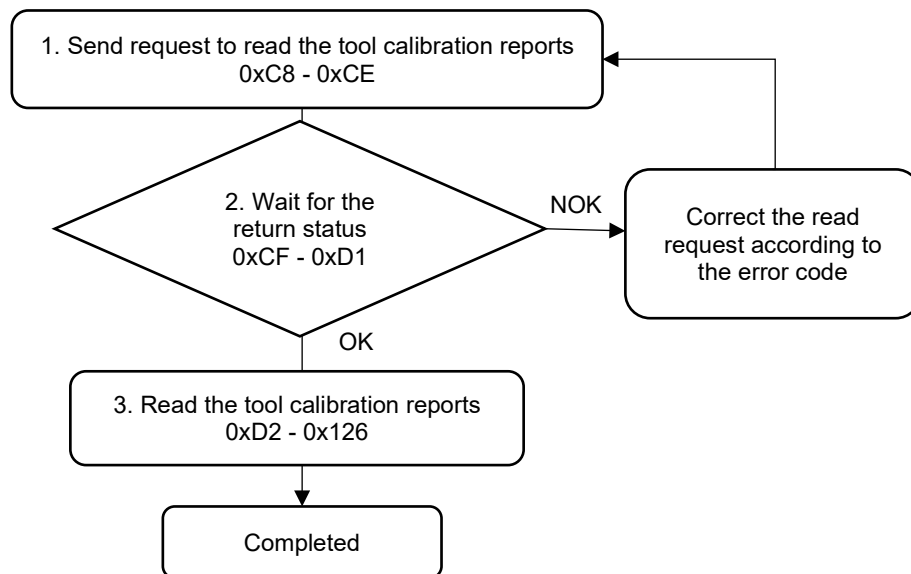
A

#661 Read the tool calibration reports

Content description:

Use function code #661 to read the tool calibration reports.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	661
C9	Version number	0
CA	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
CB	Read the calibration report page number	0 - 19
CC, CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	661
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2.

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	Read the calibration report page number	Exceeds the range of 0 - 19

3. Read the tool calibration reports in 0xD2 - 0x126.

Modbus (Hex)	Write / Read request	Read
D2	Torque unit when the first entry is saved	0: N.m; 1: kgf.cm; 2: lbf.ft; 3: lbf.in; 4: ozf.ft; 5: ozf.in; 6: cNm
D3	Torque unit when the second entry is saved	
D4	Torque unit when the third entry is saved	
D5	Torque unit when the fourth entry is saved	
D6	Torque unit when the fifth entry is saved	

Modbus (Hex)	Write / Read request	Read
D7	First entry of precision setting	Default settings: 12 kgf-cm models: 1400 30 kgf-cm models: 3500 50 kgf-cm models: 6000 75 kgf-cm models: 5705
D8	Second entry of precision setting	
D9	Third entry of precision setting	
DA	Fourth entry of precision setting	
DB	Fifth entry of precision setting	
DC	First entry of final torque in Result screen (L)	Unit: 0.001 x torque unit at present
DD	First entry of final torque in Result screen (H)	
DE	Second entry of final torque in Result screen (L)	
DF	Second entry of final torque in Result screen (H)	
E0	Third entry of final torque in Result screen (L)	
E1	Third entry of final torque in Result screen (H)	
E2	Fourth entry of final torque in Result screen (L)	
E3	Fourth entry of final torque in Result screen (H)	
E4	Fifth entry of final torque in Result screen (L)	
E5	Fifth entry of final torque in Result screen (H)	
E6	First entry of error	Unit: %
E7	Second entry of error	
E8	Third entry of error	
E9	Fourth entry of error	
EA	Fifth entry of error	
EB to F4	First entry of record date	The date is converted in to ASCII code.
F5 to FE	Second entry of record date	
FF to 108	Third entry of record date	
109 to 112	Fourth entry of record date	
113 to 11C	Fifth entry of record date	
11D	First entry of torque value calibrated by third-party device (L)	Unit: 0.001 x torque unit at present
11E	First entry of torque value calibrated by third-party device (H)	
11F	Second entry of torque value calibrated by third-party device (L)	
120	Second entry of torque value calibrated by third-party device (H)	
121	Third entry of torque value calibrated by third-party device (L)	
122	Third entry of torque value calibrated by third-party device (H)	
123	Fourth entry of torque value calibrated by third-party device (L)	
124	Fourth entry of torque value calibrated by third-party device (H)	
125	Fifth entry of torque value calibrated by third-party device (L)	
126	Fifth entry of torque value calibrated by third-party device (H)	

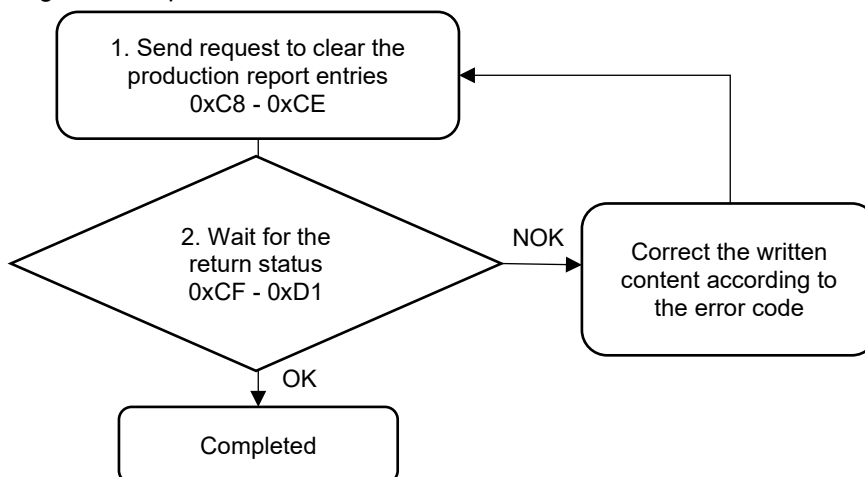
A.3.7 Reports operations

#700 Clear the production report entries

Content description:

Use function code #700 to delete all the production report entries from the controller.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	700
C9	Version number	0
CA	Confirm command	99
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	700
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

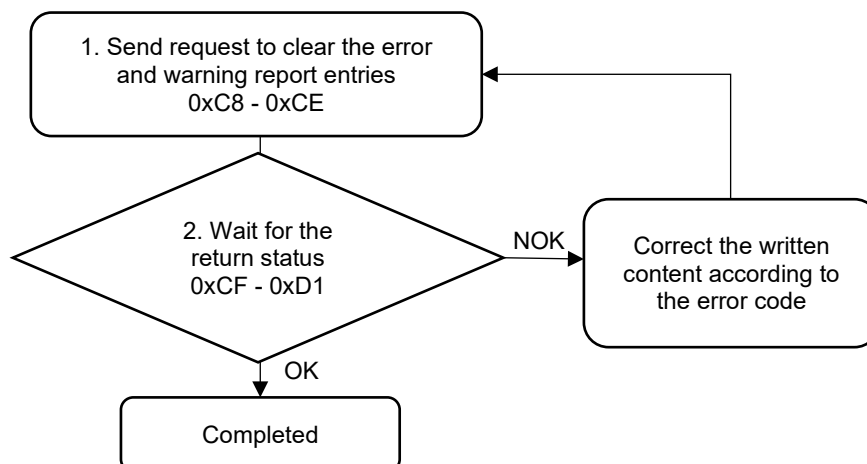
Code	Error description	Description
1	Confirm command	Must be 99

#701 Clear the error and warning report entries

Content description:

Use function code #701 to delete all the error and warning report entries from the controller.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	701
C9	Version number	0
CA	Confirm command	99
CB	Item to delete	0: clear error and warning report entries 10: clear error report entries 20: clear warning report entries
CC, CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	701
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

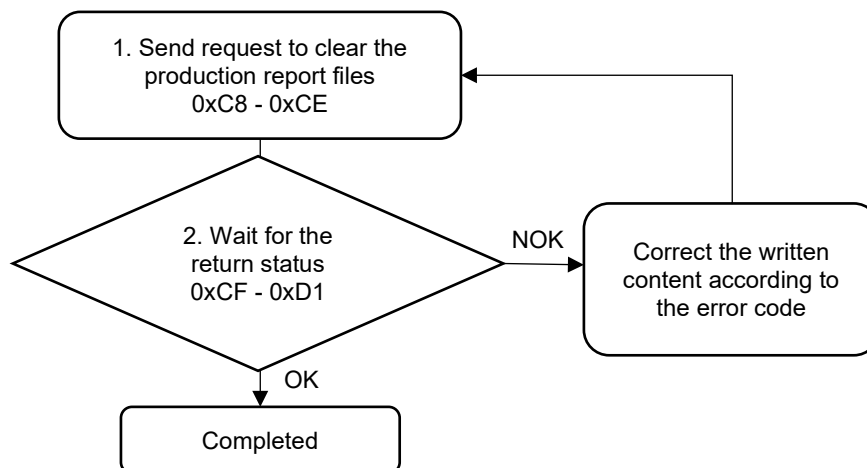
Code	Error description	Description
1	Confirm command	Must be 99

#702 Clear the production report files

Content description:

Use function code #702 to delete all the production report files exported to the HMI.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	702
C9	Version number	0
CA	Confirm command	99
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	702
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Confirm command	Must be 99

#750 Find and read the production report entries

Content description:

Use function code #750 to read the content of the production report.

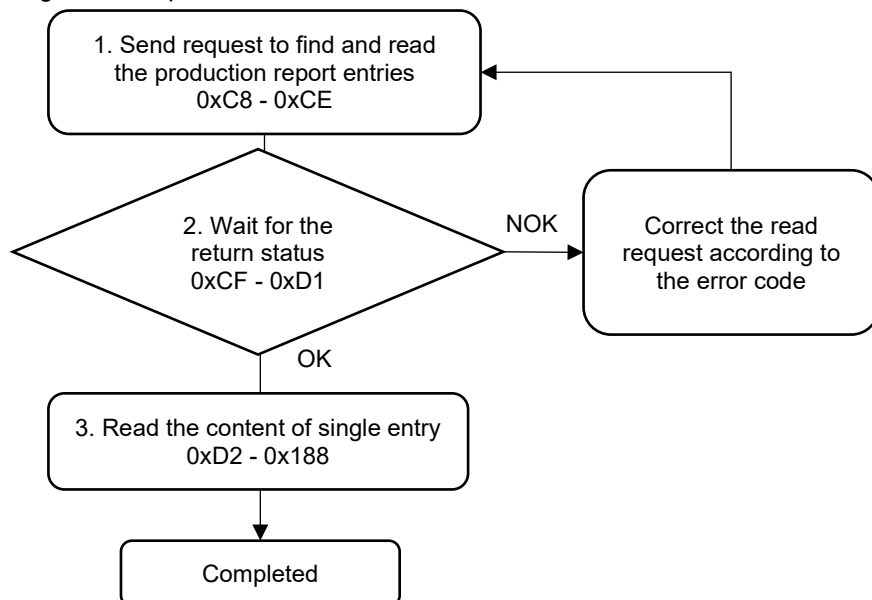
- A. Use the following Modbus address or function code #21 to read the content of the latest production report.

Modbus (Hex)	Function code #21 (Dec)	Tool 1 / Tool 2 common status	R/W
6B	108	Current No. of production report entries (L)	R
6C	109	Current No. of production report entries (H)	R

- B. Read the content of history report entries.

■ **When the version number is 0:**

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	750
C9	Version number	0
CA	Production report ID (L)	1 to 200000
CB	Production report ID (H)	0: the latest production report
CC, CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	750
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Production report ID	Exceeds the range of 0 to 200000

3. Read the content of a single production report in 0xD2 - 0x188.

Modbus (Hex)	Function code # 760 (Dec)	Function	Description
D2 to 135	11 to 110	Saved barcode string	ASCII code (100 words)
136	111	Year	-
137	112	Month	-
138	113	Day	-
139	114	Hour	-
13A	115	Minute	-
13B	116	Second	-
13C	117	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
13D	118	Screw ID (L)	Current screw progress ID
13E	119	Screw ID (H)	
13F	120	Sequence ID	Current sequence ID
140	121	Parameter ID	Current parameter ID
141	122	Target torque	Target torque set in the final stage; unit: 0.001 x torque of the current report. Refer to DWORD 0x16B and 0x16C when the torque specification of the tool is 5 N.m or higher.
142	123	Target angle	Target angle set in the final stage. If Angle Control is used through all stages, this field is the total angle of all stages.
143	124	Target torque rate	Target torque rate set in the final stage; unit: 0.0001 x torque/degree of the current report. Refer to DWORD 0x16D and 0x16E when the torque specification of the tool is 5 N.m or higher.
144	125	Final torque	Refer to DWORD 0x16F and 0x170 when the torque specification of the tool is 5 N.m or higher.
145	126	Tightening angle	Running angle of the final stage.
146	127	Rotation angle	Total angle.
147	128	Status	1: tightening OK; 2: tightening NOK; 3: loosening OK; 4: loosening NOK; 5: pass
148	129	Operation time	Total operation time of all stages (unit: 0.001 second)
149	130	AL / NG / WN code	Refer to CH12 (need to convert to Hex format)
14A	131	Max. angle of final stage	-
14B	132	Min. angle of final stage	-
14C	133	Max. torque of final stage	Refer to DWORD 0x171 and 0x172 when the torque specification of the tool is 5 N.m or higher.
14D	134	Min. torque of final stage	Refer to DWORD 0x173 and 0x174 when the torque specification of the tool is 5 N.m or higher.
14E	135	Torque unit	0: N.m; 1: kgf.cm; 2: lbf.ft; 3: lbf.in; 4: ozf.ft; 5: ozf.in; 6: cNm
14F	136	Torque specification of the tool	Tool specification; unit: 0.001 N.m

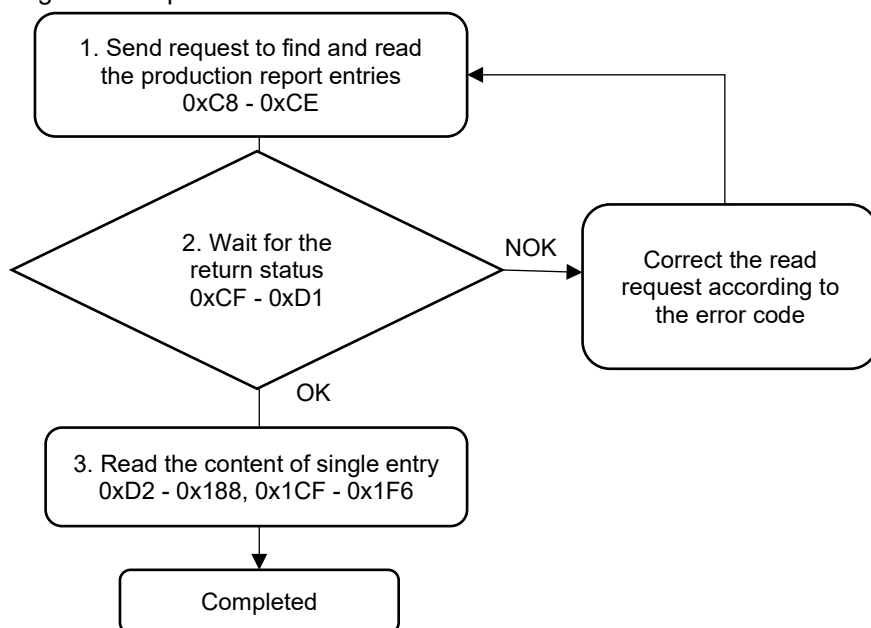
Modbus (Hex)	Function code # 760 (Dec)	Function	Description
150	137	Max. torque of the tool	Upper limit of tool specification; unit: 0.001 N.m
151	138	Pre-tightening torque	The operating torque of the third stage when the tightening is executed under Standard strategy. Refer to DWORD 0x175 and 0x176 when the torque specification of the tool is 5 N.m or higher.
152	139	The set total operation time	Timeout setting for a single tightening parameter.
153	140	Max. total angle	Set the limit of maximum total angle.
154	141	Max. torque	Torque upper limit when the current stage finishes. Refer to DWORD 0x177 and 0x178 when the torque specification of the tool is 5 N.m or higher.
155	142	Min. torque	Torque lower limit when the current stage finishes. Refer to DWORD 0x179 and 0x17A when the torque specification of the tool is 5 N.m or higher.
156	143	Max. angle	Angle upper limit when the current stage finishes.
157	144	Min. angle	Angle lower limit when the current stage finishes.
158	145	Max. operation time	Time upper limit when the current stage finishes.
159	146	Min. operation time	Time lower limit when the current stage finishes.
15A	147	Prevail torque	When the prevail torque function is enabled, the prevail torque for operation refers to this field. When the function is disabled, this field is 0. Refer to DWORD 0x17B and 0x17C when the torque specification of the tool is 5 N.m or higher.
15B	148	Final + Prevail torque	Refer to DWORD 0x17D and 0x17E when the torque specification of the tool is 5 N.m or higher.
15C	149	Actual current	The maximum current when the tightening operation is complete (unit: 0.01 Amp)
15D	150	Clamp torque	When the operation is using Torque Rate Control, this field is the final torque minus the torque when snug point is reached. When the operation is not using Torque Rate Control, this field is the final torque minus the tightening torque of the previous stage. Refer to DWORD 0x17F and 0x180 when the torque specification of the tool is 5 N.m or higher.
15E	151	Max. clamp torque	Clamp torque upper limit when the current stage finishes. Refer to DWORD 0x181 and 0x182 when the torque specification of the tool is 5 N.m or higher.
15F	152	Min. clamp torque	Clamp torque lower limit when the current stage finishes. Refer to DWORD 0x183 and 0x184 when the torque specification of the tool is 5 N.m or higher.
160	153	Clamp angle	-
161	154	Max. clamp angle	-

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Modbus (Hex)	Function code # 760 (Dec)	Function	Description
162	155	Min. clamp angle	-
163	156	Min. total angle	-
164	157	Permissions account	0: reserved 1: User1 2: User2 3: User3 4: User4 5: User5 6: Admin
165	158	Magnification of the torque recorded by system	System-specific
166	159	Target yield point	Target yield point set in the final stage; unit: 0.01%
165 to 16A	160 to 163	Reserved	-
16B	164	Target torque (L)	Unit: 0.001 x torque of the current report
16C	165	Target torque (H)	
16D	166	Target torque rate (L)	Unit: 0.0001 x torque/degree of the current report
16E	167	Target torque rate (H)	
16F	168	Final torque (L)	Unit: 0.001 x torque of the current report
170	169	Final torque (H)	
171	170	Final stage max. torque (L)	Unit: 0.001 x torque of the current report
172	171	Final stage max. torque (H)	
173	172	Final stage min. torque (L)	Unit: 0.001 x torque of the current report
174	173	Final stage min. torque (H)	
175	174	Pre-tightening torque (L)	Unit: 0.001 x torque of the current report
176	175	Pre-tightening torque (H)	
177	176	Max. torque (L)	Unit: 0.001 x torque of the current report Torque upper limit when the current stage finishes.
178	177	Max. torque (H)	
179	178	Min. torque (L)	Unit: 0.001 x torque of the current report Torque lower limit when the current stage finishes.
17A	179	Min. torque (H)	
17B	180	Prevail torque (L)	Unit: 0.001 x torque of the current report
17C	181	Prevail torque (H)	
17D	182	Final + Prevail torque (L)	Unit: 0.001 x torque of the current report
17E	183	Final + Prevail torque (H)	
17F	184	Clamp torque (L)	Unit: 0.001 x torque of the current report
180	185	Clamp torque (H)	
181	186	Max. clamp torque (L)	Unit: 0.001 x torque of the current report Clamp torque upper limit when the current stage finishes.
182	187	Max. clamp torque (H)	
183	188	Min. clamp torque (L)	Unit: 0.001 x torque of the current report Clamp torque lower limit when the current stage finishes.
184	189	Min. clamp torque (H)	
185	190	Tool spec. torque (L)	Unit: 0.001 x torque of the current report
186	191	Tool spec. torque (H)	
187	192	Tool max. torque (L)	Unit: 0.001 x torque of the current report
188	193	Tool max. torque (H)	

■ When the version number is 1:

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	750
C9	Version number	1
CA	Production report ID (L)	1 to 200000
CB	Production report ID (H)	0: the latest production report
CC, CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	750
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Production report ID	Exceeds the range of 0 to 200000

3. Read the content of a single production report in 0xD2 - 0x188 and 0x1CF - 0x1F6.

Modbus (Hex)	Function	Description
D2 to 188	Refer to the function list of version number 0	
1CF to 1E2	Sequence title	ASCII code (20 words)
1E3 to 1F6	Parameter title	ASCII code (20 words)

#751 Find and read curves

Content description:

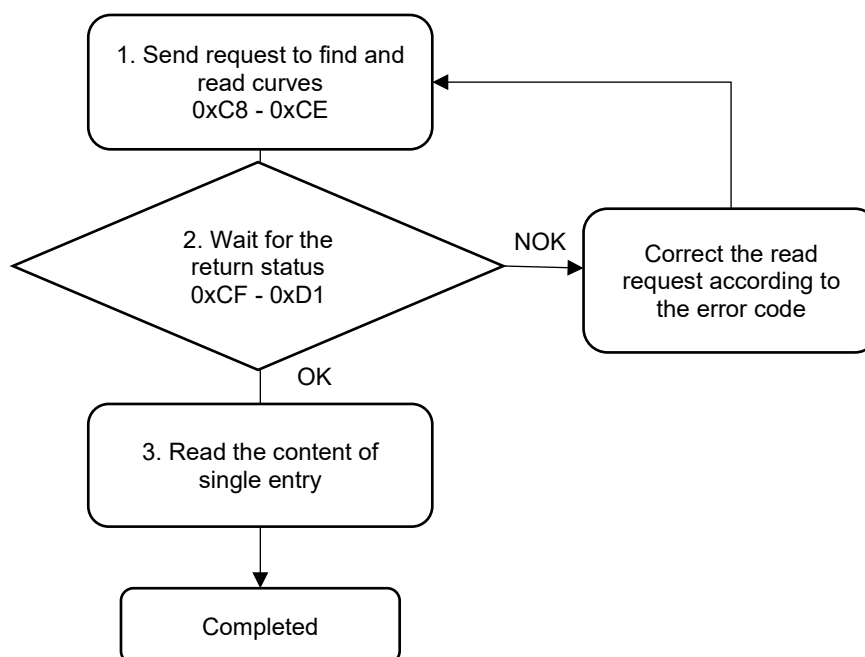
Use function code #751 to read the content of the operating curve.

- A. Use the following Modbus address or function code #21 to read the content of the latest curve.

Modbus (Hex)	Function code #21 (Dec)	Tool 1 / Tool 2 common status	R/W
6B	108	Current No. of production report entries (L)	R
6C	109	Current No. of production report entries (H)	R

- B. Check the content of history curves.
 C. Set the time as X axis and the torque as Y axis to draw a torque-time coordinate graph.
 D. Set the angle as X axis and the torque as Y axis to draw a torque-angle coordinate graph.
 E. Set the angle as X axis and the torque rate as Y axis to draw a torque rate-angle coordinate graph.
 F. Read the parameters content of the curve.

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	751
C9	Version number	0 or 1
CA	Production report ID (L)	1 to 200000
CB	Production report ID (H)	0: the latest production report

Modbus (Hex)	Write / Read request	Write
CC	Query type	0: time scale data (1 to 2000) 1: angle scale data (1 to 2000) 2: torque scale data (1 to 2000) 3: torque rate scale data (1 to 2000) 4: torque scale data DWORD (1 to 1000) 5: torque scale data DWORD (1001 to 2000) 6: torque rate scale data DWORD (1 to 1000) 7: torque rate scale data DWORD (1001 to 2000) 10: values of coordinate axis 11: parameters 20: time scale data (2001 to 4000) 21: angle scale data (2001 to 4000) 22: torque scale data (2001 to 4000) 23: torque rate scale data (2001 to 4000) 24: torque scale data DWORD (2001 to 3000) 25: torque scale data DWORD (3001 to 4000) 26: torque rate scale data DWORD (2001 to 3000) 27: torque rate scale data DWORD (3001 to 4000) 30: time scale data (4001 to 6000) 31: angle scale data (4001 to 6000) 32: torque scale data (4001 to 6000) 33: torque rate scale data (4001 to 6000) 34: torque scale data DWORD (4001 to 5000) 35: torque scale data DWORD (5001 to 6000) 36: torque rate scale data DWORD (4001 to 5000) 37: torque rate scale data DWORD (5001 to 6000) 40: time scale data (6001 to 8000) 41: angle scale data (6001 to 8000) 42: torque scale data (6001 to 8000) 43: torque rate scale data (6001 to 8000) 44: torque scale data DWORD (6001 to 7000) 45: torque scale data DWORD (7001 to 8000) 46: torque rate scale data DWORD (6001 to 7000) 47: torque rate scale data DWORD (7001 to 8000)
CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	751
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows:

Code	Error description	Description
1	Production report ID	Exceeds the range of 0 to 200000
2	Query type	Must be 0 to 7, 10, 11, 20 to 27, 30 to 37, or 40 to 47

3. Read the content of a single operating curve.

Modbus (Hex)	Query type 0	Query type 1
D2	Time scale 1 data	Angle scale 1 data
D3	Time scale 2 data	Angle scale 2 data
D4	Time scale 3 data	Angle scale 3 data
D5	Time scale 4 data	Angle scale 4 data
D6	Time scale 5 data	Angle scale 5 data
D7	Time scale 6 data	Angle scale 6 data
D8	Time scale 7 data	Angle scale 7 data
D9	Time scale 8 data	Angle scale 8 data
DA	Time scale 9 data	Angle scale 9 data
...	Time scale N data	Angle scale N data
898	Time scale 1991 data	Angle scale 1991 data
899	Time scale 1992 data	Angle scale 1992 data
89A	Time scale 1993 data	Angle scale 1993 data
89B	Time scale 1994 data	Angle scale 1994 data
89C	Time scale 1995 data	Angle scale 1995 data
89D	Time scale 1996 data	Angle scale 1996 data
89E	Time scale 1997 data	Angle scale 1997 data
89F	Time scale 1998 data	Angle scale 1998 data
8A0	Time scale 1999 data	Angle scale 1999 data
8A1	Time scale 2000 data	Angle scale 2000 data

Modbus (Hex)	Query type 2	Query type 3
D2	Torque scale 1 data	Torque rate scale 1 data
D3	Torque scale 2 data	Torque rate scale 2 data
D4	Torque scale 3 data	Torque rate scale 3 data
D5	Torque scale 4 data	Torque rate scale 4 data
D6	Torque scale 5 data	Torque rate scale 5 data
D7	Torque scale 6 data	Torque rate scale 6 data
D8	Torque scale 7 data	Torque rate scale 7 data
D9	Torque scale 8 data	Torque rate scale 8 data
DA	Torque scale 9 data	Torque rate scale 9 data
...	Torque scale N data	Torque rate scale N data
898	Torque scale 1991 data	Torque rate scale 1991 data
899	Torque scale 1992 data	Torque rate scale 1992 data
89A	Torque scale 1993 data	Torque rate scale 1993 data
89B	Torque scale 1994 data	Torque rate scale 1994 data
89C	Torque scale 1995 data	Torque rate scale 1995 data
89D	Torque scale 1996 data	Torque rate scale 1996 data
89E	Torque scale 1997 data	Torque rate scale 1997 data
89F	Torque scale 1998 data	Torque rate scale 1998 data
8A0	Torque scale 1999 data	Torque rate scale 1999 data
8A1	Torque scale 2000 data	Torque rate scale 2000 data

Modbus (Hex)	Query type 4	Query type 5
D2	Torque scale 1 data (L)	Torque scale 1001 data (L)
D3	Torque scale 1 data (H)	Torque scale 1001 data (H)
D4	Torque scale 2 data (L)	Torque scale 1002 data (L)
D5	Torque scale 2 data (H)	Torque scale 1002 data (H)
D6	Torque scale 3 data (L)	Torque scale 1003 data (L)
D7	Torque scale 3 data (H)	Torque scale 1003 data (H)
D8	Torque scale 4 data (L)	Torque scale 1004 data (L)
D9	Torque scale 4 data (H)	Torque scale 1004 data (H)
...	Torque scale N data (L)	Torque scale N data (L)
...	Torque scale N data (H)	Torque scale N data (H)
898	Torque scale 996 data (L)	Torque scale 1996 data (L)
899	Torque scale 996 data (H)	Torque scale 1996 data (H)
89A	Torque scale 997 data (L)	Torque scale 1997 data (L)
89B	Torque scale 997 data (H)	Torque scale 1997 data (H)
89C	Torque scale 998 data (L)	Torque scale 1998 data (L)
89D	Torque scale 998 data (H)	Torque scale 1998 data (H)
89E	Torque scale 999 data (L)	Torque scale 1999 data (L)
89F	Torque scale 999 data (H)	Torque scale 1999 data (H)
8A0	Torque scale 1000 data (L)	Torque scale 2000 data (L)
8A1	Torque scale 1000 data (H)	Torque scale 2000 data (H)

Modbus (Hex)	Query type 6	Query type 7
D2	Torque rate scale 1 data (L)	Torque rate scale 1001 data (L)
D3	Torque rate scale 1 data (H)	Torque rate scale 1001 data (H)
D4	Torque rate scale 2 data (L)	Torque rate scale 1002 data (L)
D5	Torque rate scale 2 data (H)	Torque rate scale 1002 data (H)
D6	Torque rate scale 3 data (L)	Torque rate scale 1003 data (L)
D7	Torque rate scale 3 data (H)	Torque rate scale 1003 data (H)
D8	Torque rate scale 4 data (L)	Torque rate scale 1004 data (L)
D9	Torque rate scale 4 data (H)	Torque rate scale 1004 data (H)
...	Torque rate scale N data (L)	Torque rate scale N data (L)
...	Torque rate scale N data (H)	Torque rate scale N data (H)
898	Torque rate scale 996 data (L)	Torque rate scale 1996 data (L)
899	Torque rate scale 996 data (H)	Torque rate scale 1996 data (H)
89A	Torque rate scale 997 data (L)	Torque rate scale 1997 data (L)
89B	Torque rate scale 997 data (H)	Torque rate scale 1997 data (H)
89C	Torque rate scale 998 data (L)	Torque rate scale 1998 data (L)
89D	Torque rate scale 998 data (H)	Torque rate scale 1998 data (H)
89E	Torque rate scale 999 data (L)	Torque rate scale 1999 data (L)
89F	Torque rate scale 999 data (H)	Torque rate scale 1999 data (H)
8A0	Torque rate scale 1000 data (L)	Torque rate scale 2000 data (L)
8A1	Torque rate scale 1000 data (H)	Torque rate scale 2000 data (H)

Modbus (Hex)	Function code #760 (Dec)	Query type 10	Description
D2	1011	Running angle of stage 1	Unit: degree
D3	1012	Running angle of stage 2	
D4	1013	Running angle of stage 3	
D5	1014	Running angle of stage 4	
D6	1015	Running angle of stage 5	
D7	1016	Running angle of stage 6	
D8	1017	Running angle of loosening stage 1	
D9	1018	Running angle of loosening stage 2	
DA	1019	Max. torque of stage 1	Unit: 0.001 x current torque unit Refer to DWORD when the torque specification of the tool is 5 N.m or higher.
DB	1020	Max. torque of stage 2	
DC	1021	Max. torque of stage 3	
DD	1022	Max. torque of stage 4	
DE	1023	Max. torque of stage 5	
DF	1024	Max. torque of stage 6	
E0	1025	Max. torque of loosening stage 1	
E1	1026	Max. torque of loosening stage 2	
E2	1027	Operation time of stage 1	Unit: ms
E3	1028	Operation time of stage 2	
E4	1029	Operation time of stage 3	
E5	1030	Operation time of stage 4	
E6	1031	Operation time of stage 5	
E7	1032	Operation time of stage 6	
E8	1033	Operation time of loosening stage 1	
E9	1034	Operation time of loosening stage 2	
EA	1035	Max. time on the scale	Unit: ms
EB	1036	Max. angle on the scale	Unit: degree
EC	1037	Max. torque on the scale	Unit: 0.001 x current torque unit Refer to DWORD when the torque specification of the tool is 5 N.m or higher.
ED	1038	Max. torque rate on the scale / Max. speed on the scale	1. Unit: 0.0001 x current torque rate unit. Refer to DWORD when the torque specification of the tool is 5 N.m or higher. 2. Unit: rpm (when the torque rate curve is replaced by the speed curve)
EE	1039	Total number of curve coordinates	Total sampling points on the curve.
EF	1040	Max. torque	Upper limit of the torque specified on the curve (unit: 0.001 x current torque unit) Refer to DWORD when the torque specification of the tool is 5 N.m or higher.
F0	1041	Min. torque	Lower limit of the torque specified on the curve (unit: 0.001 x current torque unit) Refer to DWORD when the torque specification of the tool is 5 N.m or higher.

Modbus (Hex)	Function code #760 (Dec)	Query type 10	Description
F1	1042	Max. torque rate	Target torque rated specified on the curve (unit: current 0.0001 x torque rate unit) Refer to DWORD when the torque specification of the tool is 5 N.m or higher.
F2	1043	Max. angle	Upper limit of the angle for the last operating stage (unit: degree)
F3	1044	Min. angle	Lower limit of the angle for the last operating stage (unit: degree)
F4	1045	Version of the curve scale	0: up to 2000 points can be specified on the curve. 1: the curve can support different versions of sampling rates and the use of torque rate curve 2: the curve can support different versions of sampling rates and the use of speed curve
F5	1046	Sampling rate for curves	0: 10 ms→4 ms when start torque is reached (up to 2000 points for a single curve); up to 200000 entries for the report. 1: 2 ms; max. 4 seconds per production report entry (up to 2000 points for a single curve); up to 200000 entries for the report. 2: 2 ms; max. 8 seconds per production report entry (up to 4000 points for a single curve); up to 100000 entries for the report. 3: 1 ms; max. 4 seconds per production report entry (up to 4000 points for a single curve); up to 100000 entries for the report. 4: 1 ms; max. 8 seconds per production report entry (up to 8000 points for a single curve); up to 50000 entries for the report.
F6	1047	Max. torque rate of curve	Unit: 0.0001 x current torque rate unit Refer to DWORD when the torque specification of the tool is 5 N.m or higher.
F7	1048	Min. time on the scale	Unit: ms
F8	1049	Min. angle on the scale	Unit: degree
F9	1050	Min. torque on the scale	Unit: 0.001 x current torque unit Refer to DWORD when the torque specification of the tool is 5 N.m or higher.
FA	1051	Min. torque rate on the scale / Min. speed on the scale	1. Unit: 0.0001 x current torque rate unit. Refer to DWORD when the torque specification of the tool is 5 N.m or higher. 2. Unit: rpm (when the torque rate curve is replaced by the speed curve)

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Modbus (Hex)	Function code #760 (Dec)	Query type 10	Description
FB	1052	Switching torque for stage 1 / Max. speed for stage 1	1. When the function of replacing torque rate curve with speed curve is enabled, the ending torque of the stage is recorded (unit: 0.001 x torque unit at present). Refer to DWORD when the torque specification of the tool is 5 N.m or higher. 2. When the function of replacing torque rate curve with speed curve is enabled, the maximum speed of the stage is recorded (unit: rpm).
FC	1053	Switching torque for stage 2 / Max. speed for stage 2	
FD	1054	Switching torque for stage 3 / Max. speed for stage 3	
FE	1055	Switching torque for stage 4 / Max. speed for stage 4	
FF	1056	Switching torque for stage 5 / Max. speed for stage 5	
100	1057	Switching torque for stage 6 / Max. speed for stage 6	
101 to 103	1058 to 1060	Reserved	-
104	1061	Max. torque of stage 1 (L)	Unit: 0.001 x current torque unit
105	1062	Max. torque of stage 1 (H)	
106	1063	Max. torque of stage 2 (L)	Unit: 0.001 x current torque unit
107	1064	Max. torque of stage 2 (H)	
108	1065	Max. torque of stage 3 (L)	Unit: 0.001 x current torque unit
109	1066	Max. torque of stage 3 (H)	
10A	1067	Max. torque of stage 4 (L)	Unit: 0.001 x current torque unit
10B	1068	Max. torque of stage 4 (H)	
10C	1069	Max. torque of stage 5 (L)	Unit: 0.001 x current torque unit
10D	1070	Max. torque of stage 5 (H)	
10E	1071	Max. torque of stage 6 (L)	Unit: 0.001 x current torque unit
10F	1072	Max. torque of stage 6 (H)	
110	1073	Max. torque of loosening stage 1 (L)	Unit: 0.001 x current torque unit
111	1074	Max. torque of loosening stage 1 (H)	
112	1075	Max. torque of loosening stage 2 (L)	Unit: 0.001 x current torque unit
113	1076	Max. torque of loosening stage 2 (H)	
114 to 11B	1077 to 1084	Reserved	-
11C	1085	Max. torque on the scale / Max. speed on the scale (L)	1. Unit: 0.001 x current torque rate unit
11D	1086	Max. torque on the scale / Max. speed on the scale (H)	2. Unit: rpm (the torque rate curve is replaced by the speed curve)
11E	1087	Max. torque rate on the scale (L)	Unit: 0.001 x current torque rate unit
11F	1088	Max. torque rate on the scale (H)	
120	1089	Max. torque (L)	Unit: 0.001 x current torque unit
121	1090	Max. torque (H)	
122	1091	Min. torque (L)	Unit: 0.001 x current torque unit
123	1092	Min. torque (H)	
124	1093	Max. torque rate (L)	Unit: 0.001 x current torque rate unit
125	1094	Max. torque rate (H)	
126	1095	Max. torque rate of curve (L)	Unit: 0.0001 x current torque rate unit
127	1096	Max. torque rate of curve (H)	
128	1097	Min. torque on the scale / Min. speed on the scale (L)	1. Unit: 0.001 x current torque unit 2. Unit: rpm (the torque rate curve is replaced by the speed curve)
129	1098	Min. torque on the scale / Min. speed on the scale (H)	
12A	1099	Min. torque rate on the scale (L)	Unit: 0.001 x current torque rate unit
12B	1100	Min. torque rate on the scale (H)	

Modbus (Hex)	Function code #760 (Dec)	Query type 10	Description
12C	1101	Switching torque for stage 1 / Speed for stage 1 (L)	1. Unit: 0.001 x torque unit at present 2. Unit: rpm (the torque rate curve is replaced by the speed curve)
12D	1102	Switching torque for stage 1 / Speed for stage 1 (H)	
12E	1103	Switching torque for stage 2 / Speed for stage 2 (L)	
12F	1104	Switching torque for stage 2 / Speed for stage 2 (H)	
130	1105	Switching torque for stage 3 / Speed for stage 3 (L)	
131	1106	Switching torque for stage 3 / Speed for stage 3 (H)	
132	1107	Switching torque for stage 4 / Speed for stage 4 (L)	
133	1108	Switching torque for stage 4 / Speed for stage 4 (H)	
134	1109	Switching torque for stage 5 / Speed for stage 5 (L)	
135	1110	Switching torque for stage 5 / Speed for stage 5 (H)	
136	1111	Switching torque for stage 6 / Speed for stage 6 (L)	
137	1112	Switching torque for stage 6 / Speed for stage 6 (H)	

Modbus (Hex)	#760 function code (Dec)	Query type 11 (version number 0)	Description
D2	2011	Parameter ID	1 to 500
D3	2012	Version number	0
D4	2013	Torque unit	0: N.m; 1: kgf.cm; 2: lbf.ft; 3: lbf.in; 4: ozf.ft; 5: ozf.in; 6: cNm
D5 to DB	2014 to 2020	Reserved	-
DC	2021	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
DD to E6	2022 to 2031	Reserved	-
E7	2032	Max. current for tightening	-
E8	2033	Max. tightening time	Unit: ms
E9	2034	Max. tightening angle	Unit: degree
EA	2035	Reserved	-
EB	2036	Delay before tightening starts	Unit: ms
EC	2037	Min. tightening angle	Unit: degree
ED to F1	2038 to 2042	Reserved	-
F2	2043	Max. loosening time	Unit: ms
F3	2044	Max. loosening angle	Unit: degree
F4	2045	Reserved	-
F5	2046	Delay before loosening starts	Unit: ms
F6 to FA	2047 to 2051	Reserved	-
FB	2052	ID of the prevail torque to be linked	1 to 500
FC to 103	2053 to 2060	Reserved	-

Self-defined strategy						Loosening		Query type 11 (version number 0)	Description	
Stage 1 Hex / Dec	Stage 2 Hex / Dec	Stage 3 Hex / Dec	Stage 4 Hex / Dec	Stage 5 Hex / Dec	Stage 6 Hex / Dec	1 st stage Hex / Dec	2 nd stage Hex / Dec			
104 / 2061	136 / 2111	168 / 2161	19A / 2211	1CC / 2261	1FE / 2311	230 / 2361	262 / 2411	Target On / Off (L)	Bit 0 to 1 Control method type	0: position 1: speed
									Bit 2 to 3 Operation after stage completion	0: stop 1: execute the next stage
									Bit 4 to 5 Control method	0: angle 1: torque 2: torque rate
									Bit 6 Rotation direction of the stage	0: CW 1: CCW
									Bit 7 to 8 Hold time switch of the final stage	0: Off 1: On
									Bit 9 Prevail torque	0: Off 1: On
105 / 2062	137 / 2112	169 / 2162	19B / 2212	1CD / 2262	1FF / 2312	231 / 2362	263 / 2412	Target On / Off (H)	-	
106 / 2063	138 / 2113	16A / 2163	19C / 2213	1CE / 2263	200 / 2313	232 / 2363	264 / 2413	Target torque	Unit: 0.001 N.m (used with Target On / Off Bit 4, 1: torque)	
107 / 2064	139 / 2114	16B / 2164	19D / 2214	1CF / 2264	201 / 2314	233 / 2364	265 / 2414	Rotation speed	Unit: rpm	
108 / 2065	13A / 2115	16C / 2165	19E / 2215	1D0 / 2265	202 / 2315	234 / 2365	266 / 2415	Target angle	Unit: degree (used with Target On / Off Bit 4, 0: angle)	
109 / 2066	13B / 2116	16D / 2166	19F / 2216	1D1 / 2266	203 / 2316	235 / 2366	267 / 2416	Hold / pause time after stage completion	Unit: ms	
10A / 2067	13C / 2117	16E / 2167	1A0 / 2217	1D2 / 2267	204 / 2317	236 / 2367	268 / 2417	Acceleration time	Unit: ms	
10B / 2068	13D / 2118	16F / 2168	1A1 / 2218	1D3 / 2268	205 / 2318	237 / 2368	269 / 2418	Deceleration time	Unit: ms	
10C / 2069	13E / 2119	170 / 2169	1A2 / 2219	1D4 / 2269	206 / 2319	238 / 2369	26A / 2419	Start torque for torque rate calculation	-	
10D / 2070	13F / 2120	171 / 2170	1A3 / 2220	1D5 / 2270	207 / 2320	239 / 2370	26B / 2420	Angle interval for torque rate calculation	-	
10E / 2071	140 / 2121	172 / 2171	1A4 / 2221	1D6 / 2271	208 / 2321	23A / 2371	26C / 2421	Target torque rate	Unit: 0.0001 N.m/degree (used with Target On / Off Bit 4, 2: torque rate)	
10F to 11D / 2072 to 2086	141 to 14F / 2022 to 2136	173 to 181 / 2172 to 2186	1A5 to 1B3 / 2222 to 2236	1D7 to 1E5 / 2272 to 2286	209 to 217 / 2322 to 2336	23B to 249 / 2372 to 2386	26D to 27B / 2422 to 2436	Reserved	-	
11E / 2087	150 / 2137	182 / 2187	1B4 / 2337	1E6 / 2287	218 / 2337	24A / 2387	27C / 2437	Max. angle	Used with Limit On / Off (L) Bit 2	
11F / 2088	151 / 2138	183 / 2188	1B5 / 2338	1E7 / 2288	219 / 2338	24B / 2388	27D / 2438	Min. angle	Used with Limit On / Off (L) Bit 2	
120 / 2089	152 / 2139	184 / 2189	1B6 / 2339	1E8 / 2289	21A / 2339	24C / 2389	27E / 2439	Max. clamp torque	Used with Limit On / Off (L) Bit 3	
121 / 2090	153 / 2140	185 / 2190	1B7 / 2240	1E9 / 2290	21B / 2340	24D / 2390	27F / 2440	Min. clamp torque	Used with Limit On / Off (L) Bit 3	
122 / 2091	154 / 2141	186 / 2191	1B8 / 2241	1EA / 2291	21C / 2341	24E / 2391	280 / 2441	Max. torque	Used with Limit On / Off (H) Bit 0	

Self-defined strategy						Loosening		Query type 11 (version number 0)	Description
Stage 1 Hex / Dec	Stage 2 Hex / Dec	Stage 3 Hex / Dec	Stage 4 Hex / Dec	Stage 5 Hex / Dec	Stage 6 Hex / Dec	1 st stage Hex / Dec	2 nd stage Hex / Dec		
123 / 2092	155 / 2142	187 / 2192	1B9 / 2242	1EB / 2292	21D / 2342	24F / 2392	281 / 2442	Min. torque	Used with Limit On / Off (H) Bit 0
124 to 12A / 2093 to 2099	156 to 15C / 2143 to 2149	188 to 18E / 2193 to 2199	1BA to 1C0 / 2243 to 2249	1EC to 1F2 / 2293 to 2299	21E to 224 / 2343 to 2349	250 to 256 / 2393 to 2399	282 to 288 / 2443 to 2449	Reserved	-
12B / 2100	15D / 2150	18F / 2200	1C1 / 2250	1F3 / 2300	225 / 2350	257 / 2400	289 / 2450	Max. operation time	Unit: ms Used with Limit On / Off (H) Bit 15
12C / 2101	15E / 2151	190 / 2201	1C2 / 2251	1F4 / 2301	226 / 2351	258 / 2401	28A / 2451	Min. operation time	Unit: ms Used with Limit On / Off (H) Bit 15
12D / 2102	15F / 2152	191 / 2202	1C3 / 2252	1F5 / 2302	227 / 2352	259 / 2402	28B / 2452	Reserved	-
12E / 2103	160 / 2153	192 / 2203	1C4 / 2253	1F6 / 2303	228 / 2353	25A / 2403	28C / 2453	Reserved	-
12F / 2104	161 / 2154	193 / 2204	1C5 / 2254	1F7 / 2304	229 / 2354	25B / 2404	28D / 2454	Angle range for prevail torque calculation	Set the angle range (%) of the stage to calculate the average torque
130 / 2105	162 / 2155	194 / 2205	1C6 / 2255	1F8 / 2305	22A / 2355	25C / 2405	28E / 2455	Reserved	-
131 / 2106	163 / 2156	195 / 2206	1C7 / 2256	1F9 / 2306	22B / 2356	25D / 2406	28F / 2456	Reserved	-
132 / 2107	164 / 2157	196 / 2207	1C8 / 2257	1FA / 2307	22C / 2357	25E / 2407	290 / 2457	Limit On / Off (L)	Bit 2: limit the angle
133 / 2108	165 / 2158	197 / 2208	1C9 / 2258	1FB / 2308	22D / 2358	25F / 2408	291 / 2458	Limit On / Off (H)	Bit 0: limit the torque Bit 15: limit the operation time
134 / 2109	166 / 2159	198 / 2209	1CA / 2259	1FC / 2309	22E / 2359	260 / 2409	292 / 2459	Reserved	-
135 / 2110	167 / 2160	199 / 2210	1CB / 2260	1FD / 2310	22F / 2360	261 / 2410	293 / 2460	Stage code	10: start stage 20: rundown stage (angle / torque mode) 29: rundown stage (torque rate mode) 30: pre-tightening stage 40: tightening stage 90: first stage of loosening 91: second stage of loosening

Modbus (Hex)	Function code #760 (Dec)	Query type 11 (version number 1)	Description
D2	2011	Parameter ID	1 to 500
D3	2012	Version number	1
D4	2013	Torque unit	0: N.m; 1: kgf.cm; 2: lbf.ft; 3: lbf.in; 4: o1zf.ft; 5: ozf.in; 6: cNm
D5 to D7	2014 to 2016	Reserved	-
D8	2017	1st stage loosening angle	-
D9	2018	2nd stage loosening angle	-
DA	2019	Switch for monitoring the min. loosening torque	-
DB	2020	Reserved	-
DC	2021	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
DD to E6	2022 to 2031	Reserved	-
E7	2032	Max. current for tightening	-
E8	2033	Max. tightening time	Unit: ms

Modbus (Hex)	Function code #760 (Dec)	Query type 11 (version number 1)	Description
E9	2034	Max. tightening angle	Unit: degree
EA	2035	Reserved	-
EB	2036	Delay before tightening starts	Unit: ms
EC	2037	Min. tightening angle	Unit: degree
ED to EF	2038 to 2040	Reserved	-
F0	2041	Rotation direction of loosening	-
F1	2042	Reserved	-
F2	2043	Max. loosening time	Unit: ms
F3	2044	Max. loosening angle	Unit: degree
F4	2045	Reserved	-
F5	2046	Delay before loosening starts	Unit: ms
F6	2047	Min. loosening angle	Unit: degree
F7, F8	2048, 2049	Reserved	-
F9	2050	Min. loosening torque (L)	Unit: 0.001 x torque unit
FA	2051	Min. loosening torque (H)	
FB	2052	ID of the prevail torque to be linked	1 to 500
FC to 101	2053 to 2058	Reserved	-
102	2059	1st stage loosening speed	-
103	2060	2nd stage loosening speed	-

Self-defined strategy						Loosening		Query type 11 (version number 1)	Description	
Stage 1 Hex / Dec	Stage 2 Hex / Dec	Stage 3 Hex / Dec	Stage 4 Hex / Dec	Stage 5 Hex / Dec	Stage 6 Hex / Dec	1 st stage Hex / Dec	2 nd stage Hex / Dec			
104 / 2061	136 / 2111	168 / 2161	19A / 2211	1CC / 2261	1FE / 2311	230 / 2361	262 / 2411	Target On / Off (L)	Bit 0 to 1 Control method type	0: position 1: speed
									Bit 2 to 3 Operation after stage completion	0: stop 1: execute the next stage
									Bit 4 to 5 Control method	0: angle 1: torque 2: torque rate 3: yield point
									Bit 6 Rotation direction of the stage	0: CW 1: CCW
									Bit 7 to 8 Hold time switch of the final stage	0: Off 1: On
									Bit 9 Prevail torque	0: Off 1: On
									Bit 10 Reserved	-
									Bit 11 Relative / absolute torque command	0: absolute 1: relative
									Bit 12 Timing to add the relative torque	0: beginning of a stage 1: snug point
105 / 2062	137 / 2112	169 / 2162	19B / 2212	1CD / 2262	1FF / 2312	231 / 2362	263 / 2412	Target On / Off (H)	-	
106 / 2063	138 / 2113	16A / 2163	19C / 2213	1CE / 2263	200 / 2313	232 / 2363	264 / 2413	Final target speed	-	

Self-defined strategy						Loosening		Query type 11 (version number 1)	Description
Stage 1 Hex / Dec	Stage 2 Hex / Dec	Stage 3 Hex / Dec	Stage 4 Hex / Dec	Stage 5 Hex / Dec	Stage 6 Hex / Dec	1 st stage Hex / Dec	2 nd stage Hex / Dec		
107 / 2064	139 / 2114	16B / 2164	19D / 2214	1CF / 2264	201 / 2314	233 / 2364	265 / 2414	Rotation speed	Unit: rpm
108 / 2065	13A / 2115	16C / 2165	19E / 2215	1D0 / 2265	202 / 2315	234 / 2365	266 / 2415	Target angle	Unit: degree (used with Target On / Off Bit 4, 0: angle)
109 / 2066	13B / 2116	16D / 2166	19F / 2216	1D1 / 2266	203 / 2316	235 / 2366	267 / 2416	Hold / pause time after stage completion	Unit: ms
10A / 2067	13C / 2117	16E / 2167	1A0 / 2217	1D2 / 2267	204 / 2317	236 / 2367	268 / 2417	Acceleration time	Unit: ms
10B / 2068	13D / 2118	16F / 2168	1A1 / 2218	1D3 / 2268	205 / 2318	237 / 2368	269 / 2418	Deceleration time	Unit: ms
10C / 2069	13E / 2119	170 / 2169	1A2 / 2219	1D4 / 2269	206 / 2319	238 / 2369	26A / 2419	Acceleration time of 2nd stage	-
10D / 2070	13F / 2120	171 / 2170	1A3 / 2220	1D5 / 2270	207 / 2320	239 / 2370	26B / 2420	Angle interval for torque rate calculation	-
10E / 2071	140 / 2121	172 / 2171	1A4 / 2221	1D6 / 2271	208 / 2321	23A / 2371	26C / 2421	Final pause time	Unit: ms
10F / 2072	141 / 2122	173 / 2172	1A5 / 2222	1D7 / 2272	209 / 2322	23B / 2372	26D / 2422	Reserved	-
110 / 2073	142 / 2123	174 / 2173	1A6 / 2223	1D8 / 2273	20A / 2323	23C / 2373	26E / 2423	Target torque (L)	Unit: 0.001 x torque unit (used with Target On / Off Bit 4, 1: torque)
111 / 2074	143 / 2124	175 / 2174	1A7 / 2224	1D9 / 2274	20B / 2324	23D / 2374	26F / 2424	Target torque (H)	
112 / 2075	144 / 2125	176 / 2175	1A8 / 2225	1DA / 2275	20C / 2325	23E / 2375	270 / 2425	Reserved	-
113 / 2076	145 / 2126	177 / 2176	1A9 / 2226	1DB / 2276	20D / 2326	23F / 2376	271 / 2426	Reserved	-
114 / 2077	146 / 2127	178 / 2177	1AA / 2227	1DC / 2277	20E / 2327	240 / 2377	272 / 2427	Target torque rate (L)	Unit: 0.0001 torque unit/degree (used with Target On / Off Bit 4, 2: torque rate)
115 / 2078	147 / 2128	179 / 2178	1AB / 2228	1DD / 2278	20F / 2328	241 / 2378	273 / 2428	Target torque rate (H)	
116 / 2079	148 / 2129	17A / 2179	1AC / 2229	1DE / 2279	210 / 2329	242 / 2379	274 / 2429	Torque of 1st stage (L)	Unit: 0.001 x torque unit
117 / 2080	149 / 2130	17B / 2180	1AD / 2230	1DF / 2280	211 / 2330	243 / 2380	275 / 2430	Torque of 1st stage (H)	
118 / 2081	14A / 2131	17C / 2181	1AE / 2231	1E0 / 2281	212 / 2331	244 / 2381	276 / 2431	Reserved	-
119 / 2082	14B / 2132	17D / 2182	1AF / 2232	1E1 / 2282	213 / 2332	245 / 2382	277 / 2432	Reserved	-
11A / 2083	14C / 2133	17E / 2183	1B0 / 2233	1E2 / 2283	214 / 2333	246 / 2383	278 / 2433	Max. clamp torque (L)	Unit: 0.001 x torque unit Used with Limit On / Off (L) Bit 3
11B / 2084	14D / 2134	17F / 2184	1B1 / 2234	1E3 / 2284	215 / 2334	247 / 2384	279 / 2434	Max. clamp torque (H)	
11C / 2085	14E / 2135	180 / 2185	1B2 / 2235	1E4 / 2285	216 / 2335	248 / 2385	27A / 2435	Min. clamp torque (L)	Unit: 0.001 x torque unit Used with Limit On / Off (L) Bit 3
11D / 2086	14F / 2136	181 / 2186	1B3 / 2236	1E5 / 2286	217 / 2336	249 / 2386	27B / 2436	Min. clamp torque (H)	
11E / 2087	150 / 2137	182 / 2187	1B4 / 2237	1E6 / 2287	218 / 2337	24A / 2387	27C / 2437	Max. angle	Used with Limit On / Off (L) Bit 2
11F / 2088	151 / 2138	183 / 2188	1B5 / 2238	1E7 / 2288	219 / 2338	24B / 2388	27D / 2438	Min. angle	Used with Limit On / Off (L) Bit 2
120 / 2089	152 / 2139	184 / 2189	1B6 / 2239	1E8 / 2289	21A / 2339	24C / 2389	27E / 2439	Max. torque (L)	Unit: 0.001 x torque unit Used with Limit On / Off (H) Bit 0
121 / 2090	153 / 2140	185 / 2190	1B7 / 2240	1E9 / 2290	21B / 2340	24D / 2390	27F / 2440	Max. torque (H)	

A

Self-defined strategy						Loosening		Query type 11 (version number 1)	Description
Stage 1 Hex / Dec	Stage 2 Hex / Dec	Stage 3 Hex / Dec	Stage 4 Hex / Dec	Stage 5 Hex / Dec	Stage 6 Hex / Dec	1 st stage Hex / Dec	2 nd stage Hex / Dec		
122 / 2091	154 / 2141	186 / 2191	1B8 / 2241	1EA / 2291	21C / 2341	24E / 2391	280 / 2441	Min. torque (L)	Unit: 0.001 x torque unit Used with Limit On / Off (H) Bit 0
123 / 2092	155 / 2142	187 / 2192	1B9 / 2242	1EB / 2292	21D / 2342	24F / 2392	281 / 2442	Min. torque (H)	
124 ~ 12A / 2093 ~ 2099	156 ~ 15C / 2143 ~ 2149	188 ~ 18E / 2193 ~ 2199	1BA ~ 1C0 / 2243 ~ 2249	1EC ~ 1F2 / 2293 ~ 2299	21E ~ 224 / 2343 ~ 2349	250 ~ 256 / 2393 ~ 2399	282 ~ 288 / 2443 ~ 2449	Reserved	-
12B / 2100	15D / 2150	18F / 2200	1C1 / 2250	1F3 / 2300	225 / 2350	257 / 2400	289 / 2450	Max. operation time	Unit: ms Used with Limit On / Off (H) Bit 15
12C / 2101	15E / 2151	190 / 2201	1C2 / 2251	1F4 / 2301	226 / 2351	258 / 2401	28A / 2451	Min. operation time	Unit: ms Used with Limit On / Off (H) Bit 15
12D / 2102	15F / 2152	191 / 2202	1C3 / 2252	1F5 / 2302	227 / 2352	259 / 2402	28B / 2452	Reserved	-
12E / 2103	160 / 2153	192 / 2203	1C4 / 2253	1F6 / 2303	228 / 2353	25A / 2403	28C / 2453	Reserved	-
12F / 2104	161 / 2154	193 / 2204	1C5 / 2254	1F7 / 2304	229 / 2354	25B / 2404	28D / 2454	Angle range for prevail torque calculation	Set the angle range (%) of the stage to calculate the average torque
130 / 2105	162 / 2155	194 / 2205	1C6 / 2255	1F8 / 2305	22A / 2355	25C / 2405	28E / 2455	Reserved	-
131 / 2106	163 / 2156	195 / 2206	1C7 / 2256	1F9 / 2306	22B / 2356	25D / 2406	28F / 2456	Reserved	-
132 / 2107	164 / 2157	196 / 2207	1C8 / 2257	1FA / 2307	22C / 2357	25E / 2407	290 / 2457	Limit On / Off (L)	Bit 2: limit the angle
133 / 2108	165 / 2158	197 / 2208	1C9 / 2258	1FB / 2308	22D / 2358	25F / 2408	291 / 2458	Limit On / Off (H)	Bit 0: limit the torque Bit 15: limit the operation time
134 / 2109	166 / 2159	198 / 2209	1CA / 2259	1FC / 2309	22E / 2359	260 / 2409	292 / 2459	Reserved	-
135 / 2110	167 / 2160	199 / 2210	1CB / 2260	1FD / 2310	22F / 2360	261 / 2410	293 / 2460	Stage code	10: start stage 20: rundown stage (angle / torque mode) 29: rundown stage (torque rate mode) 30: pre-tightening stage 40: tightening stage 90: 1st stage of loosening 91: 2nd stage of loosening

Modbus (Hex)	Query type 20	Query type 21
D2	Time scale 2001 data	Angle scale 2001 data
D3	Time scale 2002 data	Angle scale 2002 data
D4	Time scale 2003 data	Angle scale 2003 data
D5	Time scale 2004 data	Angle scale 2004 data
D6	Time scale 2005 data	Angle scale 2005 data
D7	Time scale 2006 data	Angle scale 2006 data
D8	Time scale 2007 data	Angle scale 2007 data
D9	Time scale 2008 data	Angle scale 2008 data
DA	Time scale 2009 data	Angle scale 2009 data
...	Time scale N data	Angle scale N data
898	Time scale 3991 data	Angle scale 3991 data
899	Time scale 3992 data	Angle scale 3992 data
89A	Time scale 3993 data	Angle scale 3993 data
89B	Time scale 3994 data	Angle scale 3994 data
89C	Time scale 3995 data	Angle scale 3995 data
89D	Time scale 3996 data	Angle scale 3996 data
89E	Time scale 3997 data	Angle scale 3997 data
89F	Time scale 3998 data	Angle scale 3998 data
8A0	Time scale 3999 data	Angle scale 3999 data
8A1	Time scale 4000 data	Angle scale 4000 data

Modbus (Hex)	Query type 22	Query type 23
D2	Torque scale 2001 data	Torque rate scale 2001 data
D3	Torque scale 2002 data	Torque rate scale 2002 data
D4	Torque scale 2003 data	Torque rate scale 2003 data
D5	Torque scale 2004 data	Torque rate scale 2004 data
D6	Torque scale 2005 data	Torque rate scale 2005 data
D7	Torque scale 2006 data	Torque rate scale 2006 data
D8	Torque scale 2007 data	Torque rate scale 2007 data
D9	Torque scale 2008 data	Torque rate scale 2008 data
DA	Torque scale 2009 data	Torque rate scale 2009 data
...	Torque scale N data	Torque rate scale N data
898	Torque scale 3991 data	Torque rate scale 3991 data
899	Torque scale 3992 data	Torque rate scale 3992 data
89A	Torque scale 3993 data	Torque rate scale 3993 data
89B	Torque scale 3994 data	Torque rate scale 3994 data
89C	Torque scale 3995 data	Torque rate scale 3995 data
89D	Torque scale 3996 data	Torque rate scale 3996 data
89E	Torque scale 3997 data	Torque rate scale 3997 data
89F	Torque scale 3998 data	Torque rate scale 3998 data
8A0	Torque scale 3999 data	Torque rate scale 3999 data
8A1	Torque scale 4000 data	Torque rate scale 4000 data

Modbus (Hex)	Query type 24	Query type 25
D2	Torque scale 2001 data (L)	Torque scale 3001 data (L)
D3	Torque scale 2001 data (H)	Torque scale 3001 data (H)
D4	Torque scale 2002 data (L)	Torque scale 3002 data (L)
D5	Torque scale 2002 data (H)	Torque scale 3002 data (H)
D6	Torque scale 2003 data (L)	Torque scale 33003 data (L)
D7	Torque scale 2003 data (H)	Torque scale 3003 data (H)
D8	Torque scale 2004 data (L)	Torque scale 3004 data (L)
D9	Torque scale 2004 data (H)	Torque scale 3004 data (H)
...	Torque scale N data (L)	Torque scale N data (L)
...	Torque scale N data (H)	Torque scale N data (H)
898	Torque scale 2996 data (L)	Torque scale 3996 data (L)
899	Torque scale 2996 data (H)	Torque scale 3996 data (H)
89A	Torque scale 2997 data (L)	Torque scale 3997 data (L)
89B	Torque scale 2997 data (H)	Torque scale 3997 data (H)
89C	Torque scale 2998 data (L)	Torque scale 3998 data (L)
89D	Torque scale 2998 data (H)	Torque scale 3998 data (H)
89E	Torque scale 2999 data (L)	Torque scale 3999 data (L)
89F	Torque scale 2999 data (H)	Torque scale 3999 data (H)
8A0	Torque scale 3000 data (L)	Torque scale 4000 data (L)
8A1	Torque scale 3000 data (H)	Torque scale 4000 data (H)

Modbus (Hex)	Query type 26	Query type 27
D2	Torque rate scale 2001 data (L)	Torque rate scale 3001 data (L)
D3	Torque rate scale 2001 data (H)	Torque rate scale 3001 data (H)
D4	Torque rate scale 2002 data (L)	Torque rate scale 3002 data (L)
D5	Torque rate scale 2002 data (H)	Torque rate scale 3002 data (H)
D6	Torque rate scale 2003 data (L)	Torque rate scale 3003 data (L)
D7	Torque rate scale 2003 data (H)	Torque rate scale 3003 data (H)
D8	Torque rate scale 2004 data (L)	Torque rate scale 3004 data (L)
D9	Torque rate scale 2004 data (H)	Torque rate scale 3004 data (H)
...	Torque rate scale N data (L)	Torque rate scale N data (L)
...	Torque rate scale N data (H)	Torque rate scale N data (H)
898	Torque rate scale 2996 data (L)	Torque rate scale 3996 data (L)
899	Torque rate scale 2996 data (H)	Torque rate scale 3996 data (H)
89A	Torque rate scale 2997 data (L)	Torque rate scale 3997 data (L)
89B	Torque rate scale 2997 data (H)	Torque rate scale 3997 data (H)
89C	Torque rate scale 2998 data (L)	Torque rate scale 3998 data (L)
89D	Torque rate scale 2998 data (H)	Torque rate scale 3998 data (H)
89E	Torque rate scale 2999 data (L)	Torque rate scale 3999 data (L)
89F	Torque rate scale 2999 data (H)	Torque rate scale 3999 data (H)
8A0	Torque rate scale 3000 data (L)	Torque rate scale 4000 data (L)
8A1	Torque rate scale 3000 data (H)	Torque rate scale 4000 data (H)

Modbus (Hex)	Query type 30	Query type 31
D2	Time scale 4001 data	Angle scale 4001 data
D3	Time scale 4002 data	Angle scale 4002 data
D4	Time scale 4003 data	Angle scale 4003 data
D5	Time scale 4004 data	Angle scale 4004 data
D6	Time scale 4005 data	Angle scale 4005 data
D7	Time scale 4006 data	Angle scale 4006 data
D8	Time scale 4007 data	Angle scale 4007 data
D9	Time scale 4008 data	Angle scale 4008 data
DA	Time scale 4009 data	Angle scale 4009 data
...	Time scale N data	Angle scale N data
898	Time scale 5991 data	Angle scale 5991 data
899	Time scale 5992 data	Angle scale 5992 data
89A	Time scale 5993 data	Angle scale 5993 data
89B	Time scale 5994 data	Angle scale 5994 data
89C	Time scale 5995 data	Angle scale 5995 data
89D	Time scale 5996 data	Angle scale 5996 data
89E	Time scale 5997 data	Angle scale 5997 data
89F	Time scale 5998 data	Angle scale 5998 data
8A0	Time scale 5999 data	Angle scale 5999 data
8A1	Time scale 6000 data	Angle scale 6000 data

Modbus (Hex)	Query type 32	Query type 33
D2	Torque scale 4001 data	Torque rate scale 4001 data
D3	Torque scale 4002 data	Torque rate scale 4002 data
D4	Torque scale 4003 data	Torque rate scale 4003 data
D5	Torque scale 4004 data	Torque rate scale 4004 data
D6	Torque scale 4005 data	Torque rate scale 4005 data
D7	Torque scale 4006 data	Torque rate scale 4006 data
D8	Torque scale 4007 data	Torque rate scale 4007 data
D9	Torque scale 4008 data	Torque rate scale 4008 data
DA	Torque scale 4009 data	Torque rate scale 4009 data
...	Torque scale N data	Torque rate scale N data
898	Torque scale 5991 data	Torque rate scale 5991 data
899	Torque scale 5992 data	Torque rate scale 5992 data
89A	Torque scale 5993 data	Torque rate scale 5993 data
89B	Torque scale 5994 data	Torque rate scale 5994 data
89C	Torque scale 5995 data	Torque rate scale 5995 data
89D	Torque scale 5996 data	Torque rate scale 5996 data
89E	Torque scale 5997 data	Torque rate scale 5997 data
89F	Torque scale 5998 data	Torque rate scale 5998 data
8A0	Torque scale 5999 data	Torque rate scale 5999 data
8A1	Torque scale 6000 data	Torque rate scale 6000 data

Modbus (Hex)	Query type 34	Query type 35
D2	Torque scale 4001 data (L)	Torque scale 5001 data (L)
D3	Torque scale 4001 data (H)	Torque scale 5001 data (H)
D4	Torque scale 4002 data (L)	Torque scale 5002 data (L)
D5	Torque scale 4002 data (H)	Torque scale 5002 data (H)
D6	Torque scale 4003 data (L)	Torque scale 5003 data (L)
D7	Torque scale 4003 data (H)	Torque scale 5003 data (H)
D8	Torque scale 4004 data (L)	Torque scale 5004 data (L)
D9	Torque scale 4004 data (H)	Torque scale 5004 data (H)
...	Torque scale N data (L)	Torque scale N data (L)
...	Torque scale N data (H)	Torque scale N data (H)
898	Torque scale 4996 data (L)	Torque scale 5996 data (L)
899	Torque scale 4996 data (H)	Torque scale 5996 data (H)
89A	Torque scale 4997 data (L)	Torque scale 5997 data (L)
89B	Torque scale 4997 data (H)	Torque scale 5997 data (H)
89C	Torque scale 4998 data (L)	Torque scale 5998 data (L)
89D	Torque scale 4998 data (H)	Torque scale 5998 data (H)
89E	Torque scale 4999 data (L)	Torque scale 5999 data (L)
89F	Torque scale 4999 data (H)	Torque scale 5999 data (H)
8A0	Torque scale 5000 data (L)	Torque scale 6000 data (L)
8A1	Torque scale 5000 data (H)	Torque scale 6000 data (H)

Modbus (Hex)	Query type 36	Query type 37
D2	Torque rate scale 4001 data (L)	Torque rate scale 5001 data (L)
D3	Torque rate scale 4001 data (H)	Torque rate scale 5001 data (H)
D4	Torque rate scale 4002 data (L)	Torque rate scale 5002 data (L)
D5	Torque rate scale 4002 data (H)	Torque rate scale 5002 data (H)
D6	Torque rate scale 4003 data (L)	Torque rate scale 5003 data (L)
D7	Torque rate scale 4003 data (H)	Torque rate scale 5003 data (H)
D8	Torque rate scale 4004 data (L)	Torque rate scale 5004 data (L)
D9	Torque rate scale 4004 data (H)	Torque rate scale 5004 data (H)
...	Torque rate scale N data (L)	Torque rate scale N data (L)
...	Torque rate scale N data (H)	Torque rate scale N data (H)
898	Torque rate scale 4996 data (L)	Torque rate scale 5996 data (L)
899	Torque rate scale 4996 data (H)	Torque rate scale 5996 data (H)
89A	Torque rate scale 4997 data (L)	Torque rate scale 5997 data (L)
89B	Torque rate scale 4997 data (H)	Torque rate scale 5997 data (H)
89C	Torque rate scale 4998 data (L)	Torque rate scale 5998 data (L)
89D	Torque rate scale 4998 data (H)	Torque rate scale 5998 data (H)
89E	Torque rate scale 4999 data (L)	Torque rate scale 5999 data (L)
89F	Torque rate scale 4999 data (H)	Torque rate scale 5999 data (H)
8A0	Torque rate scale 5000 data (L)	Torque rate scale 6000 data (L)
8A1	Torque rate scale 5000 data (H)	Torque rate scale 6000 data (H)

Modbus (Hex)	Query type 40	Query type 41
D2	Time scale 6001 data	Angle scale 6001 data
D3	Time scale 6002 data	Angle scale 6002 data
D4	Time scale 6003 data	Angle scale 6003 data
D5	Time scale 6004 data	Angle scale 6004 data
D6	Time scale 6005 data	Angle scale 6005 data
D7	Time scale 6006 data	Angle scale 6006 data
D8	Time scale 6007 data	Angle scale 6007 data
D9	Time scale 6008 data	Angle scale 6008 data
DA	Time scale 6009 data	Angle scale 6009 data
...	Time scale N data	Angle scale N data
898	Time scale 7991 data	Angle scale 7991 data
899	Time scale 7992 data	Angle scale 7992 data
89A	Time scale 7993 data	Angle scale 7993 data
89B	Time scale 7994 data	Angle scale 7994 data
89C	Time scale 7995 data	Angle scale 7995 data
89D	Time scale 7996 data	Angle scale 7996 data
89E	Time scale 7997 data	Angle scale 7997 data
89F	Time scale 7998 data	Angle scale 7998 data
8A0	Time scale 7999 data	Angle scale 7999 data
8A1	Time scale 8000 data	Angle scale 8000 data

Modbus (Hex)	Query type 42	Query type 43
D2	Torque scale 6001 data	Torque rate scale 6001 data
D3	Torque scale 6002 data	Torque rate scale 6002 data
D4	Torque scale 6003 data	Torque rate scale 6003 data
D5	Torque scale 6004 data	Torque rate scale 6004 data
D6	Torque scale 6005 data	Torque rate scale 6005 data
D7	Torque scale 6006 data	Torque rate scale 6006 data
D8	Torque scale 6007 data	Torque rate scale 6007 data
D9	Torque scale 6008 data	Torque rate scale 6008 data
DA	Torque scale 6009 data	Torque rate scale 6009 data
...	Torque scale N data	Torque rate scale N data
898	Torque scale 7991 data	Torque rate scale 7991 data
899	Torque scale 7992 data	Torque rate scale 7992 data
89A	Torque scale 7993 data	Torque rate scale 7993 data
89B	Torque scale 7994 data	Torque rate scale 7994 data
89C	Torque scale 7995 data	Torque rate scale 7995 data
89D	Torque scale 7996 data	Torque rate scale 7996 data
89E	Torque scale 7997 data	Torque rate scale 7997 data
89F	Torque scale 7998 data	Torque rate scale 7998 data
8A0	Torque scale 7999 data	Torque rate scale 7999 data
8A1	Torque scale 8000 data	Torque rate scale 8000 data

Modbus (Hex)	Query type 44	Query type 45
D2	Torque scale 6001 data (L)	Torque scale 7001 data (L)
D3	Torque scale 6001 data (H)	Torque scale 7001 data (H)
D4	Torque scale 6002 data (L)	Torque scale 7002 data (L)
D5	Torque scale 6002 data (H)	Torque scale 7002 data (H)
D6	Torque scale 6003 data (L)	Torque scale 7003 data (L)
D7	Torque scale 6003 data (H)	Torque scale 7003 data (H)
D8	Torque scale 6004 data (L)	Torque scale 7004 data (L)
D9	Torque scale 6004 data (H)	Torque scale 7004 data (H)
...	Torque scale N data (L)	Torque scale N data (L)
...	Torque scale N data (H)	Torque scale N data (H)
898	Torque scale 6996 data (L)	Torque scale 7996 data (L)
899	Torque scale 6996 data (H)	Torque scale 7996 data (H)
89A	Torque scale 6997 data (L)	Torque scale 7997 data (L)
89B	Torque scale 6997 data (H)	Torque scale 7997 data (H)
89C	Torque scale 6998 data (L)	Torque scale 7998 data (L)
89D	Torque scale 6998 data (H)	Torque scale 7998 data (H)
89E	Torque scale 6999 data (L)	Torque scale 7999 data (L)
89F	Torque scale 6999 data (H)	Torque scale 7999 data (H)
8A0	Torque scale 7000 data (L)	Torque scale 8000 data (L)
8A1	Torque scale 7000 data (H)	Torque scale 8000 data (H)

Modbus (Hex)	Query type 46	Query type 47
D2	Torque rate scale 6001 data (L)	Torque rate scale 7001 data (L)
D3	Torque rate scale 6001 data (H)	Torque rate scale 7001 data (H)
D4	Torque rate scale 6002 data (L)	Torque rate scale 7002 data (L)
D5	Torque rate scale 6002 data (H)	Torque rate scale 7002 data (H)
D6	Torque rate scale 6003 data (L)	Torque rate scale 7003 data (L)
D7	Torque rate scale 6003 data (H)	Torque rate scale 7003 data (H)
D8	Torque rate scale 6004 data (L)	Torque rate scale 7004 data (L)
D9	Torque rate scale 6004 data (H)	Torque rate scale 7004 data (H)
...	Torque rate scale N data (L)	Torque rate scale N data (L)
...	Torque rate scale N data (H)	Torque rate scale N data (H)
898	Torque rate scale 6996 data (L)	Torque rate scale 7996 data (L)
899	Torque rate scale 6996 data (H)	Torque rate scale 7996 data (H)
89A	Torque rate scale 6997 data (L)	Torque rate scale 7997 data (L)
89B	Torque rate scale 6997 data (H)	Torque rate scale 7997 data (H)
89C	Torque rate scale 6998 data (L)	Torque rate scale 7998 data (L)
89D	Torque rate scale 6998 data (H)	Torque rate scale 7998 data (H)
89E	Torque rate scale 6999 data (L)	Torque rate scale 7999 data (L)
89F	Torque rate scale 6999 data (H)	Torque rate scale 7999 data (H)
8A0	Torque rate scale 7000 data (L)	Torque rate scale 8000 data (L)
8A1	Torque rate scale 7000 data (H)	Torque rate scale 8000 data (H)

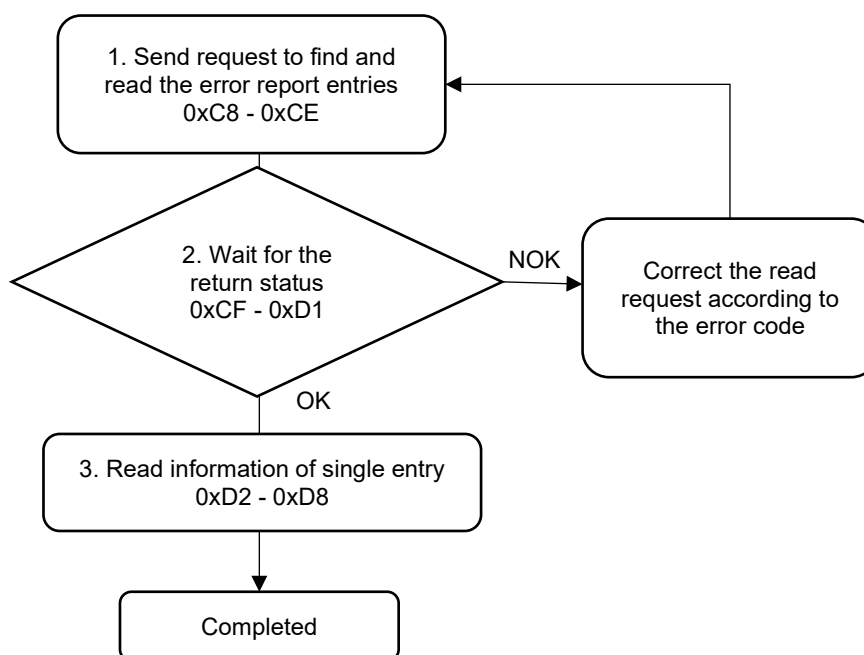
#752 Find and read the error report entries

Content description:

Use function code #752 with the following Modbus address or function code #21 to read the content of the error report.

Modbus (Hex)	Function code #21 (Dec)	Tool 1 / Tool 2 common status	R/W
69	106	Current No. of error report entries	R

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	752
C9	Version number	0
CA	Error report ID	1 to 60000 0: the latest error report
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	752
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Error report ID	Exceeds the range of 0 to 60000

3. Read the content of the error report in 0xD2 - 0xD8.

Modbus (Hex)	Function	Description
D2	Date / Time	Year
D3	Date / Time	Month
D4	Date / Time	Day
D5	Date / Time	Hour
D6	Date / Time	Minute
D7	Date / Time	Second
D8	AL / NG code	Refer to CH12*

Note: 0x1001 - 0x1999: Tool 1 device alarm (AL); 0x2000 - 0x2999: Tool 2 device alarm (AL).
0x3001 - 0x3999: Tool 1 tightening error (NG); 0x4000 - 0x4999: Tool 2 tightening error (NG).

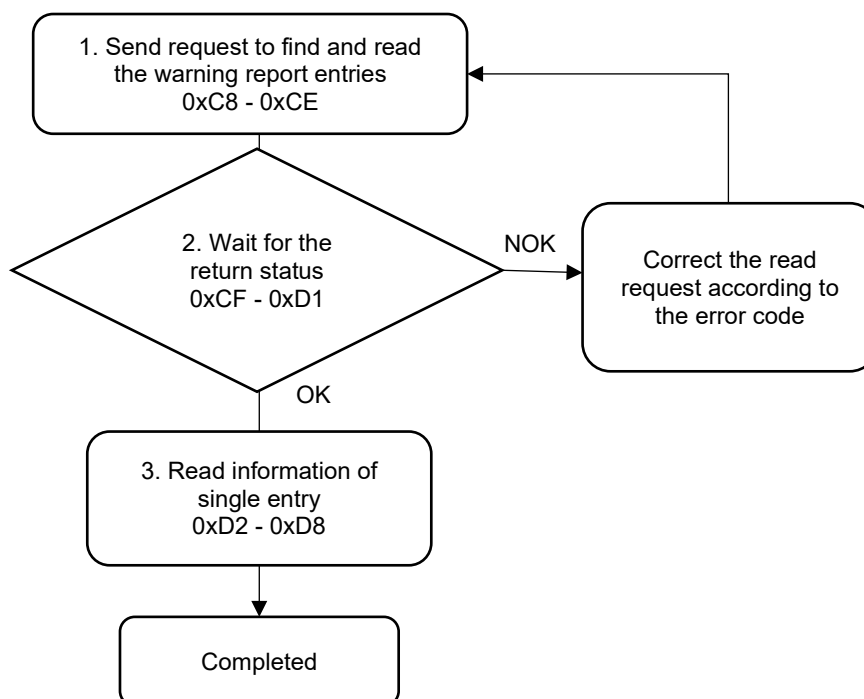
#753 Find and read the warning report entries

Content description:

Use function code #753 with the following Modbus address or function code #21 to read the content of the warning report.

Modbus (Hex)	Function code #21 (Dec)	Tool 1 / Tool 2 common status	R/W
6A	107	Current No. of warning report entries	R

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	753
C9	Version number	0
CA	Warning report ID	1 to 60000 0: the latest warning report
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	753
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Warning report ID	Exceeds the range of 0 to 60000

3. Read the content of a single warning report in 0xD2 - 0xD8.

Modbus (Hex)	Function	Description
D2	Date / Time	Year
D3	Date / Time	Month
D4	Date / Time	Day
D5	Date / Time	Hour
D6	Date / Time	Minute
D7	Date / Time	Second
D8	WN code	Refer to CH12*

Note: 0x5001 - 0x5999: Tool 1 operation warning (WN); 0x6000 - 0x6999: Tool 2 operation warning (WN)

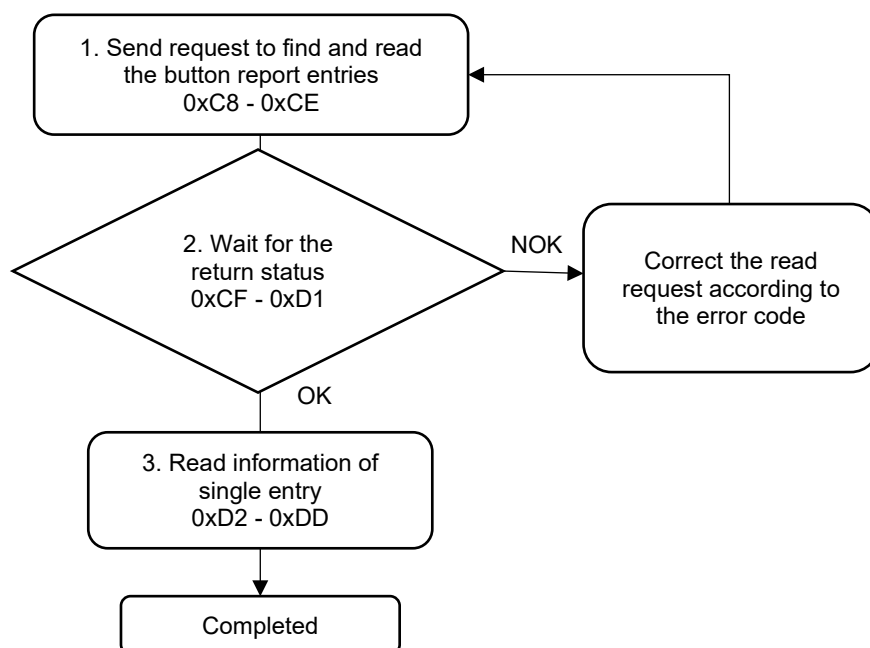
#754 Find and read the button report entries

Content description:

Use function code #754 with the following Modbus address or function code #21 to read the content of the button report.

Modbus (Hex)	Function code #21 (Dec)	Tool 1 / Tool 2 common status	R/W
6D	110	Current No. of button report entries (L)	R
6E	111	Current No. of button report entries (H)	R

Handshake signal description:



1. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	753
C9	Version number	0
CA	Button report ID	1 to 60000 0: the latest button report
CB to CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	753
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Button report ID	Exceeds the range of 0 to 60000

3. Read the content of the button report in 0xD2 - 0xDD.

Modbus (Hex)	Function	Description
D2	Date / Time	Year
D3	Date / Time	Month
D4	Date / Time	Day
D5	Date / Time	Hour
D6	Date / Time	Minute
D7	Date / Time	Second
D8	Button ID	Refer to Section 11.4 Button Report.
D9	Value before change (L)	0 to 4294967295
DA	Value before change (H)	
DB	Value after change (L)	0 to 4294967295
DC	Value after change (H)	
DD	Permissions account	0: reserved 1: User1 2: User2 3: User3 4: User4 5: User5 6: Admin

A

#760 Read the sorted production reports

Content description:

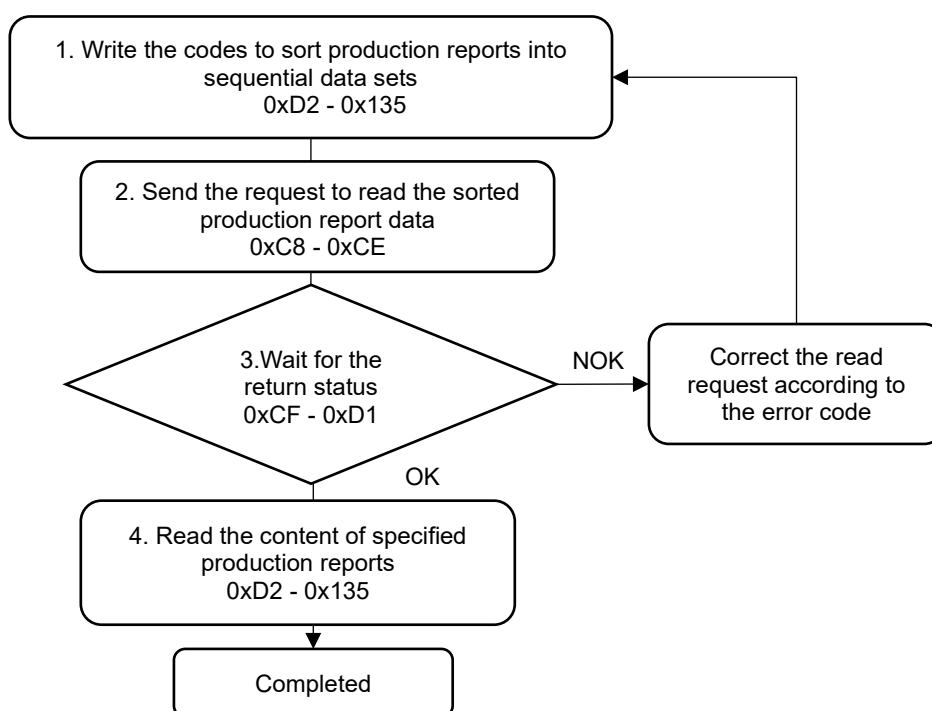
Use function code #760 to sort the data of #750 and the data of #751 query types 10 and 11 into sequential data sets and read them.

- A. Use the following Modbus address or function code #21 to read the data of the latest entry of production report.

Modbus (Hex)	Function code #21 (Dec)	Tool 1 / Tool 2 common status	R/W
6B	108	Current No. of production report entries (L)	R
6C	109	Current No. of production report entries (H)	R

- B. Read the content of history production report.

Handshake signal description:



1. Fill in the codes for sorting the production reports in 0xD2 - 0x135 (up to 100 fields can be set).

Modbus (Hex)	Function	Description
D2	Specifying query types 10 and 11 of #750 and #751 - code 1	0: Off 11 - 260: refer to function #750 for the function list. 1011 - 1260: refer to query type 10 of #751 for the function list 2011 - 2260: refer to query type 11 of #751 for the function list
D3	Specifying query types 10 and 11 of #750 and #751 - code 2	
D4	Specifying query types 10 and 11 of #750 and #751 - code 3	
...	Specifying query types 10 and 11 of #750 and #751 - code N	
134	Specifying query types 10 and 11 of #750 and #751 - code 99	
135	Specifying query types 10 and 11 of #750 and #751 - code 100	

2. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	760
C9	Version number	0
CA	Production report ID (L)	1 to 200000
CB	Production report ID (H)	0: the latest production report
CC, CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

3. Check whether the reading is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	760
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Production report ID	Exceeds the range of 0 to 200000

4. Read the content of the sorted production report data in 0xD2 - 0x135.

Modbus (Hex)	Return status	Description
D2	Corresponding data when code 1 is specified	—
D3	Corresponding data when code 2 is specified	—
D4	Corresponding data when code 3 is specified	—
...	Corresponding data when code N is specified	—
134	Corresponding data when code 99 is specified	—
135	Corresponding data when code 100 is specified	—

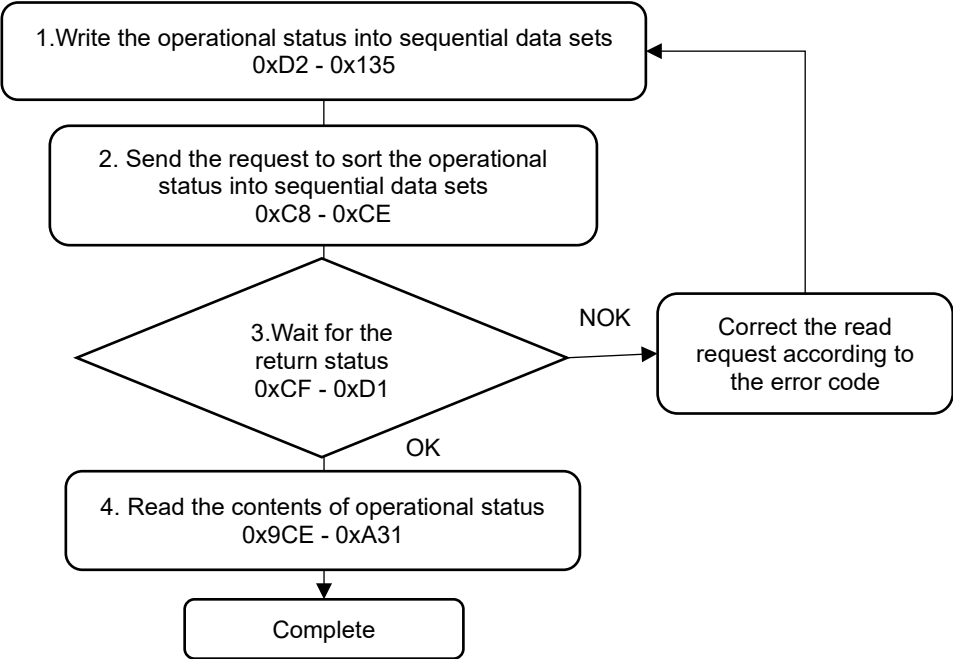
A.3.8 Others

#21 Write to sort operational status into sequential data sets

Content description:

Use function code #21 to sort the operating status in Section A.2 into sequential data sets and acquire the corresponding values.

Handshake signal description:



1. Fill in the operational status into sequential data sets to 0xD2 - 0x135 (up to 100 fields can be set).

Modbus (Hex)	Function	Description
D2	Specifying the operational status - code 1	0: Off 1 - 600: refer to the operational status table in Section A.2
D3	Specifying the operational status - code 2	
D4	Specifying the operational status - code 3	
...	Specifying the operational status - code N	
135	Specifying the operational status - code 100	

2. Fill in the read request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	21
C9	Version number	0
CA	Confirm command	99
CB ~ CD	Reserved	—
CE	Request to send command	1 (to be filled in last)

3. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	21
D0	Status	1: OK; 2: NOK
D1	Error code	–

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error code is as follows:

Code	Error description	Description
1	Confirm command	Must be 99

4. Read the corresponding value of the operational status in 0x9CE - 0xA31.

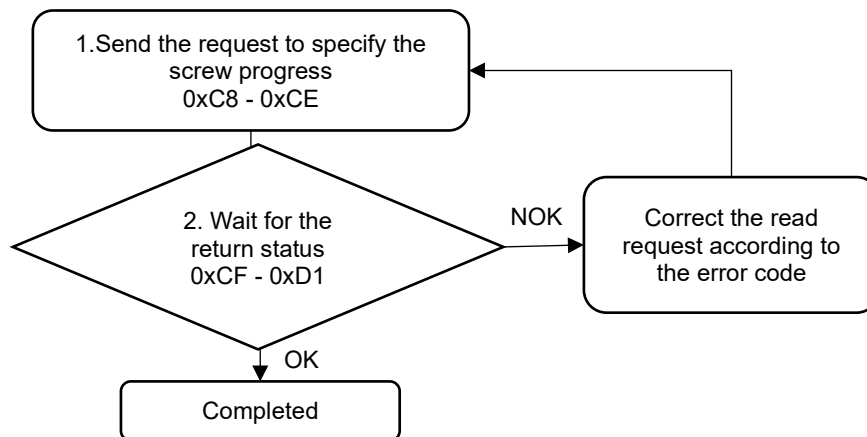
Modbus (Hex)	Return statatus	Description
9CE	Corresponding value when code 1 is specified	–
9CF	Corresponding value when code 2 is specified	–
9D0	Corresponding value when code 3 is specified	–
...	Corresponding value when code N is specified	–
A31	Corresponding value when code 100 is specified	–

#1449 Write to specify the screw progress

Content description:

Use function code #1449 to jump to the tightening parameter of the specified screw.

Handshake signal description:



1. Fill in the write request to 0xC8 - 0xCE.

Modbus (Hex)	Write / Read request	Write
C8	Function code	1449
C9	Version number	0
CA	Tool 1 / Tol 2	0: Tool 1; 1: Tool 2
CB	Specify screw progress ID (L)	–
CC	Specify screw progress ID (H)	–
CD	Reserved	–
CE	Request to send command	1 (to be filled in last)

2. Check whether the writing is successful from the return status in 0xCF - 0xD1.

Modbus (Hex)	Return status	Description
CF	Function code	1449
D0	Status	1: OK; 2: NOK
D1	Error code	-

If successful, the read status (0xD0) is 1.

If failed, the read status (0xD0) is 2. The error codes are as follows.

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	Specified screw progress ID	Exceeds the total screw quantity of the current operation

TCP/IP Communication - Function Codes

B

This chapter introduces the handshake method and information of the TCP/IP communication interface provided by the smart screwdriving controller. You can operate the controller remotely through this interface.

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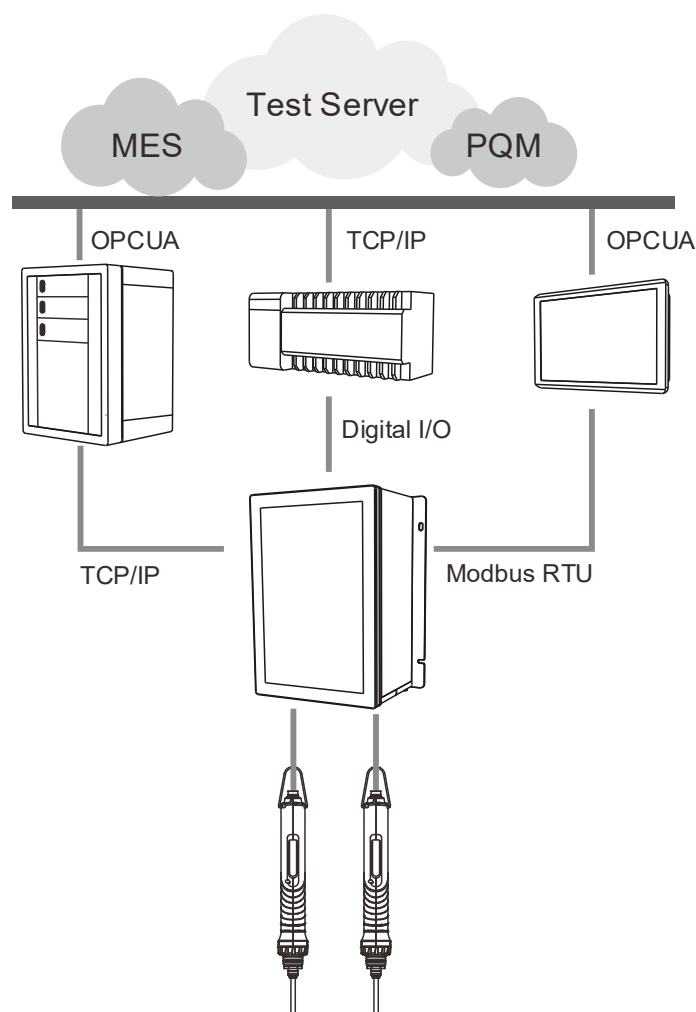
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B

B.1 System structure

B



The Delta smart screwdriving system provides multiple communication protocols: Modbus TCP (Ethernet), Modbus RTU (RS-485) and TCP/IP. It allows peripheral IIoT equipment to obtain the controller information, and then collect data to construct host computer systems, such as MES and PQM. The connection between the IIoT equipment and the smart screwdriving system is not limited by the hardware platform. Therefore, any equipment that supports the protocol (such as PC, PLC, or HMI) is able to control the smart screwdriving system and obtain tightening data.

The TCPI/P handshake information in this chapter consists of:

1. Operational status of the smart screwdriving system.
2. Handshake data of the smart screwdriving system.

TCP server

The smart screwdriving system supports TCP communication. The default IP is 192.168.1.11, and the default port is 1001. The host requires the keepalive feature to ensure it can actively re-connect to the smart screwdriving controller and obtain the correct data.

Important: to use the TCP communication, the controller firmware version must be 0.41.0112.0069(118) or later. The numbers in parentheses indicate the HMI kernel version, which must be 118 or later to support TCP communication.

B.2 Operational status of the smart screwdriving system

There are two ways of obtaining the operational status of the smart screwdriving system:

1. Use function code #10 to send the request for starting the subscription. Then, the system returns a real-time operational status (see the table of operational status in function code #50) every 0.3 seconds. To stop receiving the data, use function code #11 to send the request for canceling the subscription.
2. Use function code #50 to read the operational status value. Each time you send a request, the system returns a real-time operational status.

To obtain the result for each screw, use function codes #60, #522, and #566.

Function code table for the operational status:

Function code	Function name	Function code	Function name
#10	Write to start the subscription	#50	Read the operational status value
#11	Write to cancel the subscription	-	-
#20	Write the operational status value	-	-
#60	Write the setting of automatically transmitting the TCP data content of the result for each screw	-	-

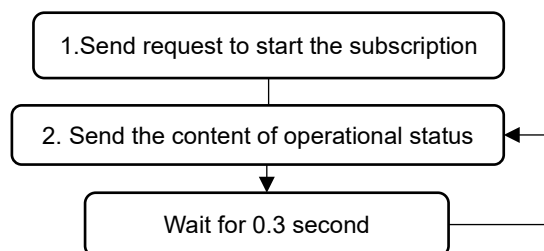
B

#10 Write to start the subscription

Content description:

Use function code #10 to have the controller automatically transmit the operational status (e.g., the operation status table of #50) every 0.3 second after the subscription is started.

Handshake signal description:



1. Send request to write the operational status value.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	10	0A00
3 - 4	Version number	-	0000
5 - 6	Mode	0: operational status table of #50 1: operational status table of #21	0000
7 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

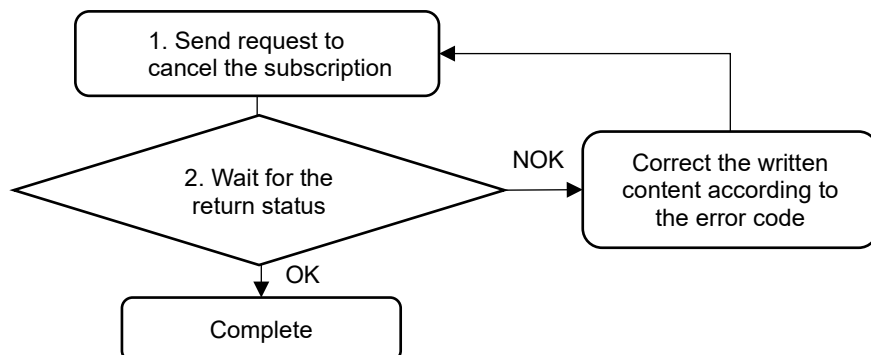
2. Automatically send the content of operational status (e.g., the operational status table of #50) every 0.3 second.

#11 Write to cancel the subscription

Content description:

Use function code #11 to disable the automatic transmission of operational status after the subscription is canceled.

Handshake signal description:



1. Send request to cancel the subscription.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	11	0B00
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	11	0B00
3 - 4	Return data length	Unit: Word	0A00
5 - 14	Reserved	-	0000
15 - 16	Return function code	11	0B00
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2.

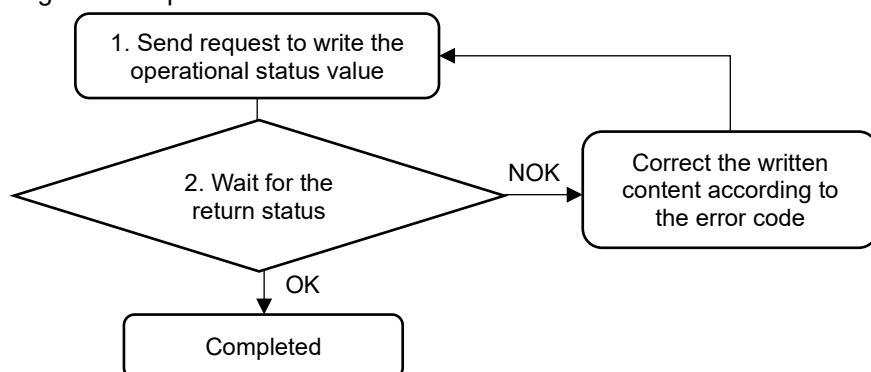
B

#20 Write the operational status value

Content description:

Use function code #20 to set the operational status value with the host controller. This function is often used to clear the flag.

Handshake signal description:



1. Send the request to write the operational status value.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	20	1400
3 - 4	Version number	-	0000
5 - 6	Data format	1: Word 2: Double Word	0100
7 - 8	Operational status	11 - 610 Refer to function code #50 for the table of operational status.	2400
9 - 10	Operational status value (L)	-	0000
11 - 12	Operational status value (H)	-	0000
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	20	1400
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Data format	1: Word 2: Double Word	0100
7 - 8	Operational status	11 - 610 Refer to function code #50 for the table of operational status.	2400
9 - 10	Operational status value (L)	-	0000
11 - 12	Operational status value (H)	-	0000
13 - 14	Reserved	-	0000
15 - 16	Return function code	20	1400
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

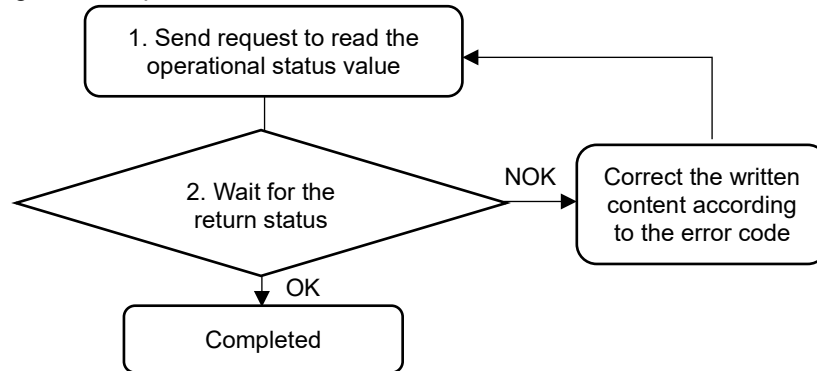
Code	Error description	Description
1	Data format	Must be 1 or 2
2	Operational status	Exceeds 11 - 610

#50 Read the operational status value

Content description:

Use function code #50 to read the operational status value with the host controller.

Handshake signal description:



1. Send the request to read the operational status value.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	50	3200
3 - 4	Version number	0 or 1	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	50	3200
3 - 4	Return data length	Unit: Word	0A00
5 - 14	Reserved	10 bytes	...
15 -16	Return function code	50	3200
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000
21 - 1220	Operational status	Refer to the following table.	-

Version 1: you can sort the return operational status into sequential data sets with function code #21.

Version 0: the table of operational status is as follows.

Byte (Dec)		Function code #20 (Hex)		Function code #21 (Hex)		Tool 1 status / Tool 2 status
Tool 1	Tool 2	Tool 1	Tool 2	Tool 1	Tool 2	
21 - 22	121 - 122	B	3D	1	33	Tightening ID set for cur. switch method
23 - 24	123 - 124	C	3E	2	34	PAR/SEQ set for cur. switch method
25 - 26	125 - 126	D	3F	3	35	SEQ ID of current switching method
27 - 28	127 - 128	E	40	4	36	PAR ID of current switching method
29 - 30	129 - 130	F	41	5	37	Current target torque / Current start torque for detecting the yield point
31 - 32	131 - 132	10	42	6	38	Current target angle
33 - 34	133 - 134	11	43	7	39	Current parameter progress
35 - 36	135 - 136	12	44	8	3A	Screw progress of current parameter (L)
37 - 38	137 - 138	13	45	9	3B	Screw progress of current parameter (H)
39 - 40	139 - 140	14	46	A	3C	Screw progress of current sequence (L)

Byte (Dec)		Function code #20 (Hex)		Function code #21 (Hex)		Tool 1 status / Tool 2 status
Tool 1	Tool 2	Tool 1	Tool 2	Tool 1	Tool 2	
41 - 42	141 - 142	15	47	B	3D	Screw progress of current sequence (H)
43 - 44	143 - 144	16	48	C	3E	Tightening OK count (L)
45 - 46	145 - 146	17	49	D	3F	Tightening OK count (H)
47 - 48	147 - 148	18	4A	E	40	Single screw tightening NOK count (L)
49 - 50	149 - 150	19	4B	F	41	Single screw tightening NOK count (H)
51 - 52	151 - 152	1A	4C	10	42	Loosening OK count (L)
53 - 54	153 - 154	1B	4D	11	43	Loosening OK count (H)
55 - 56	155 - 156	1C	4E	12	44	Single screw loosening NOK count (L)
57 - 58	157 - 158	1D	4F	13	45	Single screw loosening NOK count (H)
59 - 60	159 - 160	1E	50	14	46	Final stage max. torque
61 - 62	161 - 162	1F	51	15	47	Final stage min. torque
63 - 64	163 - 164	20	52	16	48	Current torque unit ^{*1}
65 - 66	165 - 166	21	53	17	49	Tightening / Loosening in operation ^{*2}
67 - 68	167 - 168	22	54	18	4A	Bit ID of current switching method
69 - 70	169 - 170	23	55	19	4B	Version of operational status ^{*3}
71 - 72	171 - 172	24	56	1A	4C	Clear the flag ^{*4}
73 - 74	173 - 174	25	57	1B	4D	Total screw Qty. of current sequence (L)
75 - 76	175 - 176	26	58	1C	4E	Total screw Qty. of current sequence (H)
77 - 78	177 - 178	27	59	1D	4F	Parameter Qty. of current sequence
79 - 80	179 - 180	28	5A	1E	50	Screw Qty. of current parameter (L)
81 - 82	181 - 182	29	5B	1F	51	Screw Qty. of current parameter (H)
83 - 84	183 - 184	2A	5C	20	52	All screws of current parameter finished
85 - 86	185 - 186	2B	5D	21	53	Current parameter finished
87 - 88	187 - 188	2C	5E	22	54	Current screw finished
89 - 90	189 - 190	2D	5F	23	55	Setting parameters (Waiting...)
91 - 92	191 - 192	2E	60	24	56	Final + Prevail torque
93 - 94	193 - 194	2F	61	25	57	Total angle
95 - 96	195 - 196	30	62	26	58	Tightening angle
97 - 98	197 - 198	31	63	27	59	Tightening result (1: OK; 2: NOK; 5: Pass)
99 - 100	199 - 200	32	64	28	5A	Loosening result (1: OK; 2: NOK)
101 - 102	201 - 202	33	65	29	5B	Curve creation finished
103 - 104	203 - 204	34	66	2A	5C	Restrict tightening status
105 - 106	205 - 206	35	67	2B	5D	Parameter settings OK / NOK
107 - 108	207 - 208	36	68	2C	5E	Final torque
109 - 110	209 - 210	37	69	2D	5F	Prevail torque
111 - 112	211 - 212	38	6A	2E	60	Final current
113 - 114	213 - 214	39	6B	2F	61	Cause to restrict tightening operation ^{*5}
115 - 116	215 - 216	3A	6C	30	62	Cause to restrict loosening operation ^{*6}
117 - 118	217 - 218	3B	6D	31	63	Remaining operation time (L)
119 - 120	219 - 220	3C	6E	32	64	Remaining operation time (H)
421 - 422	821 - 822	D3	19B	C9	191	Current target torque (L) / Current start torque for detecting the yield point (L)
423 - 424	823 - 824	D4	19C	CA	192	Current target torque (H) / Current start torque for detecting the yield point (H)
425 - 426	825 - 826	D5	19D	CB	193	Final stage max. torque (L)
427 - 428	827 - 828	D6	19E	CC	194	Final stage max. torque (H)
429 - 430	829 - 830	D7	19F	CD	195	Final stage min. torque (L)
431 - 432	831 - 832	D8	1A0	CE	196	Final stage min. torque (H)

Byte (Dec)		Function code #20 (Hex)		Function code #21 (Hex)		Tool 1 status / Tool 2 status
Tool 1	Tool 2	Tool 1	Tool 2	Tool 1	Tool 2	
433 - 434	833 - 834	D9	1A1	CF	197	Final + Prevail torque (L)
435 - 436	835 - 836	DA	1A2	D0	198	Final + Prevail torque (H)
437 - 438	837 - 838	DB	1A3	D1	199	Final torque (L)
439 - 440	839 - 840	DC	1A4	D2	19A	Final torque (H)
441 - 442	841 - 842	DD	1A5	D3	19B	Prevail torque (L)
443 - 444	843 - 844	DE	1A6	D4	19C	Prevail torque (H)
445 - 446	845 - 846	DF	1A7	D5	19D	Tightening OK count (L)
447 - 448	847 - 848	E0	1A8	D6	19E	Tightening OK count (H)
449 - 450	849 - 850	E1	1A9	D7	19F	Tightening NOK count (L)
451 - 452	851 - 852	E2	1AA	D8	1A0	Tightening NOK count (H)
453 - 454	853 - 854	E3	1AB	D9	1A1	Tightening OK + NOK count (L)
455 - 456	855 - 856	E4	1AC	DA	1A2	Tightening OK + NOK count (H)
457 - 458	857 - 858	E5	1AD	DB	1A3	Ready
459 - 460	859 - 860	E6	1AE	DC	1A4	Max. angle of final stage
461 - 462	861 - 862	E7	1AF	DD	1A5	Min. angle of final stage
463 - 464	863 - 864	E8	1B0	DE	1A6	Max. total angle
465 - 466	865 - 866	E9	1B1	DF	1A7	Min. total angle
467 - 468	867 - 868	EA	1B2	E0	1A8	Year
469 - 470	869 - 870	EB	1B3	E1	1A9	Month
471 - 472	871 - 872	EC	1B4	E2	1AA	Day
473 - 474	873 - 874	ED	1B5	E3	1AB	Hour
475 - 476	875 - 876	EE	1B6	E4	1AC	Minute
477 - 478	877 - 878	EF	1B7	E5	1AD	Second
479 - 480	879 - 880	F0	1B8	E6	1AE	Tightening result is 1: OK or 2: NOK; to be cleared in next operation
481 - 482	881 - 882	F1	1B9	E7	1AF	Loosening result is 1: OK or 2: NOK; to be cleared in next operation
483 - 484	883 - 884	F2	1BA	E8	1B0	Tightening result is 5: Pass; to be cleared in next operation
485 - 486	885 - 886	F3	1BB	E9	1B1	Current position of positioning arm X axis (L)
487 - 488	887 - 888	F4	1BC	EA	1B2	Current position of positioning arm X axis (H)
489 - 490	889 - 890	F5	1BD	EB	1B3	Current position of positioning arm Y axis (L)
491 - 492	891 - 892	F6	1BE	EC	1B4	Current position of positioning arm Y axis (H)
493 - 494	893 - 894	F7	1BF	ED	1B5	Current position of positioning arm Z axis (L)
495 - 496	895 - 896	F8	1C0	EE	1B6	Current position of positioning arm Z axis (H)
497 - 498	897 - 898	F9	1C1	EF	1B7	Target position of positioning arm X axis (L)
499 - 500	899 - 900	FA	1C2	F0	1B8	Target position of positioning arm X axis (H)
501 - 502	901 - 902	FB	1C3	F1	1B9	Target position of positioning arm Y axis (L)
503 - 504	903 - 904	FC	1C4	F2	1BA	Target position of positioning arm Y axis (H)
505 - 506	905 - 906	FD	1C5	F3	1BB	Target position of positioning arm Z axis (L)
507 - 508	907 - 908	FE	1C6	F4	1BC	Target position of positioning arm Z axis (H)
509 - 510	909 - 910	FF	1C7	F5	1BD	Clamp torque (L)
511 - 512	911 - 912	100	1C8	F6	1BE	Clamp torque (H)
513 - 514	913 - 914	101	1C9	F7	1BF	Clamp angle
515 - 516	915 - 916	102	1CA	F8	1C0	Snug torque (L)
517 - 518	917 - 918	103	1CB	F9	1C1	Snug torque (H)
519 - 520	919 - 920	104	1CC	FA	1C2	Tool operation prohibited ^{*7}
521 - 522	921 - 922	105	1CD	FB	1C3	Angle of Stage 1

B

Byte (Dec)		Function code #20 (Hex)		Function code #21 (Hex)		Tool 1 status / Tool 2 status
Tool 1	Tool 2	Tool 1	Tool 2	Tool 1	Tool 2	
523 - 524	923 - 924	106	1CE	FC	1C4	Angle of Stage 2
525 - 526	925 - 926	107	1CF	FD	1C5	Angle of Stage 3
527 - 528	927 - 928	108	1D0	FE	1C6	Angle of Stage 4
529 - 530	929 - 930	109	1D1	FF	1C7	Angle of Stage 5
531 - 532	931 - 932	10A	1D2	100	1C8	Angle of Stage 6
533 - 534	933 - 934	10B	1D3	101	1C9	Angle of first loosening stage
535 - 536	935 - 936	10C	1D4	102	1CA	Angle of second loosening stage
537 - 538	937 - 938	10D	1D5	103	1CB	Torque of Stage 1 (L)
539 - 540	939 - 940	10E	1D6	104	1CC	Torque of Stage 1 (H)
541 - 542	941 - 942	10F	1D7	105	1CD	Torque of Stage 2 (L)
543 - 544	943 - 944	110	1D8	106	1CE	Torque of Stage 2 (H)
545 - 546	945 - 946	111	1D9	107	1CF	Torque of Stage 3 (L)
547 - 548	947 - 948	112	1DA	108	1D0	Torque of Stage 3 (H)
549 - 550	949 - 950	113	1DB	109	1D1	Torque of Stage 4 (L)
551 - 552	951 - 952	114	1DC	10A	1D2	Torque of Stage 4 (H)
553 - 554	953 - 954	115	1DD	10B	1D3	Torque of Stage 5 (L)
555 - 556	955 - 956	116	1DE	10C	1D4	Torque of Stage 5 (H)
557 - 558	957 - 958	117	1DF	10D	1D5	Torque of Stage 6 (L)
559 - 560	959 - 960	118	1E0	10E	1D6	Torque of Stage 6 (H)
561 - 562	961 - 962	119	1E1	10F	1D7	Torque of first loosening stage (L)
563 - 564	963 - 964	11A	1E2	110	1D8	Torque of first loosening stage (H)
565 - 566	965 - 966	11B	1E3	111	1D9	Torque of second loosening stage (L)
567 - 568	967 - 968	11C	1E4	112	1DA	Torque of second loosening stage (H)
569 - 570	969 ~ 970	11D	1E5	113	1DB	Time of Stage 1
571 - 572	971 - 972	11E	1E6	114	1DC	Time of Stage 2
573 - 574	973 - 974	11F	1E7	115	1DD	Time of Stage 3
575 - 576	975 - 976	120	1E8	116	1DE	Time of Stage 4
577 - 578	977 - 978	121	1E9	117	1DF	Time of Stage 5
579 - 580	979 - 980	122	1EA	118	1E0	Time of Stage 6
581 - 582	981 - 982	123	1EB	119	1E1	Time of first loosening stage
583 - 584	983 - 984	124	1EC	11A	1E2	Time of second loosening stage
585 - 586	985 - 986	125	1ED	11B	1E3	Speed of Stage 1
587 - 588	987 - 988	126	1EE	11C	1E4	Speed of Stage 2
589 - 590	989 - 990	127	1EF	11D	1E5	Speed of Stage 3
591 - 592	991 - 992	128	1F0	11E	1E6	Speed of Stage 4
593 - 594	993 - 994	129	1F1	11F	1E7	Speed of Stage 5
595 - 596	995 - 996	12A	1F2	120	1E8	Speed of Stage 6
597 - 598	997 - 998	12B	1F3	121	1E9	Speed of first loosening stage
599 - 600	999 - 1000	12C	1F4	122	1EA	Speed of second loosening stage
601 - 602	1001 - 1002	12D	1F5	123	1EB	Total operation time
603 - 604	1003 - 1004	12E	1F6	124	1EC	Switching parameters ^{*8}
605 - 618	1005 - 1018	12F - 135	1F7 - 1FD	125 - 12B	1ED - 1F3	Reserved
619 - 620	1019 - 1020	136	1FE	12C	1F4	Current target yield point
621 - 820	1021 - 1220	137 - 19A	1FF - 262	12D - 190	1F5 - 258	Barcode string (ASCII code; 100 words)

Byte (Dec)	Function code #20 (Hex)	Function code #21 (Hex)	Tool 1 / Tool 2 common status
221 - 222	6F	65	Tool 1 AL / NG / WN code ^{*9}
223 - 224	70	66	Tool 2 AL / NG / WN code ^{*9}
225 - 226	71	67	DO status (Bit 1 - 8: Tool 1; Bit 9 - 16: Tool 2)
227 - 228	72	68	DI status (Bit 1 - 8: Tool 1; Bit 9 - 16: Tool 2)
229 - 230	73	69	DI control (Bit 1 - 8: Tool 1; Bit 9 - 16: Tool 2)
231 - 232	74	6A	Current No. of error report entries
233 - 234	75	6B	Current No. of warning report entries
235 - 236	76	6C	Current No. of production report entries (L)
237 - 238	77	6D	Current No. of production report entries (H)
239 - 240	78	6E	Current No. of button report entries (L)
241 - 242	79	6F	Current No. of button report entries (H)
243 - 244	7A	70	Production report entry creation finished
245 - 246	7B	71	Clear the common flag ^{*10}
247 - 248	7C	72	Reserved
249 - 250	7D	73	All production report entries cleared
251 - 252	7E	74	Start overwriting from production report ID 1
253 - 254	7F	75	Start overwriting from error report ID 1
255 - 256	80	76	Start overwriting from warning report ID 1
257 - 258	81	77	Start overwriting from button report ID 1
259 - 260	82	78	Keepalive
261 - 262	83	79	Single-tool / Dual-tool alternation / Dual-tool synchronization
263 - 264	84	7A	Switching method of Tool 1
265 - 266	85	7B	Switching method of Tool 2
267 - 268	86	7C	Operating curve count for Tool 1
269 - 270	87	7D	Operating curve count for Tool 2
271 - 272	88	7E	Tool No. under dual-tool alternation
273 - 274	89	7F	Notice of HMI screen refresh
275 - 276	8A	80	Flag of underlying communication error
277 - 278	8B	81	Count of underlying communication errors
279 - 280	8C	82	HMI initialization status ^{*11}
281 - 282	8D	83	Current user ID
283 - 284	8E	84	HMI main version
285 - 286	8F	85	HMI sub-version
287 - 288	90	86	Second development platform version
289 - 290	91	87	Underlying version
291 - 292	92	88	MCU version of Tool 1
293 - 294	93	89	MCU version of Tool 2
295 - 296	94	8A	Pop-up screen No.
297 - 298	95	8B	Reserved

Note:

- 0: N.m; 1: kgf.cm; 2: lbf.ft; 3: lbf.in; 4: ozf.ft; 5: ozf.in; 6: cNm
- Bit 0: tightening in progress; Bit 1: loosening in progress
- 0: does not support operational statuses bytes 211 - 410 and bytes 411 - 610
1: supports operational statuses bytes 211 - 410 and bytes 411 - 610
- Bit 0: clear the "All screws of current parameter finished" flag (bytes 42 and 92)
Bit 1: clear the "Current parameter finished" flag (bytes 43 and 93)
Bit 2: clear the "Current screw finished" flag (bytes 44 and 94)
Bit 3: clear the "Curve creation finished" flag (bytes 51 and 101)
Bit 4: clear the "Tightening result" flag (bytes 49 and 99) and "Loosening result" flag (bytes 50 and 100)

B

5. Bit 0: max. count for single screw NOK tightening
 Bit 1: prohibit tool operation when scanner string is null
 Bit 2: prohibit tool operation when screw Qty. reached
 Bit 3: max. operation time
 Bit 4: incorrect parameter sequence
 Bit 5: tightening operation restricted
 Bit 6: parameter check protection
 Bit 7: parameters do not match the tool
 Bit 8: tightening operation is prohibited by SDsoft
 Bit 9: the controller cannot operate this tool model
6. Bit 0: loosening prohibited after tightening OK
 Bit 1: loosening prohibited after tightening NOK
 Bit 2: max. count for single screw NOK loosening
 Bit 3: incorrect parameter sequence
 Bit 4: loosening operation restricted
 Bit 5: parameter range error
 Bit 6: tightening operation is prohibited by SDsoft
 Bit 7: the controller cannot operate this tool model
7. 0: remove restriction for tightening; remove restriction for loosening
 1001: restrict tightening operation; remove restriction for loosening
 1002: remove restriction for tightening; restrict loosening operation
 1003: restrict tightening operation; restrict loosening operation
8. When the system is in Single-tool operating mode and the switching method is Manual, you can set 1001 - 1500 to switch to the corresponding tightening parameters ID1 to ID500.
9. 0x1001 - 0x1999: Tool 1 device alarm (AL); 0x2000 - 0x2999: Tool 2 device alarm (AL)
 0x3001 - 0x3999: Tool 1 tightening error (NG); 0x4000 - 0x4999: Tool 2 tightening error (NG)
 0x5001 - 0x5999: Tool 1 operation warning (WN); 0x6000 - 0x6999: Tool 2 operation warning (WN)
10. Bit 0: clear the "Production report entry creation finished" flag (byte 122)
 Bit 1: clear the "All production report entries cleared" flag (byte 125)
 Bit 2: clear the "Start overwriting from production report ID 1" flag (byte 126)
 Bit 3: clear the "Start overwriting from error report ID 1" flag (byte 127)
 Bit 4: clear the "Start overwriting from warning report ID 1" flag (byte 128)
 Bit 5: clear the "Start overwriting from button report ID 1" flag (byte 129)
11. 0: underlying status is not yet initialized
 10: underlying status initialization is complete; waiting for the HMI initialization
 99: initialization of underlying status and HMI is complete

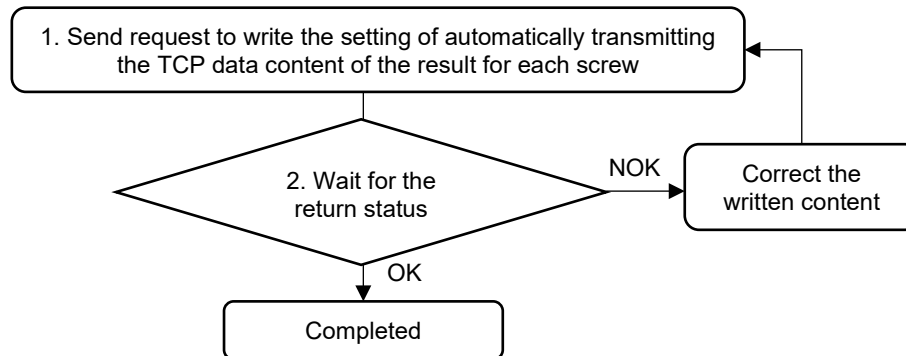
#60 Write the setting of automatically transmitting the TCP data content of the result for each screw

Content description:

Use function code #60 to write the setting of automatically transmitting the TCP data content of the result for each screw.

- Function 1: acknowledgment of receiving the TCP data of the result for each screw.

Handshake signal description:



1. Send the request to write the setting of automatically transmitting the TCP data content of the result for each screw.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	60	3C00
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Acknowledgement mode	0: acknowledgment to stop active data transmission; host controller has completed data collection 1: retransmit once; scale, time, torque 2: retransmit once; scale, angle, torque 3: retransmit once; scale, angle, torque rate 4: retransmit once; result, scale, time, torque 5: retransmit once; result, scale, angle, torque 6: retransmit; result, scale, angle, torque rate 7 - 10: reserved 11: retransmit three times; result, scale (tightening / loosening) 12: retransmit three times; result, scale (tightening) 13: retransmit three times; result, scale, all curve data (tightening / loosening) 14: retransmit three times; result, scale, all curve data (tightening)	0000
9 - 12	Reserved	4 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Bytes	Return status	Description	Example (Hex)
1 - 2	Function code	60	3C00
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Acknowledgement mode	0: acknowledgment to stop active data transmission; host controller has completed data collection 1: retransmit once; scale, time, torque 2: retransmit once; scale, angle, torque 3: retransmit once; scale, angle, torque rate 4: retransmit once; result, scale, time, torque 5: retransmit once; result, scale, angle, torque 6: retransmit; result, scale, angle, torque rate 7 - 10: reserved 11: retransmit three times; result, scale (tightening / loosening) 12: retransmit three times; result, scale (tightening) 13: retransmit three times; result, scale, all curve data (tightening / loosening) 14: retransmit three times; result, scale, all curve data (tightening)	0000
9 - 14	Reserved	6 bytes	...
15 - 16	Return function code	60	3C00
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2.

- Function 2: automatic transmission of the TCP data content of the result for each screw

This function requires using function codes #522 and #566 together. The following are the TCP data content of the result for each screw.

- (1) When the acknowledgement mode is 1 - 6, the contents of return packet are as follows.

Each time the tightening or loosening operation is complete, the TCP data is automatically transmitted once.

Byte	Read request	Description
1 - 2	Function code	60
3 - 4	Return data length	Unit: Word
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2

Byte	Read request	Description
7 - 8	Acknowledgement mode	11: scale, time point, torque point (Tables F, A, C) 12: scale, angle point, torque point (Tables F, B, C) 13: scale, angle point, torque rate point (Tables F, B, D) 21: report, scale, time point, torque point (Tables E, F, A, C) 22: report, scale, angle point, torque point (Tables E, F, B, C) 23: report, scale, angle point, torque rate point (Tables E, F, B, D) 30: time point (Table A) 31: angle point (Table B) 32: torque point (Table C) 33: torque rate point (Table D) 35: scale (Table F) 36: report + scale (Tables E, F)
9 - 10	Torque unit	0: N.m 1: kgf.cm 2: lbf.ft 3: lbf.in 4: ozf.ft 5: ozf.in 6: cNm
11 - 12	Magnification of the torque recorded by system	Non-50: torque or torque rate data divided by 1 50: torque or torque rate data divided by 50
13 - 14	Reserved	-
15 - 16	Return function code	60
17 - 18	Status	1: OK; 2: NOK
19 - 20	Error code	-
21 - N	Refer to the return content of the acknowledgement mode.	
N+1 to N+2	Suffix 1	0x11
N+3 to N+4	Suffix 2	0x99
N+5 to N+6	Suffix 3	0x33
N+7 to N+8	Suffix 4	0x66

Acknowledgement mode return contents:

Byte	Time data (return sampling points) (Table A)	Angle data (return sampling points) (Table B)
1 - 2	Time point 1 data	Angle point 1 data
3 - 4	Time point 2 data	Angle point 2 data
5 - 6	Time point 3 data	Angle point 3 data
...	Time point N data	Angle point N data
3995 - 3996	Time point 1998 data	Angle point 1998 data
3997 - 3998	Time point 1999 data	Angle point 1999 data
3999 - 4000	Time point 2000 data	Angle point 2000 data

Byte	Torque data (return sampling points) (Table C)		Torque rate data (return sampling points) (Table D)	
1 - 2	Torque point 1 data	Unit: 0.001 N.m	Torque rate point 1 data	Unit: 0.0001 N.m/degree
3 - 4	Torque point 2 data		Torque rate point 2 data	
5 - 6	Torque point 3 data		Torque rate point 3 data	
...	Torque point N data		Torque rate point N data	
3995 - 3996	Torque point 1998 data		Torque rate point 1998 data	
3997 - 3998	Torque point 1999 data		Torque rate point 1999 data	
3999 - 4000	Torque point 2000 data		Torque rate point 2000 data	

Byte	Report data (50 words in total) (Table E)	Description
1 - 2	Year, month, day	Need to be converted from 1970-01-01T00:00:00Z.
3 - 4	Hour, minute, second (L)	
5 - 6	Hour, minute, second (H)	
7 - 8	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
9 - 10	Screw ID (L)	-
11 - 12	Screw ID (H)	-
13 - 14	Sequence ID	-
15 - 16	Parameter ID	-
17 - 18	Target torque	Unit: 0.001 N.m
19 - 20	Target angle	Unit: degree
21 - 22	Target torque rate	Unit: 0.0001 N.m/degree
23 - 24	Final torque	Unit: 0.001 N.m
25 - 26	Tightening angle	Unit: degree
27 - 28	Rotation angle	Unit: degree
29 - 30	Status	1: tightening OK; 2: tightening NOK; 3: loosening OK; 4: loosening NOK; 5: pass
31 - 32	Operation time	Unit: 0.001 second
33 - 34	AL / NG / WN code	Refer to CH12
35 - 36	Max. angle of final stage	Unit: degree
37 - 38	Min. angle of final stage	Unit: degree
39 - 40	Max. torque of final stage	Unit: 0.001 Nm
41 - 42	Min. torque of final stage	Unit: 0.001 Nm
43 - 44	Torque unit	0: N.m; 1: kgf.cm; 2: lbf.ft; 3: lbf.in; 4: ozf.ft; 5: ozf.in; 6: cNm
45 - 46	Torque specification of the tool	Unit: 0.001 N.m
47 - 48	Max. torque of the tool	Unit: 0.001 N.m
49 - 50	Pre-tightening torque	Unit: 0.001 N.m
51 - 52	The set total operation time	Timeout setting for a single tightening parameter.
53 - 54	Max. total angle	-
55 - 56	Max. torque	Unit: 0.001 N.m

Byte	Report data (50 words in total) (Table E)	Description
57 - 58	Min. torque	Unit: 0.001 N.m
59 - 60	Max. angle	Angle upper limit when the current stage finishes.
61 - 62	Min. angle	Angle lower limit when the current stage finishes.
63 - 64	Max. operation time	Time upper limit when the current stage finishes. Unit: 0.01 second
65 - 66	Min. operation time	Time lower limit when the current stage finishes. Unit: 0.01 second
67 - 68	Prevail torque	Unit: 0.001 N.m
69 - 70	Final + Prevail torque	Unit: 0.001 N.m
71 - 72	Actual current	Unit: 0.01 Amp
73 - 74	Clamp torque	Unit: 0.001 N.m
75 - 76	Max. clamp torque	Unit: 0.001 N.m
77 - 78	Min. clamp torque	Unit: 0.001 N.m
79 - 80	Clamp angle	Unit: degree
81 - 82	Max. clamp angle	Unit: degree
83 - 84	Min. clamp angle	Unit: degree
85 - 86	Min. total angle	Unit: degree
87 - 88	Permissions account	0: Unknown (only available for HIM software version of 165 and before) 1: User1 2: User2 3: User3 4: User4 5: User5 6: Admin
89 - 90	Magnification of the torque recorded by system	System-specific
91 - 92	Target yield point	Unit: 0.01%
93 - 100	Reserved	-

Byte	Scale data (50 words in total) (Table F)	Description
1 - 2	Running angle of stage 1	Unit: degree
3 - 4	Running angle of stage 2	
5 - 6	Running angle of stage 3	
7 - 8	Running angle of stage 4	
9 - 10	Running angle of stage 5	
11 - 12	Running angle of stage 6	
13 - 14	Running angle of loosening stage 1	
15 - 16	Running angle of loosening stage 2	

Byte	Scale data (50 words in total) (Table F)	Description
17 - 18	Max. torque of stage 1	Unit: 0.001 N.m
19 - 20	Max. torque of stage 2	
21 - 22	Max. torque of stage 3	
23 - 24	Max. torque of stage 4	
25 - 26	Max. torque of stage 5	
27 - 28	Max. torque of stage 6	
29 - 30	Max. torque of loosening stage 1	
31 - 32	Max. torque of loosening stage 2	
33 - 34	Operation time of stage 1	Unit: 0.001 second
35 - 36	Operation time of stage 2	
37 - 38	Operation time of stage 3	
39 - 40	Operation time of stage 4	
41 - 42	Operation time of stage 5	
43 - 44	Operation time of stage 6	
45 - 46	Operation time of loosening stage 1	
47 - 48	Operation time of loosening stage 2	
49 - 50	Max. time on the scale	Unit: 0.001 second
51 - 52	Max. angle on the scale	Unit: degree
53 - 54	Max. torque on the scale	Unit: 0.001 N.m
55 - 56	Max. torque rate on the scale	Unit: 0.0001 Nm/degree
57 - 58	Total number of curve coordinates	-
59 - 60	Max. torque	Unit: 0.001 N.m
61 - 62	Min. torque	Unit: 0.001 N.m
63 - 64	Max. torque rate	Unit: 0.0001 N.m/degree
65 - 66	Max. angle	Unit: degree
67 - 68	Min. angle	Unit: degree
69 - 70	Version of the curve scale	-
71 - 72	Sampling rate for curves	0: 10 ms→4 ms when start torque is reached 1: 2 ms; max. 4 seconds per production report entry 2: 2 ms; max. 8 seconds per production report entry 3: 1 ms; max. 4 seconds per production report entry 4: 1 ms; max. 8 seconds per production report entry
73 - 74	Curve max. torque rate	Unit: 0.0001 N.m
75 - 76	Scale min. time	Unit: ms
77 - 78	Scale min. angle	Unit: degree
79 - 80	Scale min. torque	Unit: 0.001 N.m
81 - 82	Scale min. torque rate	Unit: 0.0001 N.m
83 - 84	Stage 1 switching torque	Unit: 0.001 N.m
85 - 86	Stage 2 switching torque	Unit: 0.001 N.m
87 - 88	Stage 3 switching torque	Unit: 0.001 N.m
89 - 90	Stage 4 switching torque	Unit: 0.001 N.m
91 - 92	Stage 5 switching torque	Unit: 0.001 N.m
93 - 94	Stage 6 switching torque	Unit: 0.001 N.m
95 - 100	Reserved	-

- (2) When the acknowledgement mode is 11 - 15, the contents of return packet are as follows. Each time the tightening or loosening operation is complete, the TCP data is automatically transmitted three times. If the complete data is received, use function code #60 to disable data transmission; if the data is incomplete, use function code #60 to resend the TCP data.

Byte	Read request	Description
1 - 2	Function code	60
3 - 4	Return data length	Unit: Word
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2
7 - 8	Acknowledgement mode	100: report, scale (refer to #750 table with data length of 250 words; refer to #751 type 10 table with data length of 150 words) 110: time point data (refer to #751 type 0 table) 120: angle point data (refer to #751 type 1 table) 130: torque point data (Double Word) (refer to #751 types 4 and 5 tables) 140: torque rate point data (Double Word) (refer to #751 types 6 and 7 tables)
9 - 10	Report line ID (L)	-
11 - 12	Report line ID (H)	
13 - 14	Reserved	-
15 - 16	Return function code	60
17 - 18	Status	1: OK; 2: NOK
19 - 20	Error code	-
21 - N	Refer to the return content of the acknowledgement mode.	
N+1 to N+2	Suffix 1	0x11
N+3 to N+4	Suffix 2	0x99
N+5 to N+6	Suffix 3	0x33
N+7 to N+8	Suffix 4	0x66

B

B.3 Handshake data of the smart screwdriving system

You can operate all functions of the smart screwdriving controller with the function code table through handshaking. All function codes are listed as follows:

Parameters			
Function code	Function name	Function code	Function name
#100	Write the parameter	#150	Read the parameter
#101	Write to set the tightening parameter through quick setting	#160	Read the created sets of tightening parameters
#110	Write to delete the parameter	-	-

Sequence			
Function code	Function name	Function code	Function name
#200	Write the sequence	#250	Read the sequence
#201	Write the navigator coordinates	#251	Read the navigator coordinates
#202	Write the navigator image codes	#252	Read the navigator image codes
#203	Write the coordinates of positioning arm	#253	Read the coordinates of positioning arm
#210	Write to delete the sequence	#260	Read the created sets of tightening sequences

Sources			
Function code	Function name	Function code	Function name
#300	Write the operating mode and switching method of source	#350	Read the operating mode and switching method of source
#301	Write the contents of a single source	#351	Read the contents of a single source
#302	Write to switch parameter under manual setting	-	-
#303	Write to switch sequence under manual setting	-	-
#310	Write to delete the source	-	-

Results			
Function code	Function name	Function code	Function name
#400	Write the switching method of source	#450	Read the switching method of source
#401	Write the barcode string	#451	Read the barcode string
#402	Write to clear all errors	#452	Read the scanner advanced settings
#403	Write to reset the operation progress	#453	Read the tightening status indicator
#404	Write to force execute the previous step	-	-
#405	Write to force execute the next step	-	-
#406	Write to restrict tightening operation	-	-
#407	Write to restrict loosening operation	-	-
#408	Write the scanner advanced settings	-	-

Results			
Function code	Function name	Function code	Function name
#409	Write to clear the single screw tightening NOK count	-	-
#410	Write to clear the single screw loosening NOK count	-	-
#411	Write to reset the operation time	-	-
#412	Write to reset the operational status	-	-

Controller			
Function code	Function name	Function code	Function name
#500	Write the request for permissions login	#550	Read the Ethernet settings
#501	Write the request for password change	#551	Read the page permissions
#502	Write the request for permissions logout	#552	Read the firmware version
#503	Write the page permissions	#553	Read the DI/DO functions
#504	Write the Ethernet settings	#554	Read the DI/DO conversion table
#505	Write the request for factory reset	#555	Read the default torque unit
#506	Write the buzzer sound pattern	#556	Read the default tool start condition
#507	Write the DI/DO functions	#557	Read NO / NC contact setting for a single DI/DO
#508	Write the DI/DO conversion table	#558	Read the default two-stage mode under self-defined torque control
#509	Write the default torque unit	#559	Read the display of upper limit for each stage of curves
#510	Write the default tool start condition	#560	Read the display of operation warning window
#511	Write NO / NC contact setting for a single DI/DO	#561	Read the file format of the exported result for each screw
#512	Write to export all system data	#562	Read the sampling rate for curves
#513	Write to import all system data	#563	Read the function of always monitoring the current
#514	Write the default two-stage mode under self-defined torque control	#564	Read the compensation for tool temperature rise
#515	Write the display of upper limit for each stage of curves	#565	Read the peripheral device settings
#516	Write the display of operation warning window	#566	Read the TCP data content of the result for each screw
#517	Write the file format of the exported result for each screw	#567	Read the monitor function when parameters do not match the tool
#518	Write the sampling rate for curves	#568	Read the default angle unit
#519	Write the function of always monitoring the current	#569	Read the volume of buzzer and operation screen buttons
#520	Write the compensation for tool temperature rise	#570	Read the display orientation of external screen
#521	Write the peripheral device settings	#571	Read the home screen
#522	Write the TCP data content of the result for each screw	#572	Read the RS-485 settings

B

Controller			
Function code	Function name	Function code	Function name
#523	Write the monitor function when parameters do not match the tool	#573	Read the function of converting curve data to positive values
#524	Write the default angle unit	#574	Read the second stage default loosening speed
#525	Write the volume of buzzer and operation screen buttons	#575	Read the keyboard cursor blink in Results page
#526	Write the display orientation of external screen	#576	Read whether the torque rate curve is replaced by the speed curve
#527	Write the home screen	#577	Read whether tool operation is prohibited after each tightening or loosening
#528	Write the RS-485 settings	#578	Read the DI response filter time
#529	Write the function of converting curve data to positive values	#579	Read the MAC information
#530	Write the default loosening second stage speed	#580	Read the model number
#531	Write the keyboard cursor blink in Results page	#581	Read whether the torque curve coordinates are reset to zero
#532	Write to replace torque rate curve with speed curve	#582	Read whether parameter range check at tool power-on is enabled
#533	Write to prohibit tool operation after each tightening or loosening	#583	Read the DO timer
#534	Write the DI response filter time	#584	Read the popup error message type for tightening signal that ends too early
#535	Write the model number	#585	Read the buzzer sound pattern
#536	Write to reset the torque curve coordinates to zero	#586	Read the curve cutoff point recording mode
#537	Write to check parameter ranges when tool is powered on	#587	Read the notification setting for abnormal tool temperature
#538	Write the DO timer	#588	Read whether tool operation restoration after NOK tightening is limited to alarm reset
#539	Write the popup error message type for tightening signal that ends too early	-	-
#540	Write the curve cutoff point recording mode	-	-
#541	Write to enable the notification for abnormal tool temperature	-	-
#542	Write to limit tool operation restoration after NOK tightening to alarm reset	-	-

Tool			
Function code	Function name	Function code	Function name
#600	Write to activate the tool	#650	Read the tool information
#601	Write to enable service reminder	#651	Read the lever start level
#602	Write the lever start level	#652	Read the push start level
#603	Write the push start level	#653	Read the work light brightness
#604	Write the work light brightness	#655	Read the LED light settings
#606	Write the LED light settings	#656	Read the tool calibration setting
#607	Write to calibrate the tool	#657	Read the tool firmware version
#608	Write the service reminder threshold	#658	Read the status of service reminder
-	-	#659	Read the status of tool activation
-	-	#660	Read the service reminder threshold
-	-	#661	Read the tool calibration reports

Reports			
Function code	Function name	Function code	Function name
#700	Clear the production report entries	#750	Find and read the production report entries
#701	Clear the error and warning report entries	#751	Find and read curves
#702	Clear the production report files	#752	Find and read the error report entries
-	-	#753	Find and read the warning report entries
-	-	#754	Find and read the button report entries
-	-	#760	Read the sorted production reports

Others			
Function code	Function name	Function code	Function name
#21	Write to sort operational status into sequential data sets	#1449	Write to specify the screw progress

B

B.3.1 Parameters operations

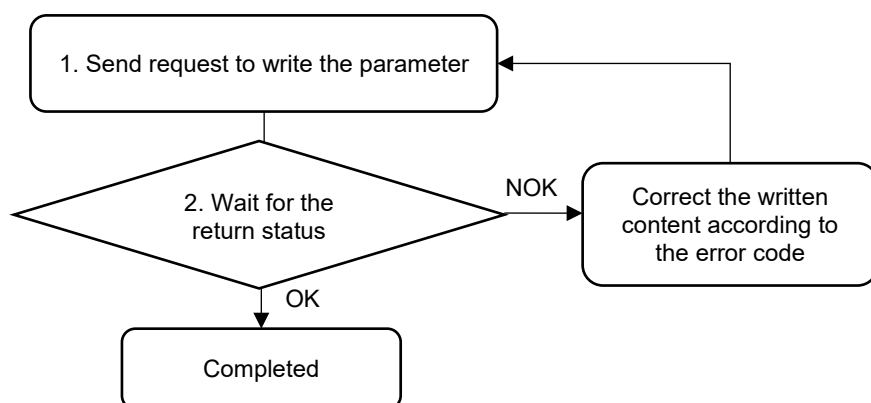
#100 Write the parameter

Content description:

Use function code #100 to set the tightening parameter.

- A. Add a tightening parameter.
- B. Modify an existing tightening parameter. First, read the parameter with function code #150. Then, modify and write the parameter with function code #100.

Handshake signal description:



1. Send the request to write the parameter.

■ Version number: 0

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	100	6400
3 - 4	Version number	0	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Parameter ID	1 to 500	0100
9 - 10	Auto identifying the strategy	0: automatically identify the strategy 1: fixed in Self-defined strategy	0000
11 - 12	Reserved	4 bytes	0000
13 - 14	Request to send command	-	0100
15 - 20	Reserved	6 bytes	...
21 - 60	Parameter title	ASCII code (20 words)	...
61 - 62	Min. tightening angle	Unit: degree (< 32767 degrees)	0000
63 - 64	Hold time switch of the final stage	Whether to stay in the Servo On state during the final stage (used with the Pause time setting)	0000
65 - 66	The prevail torque to be linked	The parameter ID to be linked	0000
67 - 68	Max. tightening time	Unit: 0.1 second (< 3276.7 seconds)	6400

Byte	Write / Read request	Description	Example (Hex)
69 - 70	Max. loosening time	Unit: 0.1 second (< 3276.7 seconds)	6400
71 - 72	Max. tightening angle	Unit: degree (< 32767 degrees)	FF7F
73 - 74	Reserved	-	0000
75 - 76	Delay before tightening starts	Unit: 0.01 second (< 327.67 seconds)	0000
77 - 78	Delay before loosening starts	Unit: 0.01 second (< 327.67 seconds)	0000
79 - 80	Start torque for switching the curve sampling rate / Start torque for cam-out detection	Unit: 0.001 N.m	F500
81 - 82	Start torque rate for snug angle calculation	Unit: 0.0001 N.m/degree	6400
83 - 84	Snug point angle correction	Unit: 0.1 degree	6400
85 - 86	Final current detection	0: Off; 1: On	0000
87 - 88	Delay before outputting the tightening result to feeder	Unit: 0.1 second	0000
89 - 90	Tool precision compensation	-10.0% to +10.0% Unit: 0.1%	0000
91 - 92	Angle delay for torque rate reached	Unit: 0.1 degree	0000
93 - 94	Trigger torque for recording cam-out angle	Unit: 0.001 N.m	0000
95 - 96	Upper limit for cam-out angle	Unit: degree	0000
97 - 98	Tolerance for cam-out	0 - 50	0000
99 - 100	Reserved	-	...

Self-defined strategy						Function	Description	Example (Hex)
Stage 1 (Hex)	Stage 2 (Hex)	Stage 3 (Hex)	Stage 4 (Hex)	Stage 5 (Hex)	Stage 6 (Hex)			
101 - 102	201 - 202	301 - 302	401 - 402	501 - 502	601 - 602	Control method	0: angle 1: torque 2: torque rate 3: clamp torque 4: clamp angle 5: yield point	0100
103 - 104	203 - 204	303 - 304	403 - 404	503 - 504	603 - 604	Tightening direction	0: CW; 1: CCW	0000
105 - 106	205 - 206	305 - 306	405 - 406	505 - 506	605 - 606	Rotation speed	Unit: rpm	6400
107 - 108	207 - 208	307 - 308	407 - 408	507 - 508	607 - 608	Target torque	Unit: 0.001 N.m	E803
109 - 110	209 - 210	309 - 310	409 - 410	509 - 510	609 - 610	Target angle	Unit: degree	0A00
111 - 112	211 - 212	311 - 312	411 - 412	511 - 512	611 - 612	Target torque rate	Unit: 0.0001 N.m/degree	0000
113 - 114	213 - 214	313 - 314	413 - 414	513 - 514	613 - 614	Angle interval for torque rate calculation	Unit: 0.1 degree	6400
115 - 116	215 - 216	315 - 316	415 - 416	515 - 516	615 - 616	Acceleration time	Unit: ms	E803
117 - 118	217 - 218	317 - 318	417 - 418	517 - 518	617 - 618	Max. angle	Unit: degree	0000

Self-defined strategy						Function	Description	Example (Hex)
Stage 1 (Hex)	Stage 2 (Hex)	Stage 3 (Hex)	Stage 4 (Hex)	Stage 5 (Hex)	Stage 6 (Hex)			
119 - 120	219 - 220	319 - 320	419 - 420	519 - 520	619 - 620	Min. angle	Unit: degree	0000
121 - 122	221 - 222	321 - 322	421 - 422	521 - 522	621 - 622	Max. torque	Unit: 0.001 N.m	F203
123 - 124	223 - 224	323 - 324	423 - 424	523 - 524	623 - 624	Min. torque	Unit: 0.001 N.m	2003
125 - 126	225 - 226	325 - 326	425 - 426	525 - 526	625 - 626	Max. operation time	Unit: 0.01 second	0000
127 - 128	227 - 228	327 - 328	427 - 428	527 - 528	627 - 628	Min. operation time	Unit: 0.01 second	0000
129 - 130	229 - 230	329 - 330	429 - 430	529 - 530	629 - 630	Prevail torque On / Off	0: Off; 1: On	0000
131 - 132	231 - 232	331 - 332	431 - 432	531 - 532	631 - 632	Angle range for prevail torque calculation	0 to 100%	0000
133 - 134	233 - 234	333 - 334	433 - 434	533 - 534	633 - 634	Pause time	Unit: ms	0000
135 - 136	235 - 236	335 - 336	435 - 436	535 - 536	635 - 636	Max. clamp torque	Unit: 0.001 Nm	0000
137 - 138	237 - 238	337 - 338	437 - 438	537 - 538	637 - 638	Min. clamp torque	Unit: 0.001 Nm	0000
139 - 140	239 - 240	339 - 340	439 - 440	539 - 540	639 - 640	Max. clamp angle	Unit: degree	0000
141 - 142	241 - 242	341 - 342	441 - 442	541 - 542	641 - 642	Min. clamp angle	Unit: degree	0000
143 - 144	243 - 244	343 - 344	443 - 444	543 - 544	643 - 644	Torque of 1st stage	Unit: 0.001 Nm	0000
145 - 146	245 - 246	345 - 346	445 - 446	545 - 546	645 - 646	Pause time after 1st stage	Unit: ms	0000
147 - 148	247 - 248	347 - 348	447 - 448	547 - 548	647 - 648	Acc. time of 2nd stage	Unit: ms	0000
149 - 150	249 - 250	349 - 350	449 - 450	549 - 550	649 - 650	Speed of final stage	Unit: rpm	0000
151 - 152	251 - 252	351 - 352	451 - 452	551 - 552	651 - 652	Deceleration time	Unit: ms	0000
153 - 154	253 - 254	353 - 354	453 - 454	553 - 554	653 - 654	Advanced settings (L)	Bit 0: synchronization through DI7/DO7 signal Bit 1: wait for another tool to complete before continuing operation Bit 2: ergo stop (0: Off; 1: On) Bit 3: release torque during pause (0: Off; 1: On) Bit 4: not included in the total angle calculation (0: Off; 1: On)	0400
155 - 156	255 - 256	355 - 356	455 - 456	555 - 556	655 - 656	Advanced settings (H)	—	0000
157 - 158	257 - 258	357 - 358	457 - 458	557 - 558	657 - 658	Max. switching torque	Unit: 0.001 N.m	0000
159 - 160	259 - 260	359 - 360	459 - 460	559 - 560	659 - 660	Min. switching torque	Unit: 0.001 N.m	0000
161 - 162	261 - 262	361 - 362	461 - 462	561 - 562	661 - 662	Target yield point	0 - 100%	0000

Self-defined strategy						Function	Description	Example (Hex)
Stage 1 (Hex)	Stage 2 (Hex)	Stage 3 (Hex)	Stage 4 (Hex)	Stage 5 (Hex)	Stage 6 (Hex)			
163 - 164	263 - 264	363 - 364	463 - 464	563 - 564	663 - 664	Start torque for yield point detection	Unit: 0.001 N.m	0000
165 - 200	265 - 300	365 - 400	465 - 500	565 - 600	665 - 700	Reserved	48 bytes	...
701 - 702						1st stage loosening angle	Unit: degree	6400
703 - 704						1st stage loosening speed	Unit: rpm	6400
705 - 706						2nd stage loosening angle	Unit: degree	100E
707 - 708						2nd stage loosening speed	Unit: rpm	F401
709 - 710						Loosening direction	0: CW; 1: CCW	0100
711 - 712						Detect loosening torque	Unit: 0.001 N.m	0000
713 - 714						Production report saving	0: Off; 1: On	0100
715 - 716						1st stage loosening acceleration time	Unit: ms	6400
717 - 718						2nd stage loosening acceleration time	Unit: ms	6400
719 - 720						Homing mode	0: Off; 1: On	0000
721 - 748						Reserved	—	
749 - 750						Bias instability of gyroscope	0 - 360 (0: Off)	0000
751 - 752						Gyroscope error	—	0000
753 - 754						Reserved	—	0000
755 - 756						Start torque for tightening angle calculation	Unit: 0.001 N.m	0000

■ Version number: 1

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	100	6400
3 - 4	Version number	1	0100
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Parameter ID	1 to 500	0100
9 - 10	Auto identifying the strategy	0: automatically identify the strategy 1: fixed in Self-defined strategy	0000
11 - 12	Reserved	-	0000
13 - 14	Request to send command	-	0100
15 - 20	Reserved	6 bytes	...
21 - 60	Parameter title	ASCII code (20 words)	...
61 - 62	Min. tightening angle	Unit: degree (< 32767 degrees)	0000
63 - 64	Hold time switch of the final stage	Whether to stay in the Servo On state during the final stage (used with the Pause time setting)	0000
65 - 66	The prevail torque to be linked	The parameter ID to be linked	0000

Byte	Write / Read request	Description	Example (Hex)
67 - 68	Max. tightening time	Unit: 0.1 second (< 3276.7 seconds)	6400
69 - 70	Max. loosening time	Unit: 0.1 second (< 3276.7 seconds)	6400
71 - 72	Max. tightening angle	Unit: degree (< 32767 degrees)	FF7F
73 - 74	Reserved	-	0000
75 - 76	Delay before tightening starts	Unit: 0.01 second (< 327.67 seconds)	0000
77 - 78	Delay before loosening starts	Unit: 0.01 second (< 327.67 seconds)	0000
79 - 80	Torque unit	0: N.m; 1: kgf.cm; 2: lb.ft; 3: lb.in; 4: ozf.ft; 5: ozf.in; 6: cN.m	0000
81 - 82	Angle interval for torque rate calculation	Unit: 0.1 degree	2C01
83 - 84	Snug point angle correction	Unit: 0.1 degree	2C01
85 - 86	Final current detection	0: Off; 1: On	0000
87 - 88	Delay before outputting the tightening result to feeder	Unit: 0.1 second	0000
89 - 90	Tool precision compensation	-10.0% to +10.0% Unit: 0.1%	0000
91 - 92	Detection level for target torque rate reached	Unit: 0.1 degree	0000
93 - 94	Start torque for switching the curve sampling rate (L) / Start torque for cam-out detection (L)	Unit: 0.001 torque unit	7701
95 - 96	Start torque for switching the curve sampling rate (H) / Start torque for cam-out detection (H)	Unit: 0.001 torque unit	0000
97 - 98	Start torque rate for snug angle calculation (L)	Unit: 0.0001 torque unit/degree	6400
99 - 100	Start torque rate for snug angle calculation (H)	Unit: 0.0001 torque unit/degree	0000
101 - 102	Trigger torque for recording cam-out angle (L)	Unit: 0.001 torque unit	0000
103 - 104	Trigger torque for recording cam-out angle (H)	Unit: 0.001 torque unit	0000
105 - 106	Upper limit for cam-out angle	Unit: degree	0000
107 - 108	Tolerance for cam-out	0 - 50	0000
109 - 110	Bias instability of gyroscope	0 - 360 (0: Off)	0000
111 - 112	Gyroscope error	-	0000
113 - 114	Gyroscope advanced settings	-	0000
115 - 116	Start torque for tightening angle calculation (L)	Unit: 0.001 torque unit	0000
117 - 118	Start torque for tightening angle calculation (H)	Unit: 0.001 torque unit	0000
119 - 140	Reserved	-	...
141 - 142	1st stage loosening angle	Unit: degree	6400
143 - 144	1st stage loosening speed	Unit: rpm	6400
145 - 146	2nd stage loosening angle	Unit: degree	201C
147 - 148	2nd stage loosening speed	Unit: rpm	2C01
149 - 150	Loosening direction	0: CW; 1: CCW	0100
151 - 152	Detect loosening torque (L)	Unit: 0.001 torque unit	0000
153 - 154	Detect loosening torque (H)	Unit: 0.001 torque unit	0000
155 - 156	Production report saving	0: Off; 1: On	0100

Byte	Write / Read request	Description	Example (Hex)
157 - 158	1st stage loosening acceleration time	Unit: ms	3200
159 - 160	2nd stage loosening acceleration time	Unit: ms	3200
161 - 162	Homing mode	0: Off; 1: On	0000
163 - 180	Reserved	—	

Self-defined strategy						Function	Description	Example (Hex)
Stage 1 (Hex)	Stage 2 (Hex)	Stage 3 (Hex)	Stage 4 (Hex)	Stage 5 (Hex)	Stage 6 (Hex)			
181 - 182	281 - 282	381 - 382	481 - 482	581 - 582	681 - 682	Control method	0: angle; 1: torque; 2: torque rate; 3: clamp torque; 4: clamp angle; 5: yield point	0100
183 - 184	283 - 284	383 - 384	483 - 484	583 - 584	683 - 684	Tightening direction	0: CW; 1: CCW	0000
185 - 186	285 - 286	385 - 386	485 - 486	585 - 586	685 - 686	Rotation speed	Unit: rpm	6400
187 - 188	287 - 288	387 - 388	487 - 488	587 - 588	687 - 688	Target torque (L)	Unit: 0.001 torque unit	E803
189 - 190	289 - 290	389 - 390	489 - 490	589 - 590	689 - 690	Target torque (H)	Unit: 0.001 torque unit	0000
191 - 192	291 - 292	391 - 392	491 - 492	591 - 592	691 - 692	Target angle	Unit: degree	3804
193 - 194	293 - 294	393 - 394	493 - 494	593 - 594	693 - 694	Target torque rate (L)	Unit: 0.0001 torque unit/degree	6300
195 - 196	295 - 296	395 - 396	495 - 496	595 - 596	695 - 696	Target torque rate (H)	Unit: 0.0001 torque unit/degree	0000
197 - 198	297 - 298	397 - 398	497 - 498	597 - 598	697 - 698	Acceleration time	Unit: ms	3200
199 - 200	299 - 300	399 - 400	499 - 500	599 - 600	699 - 700	Max. angle	Unit: degree	0000
201 - 202	301 - 302	401 - 402	501 - 502	601 - 602	701 - 702	Min. angle	Unit: degree	0000
203 - 204	303 - 304	403 - 404	503 - 504	603 - 604	703 - 704	Max. torque (L)	Unit: 0.001 torque unit	F203
205 - 206	305 - 306	405 - 406	505 - 506	605 - 606	705 - 706	Max. torque (H)	Unit: 0.001 torque unit	0000
207 - 208	307 - 308	407 - 408	507 - 508	607 - 608	707 - 708	Min. torque (L)	Unit: 0.001 torque unit	2003
209 - 210	309 - 310	409 - 410	509 - 510	609 - 610	709 - 710	Min. torque (H)	Unit: 0.001 torque unit	0000
211 - 212	311 - 312	411 - 412	511 - 512	611 - 612	711 - 712	Max. operation time	Unit: 0.01 second	0000
213 - 214	313 - 314	413 - 414	513 - 514	613 - 614	713 - 714	Min. operation time	Unit: 0.01 second	0000
215 - 216	315 - 316	415 - 416	515 - 516	615 - 616	715 - 716	Prevail torque On / Off	0: Off; 1: On	0000
217 - 218	317 - 318	417 - 418	517 - 518	617 - 618	717 - 718	Angle range for prevail torque calculation	0 - 100%	0000
219 - 220	319 - 320	419 - 420	519 - 520	619 - 620	719 - 720	Pause time	Unit: ms	0000
221 - 222	321 - 322	421 - 422	521 - 522	621 - 622	721 - 722	Max. clamp torque (L)	Unit: 0.001 torque unit	0000
223 - 224	323 - 324	423 - 424	523 - 524	623 - 624	723 - 724	Max. clamp torque (H)	Unit: 0.001 torque unit	0000
225 - 226	325 - 326	425 - 426	525 - 526	625 - 626	725 - 726	Min. clamp torque (L)	Unit: 0.001 torque unit	0000
227 - 228	327 - 328	427 - 428	527 - 528	627 - 628	727 - 728	Min. clamp torque (H)	Unit: 0.001 torque unit	0000

B

Self-defined strategy						Function	Description	Example (Hex)
Stage 1 (Hex)	Stage 2 (Hex)	Stage 3 (Hex)	Stage 4 (Hex)	Stage 5 (Hex)	Stage 6 (Hex)			
229 - 230	329 - 330	429 - 430	529 - 530	629 - 630	729 - 730	Max. clamp angle	Unit: degree	0000
231 - 232	331 - 332	431 - 432	531 - 532	631 - 632	731 - 732	Min. clamp angle	Unit: degree	0000
233 - 234	333 - 334	433 - 434	533 - 534	633 - 634	733 - 734	Torque of 1st stage (L)	Unit: 0.001 torque unit	2003
235 - 236	335 - 336	435 - 436	535 - 536	635 - 636	735 - 736	Torque of 1st stage (H)	Unit: 0.001 torque unit	0000
237 - 238	337 - 338	437 - 438	537 - 538	637 - 638	737 - 738	Pause time after 1st stage	Unit: ms	0000
239 - 240	339 - 340	439 - 440	539 - 540	639 - 640	739 - 740	Acc. time of 2nd stage	Unit: ms	E803
241 - 242	341 - 342	441 - 442	541 - 542	641 - 642	741 - 742	Speed of final stage	Unit: rpm	6400
243 - 244	343 - 344	443 - 444	543 - 544	643 - 644	743 - 744	Deceleration time	Unit: ms	3200
245 - 246	345 - 346	445 - 446	545 - 546	645 - 646	745 - 746	Advanced settings (L)	Bit 0: synchronization through DI7/DO7 signal Bit 1: wait for another tool to complete before continuing operation Bit 2: ergo stop (0: Off; 1: On) Bit 3: release torque during pause (0: Off; 1: On) Bit 4: not included in the total angle calculation (0: Off; 1: On)	0400
247 - 248	347 - 348	447 - 448	547 - 548	647 - 648	747 - 748	Advanced settings (H)	—	0000
249 - 250	349 - 350	449 - 450	549 - 550	649 - 650	749 - 750	Max. switching torque (L)	Unit: 0.001 torque unit	0000
251 - 252	351 - 352	451 - 452	551 - 552	651 - 652	751 - 752	Max. switching torque (H)	Unit: 0.001 torque unit	0000
253 - 254	353 - 354	453 - 454	553 - 554	653 - 654	753 - 754	Min. switching torque (L)	Unit: 0.001 torque unit	0000
255 - 256	355 - 356	455 - 456	555 - 556	655 - 656	755 - 756	Min. switching torque (H)	Unit: 0.001 torque unit	0000
257 - 258	357 - 358	457 - 458	557 - 558	657 - 658	757 - 758	Target yield point	0 - 100%	0000
259 - 260	359 - 360	459 - 460	559 - 560	659 - 660	759 - 760	Start torque for yield point detection (L)	Unit: 0.001 torque unit	0000
261 - 262	361 - 362	461 - 462	561 - 562	661 - 662	761 - 762	Start torque for yield point detection (H)	Unit: 0.001 torque unit	0000
263 - 280	363 - 380	463 - 480	563 - 580	663 - 680	763 - 780	Reserved	18 bytes	...

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	100	6400
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Parameter ID	1 to 500	0100
9 - 10	Auto identifying the strategy	0: automatically identify the strategy 1: fixed in Self-defined strategy	0000
11 - 14	Reserved	4 bytes	...
15 - 16	Return function code	100	6400
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

Code	Error description
1	Start: Max. torque < Min. torque
2	Start: Max. angle < Min. angle
3	Rundown: Max. torque < Min. torque
4	Rundown: Max. angle < Min. angle
5	Pre-tightening: Max. torque < Min. torque
6	Pre-tightening: Max. angle < Min. angle
7	Tightening: Max. torque < Min. torque
8	Tightening: Max. angle < Min. angle
9	Check if the stage sequence of the tightening settings is correct
10	No parameters set for this stage
11	Rundown: The set torque < Min. torque
12	Rundown: The set torque > Max. torque
13	Pre-tightening: The set torque < Min. torque
14	Pre-tightening: The set torque > Max. torque
15	Tightening: The set torque < Min. torque
16	Tightening: The set torque > Max. torque
17	The set angle > 32767
18	The set torque > Tool spec. torque The set min. torque > Tool spec. torque
19	The set torque > Tool max. torque
20	The set speed > Tool max. speed
21	Rundown: Torque cannot be 0
22	Pre-tightening: Torque cannot be 0
23	Tightening: Torque cannot be 0
24	The set rundown torque > The set pre-tightening torque
25	The set rundown torque > The set tightening torque
26	The set pre-tightening torque > The set tightening torque
27	The set clamp torque > Max. clamp torque
28	Max. clamp torque < Min. clamp torque

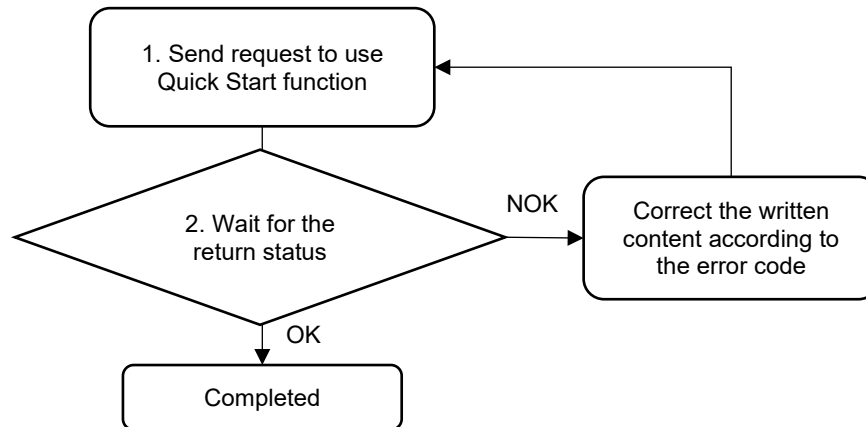
Code	Error description
29	Max. clamp torque > Tool max. torque
30	Tightening: The set speed < Min. speed
32	General: The set timeout > 32767
33	General: The set max. total angle > 32767
34	Start: Min. operation time > Max. operation time
35	Rundown: Min. operation time > Max. operation time
36	Pre-tightening: Min. operation time > Max. operation time
37	Tightening: Min. operation time > Max. operation time
38	General: The set tightening timeout > Max. operation time of the stage
39	Linked prevail torque cannot be 0
40	Cannot set multiple torque rates
41	Max. clamp angle < Min. clamp angle
42	Cannot set multiple clamp torques or angles
43	Max. switching torque < Min. switching torque
44	Max. switching torque > Tool max. torque
47	Tightening speed > Pre-tightening speed
48	Start speed > Rundown speed
49	Pre-tightening speed > Rundown speed
50	Tightening speed > Rundown speed
51	Start: Max. torque > Tightening torque
52	Start: Min. torque > Tightening torque
53	Rundown: Max. torque > Tightening torque
54	Rundown: Min. torque > Tightening torque
55	Rundown: Max. torque > Tightening torque
56	Rundown: Min. torque > Tightening torque
57	Pre-tightening: Max. torque > Tightening torque
58	Pre-tightening: Min. torque > Tightening torque
59	Parameter not applicable to this model
60	Setting multiple yield points is prohibited
61	The set torque < Min. torque
62	The set torque > Max. torque
63	Max. torque < Min. torque
64	Max. angle < Min. angle
65	Min. operation time > Max. operation time
100	Parameter ID exceeds the range of 1 to 500
101	Tool 1 / Tool 2 must be 0 or 1
102	Title string is null
103	Title already exists

#101 Write to set the tightening parameter through quick setting

Content description:

Use function code #101 to set the tightening parameter with the Quick Start function.

Handshake signal description:



1. Send request to use Quick Start function.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	101	6500
3 - 4	Version number	0	0000
5 - 6	Specifying tool and parameter	0: Tool 1; 1: Tool 2 1001 - 1500: Tool 1 parameter IDs 2001 - 2500: Tool 2 parameter IDs	0000
7 - 8	Target torque unit	0: N.m; 1: kgf.cm; 2: lbf.ft; 3: lbf.in; 4: ozf.ft; 5: ozf.in; 6: cNm	0100
9 - 10	Target torque (L)	Target torque x 1000	401F
11 - 12	Target torque (H)		0000
13 - 14	Request to send command	1 (to be filled in last)	0100

2. Check whether the writing is successful from the return status.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	101	6500
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Too 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Target torque unit	0: N.m; 1: kgf.cm; 2: lbf.ft; 3: lbf.in; 4: ozf.ft; 5: ozf.in; 6: cNm	0100
9 - 10	Target torque (L)	Target torque x 1000	401F
11 - 12	Target torque (H)		0000
13 - 14	Reserved	-	...
15 - 16	Return function code	101	6500
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

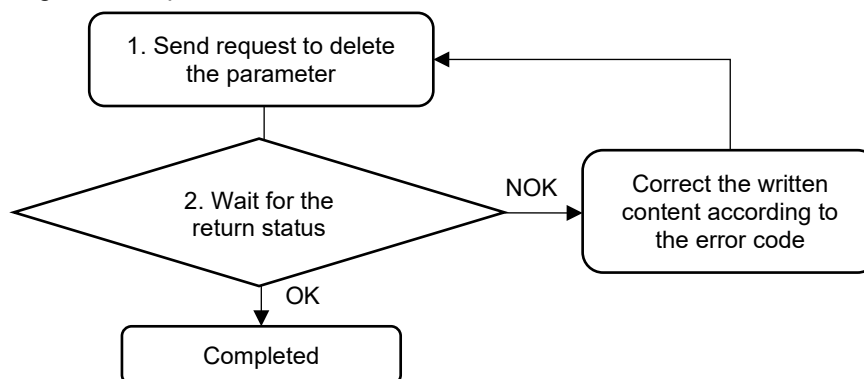
Code	Error description	Description
1	Tool 1 / Tool 2	Exceeds the range of 0 - 1, 1001 - 1500, or 2001 - 2500
2	Target torque unit	Exceeds the range of 0 - 6
3	Target torque	Exceeds the tool max. torque or lower than the tool min. torque

#110 Write to delete the parameter

Content description:

Use function code #110 to delete the tightening parameter.

Handshake signal description:



1. Send the request to delete the parameter.

Byte	Write / Read request	Description	Example (HEX)
1 - 2	Function code	110	6E00
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	The ID to be deleted	0: all the parameters 1 to 500: a single parameter	0100
9 - 12	Reserved	4 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (HEX)
1 - 2	Function code	110	6E00
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	The ID to be deleted	0: all the parameters 1 to 500: a single parameter	0100
9 - 14	Reserved	6 bytes	...
15 - 16	Return function code	110	6E00
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	The ID to be deleted	Exceeds the range of 0 to 500

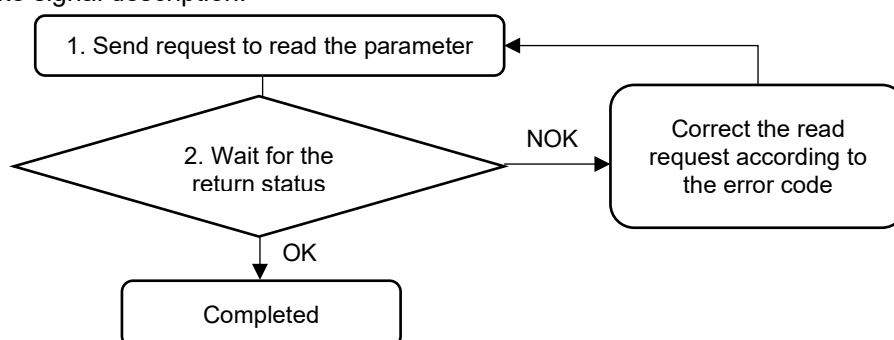
#150 Read the parameter

Content description:

Use function code #150 to read the tightening parameter.

- A. Obtain the content of an existing tightening parameter.
- B. Check whether the tightening parameter has been set.

Handshake signal description:



1. Send the request to read the tightening parameter.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	150	9600
3 - 4	Version number	0 or 1	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Parameter ID	1 to 500	0100
9 - 12	Reserved	4 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the parameter content.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	150	9600
3 - 4	Return data length	Unit: Word	3602
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Parameter ID	1 to 500	0100
9 - 14	Reserved	6 bytes	...
15 - 16	Return function code	150	9600
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000
21 - 718	Refer to function code #100 for the Write / Read request table.		

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

Code	Error description	Description
1	Parameter ID	Exceeds the range of 1 to 500
2	Tool 1 / Tool 2	Must be 0 or 1
3	Parameter content	Not set

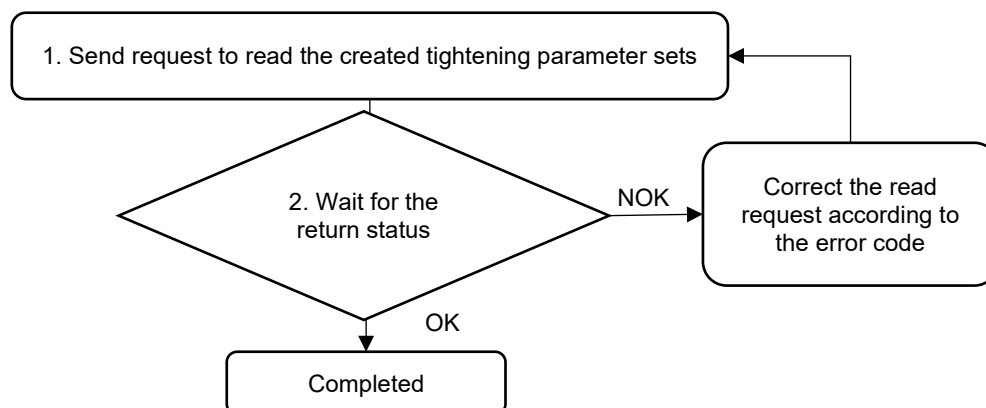
B

#160 Read the created sets of tightening parameters

Content description:

Use function code #160 to read the successfully created tightening parameters from the 1st to 500th sets.

Handshake signal description:



1. Send request to read the created tightening parameter sets.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	160	A000
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the tightening parameter contents.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	160	A000
3 - 4	Return data length	Unit: Word	FE01
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	160	A000
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000
21 - 22	1 st set of tightening parameter	0: not yet created; > 0: created	0100
23 - 24	2 nd set of tightening parameter	0: not yet created; > 0: created	0000
25 - 26	3 rd set of tightening parameter	0: not yet created; > 0: created	0000
...	nth set of tightening parameter	...	...
1019 - 1020	500 th set of tightening parameter	0: not yet created; > 0: created	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

B.3.2 Sequence operations

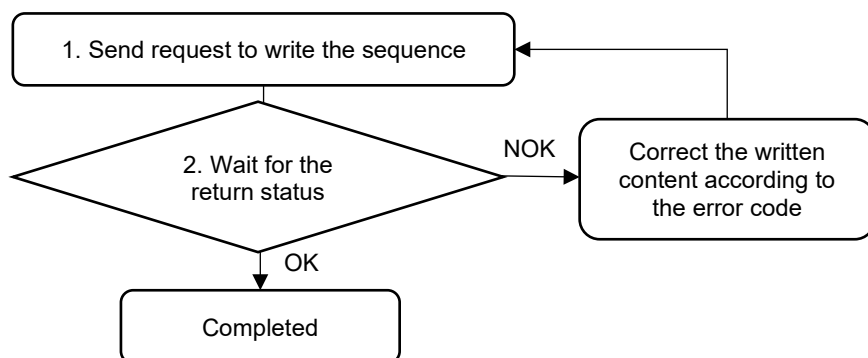
#200 Write the sequence

Content description:

Use function code #200 to set the tightening sequence.

- A. Add a tightening sequence.
- B. Modify an existing tightening sequence. First, read the sequence with function code #250. Then, modify and write the sequence with function code #200.

Handshake signal description:



1. Send the request to write the sequence.

In each sequence, you can configure up to 100 sets of parameters.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	200	C800
3 - 4	Version number	-	0000
5 - 6	Sequence ID	1 to 500	0100
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100
15 - 20	Reserved	6 bytes	...
21 - 60	Sequence title	ASCII code (20 words)	...
61 - 62	General / Navigator mode	0: General mode 1: Navigator mode (must be used with #201 and #202)	0000
63 - 64	Positioning arm function	0: Off 1: On (must be used with #203)	0000
65 - 80	Reserved	16 bytes	...
81 - 82	Tool ID for Set 1	0: Tool 1 1: Tool 2	0000
83 - 84	Tool ID for Set 2		0000
85 - 86	Tool ID for Set 3		0000
...	Tool ID for Set N		...
275 - 276	Tool ID for Set 98		0000
277 - 278	Tool ID for Set 99		0000
279 - 280	Tool ID for Set 100		0000
281 - 282	Parameter ID for Set 1	1 to 500	0100
283 - 284	Parameter ID for Set 2	1 to 500	0200

Byte	Write / Read request	Description	Example (Hex)
285 - 286	Parameter ID for Set 3	1 to 500	0000
...	Parameter ID for Set N	1 to 500	...
475 - 476	Parameter ID for Set 98	1 to 500	0000
477 - 478	Parameter ID for Set 99	1 to 500	0000
479 - 480	Parameter ID for Set 100	1 to 500	0000
481 - 482	Screw quantity for Set 1 (L)	1 to 999999	0200
483 - 484	Screw quantity for Set 1 (H)		0000
485 - 486	Screw quantity for Set 2 (L)	1 to 999999	0300
487 - 488	Screw quantity for Set 2 (H)		0000
489 - 490	Screw quantity for Set 3 (L)	1 to 999999	0000
491 - 492	Screw quantity for Set 3 (H)		0000
...	Screw quantity for Set N (L)	1 to 999999	...
...	Screw quantity for Set N (H)		...
869 - 870	Screw quantity for Set 98 (L)	1 to 999999	0000
871 - 872	Screw quantity for Set 98 (H)		0000
873 - 874	Screw quantity for Set 99 (L)	1 to 999999	0000
875 - 876	Screw quantity for Set 99 (H)		0000
877 - 878	Screw quantity for Set 100 (L)	1 to 999999	0000
879 - 880	Screw quantity for Set 100 (H)		0000
881 - 882	Bit ID for Set 1	0 to 255	0000
883 - 884	Bit ID for Set 2	0 to 255	0000
885 - 886	Bit ID for Set 3	0 to 255	0000
...	Bit ID for Set N	0 to 255	...
1075 - 1076	Bit ID for Set 98	0 to 255	0000
1077 - 1078	Bit ID for Set 99	0 to 255	0000
1079 - 1080	Bit ID for Set 100	0 to 255	0000

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	200	C800
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Sequence ID	1 to 500	0100
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	200	C800
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

Code	Error description	Description
1	Sequence ID	Exceeds the range of 1 to 500
2	The quantity is 0	Cannot be set as 0
3	Total screw quantity > 999999	Must not exceed 999999
100	Title string is null	-
101	Title already exists	-

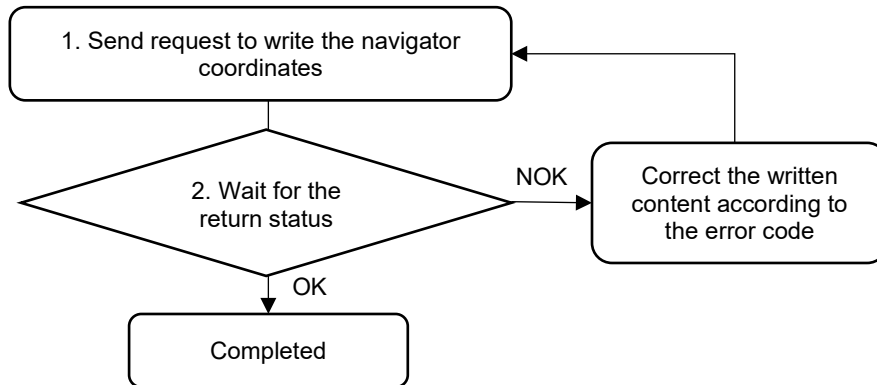
#201 Write the navigator coordinates

Content description:

Use function code #201 to set the navigator coordinates of screws.

Note: first, set the coordinates of screws with function code #201. Then, set the navigator image codes corresponding to screws with function code #202.

Handshake signal description:



1. Send the request to set the navigator coordinates of up to 100 screws.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	201	C900
3 - 4	Version number	-	0000
5 - 6	Sequence ID	1 to 500	0100
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100
15 - 20	Reserved	6 bytes	...
21 - 22	Screw ID 1 - X axis position	0 to 740	6400
23 - 24	Screw ID 1 - Y axis position	0 to 460	6400
25 - 26	Screw ID 2 - X axis position	0 to 740	C800
27 - 28	Screw ID 2 - Y axis position	0 to 460	6400
29 - 30	Screw ID 3 - X axis position	0 to 740	2C10
31 - 32	Screw ID 3 - Y axis position	0 to 460	6400
33 - 34	Screw ID 4 - X axis position	0 to 740	6400
35 - 36	Screw ID 4 - Y axis position	0 to 460	C800
37 - 38	Screw ID 5 - X axis position	0 to 740	6400
39 - 40	Screw ID 5 - Y axis position	0 to 460	2C10
...	Screw ID N - X axis position	0 to 740	0000
...	Screw ID N - Y axis position	0 to 460	0000
401 - 402	Screw ID 96 - X axis position	0 to 740	0000
403 - 404	Screw ID 96 - Y axis position	0 to 460	0000
405 - 406	Screw ID 97 - X axis position	0 to 740	0000
407 - 408	Screw ID 97 - Y axis position	0 to 460	0000
409 - 410	Screw ID 98 - X axis position	0 to 740	0000
411 - 412	Screw ID 98 - Y axis position	0 to 460	0000

Byte	Write / Read request	Description	Example (Hex)
413 - 414	Screw ID 99 - X axis position	0 to 740	0000
415 - 416	Screw ID 99 - Y axis position	0 to 460	0000
417 - 418	Screw ID 100 - X axis position	0 to 740	0000
419 - 420	Screw ID 100 - Y axis position	0 to 460	0000

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	201	C900
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Sequence ID	1 to 500	0100
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	201	C900
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

Code	Error description	Description
1	Sequence ID	Exceeds the range of 1 to 500
2	Axis position exceeds the range	X axis: exceeds the range of 0 to 740 Y axis: exceeds the range of 0 to 460

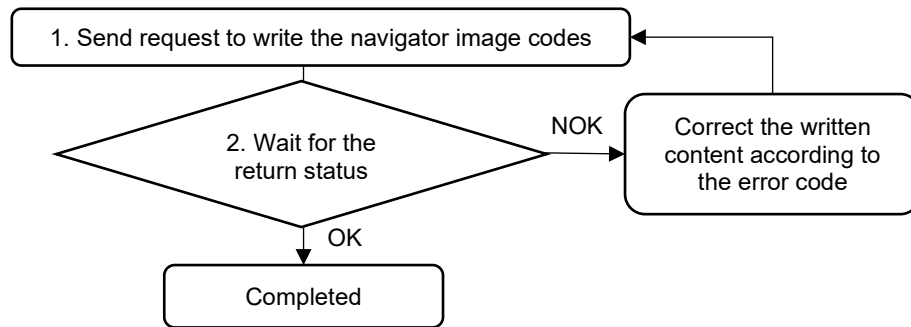
#202 Write the navigator image codes

Content description:

Use function code #202 to set the navigator image codes corresponding to the screws.

Note: first, set the coordinates of screws with function code #201. Then, set the navigator image codes corresponding to the screws with function code #202.

Handshake signal description:



1. Send the request to write the navigator image codes of up to 100 screws.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	202	CA00
3 - 4	Version number	-	0000
5 - 6	Sequence ID	1 to 500	0100
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100
15 - 20	Reserved	6 bytes	...
21 - 22	Screw ID 1 - displayed image	0: no display 1: image A 2: image B	0100
23 - 24	Screw ID 2 - displayed image	3: image C 4: image D 5: image E	0100
25 - 26	Screw ID 3 - displayed image	6: image F 7: image G 8: image H	0100
27 - 28	Screw ID 4 - displayed image	9: image I 10: image J 11: image K	0100
29 - 30	Screw ID 5 - displayed image	12: image L 13: image M 14: image N	0100
...	Screw ID N - displayed image	15: image O 16: image P 17: image Q	0000
211 - 212	Screw ID 96 - displayed image	18: image R 19: image S 20: image T	0000
213 - 214	Screw ID 97 - displayed image	21: image U 22: image V 23: image W	0000
215 - 216	Screw ID 98 - displayed image	24: image X 25: image Y 26: image Z	0000
217 - 218	Screw ID 99 - displayed image	27: image AA 28: image AB 29: image AC	0000
219 - 220	Screw ID 100 - displayed image	30: image AD	0000

B

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	202	CA00
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Sequence ID	1 to 500	0100
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	202	CA00
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

Code	Error description	Description
1	Sequence ID	Exceeds the range of 1 to 500
2	Image code exceeds the range	Exceeds the range of 0 to 30

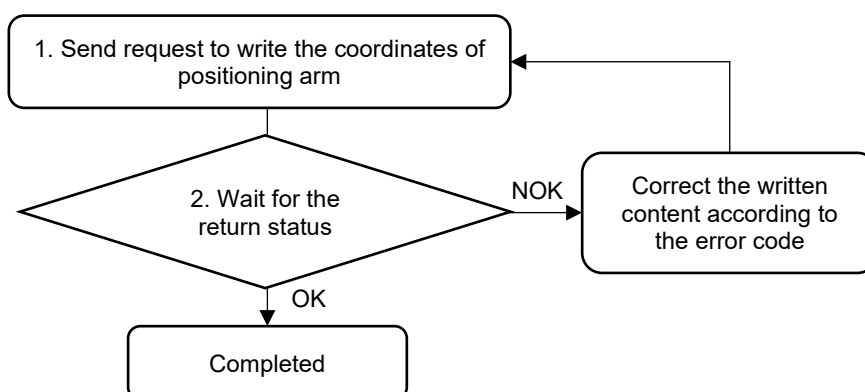
#203 Write the coordinates of positioning arm

Content description:

Use function code #203 to set the coordinates of positioning arm.

Note: first, set the tightening sequence with function code #200. Then, set the coordinates of positioning arm with function code #203.

Handshake signal description:



1. Send the request to write the coordinates of positioning arm for up to 100 screws.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	203	CB00
3 - 4	Version number	-	0000
5 - 6	Sequence ID	1 to 500	0100
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100
15 - 20	Reserved	6 bytes	...

Byte	Write / Read request	Description	Example (Hex)
21 - 22	Screw ID 1 - positioning arm X axis position (L)	Unit: mm	E803
23 - 24	Screw ID 1 - positioning arm X axis position (H)		0000
25 - 26	Screw ID 1 - positioning arm Y axis position (L)	Unit: mm	0000
27 - 28	Screw ID 1 - positioning arm Y axis position (H)		0000
29 - 30	Screw ID 1 - positioning arm Z axis position (L)	Unit: mm	E803
31 - 32	Screw ID 1 - positioning arm Z axis position (H)		0000
33 - 34	Screw ID 2 - positioning arm X axis position (L)	Unit: mm	0000
35 - 36	Screw ID 2 - positioning arm X axis position (H)		0000
37 - 38	Screw ID 2 - positioning arm Y axis position (L)	Unit: mm	0000
39 - 40	Screw ID 2 - positioning arm Y axis position (H)		0000
41 - 42	Screw ID 2 - positioning arm Z axis position (L)	Unit: mm	0000
43 - 44	Screw ID 2 - positioning arm Z axis position (H)		0000
45 - 46	Screw ID 3 - positioning arm X axis position (L)	Unit: mm	0000
47 - 48	Screw ID 3 - positioning arm X axis position (H)		0000
49 - 50	Screw ID 3 - positioning arm Y axis position (L)	Unit: mm	0000
51 - 52	Screw ID 3 - positioning arm Y axis position (H)		0000
53 - 54	Screw ID 3 - positioning arm Z axis position (L)	Unit: mm	0000
55 - 56	Screw ID 3 - positioning arm Z axis position (H)		0000
...	Screw ID N - positioning arm X axis position (L)	Unit: mm	0000
...	Screw ID N - positioning arm X axis position (H)		0000
...	Screw ID N - positioning arm Y axis position (L)	Unit: mm	0000
...	Screw ID N - positioning arm Y axis position (H)		0000
...	Screw ID N - positioning arm Z axis position (L)	Unit: mm	0000
...	Screw ID N - positioning arm Z axis position (H)		0000
1185 - 1186	Screw ID 98 - positioning arm X axis position (L)	Unit: mm	0000
1187 - 1188	Screw ID 98 - positioning arm X axis position (H)		0000
1189 - 1190	Screw ID 98 - positioning arm Y axis position (L)	Unit: mm	0000
1191 - 1192	Screw ID 98 - positioning arm Y axis position (H)		0000
1193 - 1194	Screw ID 98 - positioning arm Z axis position (L)	Unit: mm	0000
1195 - 1196	Screw ID 98 - positioning arm Z axis position (H)		0000
1197 - 1198	Screw ID 99 - positioning arm X axis position (L)	Unit: mm	0000
1199 - 1200	Screw ID 99 - positioning arm X axis position (H)		0000
1201 - 1202	Screw ID 99 - positioning arm Y axis position (L)	Unit: mm	0000
1203 - 1204	Screw ID 99 - positioning arm Y axis position (H)		0000
1205 - 1206	Screw ID 99 - positioning arm Z axis position (L)	Unit: mm	0000
1207 - 1208	Screw ID 99 - positioning arm Z axis position (H)		0000
1209 - 1210	Screw ID 100 - positioning arm X axis position (L)	Unit: mm	0000
1211 - 1212	Screw ID 100 - positioning arm X axis position (H)		0000
1213 - 1214	Screw ID 100 - positioning arm Y axis position (L)	Unit: mm	0000
1215 - 1216	Screw ID 100 - positioning arm Y axis position (H)		0000

B

Byte	Write / Read request	Description	Example (Hex)
1217 - 1218	Screw ID 100 - positioning arm Z axis position (L)	Unit: mm	0000
1219 - 1220	Screw ID 100 - positioning arm Z axis position (H)		0000

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	203	CB00
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Sequence ID	1 to 500	0100
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	203	CB00
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

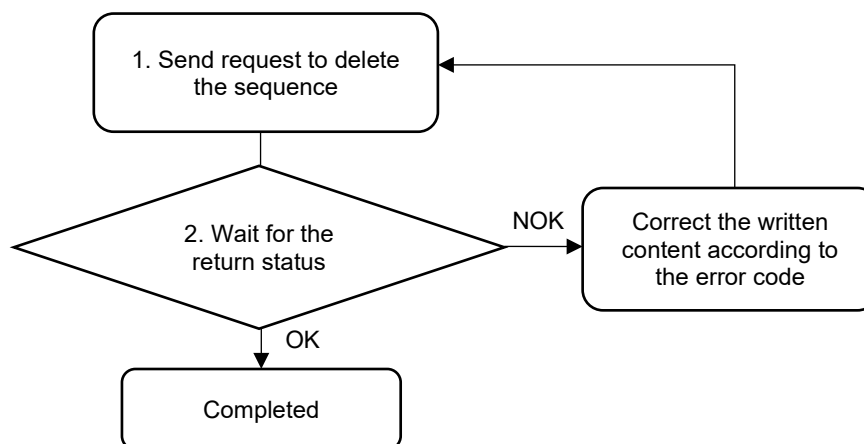
Code	Error description	Description
1	Sequence ID	Exceeds the range of 1 to 500

#210 Write to delete the sequence

Content description:

Use function code #210 to delete the tightening sequence.

Handshake signal description:



1. Send the request to delete the sequence.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	210	D200
3 - 4	Version number	-	0000
5 - 6	The ID to be deleted	0: all the sequences 1 to 500: a single sequence	0100
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	210	D200
3 - 4	Return data length	Unit: Word	0A00
5 - 6	The ID to be deleted	0: all the sequences 1 to 500: a single sequence	0100
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	210	D200
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	The ID to be deleted	Exceeds the range of 0 to 500

B

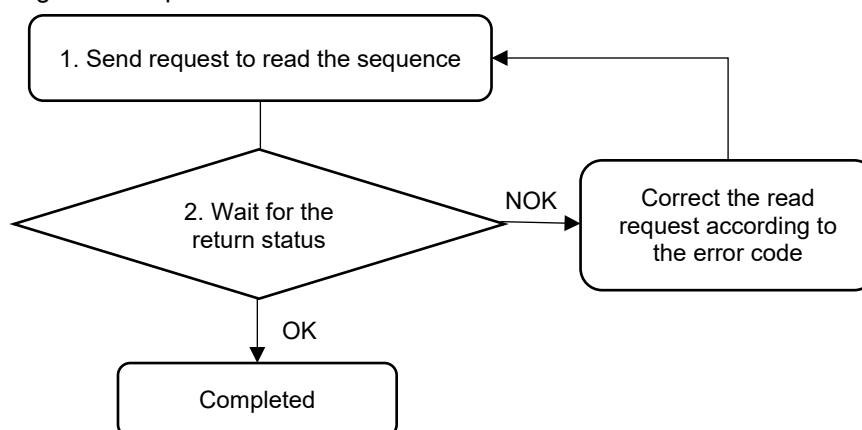
#250 Read the sequence

Content description:

Use function code #250 to read the tightening sequence.

- A. Obtain the content of an existing tightening sequence.
- B. Check whether the tightening sequence has been set.

Handshake signal description:



1. Send the request to read the sequence.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	250	FA00
3 - 4	Version number	-	0000
5 - 6	Sequence ID	1 to 500	0100
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the sequence content.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	250	FA00
3 - 4	Return data length	Unit: Word	2602
5 - 6	Sequence ID	1 to 500	0100
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	250	FA00
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000
21 - 1080	Refer to function code #200 for the Write / Read request table.		

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

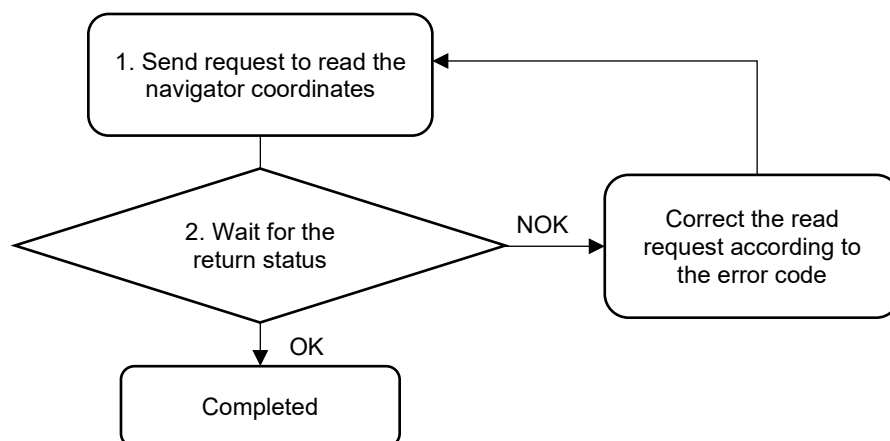
Code	Error description	Description
1	Sequence ID	Exceeds the range of 1 to 500

#251 Read the navigator coordinates

Content description:

Use function code #251 to read the navigator coordinates of screws.

Handshake signal description:



1. Send the request to read the navigator coordinates.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	251	FB00
3 - 4	Version number	-	0000
5 - 6	Sequence ID	1 to 500	0100
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the content of navigator coordinates.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	251	FB00
3 - 4	Return data length	Unit: Word	2602
5 - 6	Sequence ID	1 to 500	0100
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	251	FB00
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000
21 - 420	Refer to function code #201 for the Write / Read request table.		

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Sequence ID	Exceeds the range of 1 to 500

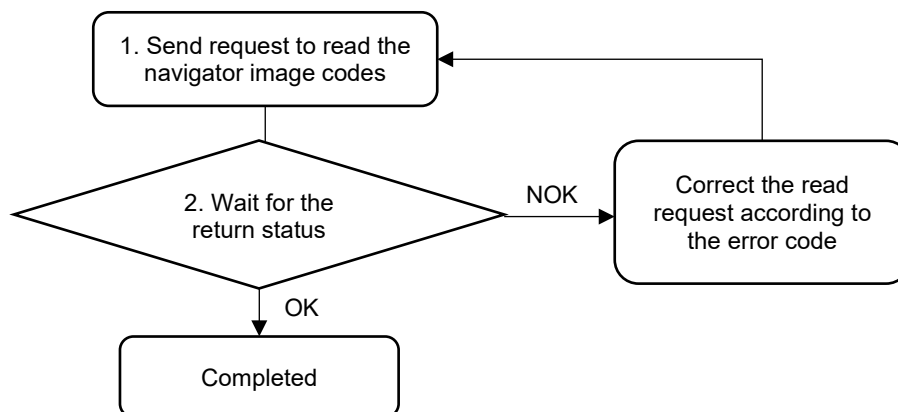
B

#252 Read the navigator image codes

Content description:

Use function code #252 to read the navigator image codes corresponding to the screws.

Handshake signal description:



1. Send the request to read the navigator image codes.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	252	FC00
3 - 4	Version number	-	0000
5 - 6	Sequence ID	1 to 500	0100
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the navigator image codes.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	252	FC00
3 - 4	Return data length	Unit: Word	6E00
5 - 6	Sequence ID	1 to 500	0100
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	252	FC00
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000
21 - 220	Refer to function code #202 for the Write / Read request table.		

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

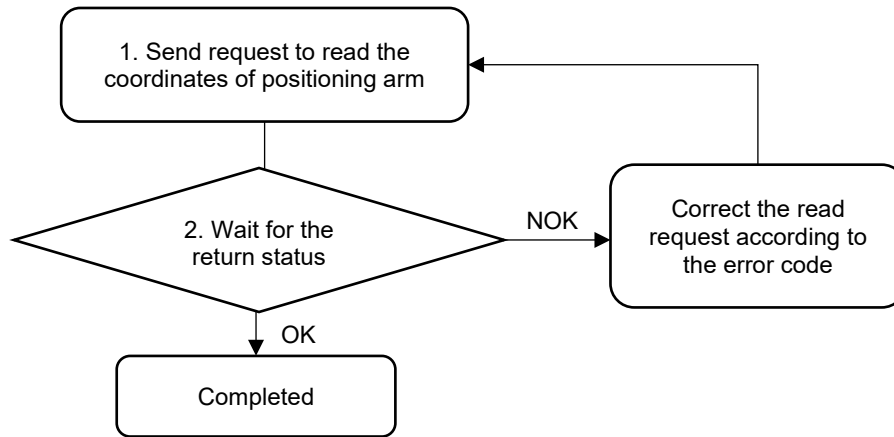
Code	Error description	Description
1	Sequence ID	Exceeds the range of 1 to 500

#253 Read the coordinates of positioning arm

Content description:

Use function code #253 to read the coordinates of positioning arm.

Handshake signal description:



1. Send the request to read the coordinates of the positioning arm.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	253	FD00
3 - 4	Version number	-	0000
5 - 6	Sequence ID	1 to 500	0100
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the coordinates of the positioning arm.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	253	FD00
3 - 4	Return data length	Unit: Word	6E00
5 - 6	Sequence ID	1 to 500	0100
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	253	FD00
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000
21 - 1220	Refer to function code #203 for the Write / Read request table.		

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Sequence ID	Exceeds the range of 1 to 500

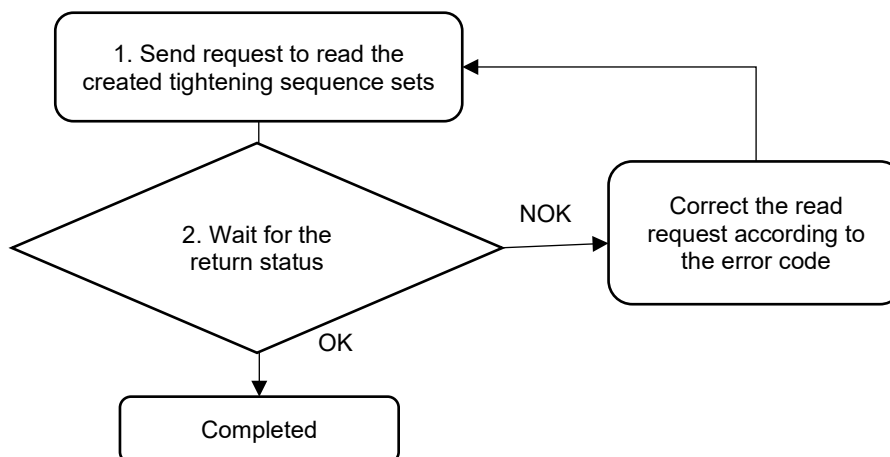
B

#260 Read the created sets of tightening sequences

Content description:

Use function code #260 to read the successfully created tightening sequences from the 1st to 500th sets.

Handshake signal description:



1. Send request to read the created tightening sequence sets.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	260	0401
3 - 4	Version number	-	0000
5 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful and read the tightening sequence creation status

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	260	0401
3 - 4	Return data length	Unit: Word	FE01
5 - 14	Reserved	10 bytes	...
15 - 16	Return function code	260	0401
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000
21 - 22	1 st set of tightening sequence	0: not yet created; > 0: created	0B00
23 - 24	2 nd set of tightening sequence	0: not yet created; > 0: created	0000
25 - 26	3 rd set of tightening sequence	0: not yet created; > 0: created	0000
...	n th set of tightening sequence	...	...
1019 - 1020	499 th set of tightening sequence	0: not yet created; > 0: created	0000

If successful, the read status is 1.

If failed, the read status is 2.

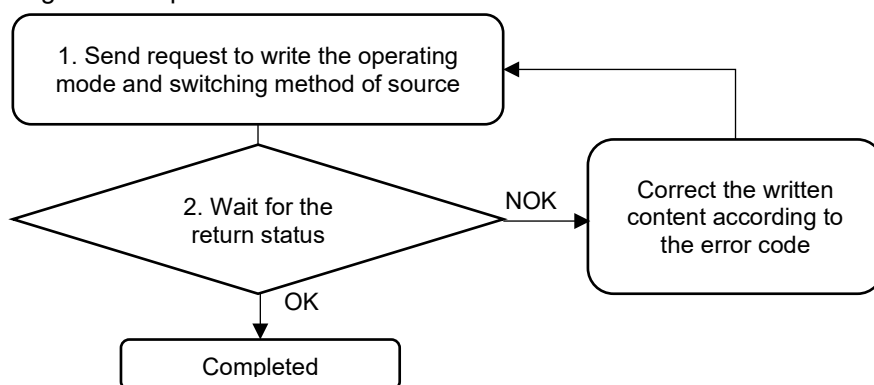
B.3.3 Sources operations

#300 Write the operating mode and switching method of source

Content description:

Use function code #300 to switch the operating mode and switching method of the controller.

Handshake signal description:



1. Send the request to write the operating mode and switching method of source.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	300	2C01
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Operating mode	0: single-tool 1: dual-tool alternation 2: dual-tool synchronization	0000
9 - 10	Switching method	0: manual 1: screw bit selector 2: barcode scanner	0000
11 - 12	Reserved	-	0000
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	300	2C01
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Operating mode	0: single-tool 1: dual-tool alternation 2: dual-tool synchronization	0000
9 - 10	Switching method	0: manual 1: screw bit selector 2: barcode scanner	0000
11 - 14	Reserved	4 bytes	...
15 - 16	Return function code	300	2C01
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

Code	Error description	Description
1	Operating mode	Exceeds the range of 0 to 2
2	Switching method	Exceeds the range of 0 to 2
3	Tool 1 / Tool 2	Must be 0 or 1
4	Cannot set to dual-tool alternation or dual-tool synchronization when only one tool is activated	-
5	Cannot use the screw bit selector mode under dual-tool alternation	-

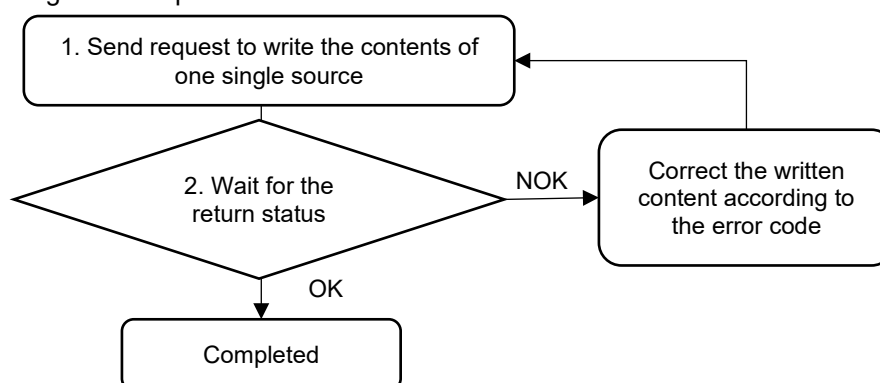
#301 Write the contents of a single source

Content description:

Use function code #301 to set the tightening source.

- A. Add a single tightening source for each tool.
- B. Modify the settings for an existing tightening source. First, read the settings of a single source with function code #351. Then, modify and write the settings with function code #301.

Handshake signal description:



1. Send the request to write the contents of one single source.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	301	2D01
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Switching method ID	Manual: 1 Screw bit selector: 1 to 255 Barcode scanner: 1 to 500	0100
9 - 12	Reserved	4 bytes	...
13 - 14	Request to send command	-	0100
15 - 20	Reserved	6 bytes	...
21 - 220	Corresponding barcode string	ASCII code (100 words); only applicable when the switching method is Barcode Scanner	...
221 - 222	Source for the switching method	0: parameter 1: sequence	0000
223 - 224	Parameter / Sequence ID	1 to 500	0100
225 - 226	Total screw quantity (L)	1 to 999999	0300
227 - 228	Total screw quantity (H)		0000
229 - 230	Bit ID	0 to 255	0000

Byte	Write / Read request	Description	Example (Hex)
231 - 232	Advanced settings (L)	Bit 0: loosening prohibited after tightening OK Bit 1: loosening prohibited after tightening NOK Bit 2: max. count for single screw NOK tightening Bit 3: max. count for single screw NOK loosening Bit 4: go to next step after tightening NOK Bit 5: go to previous step after loosening OK	0004
233 - 234	Advanced settings (H)	Bit 6: prohibit tool operation when scanner string is null Bit 7: clear scanner string when screw Qty. reached Bit 8: prohibit scanning when screw Qty. not reached Bit 9: max. operation time Bit 10: reset Qty. when screw Qty. reached Bit 11: enable reminder when tightening signal ends too early Bit 12: prohibit tool operation when scanner string length mismatch	0000
235 - 236	Single screw tightening NOK count (L)	1 to 999999	3F42
237 - 238	Single screw tightening NOK count (H)		0F00
239 - 240	Single screw loosening NOK count (L)	1 to 999999	3F42
241 - 242	Single screw loosening NOK count (H)		0F00
243 - 250	Reserved	8 bytes	...
251 - 252	Max. operation time (L)	1 to 9999999	7F96
253 - 254	Max. operation time (H)		9800
255 - 256	The parameters to be used under dual-tool alternation mode	0: use parameters of Tool 1 1: use parameters of Tool 2	0000
257 - 258	Torque unit	0: N.m; 1: kgf.cm; 2: lbf.ft; 3: lbf.in; 4: ozf.ft; 5: ozf.in; 6:cNm	0100
259 - 260	Start condition for Tool 1	0: push start 1: DI 2: lever start	0300
261 - 262	Start condition for Tool 2	3: push start or lever start 4: push start and lever start 5: DI or push start 6: DI or lever start	0300

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	301	2D01
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Switching method ID	Manual: 1 Screw bit selector: 1 to 255 Barcode scanner: 1 to 500	0100
9 - 14	Reserved	6 bytes	...
15 - 16	Return function code	301	2D01
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

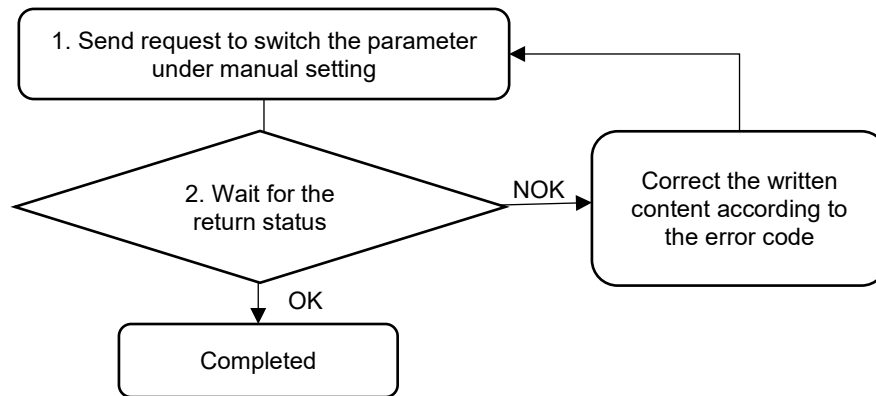
Code	Error description	Description
1	Switching method ID	Exceeds the range of 1 to 500
2	Tool 1 / Tool 2	Must be 0 or 1
10	Source for the switching method	Must be 0 or 1
11	Sequence / Parameter ID	Exceeds the range of 1 to 500
12	Total screw quantity	Exceeds the range of 1 to 999999
13	Bit ID	Exceeds the range of 0 to 255
20	Single screw tightening NOK count	Exceeds the range of 1 to 999999
21	Single screw loosening NOK count	Exceeds the range of 1 to 999999
40	Max. operation time	Exceeds the range of 1 to 9999999
50	Cannot specify the tightening parameter	Cannot select the tightening parameter under dual-tool alternation
51	Sequence error	Must select the sequence which includes tightening parameters of both Tool 1 and Tool 2 under dual-tool alternation
100	Title string is null	-
101	Title already exists	-

#302 Write to switch parameter under manual setting

Content description:

Use function code #302 to switch the parameter ID and screw quantity when the switching method is set to manual setting.

Handshake signal description:



1. Send the request to switch the parameter under manual setting.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	302	2E01
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0100
7 - 8	Parameter ID	1 to 500	0100
9 - 10	Total screw quantity (L)	0 to 999999	0300
11 - 12	Total screw quantity (H)		0000
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	302	2E01
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Parameter ID	1 to 500	0100
9 - 10	Total screw quantity (L)	0 to 999999	0300
11 - 12	Total screw quantity (H)		0000
13 - 14	Reserved	-	0000
15 - 16	Return function code	302	2E01
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

Code	Error description	Description
1	Incorrect switching method	Must be manual setting. Using function code #302 is prohibited when the switching method is set to screw bit selector or barcode scanner.
2	Tool 1 / Tool 2	Must be 0 or 1
11	Parameter ID	Exceeds the range of 1 to 500
12	Total screw quantity	Exceeds the range of 1 to 999999
50	Unsupported operating mode	Using function code #302 is prohibited under dual-tool alternation mode or dual-tool synchronization mode.

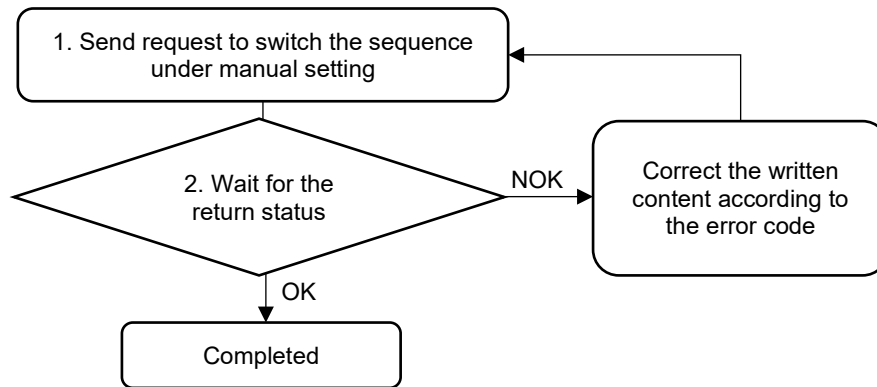
B

#303 Write to switch sequence under manual setting

Content description:

Use function code #303 to switch the sequence ID when the switching method is set to manual setting.

Handshake signal description:



1. Send the request to switch the sequence under manual setting.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	303	2F01
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0100
7 - 8	Sequence ID	1 to 500	0100
9 - 12	Reserved	4 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	303	2F01
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0100
7 - 8	Sequence ID	1 to 500	0100
9 - 14	Reserved	6 bytes	...
15 - 16	Return function code	303	2F01
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

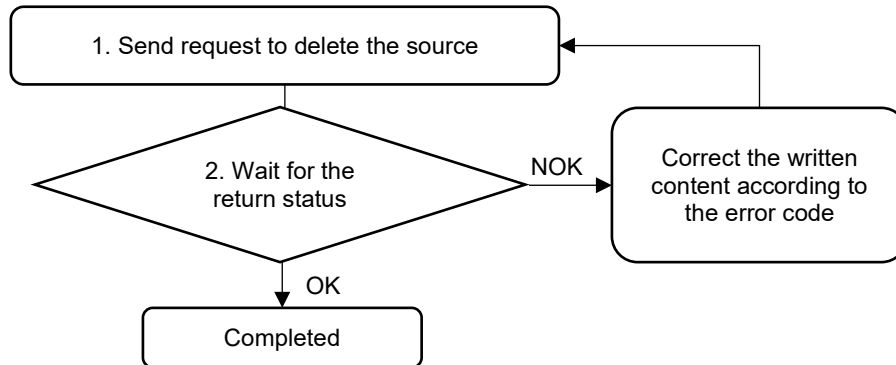
Code	Error description	Description
1	Incorrect switching method	Must be manual setting. Using function code #303 is prohibited when the switching method is set to screw bit selector or barcode scanner.
2	Tool 1 / Tool 2	Must be 0 or 1
11	Sequence ID	Exceeds the range of 1 to 500
50	Unsupported operating mode	Using function code #303 is prohibited under dual-tool alternation mode or dual-tool synchronization mode

#310 Write to delete the source

Content description:

Use function code #310 to delete the tightening source.

Handshake signal description:



1. Send the request to delete the source.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	310	3601
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Operating mode	0: single-tool 1: dual-tool alternation 2: dual-tool synchronization	0000
9 - 10	Switching method	0: manual 1: screw bit selector 2: barcode scanner	0000
11 - 12	Switching method ID to be deleted	Manual: 1 Screw bit selector: 1 to 255 Barcode scanner: 1 to 500 Delete all: 0	0100
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	310	3601
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Operating mode	0: single-tool 1: dual-tool alternation 2: dual-tool synchronization	0000
9 - 10	Switching method	0: manual 1: screw bit selector 2: barcode scanner	0000
11 - 12	Switching method ID to be deleted	Manual: 1 Screw bit selector: 1 to 255 Barcode scanner: 1 to 500 Delete all: 0	0100
13 - 14	Reserved	-	0000
15 - 16	Return function code	310	3601
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

B

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

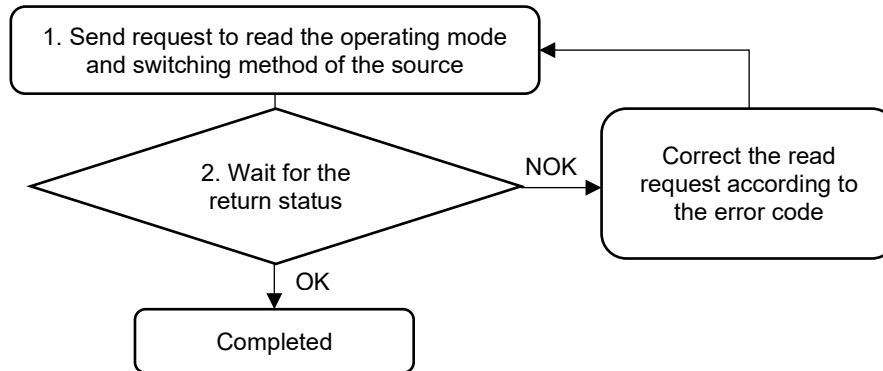
Code	Error description	Description
1	Operating mode	Exceeds the range of 0 to 2
2	Switching method	Exceeds the range of 0 to 2
3	Tool 1 / Tool 2	Must be 0 or 1
4	Cannot set to dual-tool alternation or dual-tool synchronization when only one tool is activated	-
5	Cannot use the screw bit selector mode under dual-tool alternation	-
6	Switching method ID to be deleted	Exceeds the range of 0 to 500

#350 Read the operating mode and switching method of source

Content description:

Use function code #350 to read the current operating mode and switching method of the smart screwdriving controller.

Handshake signal description:



1. Send the request to read the operating mode and switching method of the source.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	350	5E01
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the operating mode and switching method of the source.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	350	5E01
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Operating mode	0: single-tool 1: dual-tool alternation 2: dual-tool synchronization	0000
9 - 10	Switching method	0: manual 1: screw bit selector 2: barcode scanner	0000
11 - 14	Reserved	4 bytes	...
15 - 16	Return function code	350	5E01
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

B

#351 Read the contents of a single source

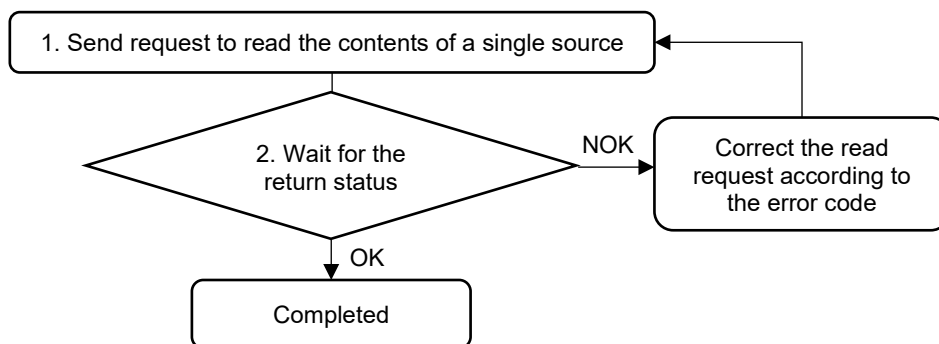
Content description:

Use function code #351 to read the settings of a single tightening source.

A. Obtain the contents of an existing tightening source.

B. Check whether the tightening source has been set.

Handshake signal description:



1. Send the request to read the contents of a single source.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	351	5F01
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Switching method ID	Manual: 1 Screw bit selector: 1 to 255 Barcode scanner: 1 to 500	0100
9 - 12	Reserved	4 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the contents of a single source.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	351	5F01
3 - 4	Return data length	Unit: Word	8D00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Switching method ID	Manual: 1 Screw bit selector: 1 to 255 Barcode scanner: 1 to 500	0100
9 - 14	Reserved	6 bytes	...
15 - 16	Return function code	351	5F01
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000
21 - 262	Refer to function code #301 for the Write / Read request table.		

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

Code	Error description	Description
1	Switching method ID	Exceeds the range of 1 to 500
2	Tool 1 / Tool 2	Must be 0 or 1

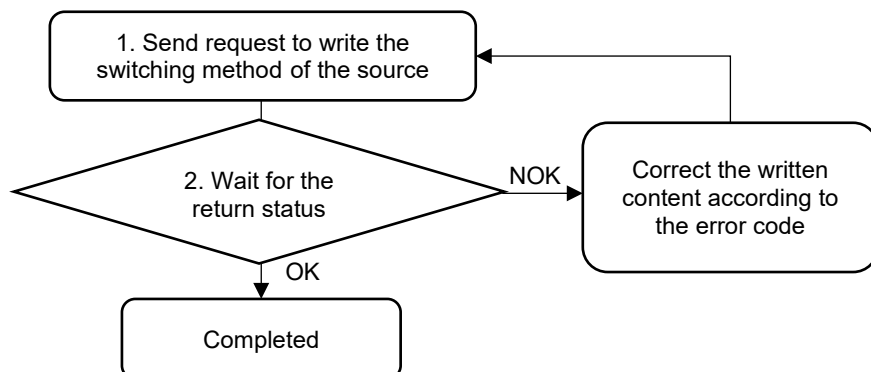
B.3.4 Results operations

#400 Write the switching method of source

Content description:

Use function code #400 to change the switching method of the tightening source in operation.

Handshake signal description:



1. Send the request to write the switching method of the source.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	400	9001
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Switching method ID	Manual: 1 Screw bit selector: 1 to 255 Barcode scanner: 1 to 500	0100
9 - 12	Reserved	4 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	400	9001
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Switching method ID	Manual: 1 Screw bit selector: 1 to 255 Barcode scanner: 1 to 500	0100
9 - 14	Reserved	6 bytes	...
15 - 16	Return function code	400	9001
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	Switching method ID	Manual: exceeds 1 Screw bit selector: exceeds the range of 1 to 255 Barcode scanner: exceeds the range of 1 to 500

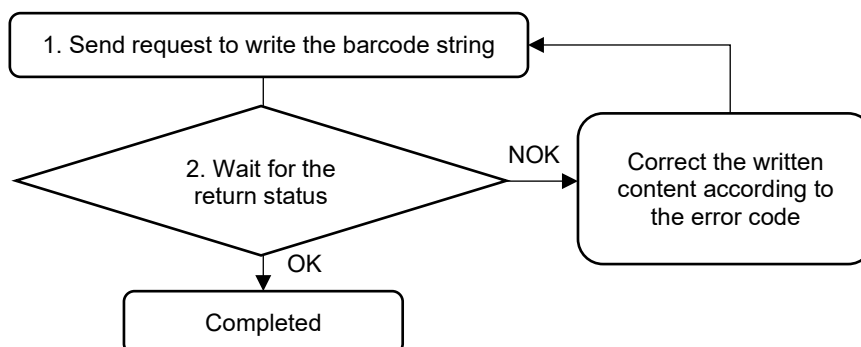
#401 Write the barcode string

Content description:

Use function code #401 to input the barcode string.

- A. Input the barcode string to the controller.
- B. If the switching method is Barcode Scanner, when you scan the barcode, the content of the string is immediately matched and switched to the corresponding parameter or sequence.

Handshake signal description:



1. Send the request to write the barcode string.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	401	9101
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100
15 - 20	Reserved	6 bytes	...
21 - 220	Barcode string	ASCII code (100 words)	...

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	401	9101
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	401	9101
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

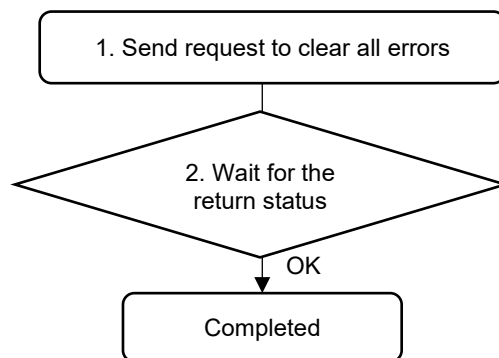
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	Matching failed	Failed to match the strings in Barcode Scanner mode

#402 Write to clear all errors

Content description:

Use function code #402 to clear all the AL, NG, and WN alarms.

Handshake signal description:



1. Send the request to clear all errors.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	402	9201
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	402	9201
3 - 4	Return data length	Unit: Word	0A00
5 - 14	Reserved	10 bytes	...
15 - 16	Return function code	402	9201
17 - 18	Status	1: OK	0100

If successful, the read status is 1.

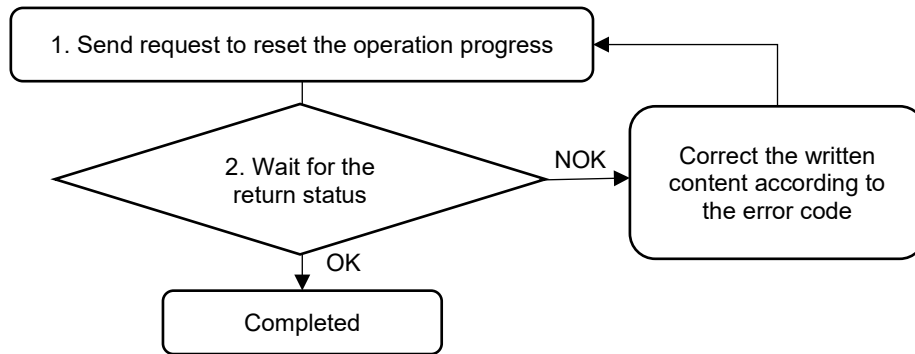
B

#403 Write to reset the operation progress

Content description:

Use function code #403 to reset the operation progress (total screw quantity, parameter quantity, and screw quantity of the parameter).

Handshake signal description:



1. Send the request to reset the operation progress.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	403	9301
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	403	9301
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	403	9301
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

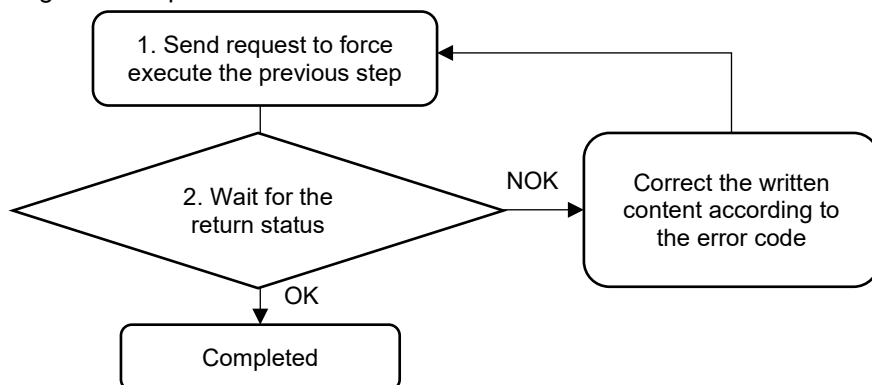
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

#404 Write to force execute the previous step

Content description:

Use function code #404 to execute the parameter of the previous screw.

Handshake signal description:



1. Send the request to force execute the previous step.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	404	9401
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	404	9401
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	404	9401
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

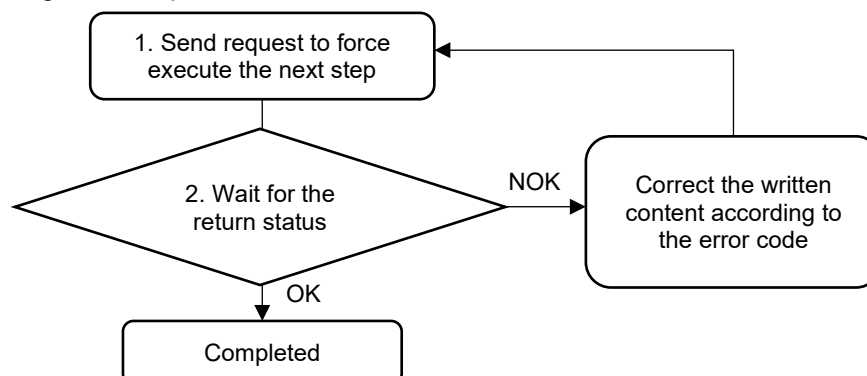
B

#405 Write to force execute the next step

Content description:

Use function code #405 to execute the parameter of the next screw.

Handshake signal description:



1. Send the request to force execute the next step.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	405	9501
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	405	9501
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	405	9501
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

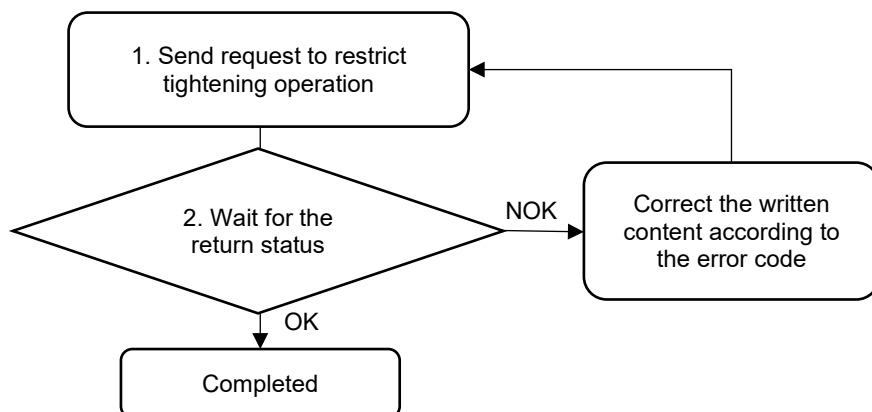
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

#406 Write to restrict tightening operation

Content description:

Use function code #406 to restrict the transducerized smart screwdriver from performing tightening operations until you use function code #406 again to remove the restriction.

Handshake signal description:



1. Send the request to restrict the tightening operation.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	406	9601
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Restrict operation / Remove restriction	0: remove restriction 1: restrict operation	0100
9 - 12	Reserved	4 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	406	9601
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Restrict operation / Remove restriction	0: remove restriction 1: restrict operation	0100
9 - 14	Reserved	6 bytes	...
15 - 16	Return function code	406	9601
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

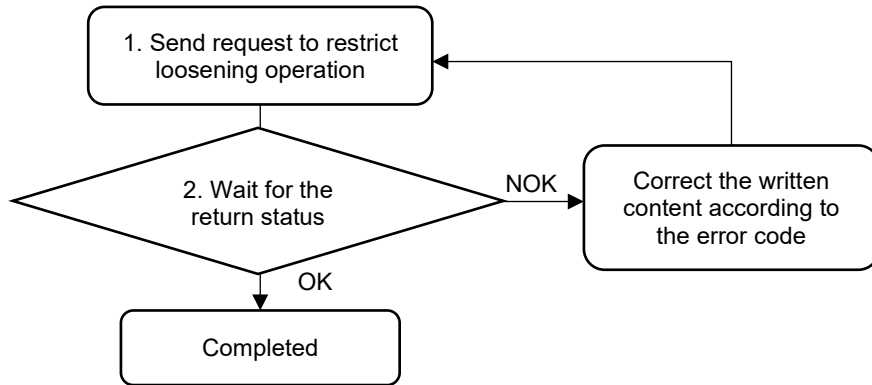
B

#407 Write to restrict loosening operation

Content description:

Use function code #407 to restrict the transducerized smart screwdriver from performing loosening operations until you use function code #407 again to remove the restriction.

Handshake signal description:



1. Send the request to restrict the loosening operation.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	407	9701
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Restrict operation / Remove restriction	0: remove restriction 1: restrict operation	0100
9 - 12	Reserved	4 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	407	9701
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Restrict operation / Remove restriction	0: remove restriction 1: restrict operation	0100
9 - 14	Reserved	6 bytes	...
15 - 16	Return function code	407	9701
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

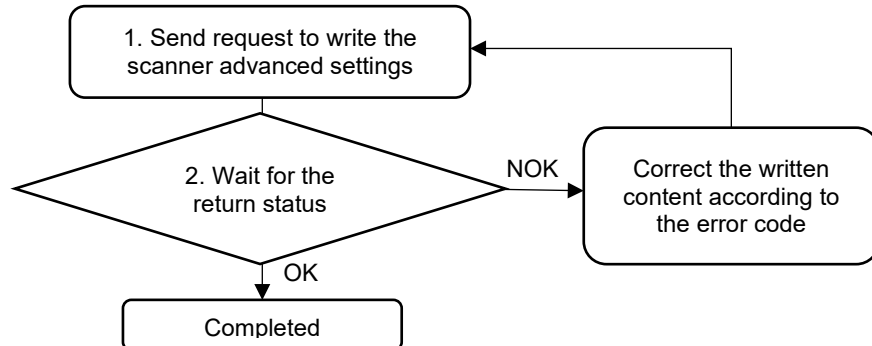
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

#408 Write the scanner advanced settings

Content description:

Use function code #408 to set the scanner advanced settings.

Handshake signal description:



1. Send the request to write the scanner advanced settings.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	408	9801
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100
15 - 20	Reserved	6 bytes	...
21 - 22	Start position (char.) for saving scanner string	1 to 200	0100
23 - 24	End position (char.) for saving scanner string		C800
25 - 26	Start position (char.) for matching scanner string		0100
27 - 28	End position (char.) for matching scanner string		C800

Example 1: when the barcode string is "ABCDE", if you set 2 to bytes 21 - 22 and 4 to bytes 23 - 24, the system then saves the 3-character string "BCD" to the Results screen.

Example 2: when the barcode string is "ABCDE" and the switching method is Barcode Scanner, if you set 2 to bytes 25 - 26 and 4 to bytes 27 - 28, the system then matches the 3-character string "BCD" against the corresponding strings of the 500 sets of parameters or sequences.

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	408	9801
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	408	9801
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	Scanner advanced setting value out of range	Exceeds the range of 1 to 200

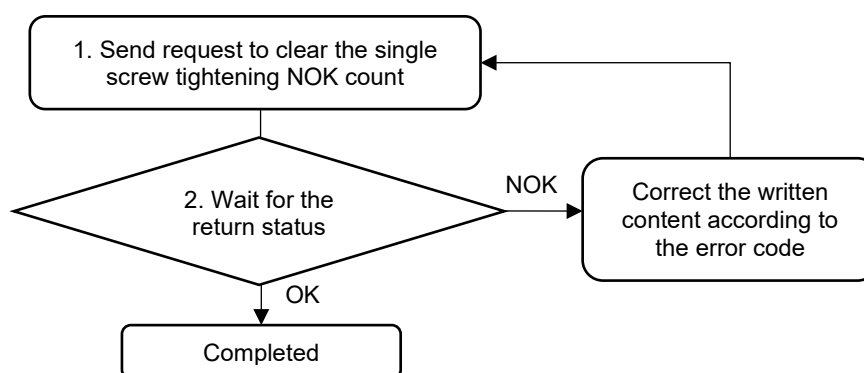
#409 Write to clear the single screw tightening NOK count

Content description:

Use function code #409 to clear the single screw tightening NOK count.

Byte (Dec)		Function code #20 (Hex)		Function code #21 (Hex)		Tool 1 status / Tool 2 status
Tool 1	Tool 2	Tool 1	Tool 2	Tool 1	Tool 2	
47 - 48	147 - 148	18	4A	E	40	Single screw tightening NOK count (L)
49 - 50	149 - 150	19	4B	F	41	Single screw tightening NOK count (H)

Handshake signal description:



1. Send the request to clear the single screw tightening NOK count.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	409	9901
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	409	9901
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	409	9901
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

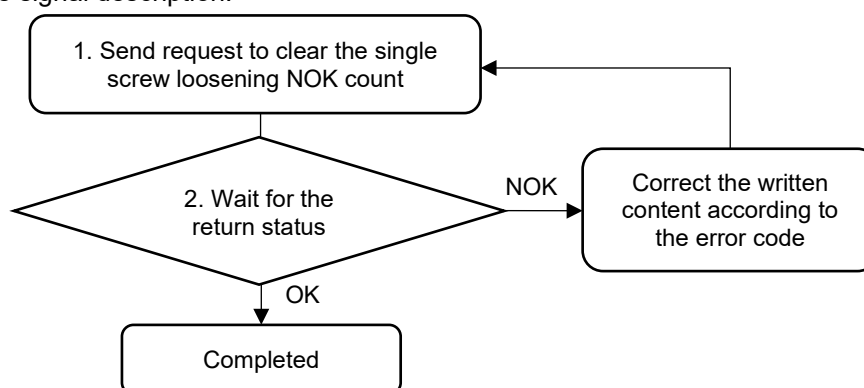
#410 Write to clear the single screw loosening NOK count

Content description:

Use function code #410 to clear the single screw loosening NOK count.

Byte		Function code #20 (Hex)		Function code #21 (Hex)		Tool 1 status / Tool 2 status
Tool 1	Tool 2	Tool 1	Tool 2	Tool 1	Tool 2	
55 - 56	155 - 156	1C	4E	12	44	Single screw loosening NOK count (L)
57 - 58	157 - 158	1D	4F	13	45	Single screw loosening NOK count (H)

Handshake signal description:



1. Send the request to clear the single screw loosening NOK count.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	410	9A01
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	410	9A01
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	410	9A01
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

B

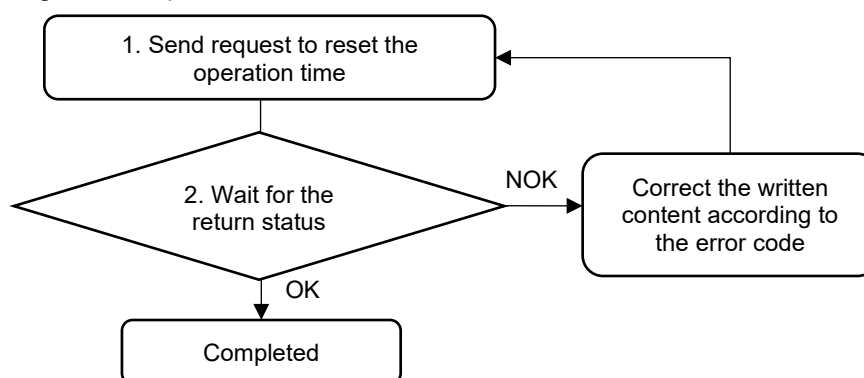
#411 Write to reset the operation time

Content description:

Use function code #411 to restart the timer for counting the remaining operation time.

Byte		Function code #20 (Hex)		Function code #21 (Hex)		Tool 1 status / Tool 2 status
Tool 1	Tool 2	Tool 1	Tool 2	Tool 1	Tool 2	
117 - 118	217 - 218	3B	6D	31	63	Remaining operation time (L)
119 - 120	219 - 220	3C	6E	32	64	Remaining operation time (H)

Handshake signal description:



1. Send the request to reset the operation time.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	411	9B01
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	411	9B01
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	411	9B01
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

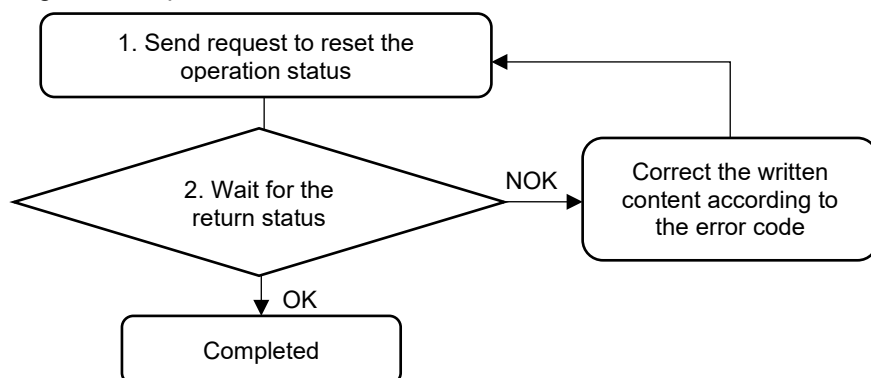
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

#412 Write to reset the operation status

Content description:

Use function code #412 to reset the operation status, including the operation progress and other tightening information.

Handshake signal description:



1. Send the request to reset the operation status.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	412	9C01
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	412	9C01
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	412	9C01
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

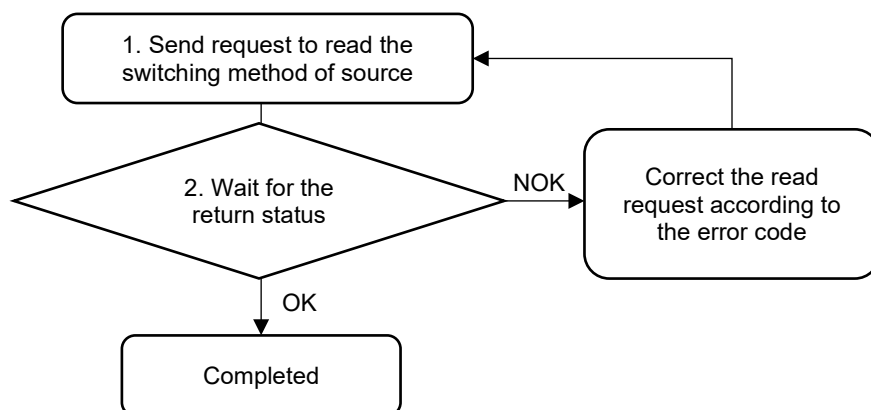
B

#450 Read the switching method of source

Content description:

Use function code #450 to read the switching method of the tightening source in operation.

Handshake signal description:



1. Send the request to read the switching method of the tightening source.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	450	C201
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the switching method of the tightening source in operation.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	450	C201
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Switching method ID	Manual: 1 Screw bit selector: 1 to 255 Barcode scanner: 1 to 500 (0: the controller is not running)	0000
9 - 14	Reserved	6 bytes	...
15 - 16	Return function code	450	C201
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

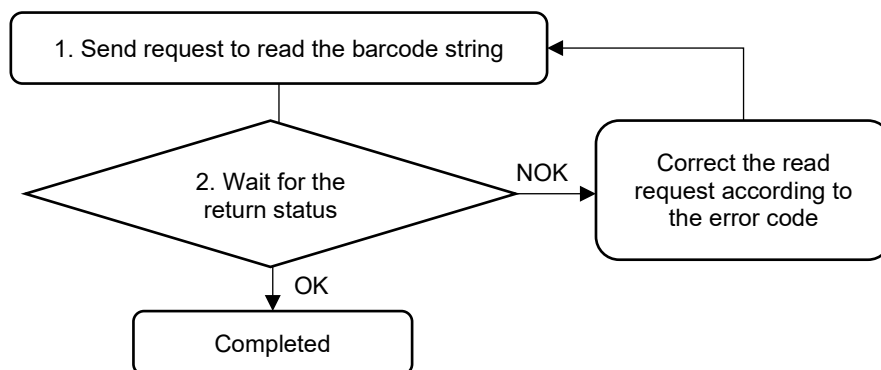
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

#451 Read the barcode string

Content description:

Use function code #451 to read the content of the barcode string.

Handshake signal description:



1. Send the request to read the barcode string.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	451	C301
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the barcode string.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	451	C301
3 - 4	Return data length	Unit: Word	6E00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	451	C301
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000
21 - 220	Barcode string	ASCII code (100 words)	...

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

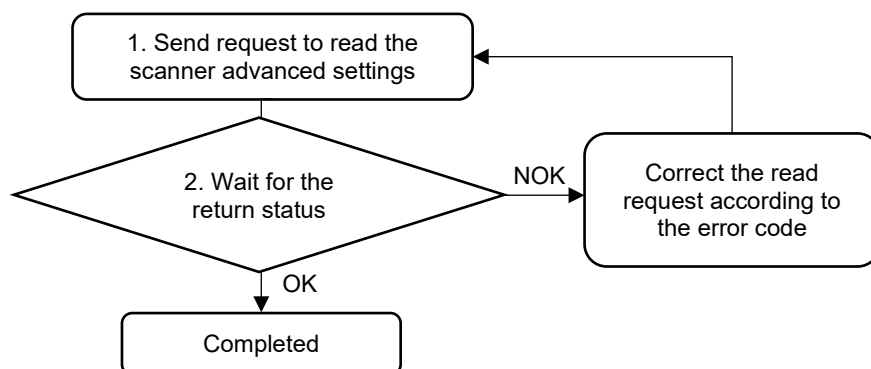
B

#452 Read the scanner advanced settings

Content description:

Use function code #452 to read the Scanner Advanced Settings.

Handshake signal description:



1. Send the request to read the scanner advanced settings.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	452	C401
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the scanner advanced settings.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	452	C401
3 - 4	Return data length	Unit: Word	0E00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	452	C401
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000
21 - 22	Start position (char.) for saving scanner string	1 to 200	0100
23 - 24	End position (char.) for saving scanner string		C800
25 - 26	Start position (char.) for matching scanner string		0100
27 - 28	End position (char.) for matching scanner string		C800

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

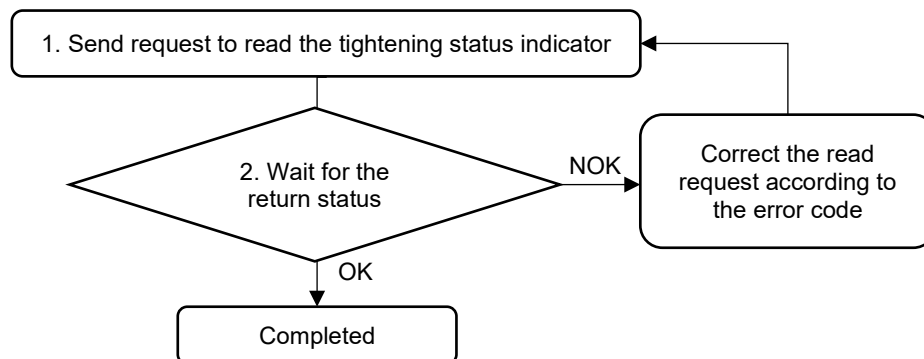
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

#453 Read the tightening status indicator

Content description:

Use function code #453 to read the tightening status indicator for up to 1000 screws.

Handshake signal description:



1. Send the request to read the tightening status indicator.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	453	C501
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the tightening status indicator.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	453	C501
3 - 4	Return data length	Unit: Word	3F02
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	453	C501
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000
21 - 22	Tightening status indicator for screw ID 1	2: green 8: red	0200
23 - 24	Tightening status indicator for screw ID 2		0200
25 - 26	Tightening status indicator for screw ID 3		0200
...	Tightening status indicator for screw ID N		...
2015 - 2016	Tightening status indicator for screw ID 998		0200
2017 - 2018	Tightening status indicator for screw ID 999		0200
2019 - 2020	Tightening status indicator for screw ID 1000		0200

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

B

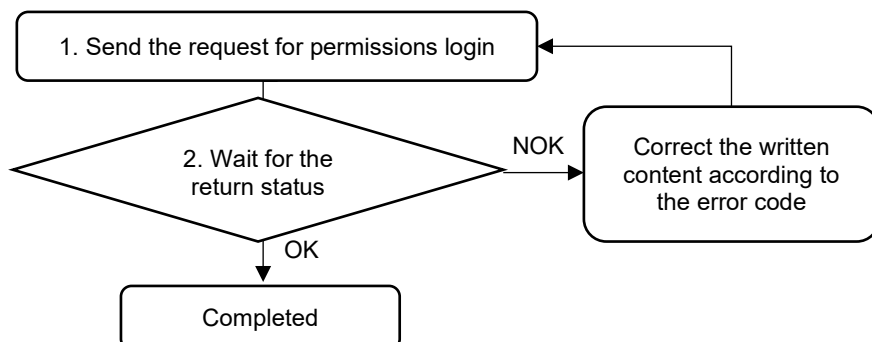
B.3.5 Controller operations

#500 Write the request for permissions login

Content description:

Use function code #500 to switch the controller permissions account through the host controller.

Handshake signal description:



1. Send the request for permissions login.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	500	F401
3 - 4	Version number	-	0000
5 - 6	Permissions account	1: User1 2: User2 3: User3 4: User4 5: User5 6: Admin (highest level)	0200
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100
15 - 20	Reserved	6 bytes	...
21 - 40	Permissions password	Numbers converted to ASCII code (10 words)	...

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	500	F401
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Permissions account	1: User1 2: User2 3: User3 4: User4 5: User5 6: Admin (highest level)	0200
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	500	F401
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

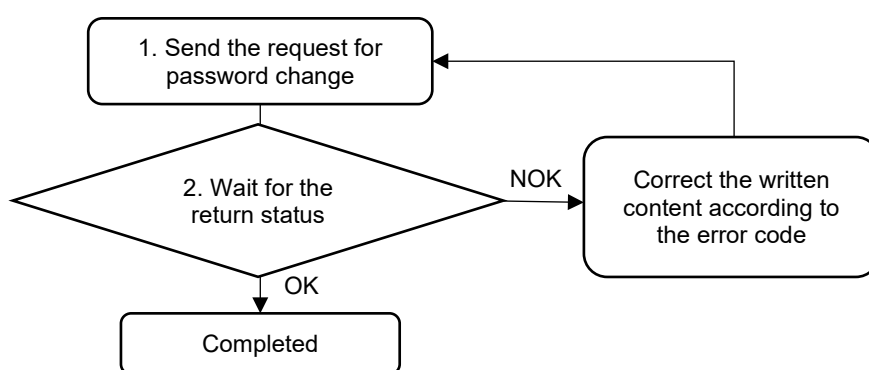
Code	Error description	Description
1	Permissions account	Exceeds the range of 1 to 6
2	Login failed	-

#501 Write the request for password change

Content description:

Use function code #501 to change the permissions password of controller through the host controller.

Handshake signal description:



1. Send the request for password change.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	501	F501
3 - 4	Version number	-	0000
5 - 6	Permissions account	1: User1 2: User2 3: User3 4: User4 5: User5 6: Admin (highest level)	0200
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100
15 - 20	Reserved	6 bytes	...
21 - 40	Old password	Numbers converted to ASCII code (10 words)	...
41 - 60	New password	Numbers converted to ASCII code (10 words)	...

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	501	F501
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Permissions account	1: User1 2: User2 3: User3 4: User4 5: User5 6: Admin (highest level)	0200
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	501	F501
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

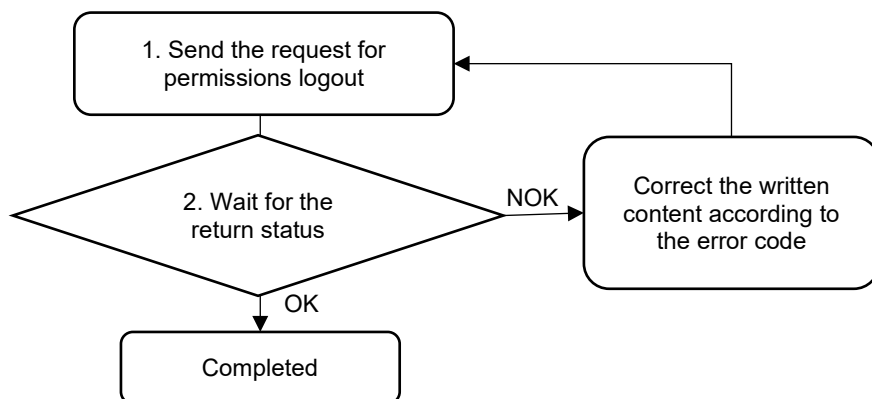
Code	Error description	Description
1	Permissions account	Exceeds the range of 1 to 6
2	Password change failed	-

#502 Write the request for permissions logout

Content description:

Use function code #502 to log out of the controller permissions account through the host controller. The default permissions account after logout is User1.

Handshake signal description:



1. Send the request for permissions logout.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	502	F601
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	502	F601
3 - 4	Return data length	Unit: Word	0A00
5 - 14	Reserved	8 bytes	...
15 - 16	Return function code	502	F601
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Logout failed	-

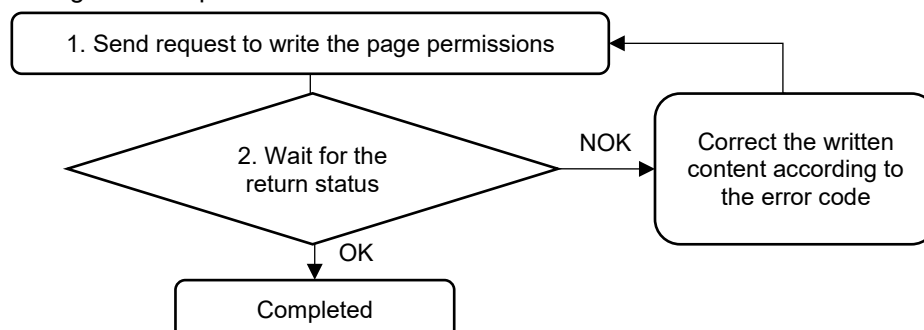
B

#503 Write the page permissions

Content description:

Use function code #503 to set the permissions of page access and edit for different accounts.

Handshake signal description:



1. Send the request to write the page permissions.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	503	F701
3 - 4	Version number	-	0000
5 - 6	Confirm command	99	6300
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100
15 - 20	Reserved	6 bytes	...
21 - 22	User1 page permissions	Bit 0: Parameters Bit 1: Sequence	00C0
23 - 24	User2 page permissions	Bit 2: Sources Bit 3: Controller	00C0
25 - 26	User3 page permissions	Bit 4: Tool Bit 5: operation for screw progress	00C0
27 - 28	User4 page permissions	Bit 6: clearing of the production report entries	00C0
29 - 30	User5 page permissions	Bit 7: clearing of the error, warning, and button report entries	00C0

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	503	F701
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Confirm command	99	6300
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	503	F701
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

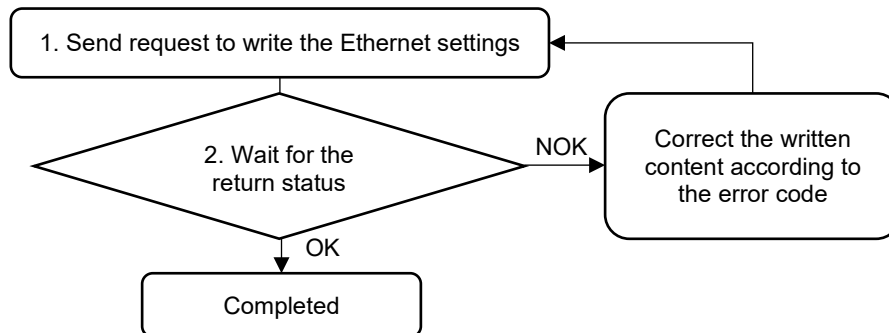
Code	Error description	Description
1	Confirm command	Must be 99

#504 Write the Ethernet settings

Content description:

Use function code #504 to write the Ethernet settings.

Handshake signal description:



1. Send the request to write the Ethernet settings.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	504	F801
3 - 4	Version number	-	0000
5 - 6	Confirm command	99	6300
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100
15 - 20	Reserved	6 bytes	...
21 - 22	IP1 address	Default: 192	C000
23 - 24	IP2 address	Default: 168	A800
25 - 26	IP3 address	Default: 1	0100
27 - 28	IP4 address	Default: 11	0B00
29 - 30	Subnet mask 1	Default: 255	FF00
31 - 32	Subnet mask 2	Default: 255	FF00
33 - 34	Subnet mask 3	Default: 255	FF00
35 - 36	Subnet mask 4	Default: 0	0000

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	504	F801
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Confirm command	99	6300
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	504	F801
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Confirm command	Must be 99

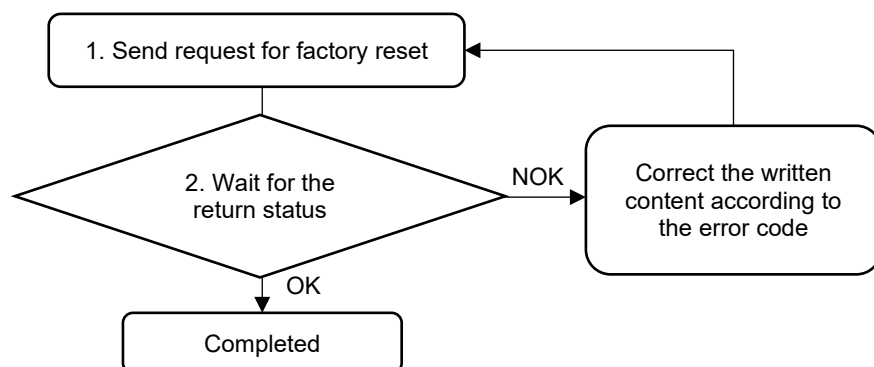
B

#505 Write the request for factory reset

Content description:

Use function code #505 to write the request for factory reset.

Handshake signal description:



1. Send the request for factory reset.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	505	F901
3 - 4	Version number	-	0000
5 - 6	Confirm command	99	6300
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

Note:

- After factory reset, all data will be deleted and all settings will be restored to default.
- If you use the account of the highest permissions (Admin) for factory reset, the Ethernet settings will be restored to default, and the error reports and warning reports will be cleared.

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	505	F901
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Confirm command	99	6300
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	505	F901
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

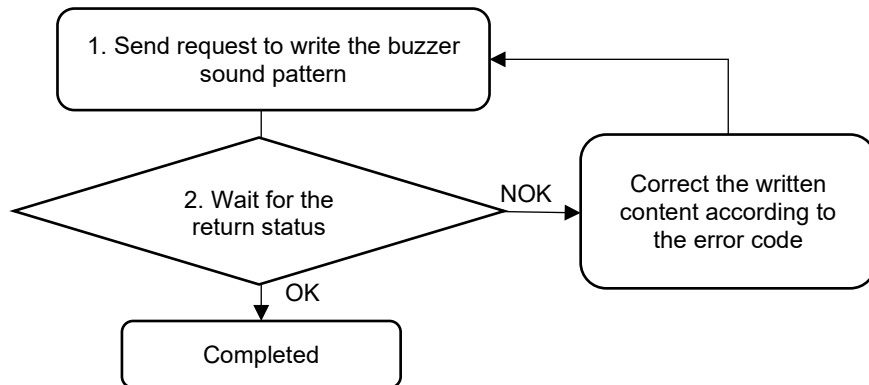
Code	Error description	Description
1	Confirm command	Must be 99

#506 Write the buzzer sound pattern

Content description:

Use function code #506 to set the buzzer sound patterns.

Handshake signal description:



1. Send the request to write the buzzer sound pattern.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	506	FA01
3 - 4	Version number	-	0000
5 - 6	Sound pattern of error signal buzzer	0: mute	0000
7 - 8	Sound pattern of finished signal buzzer for a single screw	1: one long beep 2: one short beep	0000
9 - 10	Sound pattern of finished signal buzzer for total screws	3: two short beeps	0000
11 - 12	Reserved	-	0000
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	506	FA01
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Sound pattern of error signal buzzer	0: mute	0000
7 - 8	Sound pattern of finished signal buzzer for a single screw	1: one long beep 2: one short beep	0000
9 - 10	Sound pattern of finished signal buzzer for total screws	3: two short beeps	0000
11 - 14	Reserved	4 bytes	...
15 - 16	Return function code	506	FA01
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Buzzer sound pattern	Exceeds the range of 0 to 3

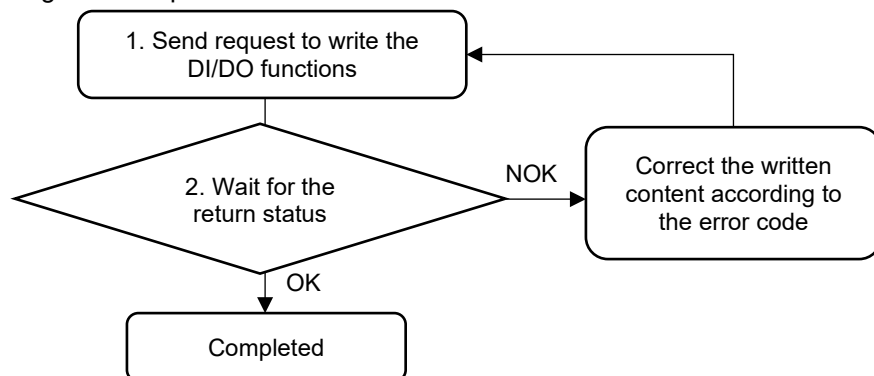
B

#507 Write the DI/DO functions

Content description:

Use function code #507 to set the DI/DO functions.

Handshake signal description:



1. Send the request to write the DI/DO functions.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	507	FB01
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100
15 - 20	Reserved	6 bytes	...
21 - 22	DO1 NO / NC	0: NO (normally open) 1: NC (normally closed)	0000
23 - 24	DO2 NO / NC		0000
25 - 26	DO3 NO / NC		0000
27 - 28	DO4 NO / NC		0000
29 - 30	DO5 NO / NC		0000
31 - 32	DO6 NO / NC		0000
33 - 34	DO7 NO / NC		0000
35 - 36	DO8 NO / NC		0000

Byte	Write / Read request	Description	Example (Hex)
37 - 38	DO1 function	0: not in use 1: ready 2: running 3: tightening OK and report is being saved	0100
39 - 40	DO2 function	4: tightening NOK and report is being saved 5: loosening OK and report is being saved 6: loosening NOK and report is being saved 7: total screw quantity reached	0200
41 - 42	DO3 function	8: device alarms or errors or warning 9: tighten to waiting stage (only available for DO7; must enable the related option in the Advanced Setting window)	0300
43 - 44	DO4 function	10: Modbus handshake end signal 11: device alarms 12: reserved	0400
45 - 46	DO5 function	13: tool reverse switch state 14: rotation direction 15: tightening result sent to feeder 16 to 23: parameter ID output (Bits 1 - 8)	0500
47 - 48	DO6 function	24 to 31: screw progress output (Bits 1 - 8) 32 to 39: sequence ID output (Bits 1 - 8) 40 to 47: bit ID output (Bits 1 - 8) 48 to 51: running tightening stage output (Bits 1 - 4)	0600
49 - 50	DO7 function	52: tightening OK and lever signal end 53: double-click the lever button 54: tightening OK and the report is saved 55: tightening NOK and the report is saved	0000
51 - 52	DO8 function	56: loosening OK and the report is saved 57: loosening NOK and the report is saved 58: busy 59 - 63: reserved	0000
53 - 54	DI1 NO / NC	0: NO (normally open) 1: NC (normally closed)	0000
55 - 56	DI2 NO / NC		0000
57 - 58	DI3 NO / NC		0000
59 - 60	DI4 NO / NC		0000
61 - 62	DI5 NO / NC		0000
63 - 64	DI6 NO / NC		0000
65 - 66	DI7 NO / NC		0000
67 - 68	DI8 NO / NC		0000

B

Byte	Write / Read request	Description	Example (Hex)
69 - 70	DI1 function	0: not in use 1: start tightening 2: start loosening	0100
71 - 72	DI2 function	3: emergency stop (dual-tool) 4: clear error and clear status (dual-tool) 5: reset total screw quantity	0200
73 - 74	DI3 function	6: next step 7: previous step 8: start tightening (pulse signal) 9: start loosening (pulse signal)	0000
75 - 76	DI4 function	10: start tightening (dual-tool) 11: start loosening (dual-tool) 12: reset operation time	0000
77 - 78	DI5 function	13: reset single screw tightening + loosening NOK counts 14: prohibit tool operation 15: clear communication flag	0000
79 - 80	DI6 function	16: wait for external signal before continuing tightening (only available for DI7; must enable the related option in the Advanced Setting window)	0000
81 - 82	DI7 function	17: emergency stop 18: prohibit tool operation (pulse signal) 19: rotation direction 20: start rotation	0000
83 - 84	DI8 function	21: clear error and clear status 22 to 39: reserved 40 to 47: bit ID input (Bit 1 - 8)	0000

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	507	FB01
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	507	FB01
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

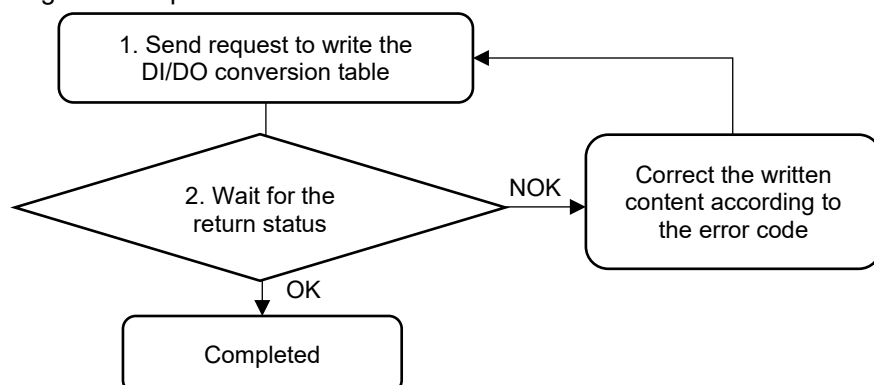
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

#508 Write the DI/DO conversion table

Content description:

Use function code #508 to set the DI/DO conversion table.

Handshake signal description:



1. Send the request to write the DI/DO conversion table.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	508	FC01
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Conversion table	0: bit ID output table 1: bit ID input table 2: parameter ID output table 4: screw progress output table 6: sequence ID output table	0100
9 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100
15 - 20	Reserved	6 bytes	...
21 - 22	DEC value corresponding to ID 0	0 to 255	0000
23 - 24	DEC value corresponding to ID 1		0100
25 - 26	DEC value corresponding to ID 2		0200
27 - 28	DEC value corresponding to ID 3		0300
29 - 30	DEC value corresponding to ID 4		0400
...	DEC value corresponding to ID N		...
523 - 524	DEC value corresponding to ID 251		FB00
525 - 526	DEC value corresponding to ID 252		FC00
527 - 528	DEC value corresponding to ID 253		FD00
529 - 530	DEC value corresponding to ID 254		FE00
531 - 532	DEC value corresponding to ID 255		FF00

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	508	FC01
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Conversion table	0: bit ID output table 1: bit ID input table 2: parameter ID output table 4: screw progress output table 6: sequence ID output table	0100
9 - 14	Reserved	6 bytes	...
15 - 16	Return function code	508	FC01
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

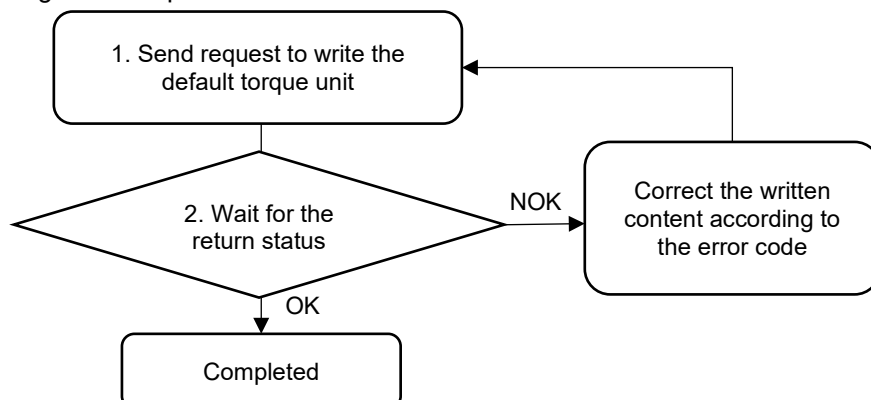
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	Conversion table	Must be 0, 1, 2, 4, or 6

#509 Write the default torque unit

Content description:

Use function code #509 to set the torque unit of the controller.

Handshake signal description:



1. Send the request to write the default torque unit.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	509	FD01
3 - 4	Version number	-	0000
5 - 6	Default torque unit	0: N.m 1: kgf.cm 2: lbf.ft 3: lbf.in 4: ozf.ft 5: ozf.in 6: cNm	0100
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	509	FD01
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Default torque unit	0: N.m 1: kgf.cm 2: lbf.ft 3: lbf.in 4: ozf.ft 5: ozf.in 6: cNm	0100
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	509	FD01
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Default torque unit	Exceeds the range of 0 to 6

B

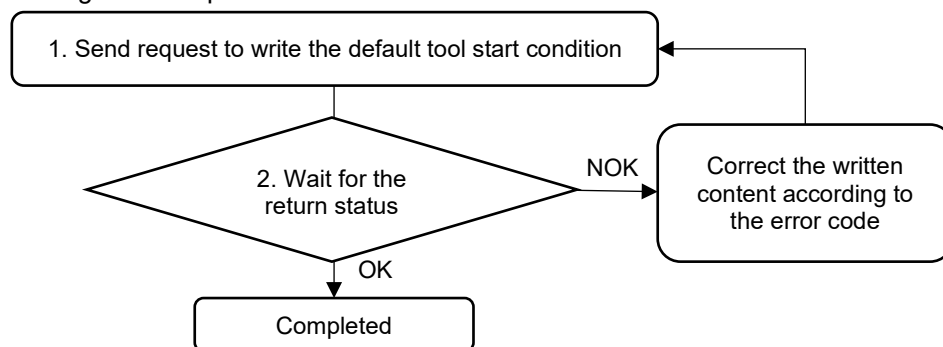
#510 Write the default tool start condition

Content description:

Use function code #510 to set the tool start condition.

Note: for example, first set the default tool start condition to "DI" with function code #510. Then, set the DI function to "start tightening" or "start loosening" with function code #507 or #511 to operate the tool through DI.

Handshake signal description:



1. Send the request to write the default tool start condition.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	510	FE01
3 - 4	Version number	-	0000
5 - 6	Default tool start condition	0: push start 1: DI 2: lever start 3: push start or lever start 4: push start and lever start 5: DI or push start 6: DI or lever start	0300
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	510	FE01
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Default tool start condition	0: push start 1: DI 2: lever start 3: push start or lever start 4: push start and lever start 5: DI or push start 6: DI or lever start	0300
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	510	FE01
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

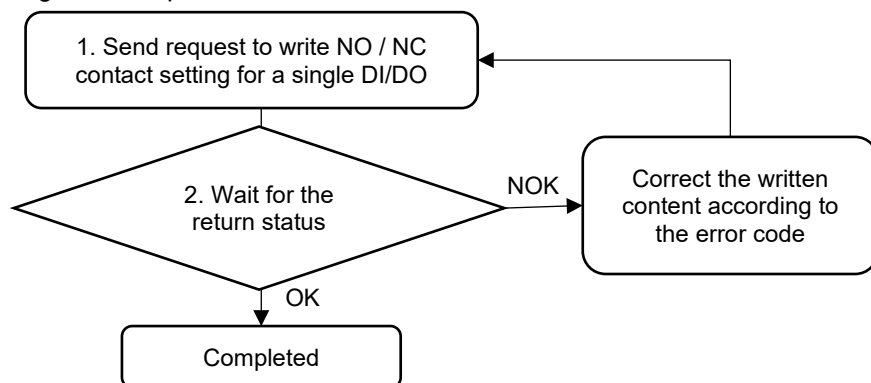
Code	Error description	Description
1	Tool start condition	Exceeds the range of 0 to 6

#511 Write NO / NC contact setting for a single DI/DO

Content description:

Use function code #511 to set a single DI/DO as an NO / NC contact.

Handshake signal description:



1. Send the request to write the NO / NC contact setting for a single DI/DO.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	511	FF01
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	DO / DI	0: DO; 1: DI	0100
9 - 10	Corresponding bit of DO / DI	0 to 7	0000
11 - 12	NO / NC	0: NO (normally open) 1: NC (normally closed)	0100
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	511	FF01
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	DO / DI	0: DO; 1: DI	0100
9 - 10	Corresponding bit of DO / DI	0 to 7	0000
11 - 12	NO / NC	0: NO (normally open) 1: NC (normally closed)	0100
13 - 14	Reserved	-	0000
15 - 16	Return function code	511	FF01
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

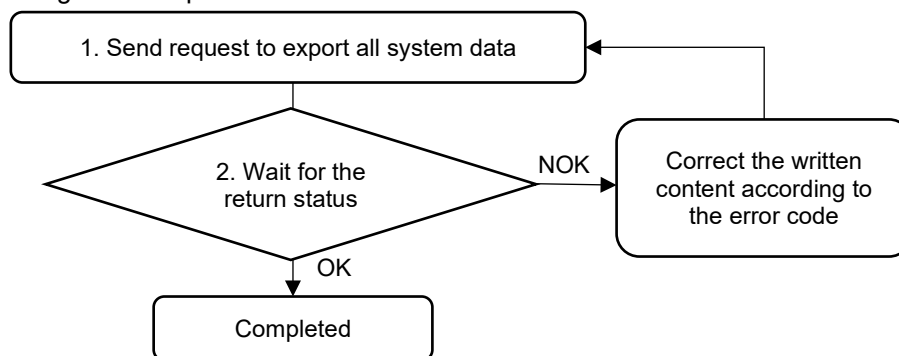
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	DO / DI	Must be 0 or 1
3	Corresponding bit of DO / DI	Exceeds the range of 0 to 7
4	NO / NC	Must be 0 or 1

#512 Write to export all system data

Content description:

Use function code #512 to export all system data, including contents of parameters, sequences, sources, controller, tool, and reports.

Handshake signal description:



1. Send the request to export all system data.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	512	0002
3 - 4	Version number	-	0000
5 - 6	Data storage location	0: USB Disk 1: HMI Disk	0100
7 - 8	Options	Bit 0: parameters Bit 1: sequences Bit 2: sources Bit 3: controller Bit 4: tool Bit 5: reports	0F00
9 - 12	Reserved	4 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	512	0002
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Data storage location	0: USB Disk 1: HMI Disk	0100
7 - 8	Options	Bit 0: parameters Bit 1: sequences Bit 2: sources Bit 3: controller Bit 4: tool Bit 5: reports	0F00
9 - 14	Reserved	6 bytes	...
15 - 16	Return function code	512	0002
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

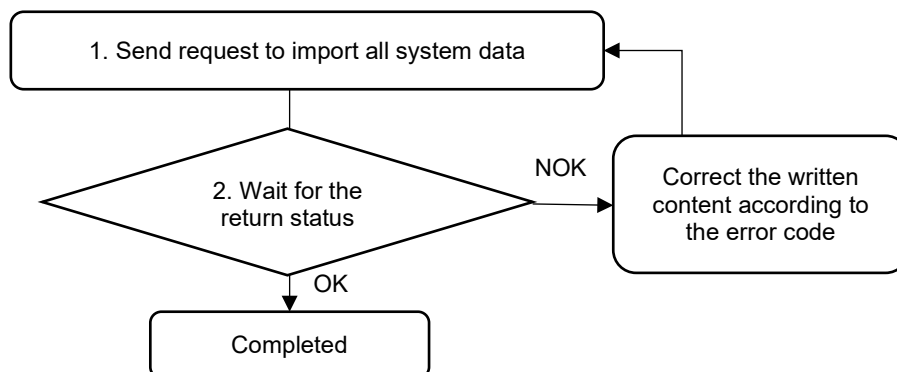
Code	Error description	Description
1	Data storage location	Must be 0 or 1

#513 Write to import all system data

Content description:

Use function code #513 to import all system data, including contents of parameters, sequences, sources, controller, and tool.

Handshake signal description:



1. Send the request to write to import all system data.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	513	0102
3 - 4	Version number	-	0000
5 - 6	Read data from	0: USB Disk 1: HMI Disk	0100
7 - 8	Options	Bit 0: parameters Bit 1: sequences Bit 2: sources Bit 3: controller Bit 4: tool	0F00
9 - 12	Reserved	4 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	513	0102
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Read data from	0: USB Disk 1: HMI Disk	0100
7 - 8	Options	Bit 0: parameters Bit 1: sequences Bit 2: sources Bit 3: controller Bit 4: tool	0F00
9 - 14	Reserved	8 bytes	...
15 - 16	Return function code	563	0102
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Read data from	Must be 0 or 1

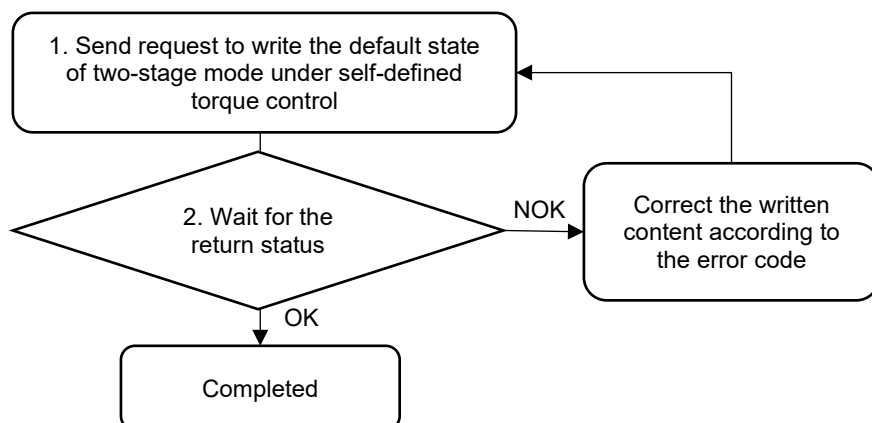
B

#514 Write the default two-stage mode under self-defined torque control

Content description:

Use function code #514 to set the default state of two-stage mode under self-defined torque control.

Handshake signal description:



1. Send the request to write the default state of two-stage mode under self-defined torque control.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	514	0202
3 - 4	Version number	-	0000
5 - 6	Two-stage mode	0: Off 1: On	0100
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	514	0202
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Two-stage mode	0: Off 1: On	0100
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	514	0202
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

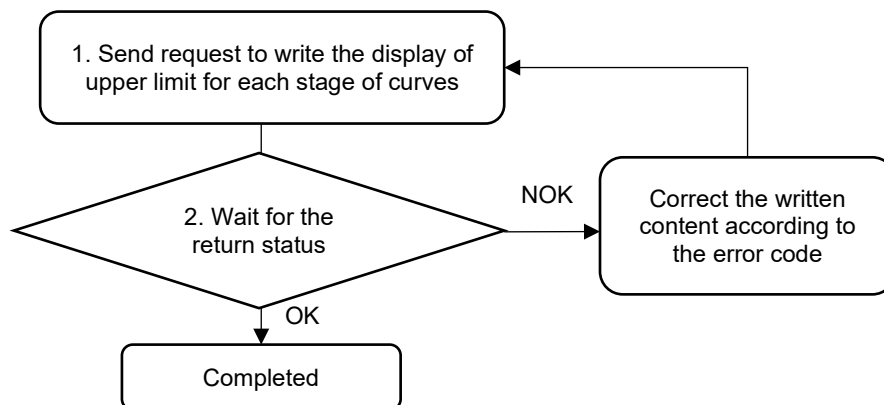
Code	Error description	Description
1	Two-stage mode	Must be 0 or 1

#515 Write the display of upper limit for each stage of curves

Content description:

Use function code #515 to set whether to display the upper limit for each stage of the curves.

Handshake signal description:



1. Send the request to write the display of upper limit for each stage of curves.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	515	0302
3 - 4	Version number	-	0000
5 - 6	Display the upper limit for each stage of curves	0: On 1: Off	0100
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	515	0302
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Display the upper limit for each stage of curves	0: On 1: Off	0100
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	515	0302
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Display the upper limit for each stage of curves	Must be 0 or 1

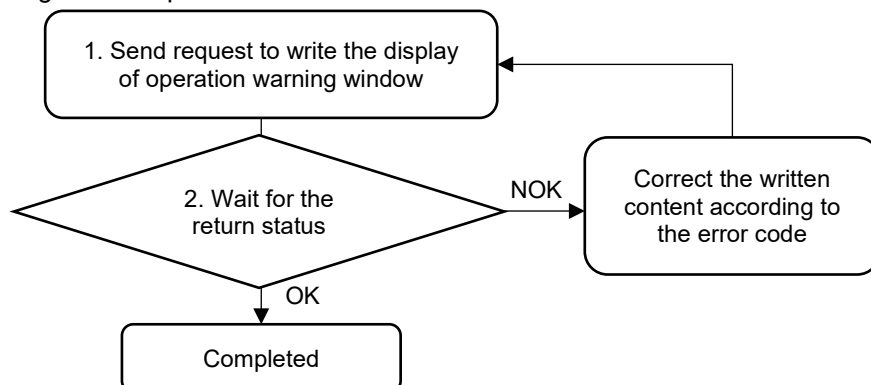
B

#516 Write the display of operation warning window

Content description:

Use function code #516 to set whether to display the operation warning window.

Handshake signal description:



1. Send the request to write the display of operation warning window.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	516	0402
3 - 4	Version number	-	0000
5 - 6	Display the operation warning window	0: On 1: Off	0100
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	516	0402
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Display the operation warning window	0: On 1: Off	0100
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	516	0402
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

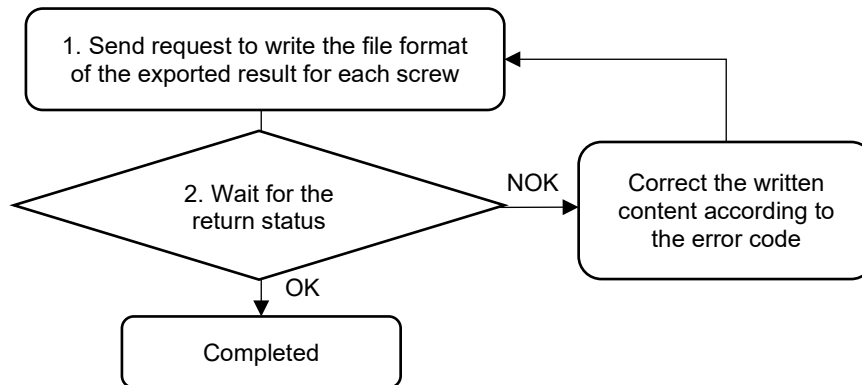
Code	Error description	Description
1	Display the operation warning window	Must be 0 or 1

#517 Write file format of the exported result for each screw

Content description:

Use function code #517 to set the file format of the exported result for each screw.

Handshake signal description:



1. Send the request to write the file format of the exported result for each screw.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	517	0502
3 - 4	Version number	-	0000
5 - 6	File format of the exported result for each screw	0: disable 1: CSV file (HMI Disk) 2: CSV file (USB Disk) 3: BIN file (HMI Disk) 4: BIN file (USB Disk)	0300
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	517	0502
3 - 4	Return data length	Unit: Word	0A00
5 - 6	File format of the exported result for each screw	0: disable 1: CSV file (HMI Disk) 2: CSV file (USB Disk) 3: BIN file (HMI Disk) 4: BIN file (USB Disk)	0300
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	517	0502
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	File format of the exported result for each screw	HMI kernel version 124 or later: exceeds the range of 0 to 4 HMI kernel version earlier than 124: exceeds the range of 0 to 2

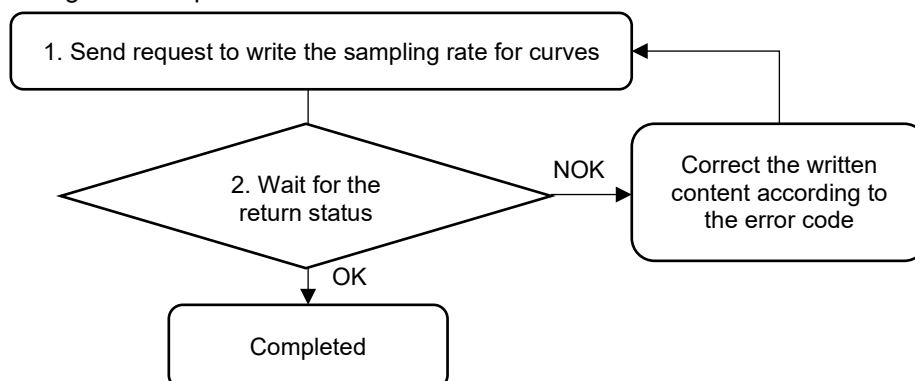
B

#518 Write the sampling rate for curves

Content description:

Use function code #518 to set the sampling rate for curves.

Handshake signal description:



1. Send the request to write the sampling rate for curves.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	518	0602
3 - 4	Version number	-	0000
5 - 6	Sampling rate for curves	0: 10 ms→4 ms when start torque is reached 1: 2 ms; max. 4 seconds per production report entry 2: 2 ms; max. 8 seconds per production report entry 3: 1 ms; max. 4 seconds per production report entry 4: 1 ms; max. 8 seconds per production report entry	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	518	0602
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Sampling rate for curves	0: 10 ms→4 ms when start torque is reached 1: 2 ms; max. 4 seconds per production report entry 2: 2 ms; max. 8 seconds per production report entry 3: 1 ms; max. 4 seconds per production report entry 4: 1 ms; max. 8 seconds per production report entry	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	518	0602
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

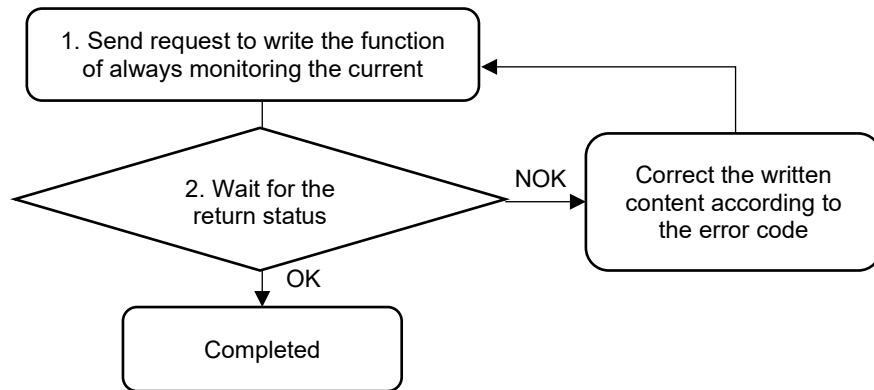
Code	Error description	Description
1	Sampling rate for curves	Exceeds the range of 0 to 4

#519 Write the function of always monitoring the current

Content description:

Use function code #519 to set whether to enable the function of always monitoring the current.

Handshake signal description:



1. Send the request to write the function of always monitoring the current.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	519	0702
3 - 4	Version number	-	0000
5 - 6	Always monitor the current	0: On 1: Off	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	519	0702
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Always monitor the current	0: On 1: Off	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	519	0702
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Always monitor the current	Must be 0 or 1

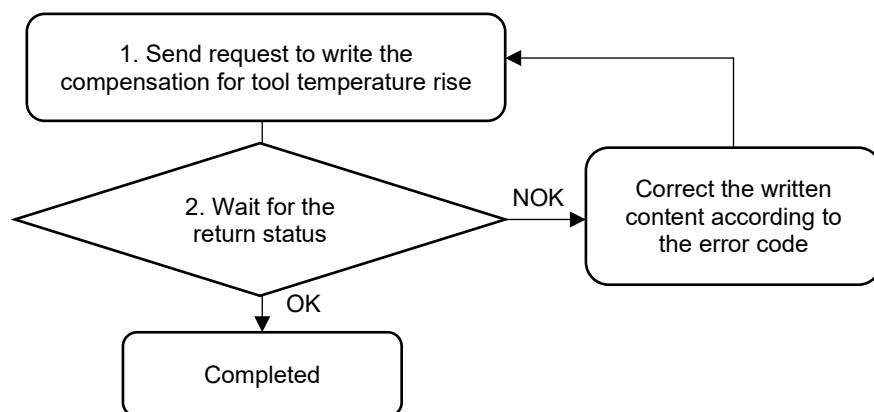
B

#520 Write the compensation for tool temperature rise

Content description:

Use function code #520 to set the compensation for tool temperature rise.

Handshake signal description:



1. Send the request to write the compensation for tool temperature rise.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	520	0802
3 - 4	Version number	-	0000
5 - 6	Compensation for tool temperature rise	0: On 1: Off	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	520	0802
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Compensation for tool temperature rise	0: On 1: Off	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	520	0802
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

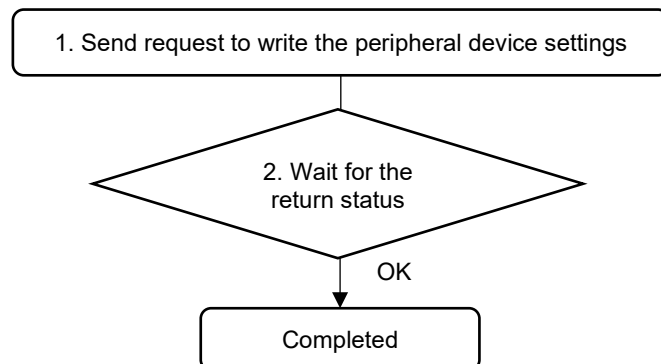
Code	Error description	Description
1	Compensation for tool temperature rise	Must be 0 or 1

#521 Write the peripheral device settings

Content description:

Use function code #521 to write the peripheral device settings.

Handshake signal description:



1. Send the request to write the peripheral device settings.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	521	0902
3 - 4	Version number	-	0000
5 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100
15 - 22	Reserved	8 bytes	...
23 - 24	RS-485 connector function	0: disabled 1: Tool 1 positioning arm 2: Tool 2 positioning arm (P65F0) 2: Tool 2 positioning arm (P65F0) 3: Tool 1 positioning arm (AM-TMTS-TA series) 4: Tool 2 positioning arm (AM-TMTS-TA series) 5: Tool 1 positioning arm (others) 6: Tool 2 positioning arm (others) 7: Tool 1 positioning arm (Modbus TCP slave) 8: Tool 2 positioning arm (Modbus TCP slave) 9: Tool 1 positioning arm (Modbus TCP master) 10: Tool 2 positioning arm (Modbus TCP master)	0000
25 - 26	Reserved	-	0000
27 - 28	Tool 1 positioning arm - target position tolerance (L)	Unit: mm	0800
29 - 30	Tool 1 positioning arm - target position tolerance (H)		0000
31 - 32	Tool 1 - X axis position offset (L)	Unit: mm	0000
33 - 34	Tool 1 - X axis position offset (H)		0000
35 - 36	Tool 1 - Y axis position offset (L)	Unit: mm	0000
37 - 38	Tool 1 - Y axis position offset (H)		0000
39 - 40	Tool 1 - Z axis position offset (L)	Unit: mm	0000

B

Byte	Write / Read request	Description	Example (Hex)
41 - 42	Tool 1 - Z axis position offset (H)		0000
43 - 44	Tool 2 positioning arm - target position tolerance (L)	Unit: mm	0800
45 - 46	Tool 2 positioning arm - target position tolerance (H)		0000
47 - 48	Tool 2 - X axis position offset (L)	Unit: mm	0000
49 - 50	Tool 2 - X axis position offset (H)		0000
51 - 52	Tool 2 - Y axis position offset (L)	Unit: mm	0000
53 - 54	Tool 2 - Y axis position offset (H)		0000
55 - 56	Tool 2 - Z axis position offset (L)	Unit: mm	0000
57 - 58	Tool 2 - Z axis position offset (H)		0000
59 - 60	L1 length of AM-TMTS-TA series	Unit: mm (default: 475 mm)	DB01
61 - 62	L2 length of AM-TMTS-TA series	Unit: mm (default: 475 mm)	DB01

2. Check whether the reading is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	521	0902
3 - 4	Return data length	Unit: Word	0A00
5 - 14	Reserved	10 bytes	...
15 - 16	Return function code	521	0902
17 - 18	Status	1: OK	0100

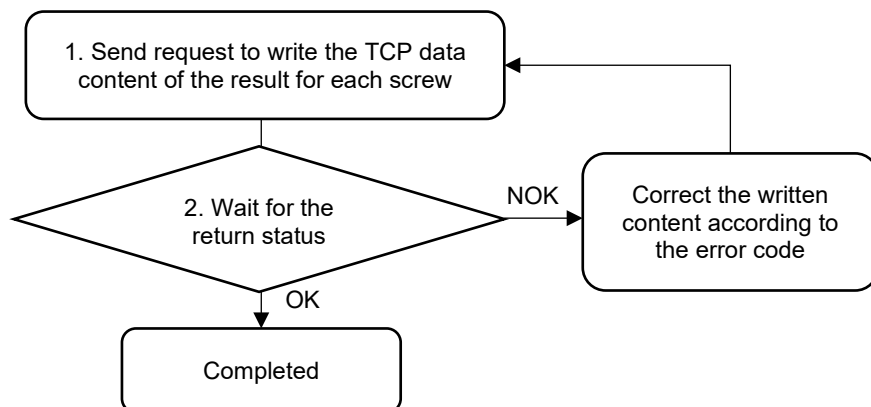
If successful, the read status is 1.

#522 Write the TCP data content of the result for each screw

Content description:

Use function code #522 to set the TCP data content of the result for each screw.

Handshake signal description:



1. Send the request to write the TCP data content of the result for each screw.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	522	0A02
3 - 4	Version number	-	0000
5 - 6	TCP data content of the result for each screw	0: disable 1: scale, time, torque 2: scale, angle, torque 3: scale, angle, torque rate 4: result, scale, time, torque 5: result, scale, angle, torque 6: result, scale, angle, torque rate 7 - 11: reserved 12: result, scale (tightening or loosening) ^{Note} 13: result, scale (tightening) ^{Note} 14: result, scale, all curve data (tightening or loosening) ^{Note} 15: result, scale, all cruve data (tightening) ^{Note}	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

Note:

1. When the data content includes result and scale, only packet 1 (Query type 10 in #750 and #751) is sent.
2. When the data content includes result, scale, and all curve data, the following packets are sent:
 Packet 1: query type 10 in #750 and #751
 Packet 2: time point data
 Packet 3: angle point data
 Packet 4: torque point data (double word)
 Packet 5: torque rate point data (double word)
3. Tightening: one packet is sent only after a tightening operation is complete.
 Loosening: one packet is sent only after a loosening opeation is complete.

B

2. Check whether the writing is successful from the return status.

Bytes	Return status	Description	Example (Hex)
1 - 2	Function code	522	0A02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	TCP data content of the result for each screw	0: disable 1: scale, time, torque 2: scale, angle, torque 3: scale, angle, torque rate 4: result, scale, time, torque 5: result, scale, angle, torque 6: result, scale, angle, torque rate 7 - 11: reserved 12: result, scale (tightening or loosening) 13: result, scale (tightening) 14: result, scale, all curve data (tightening or loosening) 15: result, scale, all cruve data (tightening)	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	522	0A02
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

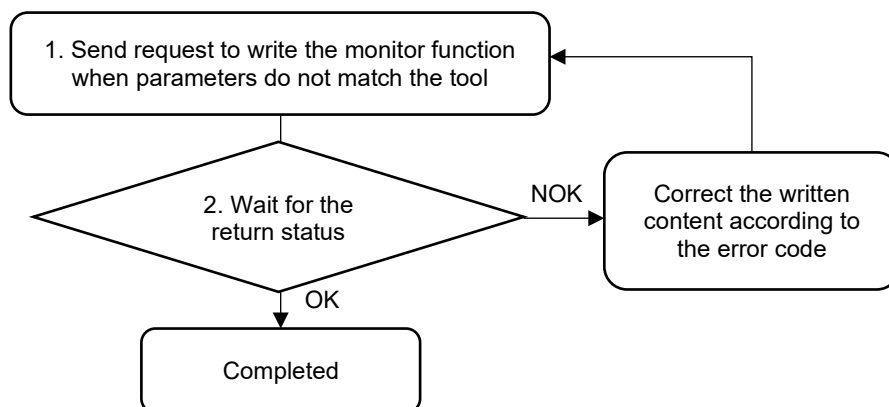
Code	Error description	Description
1	TCP data content of the result for each screw	Exceeds the range of 0 to 15

#523 Write the monitor function when parameters do not match the tool

Content description:

Use function code #523 to set the function for monitoring if the tool torque of the operating parameter matches that of the connected tool.

Handshake signal description:



1. Send the request to write the monitor function when parameters do not match the tool.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	523	0B02
3 - 4	Version number	-	0000
5 - 6	Monitor function when parameters do not match the tool	0: On 1: Off	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	523	0B02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Monitor function when parameters do not match the tool	0: On 1: Off	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	523	0B02
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Monitor function when parameters do not match the tool	Must be 0 or 1

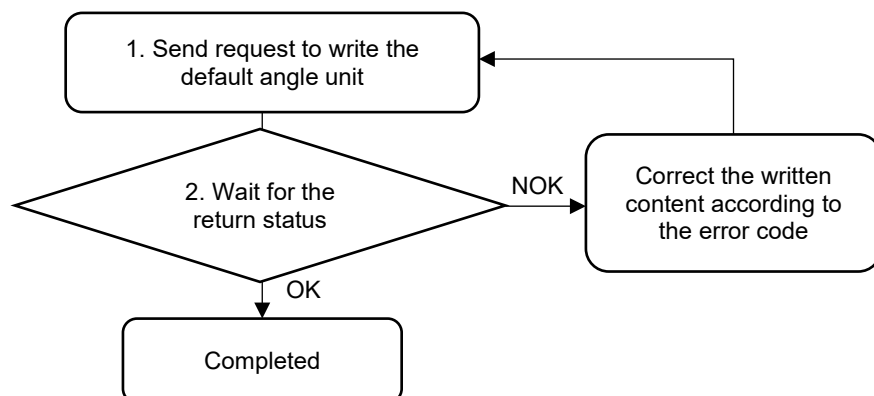
B

#524 Write the default angle unit

Content description:

Use function code #524 to set the default angle unit of the controller.

Handshake signal description:



1. Send the request to write the default angle unit.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	524	0C02
3 - 4	Version number	-	0000
5 - 6	Default angle unit	0: degree 1: turn	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	524	0C02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Default angle unit	0: degree 1: turn	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	524	0C02
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

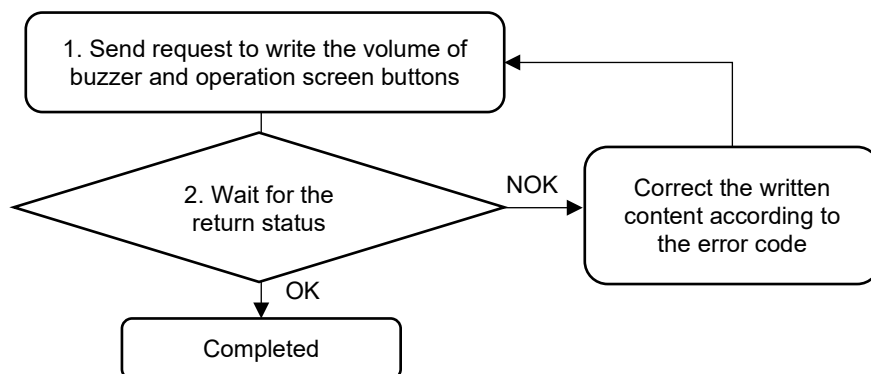
Code	Error description	Description
1	Default angle unit	Must be 0 or 1

#525 Write the volume of buzzer and operation screen buttons

Content description:

Use function code #525 to set the volume of buzzer and operation screen buttons.

Handshake signal description:



1. Send the request to write the volume of buzzer and operation screen buttons.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	525	0D02
3 - 4	Version number	-	0000
5 - 6	Volume of buzzer	0 to 100	0000
7 - 8	Volume of operation screen buttons	0 to 100	0000
9 - 12	Reserved	4 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	525	0D02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Volume of buzzer	0 to 100	0000
7 - 8	Volume of operation screen buttons	0 to 100	0000
9 - 12	Reserved	4 bytes	...
15 - 16	Return function code	525	0D02
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

Code	Error description	Description
1	Volume of buzzer	Exceeds the range of 0 to 100
2	Volume of operation screen buttons	Exceeds the range of 0 to 100

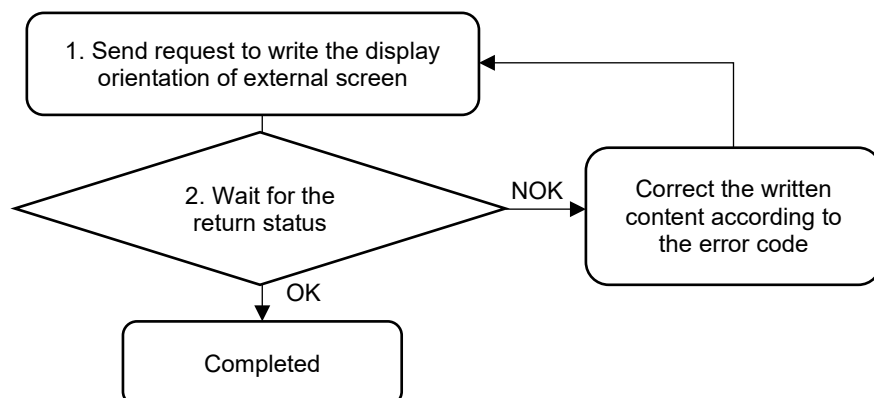
B

#526 Write the display orientation of external screen

Content description:

Use function code #526 to set the display orientation of the external screen (HDMI).

Handshake signal description:



1. Send the request to write the display orientation of external screen.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	526	0E02
3 - 4	Version number	-	0000
5 - 6	Display orientation	0: portrait 1: landscape	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	526	0E02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Display orientation	0: portrait 1: landscape	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	526	0E02
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

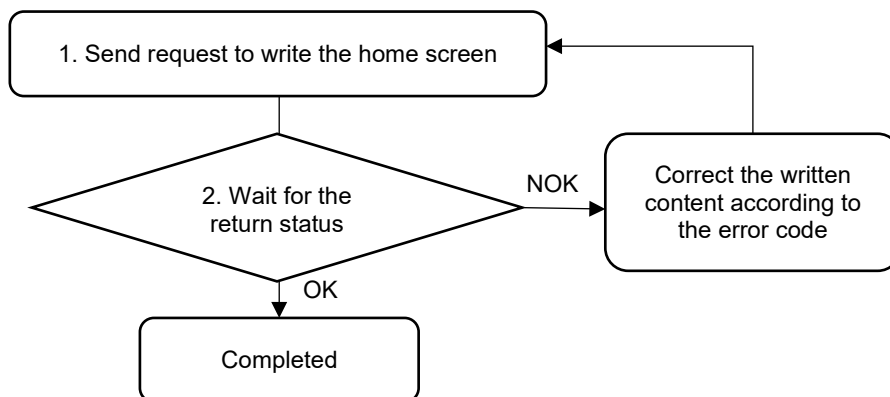
Code	Error description	Description
1	Display orientation	Must be 0 or 1

#527 Write the home screen

Content description:

Use function code #527 to set the home screen of the controller.

Handshake signal description:



1. Send the request to write the home screen.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	527	0F02
3 - 4	Version number	-	0000
5 - 6	Home screen	0: Home 1: Results	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	527	0F02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Home screen	0: Home 1: Results	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	527	0F02
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Home screen	Must be 0 or 1

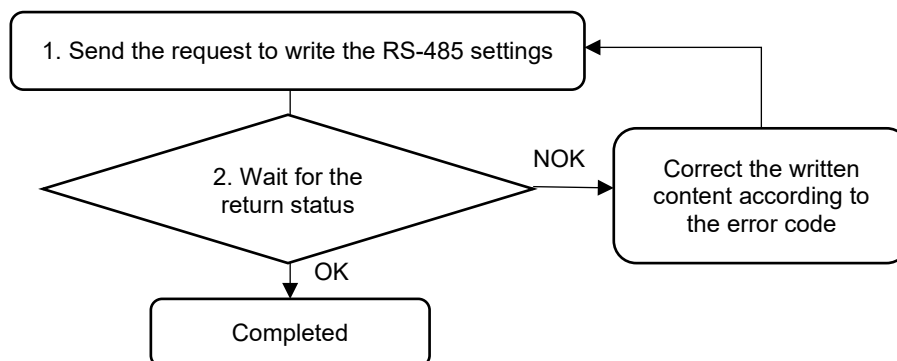
B

#528 Write the RS-485 settings

Content description:

Use function code #528 to write the RS-485 settings.

Handshake signal description:



1. Send the request to write the RS-485 settings.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	528	1002
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100
15 - 20	Reserved	6 bytes	...
21 - 22	Function On / Off	0: Off; 1: On	0100
23 - 24	Station	0 to 255	0100
25 - 26	RTU / ASCII	0: RTU; 1: ASCII	0000
27 - 28	Baud rate	0: 9600; 1: 19200; 2: 38400	0000
29 - 30	Data bit	0: 8-bit; 1: 7-bit	0000
31 - 32	Parity bit	0: None; 1: Odd; 2: Even	0000
33 - 34	Stop bit	0: 2-bit; 1: 1-bit	0000

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	528	1002
3 - 4	Return data length	Unit: Word	0A00
5 - 14	Reserved	10 bytes	...
15 - 16	Return function code	528	1002
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

Code	Error description	Description
1	Function On / Off	Must be 0 or 1
2	Station	Exceeds the range of 0 to 255
3	RTU / ASCII	Must be 0 or 1

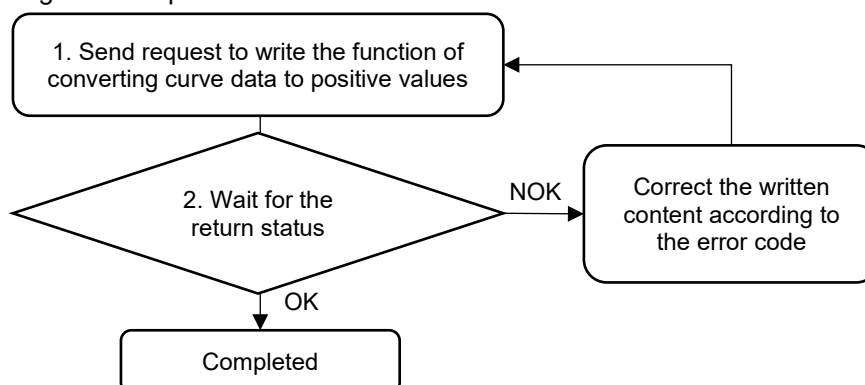
Code	Error description	Description
4	Baud rate	Exceeds the range of 0 to 2
5	Data bit	Must be 0 or 1
6	Parity bit	Exceeds the range of 0 to 2
7	Stop bit	Must be 0 or 1

#529 Write the function of converting curve data to positive values

Content description:

Use function code #529 to set the function of converting curve data to positive values.

Handshake signal description:



1. Send the request to write the function of converting curve data to positive values.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	529	1102
3 - 4	Version number	-	0000
5 - 6	Function of converting curve data to positive values	0: On 1: Off	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	529	1102
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Function of converting curve data to positive values	0: On 1: Off	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	529	1102
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

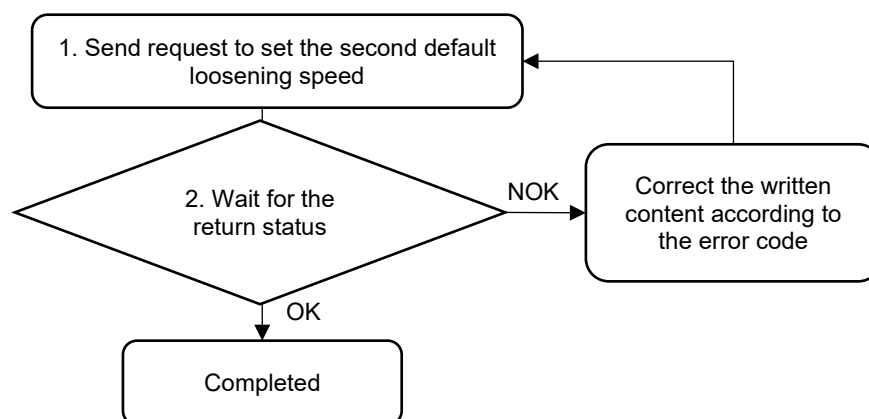
Code	Error description	Description
1	Function of converting curve data to positive values	Must be 0 or 1

#530 Write the default loosening second stage speed

Content description:

Use function code #530 to set the 2nd default loosening speed.

Handshake signal description:



1. Send request to set the second default loosening speed.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	530	1202
3 - 4	Version number	-	0000
5 - 6	2 nd default loosening speed	Unit: rpm	6400
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	530	1202
3 - 4	Return data length	Unit: Word	0A00
5 - 6	2 nd default loosening speed	Unit: rpm	6400
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	530	1202
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

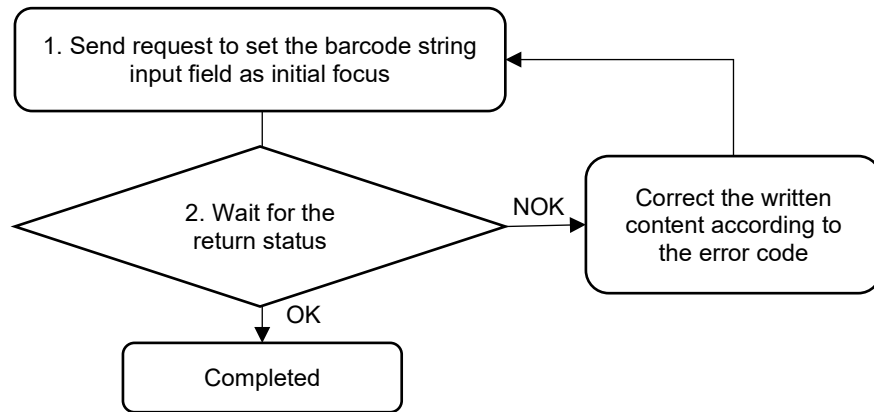
Code	Error description	Description
1	2 nd loosening speed	Exceeds the range of 10 - 2000

#531 Write the keyboard cursor blink in Results page

Content description:

Use function code #531 to set the barcode string input field as the initial focus in the **Results** tab of the **Results** page.

Handshake signal description:



1. Send request to set the barcode string input field as initial focus.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	531	1302
3 - 4	Version number	-	0000
5 - 6	Set the barcode string input field as the initial focus	0: Off 1: On	0100
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	531	1302
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Set the barcode string input field as the initial focus	0: Off 1: On	0100
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	531	1302
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Set the barcode string input field as the initial focus	Exceeds the range of 0 - 1

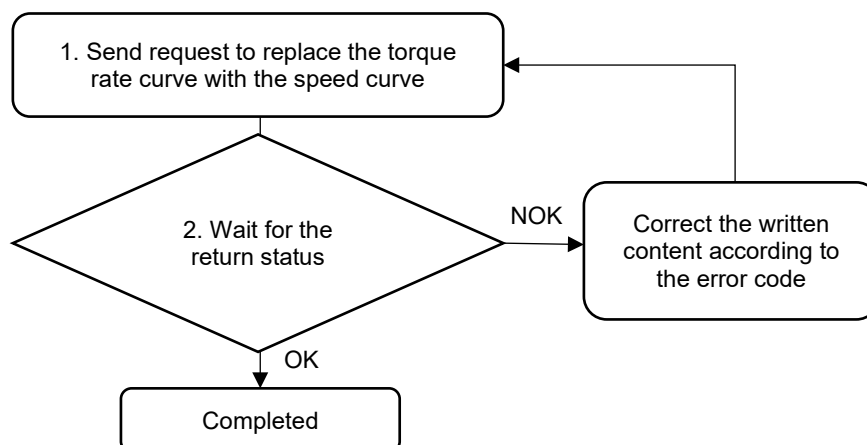
B

#532 Write to replace torque rate curve with speed curve

Content description:

Use function code #532 to show speed curve and torque curve in the **Curve** tab of the **Results** page.

Handshake signal description:



1. Send request to replace the torque rate curve with the speed curve.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	532	1402
3 - 4	Version number	-	0000
5 - 6	Replace torque rate curve with speed curve	0: show torque rate curve and torque curve (default) 1: show speed curve and torque curve	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	532	1402
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Replace torque rate curve with speed curve	0: show torque rate curve and torque curve (default) 1: show speed curve and torque curve	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	532	1402
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

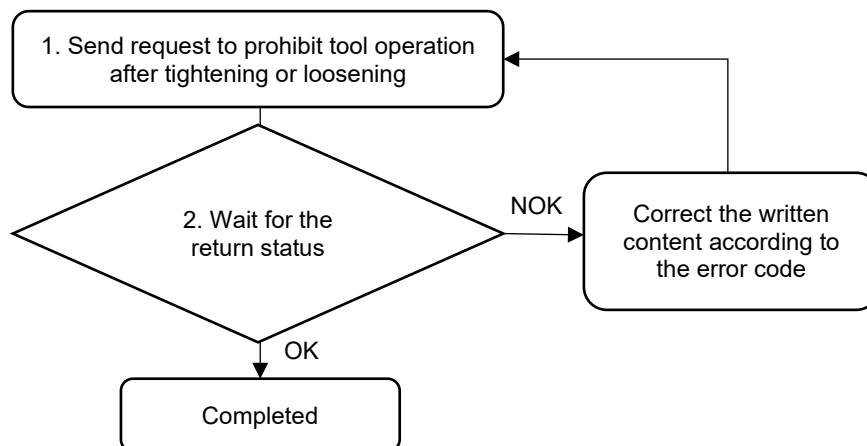
Code	Error description	Description
1	Replace torque rate curve with speed curve	Exceeds the range of 0 - 1

#533 Write to prohibit tool operation after each tightening or loosening

Content description:

Use function code #533 to prohibit tool operation after each tightening or loosening.

Handshake signal description:



1. Send request to prohibit tool operation after tightening or loosening.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	533	1502
3 - 4	Version number	-	0000
5 - 6	Prohibit tool operation after tightening or loosening	0: Off 1: prohibit operation after tightening 2: prohibit operation after loosening 3: prohibit operation after tightening or loosening	0300
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	533	1502
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Prohibit tool operation after tightening or loosening	0: Off 1: prohibit operation after tightening 2: prohibit operation after loosening 3: prohibit operation after tightening or loosening	0300
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	533	1502
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Prohibit tool operation after tightening or loosening	Exceeds the range of 0 - 3

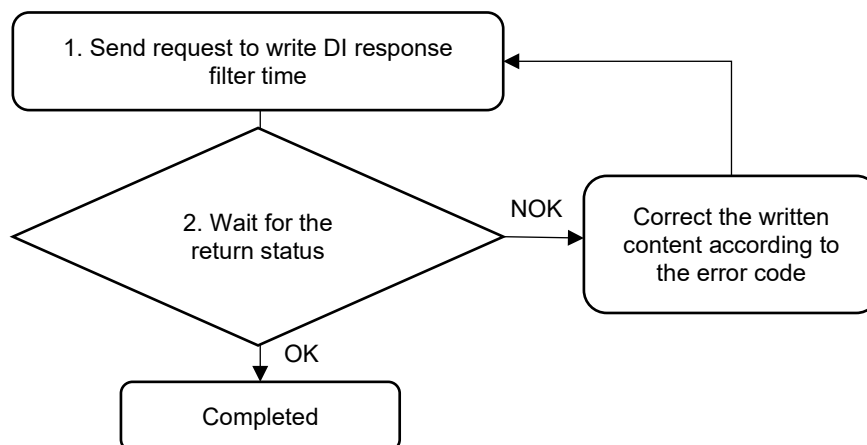
B

#534 Write the DI response filter time

Content description:

Use function code #534 to set the DI response filter time.

Handshake signal description:



1. Send request to write DI response filter time.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	534	1602
3 - 4	Version number	-	0000
5 - 6	DI response filter time	Unit: ms	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	534	1602
3 - 4	Return data length	Unit: Word	0A00
5 - 6	DI response filter time	Unit: ms	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	534	1602
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

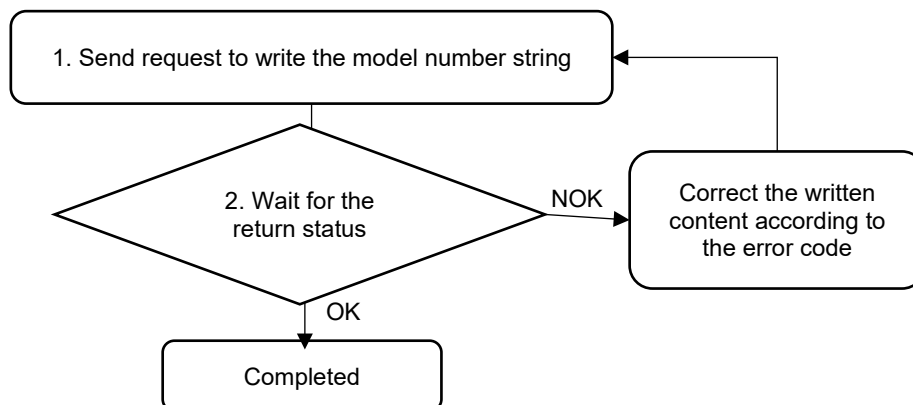
Code	Error description	Description
1	DI response filter time	Exceeds the range of 0 - 1000

#535 Write the model number

Content description:

Use function code #535 to input the model number.

Handshake signal description:



1. Send request to write the model number string.

Byte	Function	Description	Example (Hex)
1 - 2	Function code	535	1702
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100
15 - 20	Reserved	6 bytes	...
21 - 100	Model number string	ASCII code (40 words)	...

2. Check whether the writing is successful from the return status.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	535	1702
3 - 4	Return data length	Unit: Word	0A00
5 - 14	Reserved	10 bytes	...
15 - 16	Return function code	535	1702
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2.

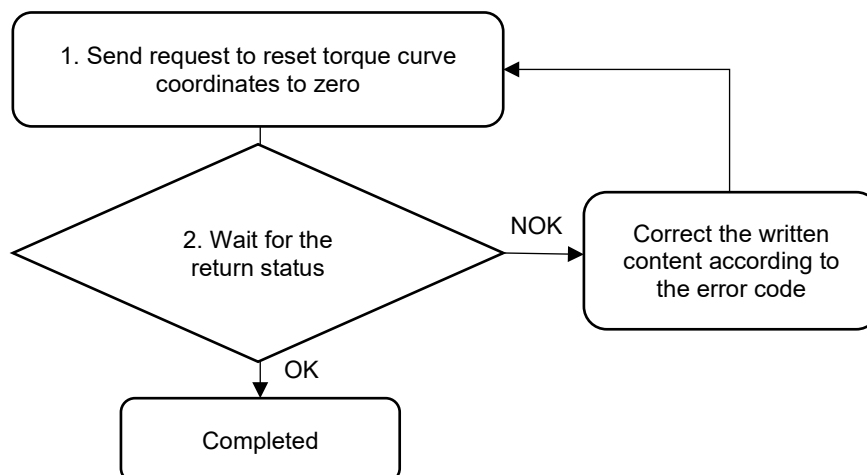
B

#536 Write to reset the torque curve coordinates to zero

Content description:

Use function code #536 to reset the torque curve coordinates to zero.

Handshake signal description:



1. Send request to reset torque curve coordinates to zero.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	536	1802
3 - 4	Version number	-	0000
5 - 6	Reset torque cruve coordinates to zero	0: Off (default) 1: On	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	536	1802
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Reset torque cruve coordinates to zero	0: Off (default) 1: On	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	536	1802
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

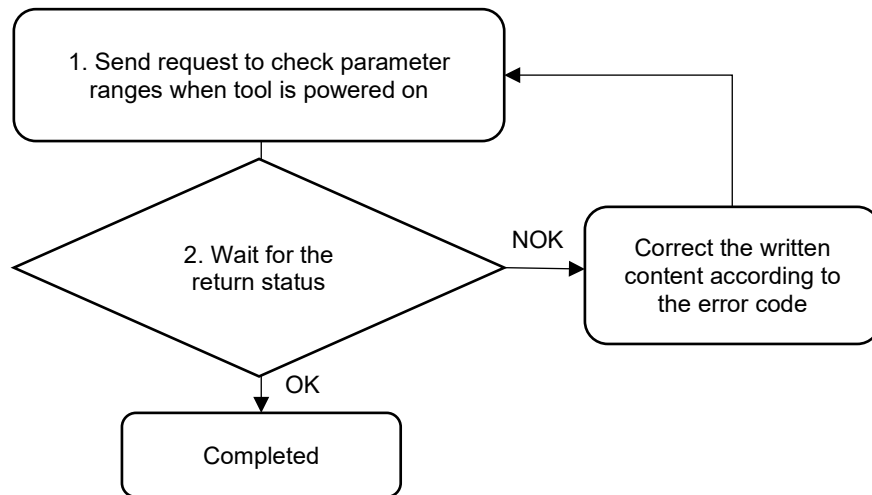
Code	Error description	Description
1	Reset torque curve coordinates to zero	Exceeds the range of 0 - 1

#537 Write to check parameter ranges when tool is powered on

Content description:

Use function code #537 to enable or disable the function of checking tool parameter ranges when the tool is powered on.

Handshake signal description:



1. Send request to check parameter ranges when tool is powered on.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	537	1902
3 - 4	Version number	-	0000
5 - 6	Check parameter ranges when tool is powered on	0: On (default) 1: Off	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	537	1902
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Check parameter ranges when tool is powered on	0: On (default) 1: Off	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	537	1902
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Check parameter ranges when tool is powered on	Exceeds the range of 0 - 1

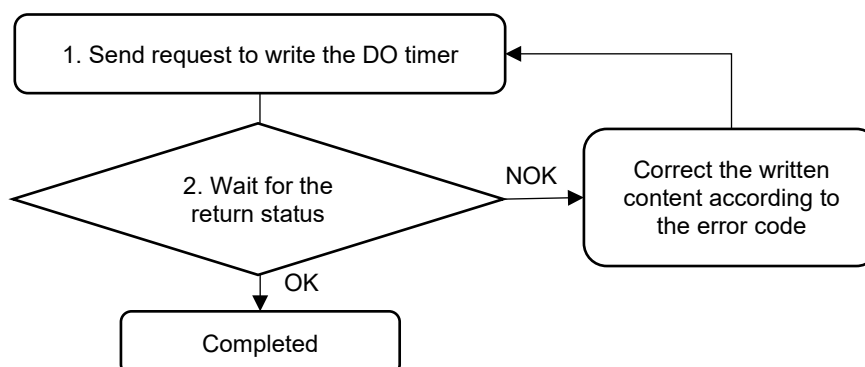
B

#538 Write the DO timer

Content description:

Use function code #538 to enable or disable the DO timer. When the DO is On, the timer starts to count; when the set duration is reached, the DO automatically switches from On to Off.

Handshake signal description:



1. Send request to write the DO timer.

Byte	Function	Description	Example (Hex)
1 - 2	Function code	538	1A02
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100
15 - 20	Reserved	6 bytes	...
21 - 22	DO1 timer	0: Off 1 - 3000: timer duration in units of millisecond	0000
23 - 24	DO2 timer		0000
25 - 26	DO3 timer		0000
27 - 28	DO4 timer		0000
29 - 30	DO5 timer		0000
31 - 32	DO6 timer		0000
33 - 34	DO7 timer		0000
35 - 36	DO8 timer		0000

2. Check whether the writing is successful from the return status.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	538	1A02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	538	1A02
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

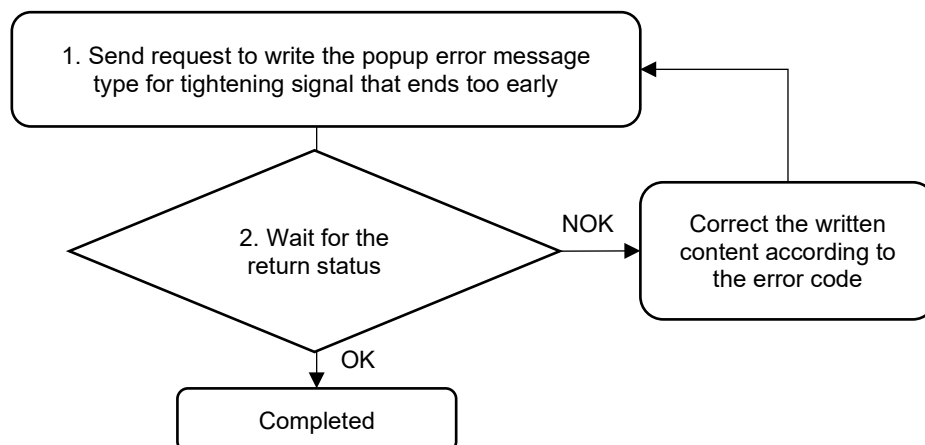
Code	Error description	Description
1	Tool 1 / Tool 2	Exceeds the range of 0 - 1

#539 Write the popup error message type for tightening signal that ends too early

Content description:

Use function code #539 to set the type of the popup error message for the tightening signal that ends too early.

Handshake signal description:



1. Send request to write the popup error message type for tightening signal that ends too early.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	539	1B02
3 - 4	Version number	-	0000
5 - 6	Popup error message type	0: error 1: warning	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	539	1B02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Popup error message type	0: error 1: warning	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	539	1B02
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Popup error message type	Exceeds the range of 0 - 1

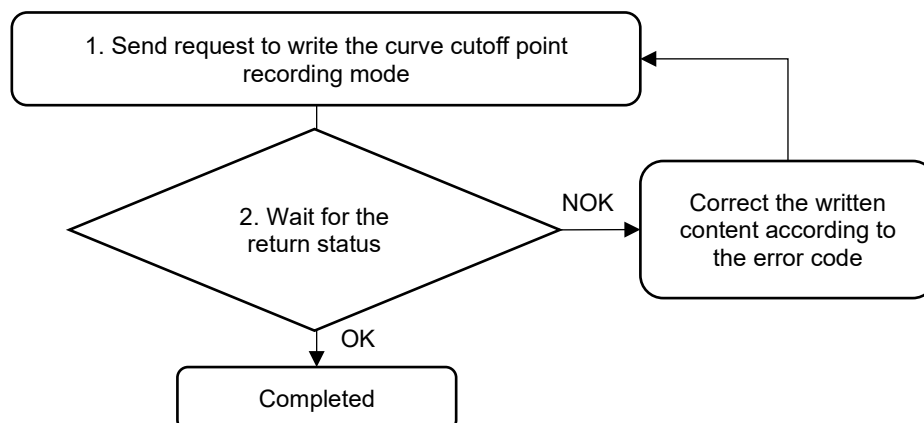
B

#540 Write the curve cutoff point recording mode

Content description:

Use function code #540 to set the curve cutoff point recording mode.

Handshake signal description:



1. Send request to write the curve cutoff point recording mode.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	540	1C02
3 - 4	Version number	-	0000
5 - 6	Curve cutoff point recording mode	0: the entire process 1: peak value	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	540	1C02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Curve cutoff point recording mode	0: the entire process 1: peak value	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	540	1C02
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

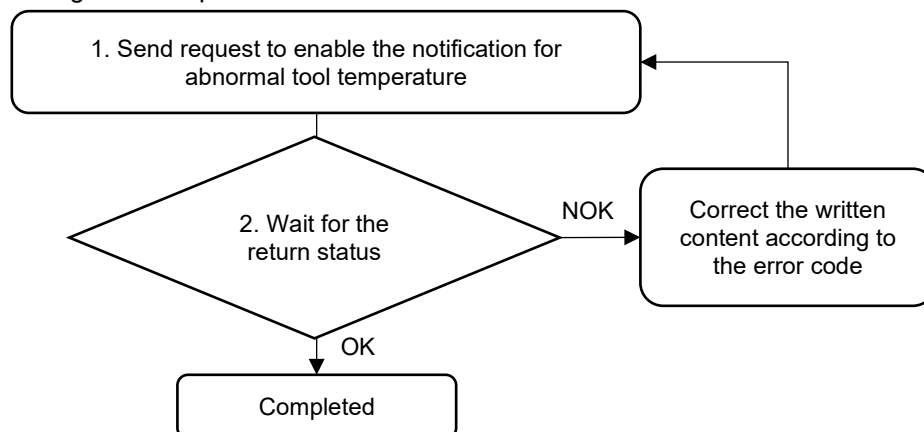
Code	Error description	Description
1	Curve cutoff point recording mode	Exceeds the range of 0 - 1

#541 Write to enable the notification for abnormal tool temperature

Content description:

Use function code #541 to enable or disable the notification for abnormal tool temperature.

Handshake signal description:



1. Send request to enable the notification for abnormal tool temperature.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	541	1D02
3 - 4	Version number	-	0000
5 - 6	Notification for abnormal tool temperature	0: On (default) 1: Off	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	541	1D02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Notification for abnormal tool temperature	0: On (default) 1: Off	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	541	1D02
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Notification for abnormal tool temperature	Exceeds the range of 0 - 1

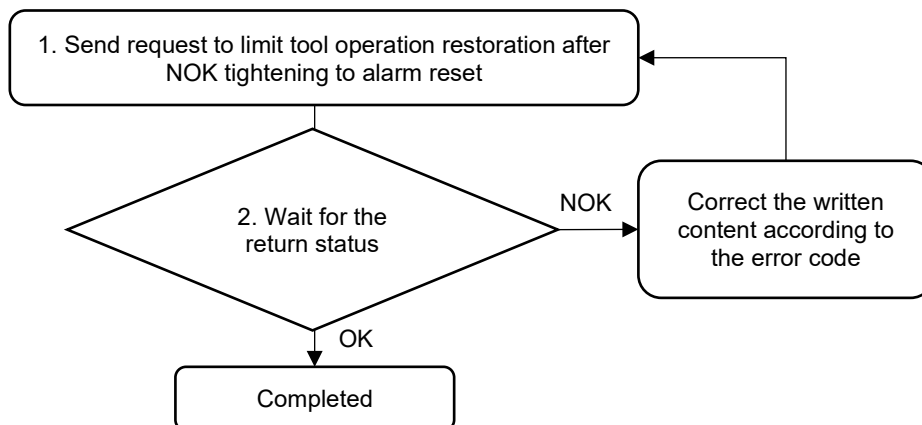
B

#542 Write to limit tool operation restoration after NOK tightening to alarm reset

Content description:

Use function code #542 to set whether the tool operation after NOK tightening can be restored by alarm reset.

Handshake signal description:



1. Send request to limit tool operation restoration after NOK tightening to alarm reset.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	542	1E02
3 - 4	Version number	-	0000
5 - 6	Limit the tool operation restoration after NOK tightening to alarm reset	0: Off 1: On	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	542	1E02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Limit the tool operation restoration after NOK tightening to alarm reset	0: Off 1: On	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	542	1E02
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

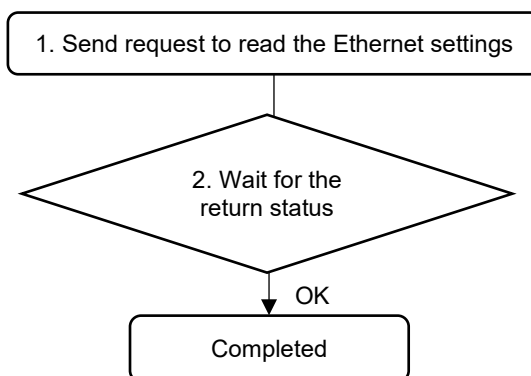
Code	Error description	Description
1	Limit the tool operation restoration after NOK tightening to alarm reset	Exceeds the range of 0 - 1

#550 Read the Ethernet settings

Content description:

Use function code #550 to read the Ethernet settings.

Handshake signal description:



1. Send the request to read the Ethernet settings.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	550	2602
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the Ethernet settings.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	550	2602
3 - 4	Return data length	Unit: Word	1200
5 - 14	Reserved	10 bytes	...
15 - 16	Return function code	550	2602
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000
21 - 22	IP1 address	Default: 192	C000
23 - 24	IP2 address	Default: 168	A800
25 - 26	IP3 address	Default: 1	0100
27 - 28	IP4 address	Default: 11	0B00
29 - 30	Subnet mask 1	Default: 255	FF00
31 - 32	Subnet mask 2	Default: 255	FF00
33 - 34	Subnet mask 3	Default: 255	FF00
35 - 36	Subnet mask 4	Default: 0	0000

If successful, the read status is 1.

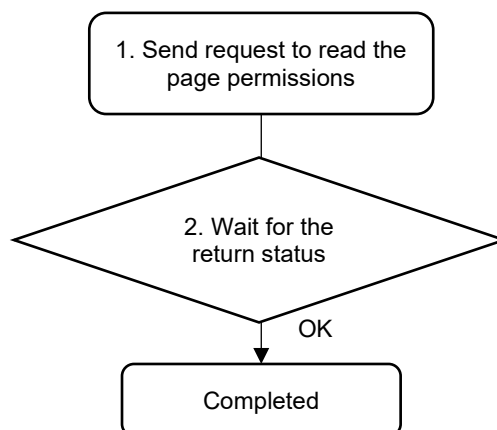
B

#551 Read the page permissions

Content description:

Use function code #551 to read the permissions of page access and edit for different accounts.

Handshake signal description:



1. Send the request to read the page permissions.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	551	2702
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the page permissions.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	551	2702
3 - 4	Return data length	Unit: Word	0F00
5 - 14	Reserved	10 bytes	...
15 - 16	Return function code	551	2702
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000
21 - 22	User1 page permissions	Bit 0: Parameters	00C0
23 - 24	User2 page permissions	Bit 1: Sequence	
25 - 26	User3 page permissions	Bit 2: Sources	
27 - 28	User4 page permissions	Bit 3: Controller	
29 - 30	User5 page permissions	Bit 4: Tool	
		Bit 5: operation for screw progress	00C0
		Bit 6: clearing of the production report entries	00C0
		Bit 7: clearing of the error, warning, and button report entries	00C0

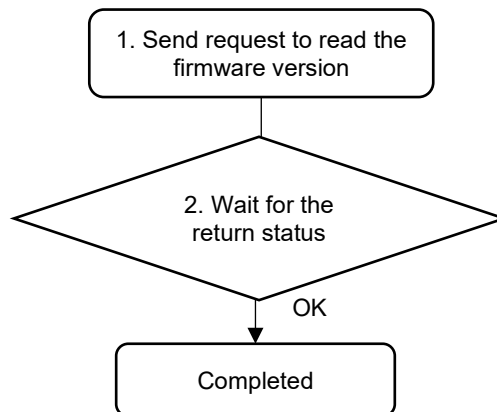
If successful, the read status is 1.

#552 Read the firmware version

Content description:

Use function code #552 to read the firmware version of the controller.

Handshake signal description:



1. Send the request to read the firmware version.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	552	2802
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the controller firmware version.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	552	2802
3 - 4	Return data length	Unit: Word	1400
5 - 14	Reserved	10 bytes	...
15 - 16	Return function code	552	2802
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000
21 - 60	Controller firmware version	ASCII code (40 words)	...

If successful, the read status is 1.

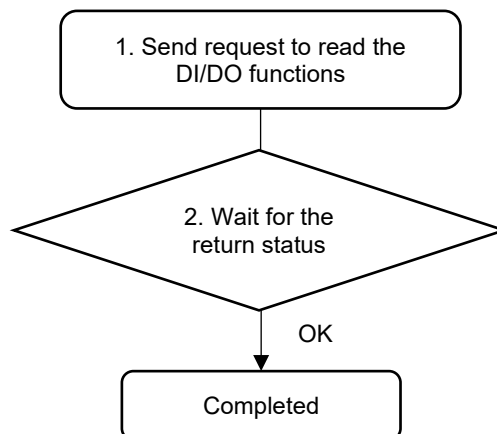
B

#553 Read the DI/DO functions

Content description:

Use function code #553 to read the DI/DO functions.

Handshake signal description:



1. Send the request to read the DI/DO functions.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	553	2902
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the content of the DI/DO functions.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	553	2902
3 - 4	Return data length	Unit: Word	2A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	553	2902
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000
21 - 84	See function code #507 for the Write / Read request table.		

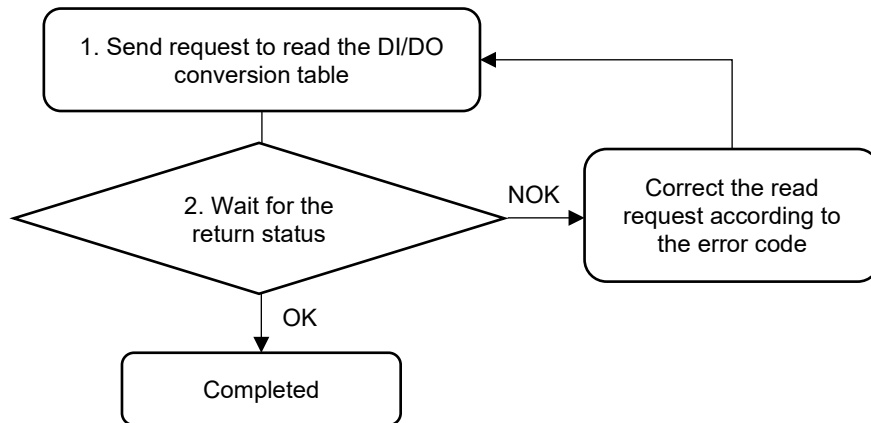
If successful, the read status is 1.

#554 Read the DI/DO conversion table

Content description:

Use function code #554 to read information about the DI/DO conversion table.

Handshake signal description:



1. Send the request to read the DI/DO conversion table.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	554	2A02
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Conversion table	0: bit ID output table 1: bit ID input table 2: parameter ID output table 4: screw progress output table 6: sequence ID output table	0100
9 - 12	Reserved	4 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the content of the DI/DO conversion table.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	554	2A02
3 - 4	Return data length	Unit: Word	0A01
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Conversion table	0: bit ID output table 1: bit ID input table 2: parameter ID output table 4: screw progress output table 6: sequence ID output table	0100
9 - 14	Reserved	6 bytes	...
15 - 16	Return function code	554	2A02
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000
21 - 22	DEC value corresponding to ID 0	0 to 255	0000

B

Byte	Return status	Description	Example (Hex)
23 - 24	DEC value corresponding to ID 1	0 to 255	0100
25 - 26	DEC value corresponding to ID 2	0 to 255	0200
27 - 28	DEC value corresponding to ID 3	0 to 255	0300
29 - 30	DEC value corresponding to ID 4	0 to 255	0400
...	DEC value corresponding to ID N	0 to 255	...
523 - 524	DEC value corresponding to ID 251	0 to 255	FB00
525 - 526	DEC value corresponding to ID 252	0 to 255	FC00
527 - 528	DEC value corresponding to ID 253	0 to 255	FD00
529 - 530	DEC value corresponding to ID 254	0 to 255	FE00
531 - 532	DEC value corresponding to ID 255	0 to 255	FF00

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

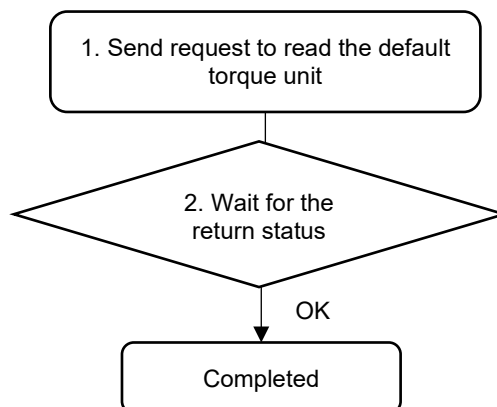
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	Conversion table	Must be 0, 1, 2, 4, or 6

#555 Read the default torque unit

Content description:

Use function code #555 to read the default torque unit of the controller.

Handshake signal description:



1. Send the request to read the default torque unit.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	555	2B02
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the default torque unit.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	555	2B02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Default torque unit	0: N.m 1: kgf.cm 2: lbf.ft 3: lbf.in 4: ozf.ft 5: ozf.in 6: cNm	0100
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	555	2B02
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000

If successful, the read status is 1.

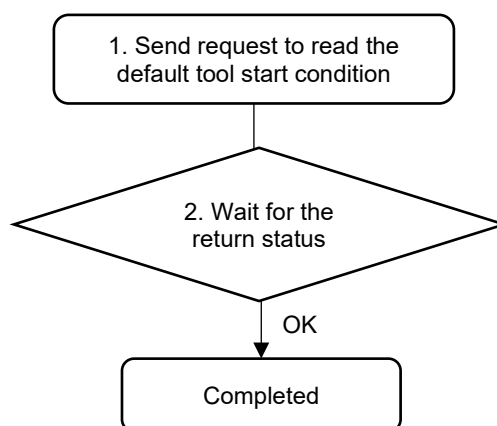
B

#556 Read the default tool start condition

Content description:

Use function code #556 to read the default tool start condition.

Handshake signal description:



1. Send the request to read the default tool start condition.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	556	2C02
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the default tool start condition.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	556	2C02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Default tool start condition	0: push start 1: DI 2: lever start 3: push start or lever start 4: push start and lever start 5: DI or push start 6: DI or lever start	0300
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	556	2C02
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000

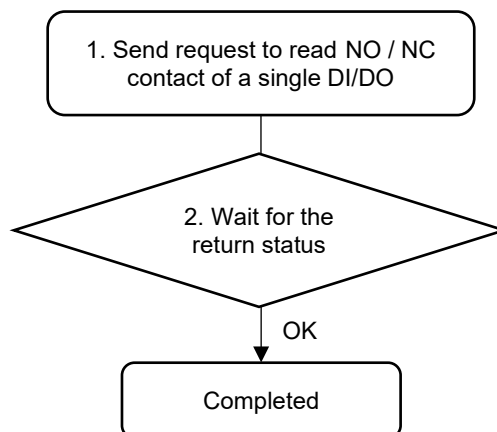
If successful, the read status is 1.

#557 Read NO / NC contact setting for a single DI/DO

Content description:

Use function code #557 to read the NO / NC contact of a single DI/DO.

Handshake signal description:



1. Send the request to read the NO / NC contact of a single DI/DO.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	557	2D02
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	DO / DI	0: DO; 1: DI	0100
9 - 10	Corresponding bit of DO / DI	0 to 7	0000
11 - 12	Reserved	-	0000
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the NO / NC contact of a single DI/DO.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	557	2D02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	DO / DI	0: DO; 1: DI	0100
9 - 10	Corresponding bit of DO / DI	0 to 7	0000
11 - 12	NO / NC	0: NO (normally open) 1: NC (normally closed)	0100
13 - 14	Reserved	-	0000
15 - 16	Return function code	557	2D02
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000

If successful, the read status is 1.

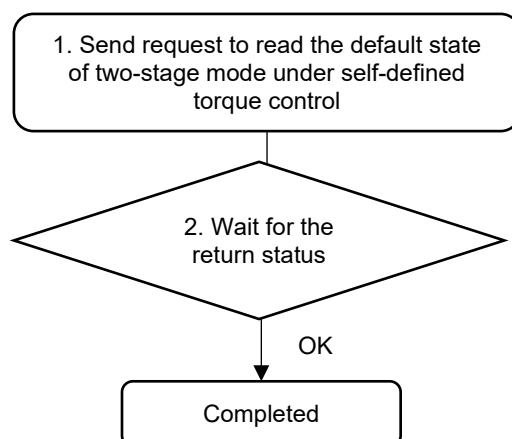
B

#558 Read the default two-stage mode under self-defined torque control

Content description:

Use function code #558 to read the default state of two-stage mode under self-defined torque control.

Handshake signal description:



1. Send the request to read the default state of two-stage mode under self-defined torque control.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	558	2E02
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the default state of two-stage mode under self-defined torque control.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	558	2E02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Two-stage mode	0: Off 1: On	0100
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	558	2E02
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000

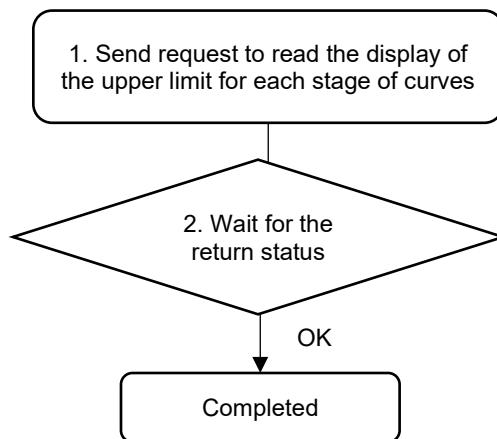
If successful, the read status is 1.

#559 Read the display of upper limit for each stage of curves

Content description:

Use function code #559 to read whether the upper limit for each stage of curves is displayed.

Handshake signal description:



1. Send the request to read the display of the upper limit for each stage of curves.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	559	2F02
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read whether the upper limit for each stage of curves is displayed.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	559	2F02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Display the upper limit for each stage of curves	0: On 1: Off	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	559	2F02
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000

If successful, the read status is 1.

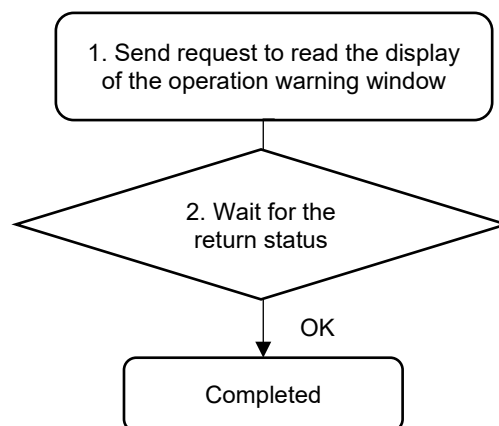
B

#560 Read the display of operation warning window

Content description:

Use function code #560 to read whether the operation warning window is displayed.

Handshake signal description:



1. Send the request to read the display of the operation warning window.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	560	3002
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read whether the operation warning window is displayed.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	560	3002
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Display the operation warning window	0: On 1: Off	0100
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	560	3002
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000

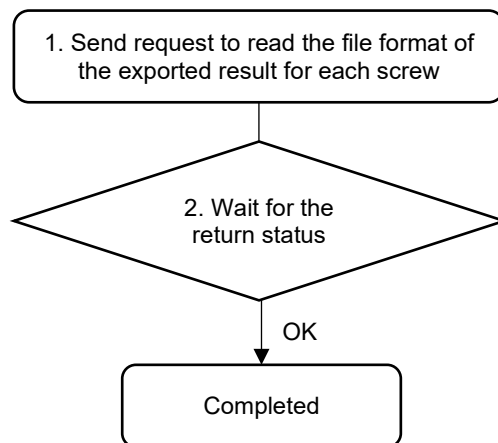
If successful, the read status is 1.

#561 Read file format of the exported result for each screw

Content description:

Use function code #561 to read the file format of the exported result for each screw.

Handshake signal description:



1. Send the request to read the file format of the exported result for each screw.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	561	3102
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the file format of the exported result for each screw.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	561	3102
3 - 4	Return data length	Unit: Word	0A00
5 - 6	File format of the exported result for each screw	0: disable 1: CSV file (HMI Disk) 2: CSV file (USB Disk) 3: BIN file (HMI Disk) 4: BIN file (USB Disk)	0300
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	561	3102
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000

If successful, the read status is 1.

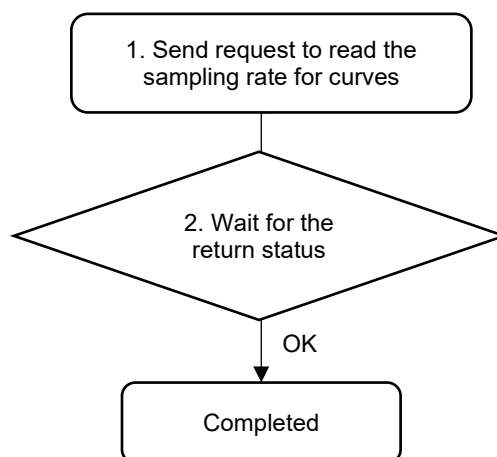
B

#562 Read the sampling rate for curves

Content description:

Use function code #562 to read the sampling rate for curves.

Handshake signal description:



1. Send the request to read the sampling rate for curves.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	562	3202
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the sampling rate for curves.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	562	3202
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Sampling rate for curves	0: 10 ms→4 ms when start torque is reached 1: 2 ms; max. 4 seconds per production report entry 2: 2 ms; max. 8 seconds per production report entry 3: 1 ms; max. 4 seconds per production report entry 4: 1 ms; max. 8 seconds per production report entry	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	562	3202
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000

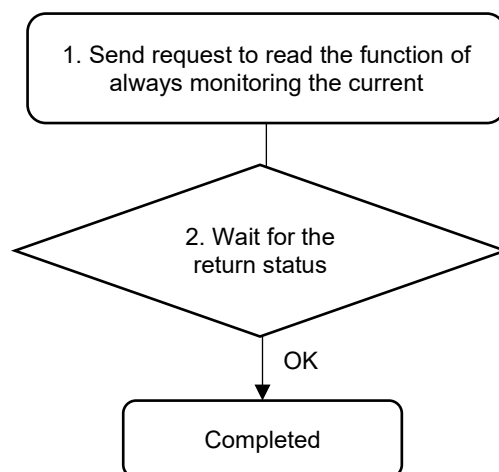
If successful, the read status is 1.

#563 Read the function of always monitoring the current

Content description:

Use function code #563 to read whether the function of always monitoring the current is enabled.

Handshake signal description:



1. Send the request to read the function of always monitoring the current.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	563	3302
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read whether the function of always monitoring the current is enabled.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	563	3302
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Always monitor the current	0: On 1: Off	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	563	3302
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000

If successful, the read status is 1.

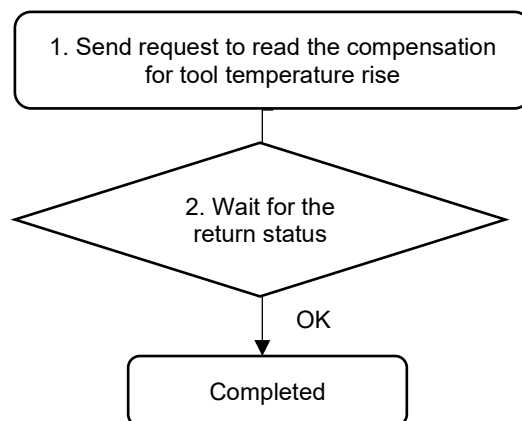
B

#564 Read the compensation for tool temperature rise

Content description:

Use function code #564 to read whether the compensation for tool temperature rise is enabled.

Handshake signal description:



1. Send the request to read the compensation for tool temperature rise.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	564	3402
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read whether the compensation for tool temperature rise is enabled.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	564	3402
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Compensation for tool temperature rise	0: On 1: Off	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	564	3402
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000

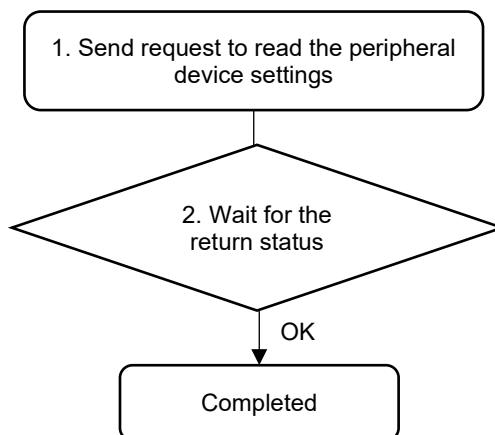
If successful, the read status is 1.

#565 Read the peripheral device settings

Content description:

Use function code #565 to read the peripheral device settings.

Handshake signal description:



1. Send the request to read the peripheral device settings.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	565	3502
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the peripheral device settings.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	565	3502
3 - 4	Return data length	Unit: Word	1D00
5 - 14	Reserved	10 bytes	...
15 - 16	Return function code	565	3502
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000
21 - 58	Refer to function code #521 for the Write / Read request table.		

If successful, the read status is 1.

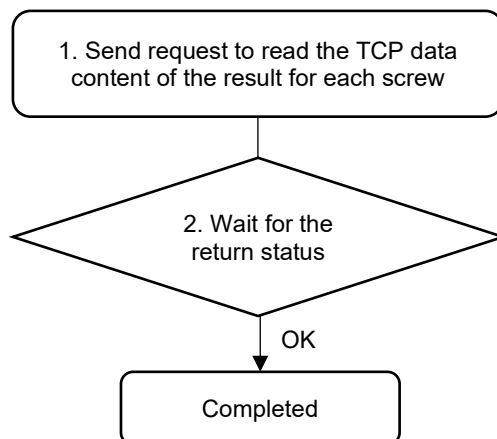
B

#566 Read the TCP data content of the result for each screw

Content description:

Use function code #566 to read the TCP data content of the result for each screw.

Handshake signal description:



1. Send the request to read the TCP data content of the result for each screw.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	566	3602
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the TCP data content of the result for each screw.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	566	3602
3 - 4	Return data length	Unit: Word	0A00
5 - 6	TCP data content of the result for each screw	0: disable 1: scale, time, torque 2: scale, angle, torque 3: scale, angle, torque rate 4: result, scale, time, torque 5: result, scale, angle, torque 6: result, scale, angle, torque rate 7 - 11: reserved 12: result, scale (tightening or loosening) 13: result, scale (tightening) 14: result, scale, all curve data (tightening or loosening) 15: result, scale, all curve data (tightening)	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	566	3602
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000

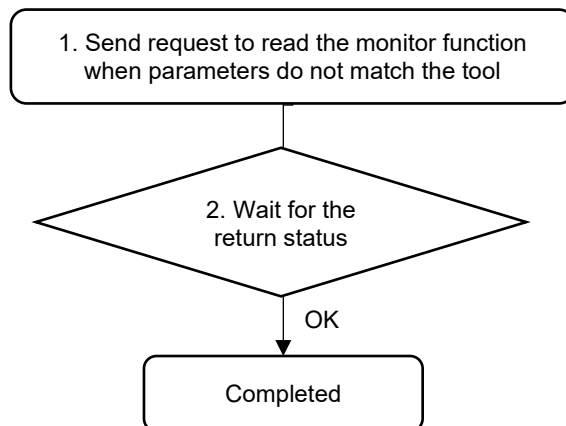
If successful, the read status is 1.

#567 Read the monitor function when parameters do not match the tool

Content description:

Use function code #567 to read the function setting for monitoring if the tool torque of the operating parameter matches that of the connected tool.

Handshake signal description:



1. Send the request to read the monitor function when parameters do not match the tool.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	567	3702
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the monitor function setting.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	567	3702
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Monitor function when parameters do not match the tool	0: On 1: Off	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	567	3702
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000

If successful, the read status is 1.

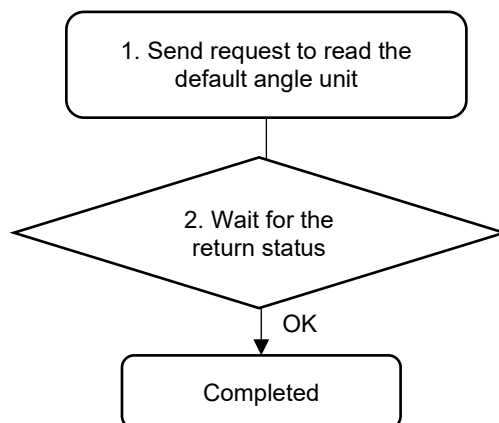
B

#568 Read the default angle unit

Content description:

Use function code #568 to read the default angle unit of the controller.

Handshake signal description:



1. Send the request to read the default angle unit.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	568	3802
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the default angle unit.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	568	3802
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Default angle unit	0: degree 1: turn	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	568	3802
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000

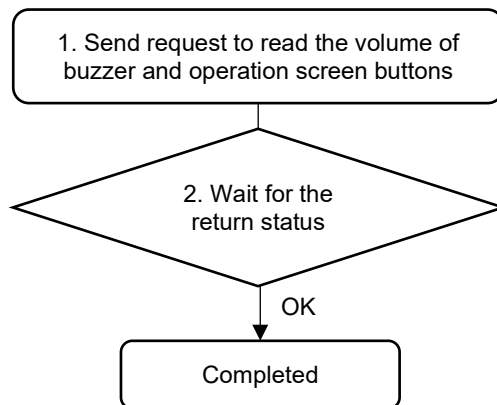
If successful, the read status is 1.

#569 Read the volume of buzzer and operation screen buttons

Content description:

Use function code #569 to read the volume of buzzer and operation screen buttons.

Handshake signal description:



1. Send the request to read the volume of buzzer and operation screen buttons.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	569	3902
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the volume of buzzer and operation screen buttons.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	569	3902
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Volume of buzzer	0 to 100	0000
7 - 8	Volume of operation screen buttons	0 to 100	0000
9 - 14	Reserved	6 bytes	...
15 - 16	Return function code	569	3902
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000

If successful, the read status is 1.

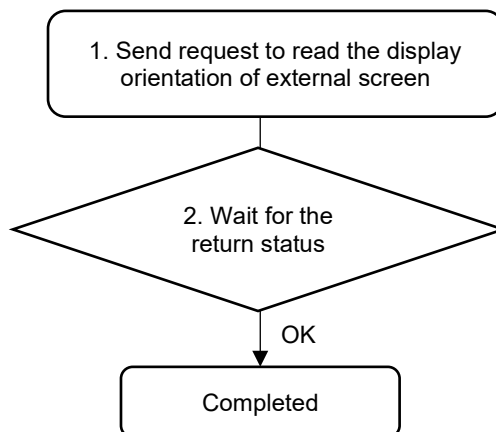
B

#570 Read the display orientation of external screen

Content description:

Use function code #570 to read the display orientation of external screen (HDMI).

Handshake signal description:



1. Send the request to read the display orientation of external screen.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	570	3A02
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the display orientation of external screen.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	570	3A02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Display orientation	0: portrait 1: landscape	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	570	3A02
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000

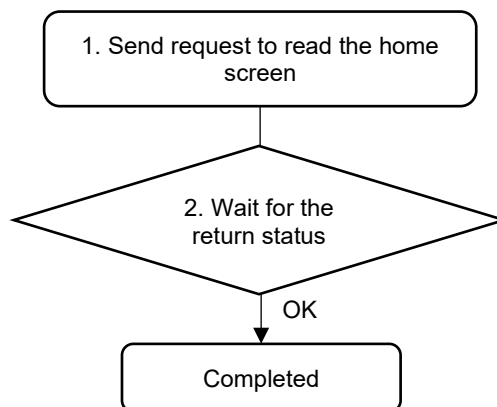
If successful, the read status is 1.

#571 Read the home screen

Content description:

Use function code #571 to read the home screen of the controller.

Handshake signal description:



1. Send the request to read the home screen.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	571	3B02
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the home screen.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	571	3B02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Home screen	0: Home 1: Results	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	571	3B02
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000

If successful, the read status is 1.

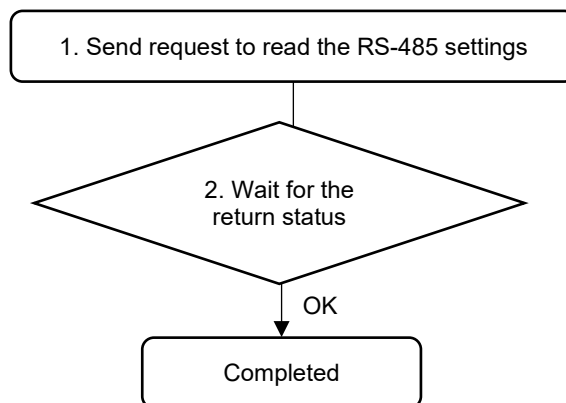
B

#572 Read the RS-485 settings

Content description:

Use function code #572 to read the RS-485 settings.

Handshake signal description:



1. Send the request to read the RS-485 settings.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	572	3C02
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the RS-485 settings.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	572	3C02
3 - 4	Return data length	Unit: Word	0A00
5 - 14	Reserved	10 bytes	...
15 - 16	Return function code	572	3C02
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000
21 - 22	Function On / Off	0: Off; 1: On	0100
23 - 24	Station	0 to 255	0100
25 - 26	RTU / ASCII	0: RTU 1: ASCII	0000
27 - 28	Baud rate	0: 9600 1: 19200 2: 38400	0000
29 - 30	Data bit	0: 8-bit 1: 7-bit	0000
31 - 32	Parity bit	0: None 1: Odd 2: Even	0000
33 - 34	Stop bit	0: 2-bit 1: 1-bit	0000

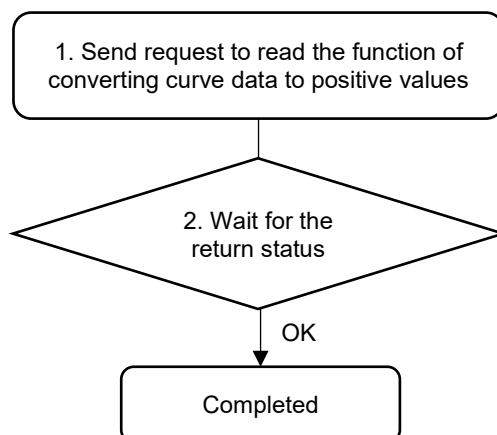
If successful, the read status is 1.

#573 Read the function of converting curve data to positive values

Content description:

Use function code #573 to read whether the function of converting curve data to positive values is enabled.

Handshake signal description:



1. Send the request to read the function of converting curve data to positive values.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	573	3D02
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read whether the function of converting curve data to positive values is enabled.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	573	3D02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Function of converting curve data to positive values	0: On 1: Off	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	573	3D02
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000

If successful, the read status is 1.

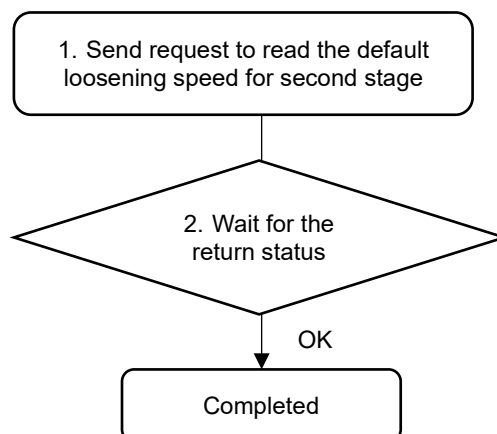
B

#574 Read the second stage default loosening speed

Content description:

Use function code #574 to read the default loosening speed for the second stage.

Handshake signal description:



1. Send request to read the default loosening speed for second stage.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	574	3E02
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	574	3E02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Default loosening speed for the second stage	Unit: rpm	6400
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	574	3E02
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000

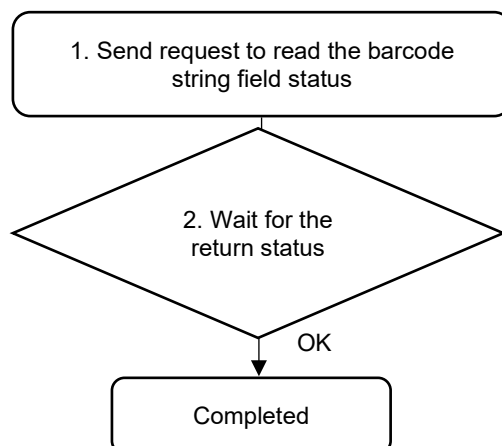
If successful, the read status is 1.

#575 Read the keyboard cursor blink in Results page

Content description:

Use function code #575 to read whether the barcode string field is set as the initial focus in the **Results** page.

Handshake signal description:



1. Send request to read the barcode string field status.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	575	3F02
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read whether the barcode string field is set as initial focus in **Results** page.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	575	3F02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Set the barcode string field as initial focus in Results page	0: Off 1: On	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	575	3F02
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000

If successful, the read status is 1.

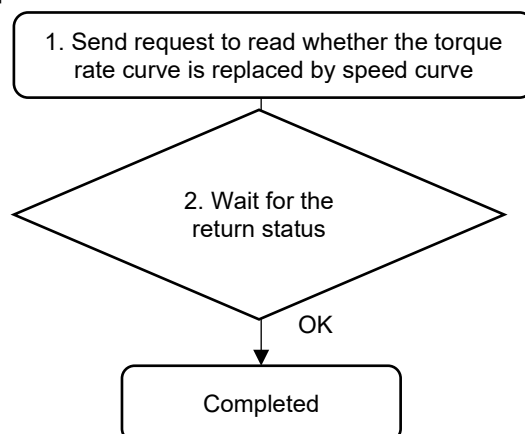
B

#576 Read whether the torque rate curve is replaced by the speed curve

Content description:

Use function code #576 to read whether the torque rate curve is replaced by the speed curve.

Handshake signal description:



1. Send request to read whether the torque rate curve is replaced by speed curve.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	576	4002
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read whether the torque rate curve is replaced by the speed curve.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	576	4002
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Whether the torque rate curve is replaced by speed curve	0: torque rate curve and torque curve are shown (default) 1: speed curve and torque curve are shown	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	576	4002
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000

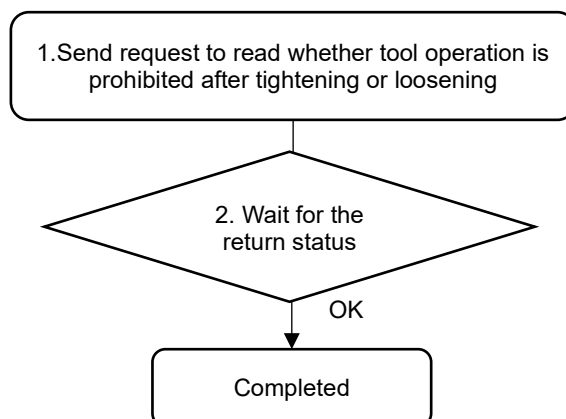
If successful, the read status is 1.

#577 Read whether tool operation is prohibited after each tightening or loosening

Content description:

Use function code #577 to read whether the tool operation is prohibited after each tightening or loosening.

Handshake signal description:



1. Send request to read whether tool operation is prohibited after tightening or loosening.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	577	4102
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read whether tool operation is prohibited after each tightening or loosening.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	577	4102
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Whether tool operation is prohibited after each tightening or loosening	0: Off 1: tool operation prohibited after tightening 2: tool operation prohibited after loosening 3: tool operation prohibited after tightening or loosening	0300
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	577	4102
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000

If successful, the read status is 1.

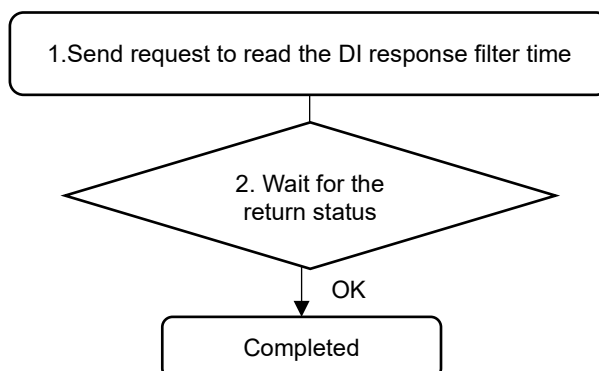
B

#578 Read the DI response filter time

Content description:

Use function code #578 to read the DI response filter time. Using this function helps mitigate the impact of noise on program execution.

Handshake signal description:



1. Send request to read the DI response filter time.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	578	4202
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the DI response filter time.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	578	4202
3 - 4	Return data length	Unit: Word	0A00
5 - 6	DI response filter time	0 - 1000	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	578	4202
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000

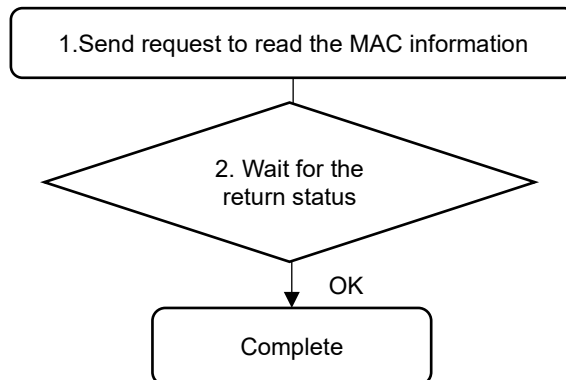
If successful, the read status is 1.

#579 Read the MAC information

Content description:

Use function code #579 to read the MAC information.

Handshake signal description:



1. Send request to read the MAC information.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	579	4302
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the MAC information.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	579	4302
3 - 4	Return data length	Unit: Word	1000
5 - 14	Reserved	10 bytes	...
15 - 16	Return function code	579	4302
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000
21 - 22	MAC1	0 - 255	0000
23 - 24	MAC2	0 - 255	1800
25 - 26	MAC3	0 - 255	2300
27 - 28	MAC4	0 - 255	EB00
29 - 30	MAC5	0 - 255	F100
31 - 32	MAC6	0 - 255	2A00

If successful, the read status is 1.

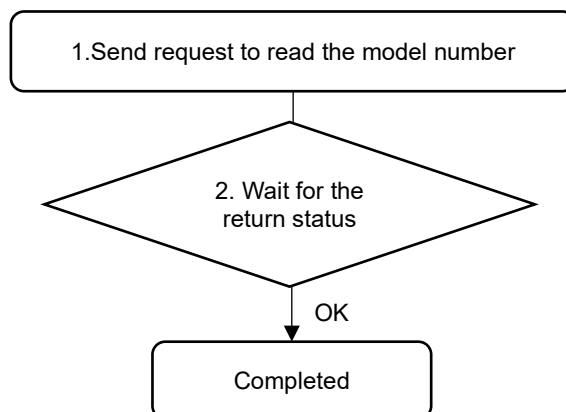
B

#580 Read the model number

Content description:

Use function code #580 to read the model number.

Handshake signal description:



1. Send request to read the model number.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	580	4402
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the model number string.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	580	4402
3 - 4	Return data length	Unit: Word	3200
5 - 14	Reserved	10 bytes	...
15 - 16	Return function code	580	4402
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000
20 - 100	Model number string	ASCII code (40 words)	0000

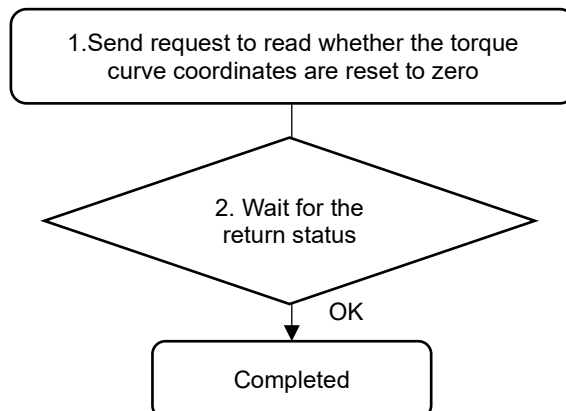
If successful, the read status is 1.

#581 Read whether the torque curve coordinates are reset to zero

Content description:

Use function code #581 to read whether the torque curve coordinates are reset to zero.

Handshake signal description:



1. Send request to read whether the torque curve coordinates are reset to zero.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	581	4502
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the status of torque curve coordinates.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	581	4502
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Reset torque curve coordinates to zero	0: Off (default) 1: On	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	581	4502
17 - 18	Status	1: OK	0100
19 - 20	Reserved	—	0000

If successful, the read status is 1.

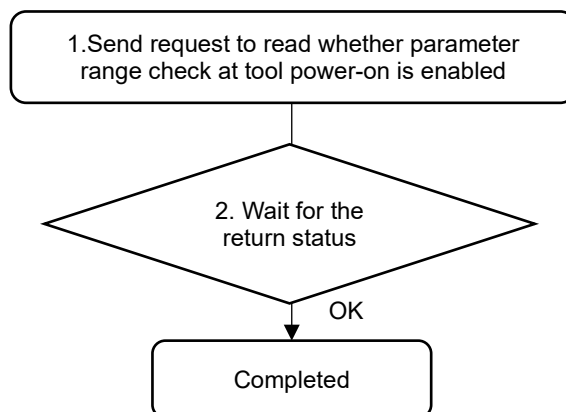
B

#582 Read whether parameter range check at tool power-on is enabled

Content description:

Use function code #582 to read whether parameter range check at tool power-on is enabled.

Handshake signal description:



1. Send request to read whether parameter range check at tool power-on is enabled.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	582	4602
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read whether parameter change check at tool power-on is enabled.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	582	4602
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Parameter range check at tool power-on	0: On (default) 1: Off	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	582	4602
17 - 18	Status	1: OK	0100
19 - 20	Reserved	—	0000

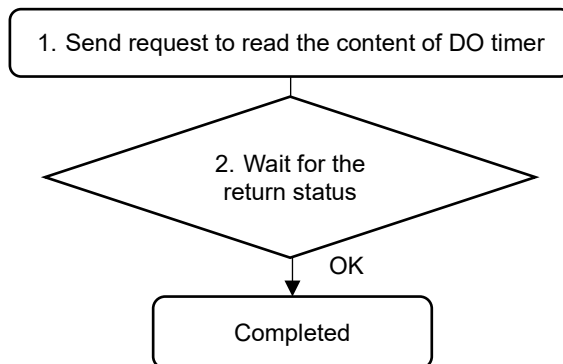
If successful, the read status is 1.

#583 Read the DO timer

Content description:

Use function code #583 to read the content of DO timer.

Handshake signal description:



1. Send request to read the content of DO timer.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	583	4702
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the content of DO timer.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	583	4702
3 - 4		Unit: Word	1200
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	583	4702
17 - 18	Status	1: OK	0100
19 - 20	Reserved	—	0000
21 - 22	DO1 timer	0: Off 1 - 3000: timer duration in units of millisecond	0000
23 - 24	DO2 timer		0000
25 - 26	DO3 timer		0000
27 - 28	DO4 timer		0000
29 - 30	DO5 timer		0000
31 - 32	DO6 timer		0000
33 - 34	DO7 timer		0000
35 - 36	DO8 timer		0000

If successful, the read status is 1.

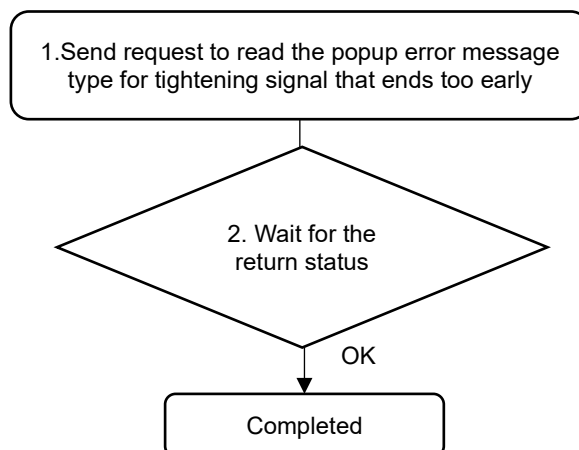
B

#584 Read the popup error message type for tightening signal that ends too early

Content description:

Use function code #584 to read the type of the popup error message for the tightening signal that ends too early.

Handshake signal description:



1. Send request to read the popup error message type for tightening signal that ends too early.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	584	4802
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the popup error message type.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	584	4802
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Popup error message type	0: error 1: warning	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	584	4802
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000

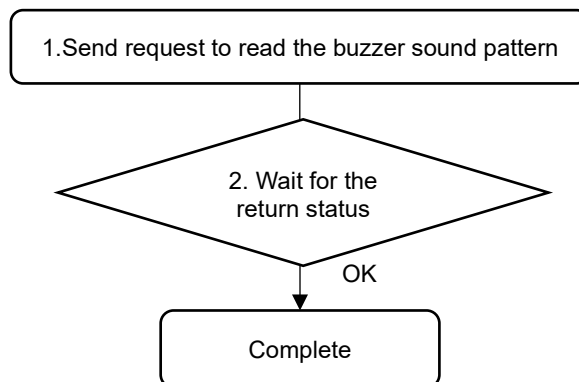
If successful, the read status is 1.

#585 Read the buzzer sound pattern

Content description:

Use function code #585 to read the buzzer sound pattern.

Handshake signal description:



1. Send request to read the buzzer sound pattern.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	585	4902
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the buzzer sound pattern.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	585	4902
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Sound pattern of error signal buzzer	0: mute 1: one long beep 2: one short beep 3: two short beeps	0000
7 - 8	Sound pattern of finished signal buzzer for a single screw		0000
9 - 10	Sound pattern of finished signal buzzer for total screws		0000
11 - 14	Reserved	4 bytes	...
15 - 16	Return function code	585	4902
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000

If successful, the read status is 1.

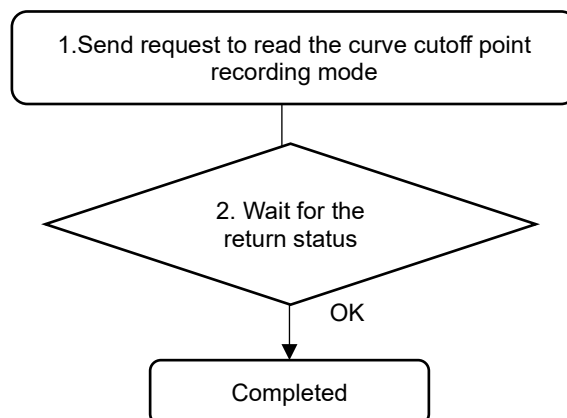
B

#586 Read the curve cutoff point recording mode

Content description:

Use function code #585 to read the curve cutoff point recording mode.

Handshake signal description:



1. Send request to read the curve cutoff point recording mode.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	586	4A02
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the curve cutoff point recording mode.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	586	4A02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Curve cutoff point recording mode	0: entire process 1: peak value	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	586	4A02
17 - 18	Status	1: OK	0100
19 - 20	Reserved	—	0000

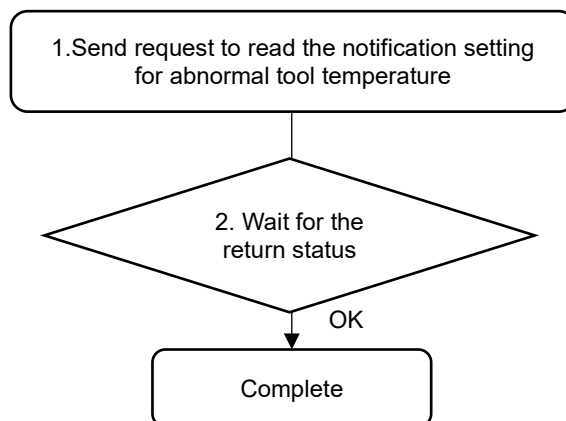
If successful, the read status is 1.

#587 Read the notification setting for abnormal tool temperature

Content description:

Use function code #587 to read whether the notification for abnormal tool temperature is enabled.

Handshake signal description:



1. Send request to read the notification setting for abnormal tool temperature.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	587	4B02
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read whether the notification for abnormal tool temperature is enabled.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	587	4B02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Notification for abnormal tool temperature	0: On (default) 1: Off	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	587	4B02
17 - 18	Status	1: OK	0100
19 - 20	Reserved	—	0000

If successful, the read status is 1.

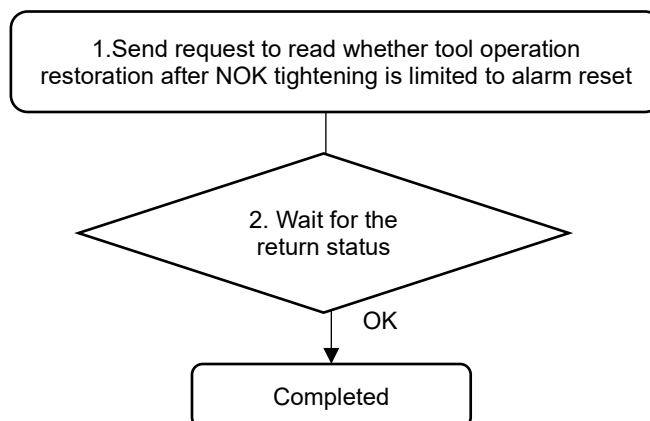
B

#588 Read whether tool operation restoration after NOK tightening is limited to alarm reset

Content description:

Use function code #588 to read if tool operation restoration after NOK tightening is limited to alarm reset.

Handshake signal description:



1. Send request to read whether tool operation restoration after NOK tightening is limited to alarm reset.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	588	4C02
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read whether tool operation restoration after NOK tightening is limited to alarm reset.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	588	4C02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Limit tool operation restoration after NOK tightening to alarm reset	0: Off 1: On	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	588	4C02
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000

If successful, the read status is 1.

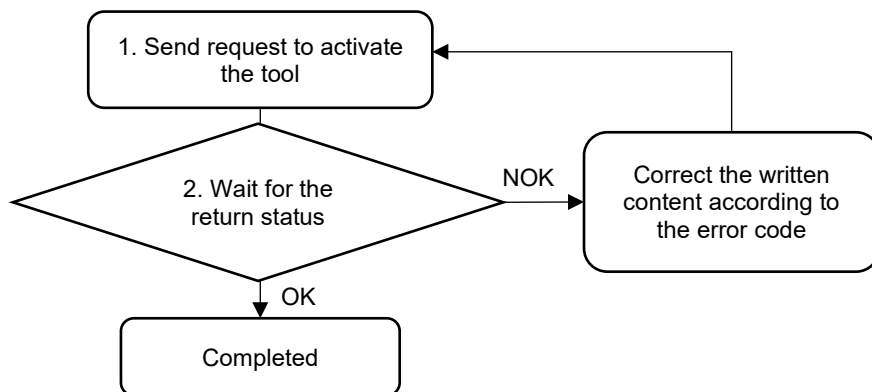
B.3.6 Tool operations

#600 Write to activate the tool

Content description:

Use function code #600 to activate Tool 1 or Tool 2.

Handshake signal description:



1. Send the request to activate the tool.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	600	5802
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Tool activation	0: Off; 1: On	0100
9 - 12	Reserved	4 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	600	5802
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Tool activation	0: Off; 1: On	0100
9 - 14	Reserved	6 bytes	...
15 - 16	Return function code	600	5802
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	Tool activation	Must be 0 or 1

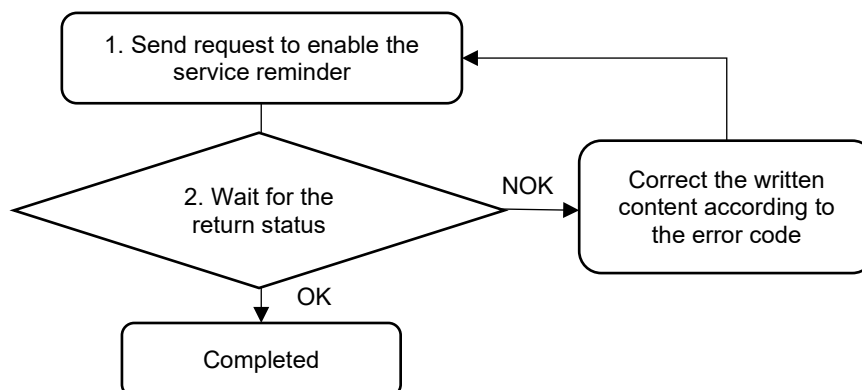
B

#601 Write to enable service reminder

Content description:

Use function code #601 to turn on or off the tool service reminder.

Handshake signal description:



1. Send the request to enable the service reminder.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	601	5902
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Service reminder	0: Off; 1: On	0100
9 - 12	Reserved	4 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	601	5902
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Service reminder	0: Off; 1: On	0100
9 - 14	Reserved	6 bytes	...
15 - 16	Return function code	601	5902
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

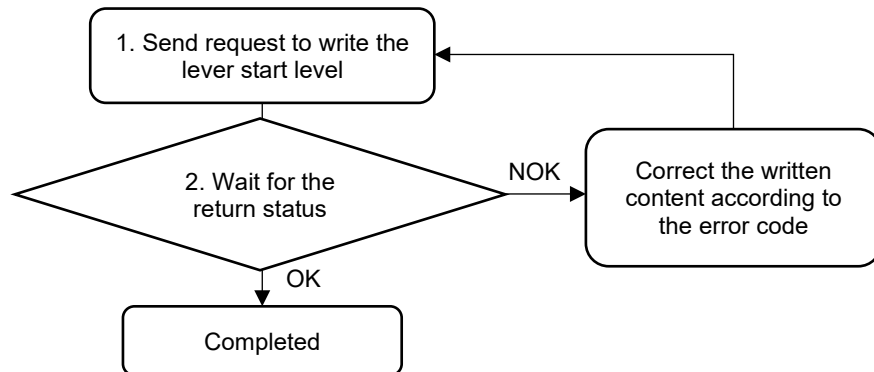
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	Service reminder	Must be 0 or 1

#602 Write the lever start level

Content description:

Use function code #602 to set the lever start level of the tool.

Handshake signal description:



1. Send the request to write the lever start level.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	602	5A02
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Threshold for On level	0 to 4095	DC05
9 - 10	Threshold for Off level	0 to 4095	4006
11 - 12	Reserved	-	0000
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	602	5A02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Threshold for On level	0 to 4095	DC05
9 - 10	Threshold for Off level	0 to 4095	4006
11 - 14	Reserved	4 bytes	...
15 - 16	Return function code	602	5A02
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	Threshold value	Exceeds the range of 0 to 4095

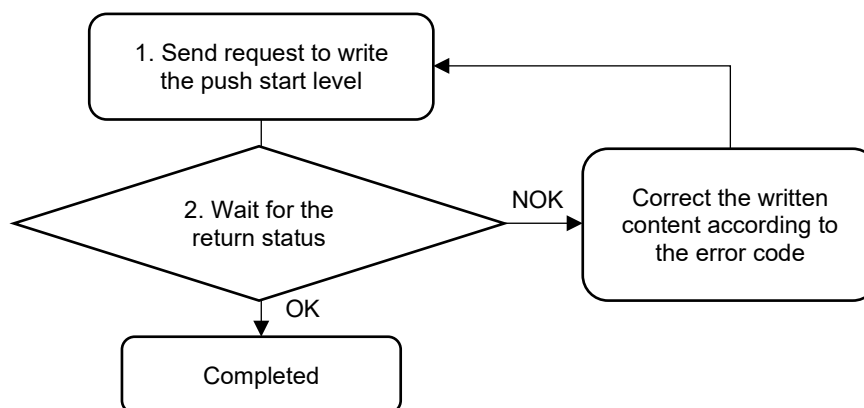
B

#603 Write the push start level

Content description:

Use function code #603 to set the push start level of the tool.

Handshake signal description:



1. Send the request to write the push start level.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	603	5B02
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Threshold for On level	0 to 4095	4C04
9 - 10	Threshold for Off level	0 to 4095	B004
11 - 12	Reserved	-	0000
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	603	5B02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Threshold for On level	0 to 4095	4C04
9 - 10	Threshold for Off level	0 to 4095	B004
11 - 14	Reserved	4 bytes	...
15 - 16	Return function code	603	5B02
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

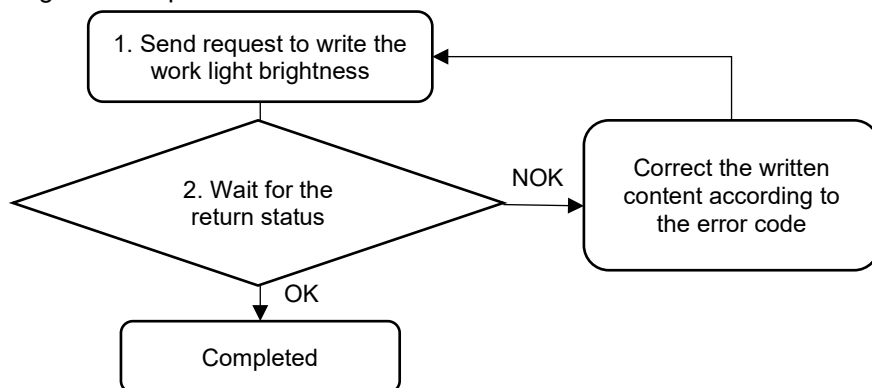
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	Threshold value	Exceeds the range of 0 to 4095

#604 Write the work light brightness

Content description:

Use function code #604 to set the work light brightness of the tool.

Handshake signal description:



1. Send the request to write the work light brightness.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	604	5C02
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Brightness	0 to 50 (0: Off; 50: max. brightness)	3200
9 - 12	Reserved	4 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	604	5C02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Brightness	0 to 50 (0: Off; 50: max. brightness)	3200
9 - 14	Reserved	6 bytes	...
15 - 16	Return function code	604	5C02
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	Brightness	Exceeds the range of 0 to 50

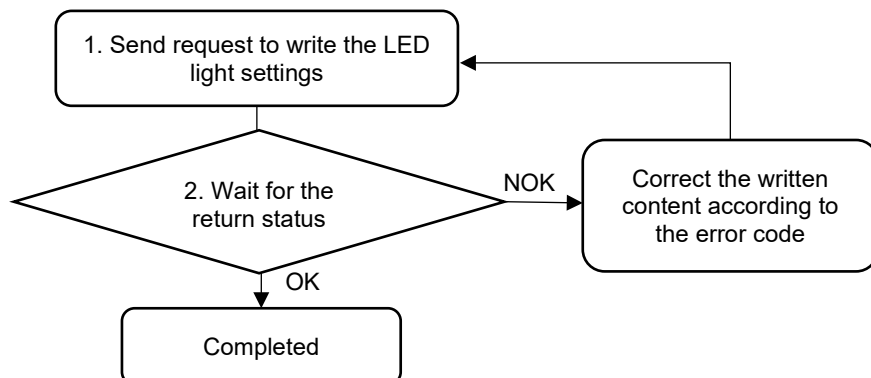
B

#606 Write the LED light settings

Content description:

Use function code #606 to set the LED light settings of the tool.

Handshake signal description:



1. Send the request to write the LED light settings.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	606	5E02
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100
15 - 20	Reserved	6 bytes	...
21 - 22	NO / NC	0: NO (normally open) 1: NC (normally closed) (Bit 1 - 8: for Tool 1; Bit 9 - 16: for Tool 2)	0000
23 - 24	Status of red indicator	0: not in use 1: ready 2: running	0400
25 - 26	Status of yellow indicator	3: tightening OK and report is being saved 4: tightening NOK and report is being saved	0100
27 - 28	Status of green indicator	5: loosening OK and report is being saved 6: loosening NOK and report is being saved 7: total screw quantity reached	0300

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	606	5E02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	606	5E02
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

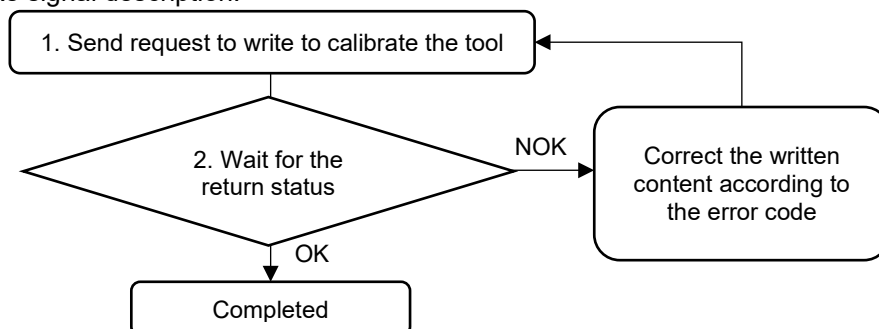
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

#607 Write to calibrate the tool

Content description:

Use function code #607 to set the precision of the tool.

Handshake signal description:



1. Send the request to write to calibrate the tool.

■ Mode 0:

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	607	5F02
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Precision	0 to 32767 (Default: 1400 for 12 kgf-cm models 3500 for 30 kgf-cm models 6000 for 50 kgf-cm models 5705 for 75 kgf-cm models)	0578
9 - 10	Reserved	-	0000
11 - 12	Mode	0	0000
13 - 14	Request to send command	-	0100

■ Mode 11:

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	607	5F02
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Final torque in Results page	Unit: 0.001 x N.m	E803
9 - 10	Torque value calibrated by third-party device	Unit: 0.001 x N.m	E903
11 - 12	Mode	11	0B00
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

■ Mode 0:

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	607	5F02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Precision	0 to 32767 (Default: 1400 for 12 kgf-cm models 3500 for 30 kgf-cm models 6000 for 50 kgf-cm models 5705 for 75 kgf-cm models)	0578
9 - 10	Reserved	-	0000
11 - 12	Mode	0	0000
13 - 14	Reserved	-	...
15 - 16	Return function code	607	5F02
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

■ Mode 11:

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	607	5F02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Final torque in Results page	Unit: 0.001 x N.m	E803
9 - 10	Torque value calibrated by third-party device	Unit: 0.001 x N.m	E903
11 - 12	Mode	0	0B00
13 - 14	Reserved	-	0000
15 - 16	Return function code	607	5F02
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

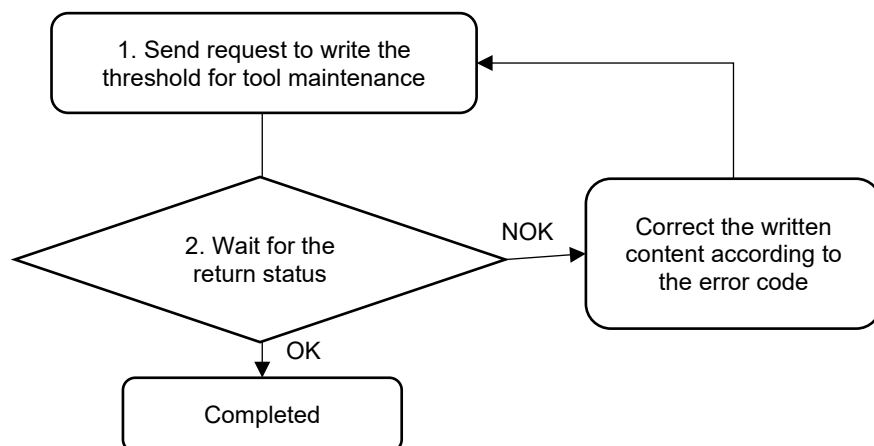
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	Precision	Exceeds the range of 0 to 32767

#608 Write the service reminder threshold

Content description:

Use function code #608 to set the threshold for tool maintenance.

Handshake signal description:



1. Send request to write the threshold for tool maintenance.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	608	6002
3 - 4	Version number	0	0000
5 - 6	Threshold for tool service reminder (L)	0 - 4294967295 (default: 250000)	0000
7 - 8	Threshold for tool service reminder (H)		3200
9 - 12	Reserved	4 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	608	6002
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Threshold for tool service reminder (L)	0 - 4294967295 (default: 250000)	0000
7 - 8	Threshold for tool service reminder (H)		3200
9 - 14	Reserved	6 bytes	...
15 - 16	Return function code	608	6002
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2.

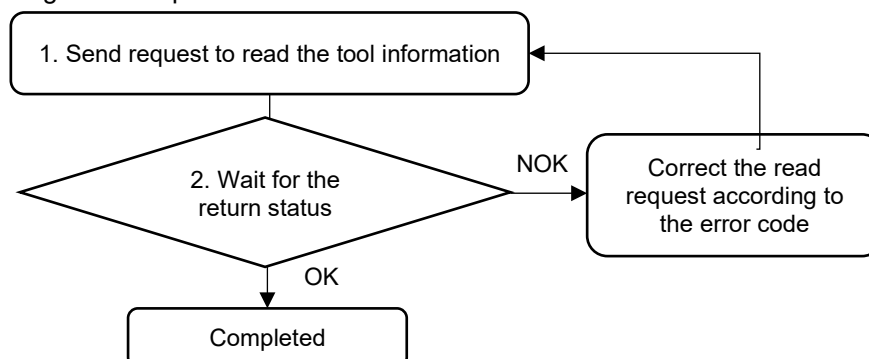
B

#650 Read the tool information

Content description:

Use function code #650 to read the tool information.

Handshake signal description:



1. Send the request to read the tool information.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	650	8A02
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the tool information.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	650	8A02
3 - 4	Return data length	Unit: Word	2300
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	650	8A02
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000
21 - 60	Model name	ASCII code (20 words)	...
61 - 62	Max. speed	Unit: rpm	D007
63 - 64	Max. torque	Unit: 0.001 Nm	B004
65 - 66	Tool temperature	Unit: °C	2100
67 - 68	Tightening + loosening count (L)	Tool life (maintenance)	0100
69 - 70	Tightening + loosening count (H)		0000
71 - 72	Tightening + loosening count (L)	Tool life (total)	0100
73 - 74	Tightening + loosening count (H)		0000
75 - 76	Recent service date - year	-	E907
77 - 78	Recent service date - month	-	0A00
79 - 80	Recent service date - day	-	0200
81 - 82	Recent service date - hour	-	1100
83 - 84	Recent service date - minute	-	1600
85 - 86	Recent service date - second	-	0500

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

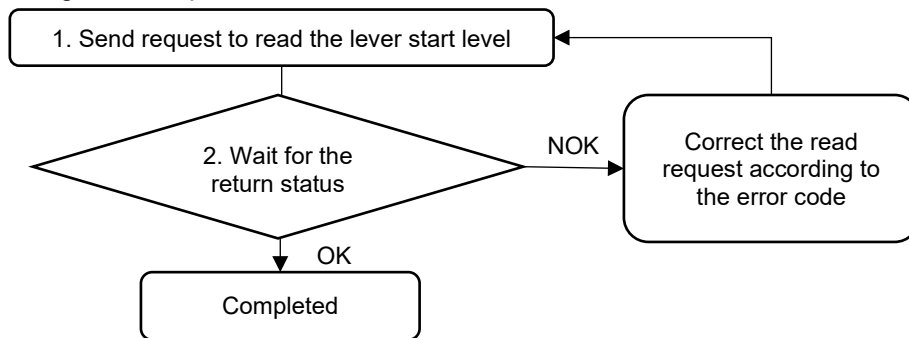
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	Data is not ready	Reread the tool information.

#651 Read the lever start level

Content description:

Use function code #651 to read the lever start level of the tool.

Handshake signal description:



1. Send the request to read the lever start level.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	651	8B02
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the contents of lever start level.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	651	8B02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Threshold of the current pressing	0 to 4095	5E06
9 - 10	Threshold for On level	0 to 4095	DC05
11 - 12	Threshold for Off level	0 to 4095	4006
13 - 14	Reserved	-	0000
15 - 16	Return function code	651	8B02
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

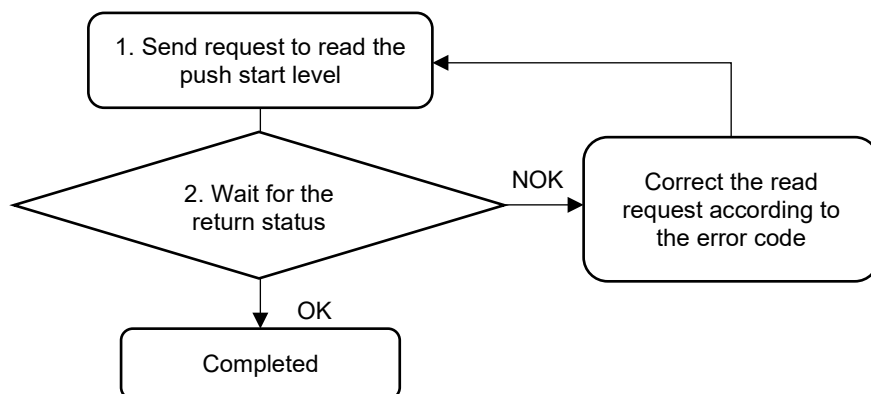
B

#652 Read the push start level

Content description:

Use function code #652 to read the push start level of the tool.

Handshake signal description:



1. Send the request to read the push start level.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	652	8C02
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the contents of push start level.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	652	8C02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Threshold of the current pressing	0 to 4095	5E06
9 - 10	Threshold for On level	0 to 4095	4C04
11 - 12	Threshold for Off level	0 to 4095	B004
13 - 14	Reserved	-	0000
15 - 16	Return function code	652	8C02
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

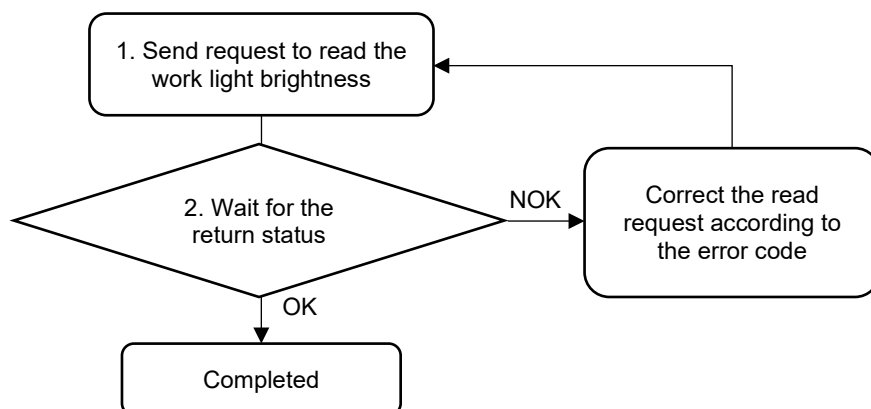
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

#653 Read the work light brightness

Content description:

Use function code #653 to read the work light brightness of the tool.

Handshake signal description:



1. Send the request to read the work light brightness.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	653	8D02
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the work light brightness.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	653	8D02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Brightness	0 to 50 (0: Off; 50: max. brightness)	3200
9 - 14	Reserved	6 bytes	...
15 - 16	Return function code	653	8D02
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

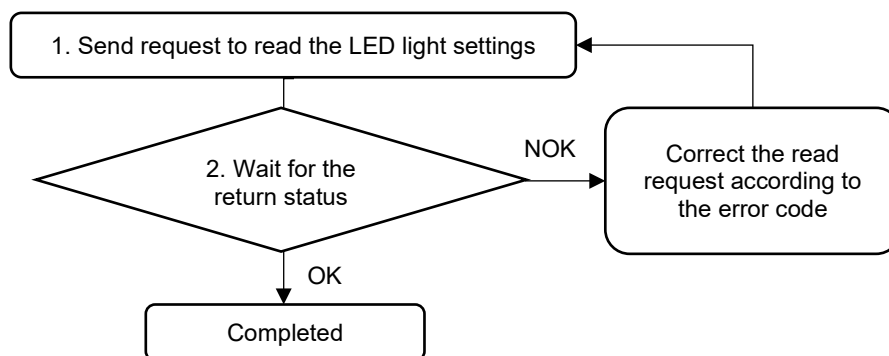
B

#655 Read the LED light settings

Content description:

Use function code #655 to read the LED light settings of the tool.

Handshake signal description:



1. Send the request to read the LED light settings.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	655	8F02
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the LED light settings.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	655	8F02
3 - 4	Return data length	Unit: Word	0E00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 14	Reserved	6 bytes	...
15 - 16	Return function code	655	8F02
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000
21 - 22	Reserved	-	0000
23 - 24	Status of red indicator	0: not in use 1: ready 2: running	0400
25 - 26	Status of yellow indicator	3: tightening OK and report is being saved 4: tightening NOK and report is being saved 5: loosening OK and report is being saved	0100
27 - 28	Status of green indicator	6: loosening NOK and report is being saved 7: total screw quantity reached	0300

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

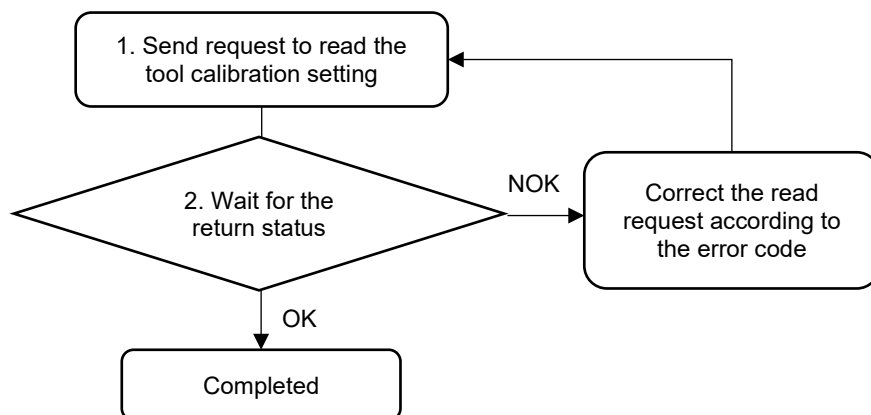
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

#656 Read the tool calibration setting

Content description:

Use function code #656 to read the precision of the tool.

Handshake signal description:



1. Send the request to read the tool calibration setting.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	656	9002
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the precision of the tool.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	656	9002
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Precision	0 to 32767 (Default: 1400 for 12 kgf-cm models 3500 for 30 kgf-cm models 6000 for 50 kgf-cm models 5705 for 75 kgf-cm models)	7805
9 - 14	Reserved	6 bytes	...
15 - 16	Return function code	656	9002
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

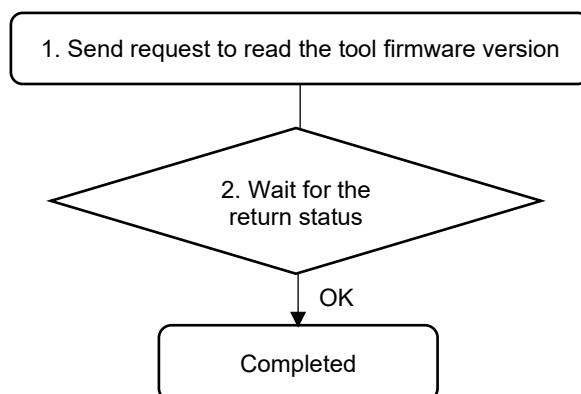
B

#657 Read the tool firmware version

Content description:

Use function code #657 to read the firmware version number of the tool.

Handshake signal description:



1. Send the request to read the tool firmware version.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	657	9102
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the firmware version number of the tool.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	657	9102
3 - 4	Return data length	Unit: Word	1400
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 14	Reserved	6 bytes	...
15 - 16	Return function code	657	9102
17 - 18	Status	1: OK	0100
19 - 20	Reserved	-	0000
21 - 60	Tool firmware version	ASCII code (20 words)	...

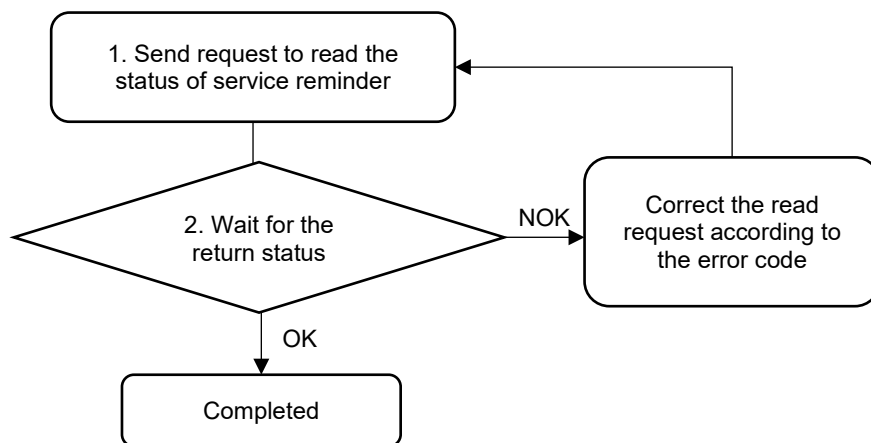
If successful, the read status is 1.

#658 Read the status of service reminder

Content description:

Use function code #658 to read the status of service reminder.

Handshake signal description:



1. Send the request to read the status of service reminder.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	658	9202
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the status of tool service reminder.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	658	9202
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Service reminder	0: Off; 1: On	0000
9 - 14	Reserved	6 bytes	...
15 - 16	Return function code	658	9202
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

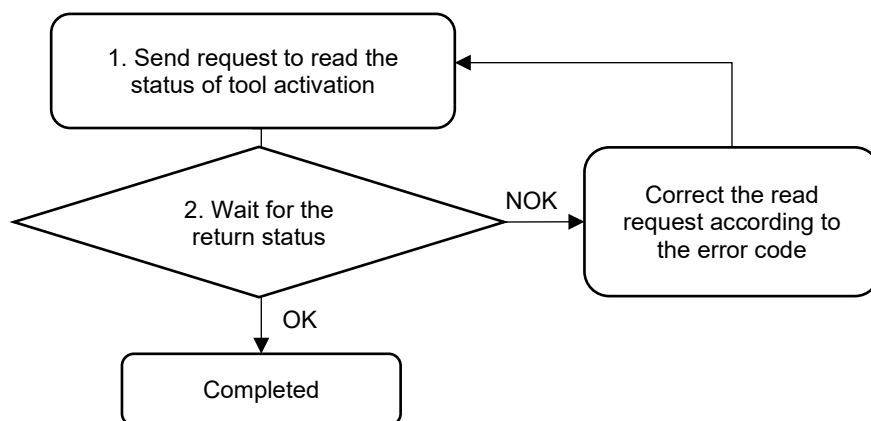
B

#659 Read the status of tool activation

Content description:

Use function code #659 to read the status of the tool activation.

Handshake signal description:



1. Send the request to read the status of tool activation.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	659	9302
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the status of tool activation.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	659	9302
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Tool activation	0: Off; 1: On	0100
9 - 14	Reserved	6 bytes	...
15 - 16	Return function code	659	9302
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

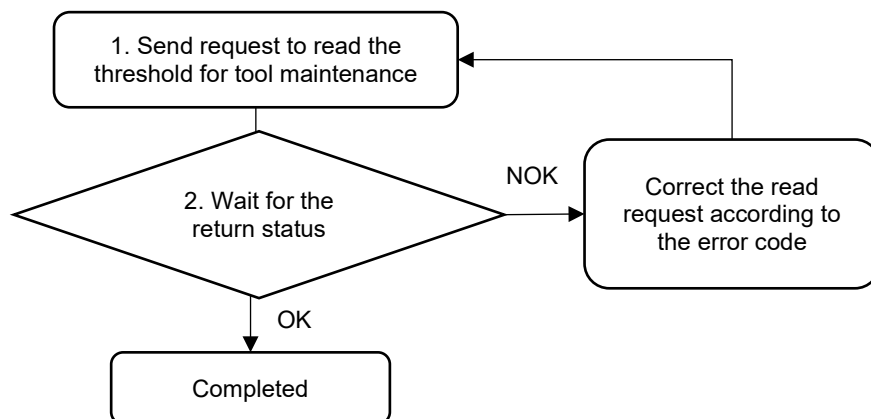
Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

#660 Read the service reminder threshold

Content description:

Use function code #660 to read the threshold for tool maintenance.

Handshake signal description:



1. Send request to read the threshold for tool maintenance.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	660	9402
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the threshold for tool service reminder.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	660	9402
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Threshold for tool service reminder (L)	Default: 250000	0000
7 - 8	Threshold for tool service reminder (H)		0000
9 - 14	Reserved	6 bytes	...
15 - 16	Return function code	660	9402
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2.

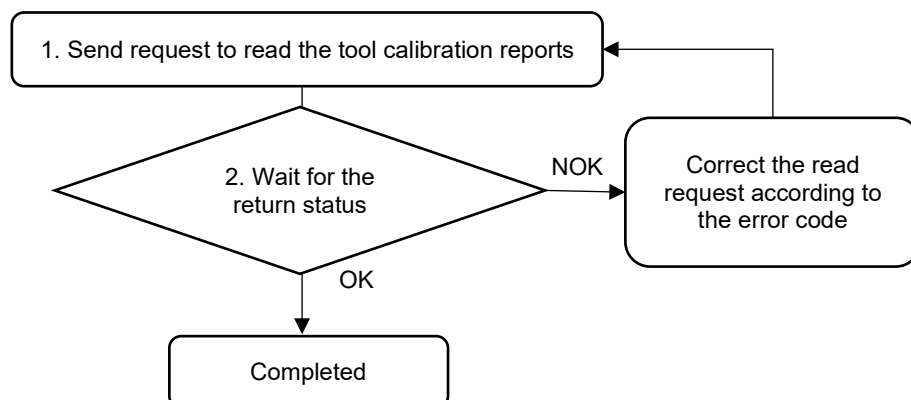
B

#661 Read the tool calibration reports

Content description:

Use function code #661 to read the tool calibration reports.

Handshake signal description:



1. Send request to read the tool calibration reports.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	661	9502
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Read the calibration report page number	0 - 19	0000
9 - 12	Reserved	4 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the tool calibration reports.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	661	9502
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Read the calibration report page number	0 - 19	0000
9 - 14	Reserved	6 bytes	...
15 - 16	Return function code	661	9502
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000
21 - 22	Torque unit when the first entry is saved	0: N.m; 1: kgf.cm; 2: lbf.ft; 3: lbf.in; 4: ozf.ft; 5: ozf.in; 6: cNm	0100
23 - 24	Torque unit when the second entry is saved		0000
25 - 26	Torque unit when the third entry is saved		0000
27 - 28	Torque unit when the fourth entry is saved		0000
29 - 30	Torque unit when the fifth entry is saved		0000
31 - 32	First entry of precision setting	Default settings: 12 kgf-cm models: 1400 30 kgf-cm models: 3500 50 kgf-cm models: 6000 75 kgf-cm models: 5705	7905
33 - 34	Second entry of precision setting		0000
35 - 36	Third entry of precision setting		0000
37 - 38	Fourth entry of precision setting		0000
39 - 40	Fifth entry of precision setting		0000

Byte	Write / Read request	Description	Example (Hex)
41 - 42	First entry of final torque in Result screen (L)	Unit: 0.001 x torque unit at present	E903
43 - 44	First entry of final torque in Result screen (H)		0000
45 - 46	Second entry of final torque in Result screen (L)		0000
47 - 48	Second entry of final torque in Result screen (H)		0000
49 - 50	Third entry of final torque in Result screen (L)		0000
51 - 52	Third entry of final torque in Result screen (H)		0000
53 - 54	Fourth entry of final torque in Result screen (L)		0000
55 - 56	Fourth entry of final torque in Result screen (H)		0000
57 - 58	Fifth entry of final torque in Result screen (L)		0000
59 - 60	Fifth entry of final torque in Result screen (H)		0000
61 - 62	First entry of error	Unit: %	0100
63 - 64	Second entry of error		0000
65 - 66	Third entry of error		0000
67 - 68	Fourth entry of error		0000
69 - 70	Fifth entry of error		0000
71 - 72	First entry of record date	The date is converted in to ASCII code.	0000
73 - 74	Second entry of record date		0000
75 - 76	Third entry of record date		0000
77 - 78	Fourth entry of record date		0000
79 - 80	Fifth entry of record date		0000
81 - 82	First entry of torque value calibrated by third-party device (L)	Unit: 0.001 x torque unit at present	DF03
83 - 84	First entry of torque value calibrated by third-party device (H)		0000
85 - 86	Second entry of torque value calibrated by third-party device (L)		0000
87 - 88	Second entry of torque value calibrated by third-party device (H)		0000
89 - 90	Third entry of torque value calibrated by third-party device (L)		0000
91 - 92	Third entry of torque value calibrated by third-party device (H)		0000
93 - 94	Fourth entry of torque value calibrated by third-party device (L)		0000
95 - 96	Fourth entry of torque value calibrated by third-party device (H)		0000
97 - 98	Fifth entry of torque value calibrated by third-party device (L)		0000
99 - 100	Fifth entry of torque value calibrated by third-party device (H)		0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1

B

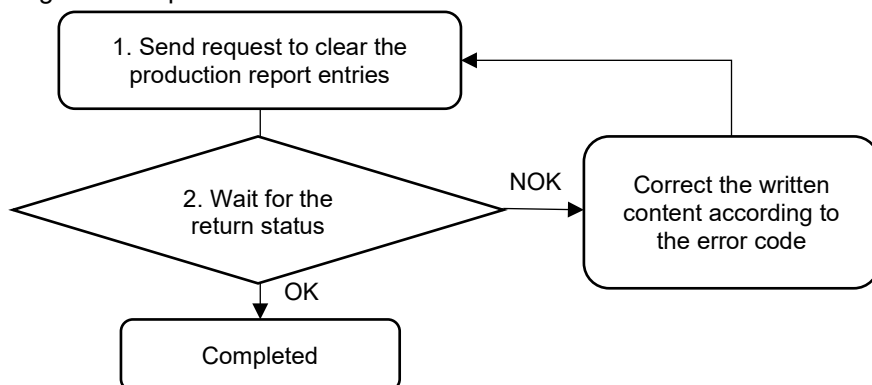
B.3.7 Reports operations

#700 Clear the production report entries

Content description:

Use function code #700 to delete all the production report entries from the controller.

Handshake signal description:



1. Send the request to clear the production report entries.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	700	BC02
3 - 4	Version number	-	0000
5 - 6	Confirm command	99	6300
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	700	BC02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Confirm command	99	6300
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	700	BC02
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

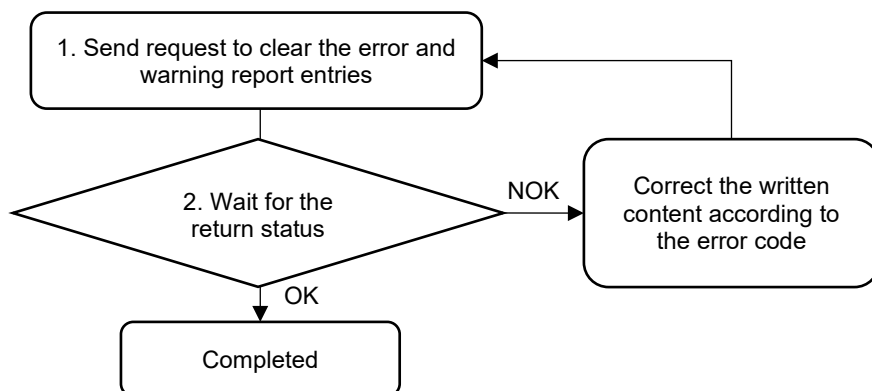
Code	Error description	Description
1	Confirm command	Must be 99

#701 Clear the error and warning report entries

Content description:

Use function code #701 to delete all the error and warning report entries from the controller.

Handshake signal description:



1. Send the request to clear the error and warning report entries.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	701	BD02
3 - 4	Version number	-	0000
5 - 6	Confirm command	99	6300
7 - 8	Item to delete	0: clear error and warning report entries 10: clear error report entries 20: clear warning report entries	0000
9 - 12	Reserved	4 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	701	BD02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Confirm command	99	6300
7 - 8	Item to delete	0: clear error and warning report entries 10: clear error report entries 20: clear warning report entries	0000
9 - 14	Reserved	6 bytes	...
15 - 16	Return function code	701	BD02
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Confirm command	Must be 99

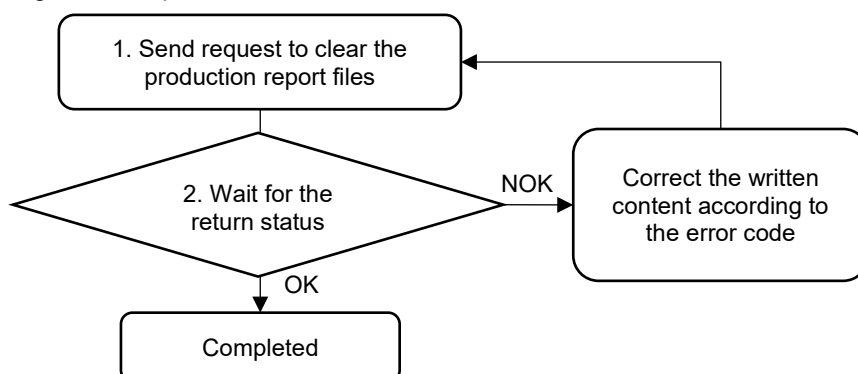
B

#702 Clear the production report files

Content description:

Use function code #702 to delete all the production report files exported to the HMI.

Handshake signal description:



1. Send the request to clear the production report files.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	702	BE02
3 - 4	Version number	-	0000
5 - 6	Confirm command	99	6300
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	702	BE02
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Confirm command	99	6300
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	702	BE02
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Confirm command	Must be 99

#750 Find and read the production report entries

Content description:

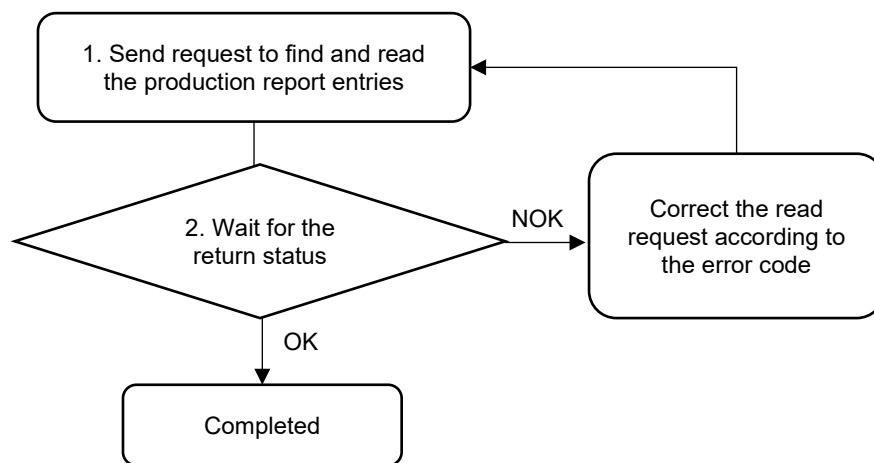
Use function code #750 to read the content of the production report.

- A. Use bytes 235 - 236 and bytes 237 - 238, function code #20, or function code #21 to read the content of the latest production report.

Byte (Dec)	Function code #20 (Hex)	Function code #21 (Hex)	Tool 1 / Tool 2 common status
235 - 236	76	6C	Current No. of production report entries (L)
237 - 238	77	6D	Current No. of production report entries (H)

- B. Read the content of history report entries.

Handshake signal description:



■ **When the version number is 0:**

1. Send the request to find and read the production report.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	750	EE02
3 - 4	Version number	0 or 1	0000
5 - 6	Production report ID (L)	1 to 200000	0100
7 - 8	Production report ID (H)	0: the latest production report	0000
9 - 12	Reserved	4 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the content of a single entry of the production report.

Byte	Function code #760 (Hex)	Return status	Description	Example (Hex)
1 - 2	-	Function code	750	EE02
3 - 4	-	Return data length	Unit: Word	CB00
5 - 6	-	Production report ID (L)	1 to 200000	0100
7 - 8	-	Production report ID (H)	0: the latest production report	0000
9 - 14	-	Reserved	6 bytes	...
15 - 16	-	Return function code	750	EE02

Byte	Function code #760 (Hex)	Return status	Description	Example (Hex)
17 - 18	-	Status	1: OK; 2: NOK	0100
19 - 20	-	Error code	-	0000
21 - 220	B - 6E	Saved barcode string	ASCII code (100 words)	...
221 - 222	6F	Year	-	E607
223 - 224	70	Month	-	0B00
225 - 226	71	Day	-	1E00
227 - 228	72	Hour	-	0900
229 - 230	73	Minute	-	0500
231 - 232	74	Second	-	1D00
233 - 234	75	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
235 - 236	76	Screw ID (L)	Current screw progress ID	0100
237 - 238	77	Screw ID (H)		0000
239 - 240	78	Sequence ID	Current sequence ID	0000
241 - 242	79	Parameter ID	Current parameter ID	0100
243 - 244	7A	Target torque	Target torque set in the final stage; unit: 0.001 x torque of the current report Refer to DWORD (bytes 327 - 330) when the torque specification of the tool is 5 N.m or higher.	1327
245 - 246	7B	Target angle	Target angle set in the final stage. If Angle Control is used through all stages, this field is the total angle of all stages.	0000
247 - 248	7C	Target torque rate	Target torque rate set in the final stage; unit: 0.0001 x torque/degree of the current report Refer to DWORD (bytes 331 - 334) when the torque specification of the tool is 5 N.m or higher.	0000
249 - 250	7D	Final torque	Refer to DWORD (bytes 335 - 338) when the torque specification of the tool is 5 N.m or higher.	2727
251 - 252	7E	Tightening angle	Running angle of the final stage.	1900
253 - 254	7F	Rotation angle	Total angle.	2F08
255 - 256	80	Status	1: tightening OK; 2: tightening NOK; 3: loosening OK; 4: loosening NOK; 5: pass	0100
257 - 258	81	Operation time	Total operation time of all stages (unit: 0.001 second)	6804
259 - 260	82	AL / NG / WN code	Refer to CH12 (need to convert to Hex format)	0000
261 - 262	83	Max. angle of final stage	-	0000
263 - 264	84	Min. angle of final stage	-	0000
265 - 266	85	Max. torque of final stage	Refer to DWORD (bytes 339 - 342) when the torque specification of the tool is 5 N.m or higher.	6F27
267 - 268	86	Min. torque of final stage	Refer to DWORD (bytes 343 - 346) when the torque specification of the tool is 5 N.m or higher.	441F
269 - 270	87	Torque unit	0: N.m; 1: kgf.cm; 2: lbf.ft; 3: lbf.in; 4: ozf.ft; 5: ozf.in; 6: cNm	0100
271 - 272	88	Torque specification of the tool	Tool specification; unit: 0.001 N.m	B004
273 - 274	89	Max. torque of the tool	Upper limit of tool specification; unit: 0.001 N.m	7805
275 - 276	8A	Pre-tightening torque	The operating torque of the third stage when the tightening is executed under Standard strategy. Refer to DWORD (bytes 347 - 350) when the torque specification of the tool is 5 N.m or higher.	C209
277 - 278	8B	The set total operation time	Timeout setting for a single tightening parameter.	6400
279 - 280	8C	Max. total angle	Set the limit of maximum total angle.	FF7F

Byte	Function code #760 (Hex)	Return status	Description	Example (Hex)
281 - 282	8D	Max. torque	Torque upper limit when the current stage finishes. Refer to DWORD (bytes 351 - 354) when the torque specification of the tool is 5 N.m or higher.	6F27
283 - 284	8E	Min. torque	Torque lower limit when the current stage finishes. Refer to DWORD (bytes 355 - 358) when the torque specification of the tool is 5 N.m or higher.	441F
285 - 286	8F	Max. angle	Angle upper limit when the current stage finishes.	0000
287 - 288	90	Min. angle	Angle lower limit when the current stage finishes.	0000
289 - 290	91	Max. operation time	Time upper limit when the current stage finishes.	0000
291 - 292	92	Min. operation time	Time lower limit when the current stage finishes.	0000
293 - 294	93	Prevail torque	When the prevail torque function is enabled, the prevail torque for operation refers to this field. When the function is disabled, this field is 0. Refer to DWORD (bytes 359 - 362) when the torque specification of the tool is 5 N.m or higher.	0000
295 - 296	94	Final + Prevail torque	Refer to DWORD (bytes 363 - 366) when the torque specification of the tool is 5 N.m or higher.	2727
297 - 298	95	Actual current	The maximum current when the tightening operation is complete (unit: 0.01 Amp)	EF02
299 - 300	96	Clamp torque	When the operation is using Torque Rate Control, this field is the final torque minus the torque when snug point is reached. When the operation is not using Torque Rate Control, this field is the final torque minus the tightening torque of the previous stage. Refer to DWORD (bytes 367 - 370) when the torque specification of the tool is 5 N.m or higher.	0000
301 - 302	97	Max. clamp torque	Clamp torque upper limit when the current stage finishes. Refer to DWORD (bytes 371 - 374) when the torque specification of the tool is 5 N.m or higher.	0000
303 - 304	98	Min. clamp torque	Clamp torque lower limit when the current stage finishes. Refer to DWORD (bytes 375 - 378) when the torque specification of the tool is 5 N.m or higher.	0000
305 - 306	99	Clamp angle	-	0000
307 - 308	9A	Max. clamp angle	-	0000
309 - 310	9B	Min. clamp angle	-	0000
311 - 312	9C	Min. total angle	-	0000
313 - 314	9D	Permissions account	0: reserved 1: User1 2: User2 3: User3 4: User4 5: User5 6: Admin	0000
315 - 316	9E	Magnification of the torque recorded by system	System-specific	0000
317 - 318	9F	Target yield point	Target yield point set in the final stage; unit: 0.01%	0000
319 - 326	A0 - A3	Reserved	8 bytes	...
327 - 328	A4	Target torque (L)	Unit: 0.001 x torque of the current report	1327
329 - 330	A5	Target torque (H)		0000
331 - 332	A6	Target torque rate (L)	Unit: 0.0001 x torque/degree of the current report	0000
333 - 334	A7	Target torque rate (H)		0000

B

Byte	Function code #760 (Hex)	Return status	Description	Example (Hex)
335 - 336	A8	Final torque (L)	Unit: 0.001 x torque of the current report	2727
337 - 338	A9	Final torque (H)		0000
339 - 340	AA	Final stage max. torque (L)	Unit: 0.001 x torque of the current report	6F27
341 - 342	AB	Final stage max. torque (H)		0000
343 - 344	AC	Final stage min. torque (L)	Unit: 0.001 x torque of the current report	441F
345 - 346	AD	Final stage min. torque (H)		0000
347 - 348	AE	Pre-tightening torque (L)	Unit: 0.001 x torque of the current report	C209
349 - 350	AF	Pre-tightening torque (H)		0000
351 - 352	B0	Max. torque (L)	Unit: 0.001 x torque of the current report Torque upper limit when the current stage finishes.	6F27
353 - 354	B1	Max. torque (H)		0000
355 - 356	B2	Min. torque (L)	Unit: 0.001 x torque of the current report Torque lower limit when the current stage finishes.	441F
357 - 358	B3	Min. torque (H)		0000
359 - 360	B4	Prevail torque (L)	Unit: 0.001 x torque of the current report	0000
361 - 362	B5	Prevail torque (H)		0000
363 - 364	B6	Final + Prevail torque (L)	Unit: 0.001 x torque of the current report	2727
365 - 366	B7	Final + Prevail torque (H)		0000
367 - 368	B8	Clamp torque (L)	Unit: 0.001 x torque of the current report	0000
369 - 370	B9	Clamp torque (H)		0000
371 - 372	BA	Max. clamp torque (L)	Unit: 0.001 x torque of the current report Clamp torque upper limit when the current stage finishes.	0000
373 - 374	BB	Max. clamp torque (H)		0000
375 - 376	BC	Min. clamp torque (L)	Unit: 0.001 x torque of the current report Clamp torque lower limit when the current stage finishes.	0000
377 - 378	BD	Min. clamp torque (H)		0000
379 - 380	BE	Tool spec. torque (L)	Unit: 0.001 x torque of the current report	CC2F
381 - 382	BF	Tool spec. torque (H)		0000
383 - 384	C0	Tool max. torque (L)	Unit: 0.001 x torque of the current report	C337
385 - 386	C1	Tool max. torque (H)		0000
387 - 526	-	Reserved	-	-

■ When the version number is 1:

1. Send the request to find and read the content of the production report.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	750	EE02
3 - 4	Version number	0 or 1	0100
5 - 6	Production report ID (L)	1 to 200000	0100
7 - 8	Production report ID (H)	0: the latest production report	0000
9 - 12	Reserved	4 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the content of a single entry of the production report.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	750	EE02
3 - 4	Return data length	Unit: Word	3001
5 - 6	Production report ID (L)	1 to 200000	0100
7 - 8	Production report ID (H)	0: the latest production report	0000
9 - 14	Reserved	6 bytes	...
15 - 16	Return function code	750	EE02
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000
21 - 526	Refer to the return status list of version number 0.		
527 - 566	Sequence title	ASCII code (20 words)	...
567 - 607	Parameter title	ASCII code (20 words)	...

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Production report ID	Exceeds the range of 0 to 200000

B

#751 Find and read curves

Content description:

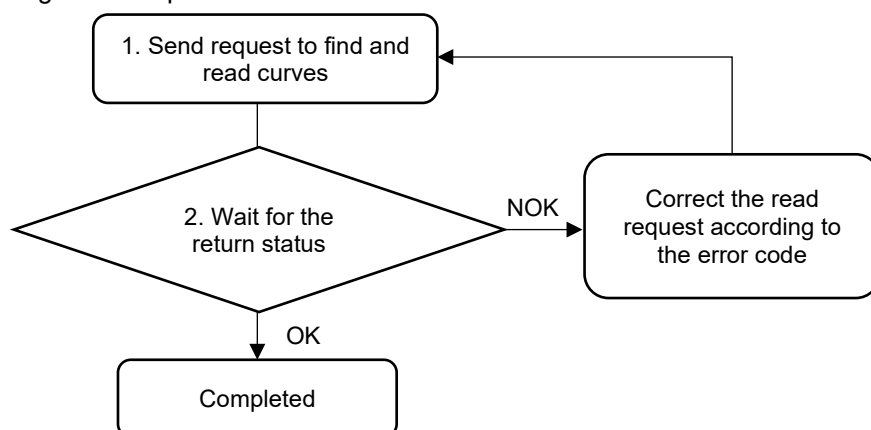
Use function code #751 to read the content of the operating curve.

- A. Use bytes 235 - 236 and bytes 237 - 238, function code #20, or function code #21 to read the content of the latest curve.

Byte (Dec)	Function code #20 (Hex)	Function code #21 (Hex)	Tool 1 / Tool 2 common status
235 - 236	76	6C	Current No. of production report entries (L)
237 - 238	77	6D	Current No. of production report entries (H)

- B. Check the content of history curves.
- C. Set the time as X axis and the torque as Y axis to draw a torque-time coordinate graph.
- D. Set the angle as X axis and the torque as Y axis to draw a torque-angle coordinate graph.
- E. Set the angle as X axis and the torque rate as Y axis to draw a torque rate-angle coordinate graph.
- F. Read the parameters content of the curve.

Handshake signal description:



1. Send the request to find and read curves.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	751	EF02
3 - 4	Version number	-	0000
5 - 6	Production report ID (L)	1 to 200000	0100
7 - 8	Production report ID (H)	0: the latest production report	0000
9 - 10	Query type	0: time scale data (1 to 2000) 1: angle scale data (1 to 2000) 2: torque scale data (1 to 2000) 3: torque rate scale data (1 to 2000) 4: torque scale data DWORD (1 to 1000) 5: torque scale data DWORD (1001 to 2000) 6: torque rate scale data DWORD (1 to 1000) 7: torque rate scale data DWORD (1001 to 2000) 10: values of coordinate axis 11: parameters 20: time scale data (2001 to 4000) 21: angle scale data (2001 to 4000) 22: torque scale data (2001 to 4000) 23: torque rate scale data (2001 to 4000) 24: torque scale data DWORD (2001 to 3000) 25: torque scale data DWORD (3001 to 4000) 26: torque rate scale data DWORD (2001 to 3000) 27: torque rate scale data DWORD (3001 to 4000) 30: time scale data (4001 to 6000) 31: angle scale data (4001 to 6000) 32: torque scale data (4001 to 6000) 33: torque rate scale data (4001 to 6000) 34: torque scale data DWORD (4001 to 5000) 35: torque scale data DWORD (5001 to 6000) 36: torque rate scale data DWORD (4001 to 5000) 37: torque rate scale data DWORD (5001 to 6000) 40: time scale data (6001 to 8000) 41: angle scale data (6001 to 8000) 42: torque scale data (6001 to 8000) 43: torque rate scale data (6001 to 8000) 44: torque scale data DWORD (6001 to 7000) 45: torque scale data DWORD (7001 to 8000) 46: torque rate scale data DWORD (6001 to 7000) 47: torque rate scale data DWORD (7001 to 8000)	0000
11 - 12	Reserved	-	0000
13 - 14	Request to send command	-	0100

B

2. Check whether the reading is successful from the return status and read the content of the operating curve.

Byte	Return status	Description		Example (Hex)
1 - 2	Function code	751		EF02
3 - 4	Return data length	Unit: Word	Query types 0 to 7 Query types 20 to 27 Query types 30 to 37 Query types 40 to 47	DA07
			Query type 10	6800
			Query type 11 (version number 0) Query type 11 (version number 1)	3002
5 - 6	Production report ID (L)	1 to 200000 0: the latest production report		0100
7 - 8	Production report ID (H)			0000
9 - 10	Query type	Refer to the description of query types in the previous page.		0000
11 - 14	Reserved	4 bytes		...
15 - 16	Return function code	751		EF02
17 - 18	Status	1: OK; 2: NOK		0100
19 - 20	Error code	-		0000
21 - ...	Curve content	Refer to the following tables.		-

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows:

Code	Error description	Description
1	Production report ID	Exceeds the range of 0 to 200000
2	Query type	Must be 0 to 7, 10, 11, 20 to 27, 30 to 37, or 40 to 47

The contents of each query type are as follows:

Byte	Query type 0	Example (Hex)	Query type 1	Example (Hex)
21 - 22	Time scale 1 data	0A00	Angle scale 1 data	0000
23 - 24	Time scale 2 data	1400	Angle scale 2 data	0300
25 - 26	Time scale 3 data	1E00	Angle scale 3 data	0800
27 - 28	Time scale 4 data	2800	Angle scale 4 data	0D00
29 - 30	Time scale 5 data	3200	Angle scale 5 data	1100
31 - 32	Time scale 6 data	3C00	Angle scale 6 data	1600
33 - 34	Time scale 7 data	4600	Angle scale 7 data	1B00
35 - 36	Time scale 8 data	5000	Angle scale 8 data	2000
37 - 38	Time scale 9 data	5A00	Angle scale 9 data	2500
...	Time scale N data	...	Angle scale N data	...
4001 - 4002	Time scale 1991 data	5604	Angle scale 1991 data	2F08
4003 - 4004	Time scale 1992 data	5804	Angle scale 1992 data	2D08
4005 - 4006	Time scale 1993 data	5A04	Angle scale 1993 data	2E08
4007 - 4008	Time scale 1994 data	5C04	Angle scale 1994 data	2C08
4009 - 4010	Time scale 1995 data	5E04	Angle scale 1995 data	2A08
4011 - 4012	Time scale 1996 data	6004	Angle scale 1996 data	2708
4013 - 4014	Time scale 1997 data	6204	Angle scale 1997 data	2508
4015 - 4016	Time scale 1998 data	6404	Angle scale 1998 data	2208
4017 - 4018	Time scale 1999 data	6604	Angle scale 1999 data	2008
4019 - 4020	Time scale 2000 data	6804	Angle scale 2000 data	1E08

Byte	Query type 2	Example (Hex)	Query type 3	Example (Hex)
21 - 22	Torque scale 1 data	0000	Torque rate scale 1 data	0000
23 - 24	Torque scale 2 data	7A00	Torque rate scale 2 data	0000
25 - 26	Torque scale 3 data	4700	Torque rate scale 3 data	0000
27 - 28	Torque scale 4 data	3D00	Torque rate scale 4 data	0000
29 - 30	Torque scale 5 data	5B00	Torque rate scale 5 data	0000
31 - 32	Torque scale 6 data	4700	Torque rate scale 6 data	1E00
33 - 34	Torque scale 7 data	7A00	Torque rate scale 7 data	3200
35 - 36	Torque scale 8 data	7A00	Torque rate scale 8 data	1400
37 - 38	Torque scale 9 data	5100	Torque rate scale 9 data	1400
...	Torque scale N data	...	Torque rate scale N data	...
4001 - 4002	Torque scale 1991 data	D725	Torque rate scale 1991 data	C409
4003 - 4004	Torque scale 1992 data	8822	Torque rate scale 1992 data	D609
4005 - 4006	Torque scale 1993 data	3F1A	Torque rate scale 1993 data	F509
4007 - 4008	Torque scale 1994 data	370F	Torque rate scale 1994 data	EB09
4009 - 4010	Torque scale 1995 data	7F05	Torque rate scale 1995 data	EB09
4011 - 4012	Torque scale 1996 data	7A00	Torque rate scale 1996 data	EB09
4013 - 4014	Torque scale 1997 data	0000	Torque rate scale 1997 data	EB09
4015 - 4016	Torque scale 1998 data	0000	Torque rate scale 1998 data	EB09
4017 - 4018	Torque scale 1999 data	6500	Torque rate scale 1999 data	EB09
4019 - 4020	Torque scale 2000 data	7000	Torque rate scale 2000 data	EB09

B

Byte	Query type 4	Example (Hex)	Query type 5	Example (Hex)
21 - 22	Torque scale 1 data (L)	0000	Torque scale 1001 data (L)	D725
23 - 24	Torque scale 1 data (H)	0000	Torque scale 1001 data (H)	0000
25 - 26	Torque scale 2 data (L)	7A00	Torque scale 1002 data (L)	8822
27 - 28	Torque scale 2 data (H)	0000	Torque scale 1002 data (H)	0000
29 - 30	Torque scale 3 data (L)	4700	Torque scale 1003 data (L)	3F1A
31 - 32	Torque scale 3 data (H)	0000	Torque scale 1003 data (H)	0000
33 - 34	Torque scale 4 data (L)	3D00	Torque scale 1004 data (L)	370F
35 - 36	Torque scale 4 data (H)	0000	Torque scale 1004 data (H)	0000
...	Torque scale N data (L)	...	Torque scale N data (L)	...
...	Torque scale N data (H)	...	Torque scale N data (H)	...
4001 - 4002	Torque scale 996 data (L)	4700	Torque scale 1996 data (L)	7F05
4003 - 4004	Torque scale 996 data (H)	0000	Torque scale 1996 data (H)	0000
4005 - 4006	Torque scale 997 data (L)	7A00	Torque scale 1997 data (L)	7A00
4007 - 4008	Torque scale 997 data (H)	0000	Torque scale 1997 data (H)	0000
4009 - 4010	Torque scale 998 data (L)	7A00	Torque scale 1998 data (L)	6700
4011 - 4012	Torque scale 998 data (H)	0000	Torque scale 1998 data (H)	0000
4013 - 4014	Torque scale 999 data (L)	5100	Torque scale 1999 data (L)	6500
4015 - 4016	Torque scale 999 data (H)	0000	Torque scale 1999 data (H)	0000
4017 - 4018	Torque scale 1000 data (L)	4A00	Torque scale 2000 data (L)	7000
4019 - 4020	Torque scale 1000 data (H)	0000	Torque scale 2000 data (H)	0000

Byte	Query type 6	Example (Hex)	Query type 7	Example (Hex)
21 - 22	Torque rate scale 1 data (L)	0000	Torque rate scale 1001 data (L)	C409
23 - 24	Torque rate scale 1 data (H)	0000	Torque rate scale 1001 data (H)	0000
25 - 26	Torque rate scale 2 data (L)	0000	Torque rate scale 1002 data (L)	D609
27 - 28	Torque rate scale 2 data (H)	0000	Torque rate scale 1002 data (H)	0000
29 - 30	Torque rate scale 3 data (L)	0000	Torque rate scale 1003 data (L)	F509
31 - 32	Torque rate scale 3 data (H)	0000	Torque rate scale 1003 data (H)	0000
33 - 34	Torque rate scale 4 data (L)	0000	Torque rate scale 1004 data (L)	EB09
35 - 36	Torque rate scale 4 data (H)	0000	Torque rate scale 1004 data (H)	0000
...	Torque rate scale N data (L)	...	Torque rate scale N data (L)	...
...	Torque rate scale N data (H)	...	Torque rate scale N data (H)	...
4001 - 4002	Torque rate scale 996 data (L)	1E00	Torque rate scale 1996 data (L)	EB09
4003 - 4004	Torque rate scale 996 data (H)	0000	Torque rate scale 1996 data (H)	0000
4005 - 4006	Torque rate scale 997 data (L)	3200	Torque rate scale 1997 data (L)	EB20
4007 - 4008	Torque rate scale 997 data (H)	0000	Torque rate scale 1997 data (H)	0000
4009 - 4010	Torque rate scale 998 data (L)	1400	Torque rate scale 1998 data (L)	EB80
4011 - 4012	Torque rate scale 998 data (H)	0000	Torque rate scale 1998 data (H)	0000
4013 - 4014	Torque rate scale 999 data (L)	1400	Torque rate scale 1999 data (L)	EB50
4015 - 4016	Torque rate scale 999 data (H)	0000	Torque rate scale 1999 data (H)	0000
4017 - 4018	Torque rate scale 1000 data (L)	1200	Torque rate scale 2000 data (L)	EB09
4019 - 4020	Torque rate scale 1000 data (H)	0000	Torque rate scale 2000 data (H)	0000

Byte	Function code #760 (Hex)	Query type 10	Description	Example (Hex)
21 - 22	3F3	Running angle of stage 1	Unit: degree	5A00
23 - 24	3F4	Running angle of stage 2		B307
25 - 26	3F5	Running angle of stage 3		0900
27 - 28	3F6	Running angle of stage 4		1900
29 - 30	3F7	Running angle of stage 5		0000
31 - 32	3F8	Running angle of stage 6		0000
33 - 34	3F9	Running angle of loosening stage 1		0000
35 - 36	3FA	Running angle of loosening stage 2		0000
37 - 38	3FB	Max. torque of stage 1	Unit: 0.001 x current torque unit. Refer to DWORD when the torque specification of the tool is 5 N.m or higher.	B700
39 - 40	3FC	Max. torque of stage 2		CC09
41 - 42	3FD	Max. torque of stage 3		380E
43 - 44	3FE	Max. torque of stage 4		2727
45 - 46	3FF	Max. torque of stage 5		0000
47 - 48	400	Max. torque of stage 6		0000
49 - 50	401	Max. torque of loosening stage 1		7000
51 - 52	402	Max. torque of loosening stage 2		7A00
53 - 54	403	Operation time of stage 1	Unit: ms	0401
55 - 56	404	Operation time of stage 2		3202
57 - 58	405	Operation time of stage 3		3200
59 - 60	406	Operation time of stage 4		0001
61 - 62	407	Operation time of stage 5		0000
63 - 64	408	Operation time of stage 6		0000
65 - 66	409	Operation time of loosening stage 1		0000
67 - 68	40A	Operation time of loosening stage 2		0000
69 - 70	40B	Max. time on the scale	Unit: ms	6804
71 - 72	40C	Max. angle on the scale	Unit: degree	3208
73 - 74	40D	Max. torque on the scale	Unit: 0.001 x current torque unit. Refer to DWORD when the torque specification of the tool is 5 N.m or higher.	D229
75 - 76	40E	Max. torque rate on the scale / Max. speed on the scale	1. Unit: 0.001 x current torque rate unit. Refer to DWORD when the torque specification of the tool is 5 N.m or higher. 2. Unit: rpm (when the torque rate curve is replaced by the speed curve)	550F
77 - 78	40F	Total number of curve coordinates	Total sampling points on the curve.	EC00
79 - 80	410	Max. torque	Upper limit of the torque specified on the curve (unit: current torque unit) Refer to DWORD when the torque specification of the tool is 5 N.m or higher.	6F27

B

B

Byte	Function code #760 (Hex)	Query type 10	Description	Example (Hex)
81 - 82	411	Min. torque	Lower limit of the torque specified on the curve (unit: current torque unit) Refer to DWORD when the torque specification of the tool is 5 N.m or higher.	441F
83 - 84	418	Max. torque rate	Target torque rated specified on the curve (unit: current torque rate unit) Refer to DWORD when the torque specification of the tool is 5 N.m or higher.	0000
85 - 86	419	Max. angle	Upper limit of the angle for the last operating stage (unit: degree)	3108
87 - 88	41A	Min. angle	Lower limit of the angle for the last operating stage (unit: degree)	1608
89 - 90	41B	Version of the curve scale	0: up to 2000 points can be specified on the curve. 1: the curve can support different versions of sampling rates and the use of torque rate curve 2: the curve can support different versions of sampling rates and the use of speed curve	0100
91 - 92	41C	Sampling rate for curves	0: 10 ms→4 ms when start torque is reached (up to 2000 points for a single curve); up to 200000 entries for the report. 1: 2 ms; max. 4 seconds per production report entry (up to 2000 points for a single curve); up to 200000 entries for the report. 2: 2 ms; max. 8 seconds per production report entry (up to 4000 points for a single curve); up to 100000 entries for the report. 3: 1 ms; max. 4 seconds per production report entry (up to 4000 points for a single curve); up to 100000 entries for the report. 4: 1 ms; max. 8 seconds per production report entry (up to 8000 points for a single curve); up to 50000 entries for the report.	0000

Byte	Function code #760 (Hex)	Query type 10	Description	Example (Hex)
93 - 94	41D	Max. torque rate of curve	Unit: 0.0001 x current torque rate unit. Refer to DWORD when the torque specification of the tool is 5 N.m or higher.	0000
95 - 96	41E	Min. time on the scale	Unit: ms	0000
97 - 98	41F	Min. angle on the scale	Unit: degree	0000
99 - 100	450	Min. torque on the scale	Unit: 0.001 x current torque unit. Refer to DWORD when the torque specification of the tool is 5 N.m or higher.	0000
101 - 102	451	Min. torque rate on the scale / Min. speed on the scale	1. Unit: 0.0001 x current torque rate unit. Refer to DWORD when the torque specification of the tool is 5 N.m or higher. 2. Unit: rpm (when the torque rate curve is replaced by the speed curve)	0000
103 - 104	452	Switching torque for stage 1 / Max. speed for stage 1	1. When the function of replacing torque rate curve with speed curve is enabled, the ending torque of the stage is recorded (unit: 0.001 x torque unit at present). Refer to DWORD when the torque specification of the tool is 5 N.m or higher. 2. When the function of replacing torque rate curve with speed curve is enabled, the maximum speed of the stage is recorded (unit: rpm).	0000
105 - 106	453	Switching torque for stage 2 / Max. speed for stage 2		0000
107 - 108	454	Switching torque for stage 3 / Max. speed for stage 3		0000
109 - 110	455	Switching torque for stage 4 / Max. speed for stage 4		0000
111 - 112	456	Switching torque for stage 5 / Max. speed for stage 5		0000
113 - 114	457 - 459	Switching torque for stage 6 / Max. speed for stage 6		0000
115 - 120	45A	Reserved	6 bytes	...
121 - 122	45B	Max. torque of stage 1 (L)	Unit: 0.001 x current torque unit	B700
123 - 124	45C	Max. torque of stage 1 (H)		0000
125 - 126	45D	Max. torque of stage 2 (L)	Unit: 0.001 x current torque unit	CC09
127 - 128	45E	Max. torque of stage 2 (H)		0000
129 - 130	45F	Max. torque of stage 3 (L)	Unit: 0.001 x current torque unit	380E
131 - 132	460	Max. torque of stage 3 (H)		0000
133 - 134	461	Max. torque of stage 4 (L)	Unit: 0.001 x current torque unit	2727
135 - 136	462	Max. torque of stage 4 (H)		0000
137 - 138	463	Max. torque of stage 5 (L)	Unit: 0.001 x current torque unit	0000
139 - 140	464	Max. torque of stage 5 (H)		0000
141 - 142	465	Max. torque of stage 6 (L)	Unit: 0.001 x current torque unit	0000
143 - 144	466	Max. torque of stage 6 (H)		0000
145 - 146	467	Max. torque of loosening stage 1 (L)	Unit: 0.001 x current torque unit	7000
147 - 148	468	Max. torque of loosening stage 1 (H)		0000

B

Byte	Function code #760 (Hex)	Query type 10	Description	Example (Hex)
149 - 150	469	Max. torque of loosening stage 2 (L)	Unit: 0.001 x current torque unit	7A00
151 - 152	46A	Max. torque of loosening stage 2 (H)		0000
153 - 168	43D - 43F	Reserved	16 bytes	...
169 - 170	43D	Max. torque on the scale (L)	Unit: 0.001 x current torque unit	D229
171 - 172	43E	Max. torque on the scale (H)		0000
173 - 174	43F	Max. torque rate on the scale / Max. speed on the scale (L)	1. Unit: 0.001 x current torque rate unit	880F
175 - 176	440	Max. torque rate on the scale / Max. speed on the scale (H)	2. 2. Unit: rpm (the torque rate curve is replaced by the speed curve)	0000
177 - 178	441	Max. torque (L)	Unit: 0.001 x current torque unit	6F27
179 - 180	442	Max. torque (H)		0000
181 - 182	443	Min. torque (L)	Unit: 0.001 x current torque unit	441F
183 - 184	444	Min. torque (H)		0000
185 - 186	445	Max. torque rate (L)	Unit: 0.001 x current torque rate unit	0000
187 - 188	446	Max. torque rate (H)		0000
189 - 190	447	Max. torque rate of curve (L)	Unit: 0.0001 x current torque rate unit	0000
191 - 192	448	Max. torque rate of curve (H)		0000
193 - 194	449	Min. torque on the scale / Min. speed on the scale (L)	1. Unit: 0.001 x current torque unit	0000
195 - 196	44A	Min. torque on the scale / Min. speed on the scale (H)	2. Unit: rpm (the torque rate curve is replaced by the speed curve)	0000
197 - 198	44B	Min. torque rate on the scale (L)	Unit: 0.001 x current torque rate unit	0000
199 - 200	44C	Min. torque rate on the scale (H)		0000
201 - 202	44D	Switching torque for stage 1 / Speed for stage 1 (L)	Unit: 0.001 x torque unit at present 2. Unit: rpm (the torque rate curve is replaced by the speed curve)	0000
203 - 204	44E	Switching torque for stage 1 / Speed for stage 1 (H)		0000
205 - 206	44F	Switching torque for stage 2 / Speed for stage 2 (L)		0000
207 - 208	450	Switching torque for stage 2 / Speed for stage 2 (H)		0000
209 - 210	451	Switching torque for stage 3 / Speed for stage 3 (L)		0000
211 - 212	452	Switching torque for stage 3 / Speed for stage 3 (H)		0000
213 - 214	453	Switching torque for stage 4 / Speed for stage 4 (L)		0000
215 - 216	454	Switching torque for stage 4 / Speed for stage 4 (H)		0000
217 - 218	455	Switching torque for stage 5 / Speed for stage 5 (L)		0000
219 - 220	456	Switching torque for stage 5 / Speed for stage 5 (H)		0000
221 - 222	457	Switching torque for stage 6 / Speed for stage 6 (L)		0000
223 - 224	458	Switching torque for stage 6 / Speed for stage 6 (H)		0000
225 - 320	-	Reserved	-	-

Byte	Function code #760 (Hex)	Query type 11 (version number 0)	Description	Example (Hex)
21 - 22	7DB	Parameter ID	1 to 500	0100
23 - 24	7DC	Version number	0	0000
25 - 26	7DD	Torque unit	0: N.m; 1: kgf.cm; 2: lbf.ft; 3: lbf.in; 4: ozf.ft; 5: ozf.in; 6: cNm	0100
27 - 40	7DE - 7E4	Reserved	14 bytes	...
41 - 42	7E5	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
43 - 62	7E6 - 7EF	Reserved	20 bytes	...
63 - 64	7F0	Max. current for tightening	-	0000
65 - 66	7F1	Max. tightening time	Unit: ms	6400
67 - 68	7F2	Max. tightening angle	Unit: degree	FF7F
69 - 70	7F3	Reserved	-	0000
71 - 72	7F4	Delay before tightening starts	Unit: ms	0000
73 - 74	7F5	Min. tightening angle	Unit: degree	0000
75 - 84	7F6 - 7FA	Reserved	10 bytes	...
85 - 86	7FB	Max. loosening time	Unit: ms	6400
87 - 88	7FC	Max. loosening angle	Unit: degree	FF7F
89 - 90	7FD	Reserved	-	0000
91 - 92	7FE	Delay before loosening starts	Unit: ms	0000
93 - 102	7FF - 803	Reserved	10 bytes	...
103 - 104	804	ID of the prevail torque to be linked	1 to 500	0000
105 - 120	805 - 80C	Reserved	16 bytes	...

Self-defined strategy						Loosening		Query type 11 (version number 0)	Description	Example (Hex)
Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	1 st stage	2 nd stage			
Byte / Hex	Byte / Hex	Byte / Hex	Byte / Hex	Byte / Hex	Byte / Hex	Hex / Dec	Hex / Dec			
121 - 122 / 80D	221 - 222 / 83F	321 - 322 / 871	421 - 422 / 8A3	521 - 522 / 8D5	621 - 622 / 907	721 - 722 / 939	821 - 822 / 96B	Target On / Off (L)	Bit 0 to 1 Control method type 0: position 1: speed Bit 2 to 3 Operation after stage completion 0: stop 1: execute the next stage Bit 4 to 5 Control method 0: angle 1: torque 2: torque rate 3: yield point Bit 6 Rotation direction of the stage 0: CW 1: CCW Bit 7 to 8 Hold time switch of the final stage 0: Off 1: On Bit 9 Prevail torque 0: Off 1: On	0500
123 - 124 / 80E	223 - 224 / 840	323 - 324 / 872	423 - 424 / 8A4	523 - 524 / 8D6	623 - 624 / 908	723 - 724 / 93A	823 - 824 / 96C	Target On / Off (H)	-	0000
125 - 126 / 80F	225 - 226 / 841	325 - 326 / 873	425 - 426 / 8A5	525 - 526 / 8D7	625 - 626 / 909	725 - 726 / 93B	825 - 826 / 96D	Target torque	Unit: 0.001 N.m (used with Target On / Off Bit 4, 1: torque)	3200

Self-defined strategy						Loosening		Query type 11 (version number 0)	Description	Example (Hex)
Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	1 st stage	2 nd stage			
Byte / Hex	Byte / Hex	Byte / Hex	Byte / Hex	Byte / Hex	Byte / Hex	Hex / Dec	Hex / Dec			
127 - 128 / 810	227 - 228 / 842	327 - 328 / 874	427 - 428 / 8A6	527 - 528 / 8D8	627 - 628 / 90A	727 - 728 / 93C	827 - 828 / 96E	Rotation speed	Unit: rpm	5000
129 - 130 / 811	229 - 230 / 843	329 - 330 / 875	429 - 430 / 8A7	529 - 530 / 8D9	629 - 630 / 90B	729 - 730 / 93D	829 - 830 / 96F	Target angle	Unit: degree (used with Target On / Off Bit 4, 0: angle)	5A00
131 - 132 / 812	231 - 232 / 844	331 - 332 / 876	431 - 432 / 8A8	531 - 532 / 8DA	631 - 632 / 90C	731 - 732 / 93E	831 - 832 / 970	Hold / pause time after stage completion	Unit: ms	0000
133 - 134 / 813	233 - 234 / 845	333 - 334 / 877	433 - 434 / 8A9	533 - 534 / 8DB	633 - 634 / 90D	733 - 734 / 93F	833 - 834 / 971	Acceleration time	Unit: ms	1E00
135 - 136 / 814	235 - 236 / 846	335 - 336 / 878	435 - 436 / 8AA	535 - 536 / 8DC	635 - 636 / 90E	735 - 736 / 940	835 - 836 / 972	Deceleration time	Unit: ms	0A00
137 - 138 / 815	237 - 238 / 847	337 - 338 / 879	437 - 438 / 8AB	537 - 538 / 8DD	637 - 638 / 90F	737 - 738 / 941	837 - 838 / 973	Start torque for torque rate calculation	-	0000
139 - 140 / 816	239 - 240 / 848	339 - 340 / 87A	439 - 440 / 8AC	539 - 540 / 8DE	639 - 640 / 910	739 - 740 / 942	839 - 840 / 974	Angle interval for torque rate calculation	-	0000
141 - 142 / 817	241 - 242 / 849	341 - 342 / 87B	441 - 442 / 8AD	541 - 542 / 8DF	641 - 642 / 911	741 - 742 / 943	841 - 842 / 975	Target torque rate	Unit: 0.0001 N.m/degree (used with Target On / Off Bit 4, 2: torque rate)	0000
143 - 172 / 818 - 826	243 - 272 / 84A - 858	343 - 372 / 87C - 88A	443 - 472 / 8AE - 8BC	543 - 572 / 8E0 - 8EE	643 - 672 / 912 - 920	743 - 772 / 944 - 952	843 - 872 / 976 - 984	Reserved	30 bytes	...
173 - 174 / 827	273 - 274 / 859	373 - 374 / 88B	473 - 474 / 8BD	573 - 574 / 8EF	673 - 674 / 921	773 - 774 / 953	873 - 874 / 985	Max. angle	Used with Limit On / Off (L) Bit 2	0807
175 - 176 / 828	275 - 276 / 85A	375 - 376 / 88C	475 - 476 / 8BE	575 - 576 / 8F0	675 - 676 / 922	775 - 776 / 954	875 - 876 / 986	Min. angle	Used with Limit On / Off (L) Bit 2	0000
177 - 178 / 829	277 - 278 / 85B	377 - 378 / 88D	477 - 478 / 8BF	577 - 578 / 8F1	677 - 678 / 923	777 - 778 / 955	877 - 878 / 987	Max. clamp torque	Used with Limit On / Off (L) Bit 3	FA00
179 - 180 / 82A	279 - 280 / 85C	379 - 380 / 88E	479 - 480 / 8C0	579 - 580 / 8F2	679 - 680 / 924	779 - 780 / 956	879 - 880 / 988	Min. clamp torque	Used with Limit On / Off (L) Bit 3	0000
181 - 182 / 82B	281 - 282 / 85D	381 - 382 / 88F	481 - 482 / 8C1	581 - 582 / 8F3	681 - 682 / 925	781 - 782 / 957	881 - 882 / 989	Max. torque	Used with Limit On / Off (H) Bit 0	D503
183 - 184 / 82C	283 - 284 / 85E	383 - 384 / 890	483 - 484 / 8C2	583 - 584 / 8F4	683 - 684 / 926	783 - 784 / 958	883 - 884 / 98A	Min. torque	Used with Limit On / Off (H) Bit 0	0000
185 - 198 / 82D - 833	285 - 298 / 85F - 865	385 - 398 / 891 - 897	485 - 498 / 8C3 - 8C9	585 - 598 / 8F5 - 8FB	685 - 698 / 927 - 92D	785 - 798 / 959 - 95F	885 - 898 / 98B - 991	Reserved	14 bytes	...
199 - 200 / 834	299 - 300 / 866	399 - 400 / 898	499 - 500 / 8CA	599 - 600 / 8FC	699 - 700 / 92E	799 - 800 / 960	899 - 900 / 992	Max. operation time	Unit: ms Used with Limit On / Off (H) Bit 15	6400
201 - 202 / 835	301 - 302 / 867	401 - 402 / 899	501 - 502 / 8CB	601 - 602 / 8FD	701 - 702 / 92F	801 - 802 / 961	901 - 902 / 993	Min. operation time	Unit: ms Used with Limit On / Off (H) Bit 15	0000
203 - 206 / 836 - 837	303 - 306 / 868 - 869	403 - 406 / 89A - 89B	503 - 506 / 8CC - 8CD	603 - 606 / 8FE - 8FF	703 - 706 / 930 - 931	803 - 806 / 962 - 963	903 - 906 / 994 - 995	Reserved	4 bytes	...
207 - 208 / 838	307 - 308 / 86A	407 - 408 / 89C	507 - 508 / 8CE	607 - 608 / 900	707 - 708 / 932	807 - 808 / 964	907 - 908 / 996	Angle range for prevail torque calculation	Set the angle range (%) of the stage to calculate the average torque.	0000
209 - 212 / 839 - 83A	309 - 312 / 86B - 86C	409 - 412 / 89D - 89E	509 - 512 / 8CF - 8D0	609 - 612 / 901 - 902	709 - 712 / 933 - 934	809 - 812 / 965 - 966	909 - 912 / 997 - 998	Reserved	4 bytes	...

Self-defined strategy						Loosening		Query type 11 (version number 0)	Description	Example (Hex)
Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	1 st stage	2 nd stage			
Byte / Hex	Byte / Hex	Byte / Hex	Byte / Hex	Byte / Hex	Byte / Hex	Hex / Dec	Hex / Dec			
213 - 214 / 83B	313 - 314 / 86D	413 - 414 / 89F	513 - 514 / 8D1	613 - 614 / 903	713 - 714 / 935	813 - 814 / 967	913 - 914 / 999	Limit On / Off (L)	Bit 2: limit the angle	0000
215 - 216 / 83C	315 - 316 / 86E	415 - 416 / 8A0	515 - 516 / 8D2	615 - 616 / 904	715 - 716 / 936	815 - 816 / 968	915 - 916 / 99A	Limit On / Off (H)	Bit 0: limit the torque Bit 15: limit the operation time	0100
217 - 218 / 83D	317 - 318 / 86F	417 - 418 / 8A1	517 - 518 / 8D3	617 - 618 / 905	717 - 718 / 937	817 - 818 / 969	917 - 918 / 99B	Reserved	-	0000
219 - 220 / 83E	319 - 320 / 870	419 - 420 / 8A2	519 - 520 / 8D4	619 - 620 / 906	719 - 720 / 938	819 - 820 / 96A	919 - 920 / 99C	Stage code	10: start stage 20: rundown stage (angle / torque mode) 29: rundown stage (torque rate mode) 30: pre-tightening stage 40: tightening stage 90: first stage of loosening 91: second stage of loosening	0A00

Byte	Function code #760 (Hex)	Query type 11 (version number 1)	Description	Example (Hex)
21 - 22	7DB	Parameter ID	1 to 500	0100
23 - 24	7DC	Version number	1	0100
25 - 26	7DD	Torque unit	0: N.m; 1: kgf.cm; 2: lbf.ft; 3: lbf.in; 4: o1zf.ft; 5: ozf.in; 6: cNm	0100
27 - 32	7DE - 7E0	Reserved	6 bytes	...
33 - 34	7E1	1st stage loosening angle	-	6400
35 - 36	7E2	2nd stage loosening angle	-	FF7F
37 - 38	7E3	Switch for monitoring the min. loosening torque	-	0000
39 - 40	7E4	Reserved	-	0000
41 - 42	7E5	Tool 1 / Tool 2	0: Tool 1; 1: Tool 2	0000
43 - 62	7E6 - 7EF	Reserved	20 bytes	...
63 - 64	7F0	Max. current for tightening	-	0000
65 - 66	7F1	Max. tightening time	Unit: ms	6400
67 - 68	7F2	Max. tightening angle	Unit: degree	FF7F
69 - 70	7F3	Reserved	-	0000
71 - 72	7F4	Delay before tightening starts	Unit: ms	0000
73 - 74	7F5	Min. tightening angle	Unit: degree	0000
75 - 80	7F6 - 7F8	Reserved	6 bytes	...
81 - 82	7F9	Rotation direction of loosening	-	0000
83 - 84	7FA	Reserved	-	0000
85 - 86	7FB	Max. loosening time	Unit: ms	6400
87 - 88	7FC	Max. loosening angle	Unit: degree	FF7F
89 - 90	7FD	Reserved	-	0000
91 - 92	7FE	Delay before loosening starts	Unit: ms	0000
93 - 94	7FF	Min. loosening angle	Unit: degree	0000
95 - 98	800, 801	Reserved	4 bytes	...
99 - 100	802	Min. loosening torque (L)	Unit: 0.001 x torque at present	0000
101 - 102	803	Min. loosening torque (H)		0000

B

Byte	Function code #760 (Hex)	Query type 11 (version number 1)	Description	Example (Hex)
103 - 104	804	ID of the prevail torque to be linked	1 to 500	0000
105 - 116	805 - 80A	Reserved	12 bytes	...
117 - 118	80B	1st stage loosening speed	-	6400
119 - 120	80C	2nd stage loosening speed	-	F401

Self-defined strategy						Loosening		Query type 11 (version number 1)	Description		Example (Hex)
Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	1 st stage	2 nd stage				
Byte / Hex	Byte / Hex	Byte / Hex	Byte / Hex	Byte / Hex	Byte / Hex	Hex / Dec	Hex / Dec				
121 - 122 / 80D	221 - 222 / 83F	321 - 322 / 871	421 - 422 / 8A3	521 - 522 / 8D5	621 - 622 / 907	721 - 722 / 939	821 - 822 / 96B	Target On / Off (L)	Bit 0 to 1 Control method type	0: position 1: speed	0500
									Bit 2 to 3 Operation after stage completion	0: stop 1: execute the next stage	
									Bit 4 to 5 Control method	0: angle 1: torque 2: torque rate 3: yield point	
									Bit 6 Rotation direction of the stage	0: CW 1: CCW	
									Bit 7 to 8 Hold time switch of the final stage	0: Off 1: On	
									Bit 9 Prevail torque	0: Off 1: On	
									Bit 10 Reserved	-	
									Bit 11 Relative / absolute torque command	0: absolute 1: relative	
									Bit 12 Timing to add the relative torque	0: beginning of a stage 1: snug point	
123 - 124 / 80E	223 - 224 / 840	323 - 324 / 872	423 - 424 / 8A4	523 - 524 / 8D6	623 - 624 / 908	723 - 724 / 93A	823 - 824 / 96C	Target On / Off (H)	-	0000	
125 - 126 / 80F	225 - 226 / 841	325 - 326 / 873	425 - 426 / 8A5	525 - 526 / 8D7	625 - 626 / 909	725 - 726 / 93B	825 - 826 / 96D	Final target speed	-	0000	
127 - 128 / 810	227 - 228 / 842	327 - 328 / 874	427 - 428 / 8A6	527 - 528 / 8D8	627 - 628 / 90A	727 - 728 / 93C	827 - 828 / 96E	Rotation speed	Unit: rpm	5000	
129 - 130 / 811	229 - 230 / 843	329 - 330 / 875	429 - 430 / 8A7	529 - 530 / 8D9	629 - 630 / 90B	729 - 730 / 93D	829 - 830 / 96F	Target angle	Unit: degree (used with Target On / Off Bit 4, 0: angle)	5A00	
131 - 132 / 812	231 - 232 / 844	331 - 332 / 876	431 - 432 / 8A8	531 - 532 / 8DA	631 - 632 / 90C	731 - 732 / 93E	831 - 832 / 970	Hold / pause time after stage completion	Unit: ms	0000	
133 - 134 / 813	233 - 234 / 845	333 - 334 / 877	433 - 434 / 8A9	533 - 534 / 8DB	633 - 634 / 90D	733 - 734 / 93F	833 - 834 / 971	Acceleration time	Unit: ms	1E00	
135 - 136 / 814	235 - 236 / 846	335 - 336 / 878	435 - 436 / 8AA	535 - 536 / 8DC	635 - 636 / 90E	735 - 736 / 940	835 - 836 / 972	Deceleration time	Unit: ms	0A00	

Self-defined strategy						Loosening		Query type 11 (version number 1)	Description	Example (Hex)
Stage 1 Byte / Hex	Stage 2 Byte / Hex	Stage 3 Byte / Hex	Stage 4 Byte / Hex	Stage 5 Byte / Hex	Stage 6 Byte / Hex	1 st stage Hex / Dec	2 nd stage Hex / Dec			
137 - 138 / 815	237 - 238 / 847	337 - 338 / 879	437 - 438 / 8AB	537 - 538 / 8DD	637 - 638 / 90F	737 - 738 / 941	837 - 838 / 973	Acceleration time of 2nd stage	-	0000
139 - 140 / 816	239 - 240 / 848	339 - 340 / 87A	439 - 440 / 8AC	539 - 540 / 8DE	639 - 640 / 910	739 - 740 / 942	839 - 840 / 974	Angle interval for torque rate calculation	-	0000
141 - 142 / 817	241 - 242 / 849	341 - 342 / 87B	441 - 442 / 8AD	541 - 542 / 8DF	641 - 642 / 911	741 - 742 / 943	841 - 842 / 975	Final pause time	Unit: ms	0000
143 - 144 / 818	243 - 244 / 84A	343 - 344 / 87C	443 - 444 / 8AE	543 - 544 / 8E0	643 - 644 / 912	743 - 744 / 944	843 - 844 / 976	Reserved	-	0000
145 - 146 / 819	245 - 246 / 84B	345 - 346 / 87D	445 - 446 / 8AF	545 - 546 / 8E1	645 - 646 / 913	745 - 746 / 945	845 - 846 / 977	Target torque (L)	Unit: 0.001 x torque at present (used with Target On / Off Bit 4, 1: torque)	3200
147 - 148 / 81A	247 - 248 / 84C	347 - 348 / 87E	447 - 448 / 8B0	547 - 548 / 8E2	647 - 648 / 914	747 - 748 / 946	847 - 848 / 978	Target torque (H)		0000
149 - 152 / 81B - 81C	249 - 252 / 84D - 84E	349 - 352 / 87F - 880	449 - 452 / 8B1 - 8B2	549 - 552 / 8E3 - 8E4	649 - 652 / 915 - 916	749 - 752 / 947 - 948	849 - 852 / 979 - 97A	Reserved	4 bytes	0000
153 - 154 / 81D	253 - 254 / 84F	353 - 354 / 881	453 - 454 / 8B3	553 - 554 / 8E5	653 - 654 / 917	753 - 754 / 949	853 - 854 / 97B	Target torque rate (L)	Unit: 0.0001 x torque at present/degree (used with Target On / Off Bit 4, 2: torque rate)	0000
155 - 156 / 81E	255 - 256 / 850	355 - 356 / 882	455 - 456 / 8B4	555 - 556 / 8E6	655 - 656 / 918	755 - 756 / 94A	855 - 856 / 97C	Target torque rate (H)		0000
157 - 158 / 81F	257 - 258 / 851	357 - 358 / 883	457 - 458 / 8B5	557 - 558 / 8E7	657 - 658 / 919	757 - 758 / 94B	857 - 858 / 97D	Torque of 1st stage (L)	Unit: 0.001 x torque at present	0000
159 - 160 / 820	259 - 260 / 852	359 - 360 / 884	459 - 460 / 8B6	559 - 560 / 8E8	659 - 660 / 91A	759 - 760 / 94C	859 - 860 / 97E	Torque of 1st stage (H)		0000
161 - 164 / 821 - 822	261 - 264 / 853 - 854	361 - 364 / 885 - 886	461 - 464 / 8B7 - 8B8	561 - 564 / 8E9 - 8EA	661 - 664 / 91B - 91C	761 - 764 / 94D - 94E	861 - 864 / 97F - 980	Reserved	4 bytes	...
165 - 166 / 823	265 - 266 / 855	365 - 366 / 887	465 - 466 / 8B9	565 - 566 / 8EB	665 - 666 / 91D	765 - 766 / 94F	865 - 866 / 981	Max. clamp torque (L)	Unit: 0.001 x torque at present Used with Limit On / Off (L) Bit 3	FA00
167 - 168 / 824	267 - 268 / 856	367 - 368 / 888	467 - 468 / 8BA	567 - 568 / 8EC	667 - 668 / 91E	767 - 768 / 950	867 - 868 / 982	Max. clamp torque (H)		0000
169 - 170 / 825	269 - 270 / 857	369 - 370 / 889	469 - 470 / 8BB	569 - 570 / 8ED	669 - 670 / 91F	769 - 770 / 951	869 - 870 / 983	Min. clamp torque (L)	Unit: 0.001 x torque at present Used with Limit On / Off (L) Bit 3	0000
171 - 172 / 826	271 - 272 / 858	371 - 372 / 88A	471 - 472 / 8BC	571 - 572 / 8EE	671 - 672 / 920	771 - 772 / 952	871 - 872 / 984	Min. clamp torque (H)		0000
173 - 174 / 827	273 - 274 / 859	373 - 374 / 88B	473 - 474 / 8BD	573 - 574 / 8EF	673 - 674 / 921	773 - 774 / 953	873 - 874 / 985	Max. angle	Used with Limit On / Off (L) Bit 2	0807
175 - 176 / 828	275 - 276 / 85A	375 - 376 / 88C	475 - 476 / 8BE	575 - 576 / 8F0	675 - 676 / 922	775 - 776 / 954	875 - 876 / 986	Min. angle	Used with Limit On / Off (L) Bit 2	0000
177 - 178 / 829	277 - 278 / 85B	377 - 378 / 88D	477 - 478 / 8BF	577 - 578 / 8F1	677 - 678 / 923	777 - 778 / 955	877 - 878 / 987	Max. torque (L)	Unit: 0.001 x torque at present Used with Limit On / Off (H) Bit 0	D503
179 - 180 / 82A	279 - 280 / 85C	379 - 380 / 88E	479 - 480 / 8C0	579 - 580 / 8F2	679 - 680 / 924	779 - 780 / 956	879 - 880 / 988	Max. torque (H)		0000
181 - 182 / 82B	281 - 282 / 85D	381 - 382 / 88F	481 - 482 / 8C1	581 - 582 / 8F3	681 - 682 / 925	781 - 782 / 957	881 - 882 / 989	Min. torque (L)	Unit: 0.001 x torque at present Used with Limit On / Off (H) Bit 0	0000
183 - 184 / 82C	283 - 284 / 85E	383 - 384 / 890	483 - 484 / 8C2	583 - 584 / 8F4	683 - 684 / 926	783 - 784 / 958	883 - 884 / 98A	Min. torque (H)		0000

B

Self-defined strategy						Loosening		Query type 11 (version number 1)	Description	Example (Hex)
Stage 1 Byte / Hex	Stage 2 Byte / Hex	Stage 3 Byte / Hex	Stage 4 Byte / Hex	Stage 5 Byte / Hex	Stage 6 Byte / Hex	1 st stage Hex / Dec	2 nd stage Hex / Dec			
185 - 198 / 82D - 833	285 - 298 / 85F - 865	385 - 398 / 891 - 897	485 - 498 / 8C3 - 8C9	585 - 598 / 8F5 - 8FB	685 - 698 / 927 - 92D	785 - 798 / 959 - 95F	885 - 898 / 98B - 991	Reserved	14 bytes	...
199 - 200 / 834	299 - 300 / 866	399 - 400 / 898	499 - 500 / 8CA	599 - 600 / 8FC	699 - 700 / 92E	799 - 800 / 960	899 - 900 / 992	Max. operation time	Unit: ms Used with Limit On / Off (H) Bit 15	6400
201 - 202 / 835	301 - 302 / 867	401 - 402 / 899	501 - 502 / 8CB	601 - 602 / 8FD	701 - 702 / 92F	801 - 802 / 961	901 - 902 / 993	Min. operation time	Unit: ms Used with Limit On / Off (H) Bit 15	0000
203 - 206 / 836 - 837	303 - 306 / 868 - 869	403 - 406 / 89A - 89B	503 - 506 / 8CC - 8CD	603 - 606 / 8FE - 8FF	703 - 706 / 930 - 931	803 - 806 / 962 - 963	903 - 906 / 994 - 995	Reserved	4 bytes	...
207 - 208 / 838	307 - 308 / 86A	407 - 408 / 89C	507 - 508 / 8CE	607 - 608 / 900	707 - 708 / 932	807 - 808 / 964	907 - 908 / 996	Angle range for prevail torque calculation	Set the angle range (%) of the stage to calculate the average torque.	0000
209 - 212 / 839 - 83A	309 - 312 / 86B - 86C	409 - 412 / 89D - 89E	509 - 512 / 8CF - 8D0	609 - 612 / 901 - 902	709 - 712 / 933 - 934	809 - 812 / 965 - 966	909 - 912 / 997 - 998	Reserved	4 bytes	...
213 - 214 / 83B	313 - 314 / 86D	413 - 414 / 89F	513 - 514 / 8D1	613 - 614 / 903	713 - 714 / 935	813 - 814 / 967	913 - 914 / 999	Limit On / Off (L)	Bit 2: limit the angle	0000
215 - 216 / 83C	315 - 316 / 86E	415 - 416 / 8A0	515 - 516 / 8D2	615 - 616 / 904	715 - 716 / 936	815 - 816 / 968	915 - 916 / 99A	Limit On / Off (H)	Bit 0: limit the torque Bit 15: limit the operation time	0100
217 - 218 / 83D	317 - 318 / 86F	417 - 418 / 8A1	517 - 518 / 8D3	617 - 618 / 905	717 - 718 / 937	817 - 818 / 969	917 - 918 / 99B	Reserved	-	0000
219 - 220 / 83E	319 - 320 / 870	419 - 420 / 8A2	519 - 520 / 8D4	619 - 620 / 906	719 - 720 / 938	819 - 820 / 96A	919 - 920 / 99C	Stage code	10: start stage 20: rundown stage (angle / torque mode) 29: rundown stage (torque rate mode) 30: pre-tightening stage 40: tightening stage 90: 1st stage of loosening 91: 2nd stage of loosening	0A00

Byte	Query type 20	Example (Hex)	Query type 21	Example (Hex)
21 - 22	Time scale 2001 data	0000	Angle scale 2001 data	0000
23 - 24	Time scale 2002 data	0000	Angle scale 2002 data	0000
25 - 26	Time scale 2003 data	0000	Angle scale 2003 data	0000
27 - 28	Time scale 2004 data	0000	Angle scale 2004 data	0000
29 - 30	Time scale 2005 data	0000	Angle scale 2005 data	0000
31 - 32	Time scale 2006 data	0000	Angle scale 2006 data	0000
33 - 34	Time scale 2007 data	0000	Angle scale 2007 data	0000
35 - 36	Time scale 2008 data	0000	Angle scale 2008 data	0000
37 - 38	Time scale 2009 data	0000	Angle scale 2009 data	0000
...	Time scale N data	...	Angle scale N data	...
4001 - 4002	Time scale 3991 data	0000	Angle scale 3991 data	0000
4003 - 4004	Time scale 3992 data	0000	Angle scale 3992 data	0000
4005 - 4006	Time scale 3993 data	0000	Angle scale 3993 data	0000
4007 - 4008	Time scale 3994 data	0000	Angle scale 3994 data	0000
4009 - 4010	Time scale 3995 data	0000	Angle scale 3995 data	0000
4011 - 4012	Time scale 3996 data	0000	Angle scale 3996 data	0000
4013 - 4014	Time scale 3997 data	0000	Angle scale 3997 data	0000
4015 - 4016	Time scale 3998 data	0000	Angle scale 3998 data	0000
4017 - 4018	Time scale 3999 data	0000	Angle scale 3999 data	0000
4019 - 4020	Time scale 4000 data	0000	Angle scale 4000 data	0000

Byte	Query type 22	Example (Hex)	Query type 23	Example (Hex)
21 - 22	Torque scale 2001 data	0000	Torque rate scale 2001 data	0000
23 - 24	Torque scale 2002 data	0000	Torque rate scale 2002 data	0000
25 - 26	Torque scale 2003 data	0000	Torque rate scale 2003 data	0000
27 - 28	Torque scale 2004 data	0000	Torque rate scale 2004 data	0000
29 - 30	Torque scale 2005 data	0000	Torque rate scale 2005 data	0000
31 - 32	Torque scale 2006 data	0000	Torque rate scale 2006 data	0000
33 - 34	Torque scale 2007 data	0000	Torque rate scale 2007 data	0000
35 - 36	Torque scale 2008 data	0000	Torque rate scale 2008 data	0000
37 - 38	Torque scale 2009 data	0000	Torque rate scale 2009 data	0000
...	Torque scale N data	...	Torque rate scale N data	...
4001 - 4002	Torque scale 3991 data	0000	Torque rate scale 3991 data	0000
4003 - 4004	Torque scale 3992 data	0000	Torque rate scale 3992 data	0000
4005 - 4006	Torque scale 3993 data	0000	Torque rate scale 3993 data	0000
4007 - 4008	Torque scale 3994 data	0000	Torque rate scale 3994 data	0000
4009 - 4010	Torque scale 3995 data	0000	Torque rate scale 3995 data	0000
4011 - 4012	Torque scale 3996 data	0000	Torque rate scale 3996 data	0000
4013 - 4014	Torque scale 3997 data	0000	Torque rate scale 3997 data	0000
4015 - 4016	Torque scale 3998 data	0000	Torque rate scale 3998 data	0000
4017 - 4018	Torque scale 3999 data	0000	Torque rate scale 3999 data	0000
4019 - 4020	Torque scale 4000 data	0000	Torque rate scale 4000 data	0000

B

Byte	Query type 24	Example (Hex)	Query type 25	Example (Hex)
21 - 22	Torque scale 2001 data (L)	0000	Torque scale 3001 data (L)	0000
23 - 24	Torque scale 2001 data (H)	0000	Torque scale 3001 data (H)	0000
25 - 26	Torque scale 2002 data (L)	0000	Torque scale 3002 data (L)	0000
27 - 28	Torque scale 2002 data (H)	0000	Torque scale 3002 data (H)	0000
29 - 30	Torque scale 2003 data (L)	0000	Torque scale 3003 data (L)	0000
31 - 32	Torque scale 2003 data (H)	0000	Torque scale 3003 data (H)	0000
33 - 34	Torque scale 2004 data (L)	0000	Torque scale 3004 data (L)	0000
35 - 36	Torque scale 2004 data (H)	0000	Torque scale 3004 data (H)	0000
...	Torque scale N data (L)	...	Torque scale N data (L)	...
...	Torque scale N data (H)	...	Torque scale N data (H)	...
4001 - 4002	Torque scale 2996 data (L)	0000	Torque scale 3996 data (L)	0000
4003 - 4004	Torque scale 2996 data (H)	0000	Torque scale 3996 data (H)	0000
4005 - 4006	Torque scale 2997 data (L)	0000	Torque scale 3997 data (L)	0000
4007 - 4008	Torque scale 2997 data (H)	0000	Torque scale 3997 data (H)	0000
4009 - 4010	Torque scale 2998 data (L)	0000	Torque scale 3998 data (L)	0000
4011 - 4012	Torque scale 2998 data (H)	0000	Torque scale 3998 data (H)	0000
4013 - 4014	Torque scale 2999 data (L)	0000	Torque scale 3999 data (L)	0000
4015 - 4016	Torque scale 2999 data (H)	0000	Torque scale 3999 data (H)	0000
4017 - 4018	Torque scale 3000 data (L)	0000	Torque scale 4000 data (L)	0000
4019 - 4020	Torque scale 3000 data (H)	0000	Torque scale 4000 data (H)	0000

Byte	Query type 26	Example (Hex)	Query type 27	Example (Hex)
21 - 22	Torque rate scale 2001 data (L)	0000	Torque rate scale 3001 data (L)	0000
23 - 24	Torque rate scale 2001 data (H)	0000	Torque rate scale 3001 data (H)	0000
25 - 26	Torque rate scale 2002 data (L)	0000	Torque rate scale 3002 data (L)	0000
27 - 28	Torque rate scale 2002 data (H)	0000	Torque rate scale 3002 data (H)	0000
29 - 30	Torque rate scale 2003 data (L)	0000	Torque rate scale 3003 data (L)	0000
31 - 32	Torque rate scale 2003 data (H)	0000	Torque rate scale 3003 data (H)	0000
33 - 34	Torque rate scale 2004 data (L)	0000	Torque rate scale 3004 data (L)	0000
35 - 36	Torque rate scale 2004 data (H)	0000	Torque rate scale 3004 data (H)	0000
...	Torque rate scale N data (L)	...	Torque rate scale N data (L)	...
...	Torque rate scale N data (H)	...	Torque rate scale N data (H)	...
4001 - 4002	Torque rate scale 2996 data (L)	0000	Torque rate scale 3996 data (L)	0000
4003 - 4004	Torque rate scale 2996 data (H)	0000	Torque rate scale 3996 data (H)	0000
4005 - 4006	Torque rate scale 2997 data (L)	0000	Torque rate scale 3997 data (L)	0000
4007 - 4008	Torque rate scale 2997 data (H)	0000	Torque rate scale 3997 data (H)	0000
4009 - 4010	Torque rate scale 2998 data (L)	0000	Torque rate scale 3998 data (L)	0000
4011 - 4012	Torque rate scale 2998 data (H)	0000	Torque rate scale 3998 data (H)	0000
4013 - 4014	Torque rate scale 2999 data (L)	0000	Torque rate scale 3999 data (L)	0000
4015 - 4016	Torque rate scale 2999 data (H)	0000	Torque rate scale 3999 data (H)	0000
4017 - 4018	Torque rate scale 3000 data (L)	0000	Torque rate scale 4000 data (L)	0000
4019 - 4020	Torque rate scale 3000 data (H)	0000	Torque rate scale 4000 data (H)	0000

Byte	Query type 30	Example (Hex)	Query type 31	Example (Hex)
21 - 22	Time scale 4001 data	0000	Angle scale 4001 data	0000
23 - 24	Time scale 4002 data	0000	Angle scale 4002 data	0000
25 - 26	Time scale 4003 data	0000	Angle scale 4003 data	0000
27 - 28	Time scale 4004 data	0000	Angle scale 4004 data	0000
29 - 30	Time scale 4005 data	0000	Angle scale 4005 data	0000
31 - 32	Time scale 4006 data	0000	Angle scale 4006 data	0000
33 - 34	Time scale 4007 data	0000	Angle scale 4007 data	0000
35 - 36	Time scale 4008 data	0000	Angle scale 4008 data	0000
37 - 38	Time scale 4009 data	0000	Angle scale 4009 data	0000
...	Time scale N data	...	Angle scale N data	...
4001 - 4002	Time scale 5991 data	0000	Angle scale 5991 data	0000
4003 - 4004	Time scale 5992 data	0000	Angle scale 5992 data	0000
4005 - 4006	Time scale 5993 data	0000	Angle scale 5993 data	0000
4007 - 4008	Time scale 5994 data	0000	Angle scale 5994 data	0000
4009 - 4010	Time scale 5995 data	0000	Angle scale 5995 data	0000
4011 - 4012	Time scale 5996 data	0000	Angle scale 5996 data	0000
4013 - 4014	Time scale 5997 data	0000	Angle scale 5997 data	0000
4015 - 4016	Time scale 5998 data	0000	Angle scale 5998 data	0000
4017 - 4018	Time scale 5999 data	0000	Angle scale 5999 data	0000
4019 - 4020	Time scale 6000 data	0000	Angle scale 6000 data	0000

Byte	Query type 32	Example (Hex)	Query type 33	Example (Hex)
21 - 22	Torque scale 4001 data	0000	Torque rate scale 4001 data	0000
23 - 24	Torque scale 4002 data	0000	Torque rate scale 4002 data	0000
25 - 26	Torque scale 4003 data	0000	Torque rate scale 4003 data	0000
27 - 28	Torque scale 4004 data	0000	Torque rate scale 4004 data	0000
29 - 30	Torque scale 4005 data	0000	Torque rate scale 4005 data	0000
31 - 32	Torque scale 4006 data	0000	Torque rate scale 4006 data	0000
33 - 34	Torque scale 4007 data	0000	Torque rate scale 4007 data	0000
35 - 36	Torque scale 4008 data	0000	Torque rate scale 4008 data	0000
37 - 38	Torque scale 4009 data	0000	Torque rate scale 4009 data	0000
...	Torque scale N data	...	Torque rate scale N data	...
4001 - 4002	Torque scale 5991 data	0000	Torque rate scale 5991 data	0000
4003 - 4004	Torque scale 5992 data	0000	Torque rate scale 5992 data	0000
4005 - 4006	Torque scale 5993 data	0000	Torque rate scale 5993 data	0000
4007 - 4008	Torque scale 5994 data	0000	Torque rate scale 5994 data	0000
4009 - 4010	Torque scale 5995 data	0000	Torque rate scale 5995 data	0000
4011 - 4012	Torque scale 5996 data	0000	Torque rate scale 5996 data	0000
4013 - 4014	Torque scale 5997 data	0000	Torque rate scale 5997 data	0000
4015 - 4016	Torque scale 5998 data	0000	Torque rate scale 5998 data	0000
4017 - 4018	Torque scale 5999 data	0000	Torque rate scale 5999 data	0000
4019 - 4020	Torque scale 6000 data	0000	Torque rate scale 6000 data	0000

Byte	Query type 34	Example (Hex)	Query type 35	Example (Hex)
21 - 22	Torque scale 4001 data (L)	0000	Torque scale 5001 data (L)	0000
23 - 24	Torque scale 4001 data (H)	0000	Torque scale 5001 data (H)	0000
25 - 26	Torque scale 4002 data (L)	0000	Torque scale 5002 data (L)	0000
27 - 28	Torque scale 4002 data (H)	0000	Torque scale 5002 data (H)	0000
29 - 30	Torque scale 4003 data (L)	0000	Torque scale 5003 data (L)	0000
31 - 32	Torque scale 4003 data (H)	0000	Torque scale 5003 data (H)	0000
33 - 34	Torque scale 4004 data (L)	0000	Torque scale 5004 data (L)	0000
35 - 36	Torque scale 4004 data (H)	0000	Torque scale 5004 data (H)	0000
...	Torque scale N data (L)	...	Torque scale N data (L)	...
...	Torque scale N data (H)	...	Torque scale N data (H)	...
4001 - 4002	Torque scale 4996 data (L)	0000	Torque scale 5996 data (L)	0000
4003 - 4004	Torque scale 4996 data (H)	0000	Torque scale 5996 data (H)	0000
4005 - 4006	Torque scale 4997 data (L)	0000	Torque scale 5997 data (L)	0000
4007 - 4008	Torque scale 4997 data (H)	0000	Torque scale 5997 data (H)	0000
4009 - 4010	Torque scale 4998 data (L)	0000	Torque scale 5998 data (L)	0000
4011 - 4012	Torque scale 4998 data (H)	0000	Torque scale 5998 data (H)	0000
4013 - 4014	Torque scale 4999 data (L)	0000	Torque scale 5999 data (L)	0000
4015 - 4016	Torque scale 4999 data (H)	0000	Torque scale 5999 data (H)	0000
4017 - 4018	Torque scale 5000 data (L)	0000	Torque scale 6000 data (L)	0000
4019 - 4020	Torque scale 5000 data (H)	0000	Torque scale 6000 data (H)	0000

Byte	Query type 36	Example (Hex)	Query type 37	Example (Hex)
21 - 22	Torque rate scale 4001 data (L)	0000	Torque rate scale 5001 data (L)	0000
23 - 24	Torque rate scale 4001 data (H)	0000	Torque rate scale 5001 data (H)	0000
25 - 26	Torque rate scale 4002 data (L)	0000	Torque rate scale 5002 data (L)	0000
27 - 28	Torque rate scale 4002 data (H)	0000	Torque rate scale 5002 data (H)	0000
29 - 30	Torque rate scale 4003 data (L)	0000	Torque rate scale 5003 data (L)	0000
31 - 32	Torque rate scale 4003 data (H)	0000	Torque rate scale 5003 data (H)	0000
33 - 34	Torque rate scale 4004 data (L)	0000	Torque rate scale 5004 data (L)	0000
35 - 36	Torque rate scale 4004 data (H)	0000	Torque rate scale 5004 data (H)	0000
...	Torque rate scale N data (L)	...	Torque rate scale N data (L)	...
...	Torque rate scale N data (H)	...	Torque rate scale N data (H)	...
4001 - 4002	Torque rate scale 4996 data (L)	0000	Torque rate scale 5996 data (L)	0000
4003 - 4004	Torque rate scale 4996 data (H)	0000	Torque rate scale 5996 data (H)	0000
4005 - 4006	Torque rate scale 4997 data (L)	0000	Torque rate scale 5997 data (L)	0000
4007 - 4008	Torque rate scale 4997 data (H)	0000	Torque rate scale 5997 data (H)	0000
4009 - 4010	Torque rate scale 4998 data (L)	0000	Torque rate scale 5998 data (L)	0000
4011 - 4012	Torque rate scale 4998 data (H)	0000	Torque rate scale 5998 data (H)	0000
4013 - 4014	Torque rate scale 4999 data (L)	0000	Torque rate scale 5999 data (L)	0000
4015 - 4016	Torque rate scale 4999 data (H)	0000	Torque rate scale 5999 data (H)	0000
4017 - 4018	Torque rate scale 5000 data (L)	0000	Torque rate scale 6000 data (L)	0000
4019 - 4020	Torque rate scale 5000 data (H)	0000	Torque rate scale 6000 data (H)	0000

Byte	Query type 40	Example (Hex)	Query type 41	Example (Hex)
21 - 22	Time scale 6001 data	0000	Angle scale 6001 data	0000
23 - 24	Time scale 6002 data	0000	Angle scale 6002 data	0000
25 - 26	Time scale 6003 data	0000	Angle scale 6003 data	0000
27 - 28	Time scale 6004 data	0000	Angle scale 6004 data	0000
29 - 30	Time scale 6005 data	0000	Angle scale 6005 data	0000
31 - 32	Time scale 6006 data	0000	Angle scale 6006 data	0000
33 - 34	Time scale 6007 data	0000	Angle scale 6007 data	0000
35 - 36	Time scale 6008 data	0000	Angle scale 6008 data	0000
37 - 38	Time scale 6009 data	0000	Angle scale 6009 data	0000
...	Time scale N data	...	Angle scale N data	...
4001 - 4002	Time scale 7991 data	0000	Angle scale 7991 data	0000
4003 - 4004	Time scale 7992 data	0000	Angle scale 7992 data	0000
4005 - 4006	Time scale 7993 data	0000	Angle scale 7993 data	0000
4007 - 4008	Time scale 7994 data	0000	Angle scale 7994 data	0000
4009 - 4010	Time scale 7995 data	0000	Angle scale 7995 data	0000
4011 - 4012	Time scale 7996 data	0000	Angle scale 7996 data	0000
4013 - 4014	Time scale 7997 data	0000	Angle scale 7997 data	0000
4015 - 4016	Time scale 7998 data	0000	Angle scale 7998 data	0000
4017 - 4018	Time scale 7999 data	0000	Angle scale 7999 data	0000
4019 - 4020	Time scale 8000 data	0000	Angle scale 8000 data	0000

Byte	Query type 42	Example (Hex)	Query type 43	Example (Hex)
21 - 22	Torque scale 6001 data	0000	Torque rate scale 6001 data	0000
23 - 24	Torque scale 6002 data	0000	Torque rate scale 6002 data	0000
25 - 26	Torque scale 6003 data	0000	Torque rate scale 6003 data	0000
27 - 28	Torque scale 6004 data	0000	Torque rate scale 6004 data	0000
29 - 30	Torque scale 6005 data	0000	Torque rate scale 6005 data	0000
31 - 32	Torque scale 6006 data	0000	Torque rate scale 6006 data	0000
33 - 34	Torque scale 6007 data	0000	Torque rate scale 6007 data	0000
35 - 36	Torque scale 6008 data	0000	Torque rate scale 6008 data	0000
37 - 38	Torque scale 6009 data	0000	Torque rate scale 6009 data	0000
...	Torque scale N data	...	Torque rate scale N data	...
4001 - 4002	Torque scale 7991 data	0000	Torque rate scale 7991 data	0000
4003 - 4004	Torque scale 7992 data	0000	Torque rate scale 7992 data	0000
4005 - 4006	Torque scale 7993 data	0000	Torque rate scale 7993 data	0000
4007 - 4008	Torque scale 7994 data	0000	Torque rate scale 7994 data	0000
4009 - 4010	Torque scale 7995 data	0000	Torque rate scale 7995 data	0000
4011 - 4012	Torque scale 7996 data	0000	Torque rate scale 7996 data	0000
4013 - 4014	Torque scale 7997 data	0000	Torque rate scale 7997 data	0000
4015 - 4016	Torque scale 7998 data	0000	Torque rate scale 7998 data	0000
4017 - 4018	Torque scale 7999 data	0000	Torque rate scale 7999 data	0000
4019 - 4020	Torque scale 8000 data	0000	Torque rate scale 8000 data	0000

B

B

Byte	Query type 44	Example (Hex)	Query type 45	Example (Hex)
21 - 22	Torque scale 6001 data (L)	0000	Torque scale 7001 data (L)	0000
23 - 24	Torque scale 6001 data (H)	0000	Torque scale 7001 data (H)	0000
25 - 26	Torque scale 6002 data (L)	0000	Torque scale 7002 data (L)	0000
27 - 28	Torque scale 6002 data (H)	0000	Torque scale 7002 data (H)	0000
29 - 30	Torque scale 6003 data (L)	0000	Torque scale 7003 data (L)	0000
31 - 32	Torque scale 6003 data (H)	0000	Torque scale 7003 data (H)	0000
33 - 34	Torque scale 6004 data (L)	0000	Torque scale 7004 data (L)	0000
35 - 36	Torque scale 6004 data (H)	0000	Torque scale 7004 data (H)	0000
37 - 38	Torque scale N data (L)	...	Torque scale N data (L)	...
...	Torque scale N data (H)	...	Torque scale N data (H)	...
4001 - 4002	Torque scale 6996 data (L)	0000	Torque scale 7996 data (L)	0000
4003 - 4004	Torque scale 6996 data (H)	0000	Torque scale 7996 data (H)	0000
4005 - 4006	Torque scale 6997 data (L)	0000	Torque scale 7997 data (L)	0000
4007 - 4008	Torque scale 6997 data (H)	0000	Torque scale 7997 data (H)	0000
4009 - 4010	Torque scale 6998 data (L)	0000	Torque scale 7998 data (L)	0000
4011 - 4012	Torque scale 6998 data (H)	0000	Torque scale 7998 data (H)	0000
4013 - 4014	Torque scale 6999 data (L)	0000	Torque scale 7999 data (L)	0000
4015 - 4016	Torque scale 6999 data (H)	0000	Torque scale 7999 data (H)	0000
4017 - 4018	Torque scale 7000 data (L)	0000	Torque scale 8000 data (L)	0000
4019 - 4020	Torque scale 7000 data (H)	0000	Torque scale 8000 data (H)	0000

Byte	Query type 46	Example (Hex)	Query type 47	Example (Hex)
21 - 22	Torque rate scale 6001 data (L)	0000	Torque rate scale 7001 data (L)	0000
23 - 24	Torque rate scale 6001 data (H)	0000	Torque rate scale 7001 data (H)	0000
25 - 26	Torque rate scale 6002 data (L)	0000	Torque rate scale 7002 data (L)	0000
27 - 28	Torque rate scale 6002 data (H)	0000	Torque rate scale 7002 data (H)	0000
29 - 30	Torque rate scale 6003 data (L)	0000	Torque rate scale 7003 data (L)	0000
31 - 32	Torque rate scale 6003 data (H)	0000	Torque rate scale 7003 data (H)	0000
33 - 34	Torque rate scale 6004 data (L)	0000	Torque rate scale 7004 data (L)	0000
35 - 36	Torque rate scale 6004 data (H)	0000	Torque rate scale 7004 data (H)	0000
37 - 38	Torque rate scale N data (L)	...	Torque rate scale N data (L)	...
...	Torque rate scale N data (H)	...	Torque rate scale N data (H)	...
4001 - 4002	Torque rate scale 6996 data (L)	0000	Torque rate scale 7996 data (L)	0000
4003 - 4004	Torque rate scale 6996 data (H)	0000	Torque rate scale 7996 data (H)	0000
4005 - 4006	Torque rate scale 6997 data (L)	0000	Torque rate scale 7997 data (L)	0000
4007 - 4008	Torque rate scale 6997 data (H)	0000	Torque rate scale 7997 data (H)	0000
4009 - 4010	Torque rate scale 6998 data (L)	0000	Torque rate scale 7998 data (L)	0000
4011 - 4012	Torque rate scale 6998 data (H)	0000	Torque rate scale 7998 data (H)	0000
4013 - 4014	Torque rate scale 6999 data (L)	0000	Torque rate scale 7999 data (L)	0000
4015 - 4016	Torque rate scale 6999 data (H)	0000	Torque rate scale 7999 data (H)	0000
4017 - 4018	Torque rate scale 7000 data (L)	0000	Torque rate scale 8000 data (L)	0000
4019 - 4020	Torque rate scale 7000 data (H)	0000	Torque rate scale 8000 data (H)	0000

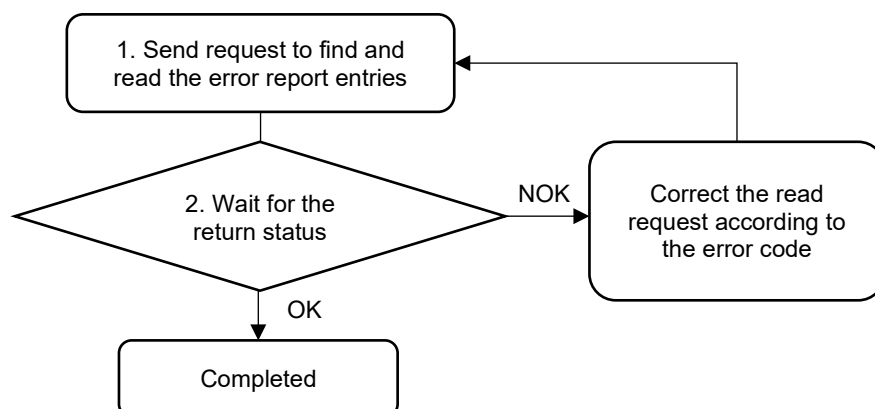
#752 Find and read the error report entries

Content description:

Use function code #752 with bytes 231 - 232, function code #20, or function code #21 to read the content of the error report.

Byte	Function code #20 (Hex)	Function code #21 (Hex)	Tool 1 / Tool 2 common status
231 - 232	74	6A	Current No. of error report entries

Handshake signal description:



1. Send the request to find and read the error report entries.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	752	F002
3 - 4	Version number	-	0000
5 - 6	Error report ID	1 to 60000 0: the latest error report	0100
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the content of the error report.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	752	F002
3 - 4	Return data length	Unit: Word	1100
5 - 6	Error report ID	1 to 60000 0: the latest error report	0100
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	752	F002
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000
21 - 22	Date / Time	Year	E607
23 - 24		Month	0B00
25 - 26		Day	1C00
27 - 28		Hour	1200
29 - 30		Minute	1100
31 - 32		Second	1000

Byte	Return status	Description	Example (Hex)
33 - 34	AL / NG code	Refer to CH12*	0230

Note: 0x1001 - 0x1999: Tool 1 device alarm (AL); 0x2000 - 0x2999: Tool 2 device alarm (AL).

0x3001 - 0x3999: Tool 1 tightening error (NG); 0x4000 - 0x4999: Tool 2 tightening error (NG).

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Error report ID	Exceeds the range of 0 to 60000

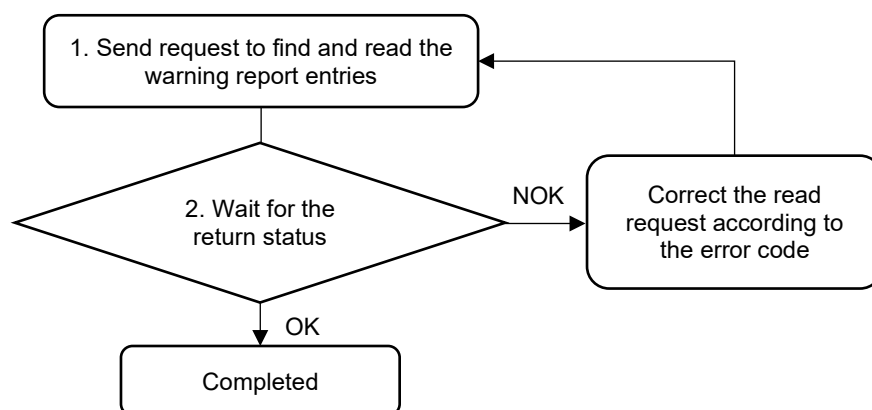
#753 Find and read the warning report entries

Content description:

Use function code #753 with bytes 233 - 234, function code #20, or function code #21 to read the content of the warning report.

Byte	Function code #20 (Hex)	Function code #21 (Hex)	Tool 1 / Tool 2 common status
233 - 234	75	6B	Current No. of warning report entries

Handshake signal description:



1. Send the request to find and read the warning report entries.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	753	F102
3 - 4	Version number	-	0000
5 - 6	Warning report ID	1 to 60000 0: the latest warning report	0100
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the content of the warning report.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	753	F102
3 - 4	Return data length	Unit: Word	1100
5 - 6	Warning report ID	1 to 60000 0: the latest warning report	0100
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	753	F102
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000
21 - 22	Date / Time	Year	E607
23 - 24		Month	0B00
25 - 26		Day	1C00
27 - 28		Hour	1200
29 - 30		Minute	1100
31 - 32		Second	1000
33 - 34	WN code	Refer to CH12*	5005

Note: 0x5001 - 0x5999: Tool 1 operation warning (WN); 0x6000 - 0x6999: Tool 2 operation warning (WN).

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Warning report ID	Exceeds the range of 0 to 60000

B

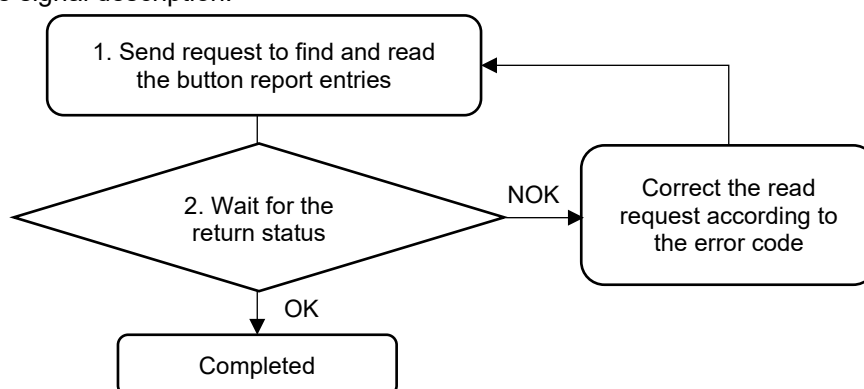
#754 Find and read the button report entries

Content description:

Use function code #754 with bytes 239 - 240 and bytes 241 - 242, function code #20, or function code #21 to read the content of the button report.

Byte	Function code #20 (Hex)	Function code #21 (Hex)	Tool 1 / Tool 2 common status
239 - 240	78	6E	Current No. of button report entries (L)
241 - 242	79	6F	Current No. of button report entries (H)

Handshake signal description:



1. Send the request to find and read the button report entries.

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	754	F202
3 - 4	Version number	-	0000
5 - 6	Button report ID	1 to 60000 0: the latest button report	0100
7 - 12	Reserved	6 bytes	...
13 - 14	Request to send command	-	0100

2. Check whether the reading is successful from the return status and read the content of the button report.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	754	F202
3 - 4	Return data length	Unit: Word	1600
5 - 6	Button report ID	1 to 60000 0: the latest button report	0100
7 - 14	Reserved	8 bytes	...
15 - 16	Return function code	754	F202
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000
21 - 22	Date / Time	Year	E607
23 - 24		Month	0B00
25 - 26		Day	1E00
27 - 28		Hour	0800
29 - 30		Minute	3500
31 - 32		Second	3300
33 - 34	Button ID	Refer to Section 11.4 Button Report.	581B
35 - 36	Value before change (L)	0 to 4294967295	0000
37 - 38	Value before change (H)		0000
39 - 40	Value after change (L)	0 to 4294967295	0100
41 - 42	Value after change (H)		0000
43 - 44	Permissions account	0: reserved 1: User1 2: User2 3: User3 4: User4 5: User5 6: Admin	0100

If successful, the read status is 1.

If failed, the read status is 2. The error code is as follows:

Code	Error description	Description
1	Button report ID	Exceeds the range of 0 to 60000

B

#760 Read the sorted production reports

Content description:

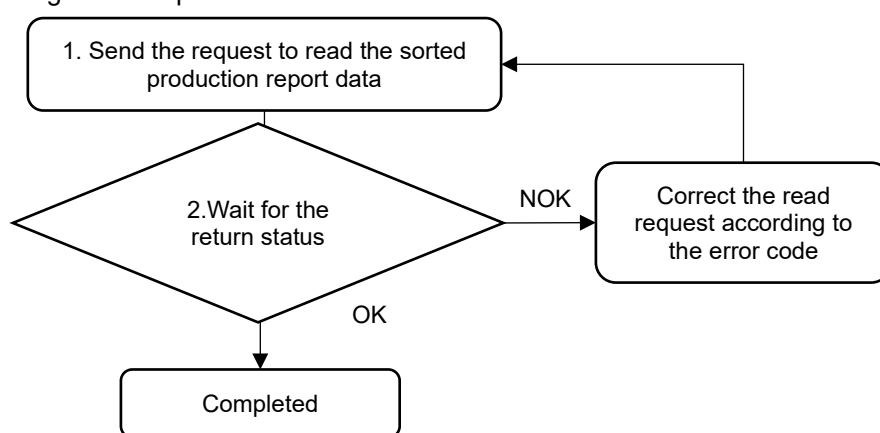
Use function code #760 to sort the data of #750 and the data of #751 query types 10 and 11 into sequential data sets and read them.

- A. Use bytes 235 - 236 and bytes 237 - 238, function code #20, or function code #21 to read the data of the latest entry of production report.

Byte	Function code #20 (Hex)	Function code #21 (Hex)	Tool 1 / Tool 2 common status
235 - 236	76	6C	Current No. of production report entries (L)
237 - 238	77	6D	Current No. of production report entries (H)

- B. Read the content of history production report.

Handshake signal description:



1. Send the request to read the sorted production report data.

Byte	Function	Description	Example (Hex)
1 - 2	Function code	760	F802
3 - 4	Version number	0	0000
5 - 6	Production report ID (L)	1 to 200000	0100
7 - 8	Production report ID (H)	0: the latest production report	0000
9 - 12	Reserved	4 bytes	...
13 - 14	Request to send command	-	0100
15 - 20	Reserved	6 bytes	...
21 - 22	Specifying query types 10 and 11 of #750 and #751 - code 1	0: Off 11 - 260: refer to function #750 for the function list. 1011 - 1260: refer to query type 10 of #751 for the function list 2011 - 2260: refer to query type 11 of #751 for the function list	6F00
23 - 24	Specifying query types 10 and 11 of #750 and #751 - code 2		F303
25 - 26	Specifying query types 10 and 11 of #750 and #751 - code 3		1008
...	Specifying query types 10 and 11 of #750 and #751 - code N		...
219 - 220	Specifying query types 10 and 11 of #750 and #751 - code 100		0000

Modbus (Hex)	Write / Read request	Write
C8	Function code	760
C9	Version number	0
CA	Production report ID (L)	1 to 200000
CB	Production report ID (H)	0: the latest production report
CC, CD	Reserved	-
CE	Request to send command	1 (to be filled in last)

2. Check whether the reading is successful from the return status and read the content of the sorted production report data.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	760	F802
3 - 4	Return data length	Unit: Word	6E00
5 - 6	Production report ID (L)	1 to 200000	0100
7 - 8	Production report ID (H)	0: the latest production report	0000
9 - 14	Reserved	6 bytes	...
15 - 16	Return function code	760	F802
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000
21 - 22	Corresponding data when code 1 is specified	Example: 0x6F corresponds to "Year" of the #750 table.	E907
23 - 24	Corresponding data when code 2 is specified	Example: 0x3F3 corresponds to "Running angle of stage 1" of query type 10 in #751.	5A00
25 - 26	Corresponding data when code 3 is specified	Example: 0x0810 corresponds to "1 st stage rotation speed" of query type 11 in #751.	6400
...	Corresponding data when code N is specified	...	...
219 - 220	Corresponding data when code 100 is specified	-	0000

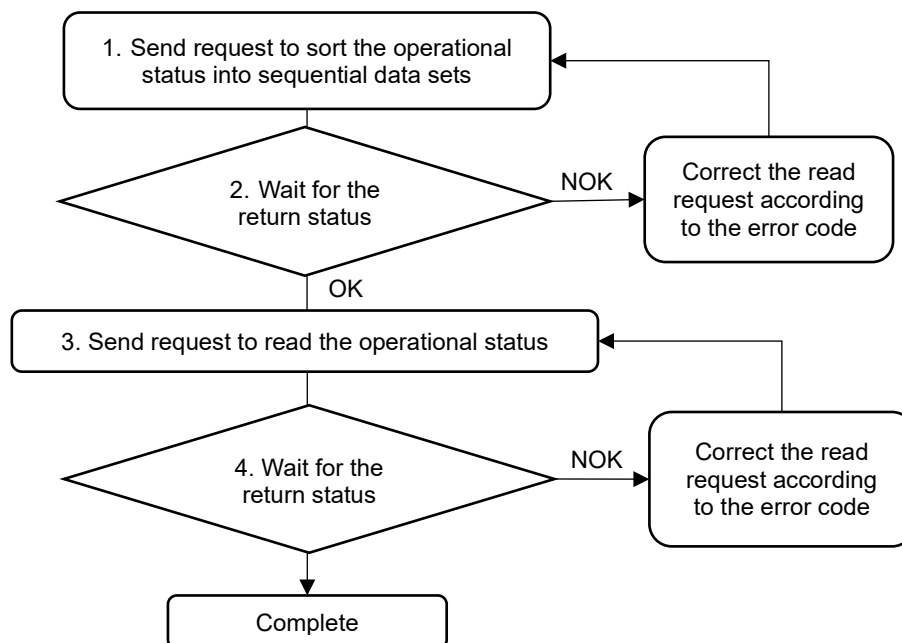
B.3.8 Others

#21 Write to sort operational status into sequential data sets

Content description:

Use function code #21 to sort the operating status in function code #50 into sequential data sets and acquire the corresponding values.

Handshake signal description:



1. Send request to sort the operational status into sequential data sets

Byte	Function	Description	Exame (Hex)
1 - 2	Function code	21	1500
3 - 4	Version number	-	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100
15 - 20	Reserved	6 bytes	...
21 - 22	Specifying the operational status - code 1	0: Off 1 - 600: refer to the operational status table in #50	D1
23 - 24	Specifying the operational status - code 2		D2
25 - 26	Specifying the operational status - code 3		DB
...	Specifying the operational status - code N		
219 - 220	Specifying the operational status - code 100		0000

2. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (hex)
1 - 2	Function code	21	1500
3 - 4	Return data length	Unit: Word	0A00
5 - 14	Reserved	10 bytes	...
15 - 16	Return function code	21	1500
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

3. Activate the subscription with function code #10 (version 1) or read the operation status with function code #50 (version 1)

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	50	3200
3 - 4	Version number	1	0000
5 - 12	Reserved	8 bytes	...
13 - 14	Request to send command	-	0100

4. Check whether the writing is successful from the return status.

Byte	Return status	Description	Example (Hex)
1 - 2	Function code	50	3200
3 - 4	Return data length	Unit: Word	6E00
5 - 14	Reserved	10 bytes	...
15 - 16	Return function code	50	3200
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	—	0000
21 - 22	Corresponding value when code 1 is specified	Example: 0xD1 corresponds to "Tool 1 final torque (L)" of the #50 table.	E903
23 - 24	Corresponding value when code 2 is specified	Example: 0xD2 corresponds to "Tool 1 final torque (H)" of the #50 table.	0000
25 - 26	Corresponding value when code 3 is specified	Example: 0xDB corresponds to "Tool 1 ready" of the #50 table.	0100
...	Corresponding value when code N is specified	—	...
219 - 220	Corresponding value when code 100 is specified	—	0000

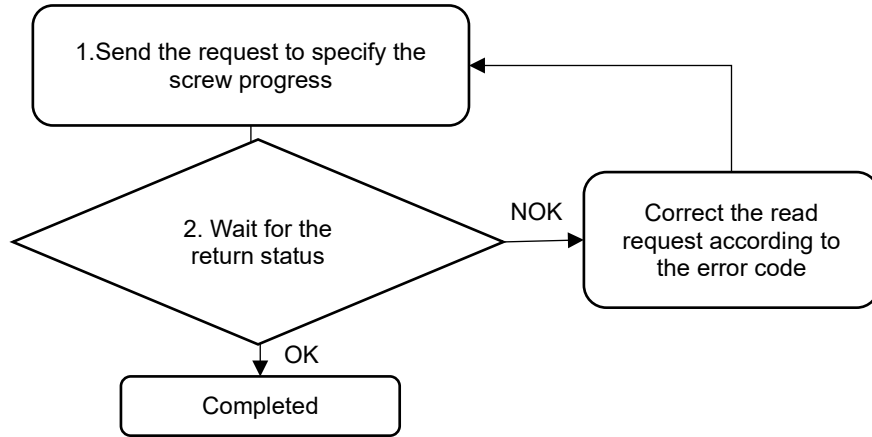
B

#1449 Write to specify the screw progress

Content description:

Use function code #1449 to jump to the tightening parameter of the specified screw.

Handshake signal description:



1. Send the request to specify the screw progress

Byte	Write / Read request	Description	Example (Hex)
1 - 2	Function code	1449	A905
3 - 4	Version number	-	0000
5 - 6	Tool 1 / Tol 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Specify screw progress ID (L)	-	0200
9 - 10	Specify screw progress ID (H)	-	0000
11 - 12	Reserved	-	0000
13 - 14	Request to send command	-	0100

2. Check whether the writing is successful from the return status.

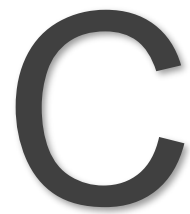
Byte	Return status	Description	Example (Hex)
1 - 2	Function code	1449	A905
3 - 4	Return data length	Unit: Word	0A00
5 - 6	Tool 1 / Tol 2	0: Tool 1; 1: Tool 2	0000
7 - 8	Specify screw progress ID (L)	-	0000
9 - 10	Specify screw progress ID (H)	-	0000
13 - 14	Reserved	-	0000
15 - 16	Return function code	1449	A905
17 - 18	Status	1: OK; 2: NOK	0100
19 - 20	Error code	-	0000

If successful, the read status is 1.

If failed, the read status is 2. The error codes are as follows.

Code	Error description	Description
1	Tool 1 / Tool 2	Must be 0 or 1
2	Specified screw progress ID	Exceeds the total screw quantity of the current operation

Positioning Arm



Follow the descriptions in this chapter to operate the positioning arm.

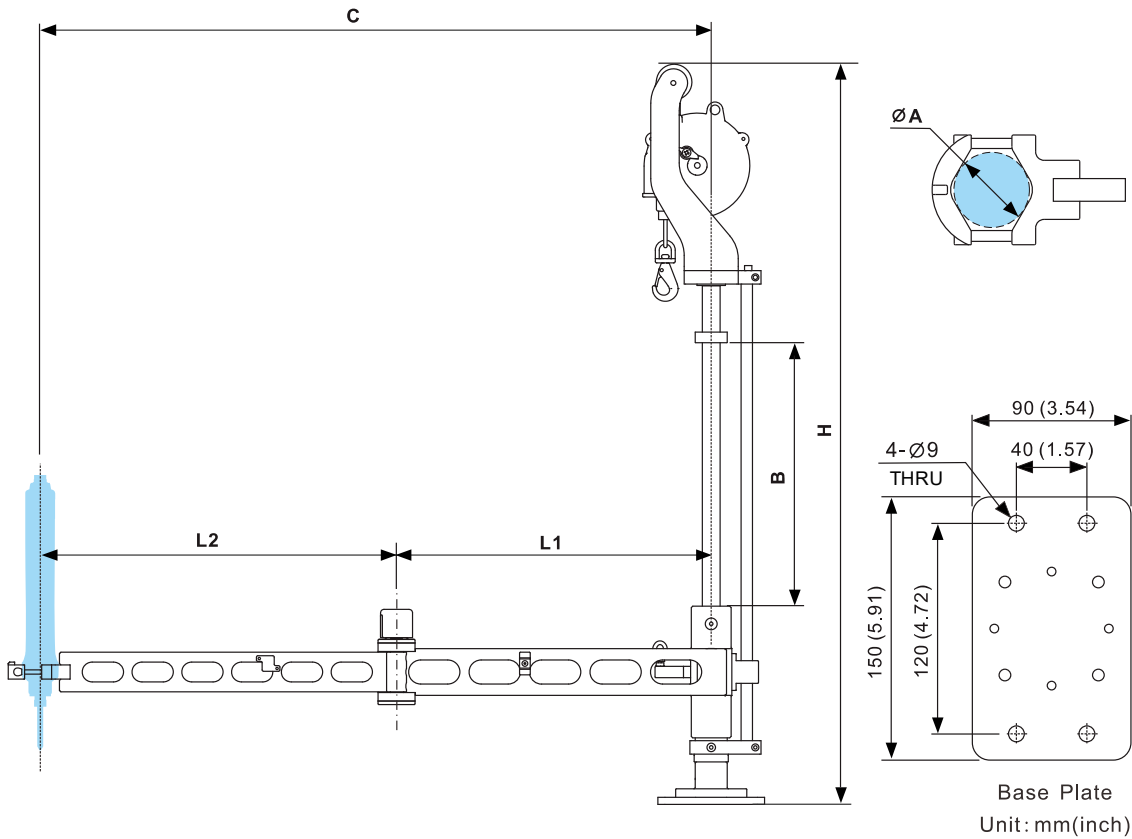


- C.1 Dimension drawings of positioning arms C-2
 - AM-TMTS-TA series C-2
 - P65F0 C-3
- C.2 Positioning arm cables and installation C-5
 - AM-TMTS-TA series C-5
 - P65F0 C-7
- C.3 Positioning arm settings and application C-8
- C.4 Positioning arm error codes C-14

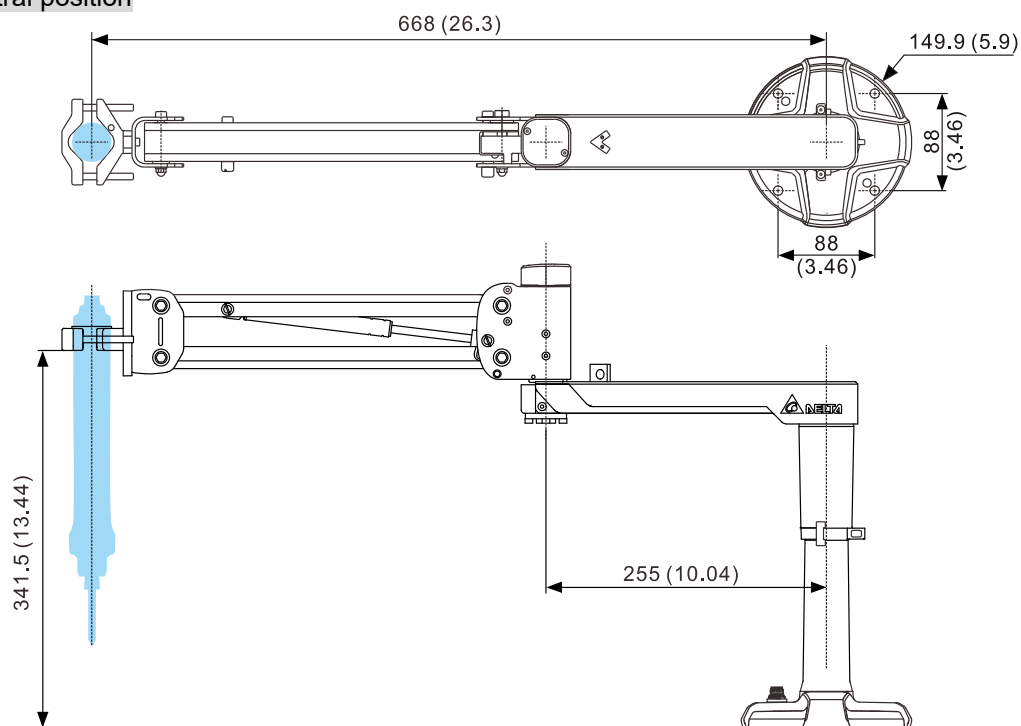
C.1 Dimension drawings of positioning arms

AM-TMTS-TA series

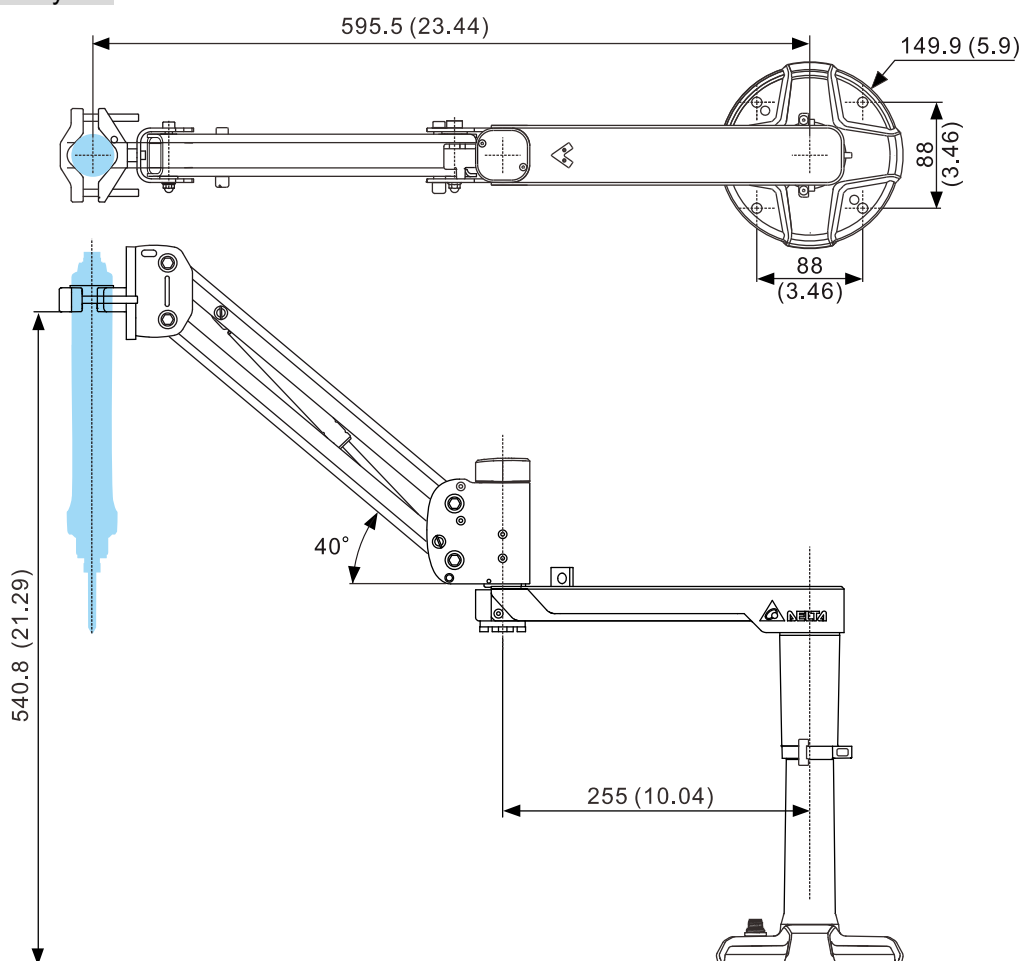
C



Specification	AM-TMTS-TA-S45B0	AM-TMTS-TA-S65B0	AM-TMTS-TA-S95B0
H	1045 mm		
ØA (tool diameter)	28 - 52 mm		
B (stroke)	400 mm		
C (max. working radius)	450 mm	650 mm	950 mm
L1	202.5 mm	312.5 mm	442.5 mm
L2	447.5 mm	337.5 mm	507.5 mm
Max. load	2 kg	1.7 kg	1.5 kg

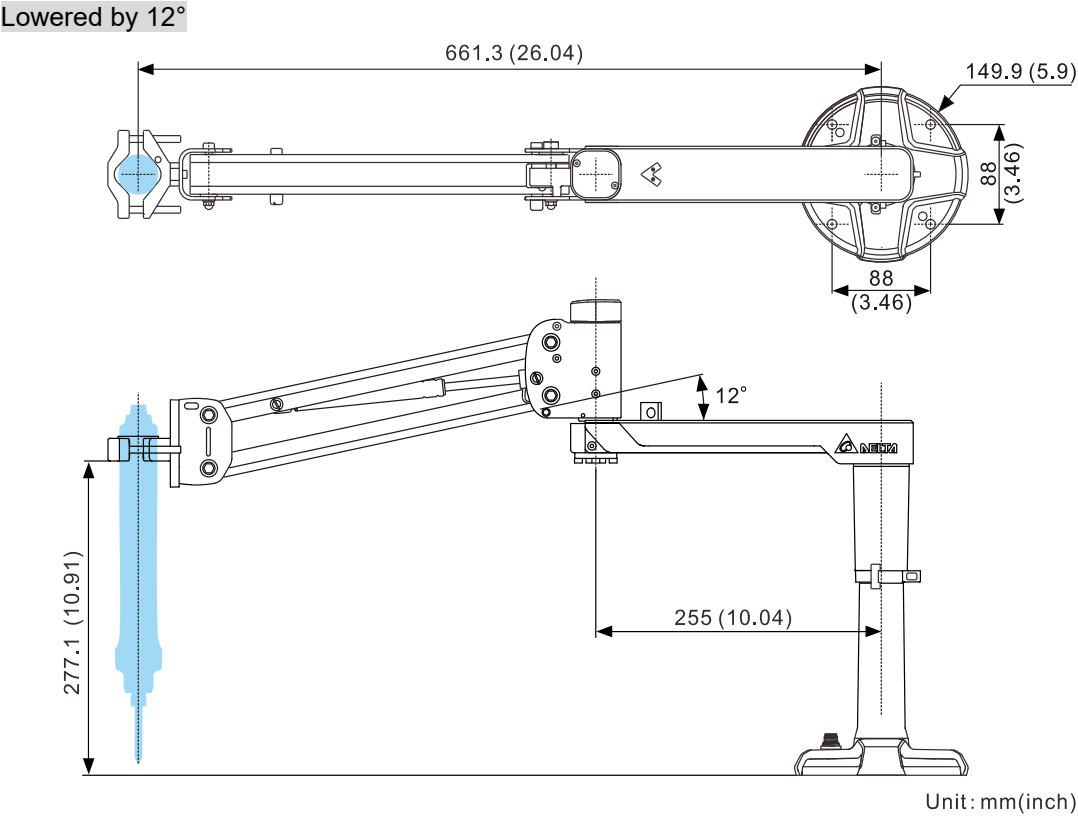
P65F0**Neutral position**

Unit: mm(inch)

Raised by 40°

Unit: mm(inch)

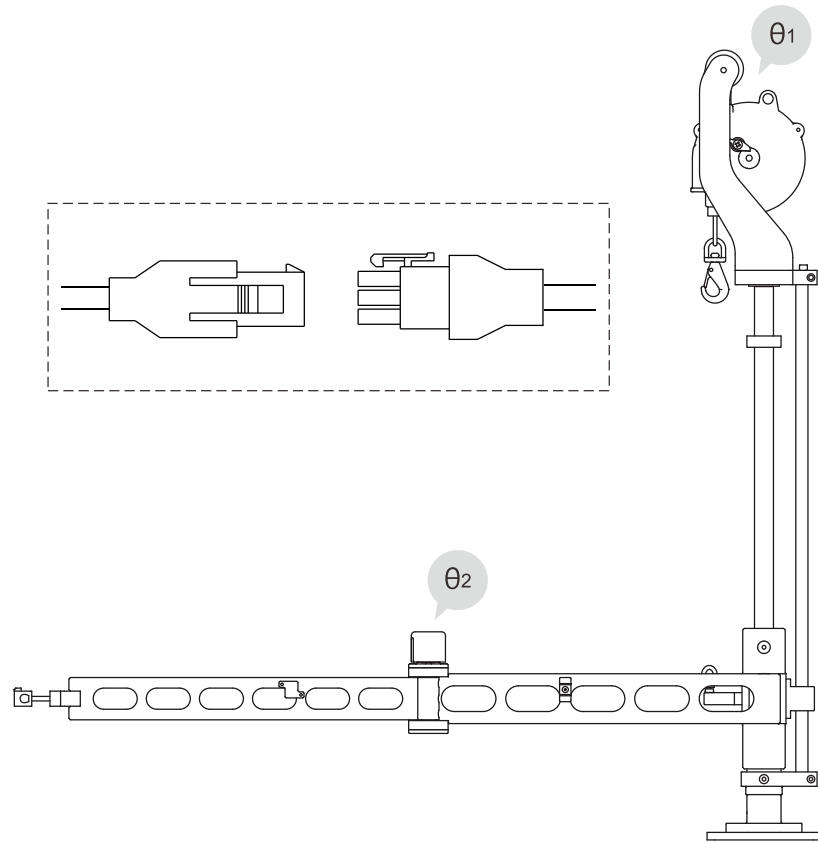
C



C.2 Positioning arm cables and installation

AM-TMTS-TA series

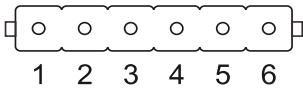
1. Connecting the encoders: connect the two encoders to $\theta 1$ and $\theta 2$ separately.



2. Connecting the controller:

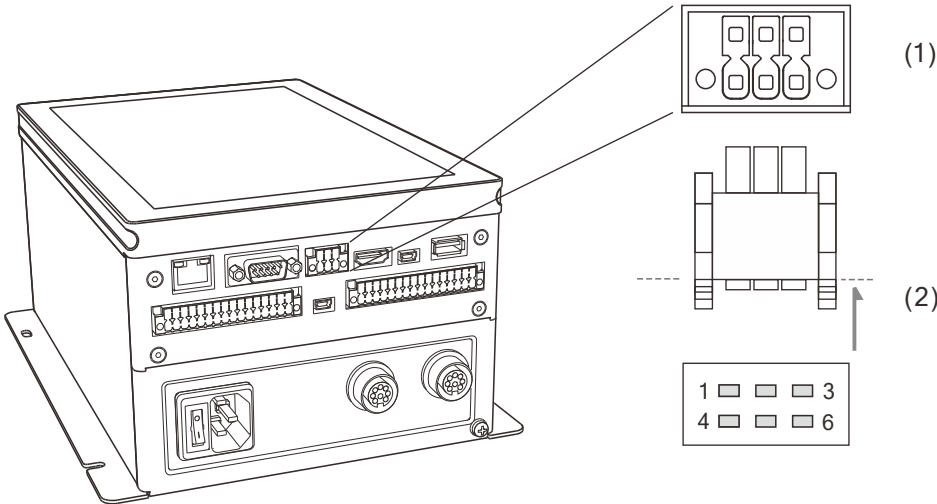
(1) Connect the positioning arm communication cable to the RS-485 port on the controller.

■ Pin assignment of positioning arm communication cable



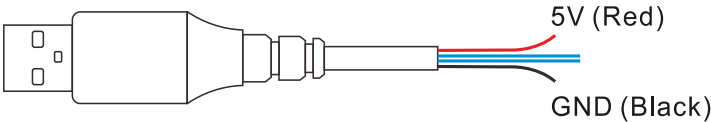
Pin	Signal	Description	Pin	Signal	Description
1	+5V	Input power 5V (+5V on the USB connector)	4	-	Reserved
2	GND	Ground of input power (GND on the USB connector)	5	D-	Differential signal Tx- / Rx- (connects to D1- on the controller)
3	-	Reserved	6	D+	Differential signal Tx+ / Rx+ (connects to D1+ on the controller)

■ Pin assignment of RS-485 port on the controller



Pin	Signal	Description	Pin	Signal	Description
1	D1-	RS485 signal Tx- / Rx-	4	D1+	RS485 signal Tx+ / Rx+
2	-	Reserved	5	-	Reserved
3	-	Reserved	6	-	Reserved

(2) Connect the positioning arm 5V power cable to the USB port on the controller:



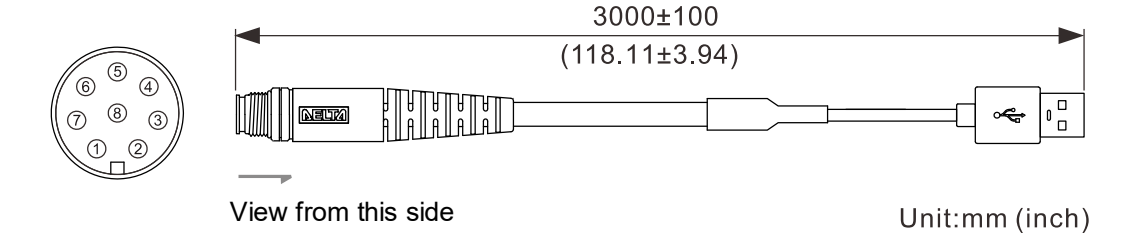
Pin	Signal	Description
1	+5V (red)	Output power (output current should at least be > 1 A)
2	-	Reserved
3	-	Reserved
4	GND (black)	Ground of output power

P65F0

1. Connecting the controller:

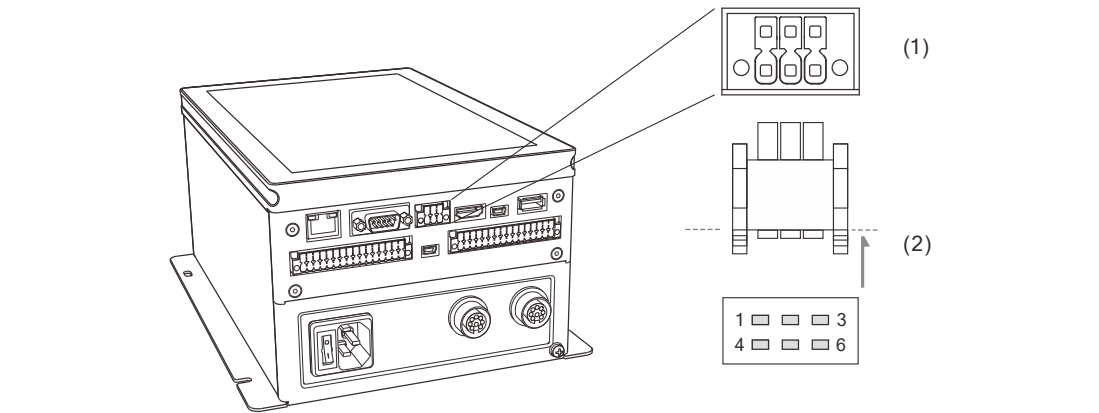
(1) Connect the positioning arm communication cable to the RS-485 port on the controller.

■ Pin assignment of positioning arm communication cable



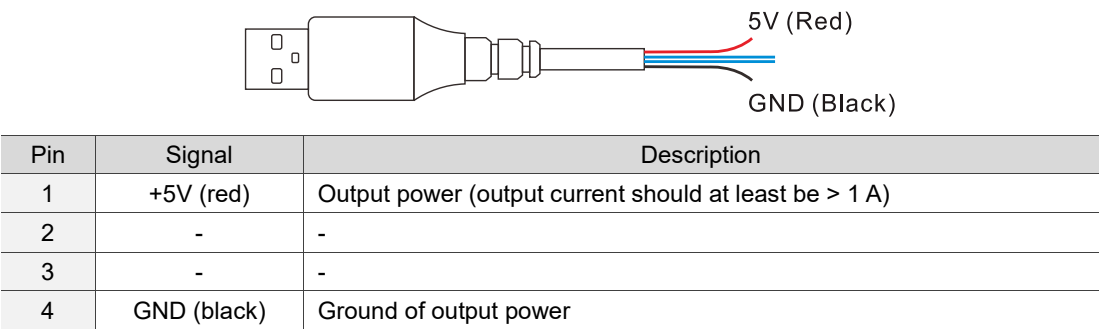
Pin	Signal	Description	Pin	Signal	Description
1	-	-	5	GND (gray / yellow & white)	Ground of input power (GND on USB connector)
2	D+ (blue / blue)	Differential signal Tx+ / Rx+ (connect to D1+ on the controller)	6	-	-
3	D- (yellow / blue & white)	Differential signal Tx- / Rx- (connect to D1- on the controller)	7	-	-
4	+5V (red / yellow)	Input power 5V (+5V on USB connector)	8	-	-

■ Pin assignment of RS-485 port on the controller



Pin	Signal	Description	Pin	Signal	Description
1	D1-	RS485 signal Tx- / Rx-	4	D1+	RS485 signal Tx+ / Rx+
2	-	Reserved	5	-	Reserved
3	-	Reserved	6	-	Reserved

(2) Connect the positioning arm 5V power cable to the USB port on the controller.



C.3 Positioning arm settings and application

1. Enable the Positioning arm:

AM-TMTS-TA series

Press **Controller > Peripheral Device > RS485_1**, and select **Tool1 Positioning Arm(AM)**.

User12024.05.1518:37

System SettingsDI/DOPeripheral DeviceCommunicationService Station



LANRS232RS485_1RS485_2HDMIHOST

DIO_1

Pin1: D-
Pin4: D+

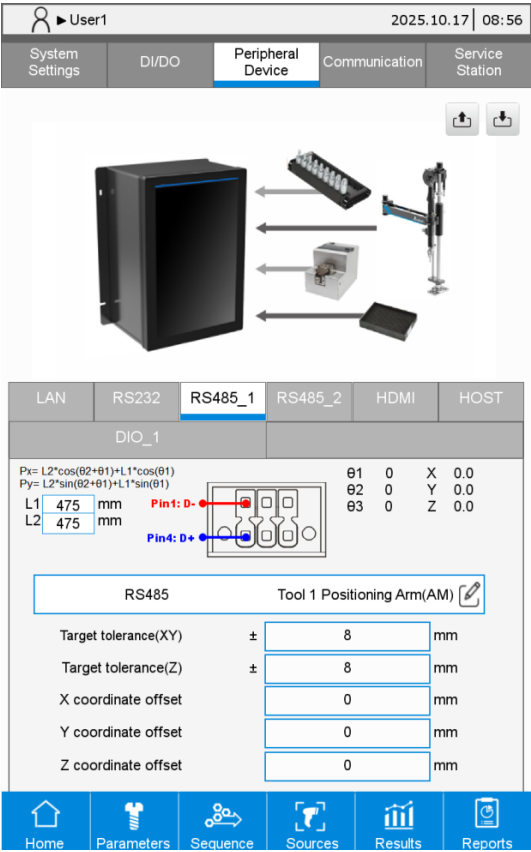
RS485

Disable

HomeParametersSequenceSourcesResultsReports

No.	Item
1	Disable
2	Tool1 Positioning Arm(P65F0)
3	Tool2 Positioning Arm(P65F0)
4	Tool1 Positioning Arm(AM)
5	Tool2 Positioning Arm(AM)

Cancel



Item	Description	Unit
Target tolerance(XY)	Positioning arm X- and Y-axis tolerance relative to the set coordinates	mm
Target tolerance(Z)	Positioning arm Z-axis tolerance relative to the set coordinate	mm
X coordinate offset	Positioning arm X-axis offset	mm
Y coordinate offset	Positioning arm Y-axis offset	mm
Z coordinate offset	Positioning arm Z-axis offset	mm
θ1	Angle recorded by Encoder 1 (0 to 360 degrees)	degree
θ2	Angle recorded by Encoder 2 (0 to 360 degrees)	degree
θ3	N/A	-
L1	Length of positioning arm L1 (default: 475 mm)	mm
L2	Length of positioning arm L2 (default: 475 mm)	mm
X	Coordinate value after conversion	mm
Y	Coordinate value after conversion	mm
Z	Coordinate value after conversion	mm

P65F0

Press **Controller > Peripheral Device > RS485_1**, and select **Tool1 Positioning arm(P65F0)**.

C

System Settings

DI/DO

Peripheral Device

Communication

Service Station

LAN

RS232

RS485_1

RS485_2

HDMI

HOST

DIO_1

Pin1: D+

Pin4: D+

RS485

Disable

Home

Parameters

Sequence

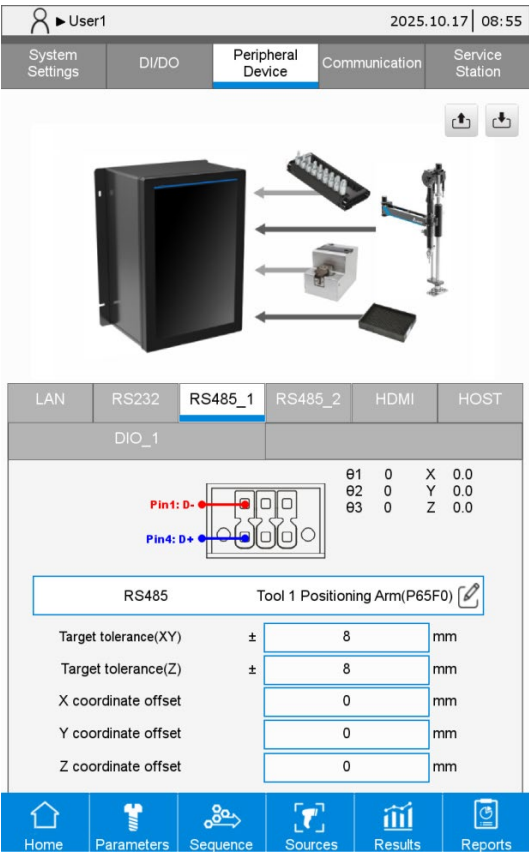
Sources

Results

Reports

No.	Item
1	Disable
2	Tool1 Positioning Arm(P65F0)
3	Tool2 Positioning Arm(P65F0)
4	Tool1 Positioning Arm(AM)
5	Tool2 Positioning Arm(AM)

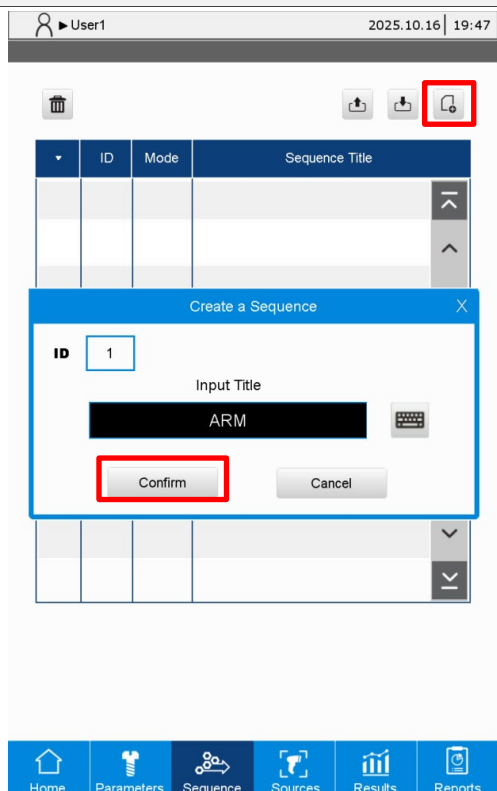
Cancel



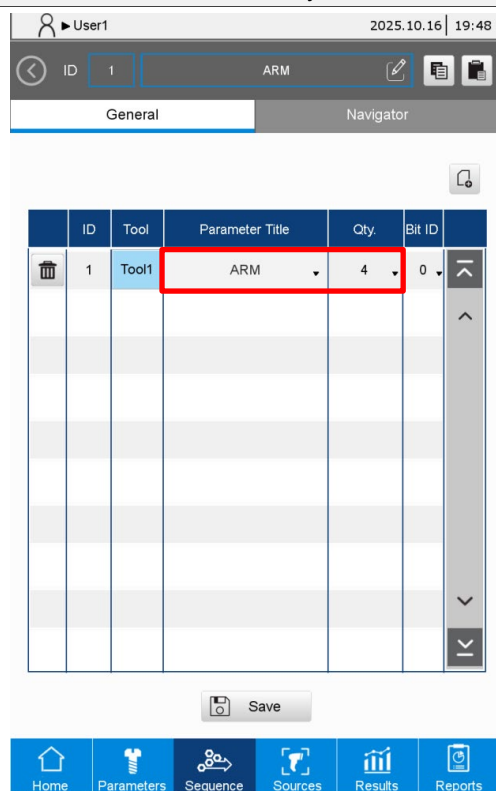
Item	Description	Unit
Target tolerance(XY)	Positioning arm X- and Y-axis tolerance relative to the set coordinates	mm
Target tolerance(Z)	Positioning arm Z-axis tolerance relative to the set coordinate	mm
X coordinate offset	Positioning arm X-axis offset	mm
Y coordinate offset	Positioning arm Y-axis offset	mm
Z coordinate offset	Positioning arm Z-axis offset	mm
θ1	Angle recorded by Encoder 1 (0 to 360 degrees)	degree
θ2	Angle recorded by Encoder 2 (0 to 360 degrees)	degree
θ3	Angle recorded by Encoder 3 (0 to 360 degrees)	-
X	Coordinate value after conversion	mm
Y	Coordinate value after conversion	mm
Z	Coordinate value after conversion	mm

2. Create a tightening sequence and edit the position coordinates of the positioning arm.

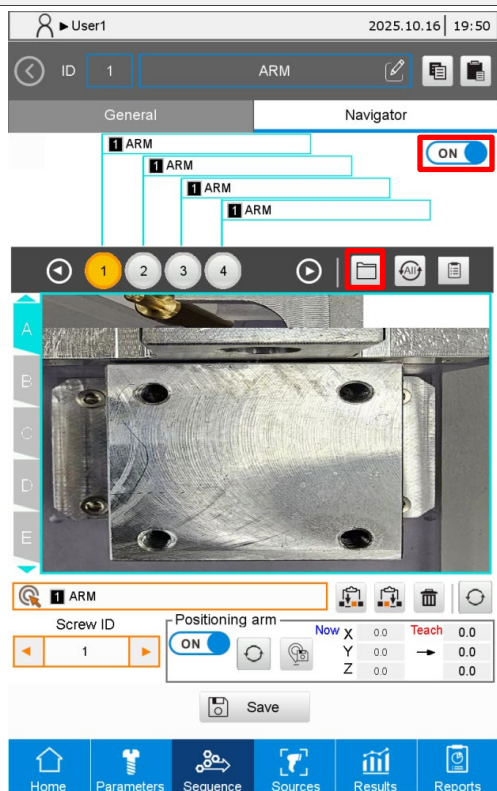
Step 1: create a new tightening sequence.



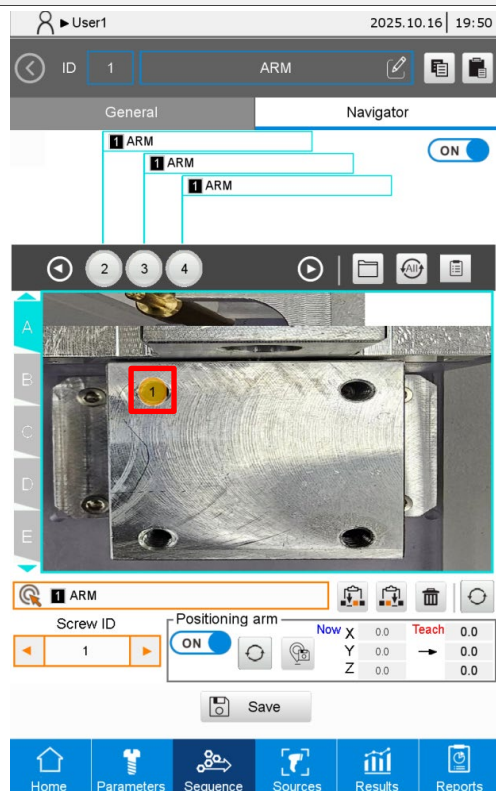
Step 2: specify the tightening parameter to execute and set the field of Qty.



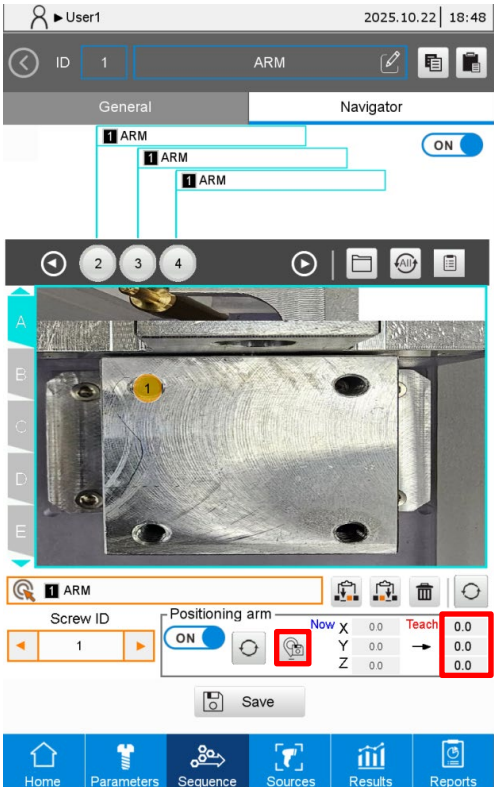
Step 3: set the Navigator function to ON and load the image.



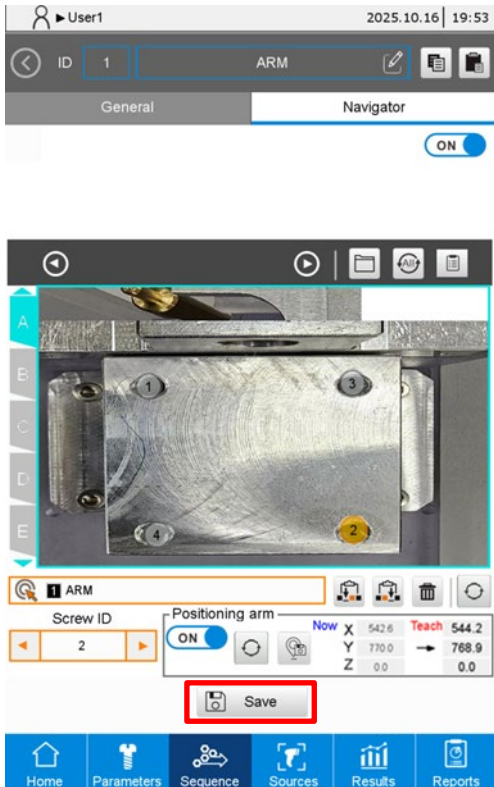
Step 4: set the navigator positions of the screw IDs.



Step 5: move the positioning arm to the screw hole and teach the point.

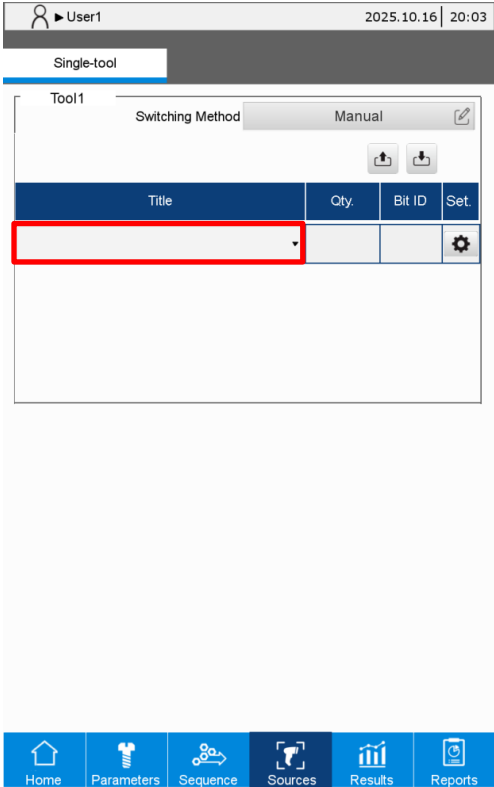


Step 6: teach all the points in sequence, and then save the tightening sequence.

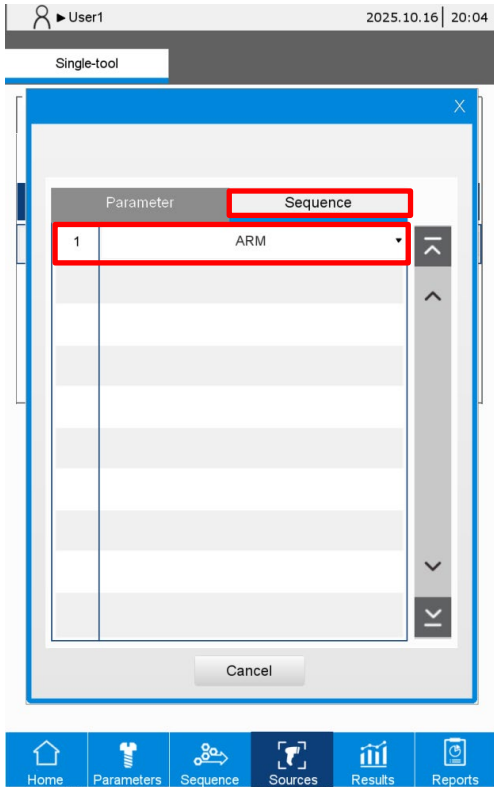


3. Go to the **Sources** page to execute the saved tightening sequence.

Step 1: select the tightening source.



Step 2: select the specific tightening sequence.



4. Go to the **Results** page to check the tightening result.

Step 1: go to the **Result** tab and press

Step 2: check the current and target position coordinates of the positioning arm.

C.4 Positioning arm error codes

WN5056 Tightening prohibited by remote control or DI

WN5008 Positioning arm interface is not configured

Revision History

Release date	Version	Chapter	Revision contents
March, 2026	V3.0 (Third edition)	1 2 3	Add the specifications of the medium torque models.
		5.3	Add the tightening strategy parameters and descriptions.
		6.2	Add the positioning arm setting function.
		7.2	Add the “Prohibit tool operation when scanner string length mismatch” item to the advanced setting of tightening source.
		8.1	Add functions to the Results tab items.
		9	Add functions in the System Settings and DI/DO tabs.
		10.1	Add the display item of “Tightening + Loosening Count (Total)”, which indicates the total tightening and total loosening counts of the smart screwdriver after shipment from the factory.
		11.2	Add the AL, NG, and Export t buttons. Added the button ID of “SystemRecord”.
		12	Update the descriptions: AL1004, AL1007, AL1013, AL1088, AL1888, AL2004, AL2007, AL2013, AL2088, AL2888, NG3002, NG4002, WN5004, WN5007, WN5052, WN5057, WN5058, WN5086, WN6004, WN6007, WN6052, WN6057, WN6058, WN6086 Add the alarms: AL1111 - AL1115, AL1896, AL1897, AL1900 - AL1903, AL2111 - AL2115, AL2896, AL2897, AL2900 - AL2903, AL1C31, AL2C31, NG3003, NG3114, NG3115, NG3214, NG3215, NG3314, NG3315, NG3414, NG3415, NG3514, NG3515, NG3981, NG3991, NG3A14, NG3A15, NG3B14, NG3B15, NG3C14, NG3C15, NG3D14, NG3D15, NG3E14, NG3E15, NG3F14, NG3F15, NG4003, NG4114, NG4115, NG4214, NG4215, NG4314, NG4315, NG4414, NG4415, NG4514, NG4515, NG4981, NG4991, NG4A14, NG4A15, NG4B14, NG4B15, NG4C14, NG4C15, NG4D14, NG4D15, NG4E14, NG4E15, NG4F14, NG4F15, WN5008 - WN500F, WN5059, WN505A, WN5087, WN5088, WN6008 - WN600F, WN6059, WN605A, WN6087, WN6088

Release date	Version	Chapter	Revision contents
March, 2026	V3.0 (Third edition)	A B	Update the function descriptions: #100, #200, #202, #301 - #303, #507, #509, #510, #512, #513, #517, #521, #522, #555, #556, #566, #606, #607, #650, #655, #701, #750, #751 Add the function codes: #21, #101, #160, #260, #530 - #542, #574 - #588, #608, #660, #661, #760, #1449
		B	Add the function codes: #10, #11, #50, #60
		C	Add the chapter: Positioning Arm
March, 2024	V2.0 (Second edition)		Optimize the page layout and the descriptions. Modify the terms: The term “smart servo screwdriver system” is changed to “smart screwdriving system”. The term “servo screwdriver controller” is changed to “smart screwdriving controller”. The term “servo screwdriver” is changed to “transducerized smart screwdriver”. Chapter rearrangement: delete the chapter “Modbus Communications”, and renumber the chapter “Troubleshooting” as CH12. Add the 75 kgf-cm smart screwdriver.
		Preface	Add the precaution for Operation: do not hit the tool.
		1.5	Modify the description for Status LED indicator.
		2.6	Add the precaution for Others: suggestion for using the screwdriver with a torque arm.
		2.7	Modify the description for maintenance cycle and measures.
		3	Delete the RS232 section.
		3.2.1	Add the note for the I/O connector pin assignment of the single-tool controller.
		5.3.1	Add the function descriptions for Advanced Setting.
		5.3.2.1 5.3.2.4	Add the Clamp Torque Control and Clamp Angle Control. Add the Two-stage mode.
		8	Add the chapters 8.3 HDMI output, 8.4 VNC remote connection.

Release date	Version	Chapter	Revision contents
March, 2024	V2.0 (Second edition)	9	Add the functions: Export / Import Two-stage Mode under Self-defined Torque Control Display the Limits of All Stages for Curves Display Operation Warning Window Export Result File for Each Screw Send Result TCP for Each Screw Sampling Rate for Curves Always Monitor the Tool Current Compensation for Tool Temperature Rise Controller Temperature
		9.3	Modify the description for Peripheral Device.
		10.2	Modify the description for Lever Start Level and Push Start Level.
		11.4	Add the chapter: Button Report
		12	Change the alarm display of tightening errors from NG1xxx to NG3xxx and operation warnings from WN1xxx to WN5xxx for Tool 1. Add the alarms for Tool 2. Add the device alarms for Tool 1: AL1016, AL1037 - AL1039, AL1083, AL1088, AL1090 - AL1095, AL1887, AL1888, AL1890 - AL1895. Add the tightening errors for Tool 1: NG3011, NG3426 - NG3429, NG3A26 - NG3A29, NG3B26 - NG3B29, NG3C26 - NG3C29, NG3D26 - NG3D29, NG3E26 - NG3E29, NG3F26 - NG3F29. Add the operation warnings for Tool 1: WN5006, WN5007, WN5058. Modify the descriptions: NG3013 & NG4013, NG3035 & NG4035, NG3881 & NG4881.
		A	Add the function codes: #110, #201 - #203, #210, #251 - #253, #302, #303, #310, #409 - #412, #453, #511 - #529, #557 - #573, #657 - #659, #702, #754. Modify the function descriptions: #100, #200, #401, #451, #507, #607, #656, #750, #751, #752, #753.
April, 2022	V1.0 (First edition)	B	Add the chapter: TCP/IP Communication - Function Codes

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